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Investigation of Torque-Angle Characteristics of Synchronous Generators for Transient Stability Analysis Using Equal Area Criterion

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Abstract—**Equal area criterion (EAC) has an equivalent concept to Lyapunov's energy function method and has been widely used for analyzing the transient behavior of synchronous generators (SGs) in power systems. This paper investigates various torque-angle characteristics of SGs that can be utilized along with the EAC approach for transient stability analysis (TSA). In addition to the two well-known transient characteristics that have been commonly used in the literature, a new transient torque-angle characteristic for SGs is introduced which uses transient reactance values in both *d* and *q* axes. Then, the error of EAC method in the calculation of critical clearing angle (CCA) and critical clearing time (CCT) is determined with different torque-angle characteristics for SGs, using computer simulations of a single-machine infinite-bus (SMIB) system. The calculated CCA and CCT values are compared and the accuracy of the subject characteristics for TSA are discussed.**

Keywords—Critical clearing angle (CCA), critical clearing time (CCT), equal area criterion (EAC), torque-angle characteristic, transient stability analysis (TSA).

I. Introduction

Power system stability is traditionally classified as 1) rotor angle, 2) voltage, and 3) frequency stability [1]. The rotor angle stability is categorized as small-signal and large-signal stability [1]. Large-signal stability, also called transient stability, refers to the ability of a power system to regain its synchronism when subjected to a severe disturbance such as a short-circuit fault on machine terminals or transmission lines [1], [2]. Transient stability analysis (TSA) deals with the ability of synchronous generators (SGs) to maintain the equilibrium between their input mechanical torque and their output electromagnetic torque [1], [2]. This can be conducted using either time-domain simulations or direct methods [3], [4]. In time-domain simulations, the dynamic response to a specific disturbance is obtained through a step-by-step integration of nonlinear equations describing the system behavior from an initial operating point [5]. Although time-domain simulations can provide very accurate response, solving the nonlinear differential equations, especially in large-scale power systems, is computationally expensive and also cannot provide enough insight into the stability margins of the system [5], [6]. Therefore, direct methods have been developed in the literature to achieve faster TSA by avoiding computation of the nonlinear differential equations and can also provide quantitative stability margins [3], [6].

At the basis of direct methods, also referred to as energy-based direct methods [7], is Lyapunov's energy function [1], [8]. Therein, the energy function is first formulated based on the summation of kinetic and potential energies, which is then compared to a critical energy threshold to evaluate the stability status [8]. Generally, establishing the energy functions is not straightforward for large-scale power systems [8]. The equal area criterion (EAC) is an equivalent approach in TSA to the energy function method, which also graphically represents the concept behind the energy functions [4], [9]. The main advantage of EAC is the fast prediction of transient stability margins in a power system with very limited computations [3], [9]. Two pivotal outputs of the TSA using the EAC are namely the critical clearing angle (CCA) and critical clearing time (CCT) [10], [11]. The CCT is defined as the maximum time duration between the instant when a fault occurs and the moment that it has to be cleared from the grid, such that the system stays transiently stable [1]. Also, the rotor angle at the instant of CCT is called CCA [10], [11]. In TSA, the CCT is an important index as it can be used to determine the time settings for protection relays and circuit breakers to be able to isolate faults before the power system becomes unstable [11]. Also, due to the wide application of the synchrophasor measurements, the pre-calculated CCAs can be employed as a warning level on the angle/time that the transmission system operators (TSOs) have to clear the fault for preventing instability [12].

Despite its merits, the EAC is typically restricted to the evaluation of transient stability of a single generator connected to an infinite bus, i.e., single-machine infinite-bus (SMIB) systems [3], [9]. However, the EAC has been extended to multi-machine power systems [9], [13]. In the extended EAC (EEAC), the generators are categorized (grouped) as stable and unstable, ultimately leading to a SMIB equivalent model for the multi-machine power systems [9]. It should also be noted that the EAC approach is inherently an approximate method for TSA [3], [10] and its accuracy highly depends on the accuracy of the torque-angle characteristic that is used along with it. Since the transient torque-angle characteristic of SGs differs markedly from the steady-state characteristic [10], obtaining an accurate transient torque-angle characteristic sometimes remains an open question, which directly impacts the accuracy of EAC method.

In this paper, two transient torque-angle characteristics that have been widely used in the literature are analyzed. Also, a new transient torque-angle characteristic is introduced where both d- and q-axes transient reactances of SGs are used in establishing the approximate transient equation of the electromagnetic torque. The performance of EAC along with these transient torque-angle characteristics is evaluated and the results are compared with the time-domain simulations. It is shown that using the new transient torque-angle characteristic with a corrected initial rotor angle (obtained from the pre-fault steady-state condition) can improve the accuracy of EAC method in calculating the CCA. However, calculation of the CCT using all the subject characteristics always accompanies some errors.

II. Torque-Angle Characteristics of SGs

To assess the EAC performance along with different torque-angle characteristics, a SMIB system is considered, wherein a synchronous generator is directly connected to an infinite bus. It is also assumed that the rotor variables are referred to the stator side, and the rotor reference frame is considered. Similar notations as in [10, Ch. 5] are assumed in this paper, except removing the primes for referred variables and superscript "r" for denoting the rotor reference frame, for simplicity. Therefore, the primes are used to denote the transient variables only.

A. Steady-State Torque-Angle Characteristic

The response of the power system to large disturbances is mainly affected by the nonlinear power/torque-rotor angle relationship of SGs [3]. In a stable steady-state condition, the electromagnetic torque of SGs should remain equal and opposite to the delivered mechanical torque. Accordingly, the transient stability is assessed via the ability of SGs to maintain this equilibrium and the convergence of their rotor angles to a stable operating point in post-fault [2], [3].

Here, a full-order model of SGs as described in [10, Ch. 5] is considered. It is also assumed that the generator speed is (roughly) equal to the synchronous speed; and the stator resistance is neglected. The steady-state electromagnetic torque characteristic (SSTAC) of the SG can be expressed in per unit as [10, Ch. 5]

$$T_e = \frac{|E_{xfd}||V_s|}{X_d}\sin(\delta) + \frac{|V_s|^2}{2}\left(\frac{1}{X_q} - \frac{1}{X_d}\right)\sin(2\delta). \tag{1}$$

The excitation voltage E_{xfd} in (1) is defined as

$$E_{xfd} = X_{md} I_{fd}, \tag{2}$$

and V_s and δ are the terminal voltage and the rotor angle of SG, respectively. Also, X_d and X_q are the d- and q-axes reactances and I_{fd} is the field current. X_{md} is the d-axis magnetizing reactance and can be calculated as

$$X_{md} = X_d - X_{ls}, \tag{3}$$

where X_{ls} is the stator leakage reactance. As seen in (1), the SSTAC includes both $\sin(\delta)$ and $\sin(2\delta)$ terms, where the latter specifies the contribution to the torque due to rotor saliency.

B. Conventional Transient Torque-Angle Characteristics

In case of a sudden disturbance to an SG, the instantaneous torque-angle characteristic of the SG differs significantly from the SSTAC [10]. Deriving an analytical expression for the actual torque of an SG with respect to the rotor angle during transients is cumbersome. Although, this can be done similarly to the SSTAC using some simplifications/approximations [10].

In many power system studies, the so-called classical model is considered in the dynamic modeling of SGs and first-swing stability analysis [8]. Thereon, it is assumed that SG consists of a d-axis transient reactance (X'_d) and a voltage source behind the transient reactance (E'). During transients, the magnitude of E' is assumed to remain constant; however, the rotor angle (δ), which represents the angle of E', will change as the rotor speed deviates from the synchronous speed [3]. In the classical model, the transient electromagnetic torque of the SG is expressed in per unit as [3, Ch. 13]

$$T_e = \frac{|E'||V_s|}{X'_d}\sin(\delta). \tag{4}$$

As seen, the electromagnetic transient torque T_e in (4) is determined only based on a $\sin(\delta)$ term. In this paper, (4) is referred to as reduced approximate transient torque-angle characteristic (RATTAC), as it only contains the $\sin(\delta)$ term and neglects the dynamic saliency of the machine.

Another transient torque-angle characteristic has also been proposed in [14] which assumes that the machine circuit is mainly inductive (due to small resistances of large generators); and subsequently, the flux linkages will remain constant in the early period of transient [10, Ch. 5]. Additionally, it is assumed that fast transients (i.e., subtransients) are neglected; and subsequently, the action of the damper windings is ignored. As a result, the d-axis flux linkages per second during transient period are expressed as [10, Ch. 5]

$$\psi_{ds} = X_d I_{ds} + X_{md} I_{fd}, \tag{5}$$

$$\psi_{fd} = X_{fd} I_{fd} + X_{md} I_{ds}, \tag{6}$$

where ψ_{ds} and I_{ds} are the stator d-axis flux linkage per second and stator current, and ψ_{fd} and X_{fd} are the field winding flux linkage per second and reactance, respectively. Substituting I_{fd} from (6) into (5), the stator d-axis flux linkage per second can be obtained as [10, Ch. 5]

$$\psi_{ds} = \left(X_d - \frac{X_{md}^2}{X_{fd}}\right) I_{ds} + \frac{X_{md}}{X_{fd}}\psi_{fd}. \tag{7}$$

Then, the d-axis transient reactance (X'_d) and the q-axis voltage behind transient reactance (E'_q) are defined as [10, Ch. 5]

$$X'_d = X_d - \frac{X_{md}^2}{X_{fd}}, \tag{8}$$

$$E'_q = \frac{X_{md}}{X_{fd}}\psi_{fd}. \tag{9}$$

Comparing (7) with the steady-state d-axis flux linkage per second that is described as [10, Ch. 5]

$$\psi_{ds} = X_d I_{ds} + E_{xfd}\,, \tag{10}$$

one can formulate the respective torque-angle characteristic based on (1) by replacing X_d with X'_d and E_{xfd} with E'_q as [10, Ch. 5]

$$T_e = \frac{|E'_q||V_s|}{X'_d}\sin(\delta) + \frac{|V_s|^2}{2}\left(\frac{1}{X_q} - \frac{1}{X'_d}\right)\sin(2\delta)\,. \tag{11}$$

In this paper, (11) is referred to as approximate transient torque-angle characteristic (ATTAC) which includes both sin(δ) and sin(2δ) terms. It should also be noticed that even in a round rotor SG, the sin(2δ) term may still appear in the ATTAC (because X'_d is not equal to X_q necessarily); although $X_d = X_q$ and the SSTAC would not have any sin(2δ) saliency term.

C. *Whole Transient Torque-Angle Characteristic*

Here, a new transient torque-angle characteristic is introduced where transient values are used for the reactance in both *d* and *q* axes. Specifically, X'_d and X'_q are used simultaneously, i.e., X_q is replaced by X'_q in the ATTAC formula (11). The rationale is that during transients, one should use transient values for machine parameters. Specifically, as mentioned in [10, Ch. 7], one of the *q*-axis damper windings, i.e., *kq1* winding, is associated with the transient reactance and the other one, i.e., *kq2* winding, is associated with the subtransient reactance. Therefore, the field *fd* winding and *kq1* winding contribute in *d*- and *q*-axes transient reactances, respectively. One can formulate the *d*- and *q*-axes transient reactances as [10, Ch. 7]

$$X'_d = X_{ls} + \frac{1}{\frac{1}{X_{md}} + \frac{1}{X_{lfd}}} = X_d - \frac{X_{md}^{\;2}}{X_{fd}}\,, \tag{12}$$

$$X'_q = X_{ls} + \frac{1}{\frac{1}{X_{mq}} + \frac{1}{X_{lkq1}}} = X_q - \frac{X_{mq}^{\;2}}{X_{kq1}}\,, \tag{13}$$

where X_{lfd} and X_{lkq1} are the field winding and *kq1* winding leakage reactances, respectively. Also, X_{mq} is the *q*-axis magnetizing reactance and can be calculated as

$$X_{mq} = X_q - X_{ls}\,. \tag{14}$$

It is noted that the middle expressions in (12) and (13) can be verified using the equivalent circuit of a three-phase synchronous machine in rotor reference frame [10, Ch. 5]. Here, by replacing X_q with X'_q in the ATTAC (11), a whole transient torque-angle characteristic (WTTAC) is obtained as

$$T_e = \frac{|E'_q||V_s|}{X'_d}\sin(\delta) + \frac{|V_s|^2}{2}\left(\frac{1}{X'_q} - \frac{1}{X'_d}\right)\sin(2\delta)\,. \tag{15}$$

It is worth mentioning that the WTTAC also includes both sin(δ) and sin(2δ) terms even for a round rotor SG (because X'_d is not equal to X'_q necessarily).

D. *Critical Clearing Angle and Time Calculation*

To visualize how the CCA and/or CCT are calculated with the EAC approach, typical SSTAC and ATTAC curves are shown in Fig. 1. Therein, T_m is the input mechanical torque of the SG and δ_o specifies the initial rotor angle corresponding to the pre-fault steady-state characteristic (i.e., SSTAC), wherein $T_e = T_m$. Once a fault occurs, the output torque of the SG shifts to the "on-fault" torque characteristic where the rotor accelerates and its angle increases to δ_c until the fault is (critically) cleared.

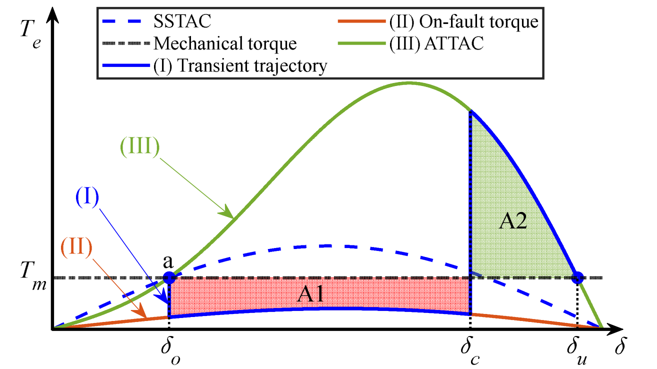

Fig. 1. Typical SSTAC and ATTAC curves for calculation of CCA and CCT using the EAC technique.

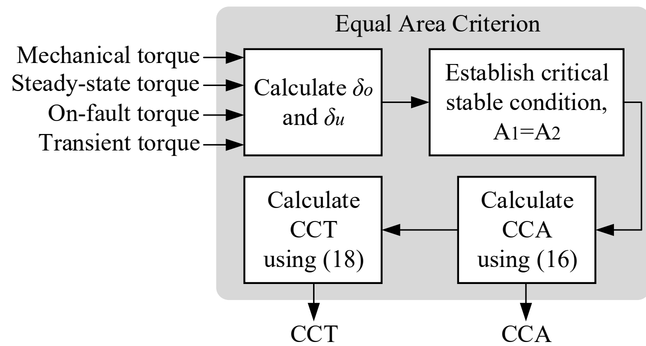

Fig. 2. Block diagram of calculating CCA and CCT with EAC approach.

After clearing the fault, the SG outputs torque according to the transient characteristic (i.e., ATTAC), and starts to decelerate while the rotor angle still increases to δ_u. The areas corresponding to the acceleration and deceleration periods are demonstrated by A_1 and A_2 in Fig. 1, where A_1 corresponds to the excessive kinetic energy that is stored in the SG during the fault and A_2 represents the energy that is released by the SG after clearing the fault. In a (critical) stable operation $A_1 = A_2$; therefore, in the EAC technique, the CCA is calculated by solving the following equation [8, Ch. 9]

$$\int_{\delta_o}^{\delta_c} (T_m - T_e^{on-fault})d\delta = \int_{\delta_c}^{\delta_u} (T_e^{ATTAC} - T_m)d\delta\,. \tag{16}$$

The analytical solution of (16) leads to a nonlinear equation for the desired angle δ_c (i.e., CCA). After calculating CCA, the CCT is obtained as follows. The swing equation for the SG is expressed as

$$\frac{2H}{\omega_b}\frac{d^2\delta}{dt^2} = T_m - T_e\,, \tag{17}$$

where H is the inertia constant and ω_b is the base (nominal) speed. Integrating both sides of (17) during the acceleration period yields [10, Ch. 5]

$$\delta_c - \delta_o = \frac{\omega_b}{2H}\int_{t=0}^{t_c}\int_{\tau=0}^{\tau}(T_m - T_e^{on-fault})dt\cdot d\tau\,. \tag{18}$$

Specifically, if $T_e^{on\text{-}fault} = 0$, the CCT (i.e., t_c in (18)) can be calculated as [10, Ch. 5]

$$t_c = \sqrt{\frac{4H(\delta_c - \delta_o)}{\omega_b T_m}}. \tag{19}$$

A generic block diagram for calculating CCA and/or CCT using the EAC approach is depicted in Fig. 2.

III. Computer Studies

Here, the accuracy of different torque-angle characteristics, discussed in Section II, is investigated when adopted for calculating CCA/CCT with the EAC approach. As the reference solution, time-domain simulations are also conducted using a full-order model of the SG in *dq* frame [10, Ch. 5]. For the TSA, a steam-turbine SG as well as a hydro-turbine SG are considered, with parameters summarized in the Appendix [also, transient reactances of the SGs are calculated using (12)–(13)]. It is assumed that the subject SG is connected to an infinite bus and initially operates at its nominal conditions (i.e., the rated MVA and power factor). Then at t = 1 s, a three-phase short-circuit occurs at the terminals of the SG; and its stability is investigated. It is also noted that the input mechanical torque (T_m) and excitation voltage (E_{xfd}) are held constant. The nominal values are T_m = 0.85 p.u. and E_{xfd} = 2.48 p.u. for the steam-turbine SG, and T_m = 0.85 p.u. and E_{xfd} = 1.6 p.u. for the hydro-turbine SG [10, Ch. 5].

A. TSA Results for the Steam-Turbine SG

Scenario–I: Here, it is assumed that the terminal voltage of the steam-turbine SG becomes zero due to the fault, and consequently, T_e = 0 during the on-fault period. Using time-domain simulations, the CCT for this scenario is found to be 0.357 s. The simulation results for a marginally stable case (fault removed after 0.357 s), and an unstable case (fault removed after 0.358 s) are shown in Figs. 3–4, respectively.

The torque-angle characteristics for the first swing of the SG are illustrated in Fig. 5 as obtained by detailed simulations as well as the subject analytical methods explained in Section II. As seen in Fig. 5, the analytical curves are aimed to approximate the average-value of the actual transient torque (obtained from the time-domain simulations) after the fault is cleared, while during fault, it is known that T_e = 0 and the curves are not used.

Sine the EAC method has been adequately demonstrated in the literature when used with ATTAC and RATTAC, here the EAC is only demonstrated with the new WTTAC as shown in Fig. 6. As opposed to the ATTAC and RATTAC that the initial rotor angles are obtained from themselves [3, Ch. 13], [10, Ch. 5], in the new WTTAC the initial rotor angle (i.e., δ_o) is calculated based on the SSTAC, as illustrated in Fig. 6 (i.e., point 'a'). The rationale is that before applying the fault, the SG was operating in steady-state; therefore, the SSTAC curve should be used for calculating the initial operating point.

The TSA results for the steam-turbine SG in scenario–I obtained from time-domain simulations as well as the EAC approach in conjunction with the subject torque-angle characteristics are summarized in Table I. Here, the results from detailed time-domain simulations are considered to be the reference.

As it can be seen in Table I, the new WTTAC and ATTAC

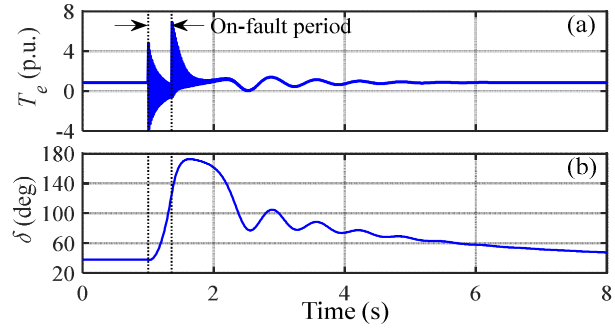

Fig. 3. Time-domain simulation results of the steam-turbine generator when the fault is cleared after 0.357 s for: (a) electromagnetic torque, and (b) rotor angle. The SG remains stable.

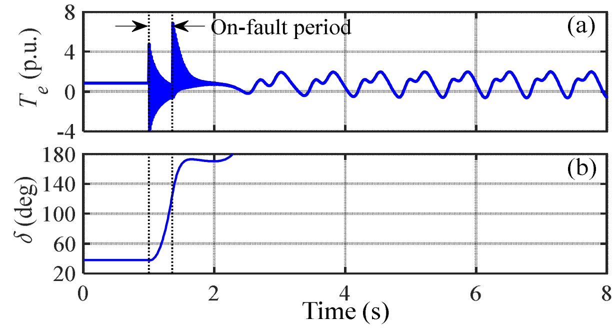

Fig. 4. Time-domain simulation results of the steam-turbine generator when the fault is cleared after 0.358 s for: (a) electromagnetic torque, and (b) rotor angle. The SG becomes unstable.

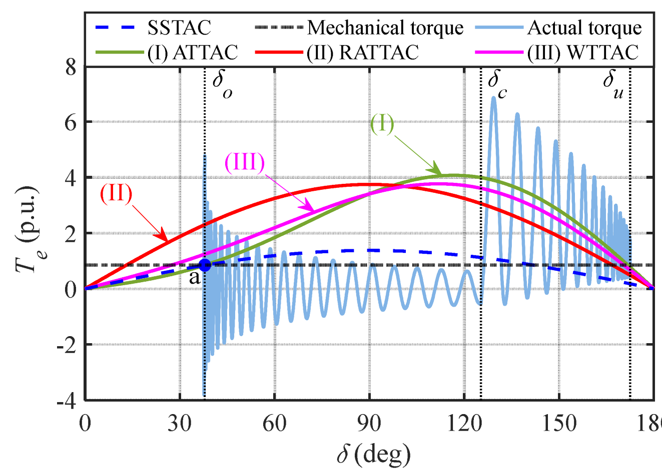

Fig. 5. Transient torque-angle characteristics for the first swing of the steam-turbine SG as obtained from detailed time-domain simulation and the analytical curves. Here, T_e = 0 during fault (scenario–I) and the system is critically stable.

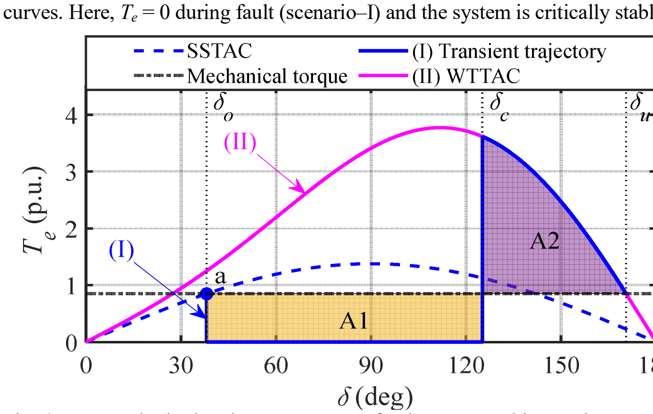

Fig. 6. EAC method using the new WTTAC for the steam-turbine SG in scenario–I when T_e = 0 during fault.

TABLE I. COMPARISON OF DIFFERENT APPROACHES FOR TRANSIENT STABILITY ASSESSMENT OF THE SUBJECT STEAM-TURBINE SG IN SCENARIO–I

	Time-domain simulation	EAC with RATTAC	EAC with ATTAC	EAC with WTTAC
$\boldsymbol{\delta_o}$	37.88°	13.10°	37.92°	38.09°
$\boldsymbol{\delta_u}$	172.66°	166.90°	171.79°	170.48°
CCA ($\boldsymbol{\delta_c}$)	125.32°	111.44°	128.90°	125.13°
CCT	0.357 s	0.346 s	0.333 s	0.325 s

are considerably more accurate than the RATTAC (with WTTAC the most accurate) in calculating CCA using the EAC method (i.e., 125.13° WTTAC vs. 128.90° ATTAC vs. 111.44° RATTAC, compared to 125.32° simulation). However, RATTAC has led to a more accurate CCT compared to ATTAC and WTTAC (i.e., 0.346 s RATTAC vs. 0.333 s ATTAC vs. 0.325 s WTTAC, compared to 0.357 s simulation).

Scenario–II: Here, it is assumed that the terminal voltage of the SG decreases to 0.1 p.u. due to the fault; therefore, $T_e \neq 0$ during the on-fault period. The torque-angle characteristic for the first swing of the SG is illustrated in Fig. 7 as obtained from detailed simulations. As seen, the average-value of the torque would not be equal to zero during the fault, as opposed to scenario–I. The EAC method with the WTTAC in scenario–II is shown in Fig. 8 where the initial operating point is obtained based on SSTAC using (1) with pre-fault parameters. It is also noted that the WTTAC curves for pre/post-fault and on-fault conditions are obtained using (15) with pre/post-fault and on-fault parameters, respectively (i.e., $|V_s|$ = 1.0 p.u. for pre/post-fault curve (III), and $|V_s|$ = 0.1 p.u. for on-fault curve (II)).

The TSA results for the steam-turbine SG in scenario–II obtained from time-domain simulations as well as the EAC approach in conjunction with the subject torque-angle characteristics are summarized in Table II. As it can be seen in Table II, consistent with the findings in Table I, WTTAC and ATTAC are much more accurate than the RATTAC (with WTTAC the most accurate) in calculating CCA using the EAC method (i.e., 133.34° WTTAC vs. 136.72° ATTAC vs. 120.95° RATTAC, compared to 132.81° simulation). However, ATTAC has slightly better accuracy in calculating CCT compared to RATTAC and WTTAC (i.e., 0.414 s ATTAC vs. 0.409 s RATTAC vs. 0.407 s WTTAC, compared to 0.419 s simulation).

B. *TSA Results for the Hydro-Turbine SG*

Here, similar TSA studies are conducted for the hydro-turbine SG in both scenarios. The corresponding results obtained from time-domain simulations as well as the EAC approach with the subject torque-angle characteristics of the hydro SG are summarized in Tables III and IV for scenarios I and II, respectively. Here, due to limited space and the general conformity of dynamic waveforms with the ones in Figs. 3–5, 7 and torque-angle characteristics with the curves in Figs. 6 and 8, only performance indices are presented in Tables III–IV. It is also noted that since the subject hydro-turbine SG does not have the *kq1* damper winding, $X_q = X'_q$ (see Appendix); therefore, the ATTAC and WTTAC characteristics are identical based on (11) and (15).

As seen in Table III for scenario–I, compared to RATTAC, the ATTAC/WTTAC provide more accurate results for CCA (i.e., 124.11° ATTAC/WTTAC vs. 115.10° RATTAC, compared to 124.62° simulation) as well as for CCT (i.e., 0.416 s ATTAC/WTTAC vs. 0.411 s RATTAC, compared to 0.463 s

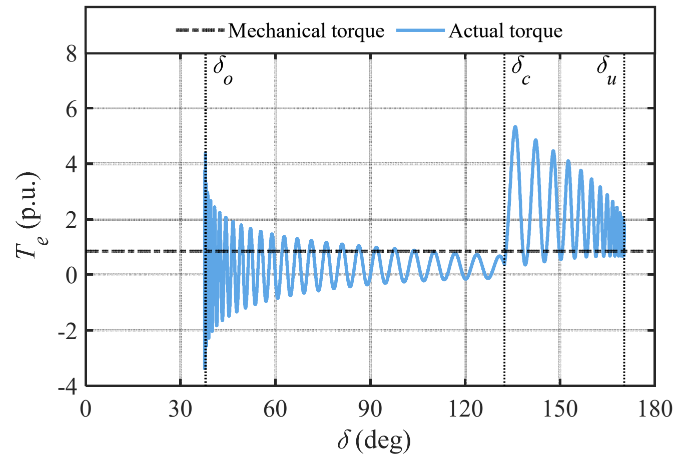

Fig. 7. Transient torque-angle characteristics for the first swing of the steam-turbine SG as obtained from detailed time-domain simulation. Here, $T_e \neq 0$ during fault (scenario–II) and the system is critically stable.

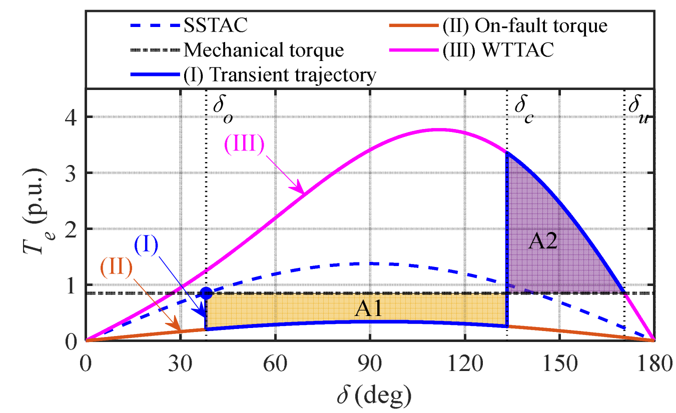

Fig. 8. EAC method using the new WTTAC for the steam-turbine SG in scenario–II when $T_e \neq 0$ during fault.

TABLE II. COMPARISON OF DIFFERENT APPROACHES FOR TRANSIENT STABILITY ASSESSMENT OF THE SUBJECT STEAM-TURBINE SG IN SCENARIO–II

	Time-domain simulation	EAC with RATTAC	EAC with ATTAC	EAC with WTTAC
$\boldsymbol{\delta_o}$	37.88°	13.10°	37.92°	38.09°
$\boldsymbol{\delta_u}$	172.35°	166.90°	171.79°	170.48°
CCA ($\boldsymbol{\delta_c}$)	132.81°	120.95°	136.72°	133.34°
CCT	0.419 s	0.409 s	0.414 s	0.407 s

TABLE III. COMPARISON OF DIFFERENT APPROACHES FOR TRANSIENT STABILITY ASSESSMENT OF THE SUBJECT HYDRO-TURBINE SG IN SCENARIO–I

	Time-domain simulation	EAC with RATTAC	EAC with ATTAC/WTTAC
$\boldsymbol{\delta_o}$	17.94°	11.72°	18.02°
$\boldsymbol{\delta_u}$	169.29°	168.28°	171.32°
CCA ($\boldsymbol{\delta_c}$)	124.62°	115.10°	124.11°
CCT	0.463 s	0.411 s	0.416 s

TABLE IV. COMPARISON OF DIFFERENT APPROACHES FOR TRANSIENT STABILITY ASSESSMENT OF THE SUBJECT HYDRO-TURBINE SG IN SCENARIO–II

	Time-domain simulation	EAC with RATTAC	EAC with ATTAC/WTTAC
$\boldsymbol{\delta_o}$	17.94°	11.72°	18.02°
$\boldsymbol{\delta_u}$	169.39°	168.28°	171.32°
CCA ($\boldsymbol{\delta_c}$)	133.98°	125.46°	133.99°
CCT	0.554 s	0.493 s	0.510 s

simulation). Consistent trend is seen in Table IV for scenario–II, showing the higher accuracy of the ATTAC/WTTAC compared to RATTAC, for CCA (i.e., 133.99° ATTAC/WTTAC vs. 125.46° RATTAC, compared to 133.98° simulation) as well as for CCT (i.e., 0.510 s ATTAC/WTTAC vs. 0.493 s RATTAC, compared to 0.554 s simulation).

According to the analysis in Sections III–A, III–B and the results in Tables I–IV, one can summarize the level of accuracy offered by each torque-angle characteristic and their corresponding complexity as presented in Table V. As seen, all the torque-angle characteristics carry some level of error in calculating CCA and/or CCT using the EAC approach, inherently due to their approximations. It was also seen that the ATTAC was more complex compared to RATTAC due to having $\sin(2\delta)$ term in addition to $\sin(\delta)$ term. Moreover, the complexity of WTTAC is higher compared to ATTAC. It is due to the fact that the initial operating point δ_o in the WTTAC method is calculated based on the steady-state SSTAC.

Therefore, one can use any of these characteristics for the calculation of a certain quantity (e.g., CCA and CCT) considering the compromise between their level of accuracy offered and their complexity.

TABLE V. LEVEL OF ACCURACY OF DIFFERENT TRANSIENT TORQUE-ANGLE CHARACTERISTICS FOR TSA WHEN USED IN THE EAC METHOD

	Accuracy for CCA	**Accuracy for CCT**	**Complexity**
RATTAC	Low	Medium	Low
ATTAC	Medium	Medium	Medium
WTTAC	High	Medium	High

IV. CONCLUSION

The equal area criterion (EAC) is beneficial and widely used for transient stability analysis of power systems which uses approximated torque-angle characteristics for synchronous generators (SGs). In this paper, the performance of several classical transient torque-angle characteristics of SGs (i.e., approximate transient torque-angle characteristic (ATTAC) and reduced approximate transient torque-angle characteristic (RATTAC)) has been investigated for application in the EAC. It has been shown that the ATTAC is generally more accurate than RATTAC for calculating the critical clearing angle (CCA) and critical clearing time (CCT). It is due to using $\sin(2\delta)$ term in addition to the $\sin(\delta)$ term in the torque-angle relation in ATTAC as opposed to RATTAC that uses the $\sin(\delta)$ term only. Although the approximate RATTAC is less complicated (due to having $\sin(\delta)$ term only), one should be aware of the compromise of the accuracy. Furthermore, a whole transient torque-angle characteristic (WTTAC) has been introduced which uses transient reactance values for both d and q axes as well as both $\sin(\delta)$ and $\sin(2\delta)$ terms. Also, in the WTTAC, the initial rotor angle is calculated based on the pre-fault steady-state torque-angle characteristic for improved accuracy. Although being fairly more complex than the ATTAC and RATTAC, the new WTTAC generally offers higher accuracy in calculating CCA and CCT for transient stability analysis using the EAC approach. The three ATTAC, RATTAC, and WTTAC are all approximations of the actual transient torque-angle characteristic of the SGs and the user can choose among them based on a compromise between their accuracy and complexity.

APPENDIX

TABLE VI. PARAMETERS OF THE STEAM-TURBINE SG

Parameter	**Value**	**Parameter**	**Value**
Rating	835 MVA	Voltage	26 kV
Poles	2	Power factor	0.85
Speed	3600 r/min	H	5.6 s
r_s	0.003 pu	X_{ls}	0.19 pu
X_d	1.80 pu	X_q	1.80 pu
r_{fd}	0.000929 pu	X_{lfd}	0.14140 pu
r_{kd}	0.013340 pu	X_{lkd}	0.08125 pu
r_{kq1}	0.001780 pu	X_{lkq1}	0.81250 pu
r_{kq2}	0.008410 pu	X_{lkq2}	0.09390 pu

TABLE VII. PARAMETERS OF THE HYDRO-TURBINE SG

Parameter	**Value**	**Parameter**	**Value**
Rating	325 MVA	Voltage	20 kV
Poles	64	Power factor	0.85
Speed	112.5 r/min	H	7.5 s
r_s	0.0019 pu	X_{ls}	0.12 pu
X_d	0.85 pu	X_q	0.48 pu
r_{fd}	0.00041 pu	X_{lfd}	0.2049 pu
r_{kd}	0.01410 pu	X_{lkd}	0.1600 pu
r_{kq2}	0.01360 pu	X_{lkq2}	0.1029 pu

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Transient Stability Analysis of Low-Inertia Shipboard Electrical Systems

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***Abstract*—Transient stability analysis (TSA) of synchronous generators (SGs) is essential for reliable operation of electrical systems. Conventionally, the equal area criterion (EAC) method is used to assess the transient stability of SGs, assuming that they are connected to an infinite bus with a fixed frequency. However, during disturbances in low-inertia systems, e.g., small microgrids and shipboard power systems, the frequency of the main ac bus may noticeably deviate from its nominal value. This paper proposes a new formulation within the EAC method for calculating the critical clearing time (CCT) of SGs in low-inertia electrical systems, considering the effect of variations in system frequency during TSA. To verify the effectiveness of the new extended EAC, the CCTs of SGs in a low-inertia multi-machine shipboard power system are obtained using the proposed formulation and compared with time-domain simulation results. It is demonstrated that the proposed EAC formulation estimates the CCTs more accurately when the main ac bus frequency varies following a disturbance.**

Keywords*—*Critical clearing time (CCT), equal area criterion (EAC), marine, shipboard, transient stability analysis (TSA).

I. Introduction

The rotor angle stability of synchronous generators (SGs) is classified into small-signal and large-signal stability based on the disturbance magnitude [1]. Large-signal stability (i.e., transient stability) refers to the ability of an SG to restore its synchronism when subjected to a severe disturbance [2], [3]. Thus, transient stability analysis (TSA) is concerned with evaluating the capacity of SGs to maintain equilibrium between their output electromagnetic torque and input mechanical torque [3]. In TSA, rotor angle stability margins are typically quantified using the critical clearing angle (CCA) and critical clearing time (CCT) [2].

The CCT is the maximum time by which a fault has to be cleared for the SG to remain marginally stable and return to its synchronous operation without slipping poles. The corresponding maximum angle attained by the SG's rotor at the instant the fault is removed is called CCA [1], [2]. The CCT is an important measure to determine the time settings for circuit breakers and other protective equipment to isolate faults and make any network reconfigurations before the SG becomes unstable [4]. Furthermore, the CCAs and CCTs may be used in monitoring and control centers as warning levels to provide situational awareness for system operators [5].

The TSA may be conducted using time-domain simulations and/or direct methods [3], [7]. In time-domain simulations, the nonlinear equations describing the system behavior are solved to determine the dynamic response to a disturbance [6]. However, time-domain simulations may be computationally expensive, especially for multi-machine systems, although they provide accurate responses [6], [7]. The direct methods are based on energy functions derived from the rotor's kinetic and potential energy, which are compared to a critical energy threshold to quantitively determine the transient stability limits [2], [8]. Nevertheless, it is challenging to decide on the energy functions of complex multi-machine systems [8].

The equal area criterion (EAC) stems from the energy function-based methods, wherein the concepts behind the energy functions are also graphically depicted [9], [10]. The EAC method allows easy and fast determination of the quantitative stability limits with limited computational resources [10], [11]. Despite the merits, the EAC is intrinsically an approximate method because the accuracy of the results depends on the approximate transient torque-angle characteristics used in its computations [12].

Although the literature contains numerous works on the TSA of SGs, to the best of the authors' knowledge, the majority of the previous works focus on the ideal cases where the SGs are connected to an infinite bus, known as single-machine infinite-bus (SMIB) systems. Additionally, for the TSA of multi-machine power systems, multiple SGs may be grouped into an equivalent SMIB [9]. However, SGs are widely used in various applications, ranging from large-scale SGs in power grids to medium- and small-scale SGs in transportation systems, such as shipboard power systems. Although the approximation of an infinite bus is consistently used in power systems modeling and simulations, it becomes a very crude approximation when applied to small electrical systems with low-inertia SGs. In these cases, the bus to which the machines are connected is not ideally an infinite bus, as its frequency may fluctuate due to load changes in the system. This is particularly relevant for islanded or shipboard electrical systems with low-inertia SGs.

In this paper, the TSA using EAC is extended to the non-ideal cases where the SG's bus frequency may change during the fault due to a change in power demand or the subsequent action of the protection system and controllers. A new formulation is derived for calculating the CCT based on the conventional EAC concept while considering the effect of changing the main bus frequency during TSA. The CCTs obtained using the proposed method are compared with the time-domain simulation results for a MW-level shipboard electrical system. It is demonstrated that the proposed EAC formulation estimates the CCTs accurately even when the main ac bus frequency changes.

II. System Modeling and Classical TSA

Although the analysis presented in this paper applies to any low-inertia power system, without loss of generality, the power system of an electrified marine vessel (EMV) is considered, as shown in Fig. 1. Therein, four identical turbine-generator pairs are connected to the main ac bus via circuit breakers. The electrical load of the EMV is divided into two parts: the electrical propulsion system and the ship's service load system. The electric propulsion system may include transformers, power electronic converters, propulsion motors, etc. The ship's service load system includes service transformers, vital and nonvital loads, etc.

This section presents the modeling of synchronous generators, their typical torque-angle characteristics used in conventional EAC for TSA, and formulations for the CCA and CCT.

A. Modeling of Synchronous Generators

In this paper, a full-order *qd*0 model of synchronous generators (SGs) is adopted for time-domain simulations to adequately capture their transient behavior. It is assumed that the SGs have one field winding and one damper winding on the *d*-axis, as well as one damper winding on the *q*-axis. It is noted that considering only one damper winding on the *q*-axis would provide a sufficient prediction of the transient behavior of small-rating, low-speed, salient-pole generators used in EMV power systems [12].

The dynamics of the stator and rotor variables are formulated (in per-unit) as [12]

$$\begin{cases} \dfrac{1}{\omega_b}\dfrac{d\psi_q}{dt} = r_s i_q - \dfrac{\omega_r}{\omega_b}\psi_d + v_q, \\ \dfrac{1}{\omega_b}\dfrac{d\psi_d}{dt} = r_s i_d + \dfrac{\omega_r}{\omega_b}\psi_q + v_d, \\ \dfrac{1}{\omega_b}\dfrac{d\psi_q}{dt} = r_s i_0 + v_0, \end{cases} \quad \begin{cases} \dfrac{1}{\omega_b}\dfrac{d\psi_{fd}}{dt} = -r_{fd} i_{fd} + v_{fd}, \\ \dfrac{1}{\omega_b}\dfrac{d\psi_{kd}}{dt} = -r_{kd} i_{kd}, \\ \dfrac{1}{\omega_b}\dfrac{d\psi_{kq}}{dt} = -r_{kq} i_{kq}. \end{cases} \tag{1}$$

Here, the state variables are the stator and rotor flux linkages in *qd*-axes. It is also conventional to use the so-called excitation voltage E_{xfd} in the field winding dynamic equation, which is defined as

$$E_{xfd} = \frac{X_{md}}{r_{fd}} v_{fd} = X_{md} i_{fd}. \tag{2}$$

The stator and rotor currents are calculated as [12]

$$\begin{cases} i_q = -\dfrac{1}{X_{ls}}\left(\psi_q - \psi_{mq}\right), \\ i_d = -\dfrac{1}{X_{ls}}\left(\psi_d - \psi_{md}\right), \\ i_0 = -\dfrac{1}{X_{ls}}\psi_0, \end{cases} \quad \begin{cases} i_{fd} = -\dfrac{1}{X_{lfd}}\left(\psi_{fd} - \psi_{md}\right), \\ i_{kd} = -\dfrac{1}{X_{lkd}}\left(\psi_{kd} - \psi_{md}\right), \\ i_{kq} = -\dfrac{1}{X_{lkq}}\left(\psi_{kq} - \psi_{mq}\right). \end{cases} \tag{3}$$

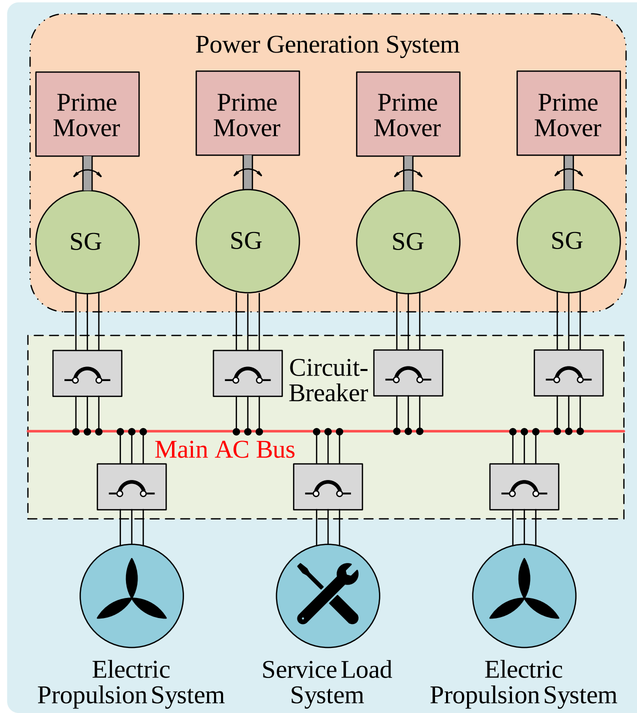

Fig. 1. Schematic of the power system of a typical electrified marine vessel.

In (3), ψ_{mq} and ψ_{md} are the magnetizing flux linkages per second, which are defined as

$$\begin{cases} \psi_{mq} = X_{aq}\left(\dfrac{\psi_q}{X_{ls}} + \dfrac{\psi_{kq}}{X_{lkq}}\right), \\ \psi_{md} = X_{ad}\left(\dfrac{\psi_d}{X_{ls}} + \dfrac{\psi_{fd}}{X_{lfd}} + \dfrac{\psi_{kd}}{X_{lkd}}\right), \end{cases} \tag{4}$$

where

$$\begin{cases} X_{aq} = \left(X_{ls} \parallel X_{mq} \parallel X_{lkq}\right), \\ X_{ad} = \left(X_{ls} \parallel X_{md} \parallel X_{lfd} \parallel X_{lkd}\right). \end{cases} \tag{5}$$

In (5), the symbol "||" denotes the parallel combination of reactances.

The electromagnetic torque produced by the generator in per-unit can be expressed as

$$T_e = \psi_d i_q - \psi_q i_d. \tag{6}$$

The dynamics of the mechanical system is also expressed in per-unit as

$$\frac{d\omega_r}{dt} = \frac{1}{2H}\left(T_m - T_e\right), \tag{7}$$

where H is the turbine-generator inertia constant, and ω_r is the rotor angular frequency in per-unit. The relative rotor angle δ is defined as

$$\delta = \int \omega_b \left(\omega_r - \omega_e\right) dt. \tag{8}$$

where ω_b is the nominal/base angular frequency, and ω_e is the angular frequency of the ac bus/terminal.

B. Steady-State Torque-Angle Characteristic

The steady-state torque-angle characteristic (SSTAC) of an SG shows its electromagnetic torque characteristics as a function of rotor angle δ at different steady-state operating conditions, which is expressed as [12]

$$T_e = \frac{\omega_b}{\omega_e}\frac{|E_{xfd}||V_s|}{X_d}\sin(\delta) + \frac{|V_s|^2}{2}\left(\frac{\omega_b}{\omega_e}\right)^2\left(\frac{1}{X_q}-\frac{1}{X_d}\right)\sin(2\delta). \quad (9)$$

In steady-state, the rotor speed ω_r is constant and equal to the angular frequency of the ac bus/terminal ω_e. It is also worth mentioning that the stator resistance r_s is typically neglected when deriving the SSTAC. In (9), ω_b is the nominal base angular frequency (e.g., 120π rad/s for a 60 Hz system), and V_s is the stator terminal voltage. Also, X_q and X_d are the q- and d-axes reactances, which can be calculated as

$$\begin{cases} X_q = X_{mq} + X_{ls}, \\ X_d = X_{md} + X_{ls}, \end{cases} \quad (10)$$

using the magnetizing reactances of the q- and d-axes (i.e., X_{mq} and X_{md}) and the stator leakage reactance X_{ls}.

To perform TSA using the EAC method, the initial rotor angle δ_0 must be determined, which is obtained using the SSTAC characteristic in (9). The initial rotor angle depends on the operating condition of the generator, which is determined by the excitation voltage E_{xfd} required to maintain the rated terminal voltage, and the input mechanical torque T_m needed to supply the power at a given operating condition.

C. Approximate Transient Torque-Angle Characteristic

The approximate transient torque-angle characteristic (ATTAC) of an SG shows the relationship between the transient torque of the generator as a function of the rotor angle. As a matter of fact, the transient torque produced by the SG following a sudden disturbance differs significantly from the steady-state value predicted by the SSTAC [12].

A variety of transient torque angle characteristics have been discussed in [13], and the ATTAC is regarded as the most widely accepted option in the field. Although the whole transient torque angle characteristic (WTTAC) described in [13] provides a better approximation, it would be identical to ATTAC for the case of this paper, where only one damper winding is considered on the q-axis. The ATTAC is formulated as [12]

$$T_e = \frac{\omega_b}{\omega_e}\frac{|E_q'||V_s|}{X_d'}\sin(\delta) + \frac{|V_s|^2}{2}\left(\frac{\omega_b}{\omega_e}\right)^2\left(\frac{1}{X_q}-\frac{1}{X_d'}\right)\sin(2\delta), \quad (11)$$

where X_d' is the d-axis transient reactance expressed as

$$X_d' = X_d - \frac{(X_{md})^2}{X_{fd}}, \quad (12)$$

and E_q' is the q-axis transient voltage behind reactance defined as

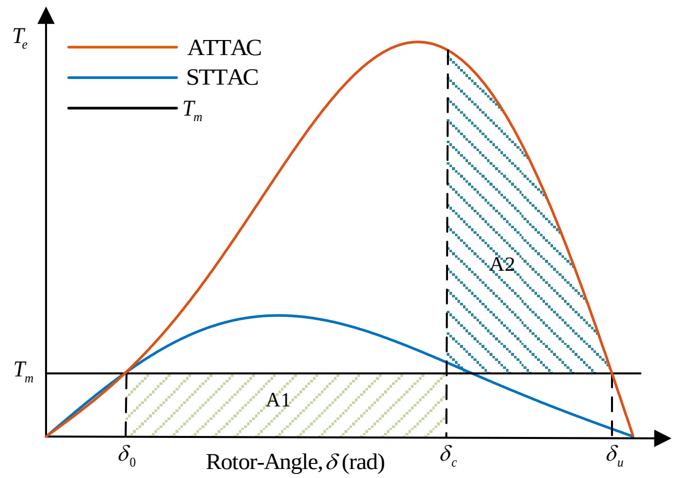

Fig. 2. Different torque-angle characteristics of synchronous generators for application in classical EAC method to calculate CCA and CCT.

$$E_q' = \frac{X_{md}}{X_{fd}}\psi_{fd}. \quad (13)$$

D. Critical Clearing Angle and Time

To quantitatively determine the transient stability limits of SGs, the CCA and CCT are calculated using the EAC method [12]. Fig. 2 shows the SG's torque-angle characteristics and how they are used graphically to calculate the CCA and CCT in the conventional EAC approach.

Specifically, in the steady-state pre-fault condition, the electromagnetic torque T_e generated by the SG equals the input mechanical torque T_m. The corresponding initial rotor angle δ_0 can also be calculated using the SSTAC based on (9). Once a three-phase fault occurs at the terminals of the SG, assuming the terminal voltage of the SG decreases to zero, the T_e also decreases to zero. Therefore, the mechanical torque accelerates the rotor, and its angle increases to δ_c until the fault is critically cleared. During the on-fault period, the rotor gains excess kinetic energy equal to area A1. Once the fault is cleared at δ_c, T_e follows the ATTAC, and the rotor decelerates while its angle keeps increasing to δ_u. During the post-fault deceleration period, the rotor loses its excess kinetic energy corresponding to area A2.

In a critically stable operation, areas A1=A2. Consequently, the CCA (i.e., critical clearing angle δ_c) can be obtained by solving the following equation [8]

$$\int_{\delta_0}^{\delta_c}(T_m)\,d\delta = \int_{\delta_c}^{\delta_u}\left(T_e^{ATTAC} - T_m\right)d\delta. \quad (14)$$

Once the CCA is calculated from (14), the corresponding CCT can be calculated from the swing equation of the SG, which can be expressed during the fault as

$$\frac{2H}{\omega_b}\frac{d^2\delta}{dt^2} = T_m. \quad (15)$$

Integrating both sides of (15) from δ_0 (corresponding to t=0) until the fault is cleared at δ_c (corresponding to t=t_c) yields [12]

$$\delta_c - \delta_0 = \frac{\omega_b}{2H} \int_{t=0}^{t_c} \int_{\tau=0}^{\tau} T_m \, d\tau \, dt \, . \tag{16}$$

Finally, the CCT (i.e., t_c) in (16) can be calculated as [12]

$$t_c = \sqrt{\frac{4H\left(\delta_c - \delta_0\right)}{\omega_b T_m}} \, . \tag{17}$$

III. Proposed Formulation for Low-Inertia Systems

Determining the frequency of the main ac bus in a low-inertia power system is crucial to performing TSA and calculating the CCA and CCT. Meanwhile, the assumption of a constant bus frequency is impractical due to the system's small number of generators with low inertia. In such systems, during a disturbance, the bus frequency may suddenly decrease or increase due to an imbalance in power generation and load demand. For example, an imbalance can be caused by a faulted SG going offline momentarily until the fault is cleared. As a result, the bus frequency may vary dynamically (at least) during the on-fault stage.

A. Low-Inertia System Bus Frequency Approximation

At the pre-fault stage, the overall system operates at a steady state; thus, the bus frequency can be reasonably assumed to be constant at the nominal frequency. During the on-fault period, the bus frequency varies due to power generation and demand imbalance and primary frequency droop control of individual SGs [8].

At the post-fault stage, it may be fair to assume a new constant bus frequency because the system begins to operate at a new (quasi) steady state as the balance between generation and load is considered to be restored. However, the post-fault bus frequency may be slightly above or below the nominal frequency. In fact, in an actual system, the secondary frequency control mechanism, i.e., the automatic generation control (AGC), may restore the frequency to the nominal value after some time [8]. Nevertheless, incorporating the AGC is beyond the scope of this paper and does not affect the TSA study.

Fig. 3 depicts three bus frequency profiles assumed for the low-inertia system of Fig. 1 during the three-phase fault study considered in this paper. Case (a) assumes that the frequency remains constant (indicated by ω_e^*) during the fault as in the conventional TSA using the EAC method. In cases (b) and (c), the frequency increases and decreases (indicated by ω_e^+ and ω_e^-, respectively), during the fault.

It is noted that in low-inertia power systems, during the on-fault period, the rate at which the frequency of the main ac bus changes depends on the remaining system's inertia and the power balance. The inertia can be calculated by the sum of the inertia of all turbine-generator pairs operating at any instant. The power system depicted in Fig. 1 consists of four identical turbine-generator pairs, so the inertia of the overall system would be four times that of each turbine-generator pair. However, during the on-fault period, assuming one machine is

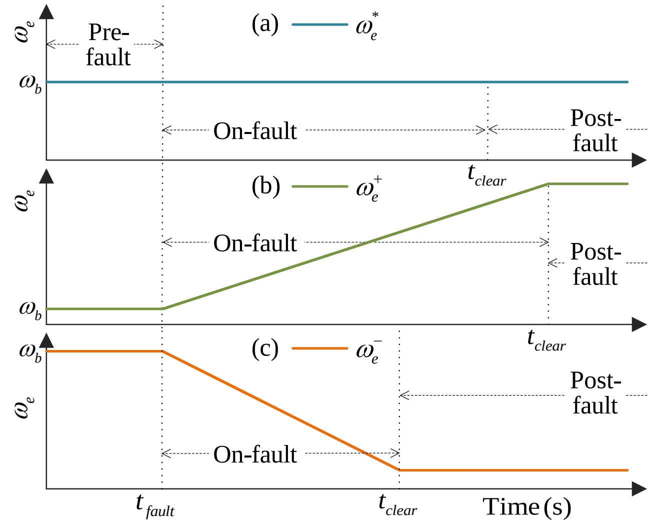

Fig. 3. The three frequency profiles considered during a three-phase fault: (a) fixed frequency as in the conventional EAC, (b) increasing frequency due to additional loss of load during the fault, and (c) decreasing frequency due to insufficient generation with the same load during the fault.

momentarily offline, the inertia of the bus is only three times that of each pair. Consequently, it would be a fair assumption to approximate the rate of change of frequency by the rate at which the rotor speed changes during its first swing following a fault. The first swing rotor acceleration during the fault can be calculated from (7) as

$$\frac{d\omega_r^{first-swing}}{dt} = \frac{T_m}{2H} \, . \tag{18}$$

Based on the rationale mentioned above, the rate at which the bus frequency ω_e changes in the system in Fig. 1 will be one-third of the first-swing rotor acceleration of an SG during the on-fault period (assuming that only one generator goes offline). Mathematically, the slope of the frequency, denoted by k, can be approximated as

$$k = \frac{d\omega_e^{on-fault}}{dt} = \frac{1}{(n-1)} \frac{d\omega_r^{first-swing}}{dt} \, , \tag{19}$$

where n is the number of identical turbine-generator pairs in the power system. The instantaneous ac bus frequency can also be expressed as

$$\omega_e^{on-fault}(t) = \omega_e^{pre-fault} + kt \, , \tag{20}$$

where t is the time in seconds after the fault has occurred.

It is also assumed that during the post-fault period, the ac bus frequency becomes constant at the value when the fault is cleared, as demonstrated in Fig. 3. Mathematically, the post-fault frequency can be formulated as

$$\omega_e^{post-fault} = \omega_e^{pre-fault} + k\, t_c \, , \tag{21}$$

where t_c is the CCT. Table I summarizes the expressions for the bus frequency profiles before, during, and after a fault, as illustrated in Fig. 3. Here, it is assumed that during the fault, the bus frequency may decrease (ω_e^-) or increase (ω_e^+) depending on whether the load remains the same or decreased.

TABLE I. THREE BUS FREQUENCY PROFILES VALUES BASED ON FIG. 3.

	Pre-fault (rad/s)	**On-fault (rad/s)**	**Post-fault (rad/s)**
ω_e^*	ω_b	ω_b	ω_b
ω_e^+	ω_b	$\omega_b + kt$	$\omega_b + kt_c$
ω_e^-	ω_b	ω_b - kt	ω_b - kt_c

B. *Critical Clearing Angle Calculation*

Since the pre-fault bus frequency is identical for the three cases shown in Fig. 3 (equal to ω_b), the SSTAC curve formulated in (9), which is a function of pre-fault frequency, can still be used to determine the initial rotor angle δ_0. However, the ATTAC, formulated in (11), depends on the post-fault operating frequency ω_e, which differs for each case in Fig. 3. The torque-angle characteristics under the three subject post-fault frequency profiles are shown in Fig. 4 for use in the EAC method. Therefore, each operating frequency in Table I corresponds to a different ATTAC curve in Fig. 4, where a higher operating frequency shifts the ATTAC curve downward and vice versa. Nevertheless, the post-fault bus frequency varies only slightly around the nominal frequency. Consequently, the changes in areas A3 and A4 in Fig. 4 are negligible compared to area A2. As a result, the CCA formulated in (14), can be used with fair accuracy to compute the CCA in all three cases.

C. *Critical Clearing Time Calculation*

The CCT expressed in (17) was derived assuming that the ac bus frequency remains constant during the fault period. Therefore, a new expression must be derived to consider the change in the ac bus frequency. Once the CCA is calculated from (14) as explained in Section III–B, the CCT can be calculated from the swing equation of the SG. Here, with the assumption of the change in the bus frequency, the swing equation needs to be derived accordingly. For this purpose, first, the relative rotor angle δ given by (8), is differentiated twice on both sides and expressed as

$$\frac{d\omega_r}{dt} = \frac{1}{\omega_b}\frac{d^2\delta}{dt^2} + \frac{d\omega_e}{dt}. \tag{22}$$

Thus, the swing equation is derived by substituting (22) in (7) as

$$\frac{1}{\omega_b}\frac{d^2\delta}{dt^2} = \frac{T_m}{2H} - \frac{d\omega_e}{dt}. \tag{23}$$

Integrating both sides of (23) during the on-fault period yields

$$\delta_c - \delta_0 = \frac{\omega_b}{2H}\int_{t=0}^{t_c}\int_{\tau=0}^{\tau} T_m\, d\tau\, dt - \omega_b \int_{t=0}^{t_c}\int_{\omega_e=0}^{\omega_e} d\omega_e\, dt. \tag{24}$$

Substituting ω_e=kt, the CCT (i.e., t_c) in (24) can be calculated as

$$t_c = \sqrt{\frac{(\delta_c-\delta_0)}{\left(\frac{\omega_b T_m}{4H} - \frac{\omega_b k}{2}\right)}}. \tag{25}$$

According to (25), the CCT increases if the bus frequency increases during the fault (i.e., positive k) and decreases if the ac bus frequency decreases (i.e., negative k). Also, when the bus frequency remains constant (k=0), the expression (25) becomes equal to the expression (17) for the classic CCT calculation.

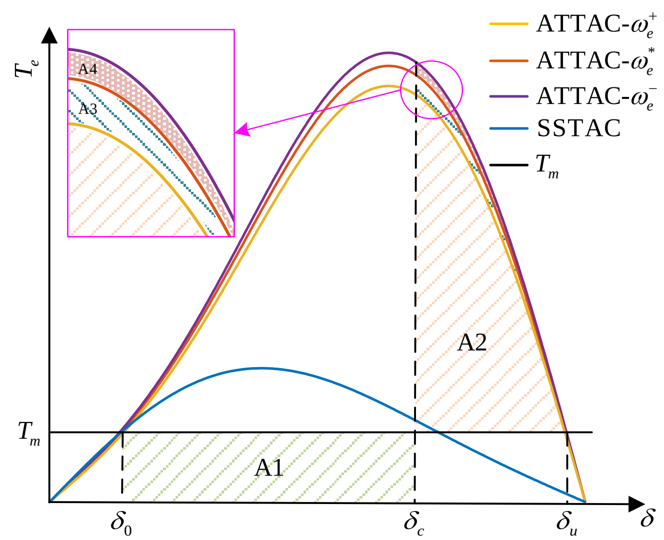

Fig. 4. Torque-angle characteristics used in the EAC method for three different post-fault operating frequencies.

To visualize this concept, the calculation of CCT using (25) is graphically illustrated in Fig. 5. In Fig. 5, the relative rotor angle δ, given by (8), is calculated by the area between the curves ω_r and ω_e. As seen in Fig. 5, the CCA is the aggregate of the initial rotor angle δ_0 and the angle accumulated during the first swing of rotor acceleration. Therein, on the one hand, when the bus frequency increases during a fault (denoted by ω_e^+ with a positive slope), the angle δ accumulates slower (highlighted in green), so the CCT (labelled t_c^+) increases compared to the infinite bus case ω_e^* (highlighted in blue) with CCT = t_c^*. On the other hand, when the bus frequency decreases (denoted by ω_e^- with a negative slope), the angle δ accumulates faster (highlighted in red), so the CCT (labelled t_c^-) decreases compared to the infinite bus case ω_e^* . Therefore, with the assumption of equal CCAs (or very close CCAs according to the discussion in Section III–B), which correspond to the three shaded equal areas (i.e., $\delta^+ = \delta^- = \delta^*$) in Fig. 5, the trends in CCT calculation using (25) is consistent with displayed CCTs in Fig. 5, i.e., $t_c^+ > t_c^* > t_c^-$

IV. COMPUTER STUDIES

To verify the effectiveness of the proposed formulation for EAC method, a low-inertia shipboard power system shown in Fig. 1 is considered for studies here, which includes four 6.4 MVA SGs [15]. Therein, the machine parameters are reported in a standard format. At the same time, the full-order qd0 model of SGs uses fundamental parameters, which can be calculated as set forth in [16] and summarized in Table III in the Appendix.

It is worth mentioning that to correctly determine the initial rotor angle δ_0, it is essential to calculate the field excitation E_{xfd} and input torque T_m corresponding to the assumed steady state condition. Without loss of generality, this paper assumes that the generators are loaded to the rated/nominal power under rated operating conditions. The steps to calculate E_{xfd} and T_m corresponding to the rated power factor are given in [12, Ch. 5]. The nominal values are found to be E_{xfd} =2.485 pu and T_m=0.8 pu.

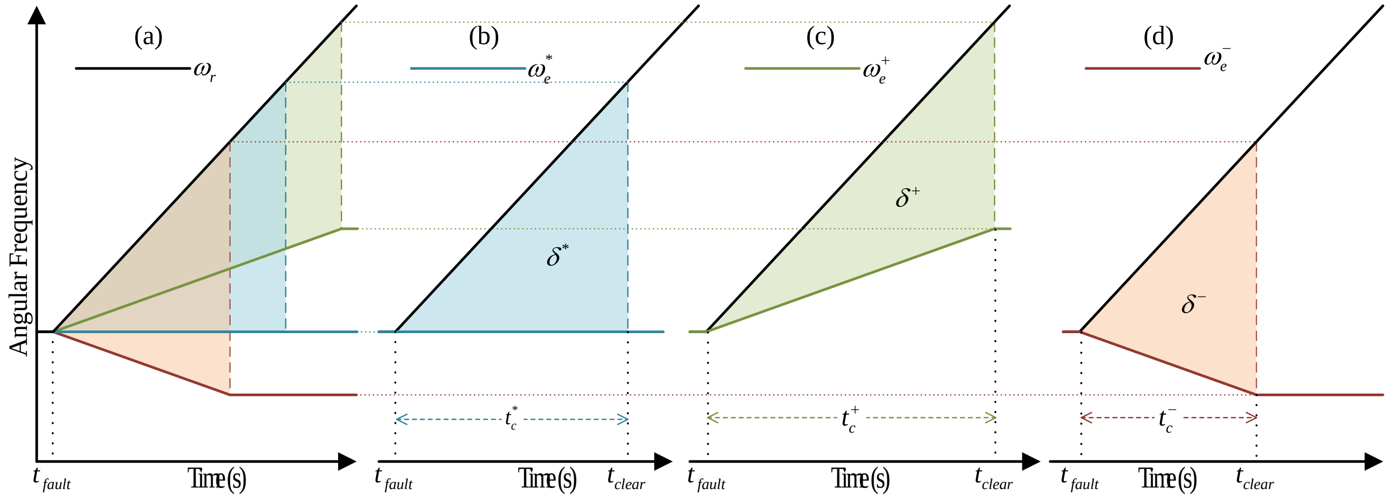

Fig. 5. Graphical representation of calculating the CCT when the ac bus frequency varies during the on-fault period: (a) all cases combined, (b) base case with constant bus frequency, (c) increasing bus frequency due to loss of load, and (d) decreasing bus frequency due to the load remaining the same.

The system is assumed to operate in a steady-state condition at pre-fault with the four generators running, and the main ac bus frequency is ω_b=377.0 rad/s. The on-fault bus frequency is calculated according to (20), whose slope is calculated to be k =28.69 rad/s^2 based on (19). The post-fault bus frequency can be calculated using (21). It is essential to acknowledge that the post-fault frequency can only be determined after finding the t_c, as shown in Table I.

A. TSA Results Using Time-Domain Simulations

The system in Fig. 1 has been implemented in MATLAB/Simulink using the full-order models of SGs with the parameters summarized in Table III. For the purpose of this paper, all shipboard loads, including the propulsion, have been lumped into a single load. The model is simulated from a steady state corresponding to the nominal condition, and at t=1s, a three-phase fault is applied by open-circuiting one SG terminal. Consequently, the electromagnetic torque generated by the machine becomes zero, i.e., T_e=0. Then, the fault is cleared at different moments, and the post-fault characteristic rotor angle δ and speed ω_r profiles are investigated for stability assessment. This is continued until the SG becomes marginally stable by trial and error. An SG is considered marginally stable if, when the fault is critically cleared, it returns to synchronous operation without slipping poles and remains stable thereafter. If the fault clearing time t_{clear} becomes any longer than CCT, the SG becomes unstable, i.e., either it does not return to synchronism or goes into asynchronous operation after slipping poles [12]. Additionally, for the SG to be first-swing stable, it must return to synchronism without slipping any poles.

The profiles of the rotor angle δ for a stable transient and an unstable case are shown in Fig. 6. The CCA and the CCT for the considered case-study with an infinite bus case (ω_e^*) is found to be 125.75° and 0.249 s, respectively. There are significant changes when the bus frequency dynamics change. The corresponding TSA results using time-domain simulations are summarized in Table II.

According to Table II, for the faulted SG in the case study of Fig. 1, it is found that when the ac bus frequency increases (ω_e^+ case), the CCT increases substantially compared to when the frequency is constant (ω_e^* case), i.e., CCT=0.332 s vs. 0.249 s. On the contrary, a decrease in bus frequency (ω_e^- case) leads to a noticeable decrease in the CCT, i.e., CCT=0.204 s vs. 0.249 s. It is also noted that according to the discussion in Section III–B, the variations in CCAs are less pronounced in the three cases as listed in Table II, i.e., CCA=130.69° and 121.42° vs. 125.75°.

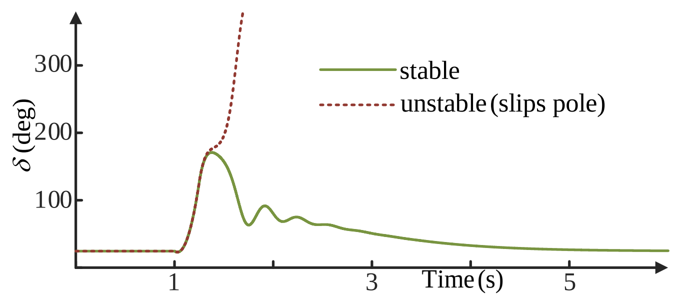

Fig. 6. Profiles of the rotor angle δ for a stable and an unstable transient.

TABLE II. Comparison of the Proposed Method with Time-Domain Simulation Results For Transient Stability Analysis

	CCA (Time-domain)	**CCA (Proposed Method)**	**CCT (Time-domain)**	**CCT (Proposed Method)**
ω_e^*	125.75°	132.01°	0.249 s	0.208 s
ω_e^+	130.69°	130.81°	0.332 s	0.253 s
ω_e^-	121.42°	132.74°	0.204 s	0.181 s

B. TSA Results Using the EAC Method

The EAC method equates energy corresponding to the rotor's acceleration and deceleration. Finding the accelerating energy requires using δ_0, which is calculated using the pre-fault bus frequency $\omega_e^{pre-fault}$ (that is the nominal frequency for all the cases in this study) and the SSTAC. However, the post-fault bus frequency $\omega_e^{post-fault}$ is used for the ATTAC curve to calculate the energy of the rotor's deceleration. The TSA results using the proposed EAC formulation are also listed in Table II.

As seen in Table II, the CCA and the CCT for the infinite bus (ω_e^*) case are 132.01° and 0.208 s, respectively. The discrepancies between the EAC and time-domain simulation results (i.e., CCA=125.75° vs. 132.01° and CCT=0.249 s vs. 0.208 s) are acknowledged and may be considered acceptable. The reader should be aware of the approximate nature of the EAC method used with the ATTAC, which makes many approximations and neglects the damping and stator resistance [12].

As expected, the results vary significantly when the on-fault bus frequency changes. As seen in Table II, when the ac bus frequency increases (ω_e^+ case), using the proposed EAC formulation method results in an increase in CCT with respect to the base case with fixed frequency, i.e., CCT=0.253 s vs. 0.208 s. Additionally, following the decrease in the ac bus frequency (ω_e^- case), the proposed EAC formulation method estimates the decrease in the CCT with respect to the base case, i.e., CCT=0.181 s vs. 0.208 s. The trend of these results is fully consistent with the time-domain simulations and the graphical representation in Fig. 5. Notably, consistent with the time-domain simulation results and the assumption in Section III–B, the variations in CCAs are relatively small across the three cases listed in Table II, i.e., CCA=130.81° and 132.74° vs. 132.01°.

V. Conclusion

Transient stability analysis is essential for coordinating protection and load balancing in low inertia power systems with synchronous machines, such as shipboard power systems and some microgrids with the presence of conventional generation. Therein, the critical clearing angles (CCAs) and critical clearing times (CCTs) of faulted generators may be used to coordinate protective relays and circuit breakers in order to maintain stable operation and balance the generation with critical/remaining loads.

This paper investigates the effect of variations in the ac bus frequency on the transient stability of synchronous generators in a low-inertia shipboard power system. It was found that the CCT is significantly affected when the ac bus frequency changes during the fault. Specifically, the CCT increases when the bus frequency increases, and the CCT decreases when the ac frequency decreases. To consider this effect, a new equal area criterion (EAC) formulation is proposed, wherein the CCT is calculated considering the variations in the ac bus frequency. The proposed method has been demonstrated and verified on a typical low-inertia shipboard power system. The results of predicting the CCTs strongly agree with the detailed simulations. Therefore, it is envisioned that the proposed method may apply to other low-inertia systems where the frequency deviations during the faults are not negligible.

Appendix

TABLE III. Fundamental Parameters of SGs

Parameter	**Value**	**Parameter**	**Value**
Rating	6.4 MVA	Voltage	6.6 KV
Power factor	0.8	Poles	6
Speed	1200 rpm	H	1.7519 s
r_s	0.0054 pu	X_{ls}	0.1050 pu
X_d	1.79 pu	X_q	0.89 pu
r_{fd}	0.0015 pu	X_{lfd}	0.1829 pu
r_{kd}	0.0203 pu	X_{kd}	0.1886 pu
r_{kq}	0.0085 pu	X_{lkq}	0.0991 pu

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A Simple Explicit Method of Representing Magnetic Saturation of Salient-Pole Synchronous Machines in Both Rotor Axes using Matlab-Simulink

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***Abstract*—This paper presents a new flux-correction method to represent magnetic saturation in classical qd machine model. By introducing new variables and reforming the saturation functions explicitly, the iterative solution of nonlinear equations is avoided. Compared with existing methods, which utilize the saturated mutual inductances in the machine state equations, the proposed method is simple and does not require calculation of the derivatives of the saturated mutual inductances. The accuracy and efficiency of the proposed method is demonstrated on a typical salient-pole synchronous-machine.**

Keywords - explict methods; magnetic saturation; Matlab-Simulink; modeling; salient pole; synchronous machines;

I. INTRODUCTION

Modeling and analysis of electrical machines has been an area of active research for many years, and a great number of approaches [1]–[9] for incorporating magnetic saturation have been proposed. The complexity and accuracy of various methods ranges quite significantly depending on underlying models and applications. The issues of particular interest include considering the saturation in one or both rotor axes, round-rotor or salient-rotor, implicit or explicit, model order and complexity, etc.

A flux-correction method first introduced by C. H. Thomas and P. C. Krause [1], [2] for implementing the magnetic saturation in rotating machines, which is currently utilized in many simulation packages, is perhaps one of the simplest and elegant techniques, which use the classical qd machine model with flux linkages as the state variables. Magnetic saturation is taken into account by correcting the mutual flux linkages of the linear machine model. Therefore, the laborious work of re-deriving the saturated machine state equations and the calculation of time derivatives of mutual inductances are avoided. The saturation characteristic may be easily represented by look-up tables without introducing the numerical noise associated with estimating derivatives.

However, a disadvantage of this method is that it results in implicit equations containing the magnetizing fluxes in the q and d axes, requiring an iterative solution [5]. This paper extends the straightforward flux-correction method by formulating the flux saturation relationship explicitly while using the same saturation information and classical qd machine model. The magnetic saturation is assumed to be along the main magnetizing flux path and is appropriately distributed along q and d axes. The saturated mutual fluxes λ_{mq}^r and λ_{md}^r are expressed explicitly in terms of the state variables without the so-called algebraic loops. The proposed machine model, as well as several existing machine models, has been implemented in Matlab-Simulink. Computer studies demonstrate that the proposed model is very fast and accurate as compared to other more complex models.

II. SYNCHRONOUS MACHINE MODEL

The dynamics of synchronous machines can be represented by the electrical and mechanical equations. The Park transformation is usually applied, which transforms the machine physical variables into qd variables in rotor reference frame. Without loss of generality, motor convention is used in the paper. The corresponding voltage equations are expressed as

$$v_{qs}^r = r_s i_{qs}^r + \omega_r \lambda_{ds}^r + p\lambda_{qs}^r \tag{1}$$

$$v_{ds}^r = r_s i_{ds}^r - \omega_r \lambda_{qs}^r + p\lambda_{ds}^r \tag{2}$$

$$v_{os}^r = r_s i_{os}^r + p\lambda_{os}^r \tag{3}$$

$$v_j = r_j i_j + p\lambda_j\,;\;\; j = kq1, kq2, fd, kd\;. \tag{4}$$

The flux linkage equations are represented as

$$\lambda_{qs}^r = L_{ls} i_{qs}^r + \lambda_{mq}^r \tag{5}$$

$$\lambda_{ds}^r = L_{ls} i_{ds}^r + \lambda_{md}^r \tag{6}$$

$$\lambda_j = L_{lj} i_j + \lambda_{mq}^r\,;\; j = kq1, kq2 \tag{7}$$

$$\lambda_j = L_{lj} i_j + \lambda_{md}^r\,;\; j = fd, kd \tag{8}$$

Assuming that magnetic circuits in the q and d axes are linear, the relationship between mutual fluxes and machine currents are given as:

$$\lambda_{md}^r = L_{md}\left(i_{ds}^r + i_{fd} + i_{kd}\right) \quad (9)$$

$$\lambda_{mq}^r = L_{mq}\left(i_{qs}^r + i_{kq1} + i_{kq2}\right) \quad (10)$$

When main flux saturation is taken into account, the mutual inductances L_{mq} and L_{md} are nonlinear functions. The electromagnetic torque is expressed as

$$T_e = \frac{3P}{4}\left(\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}\right). \quad (11)$$

III. Representing Main Flux Saturation

A. Saliency Factor

Modeling of salient-pole synchronous machine saturation requires the open-circuit saturation characteristics along both axes. However, the saturation along the q axis is usually not available. Therefore, the saliency factor approach may be applied, which re-scales the saturation in the q axis to appear similar to the d axis, as shown in Fig. 1. By doing so, the anisotropic synchronous machine is converted into isotropic one with λ_{mq}^r and i_{mq}^r scaled by the so-called saliency factor m [7]–[9] defined as

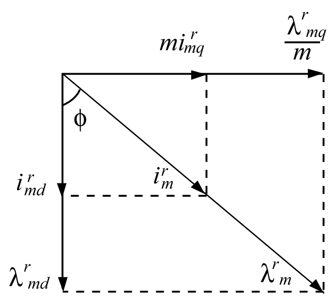

Figure 1. Fluxes and magnetizing current of a salient-pole synchronous machine

$$m = [L_{mq} / L_{md}]^{1/2}. \quad (12)$$

Then, the magnetizing current i_m^r and main flux λ_m^r may be represented as

$$i_m^r = [(m i_{mq}^r)^2 + i_{md}^{r2}]^{1/2} \quad (13)$$

$$\lambda_m^r = [(\lambda_{mq}^r / m)^2 + \lambda_{md}^{r2}]^{1/2}. \quad (14)$$

The main flux saturation is represented by means of d axis saturation function as

$$i_m^r = F_d(\lambda_m^r). \quad (15)$$

B. Saturation Modeling Approaches

Many methods [1]–[9] have been proposed to model the magnetic saturation of synchronous machines. A simple and elegant approach is the so-called flux-correction method [2] is shown in Fig. 2 for d axis.

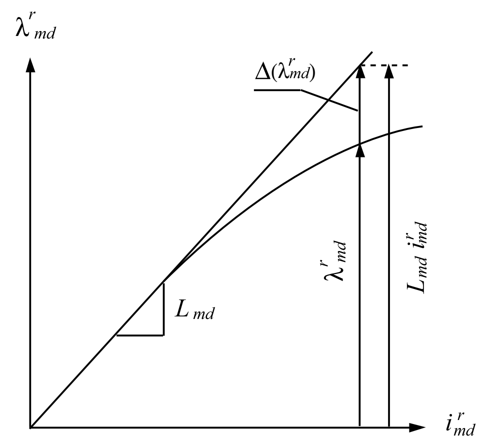

Figure 2. Saturation representation by flux correction method

According to this method, the mutual fluxes are corrected by the differences between the linear and the saturated magnetic characteristics. With the help of flux-correction method, the magnetic saturation in d and q axes can be simulated by replacing (9), (10) with (16), (17).

$$\lambda_{md}^r = L_{ad}\left(\frac{\lambda_{ds}^r}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}}\right) - \frac{L_{ad}}{L_{md}}\Delta(\lambda_{md}^r) \quad (16)$$

$$\lambda_{mq}^r = L_{aq}\left(\frac{\lambda_{qs}^r}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}}\right) - \frac{mL_{aq}}{L_{mq}}\Delta(\frac{\lambda_{mq}^r}{m}) \quad (17)$$

The inductances L_{ad} and L_{aq} are given in [2] and are not included here due to space consideration. Saturation is then taken into account by correcting the main flux with $\Delta(\lambda_m^r)$.

The flux-correction method, herein referred to as the Δ model, provides a straightforward way to include magnetic saturation in the machine model. However, the nonlinear equations (16) and (17) are implicit (which results in algebraic loops when the model is implemented in Matlab-Simulink) and have to be solved for λ_{md}^r and λ_{mq}^r iteratively in terms of the state variables, i.e. flux linkages. The simulation speed is, therefore, significantly reduced by the algebraic-loop solver.

An explicit relationship between magnetic fluxes and the state variables may be reformulated by the following procedure proposed in this paper. Herein, the derivation is presented for d axis only, whereas the same applies to q axis. The d axis magnetizing current i_{md}^r may be expressed as

$$i_{md}^r = \left(\frac{\lambda_{ds}^r}{L_{ls}} - \frac{\lambda_{md}^r}{L_{ls}}\right) + \left(\frac{\lambda_{kq1}}{L_{lkq1}} - \frac{\lambda_{md}^r}{L_{lkq1}}\right) + \left(\frac{\lambda_{kq2}}{L_{lkq2}} - \frac{\lambda_{md}^r}{L_{lkq2}}\right) \quad (18)$$

Collecting the terms with non-state variable λ_{md}^r, (18) may be rewritten as

$$i_{md}^r + \frac{\lambda_{md}^r}{L_{\Sigma ld}} = \frac{\lambda_{ds}^r}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}} \tag{19}$$

where

$$L_{\Sigma ld} = [\frac{1}{L_{ls}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lkd}}]^{-1}. \tag{20}$$

By introducing a new variable i_{zd} as

$$i_{zd} = \frac{\lambda_{ds}^r}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}}, \tag{21}$$

Equation (19) is rewritten as

$$i_{zd} = i_{md}^r + \frac{\lambda_{md}^r}{L_{\Sigma ld}} \tag{22}$$

Similarly, for q axis, the variable i_{zq} is defined as

$$i_{zq} = i_{mq}^r + \frac{\lambda_{mq}^r}{L_{\Sigma lq}} \tag{23}$$

where

$$L_{\Sigma lq} = [\frac{1}{L_{ls}} + \frac{1}{L_{lkq1}} + \frac{1}{L_{lkq2}}]^{-1}. \tag{24}$$

Based on Fig. 1 and (22), (23), it may be shown that

$$\cos\phi = \frac{i_{zd}}{i_m^r + \lambda_m^r / L_{\Sigma ld}} \tag{25}$$

$$\sin\phi = \frac{i_{zq}}{i_m^r / m + m\lambda_m^r / L_{\Sigma lq}} \tag{26}$$

As the main flux λ_m^r is saturated depending on i_m^r, the relationship between λ_m^r and i_m^r is represented with i_{zd} and i_{zq} by adding up the squares of (25) and (26). This gives

$$i_z = i_m^r + \frac{\lambda_{md}^r}{L_{\Sigma ld}} \tag{27}$$

where

$$i_z = [i_{zd}^2 + (ki_{zq})^2]^{1/2} \tag{28}$$

and

$$k = L_{\Sigma lq} /(mL_{\Sigma ld}) \tag{29}$$

The coefficient k is defined to compensate for the difference of leakage inductances in q and d axes. An assumption is made in (29) that $1/L_{\Sigma lq}$ and $1/L_{\Sigma ld}$ are much larger than $1/L_{mq}$ and $1/L_{md}$, respectively, which may be acceptable in most cases. Substituting (15) into (27), the mapping between λ_m^r and i_z is uniquely defined as

$$\lambda_m^r = G(i_z) \tag{30}$$

Eq. (30) shows that the non-state variable λ_m^r may be explicitly represented by the auxiliary variable i_z, which is by definition an explicit function of the state variables of the saturated machine model. Therefore, the algebraic loops associated with the Δ model of (16) and (17) are completely avoided. The new method is referred to as the i_z model. Another advantage of the i_z model, compared with other models that saturate the mutual inductances L_{md} and L_{mq}, is that the saturation characteristic functions (15) and (30) may be implemented by simple look-up tables. Thus, the extra work associated with fitting the saturation curves into analytical functions and calculating the derivatives of the resulting functions is avoided. The implementation of the proposed method is shown in Fig. 3.

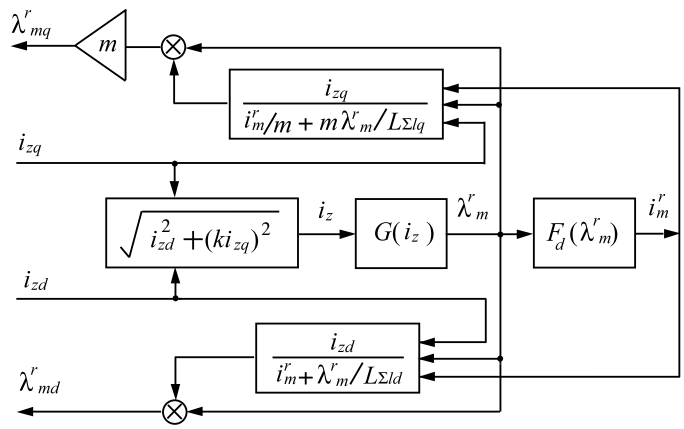

Figure 3. Implementation of the proposed saturation approach.

IV. COMPUTER STUDY

In order to verify the proposed method, a single-machine, infinite-bus system is used here. The salient-pole synchronous machine parameters are given in Appendix. The saturation characteristic was established using the methodology described in [2] and [5]. The data was then saved into appropriate lookup tables. The unsaturated machine model and the proposed i_z model were implemented using Matlab-Simulink. To validate the proposed model, the flux-correction Δ model [2] and another saturated machine model proposed in [9] (Levi's model) were implemented for comparison. The Levi's model [9] works by saturating the mutual inductances in the q and d axes, respectively.

In the following study, all models are given the same constant nominal excitation and assumed to be initially operating in steady state under no-load. As shown in Fig. 4, during the initial steady state, the saturated models are under-excited due to smaller flux, resulting in consuming reactive power as shown in Fig. 4 by positive i_{ds}^r. For the same rated excitation, the unsaturated model results in both i_{ds}^r and i_{qs}^r at zero during the initial steady state. At $t = 0.02s$, the load torque is changed from zero to 50% of the rated value. The responses produced by all models are also shown in Fig. 4.

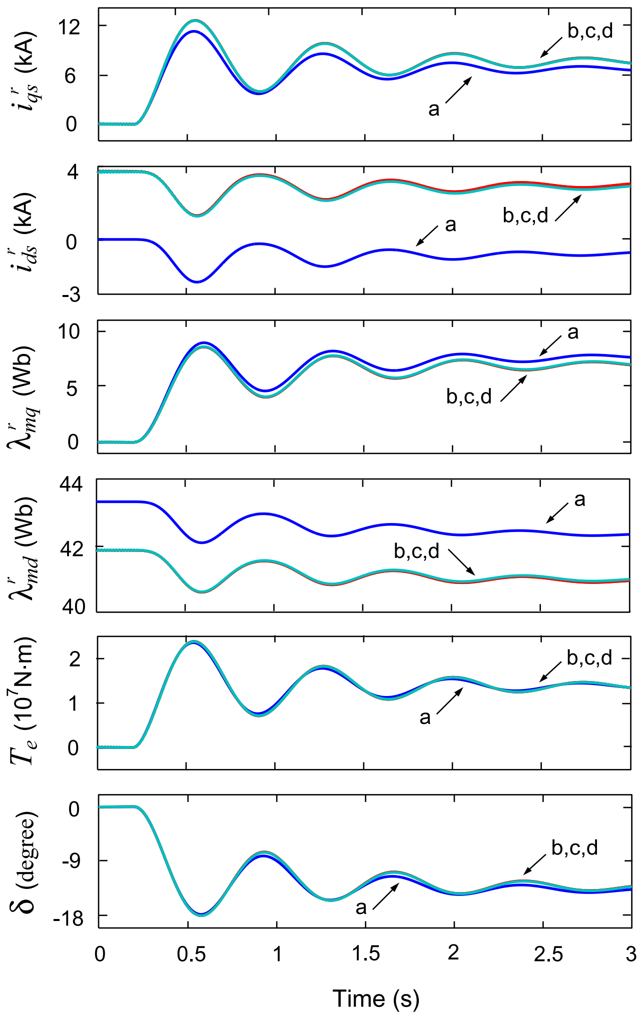

Figure 4. Transient response due to step change of mechanical torque, a- the unsaturated model; b– i_z model; c– Δ model; d–Levi's model.

As shown in Fig. 4, the i_z model, the Δ model, and Levi's model all produce the same steady-state as well as transient responses, which validates the proposed i_z model. Using the same ODE solver *ode45* and error tolerances $1e-6$ in Matlab-Simulink, the computational costs of the Δ model and the i_z model are summarized in Table I. It is shown that the explicit i_z model runs much faster than the implicit Δ model.

TABLE I. COMPUTATIONAL COSTS

Models	Δ Model	i_z Model
CPU times	1.281s	0.453s

V. CONCLUSION

This paper presented an efficient explicit method for representing the saturation along both magnetic axes in a salient-pole synchronous machine model. The d and q axes magnetic saturation is taken into account by the saliency factor approach. Simulation results show that the proposed method improves numerical efficiency and predicts the effect of saturation in both rotor axes with the same accuracy as several established models.

APPENDIX

Synchronous machine parameters: 325 MVA, 20kV, power factor 0.85, 64 poles, 112.5 r/min, $J = 35.1\times10^6\,\mathrm{J\cdot s^2}$, $T_N = 2.76\times10^7\,N\cdot m$, $r_s = 0.00234\Omega$, $L_{ls} = 0.3921mH$, $L_q = 1.6mH$, $r_{kq1} = 0.00419\Omega$, $L_{lkq1} = 2.4mH$, $r_{kq2} = 0.01675\Omega$, $L_{lkq2} = 0.3361mH$, $L_d = 2.8mH$, $r_{fd} = 0.0005\Omega$, $L_{lfd} = 0.6691mH$, $r_{kd} = 0.01736\Omega$, $L_{lkd} = 0.5226mH$.

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Efficient Explicit Representation of AC Machines Main Flux Saturation in State-Variable-Based Transient Simulation Packages

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***Abstract*—Inclusion of magnetic saturation in low-order *qd* models of induction and synchronous machines improves the accuracy of system studies. Over the years, numerous explicit and implicit methods of representing magnetic saturation have been proposed in the literature, some of which find their use in commonly used transient simulation packages. This paper presents a straightforward and improved flux correction method of modeling saturation explicitly in classical *qd* state variable ac machine models. The approach requires limited additional information and avoids the need for a dynamic inductance, which is present in many other approaches. The method is first developed for induction machines and then extended to salient-pole synchronous machines as a more general case. Computer studies demonstrate the accuracy of the proposed method for predicting the dynamic and steady-state cross-saturation phenomena, as well as its computational advantages.**

***Index Terms*—AC machines, cross-saturation, dynamic simulation, explicit methods, induction machine, magnetic saturation, synchronous machine.**

I. Introduction

MODELING of ac electrical machines has been an active area of research for a long time. Depending on degree of detail and required accuracy for machine models, one may find rich literature on finite-element analysis [1], [2], magnetic equivalent circuit models [3], and lumped-parameter coupled equivalent circuit-based models [4], [12]–[31]. This paper is focused on the so-called general-purpose *qd* machine models [4]. Although such models are derived based on many assumptions, their accuracy is generally considered sufficient for system-level transient studies. Due to their simplicity, small number of required parameters, and relative computational efficiency, similar models are found in many commercial transient simulation packages and toolboxes including the nodal-analysis-based electromagnetic transient program (EMTP-type [5]–[8]) and state-variable-based simulators (SimPowerSystems (SPS) [9], PLECS [10], ASMG [11]) as built-in component models.

The modeling of magnetic saturation is a frequent optional addition to the basic ac machine models in commercial packages [5]–[10]. For induction machines, the saturation is typically applied to the main flux, therefore taking into account the so-called qd axes cross-saturation [4]–[10]. For synchronous machines, depending on the rotor construction, the saturation is sometimes applied to the d-axis only, [8], [9], [12], [13], to the q- and d-axes independently [5], [7], or to the main flux [5]–[7], [9], [10], [13]. As suggested by experiments [14] and finite-element analysis [2], the saturation of the main flux results in better accuracy [15]. For the qd salient-pole synchronous machine model, single-saturation factor [17], [18], [21], or two-saturation factor approaches [14], [15] can be utilized depending on the availability of the q-axis and/or intermediate directions of the saturation characteristic [16], [20]. However, in practice, only the d-axis saturation characteristic can be easily obtained experimentally [16]. Therefore, a single saliency factor approach is often considered sufficient for the general-purpose models [17]–[19].

This paper is focused on the general-purpose state-variable-based models of ac machines wherein a number of approaches have previously been developed to incorporate main flux saturation using a single-saturation factor methodology. A simple approach is to modify the equivalent steady-state magnetizing inductance according to the level of saturation at the previous time step (a time step relaxation) [21], [22]. The generalized flux space-vector approach [17], [18], [23], [24] incorporates the main flux saturation and accounts for the steady state and dynamic cross-saturation. Therein, for any set of independent state variables (with the exception of all winding flux linkages), a dynamic inductance is required (adding time-varying nonzeros in matrix formulation). An approximate model can be achieved by using certain state variables and ignoring these additional terms [25], [26], as is done in PLECS [10].

The original flux correction (FC) approach [4], [27]–[29] is based on first expressing the unsaturated magnetizing fluxes and then correcting them according to the saturation characteristic. The FC approach is very straightforward, easy to append to the unsaturated model (as an option that can be turned on/off), and is therefore very convenient to use in state-variable transient simulation programs. However, in its exact formulation, this approach results in implicit equations that require iterative

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solution at every time step. An approximate FC approach was proposed in [30], which is explicit but has reduced accuracy.

The goal of this paper is to present an FC approach that is both explicit and accurate, and due to its simplicity and computational efficiency, can be easily incorporated into state-variable transient simulation programs that are widely used by engineers and researchers in industry and academia. More specifically, the properties of the proposed FC approach and the contributions of this paper can be summarized as follows.

1) First, the FC approach [4], [27]–[29] is rederived for a symmetrical induction machine in explicit form without any approximations.
2) The FC approach is then extended to a salient rotor synchronous machine using the single saliency factor approach. Several formulations are presented including implicit, approximate, and explicit (with 2-D and 1-D lookup tables).
3) The accuracy of the proposed explicit FC method is verified against several established models in commonly used simulation packages. The proposed model is shown to accurately incorporate steady state and dynamic cross-saturation, as well as to offer numerical/computational advantages.

II. Induction Machines

A. Magnetically Linear Model

To set the stage for the derivation of the proposed approach and for completeness, the magnetically linear model is briefly described first. The electrical and mechanical dynamics of a general-purpose magnetically linear squirrel-cage induction machine can be represented by a set of differential equations [4]. For the purpose of this paper, all rotor variables are referred to the stator side and the variables are expressed in the arbitrary reference frame. The motor sign convention is used. Throughout this paper, the operator p denotes the derivative with respect to time. Neglecting zero sequence, the voltage equations are as follows [4]:

$$v_{qs} = r_s i_{qs} + \omega\lambda_{ds} + p\lambda_{qs} \tag{1}$$

$$v_{ds} = r_s i_{ds} - \omega\lambda_{qs} + p\lambda_{ds} \tag{2}$$

$$0 = r_r i_{qr} + (\omega - \omega_r)\,\lambda_{dr} + p\lambda_{qr} \tag{3}$$

$$0 = r_r i_{dr} - (\omega - \omega_r)\,\lambda_{qr} + p\lambda_{dr}. \tag{4}$$

The corresponding flux linkage equations are

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \tag{5}$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \tag{6}$$

$$\lambda_{qr} = L_{lr} i_{qr} + \lambda_{mq} \tag{7}$$

$$\lambda_{dr} = L_{lr} i_{dr} + \lambda_{md}. \tag{8}$$

For the unsaturated machine, the magnetizing fluxes are

$$\lambda_{mq} = L_{mu} i_{mq} = L_{mu}\,(i_{qs} + i_{qr}) \tag{9}$$

$$\lambda_{md} = L_{mu} i_{md} = L_{mu}\,(i_{ds} + i_{dr}), \tag{10}$$

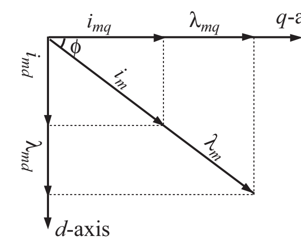

Fig. 1. Projections of the main flux linkage and magnetizing current onto qd-axes for symmetrical induction machines.

where L_{mu} is the unsaturated magnetizing inductance. The developed electromagnetic torque T_e, mechanical load torque T_m, and rotor speed ω_r are related by

$$T_e = \left(\frac{3}{2}\right)\left(\frac{P}{2}\right)(\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}) \tag{11}$$

$$T_e - T_m = J\left(\frac{2}{P}\right) p\omega_r \tag{12}$$

where P and J denote the number of poles and combined rotor-load inertia, respectively.

B. Representation of Main Flux Saturation

For smooth air-gap symmetrical induction machines, the main flux λ_m and magnetizing current i_m are aligned and are related to their qd-axes projections as depicted in Fig. 1 and expressed by the following:

$$\lambda_m = \sqrt{\lambda_{mq}^2 + \lambda_{md}^2} \tag{13}$$

$$i_m = \sqrt{i_{mq}^2 + i_{md}^2}. \tag{14}$$

The main flux saturation characteristic defines the nonlinear relationship between λ_m and i_m. It is also important to point out that, due to the choice of scaling in the reference frame transformation [4], and throughout this paper, the variables λ_m and i_m refer to peak values. The saturation characteristic can be generated experimentally from a standard no-load test. Specifically, neglecting stator and core losses, the measured rms stator line currents $I_{as,\mathrm{NL}}$ and rms line-to-neutral voltages $V_{as,\mathrm{NL}}$ are related to λ_m and i_m by

$$\lambda_m \approx \sqrt{2}\frac{(V_{as,\mathrm{NL}} - \omega_e I_{as,\mathrm{NL}} L_{ls})}{\omega_b} \tag{15}$$

$$i_m \approx \sqrt{2} I_{as,\mathrm{NL}} \tag{16}$$

where ω_e and ω_b are the operating and base frequencies, respectively. Thereafter, curve-fitting techniques such as piecewise linear interpolates, polynomials, arctangent functions [31], etc., can be used to represent the saturation characteristic. Regardless of the method, for the purpose of this paper, this characteristic is represented by the function (lookup table)

$$i_m = F(\lambda_m) \tag{17}$$

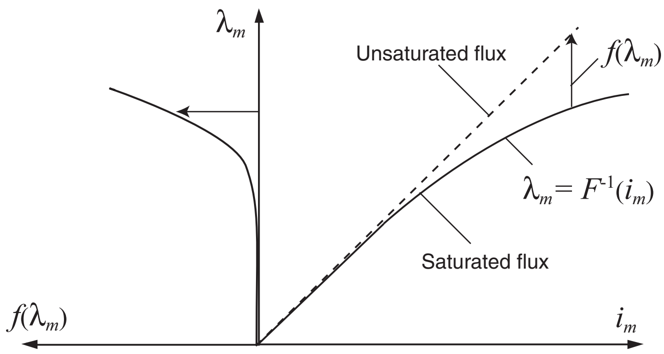

Fig. 2. Main flux saturation characteristic and FC function.

or conversely,

$$\lambda_m = F^{-1}(i_m). \tag{18}$$

The relationship between the saturated main flux λ_m and magnetizing current i_m is qualitatively depicted in Fig. 2 (right-hand side), wherein an unsaturated flux is also shown as a straight line.

C. Implicit FC Method

The basic idea of the FC approach is to define the correction function $f(\lambda_m)$ as the difference between the saturated and unsaturated curves as shown in Fig. 2 [4]. Extending this approach to qd-axes, the corrected (saturated) flux linkages in each axis become

$$\lambda_{mq} = L_{mu}(i_{qs} + i_{qr}) - f_q(\lambda_m) \tag{19}$$

$$\lambda_{md} = L_{mu}(i_{ds} + i_{dr}) - f_d(\lambda_m) \tag{20}$$

where for a symmetrical induction machine

$$f_q(\lambda_m) = \frac{\lambda_{mq}}{\lambda_m} f(\lambda_m) \tag{21}$$

$$f_d(\lambda_m) = \frac{\lambda_{md}}{\lambda_m} f(\lambda_m). \tag{22}$$

Substituting (5), (7), (21) in (19) and (6), (8), (22) in (20), and solving for λ_{mq} and λ_{md} yields

$$\lambda_{mq} = L_{aqu}\left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}}\right) - \frac{L_{aq}}{L_{mu}}\frac{\lambda_{mq}}{\lambda_m} f(\lambda_m) \tag{23}$$

$$\lambda_{md} = L_{adu}\left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}}\right) - \frac{L_{ad}}{L_{mu}}\frac{\lambda_{md}}{\lambda_m} f(\lambda_m) \tag{24}$$

where

$$L_{aqu} = L_{adu} = \left(\frac{1}{L_{mu}} + \frac{1}{L_{ls}} + \frac{1}{L_{lr}}\right)^{-1}. \tag{25}$$

Replacing (9) and (10) by (23) and (24), respectively, the classical qd induction machine model now takes into account main flux saturation in steady-state and dynamic sense exactly as defined by (17) and (18). However, λ_{mq} and λ_{md}, which are not state variables, are dependent on $f(\lambda_m)$. Therefore, (23) and (24) are implicit nonlinear equations requiring iterative solution (e.g., Newton–Raphson) and additional computing time. Although some software packages (e.g., Simulink [32]) may have the capability to solve implicit equations, it is still very desirable to reformulate the saturation model such that it requires only explicit and straightforward to calculate equations, improving the computational efficiency.

D. Explicit FC Method

It is possible to reformulate the FC method by introducing intermediate variables. In particular, considering the q-axis first, substituting (5) and (7) in (9), i_{mq} can be rewritten as a function of state variables (λ_{qs} and λ_{qr}) and the magnetizing flux linkage λ_{mq} as

$$i_{mq} = \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}}\right) - \left(\frac{\lambda_{mq}}{L_{ls}} + \frac{\lambda_{mq}}{L_{lr}}\right). \tag{26}$$

Similarly, for the d-axis, substituting (6) and (8) in (10), i_{md} is rewritten as

$$i_{md} = \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}}\right) - \left(\frac{\lambda_{md}}{L_{ls}} + \frac{\lambda_{md}}{L_{lr}}\right). \tag{27}$$

Defining the intermediate variables i_{zq} and i_{zd}, (26) and (27) can be simplified to

$$i_{zq} \equiv \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}}\right) = i_{mq} + \frac{\lambda_{mq}}{L_{\Sigma l}} \tag{28}$$

$$i_{zd} \equiv \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}}\right) = i_{md} + \frac{\lambda_{md}}{L_{\Sigma l}} \tag{29}$$

where

$$L_{\Sigma l} = \left(\frac{1}{L_{ls}} + \frac{1}{L_{lr}}\right)^{-1}. \tag{30}$$

Based on the diagram in Fig. 1 and (28), (29), we have the following relationships:

$$\cos(\phi) = \frac{i_{mq} + \lambda_{mq}/L_{\Sigma l}}{i_m + \lambda_m/L_{\Sigma l}} = \frac{i_{zq}}{i_m + \lambda_m/L_{\Sigma l}} \tag{31}$$

$$\sin(\phi) = \frac{i_{md} + \lambda_{md}/L_{\Sigma l}}{i_m + \lambda_m/L_{\Sigma l}} = \frac{i_{zd}}{i_m + \lambda_m/L_{\Sigma l}}. \tag{32}$$

Applying a basic trigonometric identity to (31) and (32), after some manipulations, we obtain

$$\left(\frac{i_{zq}}{i_m + \lambda_m/L_{\Sigma l}}\right)^2 + \left(\frac{i_{zd}}{i_m + \lambda_m/L_{\Sigma l}}\right)^2 = 1 \tag{33}$$

and

$$i_z \equiv \sqrt{i_{zq}^2 + i_{zd}^2} = i_m + \lambda_m/L_{\Sigma l}. \tag{34}$$

It is now clear that there is a unique mapping between the state-variable-dependent i_z [see (28) and (29)] and the pair of (i_m, λ_m) [see (34)]. Moreover, based on (34) and (17), it is possible to define the explicit relationship between i_z and saturated flux λ_m. This new function is denoted as

$$\lambda_m = G(i_z) \tag{35}$$

which can be readily calculated from the measured data points (17) using (34), and subsequently implemented using a 1-D lookup table. While it is not necessary for the simulation, the

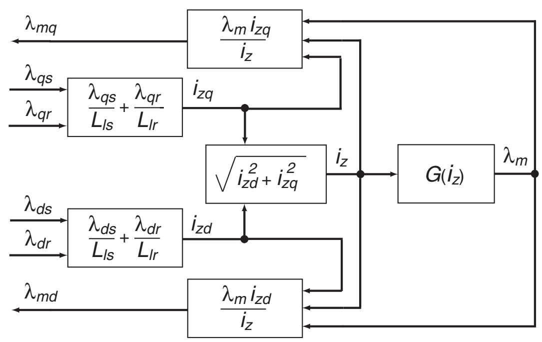

Fig. 3. Implementation of induction machine main flux saturation using the explicit FC approach.

magnetizing current i_m can also be evaluated using (34), if desired. Finally, using (34) and Fig. 1, the resulting saturated magnetizing fluxes in the q- and d-axes can be computed using

$$\lambda_{mq} = \lambda_m \frac{i_{zq}}{i_z} \text{ and } \lambda_{md} = \lambda_m \frac{i_{zd}}{i_z}. \tag{36}$$

Replacing (9) and (10) with (36) yields another method of accounting for main flux saturation in the classical qd model. A block diagram depicting all algebraic equations for this implementation of the FC method is shown in Fig. 3. As it can be seen, as opposed to the method described in Section II-C, here λ_{mq} and λ_{md} are defined in terms of intermediate variables i_{zq} and i_{zd}, which are computed from the state variables according to (28) and (29), yielding an explicit formulation.

To better understand the proposed approach, the function (35) and the saturation characteristic (18) for a 50-HP induction machine [4], [34] are plotted in Fig. 4(a), where we observe that $G(i_z)$ appears very linear, with only a small change in the slope in the saturated region. For clarity, this segment is magnified and plotted in Fig. 4(b). To explain this phenomenon, (9) and (10) are first rewritten for a saturated machine

$$\lambda_{mq} = L_{ms} i_{mq} = L_{ms}(i_{qs} + i_{qr}) \tag{37}$$

$$\lambda_{md} = L_{ms} i_{md} = L_{ms}(i_{ds} + i_{dr}) \tag{38}$$

where L_{ms} is the saturated magnetizing inductance, which depends on λ_m. Solving for i_{qs}, i_{ds}, i_{qr}, and i_{dr} in (5)–(8) and substituting the results into (37), (38), we obtain

$$\lambda_{mq} = L_{aqs}\left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{qr}}{L_{lr}}\right) = L_{aqs} i_{zq} \tag{39}$$

$$\lambda_{md} = L_{ads}\left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{dr}}{L_{lr}}\right) = L_{ads} i_{zd} \tag{40}$$

where

$$L_{aqs} = L_{ads} = \left(\frac{1}{L_{ms}} + \frac{1}{L_{ls}} + \frac{1}{L_{lr}}\right)^{-1}. \tag{41}$$

Therefore, $G(i_z)$ is equivalent to the product of L_{aqs} (or L_{ads}) and i_z. According to (41), L_{aqs} and L_{ads} appear as the parallel combination of the leakage inductances and the saturation-dependent magnetizing inductance. Since the magnetizing inductance of a rotating machine is typically much larger than its leakage inductances, L_{aqs} and L_{ads} will change

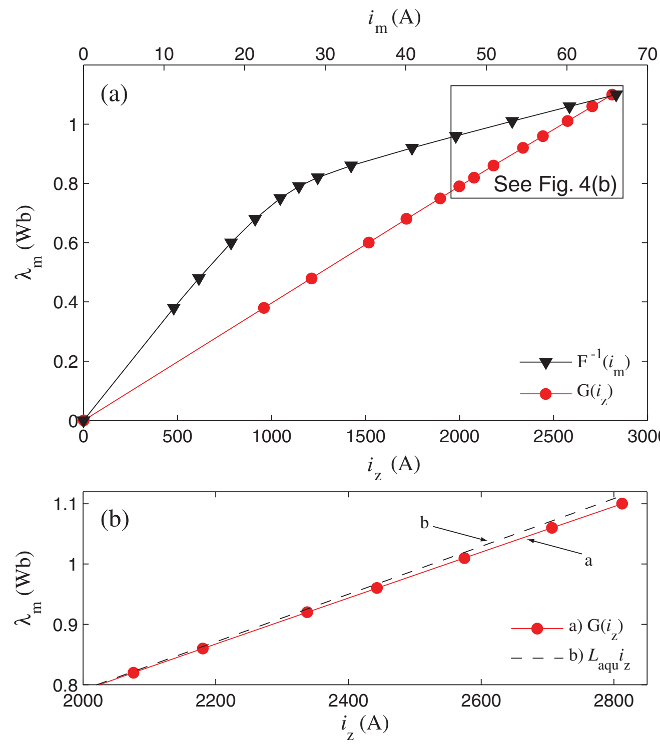

Fig. 4. (a) Saturation characteristic $F^{-1}(i_m)$ and equivalent explicit function $G(i_z)$ for a 50-HP induction machine [4], [34]; and (b) magnified view of $G(i_z)$ in the saturation region.

very little from no saturation to full saturation. As a result, it may visually appear that $G(i_z)$ remains almost constant through the operating region. However, the detailed plot in Fig. 4(b) shows a small slope with respect to a straight line produced by using the constant L_{aqu} or L_{adu}.

III. Synchronous Machines

A. Magnetically Linear Model

To extend the concept developed in Section II-D, a general-purpose qd synchronous machine model [4] is now considered. Without loss of generality, we consider one d-axis field winding fd, one d-axis damper winding kd, and two q-axis damper windings $kq1$ and $kq2$, respectively. Either round-rotor or salient-pole machines can be considered (depending on the magnetizing inductances L_{mq} and L_{md}), and if needed, damper windings can easily be added or eliminated. All rotor variables are referred to the stator side, and motor sign convention is used herein. Throughout this section, the rotor reference frame is assumed, whereas the superscript "r" is omitted for notational compactness. The corresponding voltage equations are

$$v_{qs} = r_s i_{qs} + \omega_r \lambda_{ds} + p\lambda_{qs} \tag{42}$$

$$v_{ds} = r_s i_{ds} - \omega_r \lambda_{qs} + p\lambda_{ds} \tag{43}$$

$$v_{0s} = r_s i_{0s} + p\lambda_{0s} \tag{44}$$

$$v_{kq1} = r_{kq1} i_{kq1} + p\lambda_{kq1} \text{ and } v_{kq2} = r_{kq2} i_{kq2} + p\lambda_{kq2} \tag{45}$$

$$v_{fd} = r_{fd} i_{fd} + p\lambda_{fd} \text{ and } v_{kd} = r_{kd} i_{kd} + p\lambda_{kd}. \tag{46}$$

The flux linkage equations are

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \tag{47}$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \tag{48}$$

$$\lambda_{0s} = L_{ls} i_{0s} \tag{49}$$

$$\lambda_{kq1} = L_{lkq1} i_{kq1} + \lambda_{mq} \text{ and } \lambda_{kq2} = L_{lkq2} i_{kq2} + \lambda_{mq} \tag{50}$$

$$\lambda_{fd} = L_{lfd} i_{fd} + \lambda_{md} \text{ and } \lambda_{kd} = L_{lkd} i_{kd} + \lambda_{md} \tag{51}$$

where the unsaturated magnetizing fluxes are given by

$$\lambda_{mq} = L_{mqu} i_{mq} = L_{mqu} \left(i_{qs} + i_{kq1} + i_{kq2} \right) \tag{52}$$

$$\lambda_{md} = L_{mdu} i_{md} = L_{mdu} \left(i_{ds} + i_{fd} + i_{kd} \right). \tag{53}$$

The electromagnetic torque and mechanical equations are given by (11) and (12). The rotor angle is defined as

$$\delta = \theta_r - \theta_e = \int_0^t \left(\omega_r - \omega_e \right) dt \tag{54}$$

where θ_r and θ_e are the rotor and bus voltage angles, respectively, and ω_e is the bus voltage (operating) frequency.

B. Saliency Factor Approach for Main Flux Saturation

Herein, it is assumed that the d-axis saturation characteristic of synchronous machines can be obtained experimentally under steady-state open-circuit conditions by recording the measured pairs of the field current I_{fd} (properly referred/converted to the stator side) and rms stator line-to-neutral voltages $V_{as,\mathrm{NL}}$. Based on these measurements, assuming that the core losses can be neglected, the values for the magnetizing flux λ_{md} and current i_{md} can be calculated as follows:

$$\lambda_{md} = \sqrt{2} \frac{V_{as,\mathrm{NL}}}{\omega_b} \tag{55}$$

$$i_{md} \approx I_{fd}. \tag{56}$$

As before, the variable λ_{md} refers to the peak values of the flux. Thereafter, the d-axis saturation characteristic can be defined by the function

$$i_{md} = F_d \left(\lambda_{md} \right). \tag{57}$$

Due to the shape of the salient-pole synchronous machine air gap, the q-axis saturation characteristic is also required to adequately model the main flux saturation. However, since this information is not typically provided by the manufacturers and the procedures for obtaining it experimentally are not simple [15], [16], [20], for the purpose of system-level studies, a common methodology is to use a saliency factor approach where scaling is applied to the q-axis in order to obtain an equivalent isotropic machine [17], [18], [21]. Assuming the q- and d-axes saturate at the same level, the saliency factor m is defined as follows:

$$m = \sqrt{\frac{L_{mqu}}{L_{mdu}}} = \sqrt{\frac{L_{mqs}}{L_{mds}}} \tag{58}$$

where $L_{mqu}, L_{mdu}, L_{mqs}$, and L_{mds} are the unsaturated and saturated q- and d-axis magnetizing inductances, respectively.

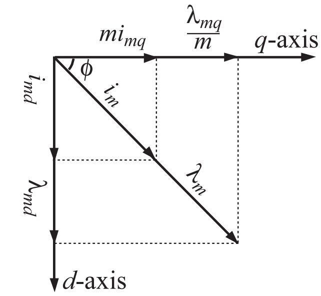

Fig. 5. Projections of the main flux linkage and magnetizing current onto qd-axes for synchronous machines considering the saliency factor approach.

The main flux and magnetizing current are then calculated according to the diagram depicted in Fig. 5 as

$$\lambda_m = \sqrt{\left(\frac{\lambda_{mq}}{m} \right)^2 + \lambda_{md}^2} \tag{59}$$

$$i_m = \sqrt{\left(i_{mq} m \right)^2 + i_{md}^2}. \tag{60}$$

After this rescaling, the d-axis saturation function (57) is applied to the main flux saturation as

$$i_m = F_d \left(\lambda_m \right) \tag{61}$$

or conversely

$$\lambda_m = F_d^{-1} \left(i_m \right). \tag{62}$$

As it can be seen from (58)–(60), this formulation automatically covers both salient-pole and round-rotor machines. For the latter case, $m = 1$, and the equations become similar to those presented in Section II-B.

C. Implicit FC Method

For the implementation of saturation in the model, the FC function $f(\lambda_m)$ depicted in Fig. 2 is redefined using the synchronous machine saturation characteristic (61). It is then possible to write the corrected (saturated) magnetizing flux for each axis as

$$\lambda_{mq} = L_{mqu} (i_{qs} + i_{kq1} + i_{kq2}) - m \cdot f_q \left(\lambda_m \right) \tag{63}$$

$$\lambda_{md} = L_{mdu} (i_{ds} + i_{fd} + i_{kd}) - f_d \left(\lambda_m \right) \tag{64}$$

where

$$f_q \left(\lambda_m \right) = \frac{\lambda_{mq}}{m \cdot \lambda_m} f \left(\lambda_m \right) \tag{65}$$

$$f_d \left(\lambda_m \right) = \frac{\lambda_{md}}{\lambda_m} f \left(\lambda_m \right). \tag{66}$$

Substituting (47), (50), and (65) into (63) and (48), (51), and (66) into (64), and solving for λ_{mq} and λ_{md} yields

$$\lambda_{mq} = L_{aqu} \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}} \right) - \frac{L_{aq}}{L_{mqu}} \frac{\lambda_{mq}}{\lambda_m} f \left(\lambda_m \right) \tag{67}$$

$$\lambda_{md} = L_{adu}\left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}}\right) - \frac{L_{ad}}{L_{mdu}}\frac{\lambda_{md}}{\lambda_m} f(\lambda_m) \tag{68}$$

where

$$L_{aqu} = \left(\frac{1}{L_{mqu}} + \frac{1}{L_{ls}} + \frac{1}{L_{lkq1}} + \frac{1}{L_{lkq2}}\right)^{-1} \tag{69}$$

$$L_{adu} = \left(\frac{1}{L_{mdu}} + \frac{1}{L_{ls}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lkd}}\right)^{-1}. \tag{70}$$

Replacing (52) and (53) by (67) and (68), respectively, the general-purpose qd synchronous machine model now takes into account the main flux saturation for both salient- and round-rotor construction by correcting the unsaturated flux using the implicit algebraic equations (67) and (68).

D. Explicit FC Method

The methodology to achieve explicit FC formulation involves similar steps to those described in Section II-D. In particular, considering the q-axis first, substituting (47) and (50) into (52), i_{mq} is expressed as a function of state variables ($\lambda_{qs}, \lambda_{kq1}$, and λ_{kq2}) and λ_{mq}. For the d-axis, substituting (48) and (51) into (53), i_{md} is rewritten as a function of state variables ($\lambda_{ds}, \lambda_{fd}$, and λ_{kd}) and λ_{md}. The result is

$$i_{mq} = \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}}\right) - \left(\frac{\lambda_{mq}}{L_{ls}} + \frac{\lambda_{mq}}{L_{lkq1}} + \frac{\lambda_{mq}}{L_{lkq2}}\right) \tag{71}$$

$$i_{md} = \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}}\right) - \left(\frac{\lambda_{md}}{L_{ls}} + \frac{\lambda_{md}}{L_{lfd}} + \frac{\lambda_{md}}{L_{lkd}}\right). \tag{72}$$

Introducing the intermediate variables i_{zq} and i_{zd}, (71) and (72) are simplified to

$$i_{zq} \equiv \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}}\right) = i_{mq} + \frac{\lambda_{mq}}{L_{\Sigma lq}} \tag{73}$$

$$i_{zd} \equiv \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}}\right) = i_{md} + \frac{\lambda_{md}}{L_{\Sigma ld}} \tag{74}$$

where

$$L_{\Sigma lq} = \left(\frac{1}{L_{ls}} + \frac{1}{L_{lkq1}} + \frac{1}{L_{lkq2}}\right)^{-1} \tag{75}$$

$$L_{\Sigma ld} = \left(\frac{1}{L_{ls}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lkd}}\right)^{-1}. \tag{76}$$

It can be observed that generally, $L_{\Sigma lq} \neq L_{\Sigma ld}$. From the vector diagram in Fig. 5 and (73), (74), we obtain

$$\cos(\phi) = \frac{i_{mq} + \lambda_{mq}/L_{\Sigma lq}}{i_m/m + m\lambda_m/L_{\Sigma lq}} = \frac{i_{zq}}{i_m/m + m\lambda_m/L_{\Sigma lq}} \tag{77}$$

$$\sin(\phi) = \frac{i_{md} + \lambda_{md}/L_{\Sigma ld}}{i_m + \lambda_m/L_{\Sigma ld}} = \frac{i_{zd}}{i_m + \lambda_m/L_{\Sigma ld}}. \tag{78}$$

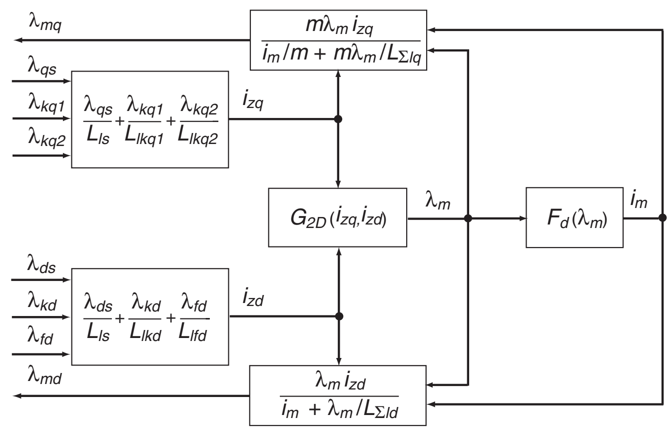

Fig. 6. Explicit implementation of synchronous machine saturation using the FC approach with a 2-D lookup table.

Applying a trigonometric identity to (77) and (78), we have

$$\left(\frac{i_{zq}}{i_m/m + m\lambda_m/L_{\Sigma lq}}\right)^2 + \left(\frac{i_{zd}}{i_m + \lambda_m/L_{\Sigma ld}}\right)^2 = 1. \tag{79}$$

Unlike in (33), the denominators of the two left-hand side terms in (79) are different due to the saliency factor m and the inequality between $L_{\Sigma lq}$ and $L_{\Sigma ld}$. Therefore, unlike it was for an induction machine in Section II-D, (79) cannot be solved easily for i_m and λ_m. Consequently, several methods to obtain the desirable explicit formulation are described here.

1) Two-Dimensional Explicit FC: Based on (73), (74), and (79), it can be observed that for any pair of intermediate variables (i_{zq}, i_{zd}), there exists a unique pair of (i_m, λ_m), as they are related by the saturation function. This unique mapping can be implemented using a 2-D lookup table with i_{zq} and i_{zd} as the inputs and λ_m as the output. Symbolically,

$$\lambda_m = G_{2\mathrm{D}}(i_{zq}, i_{zd}). \tag{80}$$

Assuming that this function (lookup table) is obtained, the corresponding explicit formulation can be realized according to Fig. 6. Therein, the intermediate variables i_{zq} and i_{zd} are calculated from the state variables (fluxes). Then, the magnetizing flux λ_m and current i_m are evaluated using the lookup tables (80) and (61), respectively. The q-axis and d-axes magnetizing fluxes are computed using a combination of (77), (79), and the projections depicted in Fig. 5 as follows:

$$\lambda_{mq} = \frac{m\lambda_m i_{zq}}{i_m/m + m\lambda_m/L_{\Sigma lq}} \text{ and } \lambda_{md} = \frac{\lambda_m i_{zd}}{i_m + \lambda_m/L_{\Sigma ld}}. \tag{81}$$

The disadvantages of this method are twofold. First, not all simulation packages provide a convenient way of implementing multidimensional lookup tables. The second disadvantage relates to the difficulty of generating $G_{2\mathrm{D}}(i_{zq}, i_{zd})$, which is done during the initialization stage. Assuming that the vectors i_{zq} and i_{zd} have a and b components, respectively, $a \times b$ values of λ_m must be calculated by solving the nonlinear equations that include (80) using an iterative approach, which may be time-consuming.

2) Approximate FC: An explicit but approximate FC method using only 1-D lookup tables was presented in [30]. For completeness, its main steps are briefly summarized here. In particular, algebraically manipulating (79) yields

$$i_z \equiv \sqrt{i_{zd}^2 + k^2 i_{zq}^2} = i_m + \lambda_m / L_{\Sigma ld} \tag{82}$$

where

$$k = \left(\frac{i_m + \lambda_m / L_{\Sigma ld}}{i_m / m + m\lambda_m / L_{\Sigma lq}} \right). \tag{83}$$

It is observed that due to the saliency factor and the asymmetry of the equivalent leakage inductances, the coefficient k is function of λ_m, making the underlying equations implicit. However, assuming $1/L_{\Sigma lq} >> 1/L_{mq}$ and $1/L_{\Sigma ld} >> 1/L_{md}$, (83) can be simplified to obtain the following approximate constant coefficient:

$$\tilde{k} = \left(\frac{L_{\Sigma lq}}{m L_{\Sigma ld}} \right). \tag{84}$$

Using (84), the intermediate variable i_z is reformulated as

$$i_z \equiv \sqrt{i_{zd}^2 + \tilde{k}^2 i_{zq}^2} \approx i_m + \lambda_m / L_{\Sigma ld} \tag{85}$$

from which a unique mapping between i_z and λ_m becomes apparent. Symbolically, this new mapping defines a function

$$\lambda_m = G\left(i_z\right). \tag{86}$$

This new lookup table (86) along with (85) is used to replace the 2-D lookup table in Fig. 6, which constitutes the approximate explicit implementation.

3) One-Dimensional Explicit FC: To avoid the errors introduced by (84) and (85) [30], a corrected methodology is proposed here. First, it is noted that (82) and (61) capture the unique relation between λ_m and i_z. Using (83), this can first be extended to obtain a mapping between k and i_z, herein defined as

$$k = H(i_z) \tag{87}$$

which can easily be realized as a 1-D lookup table. However, (87) cannot be used directly since the definition of i_z in (82) is also a function of k, which would again result in an implicit formulation. Instead, the approach presented here uses the unsaturated constant coefficient defined as

$$k_u = \left(\frac{i_{mu} + \lambda_{mu} / L_{\Sigma ld}}{i_{mu} / m + m\lambda_{mu} / L_{\Sigma lq}} \right) \tag{88}$$

where (λ_{mu}, i_{mu}) is usually the first pair of the data (λ_m, i_m) given by (61). Then, another intermediate variable i_z^* is calculated using

$$i_z^* = \sqrt{i_{zd}^2 + k_u^2 i_{zq}^2}. \tag{89}$$

If the flux λ_m is in the unsaturated region, the variable i_z^* will be close to the exact i_z as defined by (82). However, in the saturated region and more generally, it will not be the case. Thus, another step is needed, which is achieved by evaluating k as follows:

$$k = H(i_z^*). \tag{90}$$

It is then possible to compute i_z using (82) with this new k. After that, the saturated magnetizing flux λ_m is computed with (86), and the projections λ_{mq} and λ_{md} are calculated using (61) and (81).

A block diagram that shows the implementation of the proposed explicit FC approach for modeling saturation in synchronous machines is depicted in Fig. 7. As it can be seen, this approach requires three 1-D lookup tables (or two lookup tables if (61) is a fitted function), but otherwise, the formulation is very straightforward and similar to that depicted in Fig. 6.

Finally, as it was done for the induction machine, the magnetizing fluxes can be expressed in terms of state variables. Replacing L_{mqu} and L_{mdu} in (52) and (53) by L_{mqs} and L_{mds}, respectively, and substituting (50) and (51), we obtain

$$\lambda_{mq} = L_{aqs} \left(\frac{\lambda_{qs}}{L_{ls}} + \frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}} \right) \equiv L_{aqs} i_{zq} \tag{91}$$

$$\lambda_{md} = L_{ads} \left(\frac{\lambda_{ds}}{L_{ls}} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}} \right) \equiv L_{ads} i_{zd} \tag{92}$$

where

$$L_{aqs} = \left(\frac{1}{L_{mqs}} + \frac{1}{L_{ls}} + \frac{1}{L_{lkq1}} + \frac{1}{L_{lkq2}} \right)^{-1} \tag{93}$$

$$L_{ads} = \left(\frac{1}{L_{mds}} + \frac{1}{L_{ls}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lkd}} \right)^{-1}. \tag{94}$$

Therefore, the relation between λ_{mq} and i_{zq}, and also the one between λ_{md} and i_{zd}, will again be almost linear—similar to that depicted in Fig. 4. Consequently, the function $G\left(i_z\right)$ defined for synchronous machines in (86) will also visually appear close to being linear due to the fact that L_{mqs} and L_{mds}, as defined by (93) and (94), will change very little from no to full saturation.

IV. Case Studies

Three case studies are presented in this section to validate the proposed explicit FC approach and compare the induction and synchronous machine models with the reference models. For consistency, all studies were executed on a personal computer with 2.8-GHz Intel CPU and 4 GB of RAM.

A. Induction Machine

For the purpose of validation, the proposed explicit FC induction machine model (see Section II-D) has been implemented in MATLAB/Simulink environment along with the implicit FC approach (see Section II-C) and the built-in *qd* induction machine models in commercial toolboxes SPS [9] and PLECS [10]. All these models except PLECS' use winding flux linkages as the state variables. It is also important to note that PLECS' model does not include dynamic cross-saturation based on the results of [26]. A reference solution is obtained by implementing the generalized flux space-vector approach (with $\mathbf{i}_s$ and $\boldsymbol{\lambda}_s$ as the state variables [24]) in the same environment, as this approach has been previously verified experimentally and is considered to possess sufficient accuracy for the general-purpose models considered in this paper. In the reference model [24], the dynamic

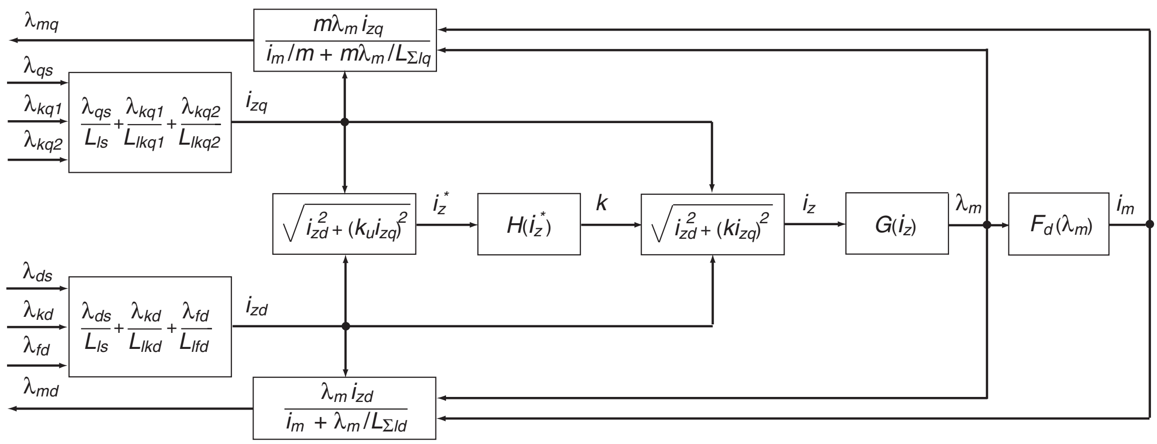

Fig. 7. Implementation of the proposed explicit FC approach for modeling saturation in synchronous machines using a saliency factor approach.

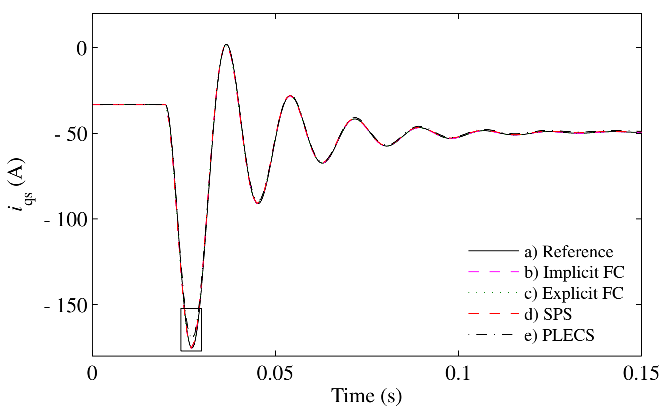

Fig. 8. Induction machine transient in stator current i_{qs} due to change in stator voltage.

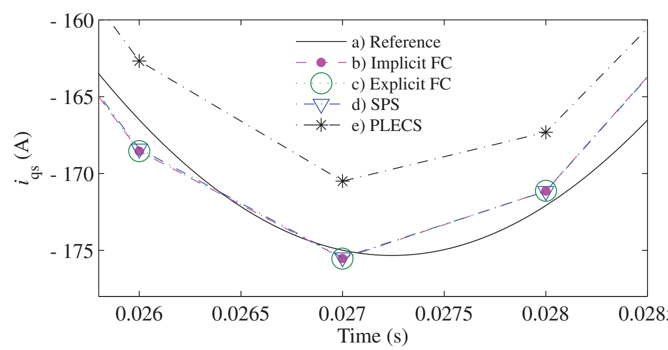

Fig. 9. Magnified view of the first peak of stator current i_{qs} transient due to change in stator voltage (induction machine).

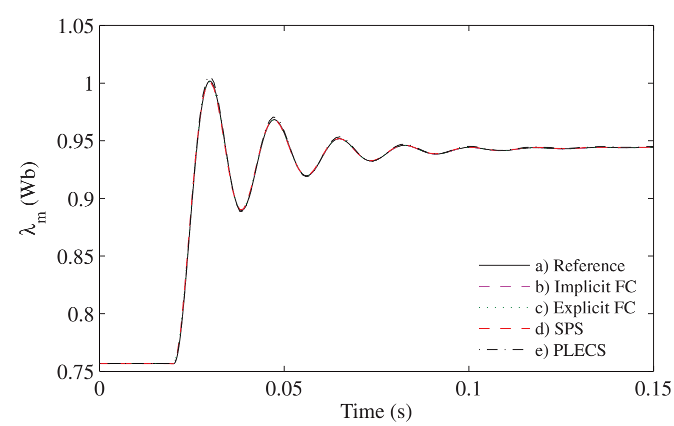

Fig. 10. Transient in main flux λ_m due to change in stator voltage (induction machine).

inductance is included, and is computed using Lagrange polynomial interpolation for nonuniformly spaced points [33]. To obtain a very accurate reference solution, the reference model was solved using the ode5 solver (Dormand–Prince) with a very small fixed time step of 1 μs. For comparison, all other models were solved using the same solver but with a larger time step of 1 ms. For consistency, all models were implemented in the synchronous reference frame.

1) Change of Stator Voltage: In this study, a 50-HP induction motor [4] (see Appendix A) is connected to an infinite bus. Initially, the machine operates in a steady state with 150-N·m mechanical load. At $t = 0.02$ s, the bus voltage is increased from 0.8 to 1 p.u. This fast electrical transient should allow the observation of the static and dynamic cross-saturation effects.

The predicted transient responses of i_{qs}, λ_m, and ω_r are plotted in Figs. 8–11. As it can be seen in Fig. 8, the models that include the dynamic cross-saturation (implicit FC; explicit FC; and SPS) predict very similar responses in the q-axis stator current. A magnified view of the first peak in i_{qs} is shown in Fig. 9. Here, the small error between the reference model, the implicit FC, explicit FC, and SPS models is mostly due to the larger step. However, the PLECS model, where the dynamic cross-saturation is neglected, shows a small but noticeable error. This error vanishes in steady state. Similar observations can be made for the transient response in main flux λ_m shown in Fig. 10. Moreover, based on Figs. 4 and 10 (and the saturation data in Appendix A), it can also be observed that the machine operates significantly in the saturated region. Finally, Fig. 11 shows the transient rotor speed ω_r, which is accurately predicted by all models.

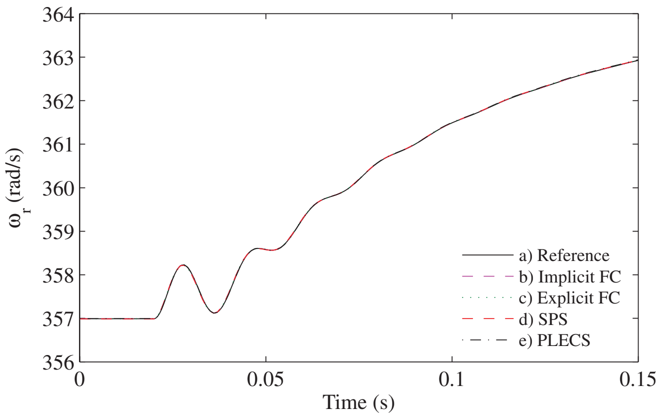

Fig. 11. Transient in electrical rotor speed ω_r due to change in stator voltage (induction machine).

B. Synchronous Machine

The proposed explicit FC synchronous machine model using 1-D lookup tables (see Section III-D3) has been implemented using MATLAB/Simulink along with the implicit FC (see Section III-C) and the approximate FC (see Section III-D2) models. For benchmarking with other existing models, the saturable synchronous machine model from the commercial toolbox SPS [9] has been implemented as well. As with the induction machine study, all these models use winding flux linkages as the state variables. The generalized flux space-vector method [18] (with $\mathbf{i}_s, i_{fd}$, and $\boldsymbol{\lambda}_m$ as the state variables) serves again as the reference model. The dynamic inductance of the reference model [18] is implemented as described in Section IV-A. The reference model is solved using the ode5 solver with a small time step of 10 μs. All other models were simulated with the same solver but using a step size of 1 ms.

To obtain a more general case, a salient-pole synchronous machine [4] has been considered. The machine parameters and saturation curve are summarized in Appendix B. To keep the focus on the accuracy of machine models, the subject machine is assumed to be connected to an infinite bus.

1) Change of Load Torque: In the following study, the machine is assumed to initially operate in a steady state under nominal excitation and terminal voltages in idle mode (without mechanical torque). At $t = 0.2$ s, the mechanical torque of 1.2 p.u. is applied. Following a relatively slow transient, the machine becomes heavily loaded and the projection of the main flux onto the q-axis and the effect of cross-saturation become significant. The corresponding transient responses in d-axis stator current i_{ds}, q-axis magnetizing flux λ_{mq}, rotor angle δ, and main flux λ_m predicted by various models are plotted in Figs. 12–16. As it can be seen in these figures, the SPS model predicts responses that are visibly different from the reference and all other methods because the SPS salient-pole synchronous machine model considers saturation along the d-axis only. Since this model neglects the q-axis saturation and cross-saturation, it therefore predicts higher rotor angle δ even in steady state, which results in higher flux projection in the q-axis λ_{mq} and higher d-axis stator current i_{ds}, respectively. All other models that utilize

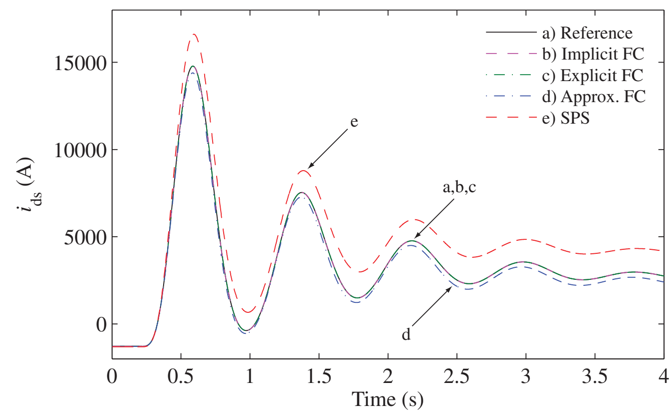

Fig. 12. Transient in stator current i_{ds} due to change of load torque (synchronous machine).

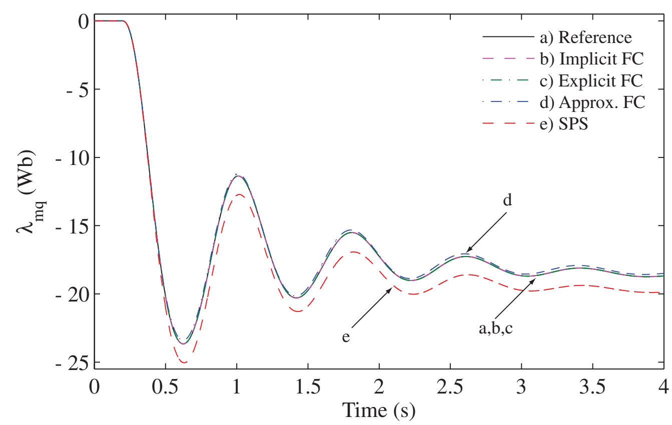

Fig. 13. Transient in magnetizing flux λ_{mq} due to change of load torque (synchronous machine).

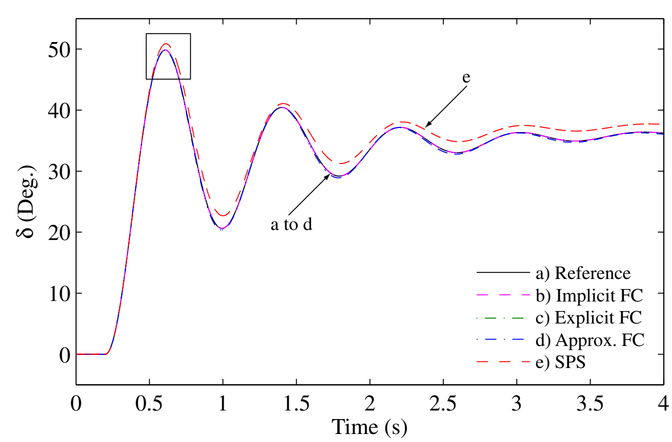

Fig. 14. Transient in rotor angle δ due to change of load torque (synchronous machine).

TABLE I
2-NORM RELATIVE ERRORS OF i_{ds}, λ_{mq}, AND δ IN CHANGE OF LOAD TORQUE TRANSIENT STUDY (SYNCHRONOUS MACHINE)

Variable	Imp. FC	Exp. FC	App. FC	SPS
i_{ds}	0.0789 %	0.0715 %	5.752 %	27.43 %
λ_{mq}	0.0173 %	0.0166 %	0.954 %	6.514 %
δ	0.0203 %	0.0198 %	0.551 %	3.964 %

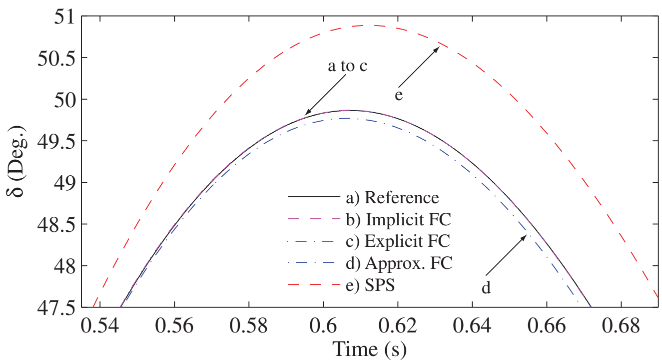

Fig. 15. Magnified view of the first peak of rotor angle δ transient due to change of load torque (synchronous machine).

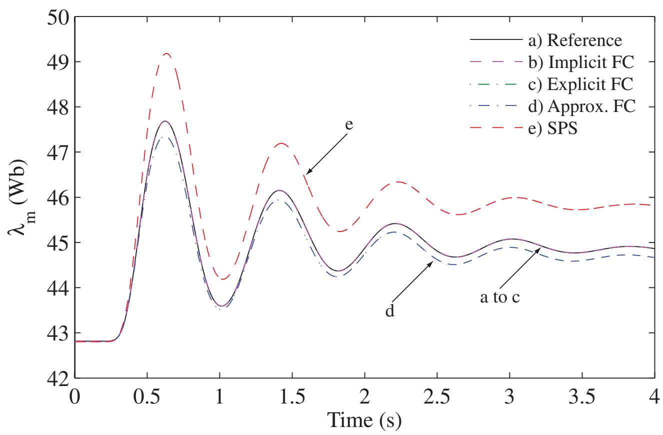

Fig. 16. Transient in main flux λ_m due to change of load torque (synchronous machine).

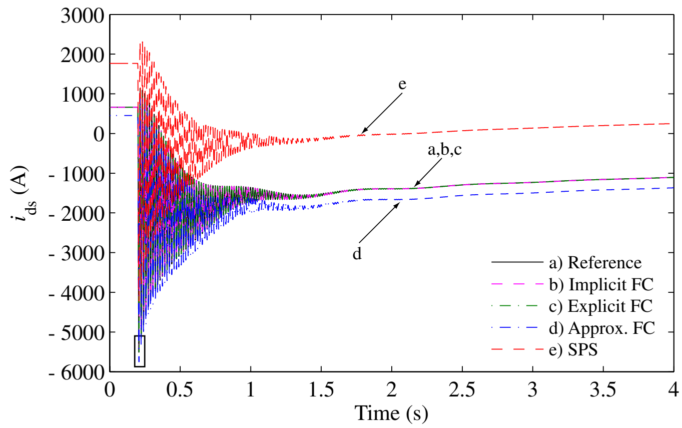

Fig. 17. Transient in stator current i_{ds} due to change of stator voltage (synchronous machine).

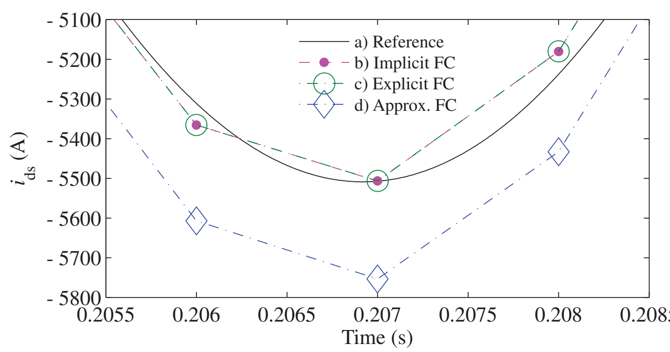

Fig. 18. Magnified view of the first peak of stator current i_{ds} transient due to change of stator voltage (synchronous machine).

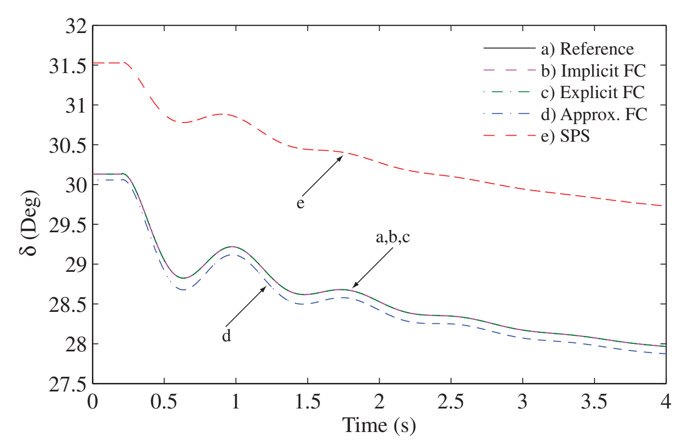

Fig. 19. Transient in rotor angle δ due to change of stator voltage (synchronous machine).

the FC approach are much closer to the reference solution, with the approximate FC method giving slightly different results. It can also be observed from Fig. 16 and the saturation data given in Appendix B that the machine is saturated in both steady state and transient. The 2-norm (cumulative) relative error [33] between the solution trajectories and the reference trajectory is used here to compare the accuracy achieved by the various models. Specifically, the 2-norm relative errors of i_{ds}, λ_{mq}, and δ with respect to the reference solution are summarized in Table I. As it can be seen in Table I, the SPS model produces by far the largest errors due to neglecting the q-axis saturation and cross-saturation. Next, the approximate FC method also gives noticeable errors (5.752%, 0.954%, and 0.551%, respectively) due to approximations in the calculation of i_z. The implicit FC method gives very small errors for all three variables (0.789%, 0.0173%, and 0.0203%, respectively). These errors are almost identical to the errors obtained with the proposed explicit FC approach, wherein the small differences are mostly caused by the large integration time step.

2) Change of Stator Voltage: In the following study, the machine is assumed to initially operate in a steady state under nominal excitation and terminal voltages, and rated mechanical torque. At $t = 0.2$ s, the stator terminal voltages are increased from 1 to 1.05 p.u., thus pushing the machine deeper into the saturated region. Similar fast electrical transients have been used in [17], [18], and [25] for investigating synchronous machine saturation and are considered very appropriate and informative. The corresponding transient responses produced by the considered models are shown in Figs. 17–19, wherein the stator current i_{ds} and the rotor angle δ are shown. For clarity, a magnified view of the first peak of i_{ds} from Fig. 17 is shown in Fig. 18, whereas the rotor angle δ is plotted in Fig. 19. As it can be seen in Figs. 17 and 19, the SPS model predicts different steady states before and after the transient as a consequence of neglecting

TABLE II
2-NORM RELATIVE ERRORS OF i_{ds} AND δ IN CHANGE OF STATOR VOLTAGE TRANSIENT STUDY (SYNCHRONOUS MACHINE)

Variable	Imp. FC	Exp. FC	App. FC	SPS
i_{ds}	1.588 %	1.584 %	19.47 %	96.86 %
δ	0.004 %	0.004 %	0.387 %	5.924 %

TABLE III
NUMERICAL EFFICIENCY OF THE MODELS FOR DIFFERENT SOLVERS

	Solver	Imp. FC	Exp. FC	App. FC	SPS
CPU time (s)	ode5	1.100	0.217	0.213	0.766
	ode15s	0.475	0.113	0.113	0.483
	ode45	0.387	0.084	0.083	0.403
No. of time steps	ode5	4001	4001	4001	4001
	ode15s	2624	2621	2621	2601
	ode45	915	906	905	881
No. of calls to sat. fun.	ode5	303 964	24 001	24 001	24 001
	ode15s	112 313	7969	7971	7899
	ode45	94 555	6338	6341	6167

the q-axis saturation and cross-saturation. The approximate FC model also shows noticeable differences in i_{ds} and δ with respect to the reference. At the same time, both explicit FC and implicit FC models predict results that are almost indistinguishable from the reference. The 2-norm relative errors with respect to the reference solution for this transient study are summarized in Table II, which shows that explicit FC and implicit FC models have almost identical small errors. Again, such small errors are mostly introduced by the larger time step.

V. COMPUTATIONAL PERFORMANCE

Presented studies demonstrate that both explicit FC and implicit FC models have the best accuracy in predicting the saturation phenomena during slow and fast transients even when using a fairly large time step of 1 ms. To investigate the computational (numerical) difference between the implicit and explicit FC formulations and the available models, the transient study of Section IV-B2 (change of synchronous machine stator voltage) is repeated using fixed-step (ode5) and variable-step (ode15s and ode45) solvers. For all variable-step solvers, the maximum and minimum time steps are 10^{-2} and 10^{-8} s, respectively, and the absolute and relative error tolerances are set to 10^{-4}. The CPU time, the number of time steps, and the number of times the saturation function was called for this study are summarized in Table III. When the fixed-step solver (ode5) is used, the number of calls to the saturation function for all explicit models (i.e., Exp. FC, App. FC, and SPS) is the same (24 001). However, the implicit FC model contains an algebraic loop that requires iterations and thus many more calls to the saturation function (303 964), resulting in the longest computing time of 1.1 s. The proposed explicit FC model runs as fast as the approximate FC model, and is about five times faster than the implicit FC model while achieving the same accuracy (see Table II). The same trend is observed when the variable-step solvers (ode15s and ode45) are used. Since the same error tolerances are used for all models, the number of time steps taken by each model when using ode15s (about 2600) and when using ode45 (about 900) is consistent and similar for each model. However, the implicit FC model requires about 12 to 15 times more calls to the saturation function, which results in slower simulation time. The SPS model is also explicit and takes approximately the same number of time steps and calls to the saturation function as the other explicit models (i.e., App. FC and Exp. FC). However, this model is slower (perhaps due to its interface/overhead) as well as less accurate (due to neglecting the q-axis saturation and cross-saturation). At the same time, the proposed explicit FC model preserves the numerical efficiency of the approximate FC and the accuracy of the implicit FC model while correctly including cross-saturation.

VI. DISCUSSION

A. Comparison With Generalized Flux Space Vector

To better understand the difference between the proposed FC and the generalized flux space-vector approaches, it is instructive to give a brief summary of the latter. In the generalized flux space-vector approach [17], [18], [23], [24], a flux space vector $\underline{\lambda}$, which may or may not physically exist, is defined as the linear combination of almost any set of independent state variables $\underline{x}_i$. Here, the underlined variables represent space vectors as in the original source references [17], [18], [23], [24]. This flux space vector is aligned with the magnetizing current $\underline{i}_m$ and both are related by

$$\underline{\lambda} = \Lambda \underline{i}_m \tag{95}$$

where Λ is a saturation-dependent equivalent inductance, which also may or may not have direct physical meaning.

To implement this approach in a state-variable-based environment, it is necessary to first write the state equations, i.e., (1)–(4) and (42)–(46) for induction and synchronous machines, respectively, in terms of state variables and inputs

$$\mathbf{A}\,(p\mathbf{x}) + \mathbf{B}\mathbf{x} = \mathbf{u} \tag{96}$$

where $\mathbf{u}$ is a vector of input, i.e., stator and rotor voltages. It is necessary to solve (96) for the state derivatives $p\mathbf{x}$ [15], [17], yielding the following standard state-space formulation:

$$p\mathbf{x} = -\mathbf{A}^{-1}\mathbf{B}\mathbf{x} + \mathbf{A}^{-1}\mathbf{u}. \tag{97}$$

On the one hand, for the specific case where all state variables are winding flux linkages (as in the proposed FC model), $\mathbf{A}$ becomes a diagonal matrix, which is easily inverted analytically. Moreover, the product term $\mathbf{A}^{-1}\mathbf{B}$ (which can also be defined analytically) will have the same number of nonzeros as $\mathbf{B}$. On the other hand, for any other choice of state variables, $\mathbf{A}$ will not be diagonal. More importantly, $\mathbf{A}$ will also be time varying due to the terms associated with the dynamic inductance. In a brute-force implementation, at every integration time step, it will be necessary to invert $\mathbf{A}$ and execute the matrix multiplication $\mathbf{A}^{-1}\mathbf{B}$ (which will result in a full matrix). Therefore, a significant overhead may result in the implementation and computation of (97) if state variables other than winding flux linkages are used.

The generalized flux space-vector approach [17], [18], [23], [24] is a systematic, theoretically sound, and experimentally verified method to obtain explicit saturated machine models using almost any set of independent state variables. By properly selecting the state variables, it is thus possible to minimize (or even eliminate) the number of algebraic equations.

It has also been shown in [17] that the number of times the time-varying dynamic inductance appears in $\mathbf{A}$ for a mixed current-flux state-space model may be lower than that for the traditional all winding current model. To simplify the model formulation, it is also possible to completely ignore the dynamic cross-saturation in the generalized flux space-vector approach [25], [26], rendering a constant (but not diagonal) matrix $\mathbf{A}$. The error due to such approximation may be large for most sets of state variables. However, in a few specific cases, the predicted transients were close to those obtained with the full algebraically exact model.

Based on these considerations, it appears that the model that uses only winding flux linkages as state variables is the most efficient and simplest to implement in commonly used simulation packages such as SPS, PLECS, etc. It was further shown in Sections IV and V that the proposed explicit FC method provides the best combination of efficiency and accuracy amongst the saturable models of the same type. The required saturation functions (see Figs. 3, 6, and 7) are also easy to append to the standard magnetically linear machine models, as these models use the same dynamic equations. This may be another practical advantage for commercial simulation programs where, depending on the required accuracy, the saturation may be turned ON/OFF by the users.

B. Generalized Current Space-Vector Approach

The proposed algebraically exact explicit FC induction machine model (see Section II-D) creates a fictitious current space vector $\underline{i}_z = i_{zd} + ji_{zq}$, which is a linear combination of the state variables, and is aligned with the main flux as

$$\Lambda \underline{i}_z = \underline{\lambda}_m. \tag{98}$$

The proposed explicit FC model may be considered a specific case of a generalized current space vector $(\underline{i}_z)$ concept, where $\underline{i}_z$ may or may not have direct physical connotation. Hence, this idea could be extended to different sets of independent state variables. However, for any choice of state variables other than winding flux linkages, a dynamic inductance would appear. In [24], it was shown that in the generalized flux space-vector approach, for some sets of state variables, $\underline{\lambda} \equiv \underline{\lambda}_m$, and consequently $\Lambda \equiv L_{ms}$. In these cases, according to (98), the generalized current and generalized flux space-vector approaches would yield the same exact set of equations. However, as shown in (39) and (40), for the pure winding flux linkage model, $\Lambda \neq L_{ms}$; therefore, the two approaches yield different (yet algebraically equivalent) equations.

VII. Conclusion

This paper presents a straightforward and improved flux correction method for modeling saturation explicitly in classical general-purpose *qd* state-variable models of induction and synchronous machines with flux linkages as state variables. The saturation characteristic is assumed to be fully defined by data obtained from simple experimental procedures, namely the no-load and open-circuit tests for induction and synchronous machines, respectively. The proposed explicit FC approach is compared to the implicit and, for synchronous machines, approximate formulations. Computer studies demonstrate that the proposed explicit FC method predicts the dynamic and steady-state cross-saturation phenomena with the same accuracy as the implicit FC approach, which represents an appreciable improvement over the approximate FC or the models that neglect the q-axis saturation and cross-saturation. At the same time, since the proposed formulation is explicit, it avoids the algebraic loop and iterative solution required by the implicit FC formulation, which represents a computational advantage. The proposed explicit FC method can therefore be used for efficient and accurate implementation of the general-purpose induction and synchronous machine component models in state-variable-based transient simulation programs.

Appendix A

1) Induction machine parameters [4]:
50 HP, 460 V, 4 poles, 60 Hz, $J = 1.662$ J·s^2, $r_s = 0.087\ \Omega$, $X_{ls} = 0.302\ \Omega$, $r_r = 0.228\ \Omega$, $X_{lr} = 0.302\ \Omega$, $X_m = 13.08\ \Omega$.
2) Saturation data (peak values) [34]

λ_m (Wb)	0.38	0.48	0.60	0.68	0.75	0.79	0.82
i_m (A)	11.20	14.33	18.33	21.31	24.42	26.74	29.07
λ_m (Wb)	0.86	0.92	0.96	1.01	1.06	1.10	
i_m (A)	33.21	40.78	46.21	53.21	60.33	66.10	

Appendix B

1) Salient-pole synchronous machine parameters [4]:
325 MVA, 20 kV, 64 poles, 60 Hz, $J = 35.1 \cdot 10^6$ J·s^2, $r_s = 0.00234\ \Omega$, $X_{ls} = 0.1478\ \Omega$, $r_{fd} = 0.0005\ \Omega$, $X_{lfd} = 0.2523\ \Omega$, $r_{kq2} = 0.01675\ \Omega$, $X_{lkq2} = 0.1267\ \Omega$, $r_{kd} = 0.01736\ \Omega$, $X_{lkd} = 0.1970\ \Omega$, $X_{md} = 0.8989\ \Omega$, $X_{mq} = 0.4433\ \Omega$.
2) Saturation data (peak values) [35]

λ_m (Wb)	10.8	22.5	31.0	35.3	38.2
i_m (kA)	4.5143	9.4981	13.255	15.261	16.707
λ_m (Wb)	40.9	42.5	45.3	47.9	50.3
i_m (kA)	18.198	19.210	21.340	23.652	25.930

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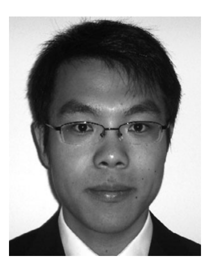

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Methods of Interfacing Rotating Machine Models in Transient Simulation Programs

IEEE Task Force on Interfacing Techniques for Simulation Tools
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***Abstract*—The electromagnetic transient programs (EMTP-like tools) are based on the nodal (or modified nodal) equations that enable an efficient numerical solution and, subsequently, fast time-domain simulations. The state-variable-based simulation programs, such as Simulink, are also used for studying the dynamics of electrical systems. Both the offline and real-time versions of these two types of simulation tools are widely used by the researchers and engineers in industry and academia to study the transient phenomena and dynamics in power systems with rotating electrical machines. This paper provides a summary of the interfacing techniques that are utilized to integrate the general-purpose models of electrical machines with the rest of the power system network for these studies. The interfacing methods are broadly classified as indirect and direct approaches. The paper also describes the numerical properties as well as limitations imposed by the interfacing of the commonly used machine models that should be considered when selecting the simulation parameters and assessing the final results.**

***Index Terms*—Electrical machines, electromagnetic transients, Electromagnetic Transients Program (EMTP), interfacing techniques, simulation, state-variable approach.**

I. Introduction

ROTATING electrical machines have been an integral part of power systems for as long as the latter have existed. The electrical machines are used as generators and motors in numerous applications in power systems in a wide range of voltage and power levels. This paper mainly considers the machine models that are based on the coupled electric circuit approach, which leads to a relatively small number of equations and have been very effectively utilized for analyzing power systems transients [1]–[4]. The validity of this type of models for both balanced and unbalanced power system operations has been confirmed [5], [6]. Over time, numerous models have been developed concurrently with the development of digital computer programs. The modeling methods were influenced by many factors such as type and size of the machine, objectives of the power system study, and the method of exchanging data between the machine and the network. While a large number of machine models may offer beneficial options to the user, it also creates challenges in terms of proper model selection. For example, the same machine modeled in two separate software programs would yield different results leading to much confusion during the implementation and validation stages. Some researchers have also questioned the applicability of the established qd model to unbalanced conditions [7]–[9]. The authors of [10], [11] made similar conclusions by comparing the simulation results of the classical qd model and the coupled-circuit phase-domain (PD) model under unbalanced operation. As the software programs matured the root causes of the modeling differences were eventually recognized and corrected [5], [12].

The objective of this task force paper is to present to the reader the various interfacing techniques used to integrate the machine models with the power system network in different simulation programs. The Electromagnetic Transients Program (EMTP) [4], [13] approach and its many derivative programs (such as ATP [14], MicroTran [15], PSCAD/EMTDC [16], EMTP-RV [17], VTB [18], etc.) are based on nodal (or modified nodal) equations and are widely used. The state variable (SV) approach is also used for the analysis of dynamic systems including electric power systems. The examples of SV-based software packages include acslX [19], Easy5 [20], Eurostag [21], MATLAB/Simulink [22], etc. Within the EMTP and SV solution methodologies, this paper broadly classifies the methods of interfacing electrical machine models with power system network into indirect and direct approaches. This paper also provides numerous examples and studies that should help many engineers and researchers in correct use of various machine models and software packages.

Another important aspect of machine modeling is the representation of magnetic saturation, which improves the accuracy and the range of applications of such models. Different approaches have been considered for various machine model structures (e.g., qd, PD) and/or simulation languages (EMTP or SV based). However, the saturation modeling methods for transient simulation programs are not discussed here as this issue requires more thorough literature survey which is beyond the scope of this paper.

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The paper is organized as follows: Section II describes the typical machine models used for power systems transients. Sections III and IV describe and demonstrate the interfacing approaches used in the EMTP and the SV programs, respectively. The final conclusions are summarized in Section V. The parameters of synchronous and induction machines used for simulation studies of this paper are summarized in the Appendix.

II. General-Purpose Machine Models

For the purpose of consistency and without loss of generality, this section considers a conventional general purpose synchronous machine model [2]–[6], whereas induction machine model is somewhat similar. The dynamics of machine can be represented by equations of corresponding electrical and mechanical subsystems. Depending on the application and the scope of studies (e.g., subsynchronous resonance, torsional dynamics, etc.), the mechanical subsystem may be represented as a lumped-parameter spring-mass system [23]. However, since the focus of this paper is on interfacing the machine models with the solution of electrical network-system, a single rigid-body mechanical subsystem is assumed here. Specifically, the equations for mechanical subsystem for all models and studies presented are

$$\frac{d\theta_r}{dt} = \omega_r \tag{1}$$

$$\frac{d\omega_r}{dt} = \frac{P}{2J}(T_e - T_m) \tag{2}$$

where θ_r and ω_r denote the rotor position and the angular electrical speed; P is the number of poles, J is the moment of inertia; T_m and T_e are the mechanical torque and electromagnetic torque, respectively. Motor convention is used for all models. Throughout this manuscript, bold capital case is used to denote matrices and bold lower case is used to denote vectors.

The original coupled-circuit phase-domain (PD) model is commonly expressed in terms of the machine's physical variables and phase coordinates. In particular, the voltage equation is expressed as

$$\begin{bmatrix} \mathbf{v}_{abcs} \\ \mathbf{v}_{qdr} \end{bmatrix} = \mathbf{R} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \boldsymbol{\lambda}_{abcs} \\ \boldsymbol{\lambda}_{qdr} \end{bmatrix} \tag{3}$$

and the flux linkages are represented as

$$\begin{bmatrix} \boldsymbol{\lambda}_{abcs} \\ \boldsymbol{\lambda}_{qdr} \end{bmatrix} = \mathbf{L}(\theta_r) \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix}. \tag{4}$$

The equations for the electrical subsystem are determined by the number of machine's windings. A typical synchronous machine model may include three stator windings $abcs$, one field winding fd, and one damper winding kd in the d axis, and two damper windings $kq1$ and $kq2$ in the q axis. $\mathbf{R}$ is a diagonal and constant matrix containing the stator and rotor resistances. Vectors $\mathbf{v}_{abcs}, \mathbf{i}_{abcs}$, and $\boldsymbol{\lambda}_{abcs}$ denote the stator voltages, current, and flux linkages, respectively. Similarly, vectors $\mathbf{v}_{qdr}, \mathbf{i}_{qdr}$, and $\boldsymbol{\lambda}_{qdr}$ denote the rotor voltages, current, and flux linkages. The combined stator and rotor self and mutual inductance matrix $\mathbf{L}(\theta_r)$ depends on the rotor position θ_r. The exact equations for the self and mutual inductances are given in [3] and are not included here due to the limited space. The electromagnetic torque is expressed in the machine variables as

$$T_e = \frac{1}{2} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix}^T \frac{\partial}{\partial \theta_r} \mathbf{L}(\theta_r) \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix}. \tag{5}$$

Although the model defined by (3)–(5) is relatively straightforward, it contains the terms (matrices) that depend on the rotor position θ_r (and, hence, changes with time), which results in an additional computational burden and complexity of the final model whether it is implemented in the EMTP or the SV approach.

To achieve a machine model with constant parameters, a change of variables can be performed. In particular, in the case of synchronous machine, all variables are typically referred to the stator side (using appropriate turns-ratios) and then transformed (projected) onto the orthogonal qd coordinates that are fixed on the rotor. This is known as the Park's $qd0$ transformation [1], [3], wherein the q axis of the rotor reference frame is assumed to be 90° leading the d axis. The corresponding transformation matrix has the following form:

$$\mathbf{K}_s = \frac{2}{3} \begin{bmatrix} \cos\theta & \cos\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) \\ \sin\theta & \sin\left(\theta - \frac{2\pi}{3}\right) & \sin\left(\theta + \frac{2\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}. \tag{6}$$

This transformation is applied to the stator circuit and variables $(abcs)$. The rotor circuit and variables (qdr) are already expressed in the qd coordinates and need not be transformed. The last row of (6) corresponds to the zero sequence (which produces the corresponding zero-sequence circuit on the stator side) and is needed to represent the unbalanced cases.

The voltage equations of the qd machine model can be represented in terms of transformed variables as

$$\begin{bmatrix} \mathbf{v}_{qd0s} \\ \mathbf{v}_{qdr} \end{bmatrix} = \mathbf{R} \begin{bmatrix} \mathbf{i}_{qd0s} \\ \mathbf{i}_{qdr} \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \boldsymbol{\lambda}_{qd0s} \\ \boldsymbol{\lambda}_{qdr} \end{bmatrix} + \omega_r \begin{bmatrix} \boldsymbol{\lambda}_{\mathrm{dq}} \\ 0 \end{bmatrix} \tag{7}$$

where vectors $\mathbf{v}_{qd0s}, \mathbf{i}_{qd0s}$ and $\boldsymbol{\lambda}_{qd0s}$ denote the stator voltages, currents, and flux linkages in $qd0$ variables, respectively. Similarly, vectors $\mathbf{v}_{qdr}, \mathbf{i}_{qdr}$ and $\boldsymbol{\lambda}_{qdr}$ denote the rotor voltages, currents, and flux linkages in qd variables. The quantity $\omega_r \boldsymbol{\lambda}_{dq}$ is the speed voltage term, where $\boldsymbol{\lambda}_{dq} = [\lambda_d \quad -\lambda_q \quad 0]^T$. This term appears in the stator voltage equation only. The flux linkage equations are given as

$$\begin{bmatrix} \boldsymbol{\lambda}_{qd0s} \\ \boldsymbol{\lambda}_{qdr} \end{bmatrix} = \mathbf{L}_{\mathrm{qd}} \begin{bmatrix} \mathbf{i}_{qd0s} \\ \mathbf{i}_{qdr} \end{bmatrix} \tag{8}$$

where the combined self and mutual inductance matrix $\mathbf{L}_{qd}$ is a constant, due to Park's transformation. The electromagnetic torque may be calculated in terms of the stator and/or the rotor currents $\mathbf{i}_{qds}$ and $\mathbf{i}_{qdr}$, the flux linkages $\boldsymbol{\lambda}_{qds}$ and $\boldsymbol{\lambda}_{qdr}$, as well as in terms of the magnetizing flux linkages λ_{mq} and λ_{md} [3]. For example

$$\begin{aligned} T_e &= \frac{3P}{4}(\lambda_{ds} i_{qs} - \lambda_{qs} i_{ds}) \\ &= \frac{3P}{4}(\lambda_{qr} i_{dr} - \lambda_{dr} i_{qr}) \\ &= \frac{3P}{4}(\lambda_{md} i_{qs} - \lambda_{mq} i_{ds}). \end{aligned} \tag{9}$$

Implementing model (7)–(8), by itself, is much easier than implementing (3)–(4). Similarly, calculating the electromagnetic torque using (9) requires significantly less effort than using (5). However, the $qd0$ model has to be interfaced with the external power network which is typically formulated in abc physical variables.

III. Interfacing Machine Models in EMTP

Although there are a number of software packages based on nodal analysis (or modified nodal analysis) including the EMTP-type and the Spice-type programs, the underlying solution approach is based on discretizing the differential equations for each circuit component by usng a particular integration rule. The EMTP [4], [13] uses an implicit trapezoidal rule for discretization and formulating the network nodal equation that has the following general form:

$$\mathbf{G}\mathbf{V}_n = \mathbf{I}_h \tag{10}$$

where $\mathbf{G}$ is the network nodal conductance matrix, and the vector $\mathbf{I}_h$ includes the so-called history current sources injected into the nodes. The nodal voltages $\mathbf{V}_n$ are unknown and calculated by solving (10) at every time step. The solution of (10) is one of the key factors that has made the EMTP languages so efficient and ultimately widely used. As the $\mathbf{G}$ matrix is usually sparse, specially designed techniques, such as optimal reordering schemes and partial LU factorization [24], [25], are often used to solve this linear system instead of inverting $\mathbf{G}$ directly. Branches with variable parameters are typically ordered such that their equivalent conductance submatrices appear in the bottom-right corner of $\mathbf{G}$ to minimize the re-factorization effort [4].

In the EMTP formulation, the rotating machines are represented outside of the network, thus requiring a special interface. This interface also represents challenges for initialization and the use of the program for harmonic analysis. A frequency-domain machine model was proposed for multiphase unbalanced harmonic load flow program in [26]. The interface between machine model and external network in frequency domain was achieved by harmonic Norton current source and admittance matrix. For time-domain machine models, significant efforts have been made by many researchers and software developers to come up with computationally efficient and numerically robust models and their respective interfaces for the network solution of the program. Based on the commonly used EMTP packages and the published literature, this paper discusses direct and indirect interfacing methods as follows:

A. Indirect Approaches

As the machine equations are usually represented in the qd coordinates/variables while the power system networks are most often expressed in physical variables and abc phase coordinates, the following indirect methods have been utilized.

1) Thevenin Prediction-Based Method: First of all, the machine qd equivalent circuits are discretized using implicit trapezoidal rule similar to other components [4], [27]–[29]. For example, the resulting discretized equivalent circuit for the d axis of synchronous machine with one damper and one field winding

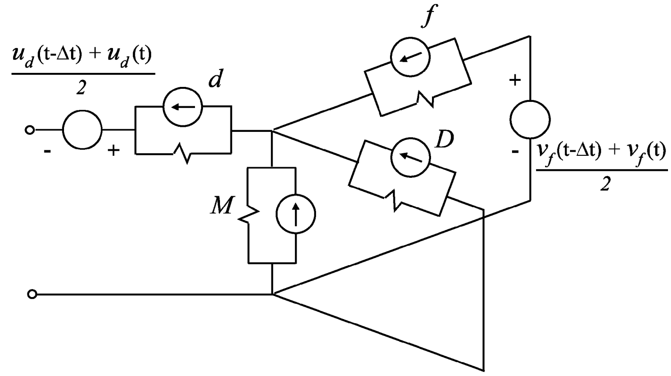

Fig. 1. Example of discretized d axis equivalent circuit for the EMTP solution.

is shown in Fig. 1, where the conventional circuit elements are replaced by the sources and resistances due to discretization. The q axis discretized circuit is similar and not shown here.

Based on the discretized $qd0$ circuit of Fig. 1, the machine's Thevenin equivalent circuits are then formulated as depicted in Fig. 2(a). As explained in [4], this step requires the prediction of electrical variables, such as the speed voltage terms $u_d(t), u_q(t)$ and the field-winding voltage $v_f(t)$, if the excitation is externally controlled. It is also observed from Fig. 2(a) that the equivalent resistances in the d and q axes, r_d and r_q, are usually not equal due to the saliency effect. If this is the case, then direct transformation of the Thevenin equivalent circuits of Fig. 2(a) into the abc coordinates will result in the time-variant conductance matrix which is not desirable. Instead, the resistances r_d and r_q may be averaged, and the new saliency terms $(r_d - r_q)/(2) \cdot i_d(t)$ and $(r_q - r_d)/(2) \cdot i_q(t)$ are then combined with the back electromotive-force terms e_d and e_q as shown in Fig. 2(b). Thus, to formulate the saliency terms using this method, the stator currents $i_d(t)$ and $i_q(t)$ are required to be predicted. Finally, the machine's qd Thevenin equivalent circuits of Fig. 2(b) are transformed back to abc coordinates using the inverse transformation (6) and the predicted rotor angle $\delta(t)$.

Assuming that the system solution at $t - \Delta t$ is known and the solution at the next time step, at time t is required, the aforementioned requires the prediction of several mechanical and electrical variables. The advantage of the prediction-based interfacing method resides in the fact that the resulting machine's equivalent resistance submatrix is constant (assuming magnetically linear machine model). This constant submatrix is then used in (10) which also avoids re-factorization of the entire network conductance $\mathbf{G}$ matrix at every time step. Programs, such as MicroTran (machine model Type 50) [15], EPRI/DCG EMTP [30] and Alternate Transients Program (ATP) (machine model Type 59) [14] utilize this method to interface the qd models with the external network.

However, the prediction of electrical variables may significantly deteriorate simulation accuracy and numerical stability (especially for large time steps) as has been documented in [31]–[34].

2) Norton Current Source Method: The qd machine model may also be interfaced with the network as a Norton current source. This type of machine-network interface is shown in Fig. 3 [16], [35], [36]. For the simplicity of the diagram, only one

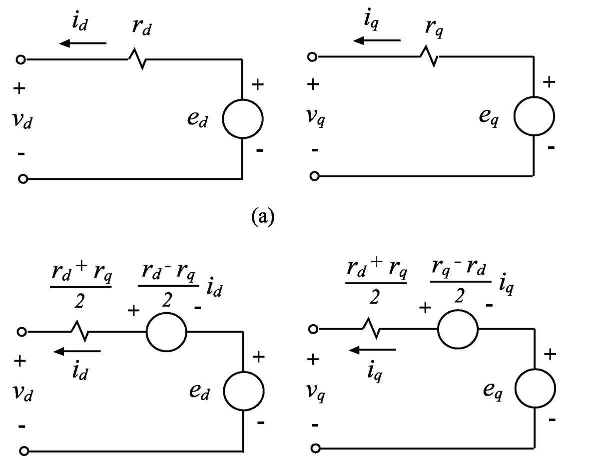

Fig. 2. Discretized qd Thevenin equivalent circuits for prediction-based interfacing method. (a) Circuit with different resistances in q and d axes due to saliency. (b) Circuit with averaged resistances to include the saliency effect.

phase is depicted. The injection current $i_m(t)$ is calculated using the terminal bus voltages from the previous time step, which implies a one time-step delay in calculating the machine variables. A problem of numerical instability may arise especially when the machine is in near open-circuit conditions. In order to improve the numerical stability and prevent the machine from being completely open-circuited, a terminating characteristic impedance/resistor r'' is introduced to the machine network interface [36]. The effect of this added resistor is then compensated by an additional current source $i_c(t)$ injected into the machine terminals as shown in Fig. 3. The value of this compensating resistor and the additional current source are calculated as follows [36]:

$$r'' = \frac{2L''}{N \cdot \Delta t} \tag{11}$$

$$i_c(t) = \frac{V(t-\Delta t)}{r''} \tag{12}$$

where L'' represents the characteristic inductance of the machine, N is the number of coherent machines in parallel, and $V(t-\Delta t)$ is the terminal voltages from the previous time step. From Fig. 3, it is observed that the actual current injected into the network is given as

$$i_{ma}(t) = i_m(t) + \frac{V(t-\Delta t) - V(t)}{r''}. \tag{13}$$

As the time step Δt is usually small, the characteristic impedance r'' is large. For small time steps, $V(t-\Delta t) \approx V(t)$, which gives $i_{ma}(t) \approx i_m(t)$.

This method is used, for example, in PSCAD/EMTDC [16] to interface all conventional $qd0$ machine models. In addition to traditional EMTP, the Norton equivalent interfacing has been adapted to the multiscale and shifted-frequency simulation approach [37]. The advantages of the Norton current source method are: 1) the machine equivalent resistance is very simple—r'' per phase and 2) the interfacing resistance/impedance is constant (given a magnetically linear machine model). However, a one-time-step delay exists in the

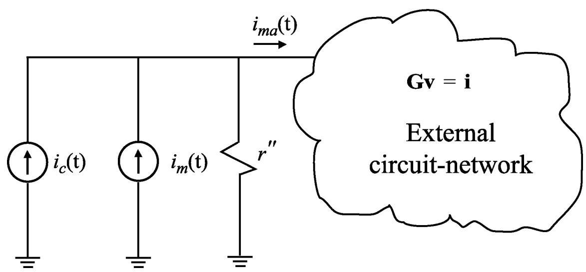

Fig. 3. Norton current-source interface of the machine model with an external network.

injecting current $i_m(t)$, and the extra interfacing error is being introduced due to the term $(V(t-\Delta t) - V(t))/(r'')$. These factors may affect the simulation accuracy and numerical stability, and one should be mindful of these properties especially when considering large time steps.

3) Compensation-Based Method: In the compensation-based method [38], the machine qd model is treated as a nonlinear device. The external network is represented as a Thevenin equivalent circuit and is interfaced with the machine model in the qd axes. The prediction of rotor speed is used at the beginning of each time step to formulate the qd transformation matrix and to facilitate the solutions of machine equations in qd coordinates. The machine stator currents are then solved and injected into the external network as ideal current sources. The complete network solutions are therefore sought according to the superposition principle which superimposes the solutions with and without the nonlinear components (i.e., the machine model). The advantage of the compensation-based method is that the subsequent numerical iterations that may be required to obtain a convergent solution with nonlinear components are confined to the machine's equations only. However, the compensation method requires the machines to be separated by the distributed-parameter transmission lines. Otherwise, artificial "stub-lines" are used which may complicate the network modeling and reduce the accuracy or require a very small time step to accommodate the models of very short lines. In ATP, the universal-machine (UM) model is implemented by using the compensation-based method [14], [38], [39].

4) Network Iterative Method: In EMTP-RV [17], the qd machine models are solved with the network equations simultaneously by using Newton's iterative method [40], [41]. In addition, inside the EMTP-RV machine models, speed and voltage iterative loops are utilized to obtain a convergent solution of the machine equations at each time step [17]. These two loops are controlled by the user-specified error tolerances. Thus, the constraint existing with the compensation method and UM models that the machines need to be separated by transmission lines is eliminated. However, iterative solutions of the machine and network equations at every time step generally require more computational resources and, therefore, reduce the overall simulation efficiency. At the same time, if a large time step is used, the system may converge at a solution that is slightly away from the correct solution even after several iterations and high tolerance settings. This effect may subsequently result in the shift of the solution trajectories relative to the correct solutions.

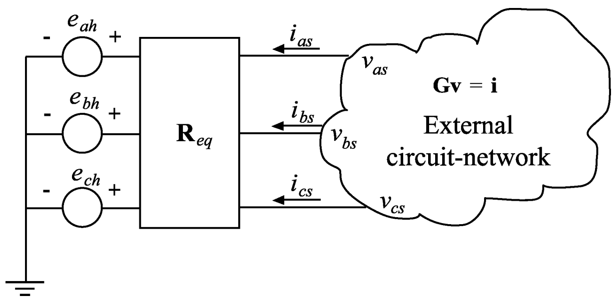

Fig. 4. Thevenin equivalent circuit for interfacing PD and VBR machine models in EMTP.

B. Direct Approaches

To improve the simulation accuracy and numerical stability, coupled-circuit PD models [31], [32], [44]–[49] and voltage-behind-reactance (VBR) models [12], [33], [34] have been proposed, respectively. Thus, the direct interface with the external network is readily achieved since the stator circuit is naturally represented in the abc coordinates. The machines are interfaced with the external network as three-phase Thevenin equivalent circuits. For example, such an interface is shown in Fig. 4, assuming a Y-connected stator winding. No electrical variables are predicted, and the slow mechanical variables (rotor speed ω_r and/or position θ_r) are predicted similar to the other EMTP interfacing methods. This achieves a simultaneous solution of machine and network electrical equations (variables) and greatly improves the numerical accuracy and stability of these models.

The generic interfacing equation for PD and VBR machine models, using the Thevenin equivalent circuit method, can be represented as

$$\mathbf{v}_{abcs} = \mathbf{R}_{\text{eq}}\mathbf{i}_{abcs} + \mathbf{e}_h \tag{14}$$

where $\mathbf{R}_{\text{eq}}$ is the equivalent resistance matrix and $\mathbf{e}_h$ is the equivalent history voltage source that contains the information from the previous time steps. This equation is obtained by discretizing the voltage equations of the respective model using the implicit trapezoidal rule and then carefully arranging the terms to achieve the compact form (14). The matrix $\mathbf{R}_{\text{eq}}$ is used to calculate the equivalent conductance submatrix and modify the network $\mathbf{G}$ matrix and the history current source vector $\mathbf{I}_h$ in (10). Although similar in terms of the interfacing equation, the PD and VBR models have significant differences in terms of their numerical properties and computational costs for calculating terms in (14).

1) Phase-Domain Model: The PD model is based on discretization of (3) and (4) and calculation of the electromagnetic torque according to (5). However, the existence of time-variant self and mutual inductances in (4) increases the computational burden of this model. Generally, matrix $\mathbf{R}_{\text{eq}}^{\text{pd}}$ and the equivalent history voltage source $\mathbf{e}_h^{\text{pd}}$ are quite costly to calculate, and it has to be performed at each time step. Moreover, as the analysis has shown, the discretized PD model may have eigenvalues that are far outside the unit circle, which further reduces the numerical accuracy per time step achievable by this model. However, due to its advantageous direct interface, this model has been well recognized. Present implementations of the PD model include Type 58 machine model [45]–[47] in ATP and the VTB machine model [48] as well as its previous implementations [31], [42]–[44], [49].

2) Voltage-Behind-Reactance Model: The VBR formulation is based on algebraically manipulating the machine equations (3)–(9) in order to achieve the following voltage equation for the stator circuit [33], [34], [50]:

$$\mathbf{v}_{abcs} = \mathbf{R}_s\mathbf{i}_{abcs} + \frac{d}{dt}[\mathbf{L}''_{abcs}(\theta_r)\mathbf{i}_{abcs}] + \mathbf{e}''_{abcs} \tag{15}$$

where the terms $\mathbf{L}''_{abcs}(\theta_r)$ and $\mathbf{e}''_{abcs}$ are the so-called subtransient inductance matrix and voltages, respectively, and the rotor is modeled by using the qd flux linkages as independent variables. The presently reported implementations include synchronous machines [50], [33] and induction machines [34], [76], [77].

The VBR model also represents the stator circuit RL branches in the abc phase coordinates. Following the same discretization procedure using implicit trapezoidal rule, the VBR models [33], [34] also enable direct interface of the machine model with the external network according to Fig. 4 and interfacing (14). Similar to the PD model, simultaneous EMTP solution of the network and machine electrical variables is achieved without the prediction of any electrical variables. However, the terms $\mathbf{R}_{\text{eq}}^{vbr}$ and $\mathbf{e}_h^{vbr}$ in (14) require less than half of the flops and are much easier to compute compared to the equivalent PD model [33], [34]. Also, the eigenvalues of the discretized VBR model tend to be better conditioned which improves the numerical accuracy.

C. Case Studies With EMTP

To demonstrate the behavior of various models and their interfaces in a clear way, a single-machine infinite-bus system with a steam-turbine generator is assumed. The corresponding parameters are summarized in the Appendix. The same system has been implemented using several popular EMTP software packages, including MicroTran, ATP, PSCAD, EMTP-RV, as well as the PD and VBR models [33].

In the transient study considered here, the machine initially operates in an idle steady-state mode with the load torque $T_m = 0$. Throughout the study, the excitation is kept constant at nominal value. At $t = 0$, a symmetric three-phase fault is applied at the machine terminals. Observing all considered models it was found that their dynamic responses are all convergent to the same solution. This is an expected result since the identical machine parameters were used in each case, which also proves the consistency of these software packages and different models. To produce a very accurate solution, the same study was also run using a small time step of $\Delta t = 1\ \mu$s. The solution trajectories produced with this small time step were then considered as a reference for responses produced by various models using different time steps.

Overall, it was observed that the accuracy of various models may be significantly affected by the time-step size. To illustrate this point, the transient solutions predicted by various models using a fairly large time step of 1 ms are shown in Figs. 5–7. Due to the limited space, the plotted variables are limited to

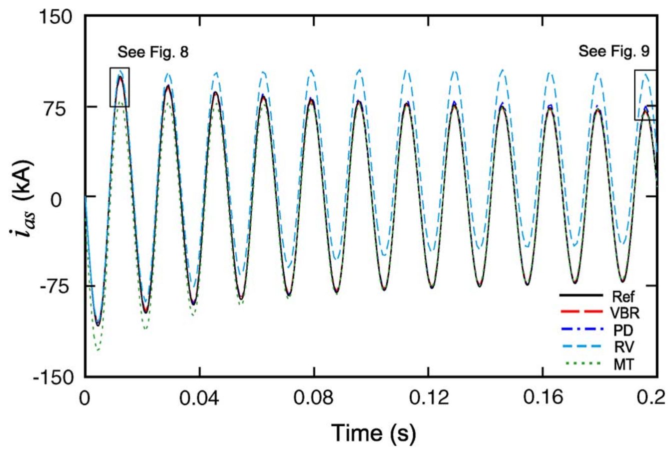

Fig. 5. Phase current as predicted by various models using a time step of 1 ms.

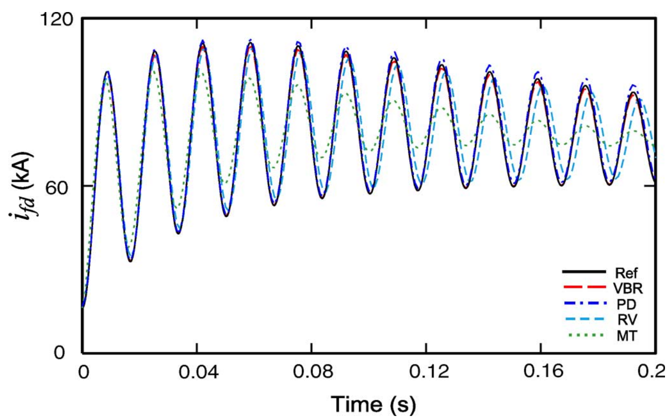

Fig. 6. Field current as predicted by various models using a time step of 1 ms.

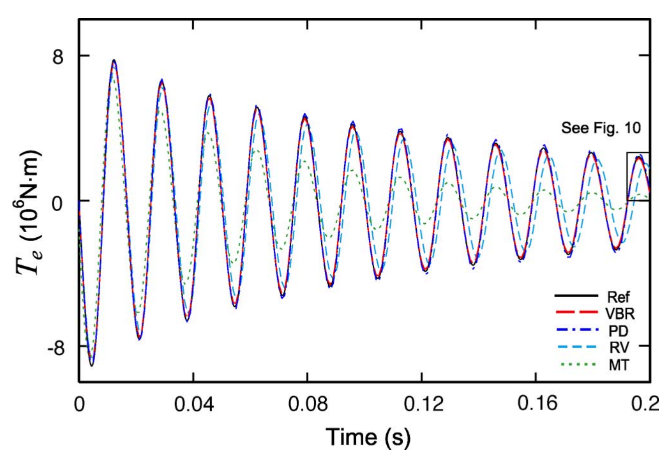

Fig. 7. Electromagnetic torque as predicted by various models using the time step of 1 ms.

the a-phase current i_{as}, the field current i_{fd}, and the electromagnetic torque T_e. The qd models of PSCAD and ATP were not convergent with the given time step and, therefore, are not shown in Figs. 5–7. The studies were conducted using the time steps of 50 μs, 100 μs, and 1 ms.

To evaluate the accuracy of each modeling interface approach, it is possible to compare the error of each model with respect to the known reference solution. Without the loss of

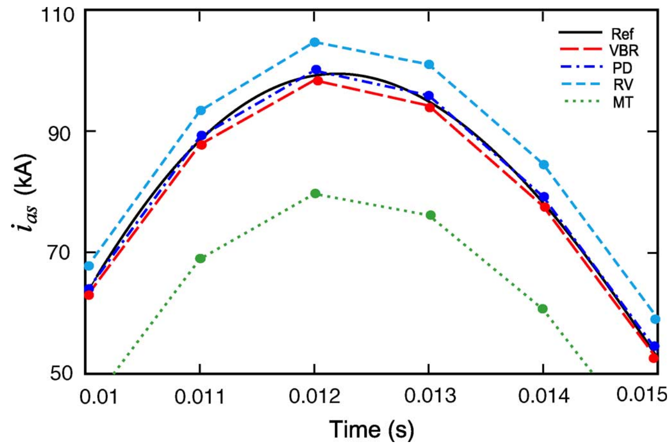

Fig. 8. Detailed view of the beginning transient of $i_{\rm as}$ with a time step of 1 ms.

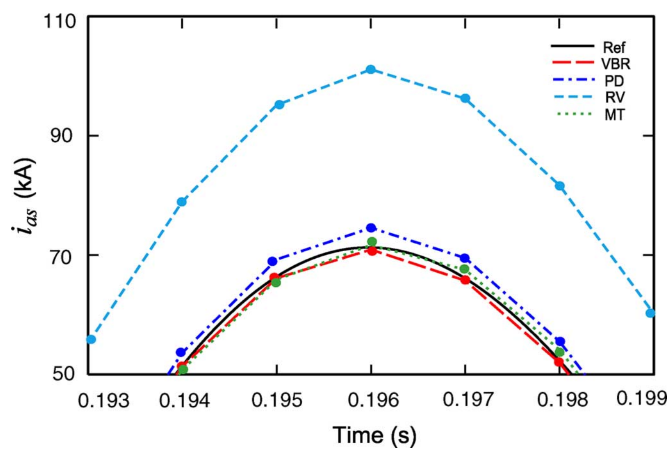

Fig. 9. Detailed view of $i_{\rm as}$ in the end of study using a time step of 1 ms.

generality, the peak of electromagnetic torque shown in Figs. 7 and 10 at time $t = 0.196$ s has been considered. The value of the torque predicted by each model has been compared with the reference value to calculate the relative error at this time. Although there are various methods of evaluating the accuracy of trajectories [33], in this paper the relative error of the peak has been considered as a sufficient and an illustrative measure. The results of error calculations for all considered models are summarized in Table I. The studies performed using a typical time step 50 μs were visually very close for all the models. This is also consistent with the errors summarized in Table I for this time step size. However, the accuracy of models with indirect interface deteriorates quickly when the step size is increased to 100 μs and above. Finally, at the large time step of 1 ms, the only stable models are MT, RV, PD, and VBR. Figs. 8–10 show the details of the phase current i_{as} and torque T_e for the remaining stable models. As can be seen in these figures, the solutions of some models clearly deviate from the reference solution. On the one hand, the very significant errors of MT and RV models (and the non-convergence of the PSCAD and ATP models) are attributed to the indirect interface of these models. Table I (third row) summarizes the errors in predicting the torque T_e corresponding to Fig. 10. Such errors are quite large, and would limit the usefulness or validity of the models

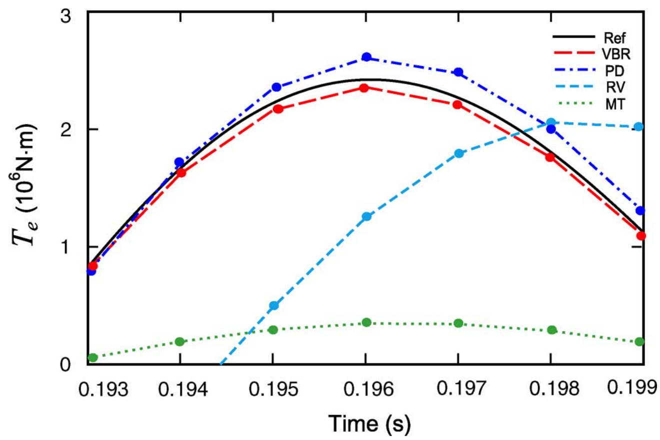

Fig. 10. Detailed view of electromagnetic torque T_e in the end of the study using a time step of 1 ms.

TABLE I
COMPARISONS OF MODEL ACCURACY USING A PEAK OF TORQUE AT 0.196 S

Models	MT	ATP	PSCAD	EMTP-RV	PD	VBR
$50\mu s$	0.037%	0.75%	0.8%	0.7%	0.42%	0.0041%
$100\mu s$	0.24%	1.5%	1.1%	1.4%	0.85%	0.037%
$1ms$	86%	N/A	N/A	49%	7.5%	2.7%

with indirect interface to the time steps of no more than 100 to 200 μs. On the other hand, the models with direct circuit interface, PD and VBR, remain stable and relatively accurate even at such a large time step of 1 ms (with the last one offering the most accuracy).

IV. Interfacing Machine Models in State Variable-Based Programs

The SV approach is also used for the analysis of dynamic systems, including electric power systems [51]–[58] and the electrical machines in particular [59]–[61]. The examples of software packages include ACSL, Easy5, Eurostag, and, of course, the MATLAB/Simulink. There are also more specialized tools, such as SimPowerSystems (SPS) [62], PLECS [63], and ASMG [64], etc., that come with circuit interfaces and built-in libraries for the simulation of transients in power and power-electronic circuits. Internally, the program engine assembles the system of differential and/or differential algebraic equations (DAEs) that constitute the state-variable-based model of the overall system. Depending upon the features of a given program, the DAEs may be converted into a first-order system of ordinary differential equations (ODE) as

$$\begin{aligned}\frac{d\mathbf{x}}{dt} &= \mathbf{f}(\mathbf{x},\mathbf{u},t)\\ \mathbf{y} &= \mathbf{g}(\mathbf{x},\mathbf{u},t)\end{aligned} \tag{16}$$

where $\mathbf{x}$ is the vector of state variables, $\mathbf{u}$ is the vector of inputs, t is a scalar to denote time, and $\mathbf{y}$ is the vector of outputs. Whenever appropriate, the linear time-invariant part of the system/circuit may then be represented by using a more compact state-space equation

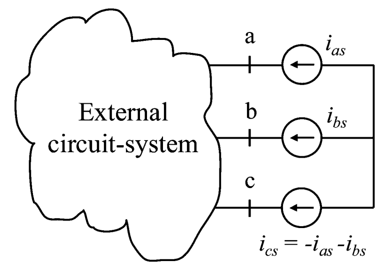

Fig. 11. Machine model interface in state-variable-based programs using voltage-controlled current sources.

$$\begin{aligned}\frac{d\mathbf{x}}{dt} &= \mathbf{Ax}+\mathbf{Bu}\\ \mathbf{y} &= \mathbf{Cx}+\mathbf{Du}\end{aligned} \tag{17}$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C}$, and $\mathbf{D}$ are the so-called state-space matrices that are computed for the given topology and parameters of the linear circuit. In some tools, such as SimPowerSystems, the circuit part is directly implemented in the form of (17), whereas the remaining part of the system (e.g., control blocks, mechanical subsystem, etc.) are in a more general form (16).

The time-domain transient responses are then calculated numerically by integrating the state-space equation (16)–(17) by using either fixed- or variable-step ODE solvers embedded in the SV program. The formulations (16)–(17) also contain useful information about the system's dynamical modes which is often utilized together with numerical linearization for the frequency-domain characterization and the design of controllers.

Similar to EMTP, the machine models in SV programs may differ depending on the choice of reference frames/coordinates and selection of the state variables. The commonly considered models here also include the classical qd, the coupled-circuit PD, and the VBR models. These models may then be interfaced with the external circuit system using either direct or indirect approaches as will be explained.

A. Indirect Approach

The qd machine models are very common and appear as the built-in library components in many SV simulation tools, including SimPowerSystems and PLECS. To interface the qd models with the external circuits, which is typically modeled in physical variables and abc phase coordinates, it is usually assumed that the machines are represented by voltage-controlled current sources [62], [63] as shown in Fig. 11. The reason for using the controlled-current-sources resides in the fact that the machine model is usually formulated as a proper state-variable model with flux linkages and/or currents as internal state variables, and the machine's terminal voltages as the inputs. Therefore, the interfacing method shown in Fig. 11 assumes a voltage-input, current-output machine model. This input–output requirement of the machine model is determined by its proper state model and results in compatible and incompatible interconnection with the external circuit system

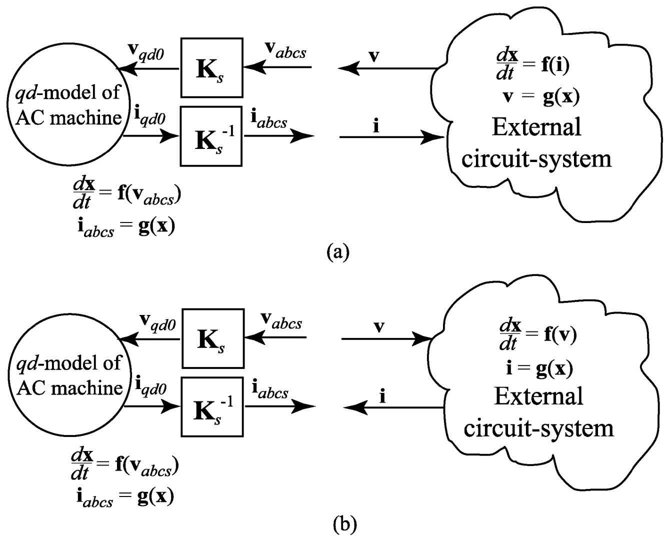

Fig. 12. Compatible and incompatible machine-network interfaces in state-variable-based programs. (a) Input–output compatible. (b) Input–output incompatible.

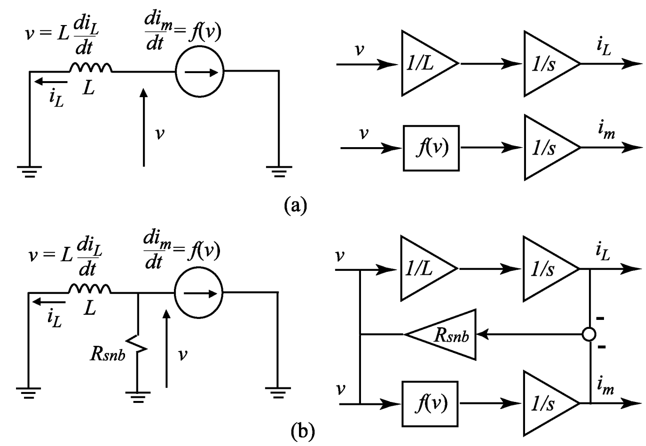

Fig. 13. Example of interface in state-variable-based programs using an artificial snubber circuit. (a) Incompatible interface. (b) Interface with snubber resistance.

as depicted in Fig. 12. Here, the external system may be a circuit and/or subsystem which is represented by its own state equations similar to (16) and (17).

As shown in Fig. 12(a), when the external circuit system has a current-input voltage-output characteristic at the interfacing terminals, it matches the machine model input and output. For example, this interface is possible whenever the external circuit system has capacitors and/or defined voltage sources that are connecting to the machine's terminals. In this case, the combined state equation for the entire system is readily formed by simply routing the respective input and output variables among the coupled subsystems models.

However, such an interface is not always available due to constraints of the external circuits system which itself may have voltage-input current-output characteristic (similar to the machine model) as shown in Fig. 12(b). For example, the external system may be another synchronous machine model which is supposed to represent a generator in the power system. In the case of an incompatible input–output interface as in Fig. 12(b), the combined state equations cannot be directly formulated as the needed input variables (voltages) are unknown.

In order to demonstrate the concept of an incompatible interface, a simpler example is shown in Fig. 13(a). Here, an inductor L is connected in series with a voltage-controlled current source $(di_m)/(\mathrm{dt}) = \mathrm{f}(v)$ (which may be another inductor). This example may be viewed as a simplified machine-network system since the qd machine models are usually represented as voltage-controlled current sources according to Fig. 11. As seen in Fig. 13(a), from the block diagram realization of these components, inductor and current source require the terminal voltage as input and produce currents as output, thus creating an incompatible interface.

1) Indirect Interfacing Using Snubbers: To enable the connection of these components and create a compatible interface, an artificial snubber circuit may be used to calculate the required input variable—the terminal voltage. For example, a snubber may be realized by using a very large resistor connected in parallel to the terminals as shown in Fig. 13(b). As can be seen in the block diagram implementation of Fig. 13(b), the added snubber is used to calculate the required terminal voltage. Here, the snubber current is calculated as the difference between the inductor current and that of the current source. This, in turn, enables the formulation of the proper state-space model of the combined system. Alternatively, one may also use a very small capacitor which will perform a similar function here.

The value of these snubbers should be selected so that the drawing current is sufficiently small and may be considered negligible within the scope of the model and studies being considered. Solving the combined system together using a single ODE solver ensures simultaneous solution of all its subsystems, albeit the solution trajectories will be affected (shifted) due to the snubbers. However, in general, these artificial snubber circuits may significantly affect the simulation accuracy and efficiency and, therefore, should be used with care. Examples of interfacing the built-in qd machine models using snubber circuits include SimPowerSystems [62] and PLECS [63].

2) Indirect Interfacing Using Time-Step Relaxation: The indirect interfacing may also be a powerful tool when the machine model and the external circuit system are formulated to have an input-output-compatible interface of the type Fig. 12(a). The numerical relaxation can be achieved if the interfacing variables are simply exchanged (updated) at each time step, allowing for the decoupled and parallel solution of each subsystem. This option may be useful, for example, when the machine model is discretized using different ODE solver and is then interfaced to the network. For example, in order to avoid forming algebraic loops, the SimPowerSystems also uses such interfacing approach when the external circuit system is discretized with the trapezoidal rule (therein referred as Tustin method) separately from the remaining Simulink blocks, while the machine models are discretized with the Forward Euler method [62]. This interface is similar to the sample-hold (zero-order-hold) that is typically attained in hardware-in-the-loop simulations [65] or the multirate simulations [66] (with or without iterations).

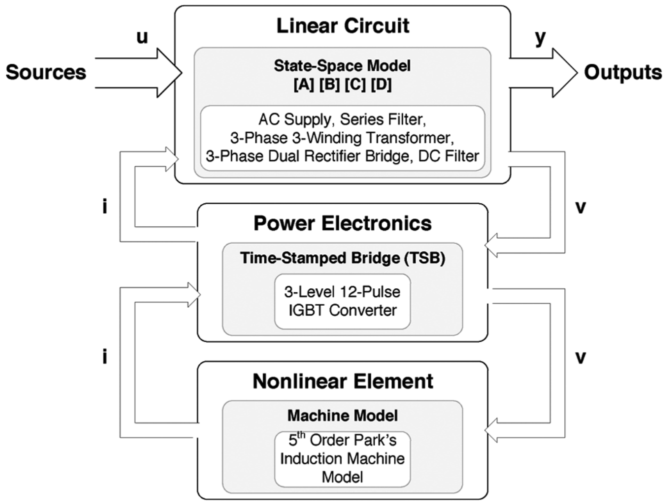

Fig. 14. Example of the machine model interfaced with a voltage-source inverter for real-time simulation [67].

This type of indirect interfacing method has also been successfully used in real-time simulations of multilevel variable frequency drives [67]. Therein, the complete electrical model was divided into three subsystems as shown in Fig. 14. All of the linear circuit elements (ac supply, series filter, transformer, the three-phase dual rectifier bridge, and the dc filter) were modeled using state space (17). The three-phase dual rectifier bridge is treated as a linear circuit since it is modeled by ideal switches. The second subsystem, the IGBT converter, was modeled by using a first-level feedback interfacing with the main state-space model through the input voltages and output currents. As depicted in Fig. 14, the IGBT converter model is formulated to provide the voltage output required by the state model of the machine, thus enabling an input-output-compatible interface. However, using time-step relaxation and solving the models separately allows the use of custom-machine models and permits using multirate integration techniques for the overall system. Moreover, accurate real-time simulation of power-electronic converters typically requires smaller time steps and correction algorithms [68] to account for the interstep switching events. An example of this interfacing approach, used with multirate integration, includes a field-programmable gate array (FPGA)-based hardware-in-the-loop simulation of a drive system [69], where the converter model and machine model were using very different time steps of 12.5 ns and 10 μs, respectively. Using sufficiently small time steps ensures convergence of this type of interfacing.

B. Direct Approaches

Whenever the external circuit system can be formulated to have an input–output compatible interface with the machine model as depicted in Fig. 12(a), the respective models can be directly connected [59]–[61] and solved together by the same ODE solver. This achieves a simultaneous solution of machine and network subsystems, which is desirable for numerical stability and good accuracy. Alternatively, the PD and VBR models can also be used in SV programs with the same goal of achieving a direct interface.

1) Phase-Domain Model: As can be seen from (3) and (4), the state model can be formed by using either the flux linkages as the state variables, which gives the following state equation:

$$\frac{d\boldsymbol{\lambda}}{\mathrm{dt}} = -\mathbf{R}[\mathbf{L}(\theta_r)]^{-1}\boldsymbol{\lambda} + \mathbf{v} \tag{18}$$

or the currents as the state variables, which gives a somewhat different structure but otherwise algebraically equivalent state equation

$$\frac{d\mathbf{i}}{\mathrm{dt}} = -[\mathbf{L}(\theta_r)]^{-1} \cdot \left[\left(\mathbf{R} + \frac{d}{\mathrm{dt}}\mathbf{L}(\theta_r)\right)\mathbf{i} - \mathbf{v}\right]. \tag{19}$$

The PD model can be implemented as a coupled circuit that is simply either a part of the overall circuit-system or a subsystem with voltage input and current output [as in Fig. 12(a) and as defined by either (18) or (19)]. To do this, one has to implement stator and rotor RL branches with all of the respective self and mutual inductances that appropriately change with the rotor position [70]. Of course, one should be mindful of the complexity of implementing the variable inductances in (4) and (5) and (18) or (19). Otherwise, the model is a straightforward inductive circuit with rotor-position-dependent magnetic coupling among the branches. The direct implementation of PD models includes [42], [70]–[73], etc., which can be readily carried out using, for example, ASMG, wherein the user has a choice of currents and/or fluxes as the state variables. Higher fidelity PD models may also be obtained by utilizing the finite-element analysis (FEA)-based programs first for constructing detailed representation of the windings [74], [75].

2) Voltage-Behind-Reactance Model: As was mentioned in Section III-B, the full-order models of either synchronous or induction machines can be formulated in the VBR form (15) [50], [76], [77]. In this formulation, the stator circuit is expressed in terms of subtransient resistances/inductances in abc coordinates using phase currents as the independent variables, and the rotor subsystem is expressed in qd variables and coordinated using flux linkages as the state variables. As a result, the equivalent stator RL branches can be readily included into the external circuit as depicted in Fig. 15, thus achieving a direct interface. Unlike the interface depicted in Fig. 11, here the current-controlled voltage sources are utilized to represent the stator subtransient voltages $\mathbf{e}''_{abcs}$. This is easily achievable as it preserves the voltage-input current-output structure of the external circuit system as in Fig. 12(b). At the same time, the rotor state model is formulated to have stator currents $\mathbf{i}_{abcs}$ as inputs and the subtransient voltages $\mathbf{e}''_{abcs}$ as the output, thus forming a direct interface. Depending on the interfacing requirements, the equivalent stator RL branches can have a different structure for synchronous machine [50], [65], [66] and for induction machine [76], [77]. For example, due to the symmetry of induction machine, there may be several implementations of the VBR model to account for possible connection of the stator windings (delta, wye, grounded/ungrounded neutral, etc.). One implementation [76, Sec. IV, VBR-III] results in constant and decoupled RL branches (diagonal resistance and inductance matrices). Therefore, interfacing machine models using this approach can be easily carried out in various simulation tools, such as Matlab/

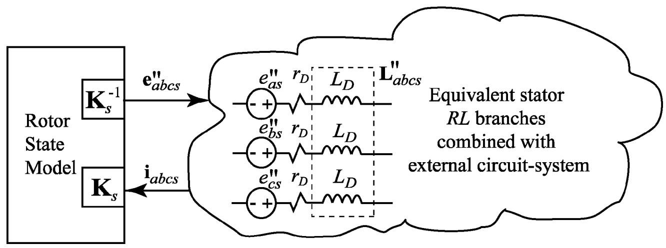

Fig. 15. Interfacing induction machine model with an external circuit using voltage-behind-reactance machine model formulation.

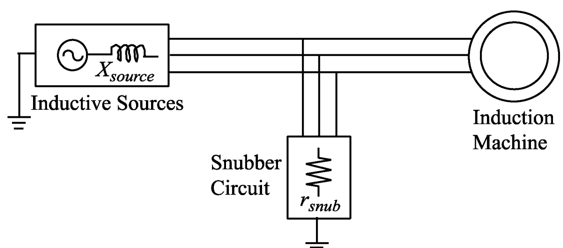

Fig. 16. Induction machine is fed from a voltage source with inductive impedance. Artificial shunt resistors are used to interface the model.

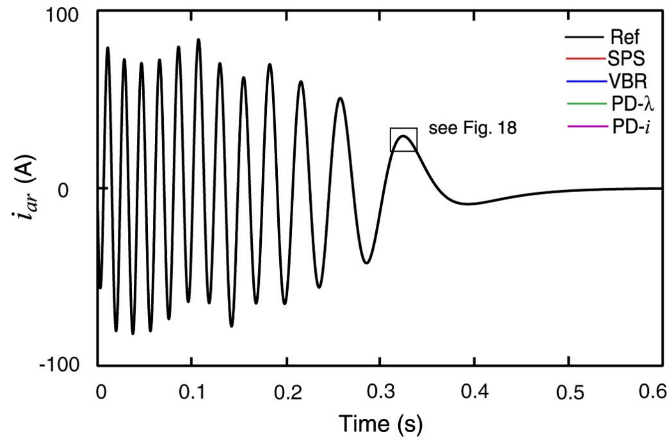

Fig. 17. Rotor current during startup transient as predicted by models with different interfacing methods.

TABLE II
MODEL EFFICIENCY COMPARISONS

Model	qd-SPS	VBR	PD-λ	PD-i
CPU Time, s	1.703	0.266	0.297	0.844
Num. of Steps	10015	2091	1732	2262
CPU Time per Step, μs	170.0	127.2	171.5	373.1

Simulink, for non-real-time applications using SimPowerSystems [77] as well as for real-time hardware-in-the-loop simulations [65].

C. Case Studies With SV Programs

To demonstrate the interfacing methods described in this section, a system composed of an induction motor, fed from an inductive source with reactance X_{source}, is considered here. The corresponding circuit diagram is depicted in Fig. 16, where the artificial snubber resistors r_{snub} required by the indirect interfacing approach are also shown. The system's parameters are summarized in the Appendix for consistency. This simple system is quite sufficient to demonstrate and explain the behavior of the underlying interfaces.

A no-load startup transient is considered here as this study spans a wide range of operating conditions in terms of mechanical and electrical variables. The time-domain responses of the rotor phase current i_{ar} predicted by the considered models are shown in Fig. 17. Other variables were consistent with previously published literature [76], [77] and are not included here due to space considerations. The models with direct and indirect interfaces have been implemented in Matlab/Simulink using the ASMG and SimPowerSystems (SPS) toolboxes. In particular, to obtain a reference solution, the VBR model was run with a very small time step of 1 μs. The built-in qd model of SimPowerSystems is used here to demonstrate the indirect interface requiring snubbers (qd-SPS). The directly interfaced PD model can be implemented by using either currents (PD-i) or flux linkages (PD-λ) as the state variables, respectively. These two models have been implemented using the ASMG toolbox which permits including branches with variable self and mutual inductances and allows for the choice of state variables. The last model with direct circuit interface is the VBR model that can be easily implemented in either SimPowerSystems or ASMG, yielding identical results [76], [77].

To provide a fair comparison, the same solver, ODE15s, was used for each model. To ensure smooth and accurate solutions that approach the reference trajectory, the relative and absolute error tolerances were set to 10^{-6}, and the maximum and minimum time-step limits were set to 10^{-3} and 10^{-10} s, respectively. As seen in Fig. 16, due to the inductance X_{source} in the source, the direct connection of the qd-SPS is not possible, thus resulting in an indirect interface which requires snubbers r_{snub}. To achieve acceptable accuracy and visibly the same results as shown in Fig. 17, we used fairly large snubbers $r_{\mathrm{snub}} = 1000\,\Omega$.

Assuming the same simulation accuracy for each model, the numerical efficiency of the interfacing methods may be evaluated by considering two factors: 1) the total number of integration steps taken to complete the study and 2) the computational load/cost per time step. The studies were run on a personal computer (Pentium 4, 2.66-GHz processor) using standard (not compiled, non-real-time) Simulink. The total CPU times, number of integration steps, and CPU times per step required to complete the study of Fig. 17 are summarized in Table II.

As can be observed in Table II, all models with a direct interface performed similarly in terms of the number of time steps taken (1732 to 2262). The longer CPU time required by the PD-i model is attributed to the rotor-position-dependent inductance matrix and its time derivative in (19), which is more costly than (18). The VBR model does not have any variable inductances and is therefore the fastest. At the same time, the qd-SPS took about five times more integration steps (10 015) to satisfy the same error tolerances.

To analyze the results of these studies, a fragment of the transient is shown in more detail in Fig. 18. The first property to notice is that the models with direct interface took fairly large time steps. Among these models, the results produced by VBR

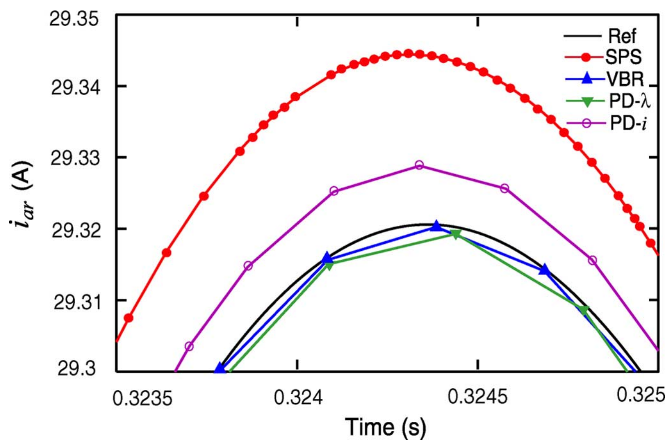

Fig. 18. Detailed view of the rotor current transient predicted by models with different interfacing methods. The relative errors are calculated by considering peak values at time $t = 0.3243$ s.

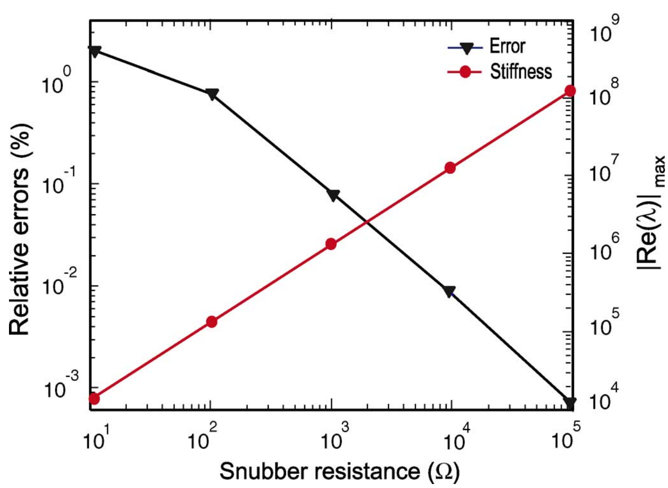

Fig. 19. Relative error and numerical stiffness for different values of the snubber circuit in the indirect interfacing method.

and PD-λ models appear to be close to the reference solution. At the same time, the qd-SPS not only required significantly more steps but it also converged to a different solution than the reference. This phenomenon is consistent with the fact that due to snubbers, the stator current (and, therefore, rotor current as well) will be somewhat different compared to the models that do not have such snubbers.

The effect of varying the snubbers is further investigated in Figs. 19 and 20. Here, the same simulation study was conducted again several times using the indirectly interfaced qd-SPS model with the snubber resistance changed in the range from 10 to 100 000 Ω, respectively. Fig. 19 (left vertical axis) shows the relative error of the rotor phase current i_{ar}. Similar to the studies in Section III-C, this error was calculated by considering the peak values at a given time instance (e.g., $t = 0.3243$ s) shown in Fig. 18. As can be seen in Figs. 18 and 19, the 1000 Ω snubber resistance results in a relative error of about 0.1%, which may be considered acceptable for many applications. If needed, this error can be reduced even further by increasing the value of the

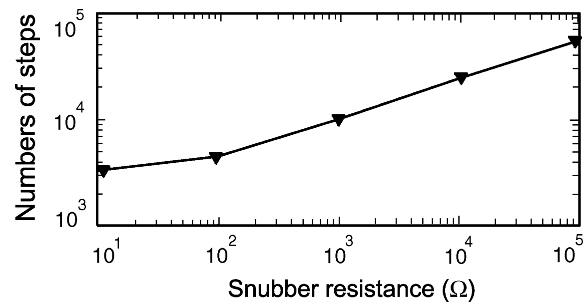

Fig. 20. Number of integration time steps taken to complete the 0.6-s study for different values of the snubber circuits in the indirect interfacing method.

snubbers, for example, to 10 000 Ω or 100 000 Ω, which may appear as a good option. However, one has to be aware of the potential challenges introduced by the snubber approach.

In general, adding the snubbers introduces the new current loops into the circuit and changes the state-space model by adding the associated dynamic modes. When the snubber resistors are made very large, the eigenvalues associated with these dynamic modes also become very large. This phenomenon is shown in Fig. 19 (right vertical axis). These very large eigenvalues contribute to the numerical stiffness of the system. As can be observed here, the additional 1000 Ω snubbers introduced the eigenvalues on the order of 10^6 to the system that was otherwise non-stiff. The consequence of numerical stiffness is that the ODE solver would generally require smaller time steps in order to satisfy the needed tolerances. This effect is demonstrated in Fig. 20, which shows the number of integration time steps taken to complete the same study for different values of the snubbers. This also explains why the qd-SPS model required significantly more time steps as summarized in Table II and can be clearly seen in Fig. 18.

V. Conclusion

This paper gives an overview of the interfacing methods that are available and used for implementing the existing models of rotating electrical machines in the commonly used simulation programs. We first discuss the machine models and their interfacing approaches that are applicable to the EMTP-type programs that are based on discretization of the circuit elements and subsequent solution of the nodal equations. Both indirect and direct interfacing approaches are demonstrated on an example system with one synchronous machine using a number of available models and EMTP software versions. It is suggested that machine models with indirect interface should be used with care especially for time steps larger than 100 to 200 μs, that is, when their accuracy may start to deteriorate.

The methods of interfacing the machine models in state-variable-based simulation programs can also be classified as direct and indirect. Similar to the EMTP, the direct interfacing of machine models offers better numerical accuracy and can be used with much larger time-step sizes. In many cases, the direct interfacing may be achieved either by appropriately formulating the systems equations or by using more contemporary machine models (i.e., PD and/or VBR). The indirect interfacing that is achieved using artificial snubbers is a simple but yet very practical way of interconnecting otherwise input-output incompatible subsystems (i.e., the machine model and the external circuit-system). When the machine model and the external circuit can form input-output compatible subsystems, interfacing them using a time-step relaxation and solving the machine model separately, is another simple method of interconnecting the machine models. When these approaches are used, one should be aware of a potential deviation of the solution trajectory, impact on the numerical stiffness of the system, and restrictions on the step size.

APPENDIX

Synchronous machine parameters [3]: Steam turbine generator, 835 MVA, 26 kV, 0.85 pf, 2 poles, 3600 r/min

$$\begin{aligned}
J &= 0.0658 \times 10^6\ \mathrm{J \cdot s^2}, \quad H = 5.6\,\mathrm{s},\\
r_s &= 0.00243\,\Omega,\\
X_{ls} &= 0.1538\,\Omega, \quad X_q = 1.457\,\Omega, \quad r_{kq1} = 0.00144\,\Omega,\\
X_{lkq1} &= 0.6578\,\Omega, \quad r_{kq2} = 0.00681\,\Omega,\\
X_{lkq2} &= 0.07602\,\Omega,\\
X_d &= 1.457\,\Omega, \quad r_{fd} = 0.00075\,\Omega, \quad X_{lfd} = 0.1145\,\Omega,\\
r_{kd} &= 0.0108\,\Omega, \quad X_{lkd} = 0.06577\,\Omega.
\end{aligned}$$

Induction machine parameters [3]: 3-HP, 220 V, 1710 r/min, 4 poles, 60-Hz, $r_s = 0.435\ \Omega, X_{ls} = 0.754\ \Omega, X_m = 26.13\,\Omega, r_r = 0.816\,\Omega, X_{lr} = 0.754\,\Omega, J = 0.089$ kg $\cdot\mathrm{m}^2$

$$\text{Source reactance}: X_{\text{source}} = 0.377\,\Omega;$$
$$\text{Snubber resistance}: r_{\text{snub}} = 1000\,\Omega.$$

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Re-examination of Synchronous Machine Modeling Techniques for Electromagnetic Transient Simulations

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***Abstract*—This paper re-examines the three synchronous machine modeling techniques used for electromagnetic transient simulations, namely, the qd model, phase-domain model, and voltage-behind-reactance model. Contrary to the claims made in several recent publications, these models are all equivalent in the continuous-time domain, as their corresponding differential equations can be algebraically derived from each other. Computer studies of a single-machine infinite-bus system demonstrate that all of these models can be used for unsymmetrical operation of power systems. The conversion of machine parameters is also discussed and is shown to have some impact on simulation accuracy, which is acceptable for most cases. When the models are discretized and interfaced with an EMTP-type network solution, the voltage-behind-reactance model is shown to be the most accurate due to its advanced structure.**

***Index Terms*—Electromagnetic Transient Programs (EMTP), phase-domain model, synchronous machine, unbalanced fault, voltage-behind-reactance model.**

I. Introduction

MODELING of synchronous machines has been an active research subject for quite a long time. Depending on the objective of studies and the required level of fidelity, the modeling approaches may be roughly divided into three categories: finite element (or difference) method [1]; equivalent magnetic circuit approach [2], [3]; and coupled electric circuit approach. This paper mainly considers the last approach, which leads to a relatively small number of equations and has been very often utilized for predicting the dynamic responses of electrical machines in power system operations. To further simplify the coupled electric circuit approach, the machine physical variables are often transformed into quadrature and direct magnetic rotor axes [4]–[7]. This approach is also known as the qd modeling of rotating machines, which is widely used in Electromagnetic Transient Programs (EMTP) [8] for power system transient simulations and analysis. Currently, there are many EMTP-type programs that use general purpose synchronous machine models based on the $qd0$ transformation. These models include the Type-50 machine model in MicroTran (MT) [9], the Type-59 machine model in EPRI/DCG EMTP [10] and ATP [11], and the synchronous machine model in PSCAD/EMTDC [12]. The validity of these models for both balanced and unbalanced power system operations has been confirmed by academic and industrial applications [13], [14].

The so-called phase-domain (PD) models, as the original form of the coupled electric circuit machine models, were proposed in [15] and [16] for the nodal analysis approach. These models achieved simultaneous solution of the machine electrical variables and the network equations, they and improve numerical accuracy and stability [17], [18]. However, the time variant self and mutual inductances of the PD model stator and rotor circuits complicate the modeling and increase the computational cost.

To integrate the advantages of the PD model with the benefit of the qd transformation, the voltage-behind-reactance (VBR) machine model was introduced in [19] for the state variable approach and extended for the EMTP-type solution in [20]. In the VBR model structure, the stator circuit is represented in abc phase coordinates while the rotor equations are expressed in qd rotor reference frame. Similar to the PD model, simultaneous solution of the network and machine electrical variables is achieved, and numerical accuracy and simulation efficiency are further improved upon.

Although all of the above-mentioned models rely on the same set of assumptions [21]–[23], some researchers have questioned the applicability of the qd model to unbalanced conditions. In [24]–[26], the authors claim that the qd machine model "assumes perfectly balanced operation conditions in machine" and "cannot accurately describe the transient and steady-state unbalanced operation of the synchronous machine." The authors of [27] and [28] made similar conclusions by comparing the simulation results of the qd and PD models under unbalanced operation. These results seem to contradict the existing machine modeling theories and undermine the long-established validity of the qd machine model for power system transient simulations.

In this paper, computer studies are conducted using the same synchronous machine as in [24]–[26] to demonstrate that the qd machine model, PD model, and VBR model all give identical simulation results for unbalanced operation, provided that the simulation time-step Δt is sufficiently small. This result is expected as all of these models can be derived from each other and are therefore equivalent in the continuous time domain. The claims made in [24]–[28] that the qd machine model cannot represent unbalanced operations are unfounded. The difference in

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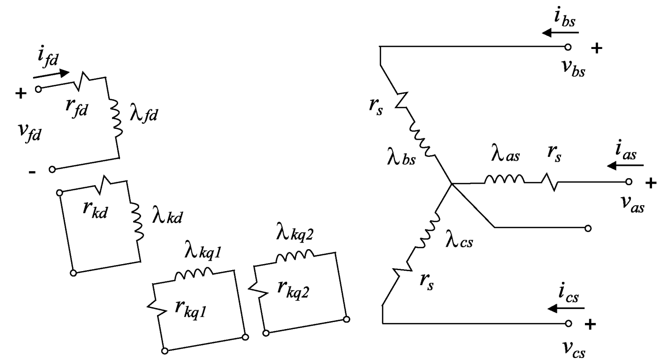

Fig. 1 Coupled-circuits model of a synchronous machine.

simulation results, however, may possibly arise from inconsistent machine parameters, problems with user interfaces, and/or model implementation. We also show that the use of equivalent circuit parameters versus the manufacturer's data (transient/subtransient inductances and time constants) should not in general lead to significant differences in simulation results. However, one should be aware of existing procedures for parameter conversion and apply them with care.

This paper also shows that numerical properties of the considered machine models differ when their respective differential equations are discretized and interfaced with the network solution. The VBR model is shown to be more accurate, even for sufficiently large integration time steps.

II. Machine Models

Prior to comparing the results, it is instructive to briefly review the various models discussed in this paper. According to the commonly used assumptions [21]–[23], a general purpose three-phase synchronous machine may be represented as a coupled circuit, depicted in Fig. 1. For the purpose of discussion and without loss of generality, the stator circuit consists of three armature windings, and the rotor circuit includes one field winding and one damper winding in d axis and two damper windings in q axis, respectively. The q axis is assumed to be leading the d axis by 90° [23]. Motor convention is used in the machine's voltage equations so that the stator currents flowing into the machine have a positive sign in the voltage equations. The flux linkage of each winding is assumed to have the same sign as the current flowing in that winding.

The mechanical subsystem is assumed to be represented as

$$p\theta_r = \omega_r \tag{1}$$

$$p\omega_r = \frac{P}{2J}(T_e - T_m). \tag{2}$$

Here, the operator $p = d/dt$, P is the number of poles, J is the moment of inertia, T_e and T_m are the developed electromagnetic torque and the mechanical torque, and θ_r and ω_r are the rotor position and speed, respectively. Although the models considered herein all are assumed to have the same mechanical subsystem represented by (1) and (2), the electromagnetic torque is expressed differently in each case.

A. The qd Model

The voltage equations of the qd machine model can be represented in terms of transformed variables as

$$\mathbf{v}_{qd0} = \mathbf{R}\mathbf{i}_{qd0} + p\lambda_{qd0} + \mathbf{u} \tag{3}$$

where

$$\mathbf{v}_{qd0} = [v_{qs} \quad v_{ds} \quad v_{0s} \quad 0 \quad 0 \quad v_{fd} \quad 0]^T \tag{4}$$

$$\mathbf{i}_{qd0} = [i_{qs} \quad i_{ds} \quad i_{0s} \quad i_{kq1} \quad i_{kq2} \quad i_{fd} \quad i_{kd}]^T \tag{5}$$

$$\mathbf{u} = [\omega_r\lambda_{ds} \quad -\omega_r\lambda_{qs} \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T \tag{6}$$

$$\boldsymbol{\lambda}_{qd0} = [\lambda_{qs} \quad \lambda_{ds} \quad \lambda_{0s} \quad \lambda_{kq1} \quad \lambda_{kq2} \quad \lambda_{fd} \quad \lambda_{kd}]^T. \tag{7}$$

Here, $\mathbf{v}_{qd0}$ and $\mathbf{i}_{qd0}$ are the winding voltages and currents, respectively, in qd rotor reference frame; v_{fd} is the field winding voltage; $\mathbf{u}$ is the speed voltage vector; and $\mathbf{R}$ is the stator and rotor resistance matrix.

The flux linkage equations are given as

$$\boldsymbol{\lambda}_{qd0} = \mathbf{L}_{qd0}\mathbf{i}_{qd0} \tag{8}$$

where the machine self and mutual inductance matrix $\mathbf{L}_{qd0}$ is constant, due to Park's transformation.

The electromagnetic torque is expressed as

$$T_e = \frac{3P}{4}(\lambda_{ds}i_{qs} - \lambda_{qs}i_{ds}). \tag{9}$$

B. Phase-Domain Model

The PD model is commonly expressed in terms of the machine's physical variables and phase coordinates. In particular, the voltage equation is expressed as

$$\begin{bmatrix}\mathbf{v}_{abcs}\\ \mathbf{v}_{qdr}\end{bmatrix} = \mathbf{R}\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}_{qdr}\end{bmatrix} + p\begin{bmatrix}\boldsymbol{\lambda}_{abcs}\\ \boldsymbol{\lambda}_{qdr}\end{bmatrix} \tag{10}$$

and the flux linkages are represented as

$$\begin{bmatrix}\boldsymbol{\lambda}_{abcs}\\ \boldsymbol{\lambda}_{qdr}\end{bmatrix} = \mathbf{L}(\theta_r)\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}_{qdr}\end{bmatrix} \tag{11}$$

where the stator and rotor self and mutual inductance matrix $\mathbf{L}(\theta_r)$ depends on the rotor position θ_r. The electromagnetic torque is expressed in the machine variables as

$$T_e = \frac{1}{2}\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}_{qdr}\end{bmatrix}^T \frac{\partial}{\partial\theta_r}\mathbf{L}(\theta_r)\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}_{qdr}\end{bmatrix}. \tag{12}$$

C. Voltage-Behind-Reactance Model

The VBR machine model represents the stator variables in abc phase coordinates and the rotor variables in transformed qd coordinates [19]. Here, the stator voltage equation is expressed as

$$\mathbf{v}_{abcs} = \mathbf{R}_s\mathbf{i}_{abcs} + p\,[\mathbf{L}''_{abcs}(\theta_r)\mathbf{i}_{abcs}] + \mathbf{v}''_{abcs} \tag{13}$$

where

$$\mathbf{L}''_{abcs}(\theta_r) = \begin{bmatrix} L_S(2\theta_r) & L_M\left(2\theta_r - \frac{2\pi}{3}\right) & L_M\left(2\theta_r + \frac{2\pi}{3}\right) \\ L_M\left(2\theta_r - \frac{2\pi}{3}\right) & L_S\left(2\theta_r - \frac{4\pi}{3}\right) & L_M(2\theta_r) \\ L_M\left(2\theta_r + \frac{2\pi}{3}\right) & L_M(2\theta_r) & L_S\left(2\theta_r + \frac{4\pi}{3}\right) \end{bmatrix} \quad (14)$$

$$L_S(\cdot) = L_{ls} + L_a - L_b \cos(\cdot) \quad (15)$$

$$L_M(\cdot) = -\frac{L_a}{2} - L_b \cos(\cdot) \quad (16)$$

$$L_a = \frac{L''_{mq} + L''_{md}}{3} \quad (17)$$

$$L_b = \frac{L''_{md} - L''_{mq}}{3}. \quad (18)$$

The inductances L''_{md} and L''_{mq} are calculated as

$$L''_{md} = \left[\frac{1}{L_{md}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lkd}}\right]^{-1} \quad (19)$$

$$L''_{mq} = \left[\frac{1}{L_{mq}} + \frac{1}{L_{lkq1}} + \frac{1}{L_{lkq2}}\right]^{-1}. \quad (20)$$

The voltage-behind-reactance term in (13) is defined as

$$\mathbf{v}''_{abcs} = [\mathbf{K}^r_s(\theta_r)]^{-1} \begin{bmatrix} v''_q & v''_d & 0 \end{bmatrix}^T \quad (21)$$

where the rotor reference frame transformation matrix is

$$\mathbf{K}^r_s(\theta_r) = \frac{2}{3} \begin{bmatrix} \cos\theta_r & \cos\left(\theta_r - \frac{2\pi}{3}\right) & \cos\left(\theta_r + \frac{2\pi}{3}\right) \\ \sin\theta_r & \sin\left(\theta_r - \frac{2\pi}{3}\right) & \sin\left(\theta_r + \frac{2\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \quad (22)$$

$$v''_q = \omega_r \lambda''_d + \frac{L''_{mq} r_{kq1} (\lambda''_q - \lambda_{kq1})}{L^2_{lkq1}} + \frac{L''_{mq} r_{kq2} (\lambda''_q - \lambda_{kq2})}{L^2_{lkq2}} + \left(\frac{r_{kq1}}{L^2_{lkq1}} + \frac{r_{kq2}}{L^2_{lkq2}}\right) L''^2_{mq} i_{qs} \quad (23)$$

$$v''_d = -\omega_r \lambda''_q + \frac{L''_{md} r_{kd} (\lambda''_d - \lambda_{kd})}{L^2_{lkd}} + \frac{L''_{md}}{L_{lfd}} v_{fd} + \frac{L''_{md} r_{fd} (\lambda''_d - \lambda_{fd})}{L^2_{lfd}} + \left(\frac{r_{fd}}{L^2_{lfd}} + \frac{r_{kd}}{L^2_{lkd}}\right) L''^2_{md} i_{ds} \quad (24)$$

with

$$\lambda''_q = L''_{mq} \frac{\lambda_{kq1}}{L_{lkq1}} + L''_{mq} \frac{\lambda_{kq2}}{L_{lkq2}} \quad (25)$$

$$\lambda''_d = L''_{md} \frac{\lambda_{fd}}{L_{lfd}} + L''_{md} \frac{\lambda_{kd}}{L_{lkd}} \quad (26)$$

The rotor state equations are represented as

$$p\lambda_j = -\frac{r_j}{L_{lj}} (\lambda_j - \lambda_{mq}); \quad j = kq1, kq2 \quad (27)$$

$$p\lambda_j = -\frac{r_j}{L_{lj}} (\lambda_j - \lambda_{md}) + v_j; \quad j = fd, kd \quad (28)$$

where

$$\lambda_{mq} = L''_{mq} \left(\frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}} + i_{qs}\right) \quad (29)$$

$$\lambda_{md} = L''_{md} \left(\frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}} + i_{ds}\right). \quad (30)$$

Here, λ_j denotes the rotor winding flux linkages; λ_{mq} and λ_{md} are the magnetizing flux linkages in the q and d axes, respectively.

The equations for rotor speed and position in the VBR machine model are identical to (1) and (2). However, the electromagnetic torque is calculated as [23]

$$T_e = \frac{3P}{4} (\lambda_{md} i_{qs} - \lambda_{mq} i_{ds}). \quad (31)$$

D. Relationship Among Models

Although the PD model is based on straightforward state equations describing the coupled circuit depicted in Fig. 1, its implementation in digital programs is complicated by the presence of the rotor-position-dependent self and mutual inductances represented in (11). The qd model is algebraically derived from the PD model using Park's transformation with no approximations [22], [23]. Because the qd model results in equations with constant coefficients, its implementation is relatively simple, which explains the wide use of this model. However, the disadvantage of the qd model is its interface with EMTP network solutions, which was shown to result in a loss of numerical accuracy and stability for large time steps [20].

The VBR model was originally derived from PD and qd models without any approximation by appropriately applying the rotor reference frame only to selected terms [19]. Therefore, the qd model, the PD model, and the VBR model can be derived from each other and are all equivalent in the continuous time domain. However, the numerical properties of these models differ when their equations are discretized using a particular integration scheme. A detailed analysis of these models in [19] and [20] shows that the VBR model is more accurate than the PD model due to rescaled and improved eigenvalues. To make a fair comparison among the models here, (1) and (2) have been discretized in the same way using implicit trapezoidal rule, and the mechanical variables θ_r and ω_r are predicted using linear extrapolation to avoid the solution of the machine nonlinear equations. This method is commonly used in EMTP-type programs and is justified on the facts that the time constant of the mechanical system is much larger than that of the electrical system.

III. Input Data Conversion

For most EMTP-type programs the self and mutual inductances of stator and rotor circuits are internally utilized to calculate the equivalent circuit parameters and solve the machine equations [22]. However, these internal circuit parameters are not directly determined from the test measurements. Instead, the transient and subtransient inductances and time constants are first defined to fit the field test data (manufacturer's data) with acceptable accuracy, from which the corresponding internal circuit parameters are calculated using a parameter conversion procedure.

There are several commonly used parameter conversion methods in the literature [29]–[32]. In [29], a classical (approximate) parameter conversion procedure is presented that assumes that the flux linkages do not change instantly following a short-circuit test, and that r_{kd} and r_{kq2} are much larger than r_{fd} and r_{kq1}, respectively, so that the subtransient period elapses much faster than the transient period. Based on these assumptions, the classical parameter conversion is summarized in Appendix A. The conversion is quite straightforward and has been widely used, although the possibility of a noticeable error was reported in [30].

A more accurate parameter conversion method proposed by Canay [30] and Dommel [31] (a refinement of Canay's data conversion) is given in Appendix B. This method is based on a set of more accurate definitions of transient and subtransient inductances and time constants in terms of machine circuit parameters. For consistency, a brief review of these definitions is included here. The detailed derivations can be found in [22, Appendix].

The transient and subtransient inductances and time constants are derived using an eigenvalue and eigenvector approach to the machine's differential equations. Without loss of generality, the procedure of deriving short-circuit time constants is given below. A similar procedure is used with the open-circuit time constants. If the armature resistance r_s is assumed to be zero, the stator state equations for the short-circuit test can be expressed as [31]

$$\frac{d\lambda_q}{dt} = -\omega_r \lambda_d \tag{32}$$

and

$$\frac{d\lambda_d}{dt} = \omega_r \lambda_q. \tag{33}$$

The solution of (32) and (33) is given by

$$\lambda_q = E_q \sin(\omega_r t) \tag{34}$$

and

$$\lambda_d = -E_q \cos(\omega_r t) \tag{35}$$

where E_q is defined by the initial condition $E_q = -\lambda_d(0)$. Here the solution is slightly different from [22, Appendix VI], as the motor convention is used in the machine voltage equations. The rotor state equations for the d axis are formulated as

$$\begin{bmatrix} \frac{d\lambda_{fd}}{dt} \\ \frac{d\lambda_{kd}}{dt} \end{bmatrix} = -\begin{bmatrix} r_{fd} & 0 \\ 0 & r_{kd} \end{bmatrix} \begin{bmatrix} i_{fd} \\ i_{kd} \end{bmatrix} + \begin{bmatrix} v_{fd} \\ 0 \end{bmatrix} \tag{36}$$

and

$$\begin{bmatrix} \lambda_d \\ \lambda_{fd} \\ \lambda_{kd} \end{bmatrix} = \begin{bmatrix} L_d & L_{md} & L_{md} \\ L_{md} & L_{fd} & L_{md} \\ L_{md} & L_{md} & L_{kd} \end{bmatrix} \begin{bmatrix} i_d \\ i_{fd} \\ i_{kd} \end{bmatrix} \tag{37}$$

where

$$L_d = L_{ls} + L_{md} \tag{38}$$

$$L_{fd} = L_{lfd} + L_{md} \tag{39}$$

$$L_{kd} = L_{lkd} + L_{md}. \tag{40}$$

After λ_{fd} and λ_{kd} are expressed in terms of i_{fd}, i_{kd}, and λ_d, the final equation has the following form [22, Appendix VI.11]

$$\begin{bmatrix} \frac{di_{fd}}{dt} \\ \frac{di_{kd}}{dt} \end{bmatrix} = \mathbf{A} \begin{bmatrix} i_{fd} \\ i_{kd} \end{bmatrix} + \mathbf{g}\left(v_{fd}, \frac{d\lambda_d}{dt}\right) \tag{41}$$

where $\mathbf{A}$ is system matrix and $\mathbf{g}$ is the forcing function vector.

The short-circuit time constants T'_d, T''_d are obtained as the negative reciprocals of the eigenvalues of the matrix $\mathbf{A}$ [22, Appendix VI.11]. The short-circuit currents i_{fd} and i_{kd} can be obtained by solving (41) analytically. The solutions of i_{fd} and i_{kd} include three parts: the steady state, the transient associated with T'_d, and the subtransient associated with T''_d. Therefore, the solution for i_d also contains three parts, since

$$i_d = \frac{\lambda_d}{L_d} - \frac{L_{md}}{L_d}(i_{fd} + i_{kd}). \tag{42}$$

After substituting the analytical solutions of i_{fd} and i_{kd} into (42), the transient part of i_d associated with T'_d can be represented as [22, Appendix VI.27a]

$$i_{d-transient} = -\frac{1}{L_d} \frac{(T'_d - T'_{d0})(T'_d - T''_{d0})}{T'_d(T'_d - T''_d)} \int_0^t e^{-\frac{(t-u)}{T'_d}} \frac{d\lambda_d}{du} du. \tag{43}$$

Using (35), the transient part of i_d can be further expressed as

$$i_{d-transient} = -\frac{E_q}{L_d} \frac{(T'_d - T'_{d0})(T'_d - T''_{d0})}{T'_d(T'_d - T''_d)} \frac{(\omega_r T'_d)^2}{1 + (\omega_r T'_d)^2} e^{-\frac{t}{T'_d}} \tag{44}$$

which can then be matched with the amplitude of the transient part of i_d as read from the short-circuit measurement. Therefore, the transient inductance can be obtained from

$$\frac{1}{L'_d} - \frac{1}{L_d} = -\frac{1}{L_d} \frac{(T'_d - T'_{d0})(T'_d - T''_{d0})}{T'_d(T'_d - T''_d)} \frac{(\omega_r T'_d)^2}{1 + (\omega_r T'_d)^2}. \tag{45}$$

Similarly, the subtransient inductance is obtained from

$$\frac{1}{L''_d} - \frac{1}{L'_d} = -\frac{1}{L_d} \frac{(T''_d - T'_{d0})(T''_d - T''_{d0})}{T''_d(T''_d - T'_d)} \frac{(\omega_r T''_d)^2}{1 + (\omega_r T''_d)^2}. \tag{46}$$

If T'_d, T''_d is much larger than the period of oscillation of $d\lambda_d/dt$ (i.e., $T'_d \gg 2\pi/\omega_r$, $T''_d \gg 2\pi/\omega_r$), the transient and subtransient inductances may be approximated as

$$\frac{1}{L'_d} - \frac{1}{L_d} = -\frac{1}{L_d} \frac{(T'_d - T'_{d0})(T'_d - T''_{d0})}{T'_d(T'_d - T''_d)} \tag{47}$$

$$\frac{1}{L''_d} - \frac{1}{L'_d} = -\frac{1}{L_d} \frac{(T''_d - T'_{d0})(T''_d - T''_{d0})}{T''_d(T''_d - T'_d)}. \tag{48}$$

When only the open-circuit or short-circuit time constants are known, the other pair can be calculated using the solution of (47) and (48) or (45) and (46), which correspond to Canay's data conversion [30] and its refinement [31], respectively. Both of these approaches are explicit and neglect the armature winding resistance to decouple the definitions of test parameters in the direct

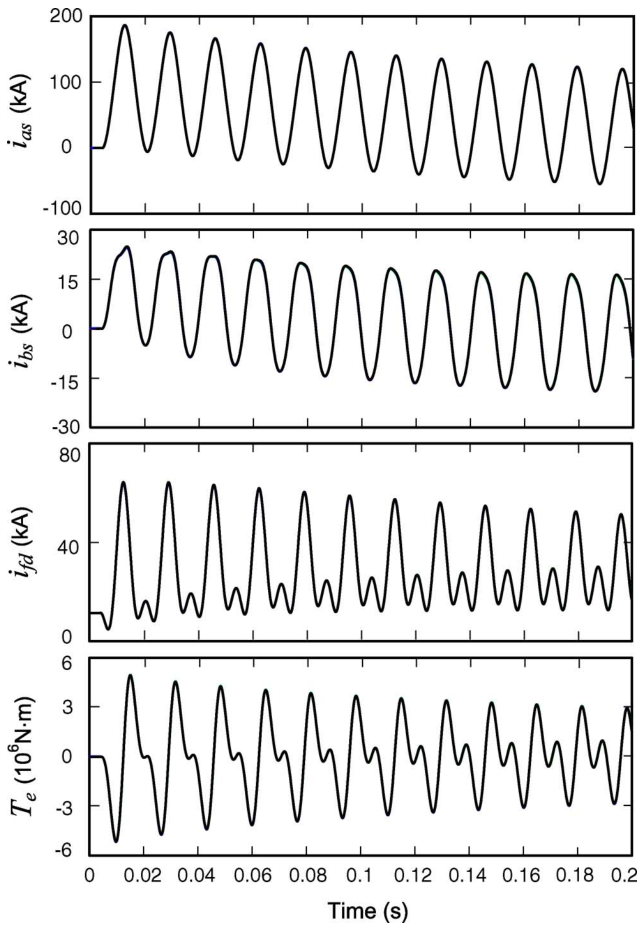

Fig. 2. Simulation results with time-step of 50 μs.

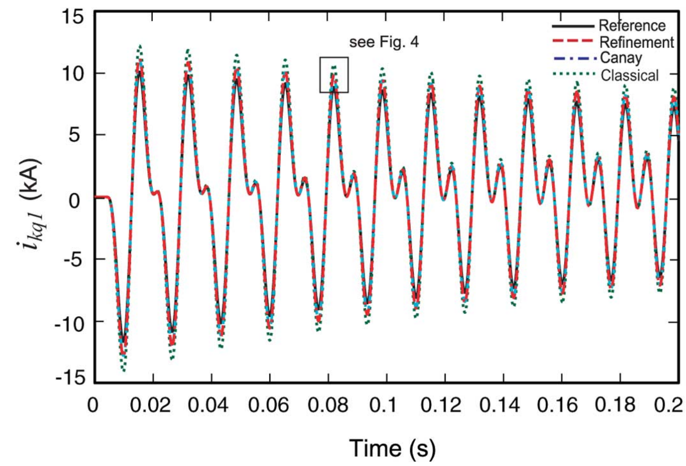

Fig. 3. Damper winding current i_{kq1} using different data conversion methods.

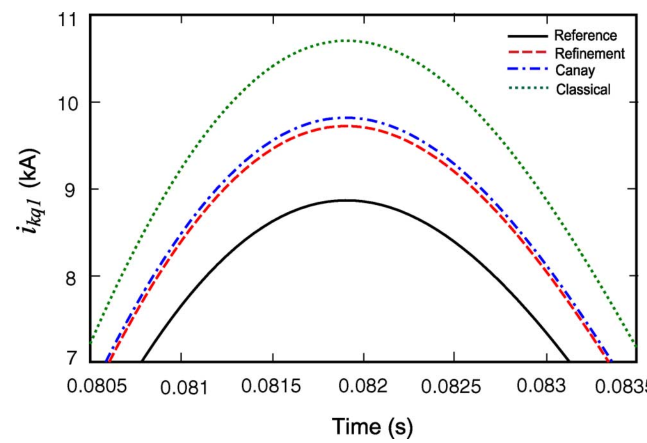

Fig. 4. Magnified plot of i_{kq1} using different data conversion methods.

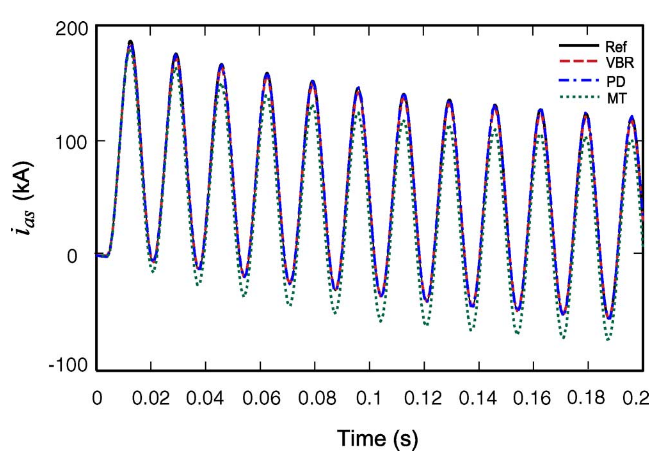

Fig. 5. Stator current i_{as} with time step of 1 ms.

and quadrature axes. In [32], the researchers removed this assumption and proposed an implicit procedure for calculating the circuit parameters from the known qd axes' transient and subtransient time constants and eigenvalues. However, the calculation results in [32] show that Canay's data conversion is quite accurate for typical machine data.

IV. Case Studies

In this paper, we used two EMTP software packages, MicroTran and ATP, both with the build-in synchronous machine models based on the traditional qd model. The PD and VBR models were also implemented using a nodal analysis approach in order to compare their performances. A single-machine infinite-bus system is used in which the synchronous machine parameters were obtained from [21] and are summarized in Appendix C. In this section, the machine reference circuit parameters were used for all machine models in Sections IV-A and IV-C. Different parameter conversion methods are used in Section IV-B, and the results are compared with the reference circuit parameters.

In the following transient study, the synchronous machine initially operates at no load. At $t = 0.004$ s, a single-phase-to-ground fault is applied at the machine terminals (phase a). The transient responses produced by the various models are plotted in Figs. 2–9. The transient studies of balanced operation have been previously considered in [20] and are not included here due to space limitations.

A. Small Time-Step Study

The considered study of an unbalanced fault is first simulated using a relatively small time step, $\Delta t = 50$ μs. The resulting transients of the stator currents i_{as}, i_{bs}, field current

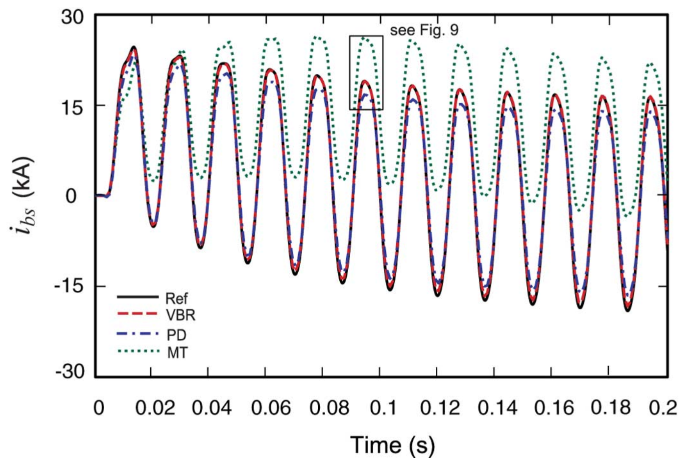

Fig. 6. Stator current i_{bs} with time step of 1 ms.

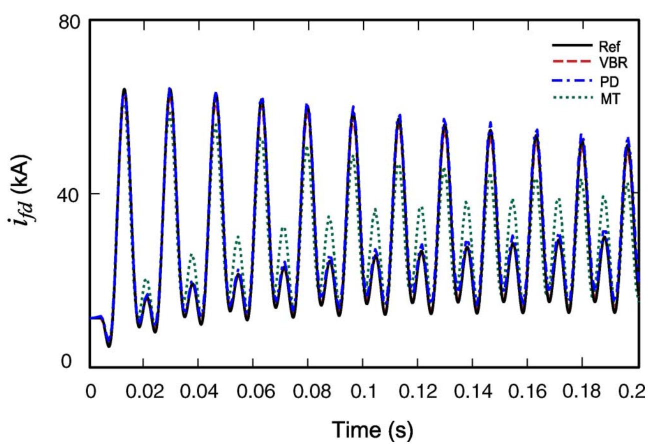

Fig. 7. Field current i_{fd} with time step of 1 ms.

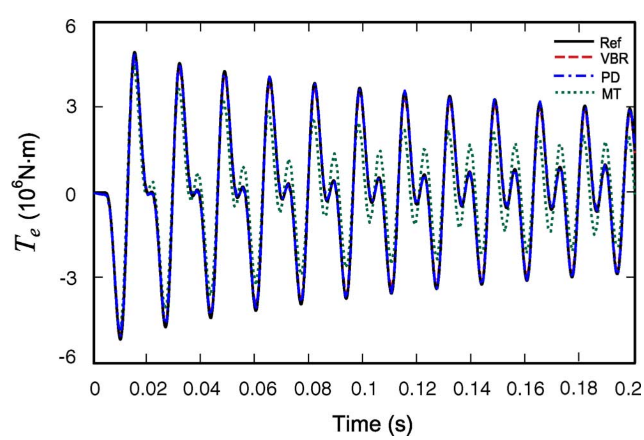

Fig. 8. Electromagnetic torque T_e with time step of 1 ms.

i_{fd}, and electromagnetic torque T_e are depicted in Fig. 2. The reference solutions were obtained using the qd machine model implemented in MATLAB/Simulink and solved with the Runge–Kutta fourth-order method using a time step of 1 μs. The simulation results obtained by the MicroTran, ATP, PD, and VBR models are overlaid with the reference solutions. As can be observed from Fig. 2, the transient responses produced by all models coincide and converge to the reference solutions.

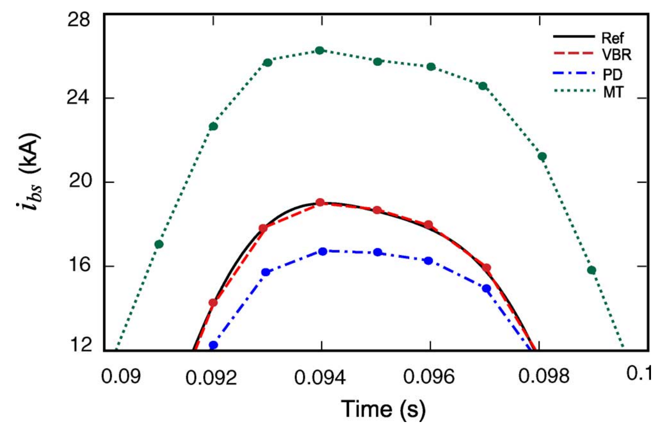

Fig. 9. Magnified plot of i_{bs} with time step of 1 ms.

This clearly demonstrates that these models are all equivalent and predict the unbalanced operations with acceptable accuracy for the given time step, which is contrary to the conclusions made in [24]–[28].

B. Impact of Parameter Conversion Methods

The authors of [24]–[26] suggested that the different simulation results obtained by the qd and PD models may be caused by "different data based on their frames of reference" [24] and listed the machine's internal circuit data and test data in the Appendices of [24], [25]. However, these two sets of data are equivalent, as is shown in [21]. To investigate this point further, we implemented the same system using MicroTran with the input of the test parameters and reference circuit parameters. Internally, MicroTran converts the test parameters into the circuit parameters using one of the three conversion methods as described in Section III. As expected, the deviation among the final circuit parameters was not very significant. The most noticeable difference was observed in the q axis parameters, which are summarized in Table I. The corresponding transient responses are visibly the same as in Fig. 2 and are not shown here due to space limitations. As Table I reveals, the most noticeable differences exists in the parameters of $kq1$ winding. Therefore, the transient observed in current i_{kq1} is shown in Fig. 3. A magnified fragment of Fig. 3 is also shown in Fig. 4 for better comparison. As can be seen in Figs. 3 and 4, the impact of various data conversion methods is small, and all transients are reasonably close to the reference solution obtained using the original circuit parameters. The classical method (dashed line) gives the most deviation from the reference (solid line), with successive improvements by Canay's method (dash-dotted line) and its minor refinement (long-dashed line), all consistent with the accuracy of the parameters summarized in Table I. Therefore, the impact of parameter conversion methods on the transient responses is insignificant. The different simulation results in [24]–[26] obtained using the qd model and the phase-domain model may be caused by inconsistent data or problematic model implementation.

C. Large Time-Step Study

To further compare the numerical properties and robustness of the machine models, a larger time step ($\Delta t = 1$ ms) is ap-

TABLE I
DATA CONVERSION FOR Q AXIS IN OHM

	x_{kq1}	x_{kq2}	r_{kq1}	r_{kq2}
Reference circuit parameters	2.423559	1.800649	0.006424216	0.02457600
Classical	2.294305	1.805382	0.005095732	0.02664194
Canay's	2.350702	1.803016	0.005886168	0.02470836
Refinement	2.350911	1.800600	0.005889885	0.02460227

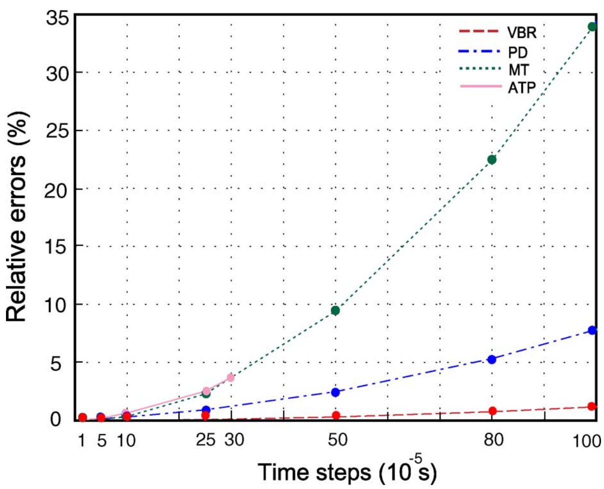

Fig. 10. Propagation of numerical error in i_{bs} for different time steps.

plied to the same test case as in Sections IV-A and IV-B. The same variables are plotted in Figs. 5–8. As can be observed in Figs. 5–8, the results produced by the various models visibly diverge from the reference solution. The transient responses obtained by MicroTran's machine model Type-50 (dotted line, MT legend) diverge from the reference solutions with the largest error among the considered models, whereas ATP's machine model Type-59 was found not convergent at such a large time step. The problem of convergence and accuracy of qd models is attributed to their interface with the external network and is discussed in detail in [17], [18], and [20]. At the same time, the VBR and PD models still produce reasonably accurate and convergent simulation results. A more detailed fragment of the transient observed in i_{bs} is shown in Fig. 9, which is a magnified plot of Fig. 6. As Fig. 9 (and Figs. 5–8) shows, the VBR model produces the most accurate results among the considered models, due to its advanced model structure [19], [20].

D. Error Behavior

In order to show the numerical accuracy of machine models at different time steps, the same study was simulated several times using integration time steps from 10 μs to 1 ms. Herein, without loss of generality, only the phase current i_{bs} is considered due to space limitation. The relative error between the reference solution $\widetilde{i}_{bs}$ (as defined in Section IV-A) and a given numerical solution trajectory i_{bs} is calculated using the 2-norm [33] as

$$\%error = \frac{||\widetilde{i}_{bs} - i_{bs}||_2}{||\widetilde{i}_{bs}||_2} \cdot 100. \tag{49}$$

The errors are calculated for different time-step sizes, and the results are shown in Fig. 10.

As can be seen in Fig. 10, the solutions of all models are convergent to the reference solution when the time step is sufficiently small. However, the simulation accuracy degrades rapidly when the time step is increased. A similar trend is observed with other variables. For $\Delta t = 1$ ms, MicroTran's qd model results in a cumulative error of about 35%, whereas ATP's Type-59 model was not convergent with a time step larger than 300 μs (under default error tolerance). Based on these studies, it appears that qd models may be used with acceptable accuracy with a time-step as large as 100 to 200 μs.

Although the PD and VBR models considered here use prediction of mechanical variables in the same way as traditional qd models (e.g., Type-50 qd model in MicroTran, Type-59, and Universal Machine models in ATP), these models achieve simultaneous solution of machine electrical variables and network variables and therefore produce stable and more accurate results. The VBR model achieves higher numerical accuracy, as shown in Figs. 9 and 10. This improved numerical accuracy is attributed to better-scaled eigenvalues of this model formulation [19], [20]. In particular, the eigenvalues of the discretized model determine the behavior and/or propagations of local errors. Interested readers may find a more detailed analysis of numerical accuracy and system eigenvalues in [34] and [35].

V. DISCUSSION

The models discussed in this paper belong to a group of general-purpose, lumped-parameter models that may be used with various EMTP-type packages to carry out simulations of typical power system transient studies with balanced and/or unbalanced conditions on the system side. However, sometimes there is a need to model and implement more detailed and/or internal phenomena, such as stator inter-turn and/or inter-windings faults.

The general purpose models (e.g., qd, PD, and VBR) are all based on the circuits that assume symmetrical stator phase windings as depicted in Fig. 1. These models are therefore not directly suitable for studying internal faults, since special consideration of the type of fault being examined is required. To model the transient phenomena associated with internal faults in a single phase, at least one of the stator windings must be represented in more detail; that is, this phase must be partitioned into smaller sections such that the fault is then applied to an appropriate portion of the winding. If the internal fault involves more than one phase, then the affected windings must be partitioned accordingly to implement the required phenomena. For example, the authors of [36] use a partitioning of the stator windings into interior and exterior sections in order to study the stator winding internal faults. Since the PD model formulation does not require any symmetry among the phases, this model formulation can be readily extended to represent the synchronous machine's internal faults, as proposed in [37]–[39], at the expense of including the required winding subsections and increasing the dimensions of the inductance matrix accordingly.

When the stator phase windings are not symmetric, the main advantage of applying the qd transformation, which previously resulted in constant matrices, is no longer present, and the transformed inductance and resistance matrices will depend on the rotor position [23]. Thereafter, straightforward transformation of stator equations into the qd rotor reference frame may offer little if any advantage over the direct coupled circuit formulation (i.e., the PD model). It might be possible to partition all three stator phases into several equal subsections and still obtain some benefit by transforming the stator equations into the qd rotor reference frame. The authors have not seen this reported in the literature, however, and more research is required to confirm this possibility.

Incorporating the effect of magnetic saturation is another important aspect of improving the accuracy and range of application of synchronous machines models. Using the qd machine models, magnetic saturation may be represented with accuracy acceptable for power-system transient analysis [21]–[23], [40]–[43]. For EMTP-type solutions, a piece-wise linear approximation is often used to represent the magnetic saturation characteristics of the qd models [22], [40]. The necessary machine parameters may be obtained using the standstill frequency response (SSFR) and/or online frequency response (OLFR) measurements [41]. Higher-order qd machine models have been proposed to fairly accurately represent the distributed-parameter nature of the rotor and saturation effects [43], with the model parameters derived from the measured saturation characteristics and the SSFR. The qd model proposed in [43] offers more accuracy than is typically required for studying power system transients at low frequencies, which comes at the expense of additional complexity.

The PD model proposed in [16] also provides a flexible and potentially accurate way to represent variable reluctance and saturation effects along different flux paths. However, the complexity of representing magnetic saturation in PD formulation is in general much higher than it is in qd models. Implementation of magnetic saturation with the VBR model has been previously considered in [44], where the authors have shown that incorporating the saturation in the d axis only may provide an adequate match with the hardware measurements. The VBR model formulation may provide different options for including magnetic saturation, with accuracy similar to that of conventional qd and PD models. However, this subject requires more thorough consideration that is beyond the scope of this paper and deserves a dedicated publication, which the authors will pursue in the near future.

On the other hand, the finite-element-based and/or magnetic-equivalent-circuit-based models are naturally much better suited for studying internal machine details from the initial design stage. Various internal faults and magnetic saturation, including very fine structural/material details, can be studied using these models. These approaches should be used instead of low-order models (i.e., the qd, PD, and VBR) whenever very high fidelity simulations of a single machine are required. The resulting models, however, are computationally very expensive.

For typical power system transient simulations, however, the accuracy achieved by low-order models has been generally considered acceptable [21]. Even a more accurate qd model [43] may appear too complex and expensive when more than one machine is being considered in the system.

VI. Conclusion

Several full-order synchronous machine models used for EMTP-type solutions are investigated in this paper. Presented computer studies of a single-phase-to-ground fault show that the qd model, PD model, and VBR model are all equivalent in the continuous time domain. They can therefore all be used for studying power system transients with unbalanced (as well as balanced) operation, provided the time step Δt is sufficiently small. For large time steps, the VBR model is shown to have good stability property and to provide the most accurate results. The input data conversion procedures discussed are shown to have a noticeable but minor impact on simulation accuracy, with results equivalent and/or acceptable for most applications. The conclusions stated in [24]–[28] with regard to the unsuitability of the qd machine model for unbalanced operations have not been confirmed.

Appendix A

Classical data conversion [45, equations (15)–(28)]: The mutual inductance in the d axis, L_{ad} is assumed to be known, or is calculated by

$$L_{ad} = L_d - L_{ls} = L_{md}. \tag{A1}$$

The unknown circuit parameters in the d axis can be calculated from the transient and subtransient inductances and time constants as

$$\frac{1}{L_{lfd}} = \frac{1}{L'_d - L_{ls}} - \frac{1}{L_{ad}} \tag{A2}$$

$$\frac{1}{L_{lkd}} = \frac{1}{L''_d - L_{ls}} - \frac{1}{L'_d - L_{ls}} \tag{A3}$$

$$r_{fd} = \frac{L_{lfd} + L_{ad}}{T'_{d0}} \tag{A4}$$

$$r_{kd} = \frac{L_{lkd}L_{ad} + L_{lkd}L_{lfd} + L_{ad}L_{lfd}}{T''_{d0}(L_{ad} + L_{lfd})}. \tag{A5}$$

Appendix B

Canay's data conversion [22]: Two time constants, T_1 and T_2, are defined as

$$T_1 = \frac{L_{lfd}}{r_{fd}} \tag{B1}$$

$$T_2 = \frac{L_{lkd}}{r_{kd}}. \tag{B2}$$

Assuming L_{ad} is known, the time constants T_1 and T_2 can be calculated using the following equations:

$$T_1 + T_2 = (T'_{d0} + T''_{d0})\frac{L_{ad} - L_d}{L_{ad}} + (T'_d + T''_d)\frac{L_d}{L_{ad}} \tag{B3}$$

$$T_1 T_2 = T'_{d0}T''_{d0}\frac{L_{ad//lfd//lkd}}{L_{ad}} \tag{B4}$$

where

$$L_{ad//lfd//lkd} = \frac{T'_d T''_d}{T'_{d0}T''_{d0}} - (L_d - L_{ad}). \tag{B5}$$

The parallel combination of L_{ad} and L_{lfd} can then be calculated as

$$L_{ad//lfd} = \frac{L_{ad}(T_1 - T_2)}{T'_{d0} + T''_{d0} - \left(1 + \frac{L_{ad}}{L_{ad//lfd//lkd}}\right)T_2}. \tag{B6}$$

The circuit parameters can be determined as

$$L_{lfd} = \frac{L_{ad//lfd} L_{ad}}{L_{ad} - L_{ad//lfd}} \tag{B7}$$

$$L_{lkd} = \frac{L_{ad//lfd//lkd} L_{ad//lfd}}{L_{ad//lfd} - L_{ad//lfd//lkd}} \tag{B8}$$

$$r_{fd} = \frac{L_{lfd}}{T_1} \tag{B9}$$

$$r_{kd} = \frac{L_{lkd}}{T_2}. \tag{B10}$$

Similar procedures apply to q axis parameters.

Appendix C

Synchronous machine parameters [21]: 555 MVA, 24 kV, 0.9 pf, 2 poles, 3600 r/min, $J = 27547.8\ \mathrm{kg \cdot m^2}$

Reference circuit parameters (p.u.):

$$r_s = 0.003,\ L_{ls} = 0.15,\ r_{kq1} = 0.00619,\ r_{kq2} = 0.02368,$$
$$r_{fd} = 0.0006,\ r_{kd} = 0.0284,\ L_d = 1.81,\ L_q = 1.76$$
$$L_{lkq1} = 0.7252,\ L_{lkq2} = 0.125,\ L_{lfd} = 0.165,\ L_{lkd} = 0.1713$$

Test parameters (p.u.):

$$X'_d = 0.3,\ \ X'_q = 0.5875,\ \ X''_d = 0.23,\ \ X''_q = 0.25,$$
$$T'_{d0} = 8.171\ \mathrm{s},\ \ T'_{q0} = 1.1943\ \mathrm{s},$$
$$T''_{d0} = 0.0294\ \mathrm{s},\ \ T''_{q0} = 0.0586\ \mathrm{s}$$

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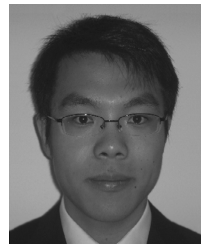

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Constant-Parameter *RL*-Branch Equivalent Circuit for Interfacing AC Machine Models in State-Variable-Based Simulation Packages

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***Abstract*—Transient simulation programs, either nodal analysis-based electromagnetic transient program (EMTP-like) or state-variable-based, are used very extensively for modeling and simulation of various power and energy systems with electrical machines. It has been shown in the literature that the method of interfacing machine models with the external electrical network plays an important role in numerical accuracy and computational performance of the overall simulation. This paper considers the state-variable-based simulation packages, and provides a constant-parameter decoupled *RL*-branch equivalent circuit for interfacing the ac induction and synchronous machine models with the external electrical network. The proposed interfacing circuit is based on the voltage-behind-reactance formulation which has been shown to have advantageous properties. For the synchronous machines, this paper describes both implicit and explicit (approximate) interfacing methods. The presented case studies demonstrate the advantages of using the proposed interfacing method over the traditional *qd*-models that are conventionally used in many simulation packages.**

***Index Terms*—AC machines, dynamic simulation, induction machine, interfacing circuit, *qd*-model, synchronous machine, voltage-behind-reactance (VBR) model.**

I. Introduction

MODELING of ac induction and synchronous machines has been an active area of research for many decades. Depending on the objectives of studies and required accuracy, various models have been proposed in the literature. This paper focuses on general-purpose models that are based on magnetically coupled circuit representation of the machine's windings which leads to a relatively small number of equations. Such models can be generally represented either in direct phase coordinates using the physical variables (phase-domain (PD) models) or in rotating qd-coordinates using the transformed variables (the classical qd-models) [1], [2]. The traditional qd-models are very common and suitable for many system-level transient studies. Such models are also widely available in many nodal analysis-based (i.e., EMTP-type [3]) and state-variable-based (e.g., Simulink [4]) simulation packages as built-in library components that are extensively used by many practicing engineers and researchers in industry and academia. Since such general-purpose machine models are widely used, improving their numerical efficiency and accuracy can have significant impact and improve many simulation packages.

To improve the interface between the machine model and the external power system network, which is typically represented in physical abc phase coordinates, the so-called PD [4]–[8] and voltage-behind-reactance (VBR) [9], [10] models have been recently proposed. These models achieve a direct interface with the external network circuit and are shown to offer some benefits for both state-variable and nodal analysis-based transient simulation programs [11]. However, the advantages of these models come at the price of having rotor-position-dependent parameters (inductances).

The main focus and goal of this paper is to propose a constant-parameter interfacing circuit (as shown in Fig. 1) for the induction and synchronous machine models particularly for the state-variable-based simulators. To the best of the authors' knowledge, such a unified (and at the same time very simple to use) interfacing circuit for both types of rotating machines has not been proposed before. Interested reader will find particularly useful the related work summarized in [9]–[15]. The present manuscript extends the previous work in this area and makes the following additional contributions;

1) A constant-parameter decoupled *RL*-branch equivalent circuit (with or without the zero-sequence) for a direct and explicit interface of the induction machine VBR models is proposed.
2) It is shown that the same generalized interface circuit can be achieved for the synchronous machine model (round or salient rotor), wherein explicit and implicit formulations are possible.
3) It is demonstrated that the proposed interfacing circuit VBR model provides advantages over conventional qd-models that require indirect (snubber circuit) interfaces.

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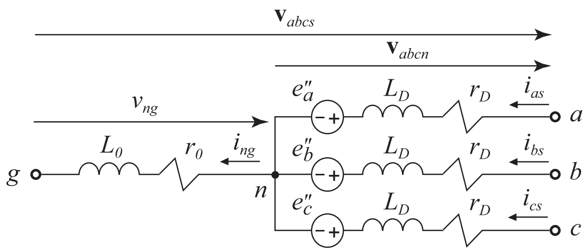

Fig. 1. New unified constant-parameter generalized VBR *RL*-branch equivalent circuit for interfacing induction and synchronous machine models with an external electrical network circuit.

4) The proposed unified interfacing circuit is very simple and easy to use, as it requires only the most basic and conventional circuit elements that are available in most simulation packages.

II. Considerations for Interfacing Machine Models

In addition to many EMTP-type packages, state-variable-based transient simulation programs are also used for the analysis of dynamic systems including electric power systems [16]–[23] together with electrical machines [24]–[26]. Examples of software packages include ACSL [27], Easy5 [28], Eurostag [29], and MATLAB/Simulink [4]. There are also more specialized tools such as SimPowerSystems (SPS) [30], RT-Lab [31], PLECS [32], and ASMG [33] that come with circuit interfaces and built-in libraries for simulation of transients in power and power-electronic circuits. Depending on the package and the circuit parameters, the circuit can be represented internally as the generalized state model (with variable parameters) or the linear time-invariant (LTI) state-space model with constant matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, and $\mathbf{D}$ as depicted in Fig. 2.

The traditional qd-state models (voltage input and current output) in these languages are generally interfaced using three-phase voltage controlled current sources as depicted in Fig. 2(a). Whenever it is possible to feed the machine model from either capacitors or the voltage sources, for example, this interface becomes input–output compatible and explicit, which would be preferred in most cases. However, the direct interfacing of machine models as current source in series to inductive branches (which is in fact very common) represents a challenge. In this case, the interfacing is typically resolved using a resistive [e.g., see Fig. 2(a)] or capacitive snubber circuit [13], [34]. The snubber is required to calculate the needed interfacing voltage and leads to many numerical disadvantages. For numerical efficiency, whenever possible, it is also desirable to form the state-space model with fixed matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, and $\mathbf{D}$.

The VBR model formulation has been introduced in the literature to achieve a direct interface of the machine models with the external (inductive) networks as depicted in Fig. 2(b). Moreover, to make the interface very effective and easy to use in most simulation packages, the interfacing circuit must be composed of very simple and constant-parameter branch elements, which may be difficult to achieve.

For a synchronous machine, which is a more general case, the interfacing circuit will contain coupled and variable inductances

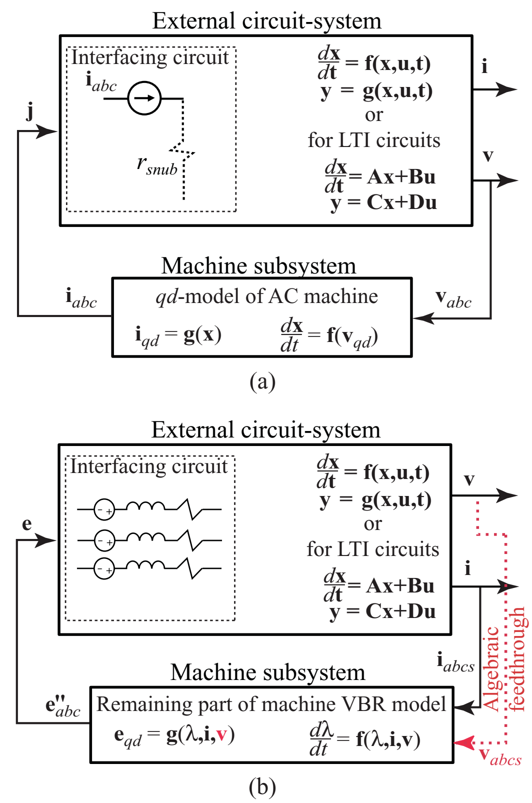

Fig. 2. Interfacing machine models with an external electrical network circuit.

(and even resistances [9]), which precludes forming the LTI state-space model with matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, and $\mathbf{D}$. Considering that a basic inductive branch that is solved within the external circuit system has the form $v = ri + p(Li) + e$, an algebraic feedthrough is being formed within the circuit solver from the input-controlled voltage sources $\mathbf{e}$ to the output branch voltages $\mathbf{v}$. In some VBR formulations [for approximation of dynamic saliency [35], for including the zero-sequence [10], for including saturation [36, etc.], in addition to the branch (stator) currents $\mathbf{i}$, the (stator) branch voltages $\mathbf{v}$ may also be required. If the subtransient voltages in a machine subsystem are calculated in terms of the rotor state variables and the stator currents, i.e., $\mathbf{e}''_{qd} = \mathbf{f}(\lambda_{qdr}, \mathbf{i}_{qds})$, then the algebraic loop is not formed as it is the case in [9] and [36], but the interfacing branches have variable impedances. However, whenever (in addition to the rotor states) the branch voltages $\mathbf{v}$ are required for calculating the controlled voltage sources $\mathbf{e}$, i.e., $\mathbf{e}''_{qd} = \mathbf{f}(\lambda_{qdr}, \mathbf{i}_{qds}, \mathbf{v}_{qds})$, an algebraic loop is being formed.

The existence of such an algebraic loop makes the overall system equation of the external circuit and the machine subsystem implicit, which generally results in invoking an iterative solution process and leads to an increase of computing time [4], [37]. It is, therefore, very desirable to have a constant-parameter decoupled interfacing circuit such that the LTI state-space model with matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, and $\mathbf{D}$ can be formed, but at the same time have explicit formulation that does not require the branch voltages for the machine subsystem (for either induction or synchronous machines).

III. VBR Model for Induction Machines

To better understand the proposed generalized interfacing circuit, it is instructive to first consider the induction machine VBR model. Without loss of generality, this paper assumes a three-phase, wye-connected, induction machine. The stator and rotor windings are assumed to be symmetric and sinusoidally distributed. Throughout this manuscript, the motor convention is used [2]. All rotor variables and parameters are referred to the stator side using an appropriate turns ratio. The corresponding coupled-circuit diagram can be found in [10] and is not repeated here due to limited space.

The derivation of the VBR model proceeds by algebraically manipulating the equations of the full-order machine model expressed in the qd arbitrary reference frame [10]. For consistency, only the key equations are given here. In particular, the voltage equation for the stator branches has the following VBR form:

$$\mathbf{v}_{abcs} = \mathbf{r}''_{abcs}\mathbf{i}_{abcs} + \mathbf{L}''_{abcs}p\mathbf{i}_{abcs} + \mathbf{e}''_{abc} \tag{1}$$

where the inductance matrix is

$$\mathbf{L}''_{abcs} = \begin{bmatrix} L_S & L_M & L_M \\ L_M & L_S & L_M \\ L_M & L_M & L_S \end{bmatrix} \tag{2}$$

with the corresponding entries defined as

$$L_S = L_{ls} + \frac{2}{3}L''_m, \quad L_M = -\frac{1}{3}L''_m. \tag{3}$$

Here, the subtransient magnetizing inductance is

$$L''_m = \left(\frac{1}{L_m} + \frac{1}{L_{lr}}\right)^{-1}. \tag{4}$$

The resistance matrix is given by

$$\mathbf{r}''_{abcs} = \begin{bmatrix} r_S & r_M & r_M \\ r_M & r_S & r_M \\ r_M & r_M & r_S \end{bmatrix} \tag{5}$$

and its entries are

$$r_S = r_s + \frac{2}{3}\frac{L''^2_m}{L^2_{lr}}r_r, \quad r_M = -\frac{1}{3}\frac{L''^2_m}{L^2_{lr}}r_r. \tag{6}$$

The so-called subtransient back EMF voltages are defined as

$$\mathbf{e}''_{abc} = [\mathbf{K}_s]^{-1} \cdot [\,e''_q \quad e''_d \quad 0\,]^T \tag{7}$$

where

$$e''_q = \omega_r\lambda''_d + \frac{L''_m r_r}{L^2_{lr}}\left(\lambda''_q - \lambda_{qr}\right) + \frac{L''_m}{L_{lr}}v_{qr} \tag{8}$$

$$e''_d = -\omega_r\lambda''_q + \frac{L''_m r_r}{L^2_{lr}}\left(\lambda''_d - \lambda_{dr}\right) + \frac{L''_m}{L_{lr}}v_{dr}. \tag{9}$$

The remaining part of the rotor subsystem includes the subtransient flux linkages in q- and d- axes as

$$\lambda''_q = L''_m\frac{\lambda_{qr}}{L_{lr}}, \quad \lambda''_d = L''_m\frac{\lambda_{dr}}{L_{lr}}. \tag{10}$$

The rotor state equations are expressed with the rotor flux linkages as the state variables as

$$p\lambda_{qr} = -\frac{r_r}{L_{lr}}(\lambda_{qr} - \lambda_{mq}) - (\omega - \omega_r)\lambda_{dr} + v_{qr} \tag{11}$$

$$p\lambda_{dr} = -\frac{r_r}{L_{lr}}(\lambda_{dr} - \lambda_{md}) + (\omega - \omega_r)\lambda_{qr} + v_{dr} \tag{12}$$

where magnetizing fluxes are

$$\lambda_{mq} = L''_m i_{qs} + \lambda''_q, \quad \lambda_{md} = L''_m i_{ds} + \lambda''_d. \tag{13}$$

Finally, the developed electromagnetic torque is calculated using the magnetizing fluxes as [2]

$$T_e = \frac{3P}{4}(\lambda_{md}i_{qs} - \lambda_{mq}i_{ds}). \tag{14}$$

Note that in (1), the inductance matrix (2) and resistance matrix (6) are independent of reference frame and constant which is a consequence of machine symmetry. These are very desirable numerical properties that make the VBR model more efficient than the coupled-circuit PD model (see VBR-I in [10]). However, using (1) directly to interface the induction machine model will require the *RL* branches with inductive and resistive coupling, which are not available in most simulation packages.

A further simplification is achieved by diagonalizing the inductance and resistance matrices (2) and (5) using the stator current and zero-sequence current as [38] and observing that

$$i_{as} + i_{bs} + i_{cs} = 3i_{0s} \text{ and } pi_{as} + pi_{bs} + pi_{cs} = 3pi_{0s}. \tag{15}$$

The off-diagonal terms in (2) and (6) may be eliminated by expressing the stator currents associated with the off-diagonal entries in terms of the zero-sequence current and the remaining phase (diagonal) currents. After algebraic manipulations, (1) can be rewritten as

$$\mathbf{v}_{abcs} = r_D\mathbf{i}_{abcs} + L_D p\mathbf{i}_{abcs} + \mathbf{e}''_{abc} + (3r_0\mathbf{i}_{0s} + 3L_0 p\mathbf{i}_{0s}) \tag{16}$$

where the zero-sequence current vector is defined as $\mathbf{i}_{0s} = [\,i_{0s} \quad i_{0s} \quad i_{0s}\,]^T$, and the remaining parameters are

$$r_D = r_s + \frac{L''^2_m}{L^2_{lr}}r_r, \quad L_D = L_{ls} + L''_m \tag{17}$$

$$r_0 = r_M = -\frac{1}{3}\frac{L''^2_m}{L^2_{lr}}r_r, \quad L_0 = L_M = -\frac{1}{3}L''_m. \tag{18}$$

If the stator winding neutral is floating, then the zero-sequence will not be present and (16) will be simplified. However, for the general-purpose interfacing circuit, the zero-sequence has to be included. The corresponding voltage equation is

$$v_{0s} = r_s i_{0s} + p\lambda_{0s}, \quad \lambda_{0s} = L_{ls}i_{0s}. \tag{19}$$

This gives the following additional state equation:

$$pi_{0s} = \frac{1}{L_{ls}}(v_{0s} - r_s i_{0s}). \tag{20}$$

Substituting (20) into (16), the following VBR formulation (see VBR-III in [10]) is achieved

$$\mathbf{v}_{abcs} = r_D \mathbf{i}_{abcs} + L_D p \mathbf{i}_{abcs} + \mathbf{e}''_{abc} + \left(3r_0 - \frac{3L_0 r_s}{L_{ls}}\right)\mathbf{i}_{0s} + \frac{3L_0}{L_{ls}}\mathbf{v}_{0s} \tag{21}$$

where $\mathbf{v}_{0s} = [\, v_{0s}\ v_{0s}\ v_{0s}\,]^T$.

Implementation of (21) requires only constant-parameter decoupled *RL*-branches and basic built-in circuit elements, which is a major advantage. However, if the zero-sequence is included, the formulation (21) becomes implicit with respect to the stator branch voltages $\mathbf{v}_{abcs}$ through the corresponding zero-sequence voltage v_{0s}, which are algebraically related. Therefore, a direct implementation of (21) with the zero-sequence (as may be required in a general case) will result in an algebraic loop [as in Fig. 2(b)] and require iterative solution [37].

To achieve a more general interfacing, it is observed that the stator branch voltages (16) may be expressed as

$$\mathbf{v}_{abcs} = \mathbf{v}_{abcn} + \mathbf{v}_{ng}. \tag{22}$$

Here, $\mathbf{v}_{abcn}$ is the phase-to-neutral voltages for the *RL* interfacing branches which is

$$\mathbf{v}_{abcn} = r_D \mathbf{i}_{abcs} + L_D p \mathbf{i}_{abcs} + \mathbf{e}''_{abc}. \tag{23}$$

It is important to distinguish $\mathbf{v}_{abcn}$ from the stator phase voltages of the original induction machine $\mathbf{v}_{abcs}$. Moreover, the voltage $\mathbf{v}_{ng} = [\, v_{ng}\ v_{ng}\ v_{ng}\,]^T$ defines a new branch for representing the optional zero-sequence of the interfacing circuit, that is

$$v_{ng} = r_0(3i_{0s}) + L_0 p(3i_{0s}) = r_0 i_{ng} + L_0 p i_{ng}. \tag{24}$$

Altogether, (22)–(24) define the four constant *RL* branches depicted in Fig. 1 that are needed for a direct and explicit interface of a machine model with an external electrical circuit. The dependent voltage sources $\mathbf{e}''_{abc}$ are functions of the branch current and determined by the rotor subsystem (8)–(9). Moreover, if the zero-sequence is not needed, the branch ng is simply removed leaving the neutral point n floating.

IV. VBR Model for Synchronous Machines

The goal of this section is to show that the VBR model for synchronous machines (with either round or salient rotor) can have the same interfacing circuit of Fig. 1. Without loss of generality, a three-phase, wye-connected synchronous machine with symmetrical and sinusoidally distributed stator windings is considered. The rotor is represented by lumped-parameter windings, and is assumed to have M damper windings in the q-axis and N damper plus one field windings in the d-axis. The rotor reference frame is used for all equations and variables expressed in qd-coordinates. For consistency with Section II, the motor convention is also used here.

A. General VBR Model With Variable Impedances

Moving the unequal resistances of q- and d-axes to the subtransient voltages [38] results in the following VBR form:

$$\mathbf{v}_{abcs} = r_s \mathbf{i}_{abcs} + p[\mathbf{L}''_{abcs}(\theta_r)\mathbf{i}_{abcs}] + \mathbf{e}''_{abcs} \tag{25}$$

where the subtransient inductance matrix is defined as

$$\mathbf{L}''_{abcs}(\theta_r) = \begin{bmatrix} L_S(2\theta_r) & L_M\left(2\theta_r - \frac{2\pi}{3}\right) & L_M\left(2\theta_r + \frac{2\pi}{3}\right) \\ L_M\left(2\theta_r - \frac{2\pi}{3}\right) & L_S\left(2\theta_r - \frac{4\pi}{3}\right) & L_M(2\theta_r) \\ L_M\left(2\theta_r + \frac{2\pi}{3}\right) & L_M(2\theta_r) & L_S\left(2\theta_r + \frac{4\pi}{3}\right) \end{bmatrix}. \tag{26}$$

Here, the entries are defined as follows:

$$L_S(\cdot) = L_s + L_a - L_b\cos(\cdot), \quad L_M(\cdot) = \frac{-L_a}{2} - L_b\cos(\cdot) \tag{27}$$

$$L_a = \frac{L''_{md} + L''_{mq}}{3}, \quad L_b = \frac{L''_{md} - L''_{mq}}{3}. \tag{28}$$

The subtransient inductances L''_{mq} and L''_{md} are defined as

$$L''_{mq} = \left(\frac{1}{L_{mq}} + \sum_{j=1}^{M}\frac{1}{L_{lkqj}}\right)^{-1} \tag{29}$$

$$L''_{md} = \left(\frac{1}{L_{md}} + \frac{1}{L_{lfd}} + \sum_{j=1}^{N}\frac{1}{L_{lkdj}}\right)^{-1}. \tag{30}$$

Finally, the subtransient VBRs are

$$\mathbf{e}''_{abc} = [\mathbf{K}_s]^{-1}\cdot[\, e''_q \quad e''_d \quad 0\,]^T \tag{31}$$

where

$$e''_q = \omega_r\lambda''_d + \sum_{j=1}^{M}\left(\frac{L''_{mq} r_{kqj}}{L^2_{lkqj}}(\lambda''_q - \lambda_{kqj})\right) + \left(\sum_{j=1}^{M}\frac{r_{kqj}}{L^2_{lkqj}}\right)L''^2_{mq} i_{qs} \tag{32}$$

$$e''_d = -\omega_r\lambda''_q + \sum_{j=1}^{N}\left(\frac{L''_{md} r_{kdj}}{L^2_{lkdj}}(\lambda''_d - \lambda_{kdj})\right) + \frac{L''_{md}}{L_{lfd}}v_{fd} + \frac{L''_{md} r_{fd}}{L^2_{lfd}}(\lambda''_d - \lambda_{fd}) + \left(\frac{r_{fd}}{L^2_{lfd}} + \sum_{j=1}^{N}\frac{r_{kdj}}{L^2_{lkdj}}\right)L''^2_{md} i_{ds}. \tag{33}$$

The subtransient flux linkages are defined as

$$\lambda''_q = L''_{mq}\left(\sum_{j=1}^{M}\frac{\lambda_{kqj}}{L_{lkqj}}\right), \quad \lambda''_d = L''_{md}\left(\frac{\lambda_{fd}}{L_{lfd}} + \sum_{j=1}^{N}\frac{\lambda_{kdj}}{L_{lkdj}}\right). \tag{34}$$

The rotor state equations are

$$p\lambda_{kqj} = -\frac{r_{kqj}}{L_{lkqj}}(\lambda_{kqj} - \lambda_{mq}); \quad j = 1, \ldots, M \tag{35}$$

$$p\lambda_{kdj} = -\frac{r_{kdj}}{L_{lkdj}}(\lambda_{kdj} - \lambda_{md}); \quad j = 1, \ldots, N$$

$$p\lambda_{fd} = -\frac{r_{fd}}{L_{lfd}}(\lambda_{fd} - \lambda_{md}) + v_{fd}. \tag{36}$$

The magnetizing fluxes similar to (13) are

$$\lambda_{mq} = L''_{mq} i_{qs} + \lambda''_q, \quad \lambda_{md} = L''_{md} i_{ds} + \lambda''_d. \tag{37}$$

The electromagnetic torque is calculated using (14).

The subtransient voltages (32)–(33) explicitly include the stator currents which contribute to the algebraic feedthrough of the rotor subsystem, whereas this is not the case for the induction machine (8)–(9). However, this does not represent a problem since most simulation packages allow current-dependent voltages sources in series with inductive branches because this combination does not result in an algebraic loop [30]. However, the variable and coupled inductances (26) are not desirable and will require recalculation of the state-space equations at every time step and result in slower simulations [30], [32], [33].

B. Approximation of Dynamic Saliency

To carry out the diagonalization of (26) similar to the induction machine formulation, the terms inside the matrix have to be constant. This can be achieved if $L''_{mq} = L''_{md}$. The subtransient inductances are also related to the operational impedances [12]. In particular, in the range of very high frequencies, the leakage inductances of the rotor windings (circuits) will dominate the impedances, and the value of the operational impedances in the q- and d-axes equivalent circuits (normalized with respect to frequency) will become equal to the subtransient inductances L''_{mq} and L''_{md}, respectively. The traditional approximation based on averaging L''_{mq} and L''_{md} was shown to reduce the model accuracy [9], [12].

An approach for approximating the dynamic saliency has been proposed in [12]. This method is based on adding one artificial damper winding to the rotor circuit with the purpose of enforcing numerical equality of L''_{mq} and L''_{md}. In doing so, the extra winding should have sufficiently high resistance as not to impact the low-frequency operational impedance of the rotor circuit. But otherwise, the added extra winding does not have any physical meaning with respect to the original parameters of a given machine. Since the subtransient inductance in the d-axis is typically smaller, the additional winding is normally added to the q-axis equivalent circuit. Therefore, denoting this added winding as $(M+1)$, its leakage inductance is readily calculated based on (29) and (30) as

$$L_{lkq(M+1)} = \left(\frac{1}{L''_{md}} - \frac{1}{L''_{mq}}\right)^{-1}. \tag{38}$$

This also adds one more state equation to (35), where $j = 1, \ldots, M+1$.

To distinguish the subtransient quantities of the approximate model that uses the $(M+1)$th winding, in this paper a triple-prime sign ($'''$) instead of double-prime sign ($''$) will be used. Thus, including this additional winding gives the desired result:

$$L'''_{mq} = L'''_{md} = L''_{md} \tag{39}$$

and makes $L_S(\cdot)$and $L_M(\cdot)$constant as

$$L_S = L_{ls} + \frac{2}{3}L''_{md}, \quad L_M = -\frac{1}{3}L''_{md}. \tag{40}$$

Now, with the constant inductance matrix, (25) will become

$$\mathbf{v}_{abcs} = r_s \mathbf{i}_{abcs} + \begin{bmatrix} L_S & L_M & L_M \\ L_M & L_S & L_M \\ L_M & L_M & L_S \end{bmatrix} p\mathbf{i}_{abcs} + \mathbf{e}''_{abcs}. \tag{41}$$

Here, the subscript M in L_M should not be confused with the winding indexing. Moreover, after the inductance matrix becomes constant, we can proceed with the diagonalization procedure. Thus, by defining

$$L_D = L_S - L_M \text{ and } L_0 = L_M \tag{42}$$

the off-diagonal elements of the constant matrix are eliminated by incorporating zero-sequence current (15) into (41). The result is

$$\mathbf{v}_{abcs} = r_s \mathbf{i}_{abcs} + L_D p\mathbf{i}_{abcs} + L_0 p(3\mathbf{i}_{0s}) + \mathbf{e}''_{abc}. \tag{43}$$

Consideration of zero-sequence for modeling of synchronous machines is more important than it was in the case of induction machines. In many instances of synchronous generators, the neutral point of the stator windings is grounded through external impedance for protection and monitoring purposes [39]. Therefore, it is important to develop a general-purpose interfacing circuit with zero-sequence that will be fully capable of predicting the unbalanced faults and grounding currents for different grounding classes (low/high resistance/impedance, etc.) and protection requirements [40].

To define a fourth branch for representing zero-sequence, the stator voltages can be expressed using (22) (as in the case of induction machine), where

$$\mathbf{v}_{abcn} = r_s \mathbf{i}_{abcs} + L_D p\mathbf{i}_{abcs} + \mathbf{e}''_{abc}. \tag{44}$$

The zero-sequence branch is defined by the voltage equation

$$v_{ng} = L_0 p(3i_{0s}) = L_0 p i_{ng}. \tag{45}$$

Finally, (22), (44), and (45) define the four constant and decoupled *RL*-branches depicted in Fig. 1. This circuit is used to interface the synchronous machine model with the external network. The circuit parameters are

$$r_D = r_s, \quad L_D = L_{ls} + L''_{md} \tag{46}$$

$$r_0 = 0, \quad L_0 = L_M = -\frac{1}{3}L''_{md}. \tag{47}$$

To summarize, the controllable sources are defined by (32)–(33), and the remainder of the rotor state model (35)–(36).

V. Implementation for Synchronous Machines

Since the induction machine VBR model lands itself directly into the circuit of Fig. 1, its implementation is explicit and can be carried out in most simulation packages using conventional branch elements and controllable (current/state-dependent) voltage sources. Therefore, this section focuses on synchronous machines only, where it is possible to have explicit (approximate) or implicit (exact) implementation of the proposed interfacing circuit.

A. Explicit Approximate Method (AVBR)

An explicit implementation method is based on dynamic approximation of saliency described in Section III-B. With the addition of the extra $M+1$ damper winding to the q-axis, the conventional VBR model is manipulated into the interfacing circuit of Fig. 1. This interfacing is explicit and straightforward to implement using conventional circuit branch elements. When adding the extra winding, the leakage inductance is defined by (38), but the user must choose the winding resistance. In general, choosing this resistance very large will improve the accuracy of this approximation at the expense of making the system numerically stiffer which will be shown in Section VI-B. Otherwise, this method gives zero error in steady state because the effect of damper windings under this condition diminishes.

B. Implicit Method (IVBR)

The artificial winding adds an additional state variable and a fast dynamic mode. The disadvantageous effect of this winding can be "undone" at the expense of making this formulation implicit, which is achieved using the singular perturbation [35].

It is observed that when the resistance of additional winding is approaching infinity, the largest eigenvalue of the system is also going to infinity. In this method, the aim is to remove the fast state from the model [35]. A new state-variable is defined as the voltage across the resistance as

$$v_{kq(M+1)} = r_{kq(M+1)} i_{kq(M+1)}. \tag{48}$$

After some algebraic manipulations, the corresponding state equation can be put into the following form [35]:

$$\begin{aligned}\varepsilon p v_{kq(M+1)} = &-\left(1-\frac{L''_{md}}{L_{lkqM}}\right) v_{kq(M+1)} \\ &- \frac{L''_{md}}{L''_d}\left(v_{qs} - \omega_r L''_d i_{ds} - e'''_q - r_s i_{qs}\right) \\ &+ \sum_{j=1}^{M}\left(\frac{L''_{md} r_{kqj}}{L^2_{lkqj}}(\lambda_{kqj} - \lambda_{mq})\right)\end{aligned} \tag{49}$$

where $\varepsilon = L_{kq(M+1)}/r_{kq(M+1)}$. The new subtransient voltages become

$$\begin{aligned}e'''_q = &\,\omega_r \lambda''_d + \sum_{j=1}^{M}\left[\frac{L''_{md} r_{kqj}}{L^2_{lkqj}}(\lambda'''_q - \lambda_{kqj})\right] \\ &- \frac{L''_{md}}{L_{lkq(M+1)}} v_{kq(M+1)} + L''^2_{md}\left(\sum_{j=1}^{M}\frac{r_{kqj}}{L^2_{lkqj}}\right) i_{qs}\end{aligned} \tag{50}$$

$$\begin{aligned}e'''_d = &-\omega_r \lambda'''_q + \sum_{j=1}^{N}\left(\frac{L''_{md} r_{kdj}}{L^2_{lkdj}}(\lambda''_d - \lambda_{kdj})\right) + \frac{L''_{md}}{L_{lfd}} v_{fd} \\ &+ \frac{L''_{md} r_{fd}}{L^2_{lfd}}(\lambda''_d - \lambda_{fd}) + L''^2_{md}\left(\frac{r_{fd}}{L^2_{lfd}} + \sum_{j=1}^{N}\frac{r_{kdj}}{L^2_{lkdj}}\right) i_{ds}.\end{aligned} \tag{51}$$

Here, the approximate q-axis subtransient flux is defined by

$$\lambda'''_q = L''_{md}\left[\sum_{j=1}^{M}\left(\frac{\lambda_{kqj}}{L_{lkqj}}\right) + \frac{\lambda_{kq(M+1)}}{L_{lkq(M+1)}}\right]. \tag{52}$$

Now, if the winding resistance approaches infinity, $r_{kq(M+1)} \to \infty$, then $\varepsilon p v_{kq(M+1)} \to 0$. Finding $v_{kq(M+1)}$ from (49) and substituting e'''_q from (50) after simplification gives

$$\begin{aligned}v_{kq(M+1)} = &\left(1 - \frac{L''_{md}}{L_{lkq(M+1)}} + \frac{L''^2_{md}}{L''_d L_{lkq(M+1)}}\right)^{-1} \\ &\Bigg\{-\frac{L''_{md}}{L''_d}\left(v_{qs} - \omega_r L''_d i_{ds} - \omega_r \lambda''_d\right) \\ &+ \sum_{j=1}^{M}\left[\frac{r_{kqj} L''_{md}}{L^2_{lkqj}}\left(\frac{L''_{md}}{L''_d} - 1\right)(\lambda_{kqj} - \lambda'''_{mq})\right] \\ &+ \sum_{j=1}^{M}\left[\frac{r_{kqj} L''^2_{md}}{L^2_{lkqj}}\left(\frac{L''_{md}}{L''_d} - 1\right) + \frac{L''_{md}}{L''_d} r_s\right] i_{qs}\Bigg\}\end{aligned} \tag{53}$$

where the approximate magnetizing flux is defined as

$$\lambda'''_{mq} = L''_{md} i_{qs} + \lambda'''_q. \tag{54}$$

To remove the additional winding parameters from the voltage equations, it is considered that as $r_{kq(M+1)} \to \infty$, then $\lambda'''_{mq} \to \lambda_{mq}$. Therefore, it can be assumed that $\lambda'''_{mq} = \lambda_{mq}$ which considering (37) and (54) gives

$$\lambda'''_q = (L''_{mq} - L''_{md}) i_{qs} + \lambda''_q. \tag{55}$$

Combining (55) with (34) results in the following expression for the flux linkage independent of $L_{kq(M+1)}$:

$$\lambda'''_q = (L''_{mq} - L''_{md}) i_{qs} + L''_{mq}\left(\sum_{j=1}^{M}\frac{\lambda_{kqj}}{L_{lkqj}}\right). \tag{56}$$

To summarize this implementation, the subtransient voltages are defined by (50), (51), and (53) and the electrical part of the circuit is defined by (46) and (47). If the machine is connected to an infinite bus, then the bus voltage v_{qs} is readily available; thus, (53) and consequently (50) do not require the solution for the branch voltages and the overall formulation becomes explicit. However, in a general case, voltage v_{qs} is calculated from the stator branch voltages $\mathbf{v}_{abcs}$ that is solved together with the external network which has the subtransient voltages $\mathbf{e}''_{abc}$ as input. This implicit relationship between $\mathbf{v}_{abcs}$ and $\mathbf{e}''_{abc}$ forms an algebraic loop that has to be solved iteratively by the program solver.

It should be pointed out that because the effect of additional (artificial) winding has been removed (up to infinite frequency), this formulation becomes algebraically equivalent to the original VBR model with rotor-position-dependant inductances. Further algebraic derivations and computer studies confirm this conclusion. An equivalent implicit implementation can also be obtained by simply moving a part of the q-axis current derivative to the subtransient voltages.

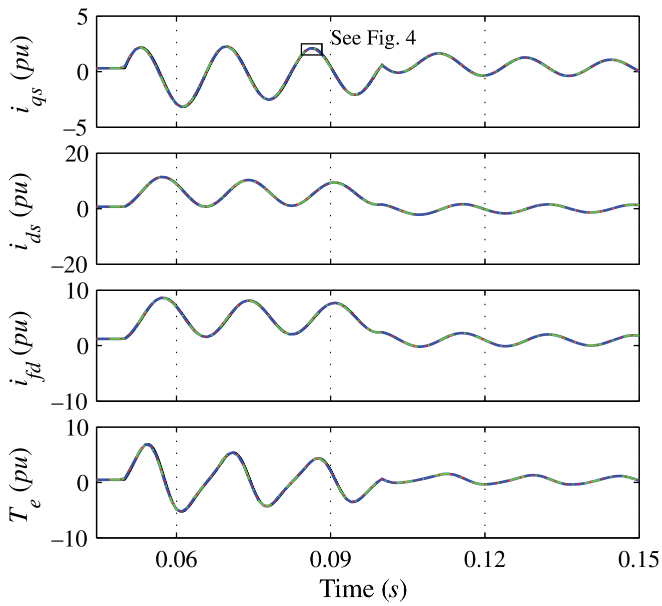

Fig. 3. Transient response to the three-phase fault predicted by various models using a fixed integration time step of 1 ms.

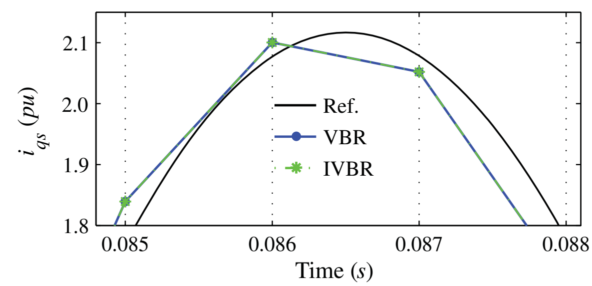

Fig. 4. Detailed view of current i_{qs} from the three-phase fault study using a fixed integration time step of 1 ms from Fig. 3.

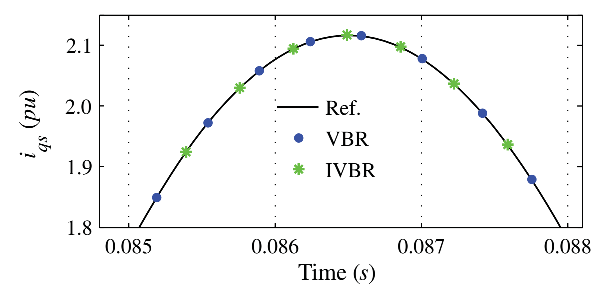

Fig. 5. Detailed view of current i_{qs} from the three-phase fault study using a variable time step for the same window shown in Fig. 4.

VI. Computer Studies for Synchronous Machine

Due to the limited space, computer studies presented in this paper are focused on applying the proposed interfacing circuit to the synchronous machines as the more general case. The parameters of the machine considered in this paper are summarized in the Appendix. The original model has one damper in the *q*-axis and one damper in the *d*-axis. Wherever appropriate, one artificial damper winding *kq*2 is added to the *q*-axis to approximate dynamic saliency. All considered models have been implemented using MATLAB/Simulink [4] and the ASMG toolbox [33], unless otherwise specified. The simulation studies are conducted on a personal computer with 2.53 GHz Intel CPU.

A. Verification of an Implicit VBR Model

In the following study, a single-machine infinite bus system is assumed in order to verify the consistency of the IVBR model. The machine is assumed to initially operate as a generator in a steady state defined by the field excitation voltage $e_{xfd} = v_{fd}X_{md}/r_{fd} = 2.15$ pu and mechanical torque $T_m = 0.5$ pu. At time $t = 0.05$ s, a three-phase fault is applied to the machine terminals. The fault is subsequently removed after three electrical cycles. The corresponding transient responses predicted by various models using fixed and variable integration time step are shown in Figs. 3–5.

The reference solution is produced by the conventional $qd0$-model implemented using standard Simulink blocks and run with a very small time step of 1 μs to ensure very high accuracy.

1) Constant Time-Step Simulation: To verify the consistency between the explicit/exact VBR model with variable inductance and the implicit model using singular perturbation (IVBR), these models were run with a large time step of 1 ms using solver ODE4. As can be seen in Fig. 3, all models produce responses that are visibly consistent and very close to each other. A magnified view of current i_{qs} shown reveals that at such a large time step there is a slight difference between the subject VBR models and the reference solution. However, as can be seen, the exact explicit VBR and IVBR produce exactly the same solution point by point. This result is expected since these models are algebraically equivalent despite their different implementation as explained in Section V.

2) Variable Time-Step Simulation: To compare the model efficiency, the same transient study was also run using variable integration time step. To achieve a reasonably accurate result, here ODE15s was used with relative and absolute error tolerances set to 10^{-4}, and the maximum and minimum time step set to 10^{-3} and 10^{-7}, respectively. The result of this study looks very similar to Fig. 3. For comparison, the same magnified view of current i_{qs} is also shown in Fig. 5 which shows that both VBR models take about the same number of time steps to satisfy the required tolerance and the solution is visibly very close to the reference.

B. Accuracy of an Approximate Explicit VBR Model (AVBR)

To evaluate the accuracy of an approximate explicit model (AVBR) that has a constant-parameter interfacing circuit, and specifically the choice of resistance in the additional artificial winding $kq2$, the same three-phase fault study is considered here. Due to the limited space, only current i_{qs} and its magnified view are shown in Fig. 6. Moreover, it was found that among other variables, current i_{qs} has the largest error due to the extra winding. The AVBR model is run with additional winding resistance r_{kq2} set to 1 and 10 pu. As is seen in

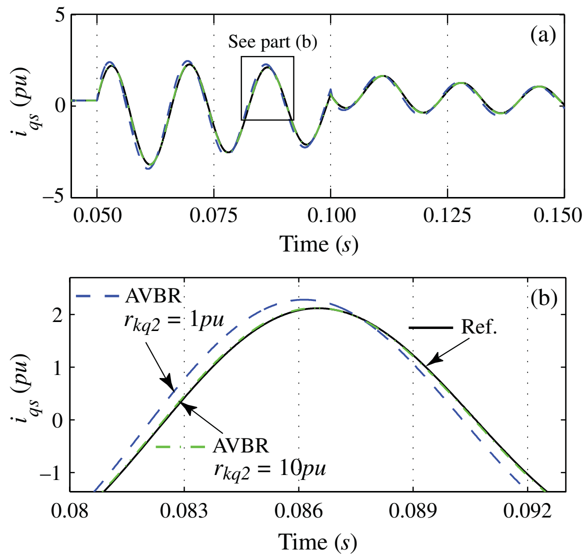

Fig. 6. Current i_{qs} from the three-phase fault study predicted by an approximate constant-parameter VBR model: (a) overall transient and (b) magnified view.

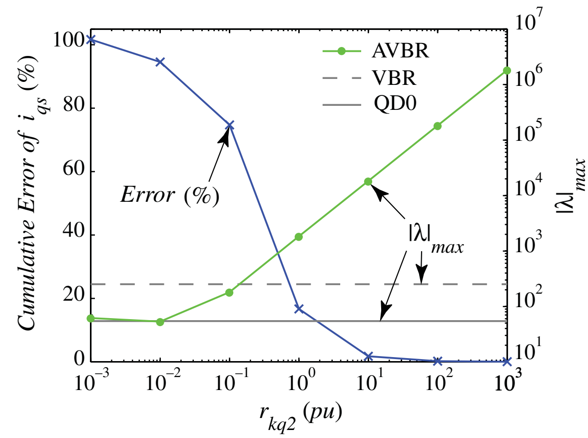

Fig. 7. Approximate model error and its largest eigenvalue versus additional winding resistance.

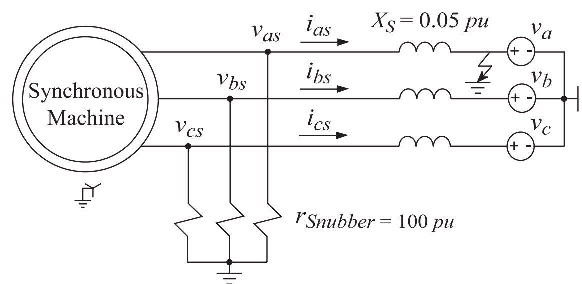

Fig. 8. Conventional *qd*0 machine model connected to an inductive network using a snubber circuit.

Fig. 6(b), when $r_{kq2} = 1$ pu, there exists a noticeable difference between the predicted transient in current i_{qs}. This difference becomes much smaller when the resistance is increased to 10 pu.

To systematically evaluate the impact of winding resistance r_{kq2} on the solution accuracy, herein the 2-norm of the cumulative error is considered [41]:

$$\varepsilon(r_{kq2}) = \frac{\left\|\tilde{i}_{qs} - i_{qs}\right\|_2}{\left\|\tilde{i}_{qs}\right\|_2} \times 100\%. \tag{57}$$

Here, $\tilde{i}_{qs}$ is the reference solution of the considered three-phase fault study (current trajectory), and i_{qs} refers to the numerical solution obtained using a given value of r_{kq2} with the approximate explicit model AVBR.

To ensure consistency with the reference solution and because the encountered error is primarily due to the saliency approximation, the time step of 1 μs was used here as well. The result of this evaluation is shown in Fig. 7. For example, for $r_{kq2} = 10$ pu, the cumulative error in i_{qs} for the chosen time span is 1.7%. As can be seen in Fig. 7, increasing the resistance of the additional winding r_{kq2} improves the solution accuracy.

However, this improvement comes at the price of increasing the magnitude of the largest eigenvalue of the system which results in numerical stiffness. In Fig. 7, the largest eigenvalue calculated as a function of resistance r_{kq2} is also plotted. For comparison, the largest eigenvalue of the conventional (time invariant) *qd*0-model and the exact VBR model are shown as well in that figure. The eigenvalues of the *qd*0-model are the smallest and the most favorable from the numerical point of view. The exact VBR model is time varying and uses stator branch currents as the independent variables, which results in higher magnitude of the largest eigenvalue. However, using $r_{kq2} = 10$ pu in the approximate VBR model raises the largest eigenvalue to the 10^4 range.

C. Verification of Models in SPS

To demonstrate the improvement achieved by the proposed constant-parameter interfacing circuit, the synchronous machine is connected to a source with 5% inductive impedance as depicted in Fig. 8. The considered source with inductive impedance represents the Thévenin-equivalent circuit of the external power system which is implemented as part of the power network using the transient simulation program and its circuit interface (and therefore should not be combined with the machine's stator leakages). To verify the models in a commonly used commercially available tool, the electrical circuit has been implemented in the SimPowerSystems (SPS) toolbox [30]. In order to interface the conventional *qd*0-model with the external network (that is represented as a physical circuits in *abc* variables/coordinates), a snubber circuit has to be used [11]. Due to lack of zero-sequence in the built-in *qd*-model in SPS, the *qd*0-model has been implemented using standard Simulink blocks and interfaced with controlled current sources as shown in Fig. 2(a). Aiming for reasonably high accuracy (on the order of 1%), a 100 pu resistive snubber is used as shown in Fig. 8. In order to emulate a severe unbalanced condition, the machine's neutral was assumed grounded, and a single-phase fault is applied to phase *a* as depicted in Fig. 8. This study

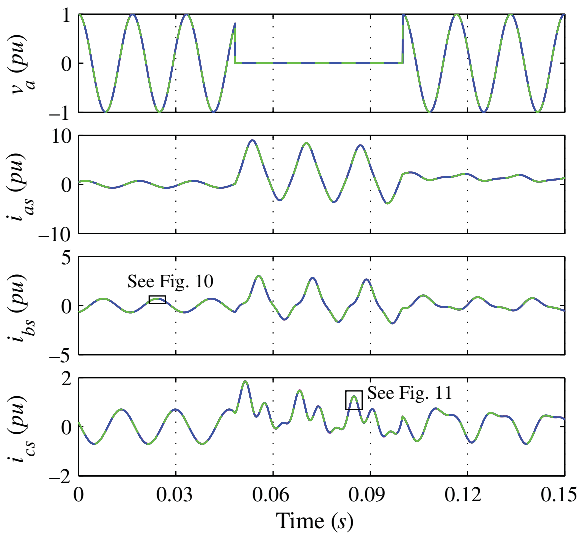

Fig. 9. Transient response to the single-phase fault predicted by the qd0-model with snubber versus constant-parameter AVBR and IVBR models.

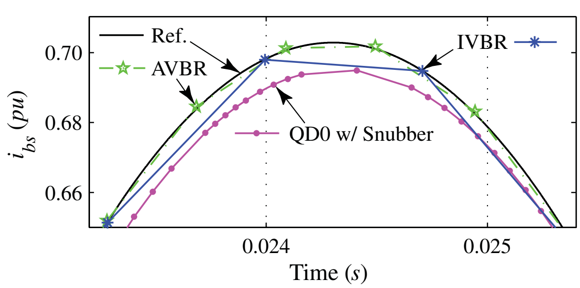

Fig. 10. Detailed view of current i_{bs} in steady state, from Fig. 9.

also verifies the applicability of the proposed interfacing circuit for modeling the unbalanced condition with the zero-sequence which may be needed particularly for modeling synchronous generators [39], [40].

The transient response of the system resulting from applying and removing this fault is shown in Fig. 9. The reference solution has been obtained using the exact VBR model (with variable inductances) that was run with a time step of 1 μs to ensure very high accuracy. Calculating the cumulative error using (57), it was found that the qd0-model with snubbers results in about 1.2% error in the phase current i_{cs}. It was further determined that the same level of accuracy (cumulative error) can be achieved by the approximate VBR model with $r_{kq2} = 15\,\mathrm{pu}$.

The magnified view of the phase current before and during the fault is shown in Figs. 10 and 11, respectively. The qd0-model has a steady-state error due to the snubbers, which can be readily seen in Fig. 10. However, the AVBR model has zero steady-state error since the effect of the damper winding disappears in steady state. Moreover, since the transient phenomenon is predominantly of low frequency, the accuracy of predicting the fault currents also remains very good as can be seen in Fig. 11. Also, as shown in Figs. 10 and 11, the IVBR model has even smaller error (about 0.1% cumulative) in transient and steady states.

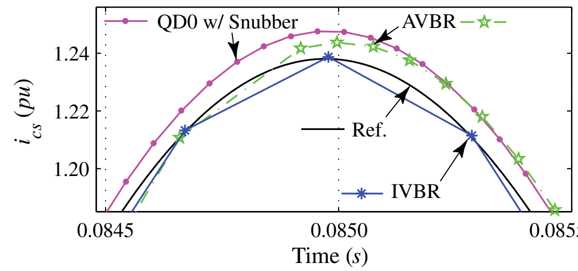

Fig. 11. Detailed view of current i_{cs} during a single-phase fault, from Fig. 9.

TABLE I
SIMULATION EFFICIENCY FOR A SINGLE-PHASE FAULT STUDY

	QD0 W/ 100 pu SNUBBER	AVBR $r_{kq2} = 15$ pu	IVBR
NUMBER OF TIME-STEPS	1871	557	400
SIMULATION TIME	0.35 s	0.20 s	0.47 s
CUMULATIVE ERROR OF i_{cs}	1.2 %	1.2 %	0.1 %
LARGEST EIGENVALUE	-1.2×10^6	-2.4×10^4	-1.7×10^2

To compare the numerical efficiency of considered models, the same solver ODE15s and settings are used here as in Section VI-A2. The time steps and CPU time taken by each model together with the magnitude of their respective largest eigenvalues are summarized in Table I. As can be seen in that table, the 100 pu snubber interface of the qd0-model results in significant numerical stiffness and the largest eigenvalue of 1.2×10^6. At the same time, the proposed interfacing circuit and the AVBR model with $r_{kq2} = 15\,\mathrm{pu}$ result in the largest eigenvalue of 2.4×10^4, which is significantly smaller. This in turn explains very large number of steps and the CPU time required by the conventional qd0-model as compared to the proposed approximate VBR model. This is also consistent with Figs. 10 and 11, wherein it can be seen that a stiffer model requires smaller time steps to satisfy the same error tolerances under the variable step integration. Table I also verifies that the IVBR model is equivalent to the exact VBR model in terms of accuracy. However, its implicit nature causes additional internal iterations at every time step which results in slower simulation overall.

VII. DISCUSSION

The approach presented in this paper focuses on interfacing the ac machines' stator circuit with the external network. In the case of synchronous machines, the rotor field winding is assumed to be fed from an external voltage source. Similarly, in the case of doubly fed induction machines, the rotor windings are assumed short-circuited (or may also be supplied from an external voltage source with some minor modification of equations). Exactly the same considerations (provisions) have been made in other previously published VBR models, e.g., [9]–[16], [25], [26], [35], [36], and [38]. Although this model formulation does not cover all possible cases and applications, the improved interfacing of the stator circuit may be very useful in many system studies where the details of the rotor circuit connection are not explicitly brought out into the power network [11].

For improved accuracy of modeling induction and synchronous machines in general, it is also desirable to include the effect of magnetic saturation. However, it has not been considered in this manuscript, as it has not been considered in many other publications, e.g., [9]–[16], [25], [26], [34], [35], [38], and even in [39] where the faults are discussed. Saturation is not always included in the power system studies [42], whereas many transient simulation packages include saturation as an option for the existing built-in models where the user may include the saturation or not depending on the availability of parameters and the required accuracy/fidelity of the study. For example, the first few cycles of the phase current during the fault (subtransient period) are mostly determined by the subtransient reactances of the machine which are small and less impacted by the saturation of the magnetizing reactance (inductance). For such studies, the unsaturated (simpler) models may, therefore, offer sufficient accuracy for estimating the fault currents magnitude [39], [42]. Implementation of saturation would require special consideration/focus and perhaps a separate dedicated publication (as it is often the case), and is outside of the scope of this manuscript. Also, when the saturation is considered, it would be difficult to have the interfacing circuit parameters and the state matrices that are constant. However, having constant state-space matrices for the overall external circuit system is very desirable.

Both explicit and implicit VBR formulations have been discussed in this paper. The implicit VBR contains an algebraic loop that is formed whenever the subtransient voltages $\mathbf{e}''_{abc}$ (in addition to the rotor states) depend on the branch voltages $\mathbf{v}_{abcs}$ (e.g., [10] and [35]). The IVBR model gives accurate results and even requires fewer time steps as shown in Table I. However, this performance comes at a price of an iterative solution and generally results in longer CPU time per step or longer overall simulation (computing) time as seen in Table I. At the same time, the AVBR model is explicit, noniterative, and for the same local error tolerances requires less than half of the CPU time as compared to the IVBR model. Therefore, considering just the number of integration time steps by itself is not an accurate measure for comparing the numerical efficiency of models.

For achieving the explicit AVBR model, an artificial damper winding has been added to the rotor circuit following the approach set forth in [12]. For either round rotor or salient pole synchronous machines, the parameters are such that $L_d \geq L_q$. However, for the subtransient inductances (reactances) this relationship is typically reversed, i.e., $L''_q \geq L''_d$. Interested reader can find the table with typical parameters for the synchronous generators (turbo and hydro), condensers, and motors in [43], p. [40], Table [2], wherein the condition $X''_q > X''_d$ holds for the low, average, and high ranges of typical parameters, respectively. The same conclusion can be made for the synchronous machines' parameters found in [44], as well as in [2] (where for the hydro generator, $X''_q \approx X''_d$). The condition $X''_q > X''_d$ follows from the fact that the field winding typically has very small leakage, which makes X''_d smaller even when X_d is larger than X_q. Based on this observation, the additional artificial winding would be normally added to the q-axis equivalent circuit and its leakage inductance calculated using (38) will have a positive value (see the Appendix). However, if for some reason one has $X''_q < X''_d$, then the winding can be added to the d-axis instead in order to remove the numerical saliency and achieve (39).

VIII. Conclusion

This paper presented VBR models with a constant-parameter decoupled *RL*-branch equivalent circuit for interfacing ac induction and synchronous machines with the external electrical network. To the best of our knowledge, such a unified and simple-to-use interfacing circuit for both types of rotating machines has not been proposed prior to this publication. The interfacing circuit is very general and allows connection of the stator winding neutral to an arbitrary external circuit (which may be required for studying unbalanced conditions, overvoltages, grounding and protection issues, etc). Such an interfacing circuit is very desirable as it is easy to implement in many commonly used simulation packages, wherein significant improvements in accuracy and efficiency can be achieved. For the induction machines, the proposed interfacing circuit and model result in explicit and exact formulation. For the synchronous machines, we describe both implicit and explicit interfacing methods. In the latter, the approximation of dynamic saliency becomes necessary for achieving explicit formulation with the same constant interfacing circuit.

The presented case studies of a synchronous machine demonstrate that such an interfacing circuit and machine models can be readily implemented in many simulation packages (e.g., SPS toolbox), where the proposed explicit interfacing is shown to give appreciable advantages over the traditional *qd*-models that are conventionally interfaced using snubber circuits. Since machine models are widely used components, the proposed method can have significant impact and improve many simulation packages.

Appendix

Synchronous machine parameters in pu [12]: 125-kW, 60 Hz, $r_s = 0.00515$, $X_{ls} = 0.0800$, $X_{mq} = 1.00$, $X_{md} = 1.77$, $r_{kq1} = 0.0610$, $r_{fd} = 0.00111$, $r_{kd1} = 0.024$, $X_{lkq1} = 0.330$, $X^*_{lkq2} = 0.14646$ (calculated for AVBR model), $X_{lfd} = 0.137$, and $X_{lkd1} = 0.334$. Subtransient reactances are $X''_q = 0.248$ and $X''_d = 0.0921$.

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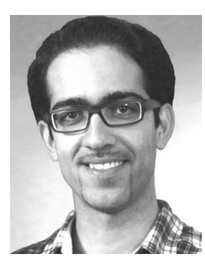

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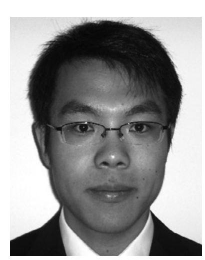

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Constant-Parameter Circuit-Based Models of Synchronous Machines

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***Abstract*—Representation of synchronous machines using constant-parameter voltage-behind-reactance (VBR) formulations improves accuracy and numerical efficiency of power systems transient simulation programs. This paper extends the VBR representation to the rotor circuit and presents two new formulations that achieve direct constant-parameter interfacing of the rotor and stator terminals with arbitrary external networks. In the first model, the entire machine is represented by constant *RL* branches that have algebraic coupling among the circuit variables. In the second model, all damper windings are implemented in state-space form to increase the numerical efficiency, while the stator and field windings are provided as constant-parameter circuits. The proposed models are validated against the commonly used and some state-of-the-art alternative models using a single machine with a simplified ac excitation system. Computer studies demonstrate the improved accuracy and efficiency of the proposed models when external rotor circuitry is considered.**

***Index Terms*—AC machines, interfacing circuit, power system modeling, power system simulation, synchronous machine, voltage-behind-reactance model.**

NOMENCLATURE

Throughout this paper, matrix and vector quantities are represented by boldface characters (e.g., $\mathbf{v}_{abcs}$), and scalar quantities are in italics (e.g., i_{fd}). All variables are assumed to be referred to the stator side using the appropriate turns ratio. The *q*-axis is 90° ahead of the *d*-axis [1]. Similar naming convention for the machine variables as in [1] is used here. The rotor reference frame is assumed, but the superscript "*r*" is omitted. The main machine quantities are listed as follows.

L_{lfd}	Field winding leakage inductance.
$L_{lkdj}, j = 1 \ldots N$	Direct-axis rotor damper winding leakage inductances.
$L_{lkqj}, j = 1 \ldots M$	Quadrature-axis rotor damper winding leakage inductances.
L_{ls}	Stator winding leakage inductance.
L_{mq}, L_{md}	Quadrature- and direct-axis magnetizing inductances.
p	Heaviside's operator (differentiation with respect to time or d/dt when *t* is time).
r_{fd}	Field winding resistance.
$r_{kdj}, j = 1 \ldots N$	Direct-axis rotor damper winding resistances.
$r_{kqj}, j = 1 \ldots M$	Quadrature-axis rotor damper winding resistances.
r_s	Stator winding resistance.
ω_r	Angular velocity of the rotor.

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I. INTRODUCTION

IN POWER systems transient studies, synchronous machines are commonly represented by lumped-parameter models [1]–[4]. The most widely used are the classical *qd*0 models [1], which are expressed in orthogonal *q*- and *d*-axis coordinates based on the rotor reference frame [5] for simpler representation and high numerical efficiency. However, since the rest of the power network and components are typically represented in physical *abc* variables, the numerical advantages of the classical *qd*0 models may deteriorate very quickly due to interfacing issues [6]. As an alternative solution to the interfacing of *qd*0 models, so-called voltage-behind-reactance (VBR) models have been recently proposed, e.g., [7]–[14] (for the state-variable-based simulators), and [15] and [16] (for the nodal-analysis-based EMTP-type programs).

This paper considers modeling of synchronous machines in state-variable-based transient simulation programs such as PLECS [17], SimPowerSystems [18], and ASMG [19], wherein classical *qd*0 models are available as built-in components usually in standard state-space form. In these programs, the *qd*0 machine models are generally interfaced to external networks as voltage-controlled current sources [6]. This can result in an incompatible network–machine interface, particularly when the machine is connected to inductive circuits (transformers, inductive filters, short transmission lines, etc.), which is very common in practical power systems. To achieve a compatible interface for modeling, fictitious snubbers are frequently used, introducing error and increasing the system's numerical stiffness [6]. Stiff systems generally require implicit solvers and/or small time-steps to ensure numerical stability [20], which undesirably increases the simulation time.

To give the reader a broad perspective on the state-of-the-art in synchronous machine modeling for power systems transient studies, the interfacing properties of various models are summarized in Table I.

As depicted in this table, the implementation of the basic coupled-circuit phase-domain model (CC-PD) [1]–[4] achieves a direct interface to arbitrary networks from the stator and rotor sides, but due to its rotor-position-dependent inductances, this

TABLE I
CLASSIFICATION OF SYNCHRONOUS MACHINE FORMULATIONS BASED ON INTERFACING WITH EXTERNAL INDUCTIVE NETWORKS

Formulation	Stator Interface	Rotor Interface	Interfacing Circuit Parameters
Classical CC-PD [1]	**Direct**	**Direct**	Variable
Classical *qd*0 [1], [5]	Indirect	**Direct**	**Constant**
Classical VBR [7], [8]	**Direct**	Indirect	Variable
Stator VBR [9]–[12]	**Direct**	Indirect	**Constant**
Stator VBR [13]	**Direct**	Indirect	Variable
Full-VBR [14]	**Direct**	**Direct**	Variable
Proposed VBR	**Direct**	**Direct**	**Constant**

model has variable parameters and is generally more computationally expensive.

The classical *qd*0 model possesses constant parameters. Its stator circuit has an indirect interface with external inductive networks (i.e., it requires snubbers), but its rotor field winding, if represented as a circuit, can be directly interfaced with the exciter circuit. This model is therefore shown in Table I as having a direct interface on the rotor side.

The VBR models [7]–[14] have been derived to achieve a direct interface of the stator terminals with an arbitrary network on the ac side. The VBR models have better scaled eigenvalues than CC-PD models and are computationally less expensive [7]. The original/classical VBR formulation [7] possesses rotor-position-dependent stator interfacing resistance and inductance matrices. It was later shown in [8] how to obtain a constant resistance matrix. Overall, the VBR models presented in [7] and [8], and also the more recent models proposed in [13] and [14], have stator interfacing branches with variable parameters (similar to the CC-PD model). Simulation of variable elements is possible in some programs (such as PLECS [17] and ASMG [19]); however, it is very computationally costly and should be avoided if possible.

The variable inductances of the stator interfacing circuit were first made constant by equating the subtransient inductances (i.e., neglecting dynamic saliency) using: 1) an additional fictitious damper winding [9]; and 2) singular perturbations [10]. A generalized VBR model and a unified interfacing circuit for both synchronous and induction machines was later proposed in [11]. This generalized formulation has a convenient constant-parameter decoupled *RL* equivalent circuit for interfacing of the machine's stator terminals with arbitrary external networks and ability to include the zero-sequence. Several numerically advantageous continuous- and discrete-time approximation techniques were recently proposed in [12] to obtain the same unified stator interfacing equivalent circuit of [11] for synchronous machines. All these models [9]–[12] achieve constant-parameter interfacing of the machine's stator terminals as shown in Table I.

The VBR models [7]–[13] were initially introduced to allow direct network–machine interfacing at the stator terminals, while assuming that a voltage source representing the exciter is connected to the rotor field winding. In these models, to interface the rotor to an arbitrary network, e.g., in ac excitation systems [21], a controlled current source is used. This also creates an incompatible interface if the field winding terminals are connected to inductive elements or a switching converter, therefore requiring a snubber. To obtain direct rotor interfacing, the model presented in [13] was recently extended to also represent the field winding using a VBR formulation [14]. Therein, the resulting equivalent interfacing circuit is defined by four coupled and variable *RL* branches, which achieves a direct interface of both stator and rotor terminals with external circuits as summarized in Table I.

The idea of modifying the subtransient voltage sources to achieve a VBR model with a constant-parameter stator interface was proposed previously in [12]. An approach to also represent the rotor circuit using a VBR formulation was presented in [14], resulting in the Full-VBR model. However, the field winding in [14] is not a constant-parameter branch (independently of saturation) due to the coupling between the d-axis stator branch and the field circuit. As seen in [14, eq. (72)], the interfacing circuit of the Full-VBR model is in part represented by a 4-by-4 inductance matrix, with rotor-position-dependent off-diagonal terms coupling the field and the stator.

This paper extends the VBR formulation proposed in [12] by adding the field and damper windings to the constant-parameter interfacing circuit. The approximation techniques used in this paper are similar to those presented in [12]. With respect to the state-of-the-art models summarized in Table I, this paper proposes new VBR models that combine the following characteristics: 1) direct stator interfacing; 2) direct rotor interfacing; and 3) constant-parameter equivalent interfacing circuit. In this regard, two formulations are presented as follows. In the first, all rotor and stator windings are represented by an equivalent circuit consisting only of constant-parameter *RL* branches and voltage sources (VBR form); this model offers direct interfacing of any rotor and/or stator windings. In the second, the damper windings are represented by standard state-space equations, while the field and stator windings remain in circuit form allowing a direct interface with external ac and exciter networks. To the best of the authors' knowledge (see Table I), this has not been achieved or established in any previous publication s (i.e., [14] and/or [12]).

II. FORMULATIONS

A. Basic Model Equations and Definitions

For the purpose of this paper, a conventional three-phase synchronous machine model [1] is considered. The effect of saturation is not considered. The loss of modeling accuracy stemming from this assumption has been thoroughly evaluated in many publications over the years, e.g., [22]–[25], but is otherwise outside of the scope of this paper. The stator is assumed to be Y-connected with access to the neutral point (which may be used for grounding). The model is formulated using an equivalent circuit representation with M damper windings in the q-axis, and N damper windings plus a field winding in the d-axis. If necessary, several algorithms are available in the literature to convert machine data given in the form of operational impedance and time constants into equivalent circuit parameters [26]. Throughout this paper, motor sign convention is used, and all variables are referred to the stator side to be consistent with [7]–[12]. With these assumptions, the *q*- and *d*-axis and zero-sequence stator and rotor voltages are expressed here as [1]

$$v_{qs} = r_s i_{qs} + \omega_r \lambda_{ds} + p\lambda_{qs} \tag{1}$$
$$v_{ds} = r_s i_{ds} - \omega_r \lambda_{qs} + p\lambda_{ds} \tag{2}$$
$$v_{0s} = r_s i_{0s} + p\lambda_{0s} \tag{3}$$
$$v_{kqj} = r_{kqj} i_{kqj} + p\lambda_{kqj}, \qquad j = 1, 2, \ldots, M \tag{4}$$
$$v_{fd} = r_{fd} i_{fd} + p\lambda_{fd} \tag{5}$$
$$v_{kdj} = r_{kdj} i_{kdj} + p\lambda_{kdj}, \qquad j = 1, 2, \ldots, N \tag{6}$$

where

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \tag{7}$$
$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \tag{8}$$
$$\lambda_{0s} = L_{ls} i_{0s} \tag{9}$$
$$\lambda_{kqj} = L_{lkqj} i_{kqj} + \lambda_{mq}, \qquad j = 1, 2, \ldots, M \tag{10}$$
$$\lambda_{fd} = L_{lfd} i_{fd} + \lambda_{md} \tag{11}$$
$$\lambda_{kdj} = L_{lkdj} i_{kdj} + \lambda_{md}, \qquad j = 1, 2, \ldots, N \tag{12}$$

and the magnetizing flux linkages are given by

$$\lambda_{mq} = L_{mq} \left(i_{qs} + \sum_{j=1}^{M} i_{kqj} \right) \tag{13}$$

$$\lambda_{md} = L_{md} \left(i_{ds} + i_{fd} + \sum_{j=1}^{N} i_{kdj} \right). \tag{14}$$

B. All-Circuit Formulation

The aim of this section is to derive a synchronous machine model consisting solely of basic circuit elements with constant parameters. Such model is easy to implement in traditional simulation packages while offering a direct interface to external circuits. The model can also be used to study special cases, e.g., broken damper windings and machines with multiple field windings [27]. This model is achieved based on the constant-parameter VBR (CP-VBR) formulation introduced in [12].

1) Rotor Voltage Equations: To implement the rotor windings as circuits, the rotor currents should be chosen as the state variables. In this regard, solving for the stator currents i_{qs} and i_{ds} in (7) and (8), substituting the result in (13) and (14), after some algebraic manipulations, the magnetizing fluxes λ_{mq} and λ_{md} are rearranged as

$$\lambda_{mq} = (L_{ls} || L_{mq}) \sum_{j=1}^{M} i_{kqj} + \frac{L_{mq}}{L_{ls} + L_{mq}} \lambda_{qs} \tag{15}$$

$$\lambda_{md} = (L_{ls} || L_{md}) \left(i_{fd} + \sum_{j=1}^{N} i_{kdj} \right) + \frac{L_{md}}{L_{ls} + L_{md}} \lambda_{ds} \tag{16}$$

where $L_a || L_b \equiv \left(L_a^{-1} + L_b^{-1} \right)^{-1}$. Inserting (15) and (16) into (10)–(12) and taking the derivative of the resulting equations gives

$$\begin{aligned} p\lambda_{kqj} = {} & p \left[L_{lkqj} i_{kqj} + (L_{ls} || L_{mq}) \sum_{a=1}^{M} i_{kqa} \right] \\ & + \frac{L_{mq}}{L_{ls} + L_{mq}} p\lambda_{qs}, \quad j = 1, 2, \ldots, M \end{aligned} \tag{17}$$

$$\begin{aligned} p\lambda_{fd} = {} & p \left[L_{lfd} i_{fd} + (L_{ls} || L_{md}) \left(i_{fd} + \sum_{j=1}^{N} i_{kdj} \right) \right] \\ & + \frac{L_{md}}{L_{ls} + L_{md}} p\lambda_{ds} \end{aligned} \tag{18}$$

$$\begin{aligned} p\lambda_{kdj} = {} & p \left[L_{lkdj} i_{kdj} + (L_{ls} || L_{md}) \left(i_{fd} + \sum_{a=1}^{N} i_{kda} \right) \right] \\ & + \frac{L_{md}}{L_{ls} + L_{md}} p\lambda_{ds}, \quad j = 1, 2, \ldots, N. \end{aligned} \tag{19}$$

Incorporating (1), (2), and (17)–(19) into (4)–(6), after algebraic manipulation, yields

$$\begin{aligned} v_{kqj} = {} & r_{kqj} i_{kqj} + pL_{lkqj} i_{kqj} + p(L_{ls} || L_{mq}) \sum_{a=1}^{M} i_{kqa} \\ & + e''_{qr}, \quad j = 1, 2, \ldots, M \end{aligned} \tag{20}$$

$$\begin{aligned} v_{fd} = {} & r_{fd} i_{fd} + pL_{lfd} i_{fd} \\ & + p(L_{ls} || L_{md}) \left(i_{fd} + \sum_{j=1}^{N} i_{kdj} \right) + e''_{dr} \end{aligned} \tag{21}$$

$$\begin{aligned} v_{kdj} = {} & r_{kdj} i_{kdj} + pL_{lkdj} i_{kdj} + p(L_{ls} || L_{md}) \left(i_{fd} + \sum_{a=1}^{N} i_{kda} \right) \\ & + e''_{dr}, \quad j = 1, 2, \ldots, N \end{aligned} \tag{22}$$

where

$$e''_{qr} = \frac{L_{mq}}{L_q} (v_{qs} - r_s i_{qs} - \omega_r \lambda_{ds}) \tag{23}$$

$$e''_{dr} = \frac{L_{md}}{L_d} (v_{ds} - r_s i_{ds} + \omega_r \lambda_{qs}). \tag{24}$$

2) Stator Voltage Equations: To build this specific stator-circuit VBR formulation, the terms involving λ_{qs} and λ_{ds} in (1) and (2) are first replaced with stator currents and rotor fluxes. To do this, the magnetizing fluxes are rewritten as [7]

$$\lambda_{mq} = L''_{mq} i_{qs} + \lambda''_q \tag{25}$$
$$\lambda_{md} = L''_{md} i_{ds} + \lambda''_d \tag{26}$$

where

$$L''_{mq} = \left(\frac{1}{L_{mq}} + \sum_{j=1}^{M} \frac{1}{L_{lkqj}} \right)^{-1} \tag{27}$$

$$L''_{md} = \left(\frac{1}{L_{md}} + \frac{1}{L_{lfd}} + \sum_{j=1}^{N} \frac{1}{L_{lkdj}} \right)^{-1} \tag{28}$$

and where the subtransient flux linkages are defined as

$$\lambda''_q = L''_{mq} \left(\sum_{j=1}^{M} \frac{\lambda_{kqj}}{L_{lkqj}} \right) \tag{29}$$

$$\lambda''_d = L''_{md} \left(\frac{\lambda_{fd}}{L_{lfd}} + \sum_{j=1}^{N} \frac{\lambda_{kdj}}{L_{lkdj}} \right). \tag{30}$$

Inserting (25) and (26) into (7) and (8), respectively, and taking the derivative of the results gives

$$p\lambda_{qs} = pL''_q i_{qs} + p\lambda''_q \tag{31}$$

$$p\lambda_{ds} = pL''_d i_{ds} + p\lambda''_d \tag{32}$$

where the subtransient inductances are defined as

$$L''_q = L_{ls} + L''_{mq}; \quad L''_d = L_{ls} + L''_{md}. \tag{33}$$

To achieve a circuit-based model, the time derivatives of the rotor subtransient flux linkages are written in terms of rotor currents and voltages using (4)–(6) as follows [7]:

$$p\lambda''_q = L''_{mq} \sum_{j=1}^{M} \frac{v_{kqj} - r_{kqj} i_{kqj}}{L_{lkqj}} \tag{34}$$

$$p\lambda''_d = L''_{md} \left(\frac{v_{fd} - r_{fd} i_{fd}}{L_{lfd}} + \sum_{j=1}^{N} \frac{v_{kdj} - r_{kdj} i_{kdj}}{L_{lkdj}} \right). \tag{35}$$

Inserting (7), (8), (13), and (14) into (1) and (2) and substituting (31), (32), (34), and (35) into the resulting equations, after algebraic manipulation, the *q*- and *d*-axis stator voltages are rewritten as

$$v_{qs} = r_s i_{qs} + \omega_r L_d i_{ds} + pL''_q i_{qs} + e''_{qs1} \tag{36}$$

$$v_{ds} = r_s i_{ds} - \omega_r L_q i_{qs} + pL''_d i_{ds} + e''_{ds1} \tag{37}$$

where the subtransient voltages e''_{qs1} and e''_{ds1} are defined as

$$e''_{qs1} = \omega_r L_{md} \left(i_{fd} + \sum_{j=1}^{N} i_{kdj} \right) + L''_{mq} \sum_{j=1}^{M} \frac{v_{kqj} - r_{kqj} i_{kqj}}{L_{lkqj}} \tag{38}$$

$$e''_{ds1} = -\omega_r L_{mq} \sum_{j=1}^{M} i_{kqj} + L''_{md} \left(\frac{v_{fd} - r_{fd} i_{fd}}{L_{lfd}} + \sum_{j=1}^{N} \frac{v_{kdj} - r_{kdj} i_{kdj}}{L_{lkdj}} \right) \tag{39}$$

and where the *q*- and *d*-axis inductances are defined as

$$L_q = L_{ls} + L_{mq}; \quad L_d = L_{ls} + L_{md}. \tag{40}$$

As shown in [12], to obtain constant-parameter stator interfacing branches in stationary *abc* phase coordinates, it is convenient to rearrange (36) and (37) as follows:

$$v_{qs} = r_s i_{qs} + \omega_r L''_d i_{ds} + pL''_d i_{qs} + e'''_{qs} \tag{41}$$

$$v_{ds} = r_s i_{ds} - \omega_r L''_d i_{qs} + pL''_d i_{ds} + e'''_{ds} \tag{42}$$

where the new subtransient voltages, denoted by three prime symbols ($'''$), are

$$e'''_{qs} = \omega_r \left[(L_d - L''_d) i_{ds} + L_{md} \left(i_{fd} + \sum_{j=1}^{N} i_{kdj} \right) \right] + L''_{mq} \sum_{j=1}^{M} \frac{v_{kqj} - r_{kqj} i_{kqj}}{L_{lkqj}} + p(L''_q - L''_d) i_{qs} \tag{43}$$

$$e'''_{ds} = -\omega_r \left[(L_q - L''_d) i_{qs} + L_{mq} \sum_{j=1}^{M} i_{kqj} \right] + L''_{md} \left(\frac{v_{fd} - r_{fd} i_{fd}}{L_{lfd}} + \sum_{j=1}^{N} \frac{v_{kdj} - r_{kdj} i_{kdj}}{L_{lkdj}} \right). \tag{44}$$

The calculation of e'''_{qs} using (43) requires access to the time derivative of i_{qs}. This current is a state variable of the stator circuit, and its derivative may be available in some simulation tools. However, for generality, it should be directly represented in terms of inputs and states. Therefore, using (36), after algebraic manipulation, the derivative term $p(L''_q - L''_d)i_{qs}$ in (43) can be expressed as

$$p(L''_q - L''_d) i_{qs} = \frac{L''_q - L''_d}{L''_q} (v_{qs} - r_s i_{qs} - \omega_r L_d i_{ds} - e''_{qs1}). \tag{45}$$

Equation (43) can then be rewritten by substituting e''_{qs1} in (45) by (38) and inserting the result back into (43), which simplifies to

$$e'''_{qs} = \omega_r \left[L''_d \left(\frac{L_d}{L''_q} - 1 \right) i_{ds} + L_{md} \frac{L''_d}{L''_q} \left(i_{fd} + \sum_{j=1}^{N} i_{kdj} \right) \right] + L''_{mq} \frac{L''_d}{L''_q} \sum_{j=1}^{M} \frac{v_{kqj} - r_{kqj} i_{kqj}}{L_{lkqj}} + \frac{L''_q - L''_d}{L''_q} \times (v_{qs} - r_s i_{qs}). \tag{46}$$

Finally, transforming (41), (42), and (3), considering (9), to *abc* phase coordinates yields the constant-parameter stator interfacing circuit presented in [11] and [12] and shown in Fig. 1 (upper part). The parameters of the stator circuit are

$$r_D = r_s; \quad L_D = L''_d; \quad L_0 = -\frac{1}{3} L''_{md}. \tag{47}$$

In summary, the all-circuit constant-parameter voltage-behind-reactance (AC-CP-VBR) machine model is shown in Fig. 1, in which the rotor circuit based on (20)–(24) is found in the lower part. The subtransient algebraic voltages are defined by (46) [or (43)], (44), (23), and (24), in addition to λ_{qs} and λ_{ds} given by (7), (8), (13), and (14). As shown in Fig. 1, the

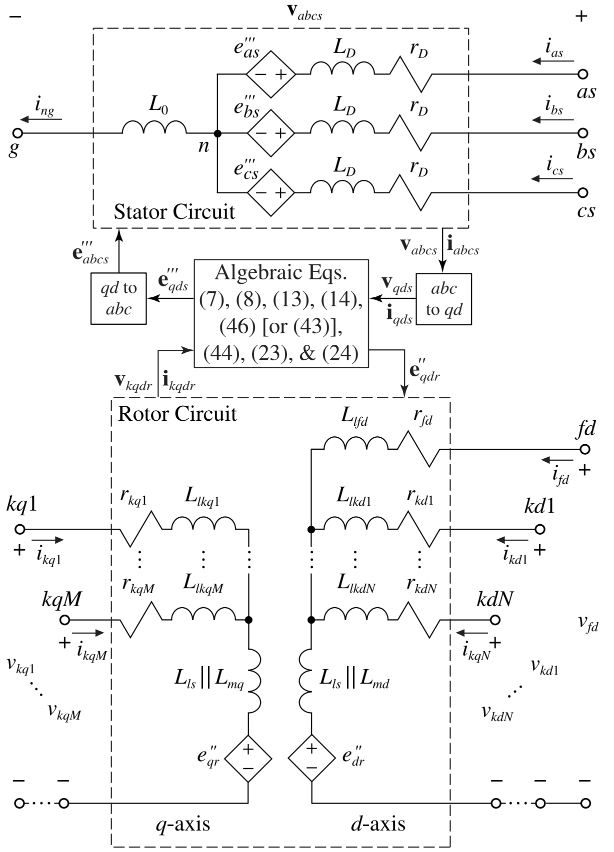

Fig. 1. All-circuit constant-parameter VBR synchronous machine model (AC-CP-VBR) with direct interfacing to arbitrary external networks.

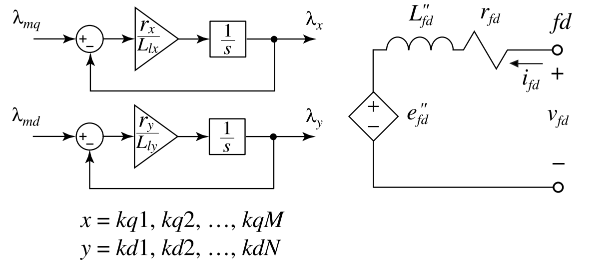

Fig. 2. Rotor subsystem for the SFC-CP-VBR model wherein the damper windings are represented as a state model and the field winding is made available as an interfacing circuit (the stator interfacing circuit is the same as in Fig. 1).

voltage terminals of the damper windings are available for connection with external circuits. This formulation may facilitate the representation of special machines with excitable auxiliary windings [27].

Using either (43) or (46) to calculate e'''_{qs} is conceptually identical; however, (46) is considered more general as it can be easily implemented in most simulation tools. In any case, these approaches may introduce an algebraic loop in the *q*-axis, which requires the use of a low-pass filter (LPF) to achieve a numerically efficient explicit model [12]. Alternatively, the time derivative of i_{qs} can be approximated using a high-pass filter [12].

C. Stator-and-Field-Circuit Formulation

In most studies, the damper windings of synchronous machines are all short-circuited. As a result, they are not interfaced with external networks, and as such there is no modeling (or numerical) advantage gained by representing them using circuit elements. In fact, representing the dynamics of all damper windings in a standard state-space form with flux linkages as state variables (as opposed to the AC-CP-VBR model) further increases the model numerical efficiency, as will be shown in Section IV.

1) Rotor Voltage Equations: To write the rotor state equations in terms of the flux linkages of the damper windings, first (10) and (12) are manipulated to rewrite the damper winding currents as [7]

$$i_{kdj} = \frac{\lambda_{kqj} - \lambda_{mq}}{L_{lkqj}}, \qquad j = 1, 2, \ldots, M \tag{48}$$

$$i_{kdj} = \frac{\lambda_{kdj} - \lambda_{md}}{L_{lkdj}}, \qquad j = 1, 2, \ldots, N. \tag{49}$$

Then, substituting (48) and (49) into (4) and (6), respectively, the damper windings state equations become

$$p\lambda_{kqj} = \frac{r_{kqj}}{L_{lkqj}}(\lambda_{mq} - \lambda_{kqj}) + v_{kqj}, \qquad j = 1, 2, \ldots, M \tag{50}$$

$$p\lambda_{kdj} = \frac{r_{kdj}}{L_{lkdj}}(\lambda_{md} - \lambda_{kdj}) + v_{kdj}, \qquad j = 1, 2, \ldots, N. \tag{51}$$

Substituting (49) into (14), after some algebraic simplifications, the *d*-axis magnetizing flux is rewritten as

$$\lambda_{md} = L'''_{md}\left(i_{ds} + i_{fd} + \sum_{j=1}^{N}\frac{\lambda_{kdj}}{L_{lkdj}}\right) \tag{52}$$

where

$$L'''_{md} = \left(\frac{1}{L_{md}} + \sum_{j=1}^{N}\frac{1}{L_{lkdj}}\right)^{-1}. \tag{53}$$

The *q*-axis magnetizing flux λ_{mq} in (50) is still computed using (25) and (29). The implementation of the damper windings state model is shown in Fig. 2 (left side).

To allow for a direct interface with external networks, i.e., an exciter, the field winding has to be represented as a circuit with input terminals. To obtain the field winding interfacing circuit, (8) is solved for i_{ds}, then the result is substituted into (52), which following algebraic manipulation gives

$$\lambda_{md} = L''''_{md}\left(i_{fd} + \frac{\lambda_{ds}}{L_{ls}} + \sum_{j=1}^{N}\frac{\lambda_{kdj}}{L_{lkdj}}\right) \tag{54}$$

where

$$L''''_{md} = \left(\frac{1}{L_{md}} + \frac{1}{L_{ls}} + \sum_{j=1}^{N} \frac{1}{L_{lkdj}} \right)^{-1}. \tag{55}$$

Substituting (54) into (11), taking the derivative of the result, and inserting (51) yields

$$p\lambda_{fd} = pL''_{fd} i_{fd} + p\frac{L''''_{md}}{L_{ls}} \lambda_{ds} + L''''_{md} \sum_{j=1}^{N} \frac{r_{kdj}(\lambda_{md} - \lambda_{kdj})}{L^2_{lkdj}} \tag{56}$$

where

$$L''_{fd} = L_{lfd} + L''''_{md}. \tag{57}$$

In the next step, (2) is solved for $p\lambda_{ds}$ and substituted into (56). The resulting equation is then inserted into (5). This procedure, considering (7) and after some algebraic simplifications, yields

$$v_{fd} = r_{fd} i_{fd} + pL''_{fd} i_{fd} + e''_{fd} \tag{58}$$

where

$$e''_{fd} = L''''_{md} \left(\frac{v_{ds} - r_s i_{ds} + \omega_r (L_{ls} i_{qs} + \lambda_{mq})}{L_{ls}} + \sum_{j=1}^{N} \frac{r_{kdj}(\lambda_{md} - \lambda_{kdj})}{L^2_{lkdj}} \right). \tag{59}$$

The resulting field winding interfacing circuit is shown in the right-hand side of Fig. 2.

2) Stator Voltage Equations: Similarly to Section II-B2, here the terms involving λ_{qs} and λ_{ds} in (1) and (2) should be replaced with the stator and field currents, and damper flux linkages. In this regard, inserting (29) and (50) into (31) first gives

$$p\lambda_{qs} = pL''_q i_{qs} + L''_{mq} \sum_{j=1}^{M} \frac{r_{kqj}(\lambda_{mq} - \lambda_{kqj})}{L^2_{lkqj}}. \tag{60}$$

Substituting (8) into (1), replacing λ_{md} with (52), and inserting (60) into the resulting equation give the q-axis stator voltage equation

$$v_{qs} = r_s i_{qs} + \omega_r L'''_d i_{ds} + pL''_q i_{qs} + e''_{qs2} \tag{61}$$

where

$$L'''_d = L_{ls} + L'''_{md} \tag{62}$$

and

$$e''_{qs2} = \omega_r L'''_{md} \left(i_{fd} + \sum_{j=1}^{N} \frac{\lambda_{kdj}}{L_{lkdj}} \right) + L''_{mq} \sum_{j=1}^{M} \frac{r_{kqj}(\lambda_{mq} - \lambda_{kqj})}{L^2_{lkqj}}. \tag{63}$$

To find the d-axis stator voltage equation, (30) is first substituted into (32). Then, the rotor flux linkage time derivatives are replaced by using (5) and (51), which yields

$$p\lambda_{ds} = pL''_d i_{ds} + L''_{md} \left(\frac{v_{fd} - r_{fd} i_{fd}}{L_{lfd}} + \sum_{j=1}^{N} \frac{r_{kdj}(\lambda_{md} - \lambda_{kdj})}{L^2_{lkdj}} \right). \tag{64}$$

Inserting (7) and (25) into (2), and substituting the result into (64), after some algebraic manipulations, the d-axis stator voltage equation becomes

$$v_{ds} = r_s i_{ds} - \omega_r L''_q i_{qs} + pL''_d i_{ds} + e''_{ds2} \tag{65}$$

where

$$e''_{ds2} = -\omega_r \lambda''_q + L''_{md} \frac{(v_{fd} - r_{fd} i_{fd})}{L_{lfd}} + L''_{md} \sum_{j=1}^{N} \frac{r_{kdj}(\lambda_{md} - \lambda_{kdj})}{L^2_{lkdj}}. \tag{66}$$

Following the approach set forth in [12] (similarly to Section II-B), the stator voltage equations (61) and (65) are rearranged as (41) and (42) where the new subtransient voltages become

$$e'''_{qs} = \omega_r \left[(L'''_d - L''_d) i_{ds} + L'''_{md} \left(i_{fd} + \sum_{j=1}^{N} \frac{\lambda_{kdj}}{L_{lkdj}} \right) \right] + L''_{mq} \sum_{j=1}^{M} \frac{r_{kqj}(\lambda_{mq} - \lambda_{kqj})}{L^2_{lkqj}} + p(L''_q - L''_d) i_{qs} \tag{67}$$

$$e'''_{ds} = -\omega_r \left[(L''_q - L''_d) i_{qs} + \lambda''_q \right] + L''_{md} \sum_{j=1}^{N} \frac{r_{kdj}(\lambda_{md} - \lambda_{kdj})}{L^2_{lkdj}} + L''_{md} \frac{v_{fd} - r_{fd} i_{fd}}{L_{lfd}}. \tag{68}$$

Note that (67) and (68) are algebraically equivalent to (43) and (44), respectively. Similarly to AC-CP-VBR, in order to replace the term $p(L''_q - L''_d) i_{qs}$ in (67) with state variables and voltages, $pL''_q i_{qs}$ is first expressed using (61) and (63) and then inserted into (67). After some algebraic manipulations, (67) becomes

$$e'''_{qs} = \omega_r \left[L''_d \left(\frac{L'''_d - L''_q}{L''_q} \right) i_{ds} + L'''_{md} \frac{L''_d}{L''_q} \left(i_{fd} + \sum_{j=1}^{N} \frac{\lambda_{kdj}}{L_{lkdj}} \right) \right] + L''_{mq} \frac{L''_d}{L''_q} \sum_{j=1}^{M} \frac{r_{kqj}(\lambda_{mq} - \lambda_{kqj})}{L^2_{lkqj}} + \frac{L''_q - L''_d}{L''_q} (v_{qs} - r_s i_{qs}). \tag{69}$$

This formulation has the same constant-parameter stator interfacing circuit as the one in the upper part of Fig. 1. Its parameters are also defined by (47).

To summarize this formulation, the complete model is obtained by replacing the rotor circuit in Fig. 1 (bottom) with Fig. 2 and changing the subtransient voltage equations. For this model, the subtransient voltages are given by (69) [or (67)], (68), and (59). The algebraic equations (25), (29), and (52) are used

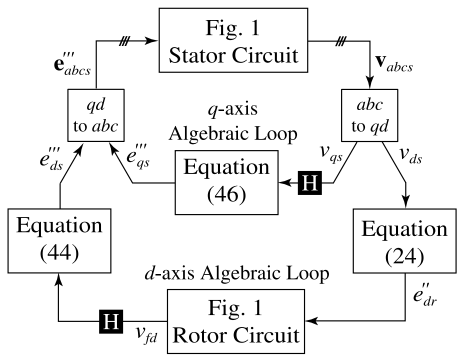

Fig. 3. Algebraic loops in AC-CP-VBR resulting in an implicit formulation. The "H" blocks indicate where the LPFs may be inserted to achieve an explicit formulation.

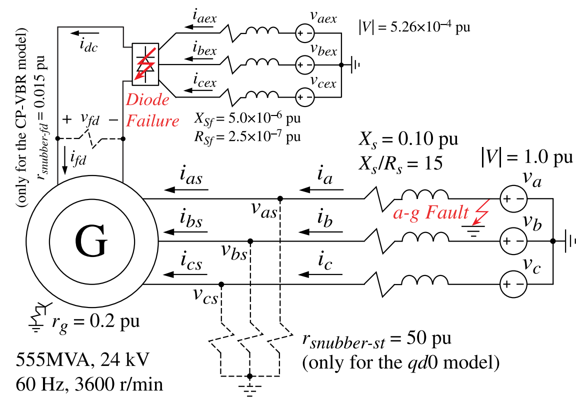

Fig. 4. Y-grounded steam turbine generator with an ac excitation system. The stator snubbers are required only for the classical $qd0$ model, and the field winding snubber is required only for the CP-VBR model [12].

to compute the magnetizing fluxes. This stator-and-field-circuit CP-VBR model is herein referred to as SFC-CP-VBR.

Similarly to the AC-CP-VBR model, algebraic loops may be introduced when the model is interfaced to an external network. The algebraic loops must be relaxed in order to achieve an efficient explicit model.

Both of the proposed models are algebraically equivalent to the lumped-parameter $qd0$ model. The mathematical manipulations were done to achieve a specific numerical property, namely constant-parameter rotor and stator interfacing circuits. Therefore, the physical meaning of parameters and performance of the model in steady state, transient, and subtransient conditions are preserved and remain identical to those of the $qd0$ and CC-PD models.

III. Implementation Considerations

The AC-CP-VBR and SFC-CP-VBR models have constant parameters. However, when connected to inductive external networks, these models will contain algebraic loops in both the d- and q-axis that can make them numerically expensive. The algebraic loops are illustrated in Fig. 3 for the AC-CP-VBR formulation. Similar algebraic loops exist for the SFC-CP-VBR model. An algebraic loop of similar nature was also encountered in [12] in the q-axis.

Models containing algebraic loops typically require the use of special solvers and additional iterations within each time-step that considerably add to the computational cost [28]. To achieve a numerically efficient solution (with acceptable accuracy but with less computational cost), it is possible to break the algebraic loops using discrete- or continuous-time LPFs [12]. For example, a first-order LPF may be considered

$$H_{\mathrm{LPF}}(s) = \frac{p_0}{s + p_0} \tag{70}$$

where $-p_0$ is the filter pole. By increasing the pole's magnitude, the filter becomes faster, improving the accuracy of the approximation. However, a very fast filter can make the system numerically stiff [12].

Inserting a LPF anywhere in the loop will relax the algebraic constraint; however, a more accurate solution can be obtained when the approximated variable is varying slowly. Herein, v_{qs} and v_{fd} are assumed to be the slowest variables in the q- and d-axis loops, respectively (see Fig. 3). Therefore, in the proposed models, the LPFs are inserted to relax these variables, as shown by the "H" blocks in Fig. 3.

If the current derivatives are not available and e'''_{qs} is calculated using (43) [or (67) for the SFC-CP-VBR model], the derivatives can be approximated using high-pass filters; however, such approach has shown to have similar computational cost to the relaxation of algebraic loops while being less accurate [12]. A more detailed discussion on the selection of the approximation technique can be found in [12].

Without loss of generality, all quantities in the proposed models are referred to the stator side. Therefore, to interface an external circuit to the rotor side, this circuit should also be referred to the stator with the appropriate turns ratio. Alternatively, it is possible to use an ideal transformer with the same turns ratio for connecting the rotor circuit to an external system.

IV. Validation of the Proposed Models

To verify the proposed models, a test system consisting of a large synchronous generator with an ac excitation system [21, Sec. 832] is considered. In this paper, a simplified model of the ac excitation system as shown in Fig. 4 is considered, wherein the alternator is represented by an inductive voltage source and the rectifier by a six-pulse diode bridge. Further details of the excitation system are omitted, as the focus of the studies is on the accuracy of machine models in predicting the electrical transients on the stator and rotor terminals. The machine parameters have been taken from [21, Sec. 3.6] and are also summarized in the appendix.

A single per-unit system based on the machine's nominal power and stator voltage is used for the generator and all external circuits. The excitation system parameters are also referred and rescaled to the stator base quantities. The steam turbine generator is assumed to be in steady state, driven by a constant mechanical torque of 0.9 p.u. The Thévenin voltage source on the stator side has a frequency of 60 Hz with nominal voltage of 1.0 p.u. On the exciter side, a six-pulse diode rectifier fed from a three-phase 60-Hz constant voltage source generates a field

TABLE II
NUMERICAL PERFORMANCE OF THE MODELS FOR THE SINGLE-PHASE-TO-GROUND FAULT STUDY

Model	No. of Time-Steps	Error $\varepsilon_{\mathrm{avg}}$		Largest Eigenvalue Magnitude $max\{\lvert\lambda\rvert\}$	Simulation Time
		Stator Source Current $\mathbf{i}_{abc}$	Exciter Source Current $\mathbf{i}_{abcex}$		
Coupled-Circuit Phase-Domain (CC-PD)	464	0.01%	0.01%	919	1.649 s
Full-VBR [14]	402	0.00%	0.00%	919	1.591 s
Classical QD0 With Stator Snubber (*qd*0)	9710 (1570)*	1.67% (1.66%)	0.07% (0.07%)	3.15×10^5	1.535 s (0.474 s)
Stator CP-VBR [12] With Rotor Snubber	18 400 (1714)	0.45% (0.47%)	3.01% (2.93%)	7.54×10^5	3.356 s (0.481 s)
All-Circuit (AC-CP-VBR)	458	2.62%	0.57%	5000	0.392 s
Stator-And-Field-Circuit (SFC-CP-VBR)	402	1.09%	0.20%	2000	0.388 s

*The values inside parentheses are for the ode15s solver.

current of approximately 1.44 p.u. The generator is assumed to be grounded through a grounding resistor ($r_g = 0.2$ p.u.), as depicted in Fig. 4.

The test system of Fig. 4 has been implemented using MATLAB/Simulink [29], [28] and the PLECS toolbox [17]. For numerically efficient and algebraic-loop-free implementation, first-order LPFs with poles of -5000 and -2000, respectively, for the AC-CP-VBR and SFC-CP-VBR models, have been used.

The existing constant-parameter models, namely *qd*0 [1] and stator CP-VBR with voltage loop relaxation [12], were also implemented. For the CP-VBR model, a first-order LPF with a pole of -5000 is used. Since PLECS' built-in *qd*0 model does not include zero-sequence and for consistency with the other models, a custom *qd*0 model has been created using conventional circuit elements [1, p. 202]. A circuit-based implementation of the *qd*0 model (as opposed to the more traditional state-space model) is required for direct connection of its field winding with the exciter. Since the stator is also implemented as a circuit, it is interfaced to the external power network using controlled current sources [6]. Additionally, since the external ac network (its Thévenin equivalent) is inductive, the *qd*0 model requires a three-phase interfacing snubber, as shown in Fig. 4. To have reasonable accuracy, a 50 p.u. stator snubber is chosen for the *qd*0 model. This value of snubber results in a system eigenvalue with a magnitude of 3.15×10^5. The system eigenvalues are found by linearizing the model around the steady-state operating point using the *linmod* and *eig* functions available in MATLAB/Simulink. The largest eigenvalue of each system is shown in Table II along with the results of a numerical performance analysis, which will be explained in this section.

The CP-VBR model [12] has a direct interface with the network through its stator terminals, but its field winding is represented as a standard state-space model with a voltage-input, current-output formulation. It is therefore interfaced to the external rotor circuit using a controlled current source. In this case, a resistive snubber must be connected to the field winding, as shown in Fig. 4. To have a reasonable accuracy, a 0.015 p.u. snubber (25 times larger than the field winding resistance) is chosen for the CP-VBR model [12]. This snubber introduces an eigenvalue with a magnitude of 7.54×10^5, as shown in Table II. Without the snubber, the largest eigenvalue magnitude of the system would be around 5000 due to the required LPF.

Finally, the Full-VBR model [14] and CC-PD model [1] are compared with the constant-parameter models. The CC-PD model is also used to generate a reference solution when executed with very small time-steps. The CC-PD model does not require any snubbers at either stator or rotor terminals, but has variable coupled inductances resulting in significant computational cost. The Full-VBR model [14] has similar properties.

To compare the accuracy of all subject models, the 2-norm cumulative relative error [30] of the entire solution trajectory is considered. For example, the 2-norm phase *a* current error is defined as

$$\varepsilon(\mathbf{i}_a) = \frac{\left\| \tilde{\mathbf{i}}_a - \mathbf{i}_a \right\|_2}{\left\| \tilde{\mathbf{i}}_a \right\|_2} \times 100\% \tag{71}$$

where $\mathbf{i}_a$ is a solution trajectory vector that includes all the calculated values over the considered period of study. Similarly, $\tilde{\mathbf{i}}_a$ is the reference solution trajectory for the same period of study. For three-phase quantities, the average of the two-norm single-phase errors is used

$$\varepsilon_{\mathrm{avg}}(\mathbf{i}_{abc}) = \frac{\varepsilon(\mathbf{i}_a) + \varepsilon(\mathbf{i}_b) + \varepsilon(\mathbf{i}_c)}{3}. \tag{72}$$

The MATLAB's general-purpose solver ode45 has been used for simulating all considered models. The *qd*0 and CP-VBR models require snubbers and are therefore stiff. For these two models, the study is repeated with the stiffly-stable solver ode15s for further comparison. For consistency among all simulation studies, the relative and absolute error tolerances are set to 10^{-4}, and the maximum and minimum time-steps are set to 10^{-3} and 10^{-7} s, respectively. To produce an accurate reference solution, the CC-PD model was used with a very small time-step of 10^{-6} s, and both error tolerances were set to 10^{-6}. All simulations were run on a personal computer with a 2.53GHz Intel CPU and Windows XP operating system. The system is assumed to operate in steady state prior to a transient event.

A. Single-Phase-to-Ground Fault in the Network

To emulate a severe unbalanced condition at the machine stator terminals, a single-phase-to-ground fault is applied at the network equivalent source. The fault is implemented by setting the phase *a* voltage to zero at $t = 10$ ms. The corresponding ac network voltages $\mathbf{v}_{abc}$ and currents $\mathbf{i}_{abc}$, the machine grounding current i_{ng}, the three-phase exciter current $\mathbf{i}_{abcex}$, the rectified exciter current i_{dc}, and the electromagnetic torque T_e predicted by various models are presented in Fig. 5. As it can be seen in Fig. 5, all the models give visibly close results. To provide a better insight, magnified views of $\mathbf{i}_{abc}$, $\mathbf{i}_{abcex}$, and i_{dc} are plotted in Figs. 6–9. Only the simulation results obtained with the ode45 solver are shown in these figures. A summary of the models' numerical performance is also given in Table II.

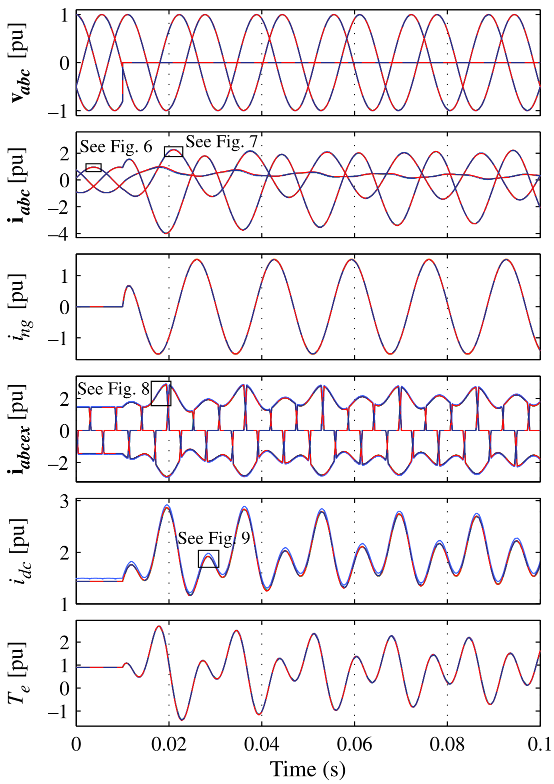

Fig. 5. Single-phase-to-ground fault case study transient responses as predicted by the considered models. From top to bottom: bus voltages $\mathbf{v}_{abc}$, bus currents $\mathbf{i}_{abc}$, machine neutral current i_{ng}, three-phase ac exciter current $\mathbf{i}_{abcex}$, rectifier current i_{dc}, and electromagnetic torque T_e.

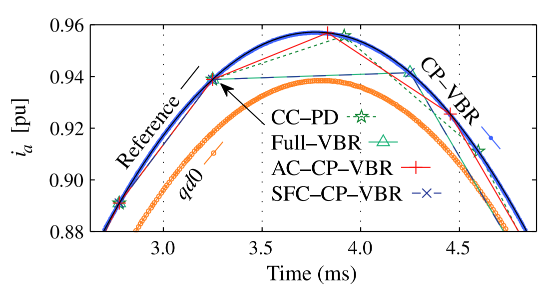

Fig. 6. Magnified fragment from Fig. 5: phase current i_a (in steady state).

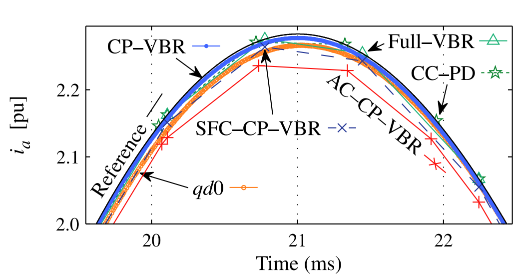

Fig. 7. Magnified fragment from Fig. 5: phase current i_a (in transient).

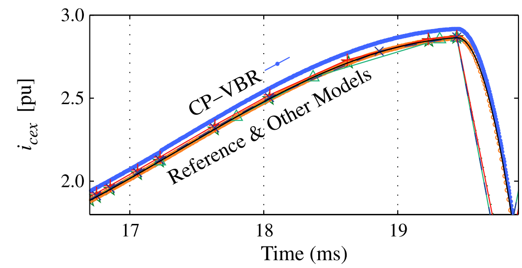

Fig. 8. Magnified fragment from Fig. 5: ac exciter phase current i_{cex}.

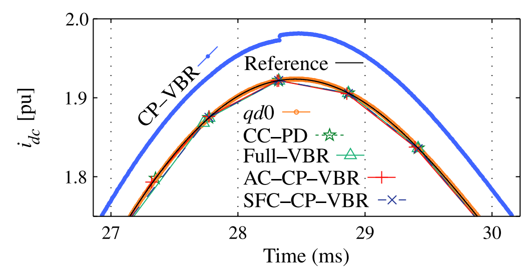

Fig. 9. Magnified fragment from Fig. 5: rectifier current i_{dc}.

After analyzing Figs. 6–9 and the second column of Table II, it is seen that when using the general-purpose solver the stiff models (i.e., *qd*0 and CP-VBR) need considerably more time-steps (9710 and 18 400) than the nonstiff models (i.e., CC-PD, Full-VBR, AC-CP-VBR, and SFC-CP-VBR) (less than 500). The CC-PD and Full-VBR models also use a small number of steps (464 and 402, respectively) while producing the most accurate results. However, due to their variable parameters, these models are by far the slowest; their simulation time is 1.649 and 1.591 s, respectively, compared to less than 0.5 s obtainable for all the other models. The high computational cost in these models results from the presence of time-varying inductances. Interested readers can find a detailed discussion on this subject in [9]. This additional computational burden is also strongly dependent on the efficiency of the state model generation algorithm, which is different for various simulation programs.

The simulation results of the *qd*0 and CP-VBR models using the stiffly-stable solver ode15s are also shown in Table II within parentheses. This reveals that using ode15s instead of the explicit solver ode45 leads to a significantly smaller number of time-steps and shorter simulation times with numerically stiff models. For example, when the CP-VBR model is solved using ode45, a total of 18 400 time-steps are required for the simulation time of 3.356 s. However, with ode15s, the number of time-steps and total simulation time significantly decrease to 1714 and 0.481 s, respectively.

Fig. 6 shows that the *qd*0 model predicts the stator currents with a noticeable error in steady state, as opposed to all other models. Similarly, as it can be seen in Figs. 8 and 9, the CP-VBR model [12] also has a noticeable error in the field winding current. Based on these figures and Table II (third and fourth columns), it can be concluded that the presence of snubbers in

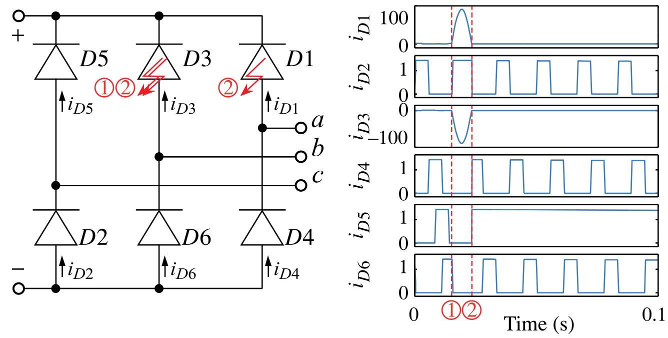

Fig. 10. Left: the diode bridge in Fig. 4. Right: simulation results of the diode currents in per unit: (2) indicates the initial failure (short-circuit) of $D3$; and (1) indicates when $D1$ and $D3$ (or corresponding protective fuses) become open due to ensuing excessive currents.

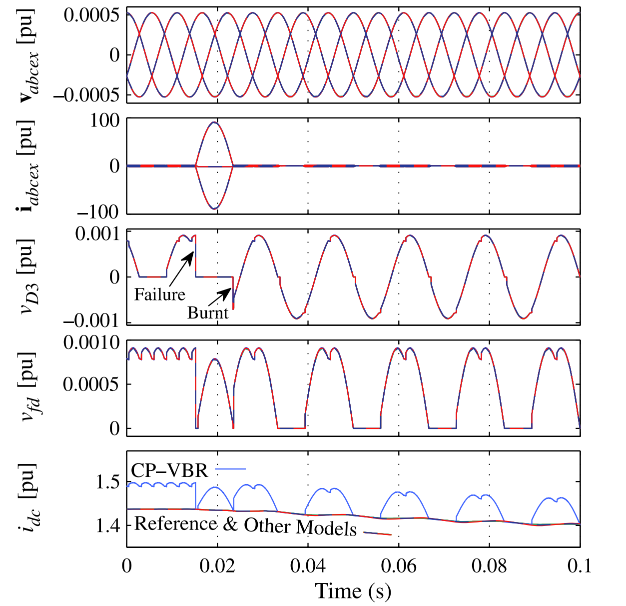

Fig. 11. Exciter diode failure case study transient responses as predicted by the considered models. From top to bottom: exciter source voltages $\mathbf{v}_{abcex}$, exciter source currents $\mathbf{i}_{abcex}$, diode $D3$ voltage v_{D3}, rectifier output voltage v_{fd}, and rectifier current i_{dc}.

the stator or rotor side increases the errors in $\mathbf{i}_{abc}$ or $\mathbf{i}_{abcex}$, respectively.

The two fastest models are the AC-CP-VBR and SFC-CP-VBR, both of which do not require snubbers and have direct stator and field windings interfacing. These two models use the LPF (70) to relax the algebraic loops. As Table II shows, the SFC-CP-VBR model offers better accuracy since its damper windings are implemented as a state-space model and do not contribute directly to the algebraic loop of the subtransient voltage e''_{dr}. Therefore, for this case, LPF poles with smaller magnitudes can be chosen (e.g., 2000). At the same time, the AC-CP-VBR model requires a filter pole with a larger magnitude (e.g., 5000) to achieve acceptable accuracy. Finally, for case studies with lower frequencies, the SFC-CP-VBR model would be able to choose significantly larger time-steps than the other constant-parameter models.

B. Diode Failure in the Exciter System

When system disturbances (such as faults) occur in the stator circuit of a synchronous generator, the field current fluctuates violently and can become negative and threaten the transient blocking voltage of the exciters' rectifiers. A computer model can be used for predicting overvoltages and overcurrents and designing the required remedies [31]. Additionally, in case of brushless exciters, testing the diode rectifiers may not be easy; however, detecting diode failures by modeling and analyzing the exciter waveforms may be a practical alternative [14].

To validate the proposed models in situations where the field winding and exciter variables are of primary interest, a diode failure scenario is simulated here. In this study, first, one of the diodes in the system shown in Fig. 4 fails and becomes short-circuited. Fig. 10 (left) illustrates the details of the three-leg diode bridge of Fig. 4, showing initial failure of diode $D3$. The third subplot of Fig. 11 shows the reverse voltage of diode $D3$ and demonstrates that it fails when its reverse voltage is at the peak (at $t = 0.0152$ s). This failure exposes the diodes $D3$ and $D1$ to excessive currents, which are more than 100 p.u. as shown in Fig. 10 (right). The fault current burns both of the diodes or their protective fuses [32]. It is assumed that

TABLE III
NUMERICAL PERFORMANCE OF THE MODELS FOR THE EXCITER DIODE FAILURE STUDY

Model	No.of Time-Steps	Error ε: Exciter Source Current $\mathbf{i}_{abcex}$	Error ε: Rectifier Current i_{dc}	Simulation Time
Coupled-Circuit Phase-Domain (CC-PD)	263	0.00%	0.00%	1.032 s
Full-VBR [14]	240	0.00%	0.01%	1.149 s
Classical QD0 With Stator Snubber ($qd0$)	9615 (1170)*	0.00% (0.09%)	0.00% (0.00%)	1.665 s (0.486 s)
Stator CP-VBR [12] With Rotor Snubber	13746 (1067)	1.24% (1.26%)	3.41% (2.88%)	2.909 s (0.456 s)
All-Circuit (AC-CP-VBR)	338	0.02%	0.06%	0.402 s
Stator-And-Field-Circuit (SFC-CP-VBR)	313	0.01%	0.02%	0.393 s

*The values inside parentheses are for ode15s solver.

the current continues to flow until it goes to zero, after which the diodes (or fuses) become open. The times of the equivalent short-circuit and open-circuit events are shown in Fig. 10 (right) by the circled numbers 1 and 2, respectively.

The exciter and field winding variables as predicted by various models for this diode failure scenario are shown in Fig. 11. This figure shows the exciter ac voltages $\mathbf{v}_{abcex}$ and currents $\mathbf{i}_{abcex}$, the reverse voltage of diode D3 v_{D3}, and the rectifier output dc voltage v_{fd} and current i_{dc}. Since the variations in the field current are relatively slow, the stator voltages and currents for this study are minimally affected and not shown for conciseness.

A summary of the performance of all considered models is given in Table III. As seen in the third and sixth columns of this table, the proposed models AC-CP-VBR and SFC-CP-VBR need fewer time-steps and are faster than the stiff models *qd*0 and CP-VBR. The CC-PD and Full-VBR models require the smallest number of time-steps (263 and 240, respectively); however, they are very slow (1.032 and 1.149 s, respectively) due to their variable inductances. Table III (third and fourth columns) and Fig. 11 indicate that all considered models have excellent accuracy except for the CP-VBR model, where the resistive snubber (see Fig. 4) adds a noticeable error to the rectifier dc current (i_{dc}) as seen in Fig. 11 (last subplot). The *qd*0 model with the stator interfacing snubber accurately predicts the field and exciter quantities since its field winding is directly interfaced and the three-phase snubber (as depicted in Fig. 4) mostly affects its stator variables.

This study demonstrates that the models with direct interface of the rotor field winding predict the rotor-exciter variables more accurately in steady state and during transients.

V. Conclusion

This paper presents two new explicit synchronous machine models with direct stator and rotor constant-parameter interfacing circuits. The proposed models are the all-circuit VBR (AC-CP-VBR) and stator-and-field-circuit VBR (SFC-CP-VBR) which are derived based on the constant-parameter VBR formulation. The new models are considerably less numerically stiff than the snubber-interfaced conventional *qd*0 and the constant-parameter stator VBR models and are more numerically efficient than the CC-PD and Full-VBR models with variable parameters.

The first model, AC-CP-VBR, is represented solely by conventional and constant circuit elements. This model is easy to implement in different simulation programs and is practical for the study of special cases where, for example, the equivalent damper windings may not be short-circuited or additional field windings are required. To further increase numerical accuracy and efficiency, the second model, SFC-CP-VBR, has been derived with short-circuited damper windings represented in state-space form (with flux linkages as the state variables).

Single-phase-to-ground fault and exciter rectifier diode failure case studies have been conducted to validate and compare the proposed AC-CP-VBR and SFC-CP-VBR models. To further emphasize the gains obtained by using the two new models, AC-CP-VBR and SFC-CP-VBR, Tables II and III show that, similar to the coupled-circuit model (CC-PD), the Full-VBR model [14] is slow due to its variable inductances. Overall, it is shown that both new models have an excellent combination of accuracy and numerical efficiency, which represents an improvement over the prior state-of-the-art CC-PD, Full-VBR, constant-parameter stator VBR (CP-VBR), and *qd*0 models.

Moreover, since variable inductances are not available in all simulation programs, the proposed new models will offer practical advantage and ease of implementation that has not been achieved by previous models.

Appendix

Steam Turbine Generator Parameters (in p.u.) [21, Sec. 3.6]: 555 MVA, 24 kV, 2-pole, 0.9 p.f., 60 Hz, $r_s = 0.003$, $X_{ls} = 0.15$, $X_{mq} = 1.61$, $X_{md} = 1.66$, $r_{kq1} = 0.00619$, $r_{kq2} = 0.02368$, $r_{fd} = 0.0006$, $r_{kd1} = 0.0284$, $X_{lkq1} = 0.7252$, $X_{lkq2} = 0.125$, $X_{lfd} = 0.165$, $X_{lkd1} = 0.1713$, $H = 5.6$ s, $X''_{mq} = 0.1$, and $X''_{md} = 0.08$. Grounding resistance $r_g = 0.2$.

Thévenin Impedance of the Network (on the Machine Base): $X_s = 10\%$ and $X/R = 15$.

Rotor-Exciter Source Impedance (Referred to the Stator, Per-Unitized on the Exciter Base):

8 MVA, 14.4 V, $X_{Sf} = 20\%$, and $R_{Sf} = 1.0\%$ (on the machine base: $X_{Sf} = 5.0 \times 10^{-6}$ p.u. and $R_{Sf} = 2.5 \times 10^{-7}$ p.u.).

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Constant-parameter synchronous machine model including main flux saturation

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Abstract: This paper presents a new general-purpose state-space synchronous machine model based on the voltage-behind-reactance (VBR) formulation which accounts for main flux saturation. The proposed model has a numerically efficient constant-parameter decoupled *RL* branch interfacing circuit comprised of the stator and field windings. Consequently, the new model can be interfaced directly with arbitrary power networks and exciter circuits. The presented studies, executed using the Simulink toolbox PLECS, demonstrate the superior combination of numerical accuracy and efficiency of the new VBR model compared with existing state-of-the-art models. As a result, the proposed model would be a desirable addition to the built-in component libraries of many industry-grade state-variable-based transient simulators.

Nomenclature

The main machine variables are as follows:

i_{as}, i_{bs}, and i_{cs}:	stator currents on phase a, b, and c
i_{fd}, v_{fd}, and λ_{fd}:	field winding current, voltage, and flux linkage
i_{kqj} and i_{kdj}:	current of the jth q-axis and d-axis damper windings
i_{m} and λ_{m}:	magnetising current and main flux
$i_{\mathrm{m}q}$ and $i_{\mathrm{m}d}$:	q-axis and d-axis magnetising currents
i_{qs}, i_{ds}, and $i_{0\mathrm{s}}$:	q-axis, d-axis, and zero-sequence stator currents
v_{as}, v_{bs}, and v_{cs}:	stator voltages on phase a, b, and c
v_{qs}, v_{ds}, and $v_{0\mathrm{s}}$:	q-axis, d-axis, and zero-sequence stator voltages
θ_{r}:	electrical rotor position
λ_{kqj} and λ_{kdj}:	flux linkage of the jth q-axis and d-axis damper windings
$\lambda_{\mathrm{m}q}$ and $\lambda_{\mathrm{m}d}$:	q-axis and d-axis magnetising fluxes
λ_{qs}, λ_{ds}, and $\lambda_{0\mathrm{s}}$:	q-axis, d-axis, and zero-sequence stator flux linkages
ω_{r}:	electrical rotor speed.

Stator variables are sometimes written in compact vector form as $\boldsymbol{f}_{qd0\mathrm{s}} \equiv [\, f_{qs} \quad f_{ds} \quad f_{0\mathrm{s}} \,]^{\mathrm{T}}$ and $\boldsymbol{f}_{abcs} \equiv [\, f_{as} \quad f_{bs} \quad f_{cs} \,]^{\mathrm{T}}$, where $\boldsymbol{f}$ may represent voltages or currents.

The main machine parameters are as follows:

L_{lfd} and r_{fd}:	field winding leakage inductance and resistance
$L_{\mathrm{l}kdj}$ and r_{kdj}:	leakage inductance and resistance of the jth d-axis damper winding
$L_{\mathrm{l}kqj}$ and r_{kqj}:	leakage inductance and resistance of the jth q-axis damper winding
L_{ls} and r_{s}:	stator leakage inductance and resistance
$L_{\mathrm{m}q}$ and $L_{\mathrm{m}d}$:	q-axis and d-axis magnetising inductances
S_{F}:	saliency factor.

1 Introduction

The general-purpose lumped-parameter synchronous machine models [1] available as standard built-in components in state-variable-based transient simulators (e.g. PLECS [2] and SimPowerSystems [3]) are widely used by system analysts to execute a variety of single- and multi-machine case studies. These general-purpose models can be implemented using different combinations of state variables [2–6]. Although the resulting formulations are algebraically equivalent, their numerical properties may greatly differ [4]. A common approach is to represent the machine equations in transformed *qd* coordinates [1]. However, the interface of classical *qd* models with inductive circuits requires snubbers, which introduces considerable numerical error and increases the computational cost of the entire simulation [4]. An alternative is to represent the machine directly using coupled inductances and resistances in the phase domain (*abc* coordinates) [4]. This coupled-circuit phase-domain model (CC-PD) can be seamlessly interfaced with circuit elements, for example, cables, transformers, and power electronic converters [4]. The CC-PD model's main downside is its high computational cost resulting from its time-varying inductances and poorly scaled eigenvalues [5]. Another possibility is to use a voltage-behind-reactance (VBR) formulation [5]. In the original VBR model [5], the stator is represented in phase variables using circuit elements, while the rotor is modelled in standard state-space form using *qd*-coordinate fluxes as state variables. Its major drawback is that its interfacing circuit contains rotor-position-dependent inductances due to dynamic saliency [5, 7].

Several methods have been proposed to obtain a magnetically linear synchronous machine VBR model with a constant-parameter interfacing circuit [7–9]. One such approach consists of transferring the time-varying parts of the interfacing circuit's inductances to its voltage sources [9]. Several filters can then be used to obtain a proper and explicit state model [9]. Explicit constant-parameter VBR models significantly reduce CPU time compared with variable-parameter models [7, 9].

Representation of main flux saturation is a common option in built-in synchronous machine models [2, 3]. The improvement in modelling accuracy resulting from incorporating main flux saturation is presented in many classical references such as [5, 10, 11], and is considered beneficial for many system-level studies. A synchronous machine VBR model with magnetic saturation was first proposed in [12]. Therein, main flux saturation was only considered along the *d*-axis. A VBR model representing saturation in both axes (and including cross saturation) was later proposed in [13]. Both models have time-varying interfacing inductances due to saturation and dynamic saliency. A constant-parameter induction machine VBR model with main flux saturation was recently proposed in [14], whose derivation was facilitated by the machine's symmetry. No such model exists for saturable synchronous machines, which is the main motivation behind this paper.

One limitation of the VBR models proposed in [5–13] is that only the stator is represented using circuit elements. Consequently, similarly to the *qd* model's stator windings [4], the field winding of these VBR models has a voltage-input, current-output formulation. Fortunately, in many simulation studies, the field winding is connected to an equivalent controlled voltage source, for example, the output of a transfer function (TF)-based equivalent excitation system [15]. Consequently, such sources can be directly interfaced with the rotor subsystem state-space model. However, it is sometimes necessary and/or convenient to model the exciter using circuit components, for example to analyse diode failures in brushless exciters [16, 17], or to study the impact of field discharge resistors and exciter breakers [18, 19]. Such detailed exciter models may result in an incompatible interface with the VBR models presented in [5–13], requiring a snubber at the field winding terminals. To overcome this limitation, the model of [13] was later extended to add the field winding to its interfacing circuit [16]; minor variations of these two models ([13, 16]) are now available in PLECS' component library. Two unsaturated VBR models with constant-parameter stator and field windings were also recently proposed in [17].

The main objective of this paper is to present a new general-purpose lumped-parameter synchronous machine model suitable for inclusion in the built-in component libraries of many industry-grade transient simulators. Unlike all other state-of-the-art synchronous machine models ([1–13, 16, 17]), the proposed model combines representation of main flux saturation with a constant-parameter interfacing circuit. Moreover, to provide a direct interface with arbitrary external power networks and exciter models, the interfacing circuit of the proposed model includes both the stator and field windings. The new model is achieved by extending the stator-and-field-circuit VBR model presented in [17] to account for main flux saturation while maintaining a constant-parameter interfacing circuit. In addition, a procedure to select the filter parameters of the proposed model is also presented in this paper. Such an approach was not included in [17].

2 General-purpose synchronous machine model

In this paper, a general-purpose lumped-parameter wye-connected synchronous machine model with M damper windings in the q-axis and N damper windings in the d-axis is considered [1]. The notations and assumptions are the same as in [1], for example, differential leakage inductances are neglected. Motor sign convention is considered, and all parameters/variables are referred to the stator side.

2.1 qd model

The starting point for the derivation of VBR models is the traditional *qd* model [1]. Its voltage equations are

$$v_{qs} = r_s i_{qs} + \omega_r \lambda_{ds} + p\lambda_{qs} \tag{1}$$

$$v_{ds} = r_s i_{ds} - \omega_r \lambda_{qs} + p\lambda_{ds} \tag{2}$$

$$v_{0s} = r_s i_{0s} + p\lambda_{0s} \tag{3}$$

$$0 = r_{kqj} i_{kqj} + p\lambda_{kqj}, \quad j = 1, \ldots, M \tag{4}$$

$$0 = r_{kdj} i_{kdj} + p\lambda_{kdj}, \quad j = 1, \ldots, N \tag{5}$$

$$v_{fd} = r_{fd} i_{fd} + p\lambda_{fd} \tag{6}$$

where p is Heaviside's operator (d/dt). The flux linkages defined in (1)–(6) are expressed as

$$\lambda_{qs} = L_{ls} i_{qs} + \lambda_{mq} \tag{7}$$

$$\lambda_{ds} = L_{ls} i_{ds} + \lambda_{md} \tag{8}$$

$$\lambda_{0s} = L_{ls} i_{0s} \tag{9}$$

$$\lambda_{kqj} = L_{lkqj} i_{kqj} + \lambda_{mq}, \quad j = 1, \ldots, M \tag{10}$$

$$\lambda_{kdj} = L_{lkdj} i_{kdj} + \lambda_{md}, \quad j = 1, \ldots, N \tag{11}$$

$$\lambda_{fd} = L_{lfd} i_{fd} + \lambda_{md}. \tag{12}$$

The magnetising fluxes λ_{mq} and λ_{md} are defined in Section 2.2. The torque and mechanical equations can be found in many references, for example [1, 20].

2.2 Main flux saturation

Unlike in induction machines [14], the main flux λ_m and magnetising current i_m of synchronous machines are not necessarily aligned [21], making the representation of saturation more challenging. To simplify modelling, a commonly used approach is to define an equivalent isotropic machine in which λ_m and i_m are aligned [5, 20, 21]. The main flux and magnetising current of this equivalent machine are related to their q- and d-axis projections by [20, see Fig. 5]

$$\lambda_m = \sqrt{\left(\lambda_{mq}/S_F\right)^2 + \lambda_{md}^2} \text{ and } i_m = \sqrt{\left(S_F i_{mq}\right)^2 + i_{md}^2} \tag{13}$$

where S_F is the saliency factor. The definition of S_F varies; for instance, it can be a function of the magnetising current [21]. In this paper we use the single saturation factor approach [5, 20], wherein S_F is constant and defined as

$$S_F = \sqrt{L_{mqu}/L_{mdu}} \tag{14}$$

where L_{mqu} and L_{mdu} are the unsaturated q- and d-axis magnetising inductances, respectively. This simple approach is deemed sufficiently accurate for most system-level studies [5], and as such is also used in PLECS' built-in models. Thereafter, the readily available d-axis open-circuit saturation curve $i_{md} = F_d(\lambda_{md})$ can be applied to the main flux as [20]

$$i_m = F_d(\lambda_m). \tag{15}$$

Consequently, the magnetising fluxes can be written as

$$\lambda_{mq} = L_{mq}(\lambda_m) i_{mq} \equiv L_{mq}(\lambda_m)\left(i_{qs} + \sum\nolimits_{j=1}^{M} i_{kqj}\right) \tag{16}$$

$$\lambda_{md} = L_{md}(\lambda_m) i_{md} \equiv L_{md}(\lambda_m)\left(i_{ds} + i_{fd} + \sum\nolimits_{j=1}^{N} i_{kdj}\right) \tag{17}$$

where $L_{md}(\lambda_m)$ is computed from (15) and $L_{mq}(\lambda_m) = S_F^2 L_{md}(\lambda_m)$.

3 Variable-parameter-VBR model (VP-VBR)

This section presents the equations of the saturable stator-and-field-circuit variable-parameter VBR model (VP-VBR) with main flux saturation [16]. A similar model is available in PLECS [2].

3.1 Rotor state equations

The state variables of the VP-VBR model's state-space rotor subsystem are the magnetising fluxes λ_{mq} and λ_{md}, and the rotor fluxes λ_{kqj} ($j=2,\ldots,M$) and λ_{kdj} ($j=2,\ldots,N$). This means that λ_{kq1} and λ_{kd1} are auxiliary algebraic variables. This is different from the magnetically linear stator-and-field-winding model presented in [17], wherein all the rotor flux linkages are state variables.

Differentiating (16) and (17) with respect to time gives

$$p\begin{bmatrix} i_{mq} \\ i_{md} \end{bmatrix} = \boldsymbol{\Gamma}_{mi}(\boldsymbol{\lambda}_{mqd}) p \begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} \tag{18}$$

where
see (19)
and where the reluctances $\Gamma_{mq}(\lambda_m)$ and $\Gamma_{md}(\lambda_m)$ are equal to $1/L_{mq}(\lambda_m)$ and $1/L_{md}(\lambda_m)$, respectively.

Substituting (7), (8), and (10)–(12) into (16) and (17), the magnetising currents can be rewritten in terms of fluxes as

$$i_{mq} = \frac{\lambda_{qs} - \lambda_{mq}}{L_{ls}} + \sum_{j=1}^{M} \frac{\lambda_{kqj} - \lambda_{mq}}{L_{lkqj}} \tag{20}$$

$$i_{md} = \frac{\lambda_{ds} - \lambda_{md}}{L_{ls}} + \frac{\lambda_{fd} - \lambda_{md}}{L_{lfd}} + \sum_{j=1}^{N} \frac{\lambda_{kdj} - \lambda_{md}}{L_{lkdj}}. \tag{21}$$

Inserting (20) and (21) into the left-hand side of (18) and solving for $p\boldsymbol{\lambda}_{mqd}$ yields

$$p\begin{bmatrix} \lambda_{mq} \\ \lambda_{md} \end{bmatrix} = \boldsymbol{L}''_{\Sigma}(\boldsymbol{\lambda}_{mqd}) \left(\begin{bmatrix} \dfrac{p\lambda_{qs}}{L_{ls}} \\ \dfrac{p\lambda_{ds}}{L_{ls}} \end{bmatrix} + \begin{bmatrix} 0 \\ \dfrac{p\lambda_{fd}}{L_{lfd}} \end{bmatrix} + \begin{bmatrix} \sum_{j=1}^{M} \dfrac{p\lambda_{kqj}}{L_{lkqj}} \\ \sum_{j=1}^{N} \dfrac{p\lambda_{kdj}}{L_{lkdj}} \end{bmatrix} \right) \tag{22}$$

where

$$\boldsymbol{L}''_{\Sigma}(\boldsymbol{\lambda}_{mqd}) = \left(\boldsymbol{\Gamma}_{mi}(\boldsymbol{\lambda}_{mqd}) + \begin{bmatrix} \dfrac{1}{L_{\Sigma lq}} & 0 \\ 0 & \dfrac{1}{L_{\Sigma ld}} \end{bmatrix} \right)^{-1} \tag{23}$$

and

$$L_{\Sigma lq} = \left(\frac{1}{L_{ls}} + \sum_{j=1}^{M} \frac{1}{L_{lkqj}} \right)^{-1} \text{ and } L_{\Sigma ld} = \left(\frac{1}{L_{ls}} + \frac{1}{L_{lfd}} + \sum_{j=1}^{N} \frac{1}{L_{lkdj}} \right)^{-1}. \tag{24}$$

The right-hand side of (22) can be computed as follows. First, using (1), (2), and (6)–(8), $p\lambda_{qs}$, $p\lambda_{ds}$, and $p\lambda_{fd}$ can be formulated as a function of states and inputs (stator and field currents and voltages, and electrical rotor speed) as

$$p\lambda_{qs} = v_{qs} - r_s i_{qs} - \omega_r \left(L_{ls} i_{ds} + \lambda_{md} \right) \tag{25}$$

$$p\lambda_{ds} = v_{ds} - r_s i_{ds} + \omega_r \left(L_{ls} i_{qs} + \lambda_{mq} \right) \tag{26}$$

$$p\lambda_{fd} = v_{fd} - r_{fd} i_{fd}. \tag{27}$$

The rotor flux linkage time derivatives can also be expressed as a function of rotor and magnetising fluxes by inserting (10) and (11) into (4) and (5), yielding

$$p\lambda_{kqj} = -\frac{r_{kqj}}{L_{lkqj}} \left(\lambda_{kqj} - \lambda_{mq} \right), \quad j = 1, \ldots, M \tag{28}$$

$$p\lambda_{kdj} = -\frac{r_{kdj}}{L_{lkdj}} \left(\lambda_{kdj} - \lambda_{md} \right), \quad j = 1, \ldots, N. \tag{29}$$

Since (28) and (29) are not used as state equations for $j=1$, it is necessary to write $\boldsymbol{\lambda}_{kqd1}$ as a function of states and inputs. Substituting (10) and (11) into (16) and (17) and solving for λ_{kq1} and λ_{kd1} yields

$$\lambda_{kq1} = L_{lkq1} \left(\frac{\lambda_{mq}}{L''_{mq}(\lambda_m)} - i_{qs} - \sum_{j=2}^{M} \frac{\lambda_{kqj}}{L_{lkqj}} \right) \tag{30}$$

$$\lambda_{kd1} = L_{lkd1} \left(\frac{\lambda_{md}}{L'''_{md}(\lambda_m)} - i_{ds} - i_{fd} - \sum_{j=2}^{N} \frac{\lambda_{kdj}}{L_{lkdj}} \right) \tag{31}$$

where $L''_{mq}(\lambda_m)$ and $L'''_{md}(\lambda_m)$ are defined as

$$L''_{mq}(\lambda_m) = \left(\frac{1}{L_{mq}(\lambda_m)} + \sum_{j=1}^{M} \frac{1}{L_{lkqj}} \right)^{-1} \tag{32}$$

$$L'''_{md}(\lambda_m) = \left(\frac{1}{L_{md}(\lambda_m)} + \sum_{j=1}^{N} \frac{1}{L_{lkdj}} \right)^{-1}. \tag{33}$$

In summary, the state equations of the VP-VBR model's rotor subsystem are given by (22), (28), and (29).

3.2 Variable-parameter interfacing circuit equations

The derivation of the interfacing circuit equations starts by inserting (7), (8), and (12) into (1), (2), and (6), yielding

$$v_{qs} = r_s i_{qs} + \omega_r \left(L_{ls} i_{ds} + \lambda_{md} \right) + L_{ls} p i_{qs} + p\lambda_{mq} \tag{34}$$

$$v_{ds} = r_s i_{ds} - \omega_r \left(L_{ls} i_{qs} + \lambda_{mq} \right) + L_{ls} p i_{ds} + p\lambda_{md} \tag{35}$$

$$v_{fd} = r_{fd} i_{fd} + L_{lfd} p i_{fd} + p\lambda_{md}. \tag{36}$$

To form a proper circuit model, the time derivatives of the magnetising fluxes must be eliminated from (34)–(36). A possibility is to substitute (22); however, the right-hand sides of (34)–(36) would be function of stator and field voltages due to the presence of $p\lambda_{qs}$, $p\lambda_{ds}$, and $p\lambda_{fd}$, which would in turn require solving a 3×3 system of equations. Instead, we choose to reformulate $p\boldsymbol{\lambda}_{mqd}$ so that it becomes function of pi_{qs}, pi_{ds}, and pi_{fd}. This is achieved by redefining (20) and (21) as

$$i_{mq} = i_{qs} + \sum_{j=1}^{M} \frac{\lambda_{kqj} - \lambda_{mq}}{L_{lkqj}} \tag{37}$$

$$i_{md} = i_{ds} + i_{fd} + \sum_{j=1}^{N} \frac{\lambda_{kdj} - \lambda_{md}}{L_{lkdj}} \tag{38}$$

$$\boldsymbol{\Gamma}_{mi}(\boldsymbol{\lambda}_{mqd}) = \begin{bmatrix} \left(\dfrac{di_m}{d\lambda_m} - \Gamma_{md}(\lambda_m) \right) \dfrac{\lambda_{mq}^2}{S_F^4 \lambda_m^2} + \Gamma_{mq}(\lambda_m) & \left(\dfrac{di_m}{d\lambda_m} - \Gamma_{md}(\lambda_m) \right) \dfrac{\lambda_{mq}\lambda_{md}}{S_F^2 \lambda_m^2} \\ \left(\dfrac{di_m}{d\lambda_m} - \Gamma_{md}(\lambda_m) \right) \dfrac{\lambda_{mq}\lambda_{md}}{S_F^2 \lambda_m^2} & \left(\dfrac{di_m}{d\lambda_m} - \Gamma_{md}(\lambda_m) \right) \dfrac{\lambda_{md}^2}{\lambda_m^2} + \Gamma_{md}(\lambda_m) \end{bmatrix} \tag{19}$$

and substituting (37) and (38) into (18), yielding

$$p\begin{bmatrix}\lambda_{mq}\\ \lambda_{md}\end{bmatrix} = \boldsymbol{L}'''_{mi}(\boldsymbol{\lambda}_{mqd})\left(p\begin{bmatrix}i_{qs}\\ i_{ds}\end{bmatrix} + p\begin{bmatrix}0\\ i_{fd}\end{bmatrix} + \begin{bmatrix}\sum_{j=1}^{M}\dfrac{p\lambda_{kqj}}{L_{lkqj}}\\ \sum_{j=1}^{N}\dfrac{p\lambda_{kdj}}{L_{lkdj}}\end{bmatrix}\right) \tag{39}$$

where the modified inductance matrix is

$$\boldsymbol{L}'''_{mi}(\boldsymbol{\lambda}_{mqd}) \equiv \begin{bmatrix}L'''_{mqq} & L'''_{mqd}\\ L'''_{mqd} & L'''_{mdd}\end{bmatrix} = \left(\boldsymbol{\Gamma}_{mi}(\boldsymbol{\lambda}_{mqd}) + \begin{bmatrix}\sum_{j=1}^{M}\dfrac{1}{L_{lkqj}} & 0\\ 0 & \sum_{j=1}^{N}\dfrac{1}{L_{lkdj}}\end{bmatrix}\right)^{-1}. \tag{40}$$

The machine-network interfacing equation (in qd coordinates) is then obtained by inserting (39) into (34)–(36), and combining the results with (3) and (9) as

$$\begin{bmatrix}\boldsymbol{v}_{qd0s}\\ v_{fd}\end{bmatrix} = \left(\boldsymbol{R}_c + \omega_r\boldsymbol{L}_{\omega_r}(\boldsymbol{\lambda}_{mqd})\right)\begin{bmatrix}\boldsymbol{i}_{qd0s}\\ i_{fd}\end{bmatrix} + \boldsymbol{L}'''_{qd0sf}(\boldsymbol{\lambda}_{mqd})p\begin{bmatrix}\boldsymbol{i}_{qd0s}\\ i_{fd}\end{bmatrix} + \begin{bmatrix}\boldsymbol{e}'''_{qd0s}\\ e'''_{fd}\end{bmatrix} \tag{41}$$

where

$$\boldsymbol{R}_c = \mathrm{diag}\begin{bmatrix}r_s & r_s & r_s & r_{fd}\end{bmatrix} \tag{42}$$

$$\boldsymbol{L}_{\omega_r}(\boldsymbol{\lambda}_{mqd}) = \begin{bmatrix}-L'''_{mqd} & L'''_{qq} & 0 & 0\\ -L'''_{dd} & L'''_{mqd} & 0 & 0\\ 0 & 0 & 0 & 0\\ -L'''_{mdd} & L'''_{mqd} & 0 & 0\end{bmatrix} \tag{43}$$

$$\boldsymbol{L}'''_{qd0sf}(\boldsymbol{\lambda}_{mqd}) = \begin{bmatrix}L'''_{qq} & L'''_{mqd} & 0 & L'''_{mqd}\\ L'''_{mqd} & L'''_{dd} & 0 & L'''_{mdd}\\ 0 & 0 & L_{ls} & 0\\ L'''_{mqd} & L'''_{mdd} & 0 & L'''_{fdd}\end{bmatrix} \tag{44}$$

$$\boldsymbol{e}'''_{qd0s} = \begin{bmatrix}e'''_{qs} & e'''_{ds} & 0\end{bmatrix}^T. \tag{45}$$

The modified subtransient inductances appearing in (43) and (44) are defined as

$$L'''_{qq} = L_{ls} + L'''_{mqq},\ L'''_{dd} = L_{ls} + L'''_{mdd},\ \text{and } L'''_{fdd} = L_{lfd} + L'''_{mdd} \tag{46}$$

while the modified subtransient voltages are given by

$$\begin{bmatrix}e'''_{qs}\\ e'''_{ds}\\ e'''_{fd}\end{bmatrix} = \omega_r\left(\begin{bmatrix}\lambda_{md}\\ -\lambda_{mq}\\ 0\end{bmatrix} + \begin{bmatrix}L'''_{mqd} & -L'''_{mqq}\\ L'''_{mdd} & -L'''_{mqd}\\ L'''_{mdd} & -L'''_{mqd}\end{bmatrix}\begin{bmatrix}i_{qs}\\ i_{ds}\end{bmatrix}\right) + \begin{bmatrix}L'''_{mqq} & L'''_{mqd}\\ L'''_{mqd} & L'''_{mdd}\\ L'''_{mqd} & L'''_{mdd}\end{bmatrix}\begin{bmatrix}\sum_{j=1}^{M}\dfrac{p\lambda_{kqj}}{L_{lkqj}}\\ \sum_{j=1}^{N}\dfrac{p\lambda_{kdj}}{L_{lkdj}}\end{bmatrix}. \tag{47}$$

The time derivatives of the rotor fluxes can be computed using (28)–(31). Finally, applying Park's transformation to the stator variables in (41) [16, see (66) and (67)] yields the final machine-network interfacing equation of the VP-VBR model

$$\begin{bmatrix}\boldsymbol{v}_{abcs}\\ v_{fd}\end{bmatrix} = \boldsymbol{R}_c\begin{bmatrix}\boldsymbol{i}_{abcs}\\ i_{fd}\end{bmatrix} + \left(\boldsymbol{L}_{lcsf} + \boldsymbol{L}'''_{vsf}(\theta_r, \boldsymbol{\lambda}_{mqd})\right)p\begin{bmatrix}\boldsymbol{i}_{abcs}\\ i_{fd}\end{bmatrix} + \begin{bmatrix}\boldsymbol{e}'''_{abcs}\\ e'''_{fd}\end{bmatrix} \tag{48}$$

where

$$\boldsymbol{L}_{lcsf} = \mathrm{diag}\begin{bmatrix}L_{ls} & L_{ls} & L_{ls} & L_{lfd}\end{bmatrix} \tag{49}$$

see (50)

$$\boldsymbol{e}'''_{abcs} = \boldsymbol{K}_s^{-1}\begin{bmatrix}e'''_{qs} & e'''_{ds} & 0\end{bmatrix}^T \tag{51}$$

and where θ_r and $\boldsymbol{K}_s^{-1}$ are the electrical rotor position and the inverse of Park's transformation matrix [1], respectively. The inductances necessary to compute (50) are defined as

$$L_1(\varphi) = L_b\cos(2\varphi) + L_c\sin(2\varphi) \tag{52}$$

$$L_2(\varphi) = L_c\cos\varphi + L_d\sin\varphi \tag{53}$$

$$L_a = \frac{L'''_{mqq} + L'''_{mdd}}{3} \text{ and } L_b = \frac{L'''_{mqq} - L'''_{mdd}}{3} \tag{54}$$

$$L_c = \frac{2}{3}L'''_{mqd} \text{ and } L_d = \frac{2}{3}L'''_{mdd} \tag{55}$$

where $\varphi = \{\theta_r, \theta_r + 120°, \theta_r - 120°\}$.

4 Constant-parameter-VBR (CP-VBR)

4.1 Constant-parameter interfacing circuit equations

Analysing (48), (50), and (52)–(55), it is observed that the interfacing circuit of the VP-VBR model is independent of the rotor position and the level of saturation if L_a is constant and $L_b = L_c = L_d = 0$. Tracing these terms back to (41), it can be seen that a constant-parameter model can be achieved if $\boldsymbol{L}'''_{qd0sf}$ is a diagonal matrix of the form

$$\boldsymbol{L} = \mathrm{diag}\begin{bmatrix}L_\alpha & L_\alpha & L_\beta & L_\chi\end{bmatrix} \tag{56}$$

$$\boldsymbol{L}'''_{vsf}(\theta_r, \boldsymbol{\lambda}_{mqd}) = \begin{bmatrix}L_a & -0.5L_a & -0.5L_a & 0\\ -0.5L_a & L_a & -0.5L_a & 0\\ -0.5L_a & -0.5L_a & L_a & 0\\ 0 & 0 & 0 & L'''_{mdd}\end{bmatrix} + \begin{bmatrix}L_1(\theta_r) & L_1(\theta_r+120°) & L_1(\theta_r-120°) & 1.5L_2(\theta_r)\\ L_1(\theta_r+120°) & L_1(\theta_r-120°) & L_1(\theta_r) & 1.5L_2(\theta_r-120°)\\ L_1(\theta_r-120°) & L_1(\theta_r) & L_1(\theta_r+120°) & 1.5L_2(\theta_r+120°)\\ L_2(\theta_r) & L_2(\theta_r-120°) & L_2(\theta_r+120°) & 0\end{bmatrix} \tag{50}$$

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where L_α, L_β, and L_χ are arbitrary constants. This structure can be achieved by transferring select terms to the subtransient voltage sources and modifying the speed voltage terms accordingly [9, 14, 17]. Applying this approach to (41) yields the following machine-network interfacing equation

$$\begin{bmatrix} \boldsymbol{v}_{qd0s} \\ v_{fd} \end{bmatrix} = \left(\boldsymbol{R}_c + \omega_r \boldsymbol{L}_{\omega_r u}\right)\begin{bmatrix} \boldsymbol{i}_{qd0s} \\ i_{fd} \end{bmatrix} + \boldsymbol{L}'''_{qd0sfu} p \begin{bmatrix} \boldsymbol{i}_{qd0s} \\ i_{fd} \end{bmatrix} + \begin{bmatrix} \boldsymbol{e}''''_{qd0s} \\ e''''_{fd} \end{bmatrix} \tag{57}$$

where

$$\boldsymbol{L}_{\omega_r u} = \begin{bmatrix} 0 & L_{ls} + L'''_{mddu} & 0 & 0 \\ -L_{ls} - L'''_{mddu} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{58}$$

$$\boldsymbol{L}'''_{qd0sfu} = \mathrm{diag}\left[L_{ls} + L'''_{mddu} \quad L_{ls} + L'''_{mddu} \quad L_{ls} \quad L_{lfd} + L'''_{mddu} \right] \tag{59}$$

$$\boldsymbol{e}''''_{qd0s} = \left[e''''_{qs} \quad e''''_{ds} \quad 0 \right]^{\mathrm{T}}. \tag{60}$$

The newly introduced inductance L'''_{mddu} refers to the unsaturated value of L'''_{mdd}, while the modified voltage sources e''''_{qs}, e''''_{ds}, and e''''_{fd} are expressed as (see (61))
where

$$\Delta L'''_{mdd} = L'''_{mdd} - L'''_{mddu}. \tag{62}$$

The inductance matrix of (61) will contain negative terms in the saturated region. This is not a source of numerical instability since this matrix is in series with the much larger and positive $\boldsymbol{L}'''_{qd0sfu}$. Finally, converting the stator variables of (57) to abc coordinates yields the CP-VBR machine-network interfacing equation

$$\begin{bmatrix} \boldsymbol{v}_{abcs} \\ v_{fd} \end{bmatrix} = \boldsymbol{R}_c \begin{bmatrix} \boldsymbol{i}_{abcs} \\ i_{fd} \end{bmatrix} + \boldsymbol{L}'''_{csf} p \begin{bmatrix} \boldsymbol{i}_{abcs} \\ i_{fd} \end{bmatrix} + \begin{bmatrix} \boldsymbol{e}''''_{abcs} \\ e''''_{fd} \end{bmatrix} \tag{63}$$

where the constant inductance matrix $\boldsymbol{L}'''_{csf}$ is expressed as

$$\boldsymbol{L}'''_{csf} = \boldsymbol{L}_{lcsf} + \frac{2}{3} L'''_{mddu} \begin{bmatrix} 1 & -0.5 & -0.5 & 0 \\ -0.5 & 1 & -0.5 & 0 \\ -0.5 & -0.5 & 1 & 0 \\ 0 & 0 & 0 & 1.5 \end{bmatrix} \tag{64}$$

and $\boldsymbol{e}''''_{abcs}$ is computed similarly to (51). If desired, (63) can be decoupled by introducing a zero-sequence branch to the stator [22], resulting in

$$\begin{bmatrix} \boldsymbol{v}_{abcs} \\ v_{fd} \end{bmatrix} = \boldsymbol{R}_c \begin{bmatrix} \boldsymbol{i}_{abcs} \\ i_{fd} \end{bmatrix} + \boldsymbol{L}_D p \begin{bmatrix} \boldsymbol{i}_{abcs} \\ i_{fd} \end{bmatrix} + \boldsymbol{L}_0 p \begin{bmatrix} \boldsymbol{i}_{ng} \\ 0 \end{bmatrix} + \begin{bmatrix} \boldsymbol{e}''''_{abcs} \\ e''''_{fd} \end{bmatrix} \tag{65}$$

where

$$\boldsymbol{i}_{ng} = \left[i_{ng} \quad i_{ng} \quad i_{ng} \right]^{\mathrm{T}} \text{ and } i_{ng} = i_{as} + i_{bs} + i_{cs} \tag{66}$$

$$\begin{aligned} \boldsymbol{L}_D &= \mathrm{diag}\left[L_{Ds} \quad L_{Ds} \quad L_{Ds} \quad L_{Df} \right] \text{ and } \boldsymbol{L}_0 \\ &= \mathrm{diag}\left[L_0 \quad L_0 \quad L_0 \quad 0 \right] \end{aligned} \tag{67}$$

$$L_{Ds} = L_{ls} + L'''_{mddu}, \; L_{Df} = L_{lfd} + L'''_{mddu}, \text{ and } L_0 = -\frac{L'''_{mddu}}{3}. \tag{68}$$

The state equations of the rotor subsystem are still given by (22), (28), and (29).

4.2 Numerical approximation

The modified subtransient voltages (61) contain current time derivatives, resulting in an improper state model. To form a proper state model, it is possible to approximate them using high-pass filters [9, 14]. A more accurate option is to eliminate the time derivatives algebraically [9, 17]. This is done by solving for pi_{qs}, pi_{ds}, and pi_{fd} in (41), yielding

$$p\begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{fd} \end{bmatrix} = \boldsymbol{\Gamma}'''_{qdsf}\left(\begin{bmatrix} v_{qs} \\ v_{ds} \\ v_{fd} \end{bmatrix} - \begin{bmatrix} r_s & \omega_r L_{ls} & 0 \\ -\omega_r L_{ls} & r_s & 0 \\ 0 & 0 & r_{fd} \end{bmatrix}\begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{fd} \end{bmatrix} - \omega_r \begin{bmatrix} \lambda_{md} \\ -\lambda_{mq} \\ 0 \end{bmatrix} - \begin{bmatrix} L'''_{mqq} & L'''_{mqd} \\ L'''_{mqd} & L'''_{mdd} \\ L'''_{mqd} & L'''_{mdd} \end{bmatrix}\begin{bmatrix} \sum_{j=1}^{M} \frac{p\lambda_{kqj}}{L_{lkqj}} \\ \sum_{j=1}^{N} \frac{p\lambda_{kdj}}{L_{lkdj}} \end{bmatrix} \right) \tag{69}$$

where the symmetric matrix $\boldsymbol{\Gamma}'''_{qdsf}$ is given by

$$\boldsymbol{\Gamma}'''_{qdsf} \equiv \begin{bmatrix} \Gamma_{11} & \Gamma_{12} & \Gamma_{13} \\ \Gamma_{21} & \Gamma_{22} & \Gamma_{23} \\ \Gamma_{31} & \Gamma_{32} & \Gamma_{33} \end{bmatrix} = \begin{bmatrix} L'''_{qq} & L'''_{mqd} & L'''_{mqd} \\ L'''_{mqd} & L'''_{dd} & L'''_{mdd} \\ L'''_{mqd} & L'''_{mdd} & L'''_{fdd} \end{bmatrix}^{-1}. \tag{70}$$

Substituting (69) into (61), it is observed that the modified subtransient voltages e''''_{qs}, e''''_{ds}, and e''''_{fd} (which are outputs of the rotor state-space subsystem) are now function of the terminal voltages. This can result in algebraic loops when the terminal voltages are neither states nor inputs, for example, when the machine windings are connected to inductive elements [9, 14]. Algebraic loops are undesirable as they cause a sharp decrease in numerical robustness and a notable increase in computational cost [9, 14]. Certain commercial programs are also unable to solve algebraic loops [2].

Using an approach similar to [9, 17], algebraic loops can be prevented by approximating the terminal voltages found in (61) and (69) using first-order low-pass filters of the form

$$H_j(s) = \frac{p_{0j}}{s + p_{0j}} \tag{71}$$

Where $-p_{0j}$ is the filter pole and $j = \{qs, ds, fd\}$, which correspond to the filters of v_{qs}, v_{ds}, and v_{fd}, respectively. A block diagram of the

$$\begin{bmatrix} e''''_{qs} \\ e''''_{ds} \\ e''''_{fd} \end{bmatrix} = \omega_r \left(\begin{bmatrix} \lambda_{md} \\ -\lambda_{mq} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & -L'''_{mddu} \\ L'''_{mddu} & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} i_{qs} \\ i_{ds} \end{bmatrix} \right) + \begin{bmatrix} L'''_{mqq} & L'''_{mqd} \\ L'''_{mqd} & L'''_{mdd} \\ L'''_{mqd} & L'''_{mdd} \end{bmatrix}\begin{bmatrix} \sum_{j=1}^{M} \frac{p\lambda_{kqj}}{L_{lkqj}} \\ \sum_{j=1}^{N} \frac{p\lambda_{kdj}}{L_{lkdj}} \end{bmatrix} + \begin{bmatrix} L'''_{mqq} - L'''_{mddu} & L'''_{mqd} & L'''_{mqd} \\ L'''_{mqd} & \Delta L'''_{mdd} & L'''_{mdd} \\ L'''_{mqd} & L'''_{mdd} & \Delta L'''_{mdd} \end{bmatrix} p \begin{bmatrix} i_{qs} \\ i_{ds} \\ i_{fd} \end{bmatrix} \tag{61}$$

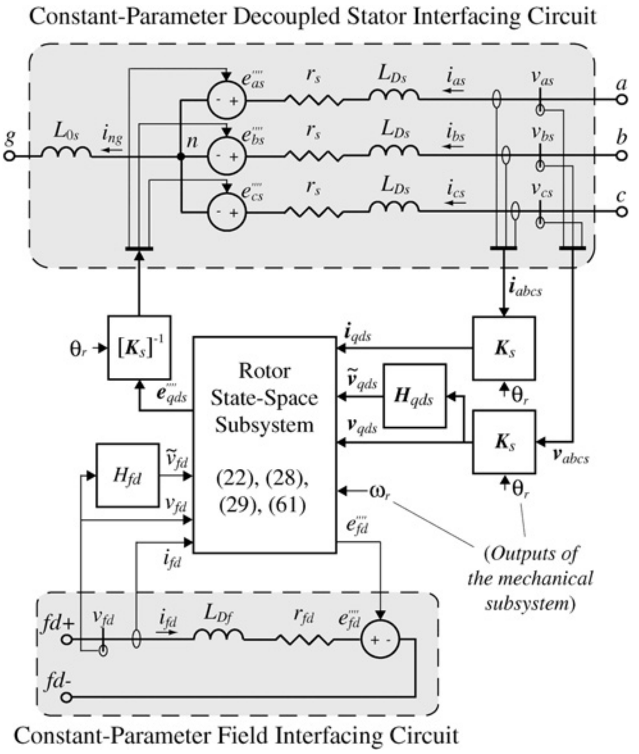

Fig. 1 *Block diagram of the proposed saturable CP-VBR model*

CP-VBR model is shown in Fig. 1. Therein, the tilde (~) symbol indicates filtered values.

5 Pole selection procedure

The low-pass filters (71) introduce numerical error, which can be alleviated by selecting large values of p_{0j}. However, this will increase stiffness, and possibly force the use of smaller integration step sizes to ensure numerical stability and/or sufficient accuracy. A procedure to select the smallest values of p_{0j} which achieve a given accuracy up to a defined frequency is presented here. This approach with minor changes can also be used for the unsaturated stator-and-field-circuit CP-VBR model previously proposed in [17].

One of the unsaturated constant-parameter VBR formulation presented in [9] requires approximating v_{qs} using a low-pass filter. A procedure to select the value of its pole based on the comparison of the original and approximate q-axis operational inductance $L_q(s)$ was recently proposed in [23]. The q-axis operational inductance, defined as the ratio of λ_{qs} over i_{qs}, is related to stator currents and voltages by [1]

$$Z_q(s) \equiv r_s + sL_q(s) = \frac{v_{qs}}{i_{qs}} \tag{72}$$

wherein Heaviside's operator p is replaced by Laplace's operator s to follow standard convention [1]. The approach presented in [23] is extended in here to choose not only the value of p_{0qs} but also those of p_{0ds} and p_{0fd}.

Unlike the model presented in [9] and analysed in [23], the CP-VBR model considered in this paper includes saturation. Consequently, the operational inductances will be saturation-dependent. Moreover, it can be seen from (57), (61), and (69) that there is now coupling between the q and d axes due to the cross-saturation term L'''_{mqd}.

However, as explained in [24], dynamic cross-saturation can be neglected while only incurring a small loss in numerical accuracy (for this specific model). Consequently, for the purpose of selecting the filter poles, the inductance L'''_{mqd} is set to 0, thereby effectively decoupling the q and d axes. Moreover, at higher frequencies, the operational inductances converge towards the subtransient inductances, which depend only lightly on saturation [22]. Because the proposed model contains low-pass filters, only the higher frequencies will be distorted. Therefore, saturation has a limited effect on the CP-VBR model's numerical accuracy, and the pole selection procedure can be executed only once (at any level of saturation). This assumption will be verified with an example in Section 5.3.

5.1 q-axis filter

The original q-axis operational inductance $L_{qo}(s)$ can be obtained as follows. First, consider an unsaturated machine and set $\omega_r = 0$. Then, insert (47) into (41) (considering only the q-axis), and replace all other variables by q-axis stator voltages and currents using (1), (7), and (28). Finally, rearrange the resulting equation in the form of (72), yielding

$$L_{qo}(s) = \frac{L'''_{qq}\alpha_1(s) - L_{ls}\alpha_2(s)}{\alpha_1(s) - \alpha_2(s)}. \tag{73}$$

Under the same assumptions, the operational inductance of the CP-VBR model $L_{qc}(s)$ is found by substituting (69) into (61) (using the filtered value of v_{qs}), and inserting the resulting equation into (57), which after elimination of internal variables gives

$$L_{qc}(s) = \frac{\left(s + p_{0qs}\right)\left(L_{Ds}\alpha_1(s) - \beta_1 L_{ls}\alpha_2(s)\right)}{\left(s + p_{0qs}\right)\left(\alpha_1(s) - \beta_1\alpha_2(s)\right) - p_{0qs}\beta_2\alpha_1(s)}. \tag{74}$$

The functions $\alpha_1(s)$ and $\alpha_2(s)$ along with the coefficients β_1 and β_2 are given in the Appendix (Section 9.1).

The smallest value of p_{0qs} yielding a relative error smaller than ε up to a fit frequency $f_{\rm fit}$ (in terms of the magnitude of the q-axis operational inductance) can be found using the iterative procedure outlined in [23]. A flow chart describing the procedure is also presented in Fig. 2*a*.

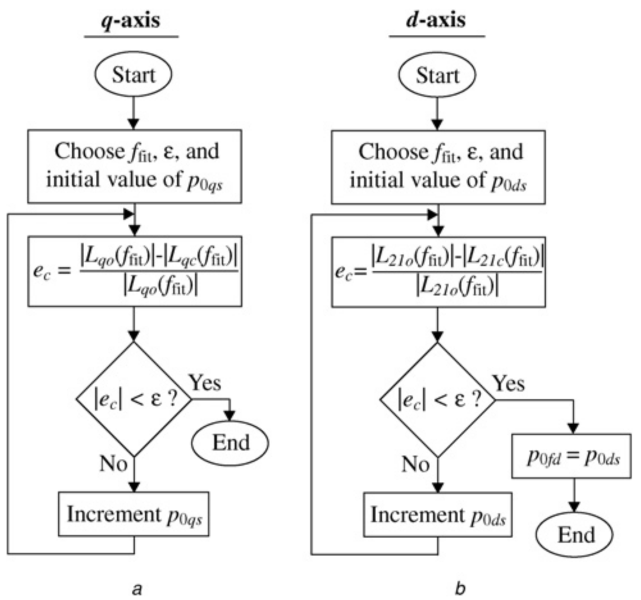

Fig. 2 *Flow chart of the pole selection procedure*

a q-axis filter's p_{0qs}
b d-axis filters' p_{0ds} and p_{0fd}

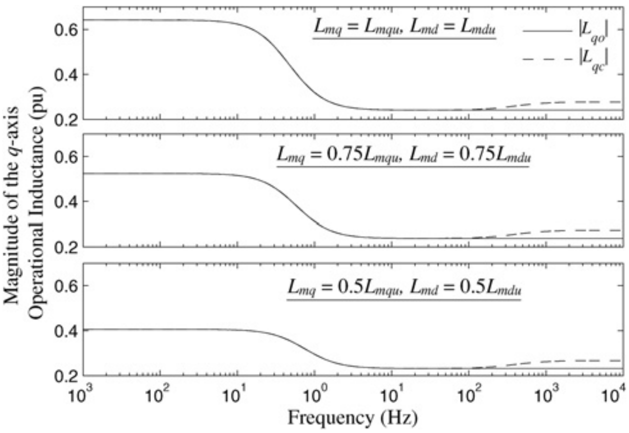

Fig. 3 *Magnitude of the original and approximate q-axis operational inductances L_{qo} and L_{qc}. From top to bottom: determined with $L_{mq} = L_{mqu}$ and $L_{md} = L_{mdu}$ (no saturation); $L_{mq} = 0.75L_{mqu}$ and $L_{md} = 0.75L_{mdu}$ (medium saturation); and $L_{mq} = 0.5L_{mqu}$ and $L_{md} = 0.5L_{mdu}$ (high saturation)*

5.2 d-axis filters

The machine's d-axis can be described by a two-port network whose terminals are connected to the stator and field windings [25]. From circuit theory, it is known that the input-output characteristic of a two-port network can be represented using equivalent network parameters, for example, impedance, admittance, or hybrid parameters. Admittance parameters, that is

$$\begin{bmatrix} i_{ds} \\ i_{fd} \end{bmatrix} = \boldsymbol{Y}_d(s)\begin{bmatrix} v_{ds} \\ v_{fd} \end{bmatrix} \equiv \begin{bmatrix} Y_{11}(s) & Y_{12}(s) \\ Y_{21}(s) & Y_{22}(s) \end{bmatrix}\begin{bmatrix} v_{ds} \\ v_{fd} \end{bmatrix} \tag{75}$$

are sometimes referred to as short-circuit parameters since they are determined by shorting one pair of terminals. Specifically, $Y_{11}(s)$ and $Y_{21}(s)$ are obtained with $v_{fd} = 0$, and $Y_{12}(s)$ and $Y_{22}(s)$ are found with $v_{ds} = 0$. It is therefore convenient to work with admittance parameters in this analysis since the effect of p_{0ds} and p_{0fd} is decoupled.

The original d-axis admittance parameters of the machine can be obtained similarly to the q-axis operational inductance. Using the same assumptions (i.e. neglecting saturation and setting ω_r to 0), $\boldsymbol{Y}_{do}(s)$ is found by inserting (47) into (41) (this time only considering the d-axis) and eliminating internal variables using (2), (6), (8), (12), and (29), yielding

$$\boldsymbol{Y}_{do}(s) = \left[\alpha_3(s)\boldsymbol{\alpha}_5(s) - \alpha_4(s)\boldsymbol{\alpha}_6(s)\right]^{-1}\left[\left(\alpha_3(s) - \alpha_4(s)\right)\boldsymbol{I}_2\right] \tag{76}$$

where $\boldsymbol{I}_2$ is the 2×2 identity matrix. Similarly, to compute the admittance matrix $\boldsymbol{Y}_{dc}(s)$ of the CP-VBR model, (69) is inserted into (61) (using the filtered values of v_{ds} and v_{fd}), and the resulting equation is substituted into (57). After elimination of the internal variables, $\boldsymbol{Y}_{dc}(s)$ can be formulated as (see (77))

The functions and coefficients appearing in (76) and (77) are given in the Appendix (Section 9.1).

Here, four TFs can be considered to select p_{0ds} and p_{0fd} (two each). An option is to apply the method used in Section 5.1 to all four TFs, and subsequently select the most constraining results. However, this proves unnecessary as explained below and validated in Section 6.

It can be seen that at higher frequencies, where the d-axis circuit is predominantly inductive, the order of magnitude of the four TFs is similar. Since v_{fd} is typically a few orders of magnitude smaller than the stator voltage, it can be concluded that its high-frequency content – which is distorted by H_{fd} – has a marginal effect on i_{ds} and i_{fd}. Consequently, there is no need to determine p_{0fd} meticulously. Moreover, $Y_{11}(s)$ and $Y_{21}(s)$ are both function of p_{0ds} (but not of p_{0fd}). Our experiments with several machines have shown that for any given frequency, the relative error of $Y_{21}(s)$ is usually greater than that of $Y_{11}(s)$. As a result, a larger value of p_{0ds} (i.e. more constraining) is obtained by applying the approach of Section 5.1 to $Y_{21}(s)$ instead of $Y_{11}(s)$.

The proposed approach to determine p_{0ds} and p_{0fd} is presented in Fig. 2*b*. The value of p_{0ds} is obtained by comparing the magnitude of $L_{21\text{o}}(s)$ and $L_{21\text{c}}(s)$, which correspond, respectively, to the original and approximate short-circuit inductances. The short-circuit inductance $L_{21j}(s)$ is related to $Y_{21j}(s)$ by

$$L_{21j}(s) = \frac{1}{sY_{21j}(s)} \tag{78}$$

where $j = \{o, c\}$. Finally, p_{0fd} is assumed equal to p_{0ds}, which is a conservative assumption but does not worsen stiffness.

The procedure outlined in Fig. 2 can be easily implemented in transient simulators, wherein users would only be required to enter the fit frequency f_{fit} and the error tolerance ε.

5.3 Example

Since saturable synchronous machine models are represented by a set of non-linear time-varying ordinary differential equations, it would be very difficult to analytically define a precise error bound. We therefore opted to assess the numerical accuracy of the proposed model (which uses approximating filters) in the frequency domain. Consequently, comparing the frequency responses of the proposed model with the original ones (produced by the reference model) gives a good indication of the numerical error/distortion introduced by the proposed model.

To better illustrate the effect of the low-pass filters on the machine frequency response, the procedure described in the flow chart of Fig. 2 is executed for a 120-MVA synchronous generator (SG) [26] (see Section 9.2 for the machine parameters). For conciseness, only the q-axis is considered in this example; however, the observations extend naturally to the d-axis.

Setting f_{fit} to 100 Hz and ε to 1%, the procedure of Fig. 2*a* yields $p_{0qs} = 2240$ for the subject machine. The magnitude of the original and approximate operational inductances $L_{qo}(s)$ and $L_{qc}(s)$ are plotted in the upper box of Fig. 3, which validates that the proposed approach introduces little error up to around the user-selected fit frequency f_{fit}. Larger values of f_{fit} can be used to increase the accuracy range if needed, at the expense of an increase in numerical stiffness and thus a decrease in numerical efficiency.

As mentioned at the beginning of Section 5, the procedure of Fig. 2 is executed using unsaturated machine parameters. To verify the soundness of this approach, the pole selection procedure is repeated twice, this time assuming magnetising inductances L_{mq} and L_{md} that are, respectively, equal to 75 and 50% of their unsaturated values L_{mqu} and L_{mdu}. Using the same criteria ($f_{\text{fit}} =$ 100 Hz and $\varepsilon = 1\%$), the proposed approach yields $p_{0qs} = 2230$ and 2210, respectively. Consequently, the level of saturation has a negligible impact on the outcome of the pole selection procedure. This can be better visualised in the middle and lower boxes of Fig. 3, which present the original and approximate operational inductances $L_{qo}(s)$ and $L_{qc}(s)$ for the subject cases. As seen in Fig. 3, the lower-frequency content varies greatly with the level of saturation, but is otherwise approximated with virtually no error by the proposed approach at any level of saturation. At the same time, Fig. 3 shows that the higher-frequency content is almost independent of the degree of saturation. Consequently, the proposed pole selection procedure is valid for any practical level of saturation.

$$\boldsymbol{Y}_{dc}(s) = \left[\alpha_3(s)\left(\boldsymbol{\alpha}_7(s) - \boldsymbol{\beta}_5\right) - \alpha_4(s)\boldsymbol{\beta}_4\boldsymbol{\alpha}_6(s)\right]^{-1}\left[\alpha_3(s)\left(\boldsymbol{I}_2 - \boldsymbol{\beta}_3\boldsymbol{F}(s)\right) - \alpha_4(s)\boldsymbol{\beta}_4\right]. \tag{77}$$

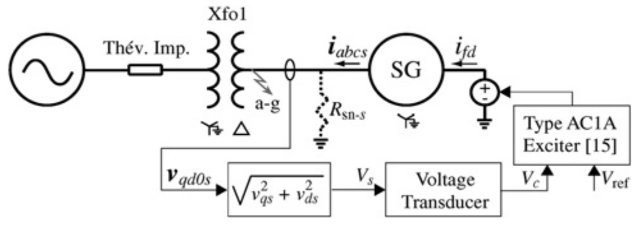

Fig. 4 *One-line-diagram of the test system with the TF-based exciter model*

6 Computer studies

The CP-VBR (see Section 4), VP-VBR (see Section 3), and qd (see [20]) models have been implemented in MATLAB/Simulink using the PLECS toolbox [2]. The CC-PD model is not considered due to space constraints; regardless, its numerical performances have been shown to be similar to those of the stator-and-field-circuit VP-VBR model in modern state-variable-based packages [17]. The efficient qd model [20] is implemented using state-space equations, that is, without circuit elements. Consequently, it cannot be interfaced directly with some exciter circuits. Single-machine studies are considered for clarity and conciseness; nevertheless, the proposed CP-VBR model can also be used with multi-machine systems. A per unit representation is used in all models and studies. All parameters are given on the generator's base power. As with the rotor circuit, the exciter parameters are referred to the machine's stator, resulting in very small voltages. For the sole purpose of visualisation, the field voltage (or exciter output voltage) will be re-scaled using $e_{xfd} = (X_{md}/r_{fd})v_{fd}$ [1].

Unless otherwise stated, the subject models are simulated using Simulink's explicit ode45 solver. The maximum and minimum step sizes are 1 ms and 0.1 μs, respectively, and the relative and absolute error tolerances are 1×10^{-4}. Reference solution trajectories are obtained using the algebraically exact VP-VBR model with the same solver but with stricter settings: maximum and minimum step sizes of 1 μs and 10 ns, respectively, and relative and absolute tolerances of 1×10^{-5}. Consequently, this reference solution will contain negligible numerical error. Experimental measurements are not considered since only the numerical accuracy of the subject models is of interest in this paper. All simulations are executed on a PC with Windows XP and a 2.83-GHz Intel CPU.

6.1 TF-based exciter model

In the first study, a 120-MVA wye-grounded SG [26] is connected through a step-up transformer (Xfo1) to a Thévenin representation of a transmission system as shown in Fig. 4. The machine and network parameters are summarised in the Appendix (Sections 9.2 and 9.3, respectively). The source voltage is set to 1 pu.

The SG is fed by an ac excitation system with a diode bridge rectifier [15, 27]. The exciter is represented using the TF-based IEEE AC1A model [15, Sections 4 and 6.1]. The sample parameters presented in [15, Annex H.5] are used, with the exception of the over-excitation and under-excitation limiters which are omitted. The reference voltage V_{ref} (controlling the voltage at the stator terminals) is set to 1.05 pu.

As explained in [4], resistive snubbers are added to the stator terminals ($R_{sn\text{-}s}$) of the qd model to achieve a compatible machine-network interface. To highlight the resulting compromise between numerical accuracy and efficiency, two snubbers are considered: $R_{sn\text{-}s} = 10$ pu and $R_{sn\text{-}s} = 1$ pu. The resulting formulations are referred to as $qd1$ and $qd2$, respectively. Since the TF-based exciter model outputs a voltage, no snubber is required at the field terminals of the machine. The filter poles of the CP-VBR model are found using the procedure described in Section 5 with $f_{fit} = 100$ Hz and $\varepsilon = 1\%$. They are $p_{0qs} = 2240$ and $p_{0ds} = p_{0fd} = 4900$.

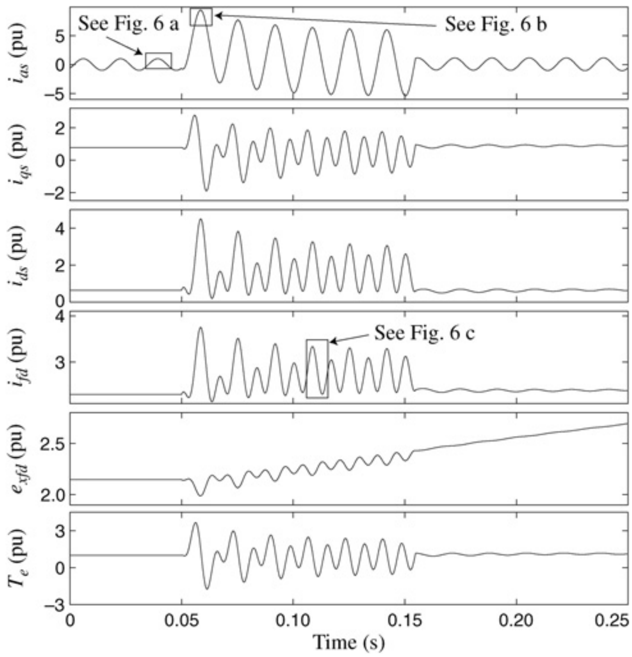

Fig. 5 *Reference trajectories for a single-phase-to-ground fault using a TF-based exciter model. From top to bottom: stator currents i_{as}, i_{qs}, and i_{ds}, field current i_{fd}, field voltage e_{xfd}, and electromagnetic torque T_e*

In this case study, the SG initially operates in steady state with nominal torque. At $t = 0.05$ s, a single-phase-to-ground fault occurs on phase a of the stator terminals. A small fault resistance of 0.01 pu is considered. It is assumed that the fault is cleared after 100 ms.

The reference trajectories of the stator currents i_{as}, i_{qs}, and i_{ds}, the field current i_{fd}, the field voltage e_{xfd}, and the electromagnetic torque T_e are presented in Fig. 5. The peak value of i_{as} (which contains a dc offset [1]) is more than 9 times larger than its nominal value. Since a TF-based exciter model is used (i.e. the switching harmonics are neglected), Fig. 5 is devoid of high frequencies.

Magnified views of the stator current i_{as} as predicted by the subject models before and during the fault are reproduced in Figs. 6*a* and *b*, respectively. The $qd2$ model is shown to predict the phase and magnitude of i_{as} with notable errors which are caused by the presence of a small snubber ($R_{sn\text{-}s} = 1$ pu). All other models calculate i_{as} with much better accuracy. In particular, it is seen in Fig. 6*a* that the CP-VBR model introduces no error in steady state since the filters do not distort the low-frequency content. This is unlike in Fig. 6*b*, where the low-pass filters create some numerical error during the fault condition. An enlarged view of the field current i_{fd} during the fault condition is also presented in Fig. 6*c*. The CP-VBR model again introduces some numerical error, which could be alleviated if necessary by increasing the magnitude of the filter poles. The $qd2$ model also introduces some error in i_{fd}.

To evaluate the models' accuracy, the 2-norm relative errors [14] of $\boldsymbol{i}_{abcs}$, i_{fd}, and T_e for $t = 0$ to 0.25 s are presented in Table 1. The 2-norm error of $\boldsymbol{i}_{abcs}$ refers to the mean of the errors of i_{as}, i_{bs}, and i_{cs}. The VP-VBR model is virtually errorless since it has no approximations and a higher-order solver with small error tolerances is used. The 2-norm errors of the proposed saturable CP-VBR model vary between 1 and 1.5% (e.g. 1.031% for $\boldsymbol{i}_{abcs}$), which is adequate for most studies. These results also support the validity of the pole selection procedure. The numerical accuracy of the $qd1$ model is similar to that of CP-VBR. Finally, as seen in Fig. 6, the 2-norm errors of $qd2$ are very large, for example, 14.52% for T_e.

The number of integration time steps and the CPU time taken by each model for this case study (simulated from $t = 0$ to 3 s) are summarised in Table 2. The VP-VBR model takes the fewest number of steps (3034). However, due to its variable-parameter circuit, it is more than 2.5 times slower than the CP-VBR model

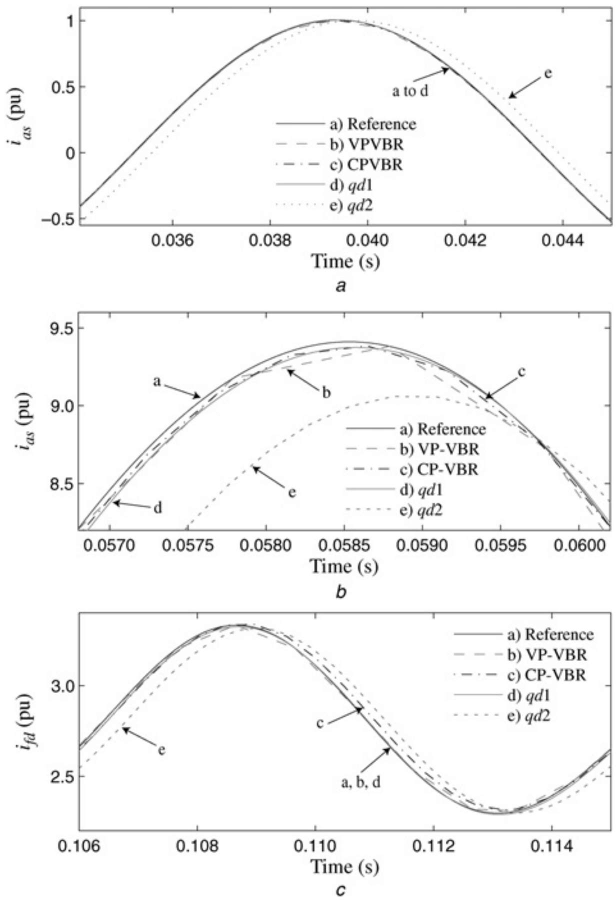

Fig. 6 *Magnified views of machine currents before and during a single-phase-to-ground fault using a TF-based exciter model*

a Phase a stator current i_{as} (before the fault)
b Phase a stator current i_{as} (during the fault)
c Field current i_{fd} (during the fault)

(3.46 s vs. 1.32 s, respectively). Table 2 demonstrates that the good accuracy of the *qd*1 model (see Table 1) is achieved at the expense of a high computational cost (8.35 s with ode45). Its CPU time can be decreased if a stiffly-stable solver (e.g. ode15s) is used; however, it remains slower than both VBR models. Finally, the *qd*2 model, which has poor numerical accuracy, is nevertheless slower than the proposed CP-VBR model.

To further assess the effect of the low-pass filters, the study is re-executed twice with the CP-VBR model using different filter parameters: (i) $p_{0qs} = 1170$ and $p_{0ds} = p_{0fd} = 2710$; and (ii) $p_{0qs} = 3310$ and $p_{0ds} = p_{0fd} = 7110$. These values were obtained by running the pole selection procedure of Section 5 with $f_{\text{fit}} = 50$ and 150 Hz, respectively (the error tolerance ε is set to 1% in both cases). The resulting 2-norm errors, number of time steps, and CPU time for these two cases are compiled in Table 3 along with those from the original case study (i.e. with $f_{\text{fit}} = 100$ Hz). As predicted, increasing the magnitude of the poles improves numerical accuracy; however, it comes at the expense of a decrease in numerical efficiency. Table 3 therefore clearly demonstrates the trade-off involved with the proposed model.

Table 1 2-norm relative error (%) of $\boldsymbol{i}_{abcs}$, i_{fd}, and T_e for a single-phase-to-ground fault with a TF-based exciter model

Model	$\boldsymbol{i}_{abcs}$	i_{fd}	T_e
VP-VBR	0.001	0.001	0.001
CP-VBR	1.031	1.227	1.435
qd1 ($R_{sn\text{-}s} = 10$ pu)	1.923	0.371	1.388
qd2 ($R_{sn\text{-}s} = 1$ pu)	20.27	2.391	14.52

Table 2 Numerical performance of the subject models for a single-phase-to-ground fault with a detailed exciter model

Model	Solver	No. time steps	CPU time, s
VP-VBR	ode45	3034	3.46
CP-VBR	ode45	4960	1.32
*qd*1 ($R_{sn\text{-}s} = 10$ pu)	ode45	45,092	8.35
	ode15s	9187	3.82
*qd*2 ($R_{sn\text{-}s} = 1$ pu)	ode45	7756	1.55
	ode15s	4599	1.55

Table 3 Numerical accuracy and efficiency of the proposed CP-VBR model for different filter parameters

f_{fit}, Hz	2-norm error, %			No. time steps	CPU time, s
	$\boldsymbol{i}_{abcs}$	i_{fd}	T_e		
50	1.765	1.992	2.585	3094	0.93
100	1.031	1.227	1.435	4960	1.32
150	0.715	0.874	0.966	6939	2.01

6.2 Detailed exciter model

In this section, the TF-based ac exciter is replaced with a detailed circuit-based potential-source static excitation system [27]. The new system is shown in Fig. 7. The simplified control system consists of a voltage transducer, a PI controller, a slew rate limiter, and SimPowerSystems' built-in six-pulse generator [3]. Its parameters are written in the Appendix (Section 9.4).

The *qd* models now require snubbers at the stator and field terminals. For *qd*1, the field snubber $R_{\text{sn-f}}$ is set to $100r_{fd}$; for *qd*2, we have $R_{\text{sn-f}} = 10r_{fd}$. The VBR models do not require snubbers at either set of terminals.

The fault study from Section 6.1 is repeated with the new system/exciter model. The reference trajectories of the field current i_{fd}, field voltage e_{xfd}, and ac side exciter currents $\boldsymbol{i}_{abc\text{ex}}$ are presented in Fig. 8. The trajectories of the stator currents and electromagnetic torque are similar to those of Fig. 5 and are not presented here due to space constraints. As seen in Fig. 8, the field voltage is highly oscillatory; however, current i_{fd} still contains only lower frequencies. This demonstrates the low impact of the higher frequency content of v_{fd}.

An enlarged view of i_{fd} is reproduced in Fig. 9*a*. This time, the *qd*1 model exhibits a small but notable error (similarly to the CP-VBR model), which is a result of its field snubber. The *qd*2 model is considerably worse. Fig. 9*b* contains a magnified view of the phase *b* exciter current $i_{b\text{ex}}$. Both VBR models predict $i_{b\text{ex}}$ very accurately. This is not the case of the *qd* models, which exhibit notable phase shifts.

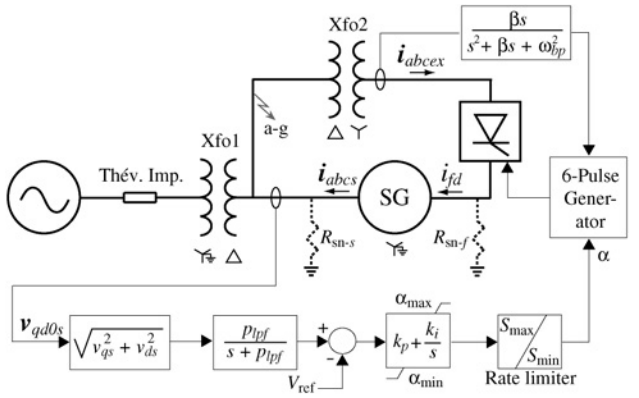

Fig. 7 *One-line-diagram of the test system with a circuit-based exciter model*

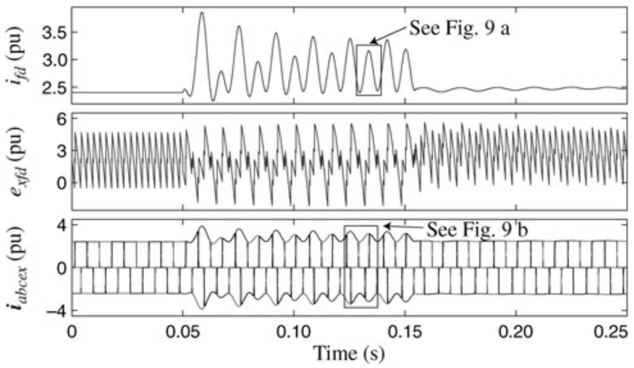

Fig. 8 *Reference trajectories for a single-phase-to-ground fault using a detailed exciter model. From top to bottom: field current i_{fd}, field voltage e_{xfd}, and ac exciter currents $\boldsymbol{i}_{abcex}$*

The 2-norm relative errors for this case study (for $t = 0$ to 0.25 s) are summarised in Table 4. These results confirm the observations made in Fig. 9. In particular, the numerical errors of the CP-VBR model are very similar for both case studies (see Tables 1 and 4). Conversely, the 2-norm error of i_{fd} as predicted by $qd1$ is considerably worse in the second case study (1.645%) than in the first (0.371%).

To validate the lesser impact of p_{0fd} on numerical accuracy, this case study has been repeated with the CP-VBR model using a reduced value of p_{0fd} (490 instead of 4900). The 2-norm errors of $\boldsymbol{i}_{abcs}$, i_{fd}, and T_e are 1.035, 1.151, and 1.532%, respectively, which verifies our assumption.

Finally, Table 5 contains the number of integration steps and the CPU time of the subject models for $t = 0$ to 3 s. Unlike in the first case study, the two VBR models use a similar number of steps (around 21,000), showing that the higher stiffness of the CP-VBR model has no impact with this system. As a result, the CP-VBR model is almost 10 times faster than the VP-VBR model (6.97 and 69 s, respectively), which is a substantial improvement. The CP-VBR model is also almost twice as fast as the $qd1$ model (with ode15s), while introducing less error (see Table 4). Moreover, the CP-VBR model is faster than the less accurate $qd2$ model (which requires more than 9 s). Consequently, the proposed CP-VBR model clearly exhibits the best combination of numerical accuracy and efficiency.

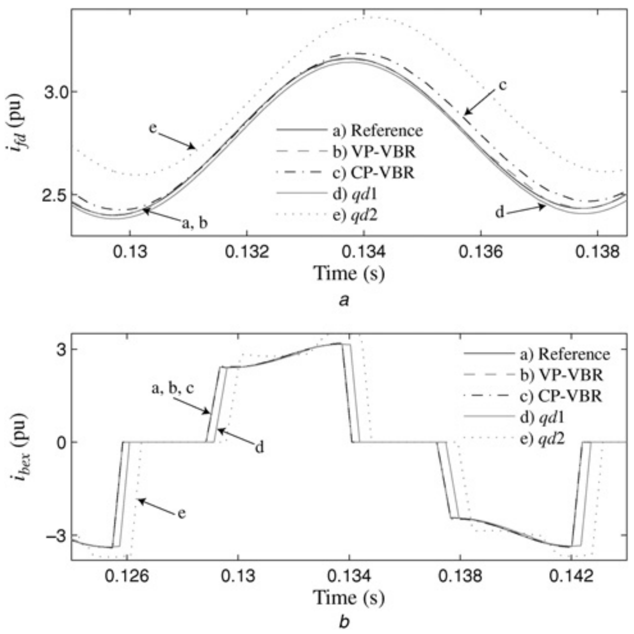

Fig. 9 *Magnified views of currents during a single-phase-to-ground fault using a detailed exciter model*

a Field current i_{fd}
b Phase b exciter current i_{bex}

Table 4 2-norm relative error (%) of $\boldsymbol{i}_{abcs}$, i_{fd}, and T_e for a single-phase-to-ground fault with a detailed exciter model

Model	$\boldsymbol{i}_{abcs}$	i_{fd}	T_e
VP-VBR	0.001	0.006	0.007
CP-VBR	1.040	1.132	1.531
$qd1$ ($R_{sn\text{-}s}$ = 10 pu, $R_{sn\text{-}f} = 100r_{fd}$)	1.913	1.645	1.635
$qd2$ ($R_{sn\text{-}s}$ = 1 pu, $R_{sn\text{-}f} = 10r_{fd}$)	21.60	7.012	16.74

Table 5 Numerical performance of the subject models for a single-phase-to-ground fault with a TF-based exciter model

Model	Solver	No. time steps	CPU time, s
VP-VBR	ode45	21,111	69.0
CP-VBR	ode45	21,198	6.97
$qd1$ ($R_{sn\text{-}s}$ = 10 pu, $R_{sn\text{-}f} = 100r_{fd}$)	ode45	177,193	42.6
	ode15s	45,271	13.8
$qd2$ ($R_{sn\text{-}s}$ = 1 pu, $R_{sn\text{-}f} = 10r_{fd}$)	ode45	37,736	9.52
	ode15s	43,390	9.35

7 Conclusion

A new state-space synchronous machine model representing the stator and field windings using a constant-parameter VBR formulation and incorporating main flux saturation is presented in this paper. The proposed model can therefore be interfaced directly with arbitrary power networks and exciter circuits. This numerically efficient model is achieved through the use of low-pass filters. An approach to select the filter poles to achieve a specified level of numerical accuracy is also presented. The proposed model is compared with other state-of-the-art models in two case studies. The new VBR model is shown to offer the best combination of numerical efficiency and accuracy among all subject models. It is envisioned that the proposed model will be included into the built-in libraries of many industry-grade transient simulators, thereby offering considerable computational savings to power system analysts worldwide.

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9 Appendices

9.1 Pole selection functions and coefficients

The functions $\alpha_1(s)$ and $\alpha_2(s)$ and the coefficients β_1 and β_2 introduced in Section 5.1 are expressed as

$$\alpha_1(s) = \prod_{j=1}^{M} \left(r_{kqj} + L_{lkqj} s \right) \tag{79}$$

$$\alpha_2(s) = L'''_{mqqu} \sum_{j=1}^{M} \left[\frac{r_{kqj}}{L_{lkqj}} \prod_{i=1, i \neq j}^{M} \left(r_{kqi} + L_{lkqi} s \right) \right] \tag{80}$$

$$\beta_1 = \frac{L'''_{ddu}}{L'''_{qqu}} \text{ and } \beta_2 = 1 - \beta_1. \tag{81}$$

The functions and coefficients used in Section 5.2 are defined as

$$\alpha_3(s) = \prod_{j=1}^{N} \left(r_{kdj} + L_{lkdj} s \right) \tag{82}$$

$$\alpha_4(s) = L'''_{mddu} \sum_{j=1}^{N} \left[\frac{r_{kdj}}{L_{lkdj}} \prod_{i=1, i \neq j}^{N} \left(r_{kdi} + L_{lkdi} s \right) \right] \tag{83}$$

$$\boldsymbol{\alpha}_5(s) = \begin{bmatrix} r_s + L'''_{ddu} s & L'''_{mddu} s \\ L'''_{mddu} s & r_{fd} + L'''_{fddu} s \end{bmatrix} \tag{84}$$

$$\boldsymbol{\alpha}_6(s) = \text{diag}\left[r_s + L_{ls} s \quad r_{fd} + L_{lfd} s \right] \tag{85}$$

$$\boldsymbol{\alpha}_7(s) = \text{diag}\left[r_s + L'''_{ddu} s \quad r_{fd} + L'''_{fddu} s \right] \tag{86}$$

$$\boldsymbol{\beta}_3 = \begin{bmatrix} 0 & L'''_{mddu} \\ L'''_{mddu} & 0 \end{bmatrix} \begin{bmatrix} \Gamma_{22} & \Gamma_{23} \\ \Gamma_{32} & \Gamma_{33} \end{bmatrix} \text{ and } \boldsymbol{\beta}_4 = \boldsymbol{I}_2 - \boldsymbol{\beta}_3 \tag{87}$$

$$\boldsymbol{\beta}_5 = \boldsymbol{\beta}_3 \text{diag}\left[r_s \quad r_{fd} \right] \text{ and } \boldsymbol{F}(s) = \text{diag}\left[H_{ds}(s) \quad H_{fd}(s) \right]. \tag{88}$$

The inductances L'''_{qqu}, L'''_{mqqu}, L'''_{ddu}, L'''_{fddu}, and L'''_{mddu} refer to the unsaturated values of L'''_{qq}, L'''_{mqq}, L'''_{dd}, L'''_{fdd}, and L'''_{mdd}, respectively.

9.2 120-MVA hydro generator parameters

Machine parameters [26]:

S_b = 120 MVA	V_{lls} = 13.8 kV	f = 60 hz	H = 3.42 s
X_{ls} = 0.17 pu	r_s = 0.0037 pu	X_{mq} = 0.472 pu	X_{md} = 0.936 pu
X_{lkq1} = 0.0835 pu	r_{kq1} = 0.0035 pu	X_{lkd1} = 0.122 pu	r_{kd1} = 0.0035 pu
X_{lfd} = 0.152 pu	r_{fd} = 0.0007 pu		

Open-circuit saturation characteristic (d-axis):

i_{md} (pu)	0.75	0.88	1.04	1.15	1.29	1.60	1.98
λ_{md} (pu)	0.70	0.80	0.90	0.95	1.00	1.10	1.20

9.3 Power system parameters (on the machine base)

Thévenin equivalent: $Z_0 = 0.00233 + j0.03$ pu, $Z_1 = 0.00233 + j0.02$ pu.
Step-up transformer Xfo1: $V_H = V_L = 1$ pu, $Z_0 = Z_1 = 0.005 + j0.12$ pu.

9.4 Detailed exciter model

Low-pass filter: $p_{lpf} = 100$.
PI controller: $k_p = k_i = 200$, $\alpha_{min} = 10$, $\alpha_{max} = 150$.
Rate limiter: $S_{min} = -100$, $S_{max} = 100$.
Excitation transformer Xfo2: $S_b = 1$ MVA, $V_H = 13.8$ kV, $V_L = 0.0025 V_H$, $Z_0 = Z_1 = 0.01 + j0.10$ pu.
Band-pass filter: $\beta = 30$, $\omega_{bp} = 377$ rad/s.

A Voltage-Behind-Reactance Synchronous Machine Model for the EMTP-Type Solution

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***Abstract*—A full-order, voltage-behind-reactance synchronous machine model has recently been proposed in the literature. This paper extends the voltage-behind-reactance formulation for the electromagnetic transient program (EMTP)-type solution, in which the rotor subsystem is expressed in qd coordinates and the stator subsystem is expressed in abc phase coordinates. The model interface with the nodal-analysis network solution is non-iterative and simultaneous. An example of a single-machine, infinite-bus system shows that the proposed model is more accurate and efficient than several existing EMTP machine models.**

***Index Terms*—Computational techniques, electromagnetic transient program (EMTP), phase-domain (PD) model, synchronous machine, voltage-behind-reactance (VBR) model.**

I. Introduction

THERE have been numerous models proposed to represent synchronous machines for power-system analysis. For transient stability studies, reduced-order machine models that neglect stator transients are commonly used [1]. However, for electromagnetic transient studies [2], full-order models are often used, due to their greater accuracy. Depending on the modeling languages, various machine models have been developed using the state-variable approach [3]–[5] (wherein the discretization is performed at the system level by the ODE solver) and the nodal-analysis approach [6], [7] (wherein the discretization is done at the component/branch level using a particular integration rule).

In this paper, synchronous machine models suitable for the electromagnetic transient program (EMTP) solution are investigated. The EMTP and its derivative programs are extensively used by industry and academia as powerful and standard simulation tools, wherein the classical full-order qd synchronous machine model is available.

In the existing EMTP-type software packages, there are three commonly used methods for interfacing the qd machine model with the external network. In the first approach, the machine is represented internally in qd coordinates, but it is interfaced with the external network using a Thevenin equivalent circuit in phase coordinates. The voltage sources of the Thevenin equivalent contain predicted machine electrical and mechanical variables, and the equivalent resistances are averaged to incorporate rotor saliency [8]. A prediction-correction scheme is used for the machine variables, whereas the network variables are solved without iterations. The synchronous machine models Type-50 in MicroTran [9] and Type-59 in DCG/EPRI EMTP [10] fall into this category.

The second approach is based on the compensation method, in which the external ac system is represented as a Thevenin equivalent circuit and is interfaced with the synchronous machine in qd axes. Linear extrapolation of the rotor speed is used at the beginning of each time step. To obtain the solution, iterations are applied to both machine's electrical and mechanical variables. The compensation method works well as long as the distributed-parameter transmission lines separate the machines. When this is not the case, artificial "stub lines" are sometimes used, complicating the network modeling and reducing its accuracy. The universal machine model in the ATP is implemented using this method [11].

In the third method, used in PSCAD/EMTDC [12], the machine model is interfaced with the network as a compensation current source and a special terminating impedance [13]. The machine model is represented as a Norton current source that is calculated using the terminal bus voltages from the previous time step. Therefore, a one-time-step delay exists in this type of machine model.

The key advantage of the qd model is that it uses a constant inductance matrix, making it numerically efficient. Also, the qd model structure makes it possible to incorporate the magnetic saturation in a number of convenient ways [2], [11]. However, the three methods of interfacing the traditional qd machine model with the EMTP network solution artificially reduce simulation efficiency by requiring a small time-step Δt to keep the interfacing error under certain tolerance. When larger time-steps are used, the accuracy of the simulation deteriorates. Moreover, the existing interfacing methods have been shown to cause numerical instability problems [14], [15], which is especially undesirable in real-time simulations [16].

As an alternative to the classical qd model, some researchers have turned to the original coupled-circuit machine model formulation, in which the model is expressed in physical variables and phase coordinates [17]–[21]. In the EMTP community, this approach is known as the phase-domain (PD) model and may provide more accurate representation of machine internal phenomena, such as internal machine faults [20]. Because the stator circuit is directly interfaced to the rest of the network, no predictions and/or iterations of electrical variables are needed and the numerical stability is improved. However, the existence of time-

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variant self and mutual inductances increases the computational burden of this model. Present implementations include Type-58 in ATP [21] (which requires re-triangulation at each time step [22], [23]) and the recently developed VTB model [24].

In order to improve simulation efficiency, a so-called voltage-behind-reactance (VBR) synchronous machine model was proposed in [25] for the state-variable approach. In [26], the VBR synchronous machine model was demonstrated in the hardware-in-the-loop application. This paper extends the VBR model formulation for the EMTP-type solution that is the basis of the real-time power systems simulator at the University of British Columbia [27]. The simulation studies presented provide CPU times and an error analysis that demonstrate the improvement of numerical efficiency and accuracy of the proposed VBR model over several commonly used models, including the PD model. The advantages of the proposed VBR model include the following.

1) Similar to the PD model, the stator circuit is expressed in phase coordinates using the physical currents as the independent variables and is directly interfaced with the external network. A simultaneous EMTP solution is thereby achieved.
2) The rotor equations are expressed in qd-rotor reference frame using flux linkages as the independent variables, which significantly reduces the computational burden as compared to the PD model. The VBR model is also shown to have improved numerical accuracy due to better scaled eigenvalues.
3) The model formulation is very flexible and can be readily extended to include an arbitrary number of electrical phases and/or damper windings.
4) Partitioning of the stator and rotor equations provides a natural decoupling of the time scales. Simulation speed may be further improved through the multi-rate integration [28] and latency techniques [29], which we will investigate in future studies.

II. Synchronous Machine Models

The dynamics of synchronous machines can be represented by equations of corresponding electrical and mechanical subsystems. Without loss of generality, this paper considers a three-phase synchronous machine with one field winding, fd, and one damper winding, kd, in the d-axis, and two damper windings, kq1 and kq2, in the q-axis. Motor convention is used for all models, wherein the q-axis of the rotor reference frame is assumed to be 90° leading the d-axis [30]. Throughout this paper, bold uppercase is used to denote matrices, and bold lowercase is used to denote vectors. Also, the operator $p = d/dt$. The equations for mechanical subsystems are assumed to be the same for all models considered here, specifically

$$p\theta_r = \omega_r \tag{1}$$
$$p\omega_r = \frac{P}{2J}(T_e - T_m). \tag{2}$$

Here, θ_r and ω_r are the rotor position and the angular electrical speed; T_m and T_e are the mechanical torque and electromagnetic torque, respectively. For the purpose of consistency and further discussion, the relevant machine models in their general form are briefly reviewed below.

A. Traditional qd Model

To obtain the qd model, the equations of the coupled-circuit machine model in physical variables are transformed to the rotor reference frame (Park's transformation). The resulting voltage equation, which includes the stator and rotor, may be compactly expressed as

$$\mathbf{v}_{qd0} = \mathbf{R}\mathbf{i}_{qd0} + p\boldsymbol{\lambda}_{qd0} + \mathbf{u} \tag{3}$$

where

$$\mathbf{v}_{qd0} = [\, v_{qs} \quad v_{ds} \quad v_{0s} \quad 0 \quad 0 \quad v_{fd} \quad 0]^T \tag{4}$$
$$\mathbf{i}_{qd0} = [\, i_{qs} \quad i_{ds} \quad i_{0s} \quad i_{kql} \quad i_{kq2} \quad i_{fd} \quad i_{kd}]^T \tag{5}$$
$$\mathbf{u} = [\, \omega_r\lambda_{ds} \quad -\omega_r\lambda_{qs} \quad 0 \quad 0 \quad 0 \quad 0 \quad 0]^T \tag{6}$$
$$\boldsymbol{\lambda}_{qd0} = [\, \lambda_{qs} \quad \lambda_{ds} \quad \lambda_{0s} \quad \lambda_{kq1} \quad \lambda_{kq2} \quad \lambda_{fd} \quad \lambda_{kd}]^T \tag{7}$$
$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_s & \\ & \mathbf{R}_r \end{bmatrix}. \tag{8}$$

Here, $\mathbf{v}_{qd0}$ and $\mathbf{i}_{qd0}$ are the vectors of voltages and currents, respectively; v_{fd} represents the field winding voltage; $\mathbf{u}$ includes the speed voltage terms; and $\mathbf{R}$ is a constant diagonal matrix containing the stator and rotor resistances. The flux linkage equations may be expressed as

$$\boldsymbol{\lambda}_{qd0} = \mathbf{L}_{qd0}\mathbf{i}_{qd0} \tag{9}$$

where

$$\mathbf{L}_{qd0} = \begin{bmatrix} \mathbf{L}_{qd0s} & \mathbf{L}_{qd0sr} \\ \mathbf{L}_{qd0rs} & \mathbf{L}_{qd0r} \end{bmatrix}. \tag{10}$$

Note that the inductance matrix, $\mathbf{L}_{qd0}$, does not depend on the rotor position. Finally, the electromagnetic torque is expressed as

$$T_e = \frac{3P}{4}(\lambda_{ds}i_{qs} - \lambda_{qs}i_{ds}). \tag{11}$$

B. Phase-Domain Model

The general form of the PD synchronous machine model is the coupled-circuit model, expressed in physical variables and coordinates. In particular, the voltage equation can be expressed as

$$\begin{bmatrix} \mathbf{v}_{abcs} \\ \mathbf{v}_{qdr} \end{bmatrix} = \mathbf{R}\begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix} + p\begin{bmatrix} \boldsymbol{\lambda}_{abcs} \\ \boldsymbol{\lambda}_{qdr} \end{bmatrix}. \tag{12}$$

The flux linkages are given as

$$\begin{bmatrix} \boldsymbol{\lambda}_{abcs} \\ \boldsymbol{\lambda}_{qdr} \end{bmatrix} = \mathbf{L}(\theta_r)\begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix} \tag{13}$$

with the inductance matrix now depending on the position of the rotor

$$\mathbf{L}(\theta_r) = \begin{bmatrix} \mathbf{L}_s(\theta_r) & \mathbf{L}_{sr}(\theta_r) \\ \mathbf{L}_{rs}(\theta_r) & \mathbf{L}_r \end{bmatrix}. \tag{14}$$

The developed electromagnetic torque is

$$T_e = \frac{1}{2}\begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix}^T \frac{\partial}{\partial \theta_r}\mathbf{L}(\theta_r)\begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix}. \tag{15}$$

The main advantage of this model for the EMTP solution is that the stator circuit is directly integrated with the electrical network, thereby avoiding the interfacing and stability problems common in the qd model. However, the terms dependent on rotor position in (13)–(15) result in an additional computational burden and complexity of the final discretized model.

C. Voltage-Behind-Reactance Model

The VBR formulation decouples the synchronous machine model into stator and rotor subsystems. Since the stator network is interfaced with the EMTP solution, the stator phase currents are used as the independent variables. However, the rotor subsystem is expressed in qd rotor reference frame, with the flux linkages used as the independent variables. A detailed derivation of the VBR model can be found in [25]. For completeness, only the final form suitable for the EMTP solution is given here. In particular, the stator voltage equation can be expressed as

$$\mathbf{v}_{abcs} = \mathbf{R}_s\mathbf{i}_{abcs} + p\left[\mathbf{L}''_{abcs}(\theta_r)\mathbf{i}_{abcs}\right] + \mathbf{v}''_{abcs} \tag{16}$$

where $\mathbf{R}_s$ is a constant diagonal matrix representing stator resistances [26], and $\mathbf{L}''_{abcs}(\theta_r)$ is the so-called subtransient inductance matrix, defined as in (17), shown at the bottom of the page, where

$$L_S(\cdot) = L_{ls} + L_a - L_b\cos(\cdot) \tag{18}$$

$$L_M(\cdot) = -\frac{L_a}{2} - L_b\cos(\cdot) \tag{19}$$

$$L_a = \frac{L''_{mq} + L''_{md}}{3} \tag{20}$$

$$L_b = \frac{L''_{md} - L''_{mq}}{3}. \tag{21}$$

The inductances L''_{md} and L''_{mq} are calculated as

$$L''_{md} = \left[\frac{1}{L_{md}} + \frac{1}{L_{lfd}} + \frac{1}{L_{lkd}}\right]^{-1} \tag{22}$$

$$L''_{mq} = \left[\frac{1}{L_{mq}} + \frac{1}{L_{lkq1}} + \frac{1}{L_{lkq2}}\right]^{-1}. \tag{23}$$

The subtransient voltages $\mathbf{v}''_{abcs}(t)$ in (16) are defined as

$$\mathbf{v}''_{abcs} = \left[\mathbf{K}_s^r(\theta_r)\right]^{-1}\left[v''_q \quad v''_d \quad 0\right]^T \tag{24}$$

where

$$\begin{aligned} v''_q =& \omega_r\lambda''_d + \frac{L''_{mq}r_{kq1}\left(\lambda''_q - \lambda_{kq1}\right)}{L^2_{lkq1}} \\ &+ \frac{L''_{mq}r_{kq2}\left(\lambda''_q - \lambda_{kq2}\right)}{L^2_{lkq2}} \\ &+ \left(\frac{r_{kq1}}{L^2_{lkq1}} + \frac{r_{kq2}}{L^2_{lkq2}}\right)L''^2_{mq}i_{qs} \end{aligned} \tag{25}$$

$$\begin{aligned} v''_d =& -\omega_r\lambda''_q + \frac{L''_{md}r_{kd}\left(\lambda''_d - \lambda_{kd}\right)}{L^2_{lkd}} + \frac{L''_{md}}{L_{lfd}}v_{fd} \\ &+ \frac{L''_{md}r_{fd}\left(\lambda''_d - \lambda_{fd}\right)}{L^2_{lfd}} \\ &+ \left(\frac{r_{fd}}{L^2_{lfd}} + \frac{r_{kd}}{L^2_{lkd}}\right)L''^2_{md}i_{ds} \end{aligned} \tag{26}$$

with

$$\lambda''_q = L''_{mq}\frac{\lambda_{kql}}{L_{lkq1}} + L''_{mq}\frac{\lambda_{kq2}}{L_{lkq2}} \tag{27}$$

$$\lambda''_d = L''_{md}\frac{\lambda_{fd}}{L_{lfd}} + L''_{md}\frac{\lambda_{kd}}{L_{lkd}}. \tag{28}$$

Rotor dynamics are represented by the following equations:

$$p\lambda_j = -\frac{r_j}{L_{lj}}(\lambda_j - \lambda_{mq}); \quad j = kq1, kq2 \tag{29}$$

$$p\lambda j = -\frac{r_j}{L_{lj}}(\lambda_j - \lambda_{md}) + v_j; \quad j = fd, kd \tag{30}$$

where

$$\lambda_{mq} = L''_{mq}\left(\frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}} + i_{qs}\right) \tag{31}$$

$$\lambda_{md} = L''_{md}\left(\frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}} + i_{ds}\right). \tag{32}$$

$$\mathbf{L}''_{abcs}(\theta_r) = \begin{bmatrix} L_S(2\theta_r) & L_M\left(2\theta_r - \frac{2\pi}{3}\right) & L_M\left(2\theta_r + \frac{2\pi}{3}\right) \\ L_M\left(2\theta_r - \frac{2\pi}{3}\right) & L_S\left(2\theta_r - \frac{4\pi}{3}\right) & L_M(2\theta_r) \\ L_M\left(2\theta_r + \frac{2\pi}{3}\right) & L_M(2\theta_r) & L_S\left(2\theta_r + \frac{4\pi}{3}\right) \end{bmatrix} \tag{17}$$

Here, λ_j denotes the rotor flux linkages, and λ_{mq} and λ_{md} are the magnetizing flux linkages in qd axes, respectively.

The mechanical equations for the VBR model are identical to (1) and (2). The electromagnetic torque may be calculated as [30]

$$T_e = \frac{3P}{4}(\lambda_{md} i_{qs} - \lambda_{mq} i_{ds}). \tag{33}$$

III. Discrete-Time Model Representations

In order to obtain numerical solutions of the synchronous machine model within the EMTP, the implicit trapezoidal rule is applied to obtain the corresponding difference equations. In particular, discretizing (1) and (2), the difference equations for the rotor position and speed are obtained as

$$\theta_r(t) = \theta_r(t-\Delta t) + \frac{\Delta t}{2}(\omega_r(t) + \omega_r(t-\Delta t)) \tag{34}$$

$$\omega_r(t) = \omega_r(t-\Delta_t) + \frac{\Delta t P}{4J}(T_e(t) + T_e(t-\Delta t)) - \frac{\Delta t P}{2J} T_m \tag{35}$$

which are common to all models considered here. The discretized forms of the qd synchronous machine model [8] are not included in this paper due to space considerations. Instead, the PD and the proposed VBR models are compared, since these two models have similar interfaces with the network. In particular, the general form of these models interfaced into the external network for the EMTP solution can be represented as

$$\mathbf{v}_{abcs}(t) = \mathbf{R}_{eq}(t)\mathbf{i}_{abcs}(t) + \mathbf{e}_h(t) \tag{36}$$

where $\mathbf{R}_{eq}(t)$ is the equivalent resistance matrix (which may need to be inverted) and $\mathbf{e}_h(t)$ is the final history source term.

A. Discrete-Time Phase-Domain Model

Applying the implicit trapezoidal rule with time-step Δt to the voltage (12) gives the following difference equation for the stator voltages of the PD model:

$$\mathbf{v}_{abcs}(t) = \left(\mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}_s(t)\right)\mathbf{i}_{abcs}(t) + \frac{2}{\Delta t}\mathbf{L}_{sr}(t)\mathbf{i}_{qdr}(t) + \mathbf{e}_{sh}^{pd}(t) \tag{37}$$

where

$$\mathbf{e}_{sh}^{pd}(t) = \left(\mathbf{R}_s - \frac{2}{\Delta t}\mathbf{L}_s(t-\Delta t)\right)\mathbf{i}_{abcs}(t-\Delta t) - \frac{2}{\Delta t}\mathbf{L}_{sr}(t-\Delta t)\mathbf{i}_{qdr}(t-\Delta t) - \mathbf{v}_{abcs}(t-\Delta t). \tag{38}$$

The rotor difference equation can be expressed as

$$\mathbf{i}_{qdr}(t) = \left(\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r\right)^{-1} \times \left(\mathbf{v}_{qdr}(t) - \frac{2}{\Delta t}\mathbf{L}_{rs}(t)\mathbf{i}_{abcs}(t) - \mathbf{e}_{rh}^{pd}(t)\right) \tag{39}$$

where

$$\mathbf{e}_{rh}^{pd}(t) = \left(\mathbf{R}_r - \frac{2}{\Delta t}\mathbf{L}_r\right)\mathbf{i}_{qdr}(t-\Delta t) - \frac{2}{\Delta t}\mathbf{L}_{rs}(t-\Delta t)\mathbf{i}_{abcs}(t-\Delta t) - \mathbf{v}_{qdr}(t-\Delta t). \tag{40}$$

Substituting (39) into (37), the PD synchronous machine model can be finally interfaced into the external network as

$$\mathbf{v}_{abcs}(t) = \mathbf{R}_{eq}^{pd}(t)\mathbf{i}_{abcs}(t) + \mathbf{e}_h^{pd}(t) \tag{41}$$

where

$$\mathbf{R}_{eq}^{pd}(t) = \mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}_s(t) - \frac{4}{\Delta t^2}\mathbf{L}_{sr}(t)\left(\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r\right)^{-1}\mathbf{L}_{rs}(t) \tag{42}$$

and

$$\mathbf{e}_h^{pd}(t) = \mathbf{e}_r^{pd}(t) + \mathbf{e}_{sh}^{pd}(t) \tag{43}$$

with

$$\mathbf{e}_r^{pd}(t) = \frac{2}{\Delta t}\mathbf{L}_{sr}(t)\left(\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r\right)^{-1}\left(\mathbf{v}_{qdr}(t) - \mathbf{e}_{rh}^{pd}(t)\right). \tag{44}$$

The electromagnetic torque is calculated in phase variables according to (15) or its expended form [30, eq. 5.3-4]. The mechanical subsystem is solved using (34) and (35).

B. Discrete-Time Voltage-Behind-Reactance Model

Discretizing the VBR stator voltage (16) using the implicit trapezoidal rule gives the following equation:

$$\mathbf{v}_{abcs}(t) = \left(\mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}''_{abcs}(t)\right)\mathbf{i}_{abcs}(t) + \mathbf{v}''_{abcs}(t) + \mathbf{e}_{sh}^{vbr}(t) \tag{45}$$

where

$$\mathbf{e}_{sh}^{vbr}(t) = \left(\mathbf{R}_s - \frac{2}{\Delta t}\mathbf{L}''_{abcs}(t-\Delta t)\right)\mathbf{i}_{abcs}(t-\Delta t) + \mathbf{v}''_{abcs}(t-\Delta t) - \mathbf{v}_{abcs}(t-\Delta t). \tag{46}$$

To interface the VBR model into the external network, (45) should be put into form of (36). Therefore, $\mathbf{v}''_{abcs}$ should be expressed in terms of $\mathbf{i}_{abcs}$. This step can be achieved by discretizing the rotor state equations and solving for the rotor subsystem output variables. After some algebraic manipulation, the difference equations for the rotor flux linkages may be expressed as

$$\begin{bmatrix}\lambda_{kq1}(t)\\ \lambda_{kq2}(t)\end{bmatrix} = \mathbf{E}_1 i_{qs}(t) + \mathbf{E}_2\begin{bmatrix}\lambda_{kq1}(t-\Delta t)\\ \lambda_{kq2}(t-\Delta t)\end{bmatrix} + \mathbf{E}_1 i_{qs}(t-\Delta t) \tag{47}$$

$$\begin{bmatrix}\lambda_{fd}(t)\\ \lambda_{kd}(t)\end{bmatrix} = \mathbf{F}_1 i_{ds}(t) + \mathbf{F}_2\begin{bmatrix}\lambda_{fd}(t-\Delta t)\\ \lambda_{kd}(t-\Delta t)\end{bmatrix} + \mathbf{F}_1 i_{ds}(t-\Delta t) + \mathbf{F}_3 v_{fd}. \tag{48}$$

Here, constant matrices $\mathbf{E}_1, \mathbf{E}_2, \mathbf{F}_1, \mathbf{F}_2$, and $\mathbf{F}_3$ are due to the qd transformation and are given in Appendix A. Further, substituting (47), (48), (27), and (28) into the rotor output (25) and (26), $\mathbf{v}''_{qd}$ may be expressed as

$$\mathbf{v}''_{qd}(t) = (\, \mathbf{k}_1(\omega_r) \quad \mathbf{k}_2(\omega_r)\,)\mathbf{i}_{qds}(t) + \mathbf{h}_{qdr}(t). \tag{49}$$

Here, $\mathbf{k}_1(\omega_r)$ and $\mathbf{k}_2(\omega_r)$ are vectors that depend on the rotor speed ω_r; $\mathbf{h}_{qdr}$ is equivalent history source, including the excitation voltage and the history values of the stator currents and the rotor flux linkages, respectively. These variables are also defined in Appendix A.

After $\mathbf{v}''_{qd}$ and $\mathbf{i}_{qdr}$ are transformed into abc phase coordinates, the subtransient voltages $\mathbf{v}''_{abcs}$ are expressed as

$$\mathbf{v}''_{abcs}(t) = \mathbf{K}(t)\mathbf{i}_{abcs}(t) + \mathbf{e}_r^{vbr}(t) \tag{50}$$

where

$$\mathbf{K}(t) = [\mathbf{K}_s^r(\theta_r)]^{-1} \begin{bmatrix} \mathbf{k}_1(\omega_r) & \mathbf{k}_2(\omega_r) & 0_{2\times 1} \\ 0 & 0 & 0 \end{bmatrix} \mathbf{K}_s^r(\theta_r) \tag{51}$$

and

$$\mathbf{e}_r^{vbr}(t) = [\mathbf{K}_s^r(\theta_r)]^{-1} \begin{bmatrix} \mathbf{h}_{qdr}(t) \\ 0 \end{bmatrix}. \tag{52}$$

Finally, substituting (50) into (45), the VBR model can be interfaced into the external network as

$$\mathbf{v}_{abcs}(t) = \mathbf{R}_{eq}^{vbr}(t)\mathbf{i}_{abcs}(t) + \mathbf{e}_h^{vbr}(t) \tag{53}$$

where

$$\mathbf{R}_{eq}^{vbr}(t) = \mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}''_{abcs}(t) + \mathbf{K}(t) \tag{54}$$

and

$$\mathbf{e}_h^{vbr}(t) = \mathbf{e}_r^{vbr}(t) + \mathbf{e}_{sh}^{vbr}(t). \tag{55}$$

Similar to the PD model, the rotor position and speed are calculated using (34) and (35). However, the electromagnetic torque, T_e, is calculated using (33).

C. Model Complexity

The difference equations of the PD and VBR models have similar forms, since (41)–(43) are analogous to (53)–(55). However, the computational cost associated with these equations is substantially different. The number of floating point operations (flops) required to complete the calculations is often used as a measure of numerical complexity and/or efficiency of a given algorithm. Here, we use the definition of a flop as one addition, subtraction, multiplication, or division of two floating-point numbers [31]. The number of flops for one trigonometric function (trig) evaluation (cos or sin) depends on the floating-point-unit (FPU) processor and/or internal implementation and may cost several flops. After very careful evaluation of the model equations (taking into account that many coefficients and terms can be precalculated and stored for better speed), the number of flops and trigs for the PD and VBR models are summarized in Table I. The total number of flops is also roughly divided among the different terms/equations to better understand where the computational enhancement is achieved.

TABLE I
FLOPS AND TRIG FUNCTIONS COUNT PER TIME STEP

PD Model			VBR Model		
$\mathbf{L}_s(\theta_r)$, $\mathbf{L}_{sr}(\theta_r)$, $\mathbf{L}_{rs}(\theta_r)$	flop: 33	trig: 9	$\mathbf{L}''_{abcs}(\theta_r)$, $\mathbf{K}_s^r(\theta_r)$, $[\mathbf{K}_s^r(\theta_r)]^{-1}$	flop: 20	trig: 9
$\mathbf{R}_{eq}^{pd}$	96	-	$\mathbf{R}_{eq}^{vbr}$	76	-
$\mathbf{e}_h^{pd}$	280	-	$\mathbf{e}_h^{vbr}$	105	-
$\mathbf{i}_{qdr}$	76	-	$\boldsymbol{\lambda}_{qdr}$	46	-
T_e^{pd}	62	1	T_e^{vbr}	4	-
Total	546	10	Total	241	9

As can be seen in Table I, a significant number of flops is spent on computing the history terms $\mathbf{e}_h^{pd}$ and $\mathbf{e}_h^{vbr}$. Because the VBR model utilizes qd transformation for the rotor part, most of the terms and/or coefficients in the discretized rotor (47)–(49) are constant and therefore precalculated outside and before the major time step loop. At the same time, many of the terms and/or equations in the PD model contain time-variant coefficients that must be recalculated due to changing inductances. Another significant saving is achieved in calculating the electromagnetic torque using (33) instead of (15). The difference in the total number of flops will further increase to the benefit of VBR model if one considers a synchronous machine with larger number of damper windings (e.g., 6 in [25]). This can be clearly observed as the dimensions of matrices $\mathbf{L}_{sr}(\theta_r)$, $\mathbf{L}_{rs}(\theta_r)$, and $\mathbf{L}_r$ will increase for the PD model, which further increases the computational costs. However, the dimensions of $\mathbf{L}''_{abcs}(\theta_r)$, $\mathbf{K}_s^r(\theta_r)$, $[\mathbf{K}_s^r(\theta_r)]^{-1}$ do not change, and there will be less of an increase in the number of flops for the VBR model.

IV. INTERFACE PROCEDURE

The method used for interfacing the VBR synchronous machine model is similar to that of the PD model. In particular, the machine is connected to the external network as a three-phase Thevenin equivalent circuit, as shown in Fig. 1. Without loss of generality, the machine windings are assumed to be Y-connected, with the neutral grounded, although any other connection of windings is possible. The sequence of calculation steps in interfacing the VBR model is briefly described here, assuming that the solution at time-step $t - \Delta t$ is known and that the solution at t is to be found.

1) *Predict the mechanical variables:* As the mechanical equations are nonlinear, the exact and simultaneous solution of the mechanical and electrical variables, in general, would require iterations. However, since the me-

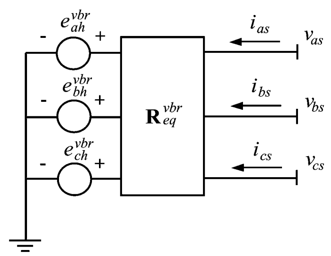

Fig. 1. Thevenin equivalent circuit of the proposed VBR model.

chanical variables change relatively slowly compared to the electrical variables, the linear extrapolation of θ_r and ω_r used in [2] and [20] is also applied here as

$$\theta_r(t) = 2\theta_r(t - \Delta t) - \theta_r(t - 2\Delta t) \tag{56}$$

$$\omega_r(t) = 2\omega_r(t - \Delta t) - \omega_r(t - 2\Delta t). \tag{57}$$

2) *Form the Thevenin equivalent circuit:* (54) and (55) are evaluated to assemble the Thevenin equivalent circuit of the synchronous machine.
3) *Solve the network equations:* The network conductance matrix $\mathbf{G}$ is triangularized, and the network variables are solved.
4) *Update machine's stator and rotor variables:* Stator currents and rotor flux linkages are calculated according to (53), (47), and (48). Subtransient voltages $\mathbf{v}''_{abcs}$ and equivalent history terms are also calculated by (50), (46), and (52).
5) *Update the machine's mechanical part:* The electromagnetic torque T_e is calculated using (33). Then, the rotor displacement θ_r and speed ω_r are recalculated using (34) and (35).

V. Computer Studies

A single-machine, infinite-bus case system is assumed here to compare the different models. The machine parameters obtained from [30] are summarized in Appendix B. To validate the proposed VBR model, the case system has been implemented using various simulation packages, including MicroTran, ATP, and MATLAB/Simulink. In the transient study considered here, the machine initially operates in an idle steady-state mode with load torque $T_m = 0$, and the nominal excitation is kept constant. At $t = 0$, a symmetric three-phase fault is applied at the machine terminals. The dynamic responses produced by various models using different time steps are plotted in Figs. 2 and 3. The studies of unbalance operations are discussed in [32].

A. Model Verification

Fig. 2 depicts the fault transient observed in the field current i_{fd}, a-phase current i_{as}, and the electromagnetic torque T_e. Other variables are not shown due to space limitations. Since the analytical solution is not available, a reference solution was obtained using the qd model implemented in MATLAB/Simulink (state-variable approach) and solved with the Runge–Kutta fourth-order method with an integration time-step $\Delta t = 1\ \mu$s. This solution is considered a trustworthy reference because it was obtained with a high-order method

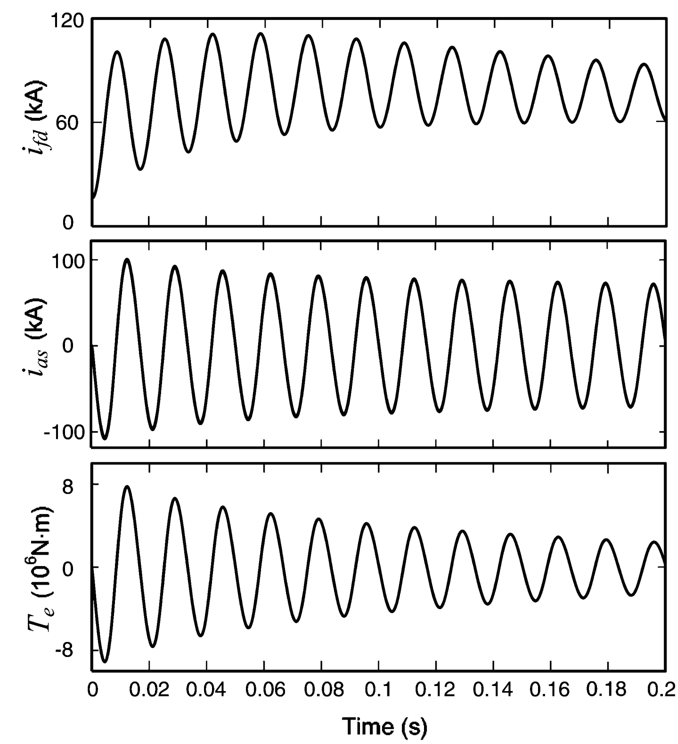

Fig. 2. Simulation results with time-step of 50 μs.

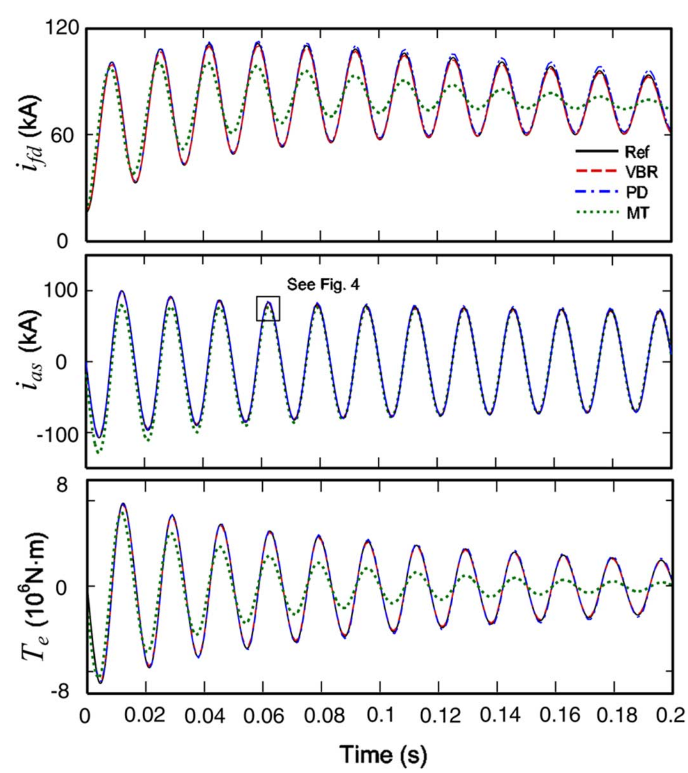

Fig. 3. Simulation results with time-step of 1 ms.

using a very small time-step. The same transient study was reproduced by different models using the time-step $\Delta t = 50\ \mu$s. The corresponding results are superimposed with the reference solutions in Fig. 2, wherein it is shown that the responses

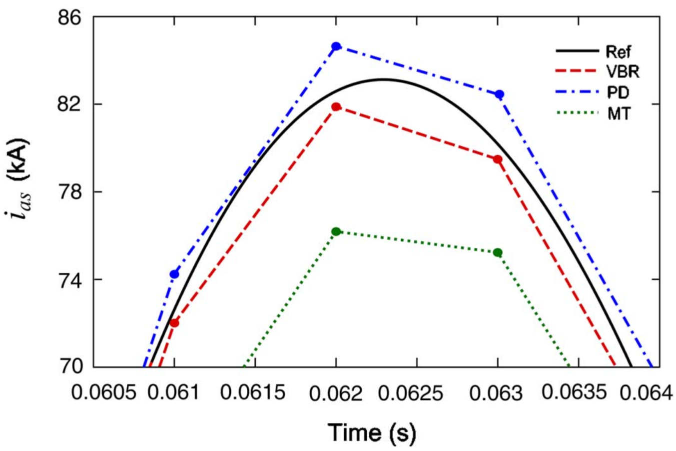

Fig. 4. Detailed view of the portion of i_{as} with the time-step of 1 ms.

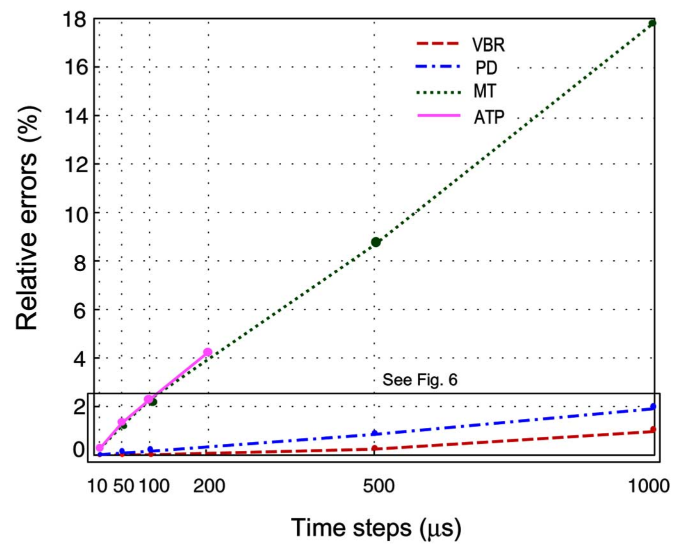

Fig. 5. Propagation of numerical errors in i_{as}.

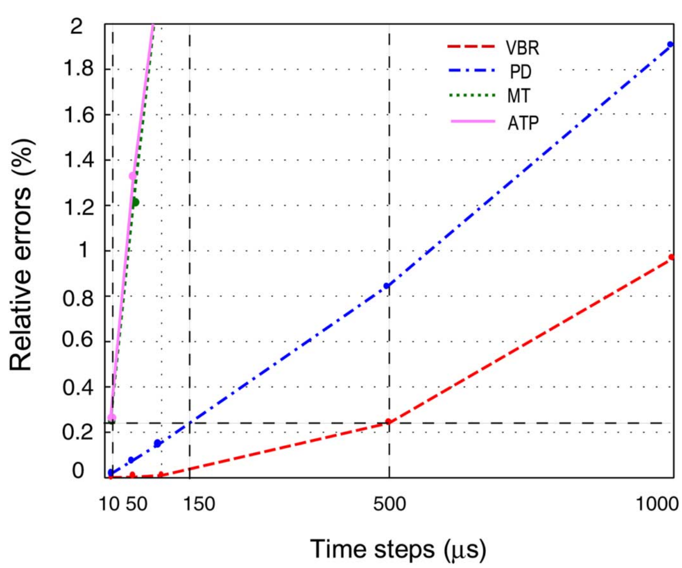

Fig. 6. Comparison of time-steps and numerical errors in i_{as}.

predicted by the above-mentioned models are visibly indistinguishable from the reference solution and each other. This result indicates that the qd model, the PD model, and the VBR model are all equivalent despite the different simulation languages and integration methods used. This study also validates the proposed VBR model.

B. Numerical Accuracy

Although the simulation results obtained by different models are all convergent to the reference solution when the step size is small, error between the solutions still exists. To study the error behavior and stability of different models, the same simulation study was run with different integration time steps. An example study performed with a larger time-step ($\Delta t = 1$ ms) is shown in Fig. 3, wherein the general trend of error produced by different models can be observed. Here, and in other figures, the legend MT denotes the results obtained using MicroTran. Studies with other time-steps are not included due to space limitations. Fig. 3 shows that at such a large time-step, the MicroTran's Type-50 machine model (see dotted-line MT) has a large error, while the ATP Type-59 model was no longer convergent. This behavior is mainly due to the interface of the qd model with the external network, which quickly deteriorates the simulation accuracy as the time-step increases. At the same time, the PD and VBR models remain stable at this large Δt and still produce results reasonably close to the reference solution. To see the details among the simulation results produced by the stable models, a fragment of the peak of current i_{as} is shown in Fig. 4. As can be seen, the solution points obtained by the VBR model are closer to the reference solution than the results of either the PD model or the MT model.

To evaluate the accuracy of different numerical solutions, a relative error between the reference solution trajectory and a given numerical solution may be considered. The relative error is calculated here using the 2-norm [33] as

$$\%\text{error} = \frac{\|\widetilde{\mathbf{f}} - \mathbf{f}\|_2}{\|\mathbf{f}\|_2} \times 100 \tag{58}$$

where $\mathbf{f}$ denotes the reference solution and $\widetilde{\mathbf{f}}$ is the numerical solution. Without loss of generality, the relative error was calculated for one variable only, stator current i_{as}, since the error of the other variables is similar. The error was calculated for different time-step sizes. The results for the different models are shown in Figs. 5 and 6.

As can be seen in Fig. 5, the PD and the VBR models are all noticeably more accurate than the qd models of MicroTran and ATP. In particular, when the time-step Δt is larger than 0.2 ms, the MicroTran's qd model will have an error exceeding 4%, and the ATP's qd model is no longer convergent. Although the qd model implemented in MicroTran appears to be stable even after $\Delta t = 0.2$ ms, the large error observed in other variables (see Fig. 3, i_{fd} and T_e) supports the conclusion that the traditional qd models should preferably be used with small step size and with caution.

TABLE II
COMPARISON OF CPU TIMES

CPU times	PD Model	VBR Model
0.2s study	0.031s	0.015s
Per time-step	7.75μs	3.75μs

C. Model Efficiency

The proposed VBR and PD models were both implemented in standard C language and compiled for the purpose of benchmark comparisons. The compiled models were executed on a personal computer (PC) with a Pentium 4, 2.66-GHz processor and 512M RAM. The CPU times required by the two models for the 0.2-s case study with the integration time-step $\Delta t = 50\ \mu\mathrm{s}$ are summarized in Table II. It can be seen that although both models can realize a faster than real-time simulation speed, the proposed VBR model achieves a roughly 200% improvement of simulation speed over the PD model. This result is very much consistent with the flops count provided in Table I for the two models.

To compare the overall simulation efficiency, some common error tolerance should be assumed. Herein, a relative error tolerance of 0.25% is considered. To achieve this tolerance, MicroTran's and ATP's qd models require a time-step $\Delta t = 10\ \mu\mathrm{s}$, and the PD model needs a time-step $\Delta t = 150\ \mu\mathrm{s}$, as shown in Fig. 6. However, the proposed VBR model may use the time-step as large as 500 μs, which outperforms the PD model by 3.3 times and the traditional qd models by about 50 times. Altogether, taking into account the CPU times in Table II, the VBR model demonstrates a 660% improvement over the PD model for the same relative tolerance of 0.25%.

VI. ERROR ANALYSIS

It is important to point out that the standard qd model, the PD model, and the VBR model are all equivalent for continuous-time analysis since no approximation is made when these models are algebraically derived from each other. However, when these models are discretized using a specific integration rule and interfaced into the EMTP network solution, their numerical properties are different.

As shown in Figs. 5 and 6, the qd models (MT and ATP) quickly lose accuracy compared to the PD and the VBR models. Such behavior is due to the interface of the qd models and the prediction of fast electrical variables such as stator currents and speed voltages [2].

The accuracy of PD and VBR models should be examined in more detail. As was shown in [25], the choice of independent variables and the structure of the model change the system's eigenvalues, which are linked to the accuracy and performance of different numerical solvers.

To achieve the EMTP solution, the PD and VBR models are discretized using the same implicit trapezoidal integration rule as described in Section III. To further compare the numerical property of these two models, the corresponding difference equations should be analyzed. Since the trapezoidal rule is a one-step integration scheme, the model equations may be expressed in the form of a one-step update formula as [33]

$$\mathbf{x}_n = \mathbf{\Phi}(\Delta t, t, \mathbf{x}_{n-1})\mathbf{x}_{n-1} + \mathbf{f}(\Delta t, t). \tag{59}$$

TABLE III
EIGENVALUES OF MACHINE MODELS

Continuous Time		Discrete Time	
PD	VBR	PD	VBR
954.2	32.0+j26.4	0.395	0.945-j0.028
890.3	32.0-j26.4	0.418	0.945+j0.028
-4.2	-0.88	0.976	0.994
-6.00	-5.39	0.994	0.995
-24.2	-5.96	0.996	0.999
-904.6	-57.3+j30.6	2.358	1.031-j0.027
-968.3	-57.3-j30.6	2.498	1.031+j0.027

Here, the vector $\mathbf{x}$ contains combined independent variables (currents and/or flux linkages); $\mathbf{f}(\Delta t, t)$ represents the forcing function that includes the terminal voltages, $\mathbf{v}_{abcs}$, and the excitation voltage, v_{fd}. Thereafter, the numerical behavior and the local error propagation can be related to the eigenvalues of the update matrix $\mathbf{\Phi}(\Delta t, t, \mathbf{x}_{n-1})$, as this matrix directly relates the present step solution (error) to the next step solution (error). After extensive algebraic manipulations with the equations of Section III, both PD and VBR models can be put into the form of (59). Only the final update matrices $\mathbf{\Phi}^{pd}\left(\Delta t, t, \mathbf{x}_{n-1}^{pd}\right)$ and $\mathbf{\Phi}^{vbr}\left(\Delta t, t, \mathbf{x}_{n-1}^{vbr}\right)$ are given in Appendix A. The corresponding discrete-time eigenvalues are calculated for $\Delta t = 1\ \mathrm{ms}$ and summarized in Table III. For consistency, the continuous-time eigenvalues were also calculated using the methodology described in [25] and are included as well.

It is interesting to note that although both PD and VBR models have rotor-position-dependent terms, the respective systems have time-invariant eigenvalues due to the machine symmetry. Another important observation is that both models, when expressed in continuous time, have some eigenvalues with positive real parts. Contrary to the linear time-invariant systems, positive eigenvalues do not imply instability of the time-varying systems [34]. Although it is preferable to have eigenvalues with negative real part, the existence of positive eigenvalues may increase the propagation of local errors when a particular numerical solution scheme is used [35]. In this regard, the positive eigenvalues of the VBR model are much smaller than those of the PD model, which indicates a potentially better numerical conditioning of the VBR model.

When the models are discretized, the eigenvalues of the corresponding difference equation (59) should be compared against the unit circle on the complex plane. Here, by analogy with continuous-time systems, the eigenvalues outside of the unit circle do not imply the system's instability, because both models are time-varying. However, for numerical considerations, it is desirable to have the eigenvalues inside the unit circle (or closer to the origin). In this regard, the magnitude of the largest eigenvalue of the VBR model (1.031) is roughly two times smaller than the largest eigenvalue of the PD model (2.498). This observation again indicates a better numerical conditioning of the VBR model and is completely consistent with the relative error results shown in Figs. 5 and 6.

VII. DISCUSSION OF NETWORK SOLUTION

Although the focus of this paper is the VBR synchronous machine model, it is also of interest and importance to develop an efficient network solution approach to integrate the VBR machine model in EMTP-type programs. Similar to the PD model, the VBR machine model introduces time-varying elements in the network conductance matrix, which may lead to an increase of the execution time if the entire network conductance matrix is re-factorized at each time step (see step 3, Section IV). Some case studies presented in [14] compare the calculation times and the number of triangulations when using Type-58 (PD model) and Type-59 (qd model) synchronous machine models, respectively. For a 190-bus three-generator system, the ratio of 308.3/70.1 (about 4.4 times) of increased calculation time due to re-triangulation was reported [14, Table I].

The impact of time-varying elements on the efficiency of EMTP solution depends on the relative proportion of machines verses other components and the overall system size [36]. If there are only one or a few synchronous machines in the network, the extra computation effort can be reduced by using the compensation method [2]. To get an idea about the increase in execution time with the compensation method, the IEEE First Benchmark Model for SSR [37] was considered, wherein an increase by about 15% per-time-step was observed. Another method is to place the nodes with synchronous machines as the last nodes in the nodal equations and triangularize first the upper part of the conductance matrix before the time-step loop starts. That will produce a small lower sub-matrix [38, Fig. 7], which can then be modified with the time-varying values and solved at each time step to minimize computational overhead of the entire system. The use of advanced factorization techniques [38], [39] will become particularly important for larger networks. On the positive side, the solution obtained by the VBR model using the same time step will be more accurate, which may often justify the extra computational effort.

A more detailed analysis of the impact of using the VBR or PD models may be carried out using several typical systems (perhaps at least two—one with small proportion on machines and another with many machines). When conducting such studies, in addition to comparing the CPU cost per-time-step, one should also consider that both PD and VBR models permit much larger time-step and/or better accuracy (see Fig. 6), which is an advantage that was not utilized in [14] and should be fully exploited. Further investigation into this matter is of definite interest and significance that may be properly addressed in a dedicated publication.

VIII. Conclusion

This paper presented a VBR synchronous machine model for the nodal analysis method and EMTP-type programs. The VBR model interface with the electric network is non-iterative, and simultaneous solution of the machine variables and the network variables is achieved similar to the known PD model. Case studies of a single-machine, infinite-bus system demonstrate that the proposed model has computational advantages over the existing EMTP machine models.

Appendix A

See (A1)–(A10) at the bottom of this page and the top of the following page.

$$\mathbf{E}_1 = \begin{bmatrix} 2-\Delta t b_{11} & -\Delta t b_{12} \\ -\Delta t b_{21} & 2-\Delta t b_{22} \end{bmatrix}^{-1} \begin{bmatrix} \Delta t b_{13} \\ \Delta t b_{23} \end{bmatrix} \tag{A1}$$

$$\mathbf{E}_2 = \begin{bmatrix} 2-\Delta t b_{11} & -\Delta t b_{12} \\ -\Delta t b_{21} & 2-\Delta t b_{22} \end{bmatrix}^{-1} \begin{bmatrix} 2+\Delta t b_{11} & \Delta t b_{12} \\ \Delta t b_{21} & 2+\Delta t b_{22} \end{bmatrix} \tag{A2}$$

$$\mathbf{F}_1 = \begin{bmatrix} 2-\Delta t b_{31} & -\Delta t b_{32} \\ -\Delta t b_{41} & 2-\Delta t b_{42} \end{bmatrix}^{-1} \begin{bmatrix} \Delta t b_{33} \\ \Delta t b_{43} \end{bmatrix} \tag{A3}$$

$$\mathbf{F}_2 = \begin{bmatrix} 2-\Delta t b_{31} & -\Delta t b_{32} \\ -\Delta t b_{41} & 2-\Delta t b_{42} \end{bmatrix}^{-1} \begin{bmatrix} 2+\Delta t b_{31} & \Delta t b_{32} \\ \Delta t b_{41} & 2+\Delta t b_{42} \end{bmatrix} \tag{A4}$$

$$\mathbf{F}_3 = \begin{bmatrix} 2-\Delta t b_{31} & -\Delta t b_{32} \\ -\Delta t b_{41} & 2-\Delta t b_{42} \end{bmatrix}^{-1} \begin{bmatrix} 2\Delta t \\ 0 \end{bmatrix} \tag{A5}$$

$$b_{11} = \frac{r_{kq1}}{L_{lkq1}}\left(\frac{L''_{mq}}{L_{lkq1}} - 1\right); \quad b_{12} = \frac{r_{kq1}L''_{mq}}{L_{lkq1}L_{lkq2}}; \quad b_{13} = \frac{r_{kq1}L''_{mq}}{L_{lkq1}}$$

$$b_{21} = \frac{r_{kq2}L''_{mq}}{L_{lkq1}L_{lkq2}}; \quad b_{22} = \frac{r_{kq2}}{L_{lkq2}}\left(\frac{L''_{mq}}{L_{lkq2}} - 1\right); \quad b_{23} = \frac{r_{kq2}L''_{mq}}{L_{lkq2}}$$

$$b_{31} = \frac{r_{fd}}{L_{lfd}}\left(\frac{L''_{md}}{L_{lfd}} - 1\right); \quad b_{32} = \frac{r_{fd}L''_{md}}{L_{lfd}L_{lkd}}; \quad b_{33} = \frac{r_{fd}L''_{md}}{L_{lfd}}$$

$$b_{41} = \frac{r_{kd}L''_{md}}{L_{lfd}L_{lkd}}; \quad b_{42} = \frac{r_{kd}}{L_{lkd}}\left(\frac{L''_{md}}{L_{lkd}} - 1\right); \quad b_{43} = \frac{r_{kd}L''_{md}}{L_{lkd}}$$

$$\mathbf{k}_1(\omega_t) = \begin{bmatrix} c_{11} & c_{12} \\ c_{21}(\omega_r) & c_{22}(\omega_r) \end{bmatrix} \mathbf{E}_1 + \begin{bmatrix} c_{15} \\ 0 \end{bmatrix} \tag{A6}$$

$$\mathbf{k}_2(\omega_r) = \begin{bmatrix} c_{13}(\omega_r) & c_{14}(\omega_r) \\ c_{23} & c_{24} \end{bmatrix} \mathbf{F}_1 + \begin{bmatrix} 0 \\ c_{25} \end{bmatrix} \tag{A7}$$

$$\begin{aligned} \mathbf{h}_{qdr}(t) =& \begin{bmatrix} c_{11} & c_{12} \\ c_{21}(\omega_r) & c_{22}(\omega_r) \end{bmatrix} \mathbf{E}_2 \begin{bmatrix} \lambda_{kq1}(t-\Delta t) \\ \lambda_{kq2}(t-\Delta t) \end{bmatrix} \\ &+ \begin{bmatrix} c_{11} & c_{12} \\ c_{21}(\omega_r) & c_{22}(\omega_r) \end{bmatrix} \mathbf{E}_1 i_{qs}(t-\Delta t) \\ &+ \begin{bmatrix} c_{13}(\omega_r) & c_{14}(\omega_r) \\ c_{23} & c_{24} \end{bmatrix} \mathbf{F}_2 \begin{bmatrix} \lambda_{fd}(t-\Delta t) \\ \lambda_{kd}(t-\Delta t) \end{bmatrix} \\ &+ \begin{bmatrix} c_{13}(\omega_r) & c_{14}(\omega_r) \\ c_{23} & c_{24} \end{bmatrix} \mathbf{F}_1 i_{ds}(t-\Delta t) \\ &+ \left(\begin{bmatrix} c_{13}(\omega_r) & c_{14}(\omega_r) \\ c_{23} & c_{24} \end{bmatrix} \mathbf{F}_3 + \begin{bmatrix} 0 \\ c_{26} \end{bmatrix} \right) v_{fd} \end{aligned} \tag{A8}$$

$$c_{11} = \frac{L''_{mq} r_{kq1}}{L^2_{lkq1}} \left(\frac{L''_{mq}}{L_{lkq1}} - 1 \right) + \frac{L''_{mq} r_{kq2}}{L^2_{lkq2} L_{lkq1}}$$

$$c_{12} = \frac{L''_{mq} r_{kq2}}{L^2_{lkq2}} \left(\frac{L''_{mq}}{L_{lkq2}} - 1 \right) + \frac{L''^2_{mq} r_{kq1}}{L^2_{lkq1} L_{lkq2}}$$

$$c_{21}(\omega_r) = \frac{-\omega_r L''_{mq}}{L_{lkq1}}; \quad c_{22}(\omega_r) = \frac{-\omega_r L''_{mq}}{L_{lkq2}}$$

$$c_{13}(\omega_r) = \frac{\omega_r L''_{md}}{L_{lfd}}; \quad c_{14}(\omega_r) = \frac{\omega_r L''_{md}}{L_{lkd}}$$

$$c_{23} = \frac{L''_{md} r_{fd}}{L^2_{lfd}} \left(\frac{L''_{md}}{L_{lfd}} - 1 \right) + \frac{L''^2_{md} r_{kd}}{L^2_{lkd} L_{lfd}}$$

$$c_{24} = \frac{L''_{md} r_{kd}}{L^2_{lkd}} \left(\frac{L''_{md}}{L_{lkd}} - 1 \right) + \frac{L''^2_{md} r_{fd}}{L^2_{lfd} L_{lkd}}$$

$$c_{15} = \left(\frac{r_{kq1}}{L^2_{lkq1}} + \frac{r_{kq2}}{L^2_{lkq2}} \right) L''^2_{mq}; \quad c_{25} = \left(\frac{r_{fd}}{L^2_{lfd}} + \frac{r_{kd}}{L^2_{lkd}} \right) L''^2_{md}$$

$$c_{26} = \frac{L''_{md}}{L_{lfd}}$$

$$\mathbf{\Phi}^{pd}\left(\Delta t, t, \mathbf{x}^{pd}_{n-1}\right) = \begin{bmatrix} \mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}_s(t) & \frac{2}{\Delta t}\mathbf{L}_{sr}(t) \\ \frac{2}{\Delta t}\mathbf{L}_{rs}(t) & \mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r \end{bmatrix}^{-1} \begin{bmatrix} -\mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}_s(t-\Delta t) & \frac{2}{\Delta t}\mathbf{L}_{sr}(t-\Delta t) \\ \frac{2}{\Delta t}\mathbf{L}_{rs}(t-\Delta t) & -\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r \end{bmatrix} \tag{A9}$$

$$\begin{aligned} \mathbf{\Phi}^{vbr}\left(\Delta t, t, \mathbf{x}^{vbr}_{n-1}\right) =& \begin{bmatrix} \mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}''_{abcs}(t) + \mathbf{M}_2(t) & \mathbf{M}_1(t) \\ -\begin{bmatrix} \mathbf{E}_1 & 0_{2\times 1} & 0_{2\times 1} \\ 0_{2\times 1} & \mathbf{F}_1 & 0_{2\times 1} \end{bmatrix} \mathbf{K}^r_s(t) & \mathbf{I}_{4\times 4} \end{bmatrix}^{-1} \\ &+ \begin{bmatrix} -\mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}_s(t-\Delta t) - \mathbf{M}_2(t-\Delta t) & -\mathbf{M}_1(t-\Delta t) \\ \begin{bmatrix} \mathbf{E}_1 & 0_{2\times 1} & 0_{2\times 1} \\ 0_{2\times 1} & \mathbf{F}_1 & 0_{2\times 1} \end{bmatrix} \mathbf{K}^r_s(t-\Delta t) & \begin{bmatrix} \mathbf{E}_1 & 0_{2\times 1} \\ 0_{2\times 1} & \mathbf{F}_2 \end{bmatrix} \end{bmatrix} \end{aligned} \tag{A10}$$

$$\mathbf{M}_1(t) = \left[\mathbf{K}^r_s(t)\right]^{-1} \begin{bmatrix} c_{11} & c_{12} & c_{13}(\omega_r) & c_{14}(\omega_r) \\ c_{21}(\omega_r) & c_{22}(\omega_r) & c_{23} & c_{24} \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{M}_2(t) = \left[\mathbf{K}^r_s(t)\right]^{-1} \begin{bmatrix} c_{15} & 0 & 0 \\ 0 & c_{25} & 0 \\ 0 & 0 & 0 \end{bmatrix} \mathbf{K}^r_s(t).$$

APPENDIX B

Synchronous machine parameters [30]: 835 MVA, 26 kV, 0.85 pf, 2 poles, 3600 r/min, $J = 0.0658 \times 10^6$ J $\cdot$ s^2, $r_s = 0.00243\ \Omega$, $X_{ls} = 0.1538\ \Omega$, $X_q = 1.457\ \Omega$, $r_{kq1} = 0.00144\ \Omega$, $X_{lkq1} = 0.6578\ \Omega$, $r_{kq2} = 0.00681\ \Omega$, $X_{lkq2} = 0.07602\ \Omega$, $X_d = 1.457\ \Omega$, $r_{fd} = 0.00075\ \Omega$, $X_{lfd} = 0.1145\ \Omega$, $r_{kd} = 0.0108\ \Omega$, and $X_{lkd} = 0.06577\ \Omega$.

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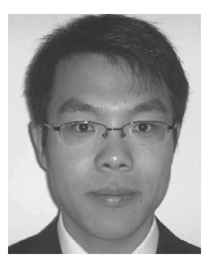

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Magnetically-Saturable Voltage-Behind-Reactance Synchronous Machine Model for EMTP-Type Solution

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***Abstract*—A so-called voltage-behind-reactance (VBR) machine model has recently been proposed for the electro-magnetic transient programs (EMTP) as an advantageous alternative to the conventional qd and phase-domain models. This paper extends the previous research and proposes a magnetically saturable VBR synchronous machine model for EMTP-type solutions. The proposed saturable VBR model utilizes the saliency factor approach to represent main-flux saturation for the salient-pole synchronous machines with the qd axes static and dynamic cross saturation included. An efficient piecewise-linear method is used for representing the nonlinear saturation characteristic within the discretized EMTP solution. Case studies verify that the new model maintains the improved numerical accuracy in steady state and transients even with large time step.**

***Index Terms*—Electro-magnetic transient programs (EMTP), magnetic saturation, phase-domain model, synchronous machine modeling, voltage-behind-reactance model.**

I. INTRODUCTION

NUMEROUS machine models [1]–[17] were proposed depending on required modeling fidelity and applications. This paper focuses on the synchronous machine models that are suitable for the power systems electro-magnetic transient programs (EMTP) [18]. EMTP-like programs are extensively used in industry and academia as powerful and standard transient simulation tools. Therein, the so-called general-purpose models of electrical machines are based on the classical qd reference frame theory [19] and are readily available as built-in library components.

For the salient-pole synchronous machines, the existence of anisotropic rotor structure causes difficulties for the determination of the q-axis saturation characteristic and generally complicates the modeling. Consequently, depending on the required accuracy and acceptable model complexity, in typical models used in EMTP, magnetic saturation may be included along d-axis only; or along both qd axes separately ignoring cross saturation; or more accurately along the main flux path taking qd axes cross saturation into account. In order to incorporate magnetic saturation into the qd salient-pole synchronous machine model, single-saturation factor [1], [7], [9]–[11] or two-saturation factor approaches [4], [5], [8] can be utilized depending on the availability of the q-axis and/or intermediate directions of the saturation characteristic [14], [17].

Magnetic saturation in qd machine models has been implemented for various EMTP software packages [20]–[23]. These implementations include the saturable Type-59 machine model [24], where the main flux saturation is accounted for round-rotor and salient-pole synchronous machines with cross saturation included. The Universal Machine (UM) models [25] in ATP also includes magnetic saturation, wherein for the salient-pole synchronous machines, the saturation along qd axes is considered separately, and for the round-rotor synchronous machines, the saturation of the main flux is considered. In PSCAD/EMTDC, a typical built-in synchronous machine model permits only d-axis saturation [22]. In EMTP-RV [23], the main flux saturation is assumed for the round-rotor synchronous machines, while for the salient-pole machines, the saturation is represented along q and d axes separately.

As an alternative to conventional qd model, the so-called phase-domain (PD) models were also proposed for simulations of electromagnetic transients [12], [26]–[28]. However, the complexity of representing magnetic saturation using PD formulation is much higher than that of the classical qd models. The voltage-behind-reactance (VBR) machine models have been proposed recently for state-variable-based simulation languages [29], [30] and EMTP-type programs [31]–[33] as a computationally efficient alternative to the PD models. As has been documented in a number of recent publications, the VBR model formulation achieves many advantageous properties such as direct model interface with the external network, significantly improved numerical efficiency and accuracy. The interested reader will find detailed analysis and comparisons of qd, PD, and VBR models in [29]–[33] which is not repeated here due to limited space.

A VBR synchronous machine model that includes the d-axis saturation was implemented in [34] using state-space modeling approach. A high-order VBR synchronous machine model for representing arbitrary rotor network and saturation effects was proposed in [35]. The VBR induction machine model that includes main flux saturation was recently implemented for the

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EMTP-type solution [36]. This paper extends the previous work on VBR model formulation [31] and representation of magnetic saturation [36] to provide a new saturable VBR salient-pole synchronous machine model for the EMTP-type programs. The overall contributions of this paper can be summarized as follows.

- We present VBR model of a salient-pole synchronous machine that includes saturation of the main flux path using the saliency factor approach. The new VBR model accurately represents the qd axes cross saturation in steady state and transients.
- We utilize a piecewise-linear representation of the nonlinear saturation characteristic, which makes the proposed model very efficient for the discretized EMTP solution. The proposed model allows an arbitrary number of piecewise-linear segments to approximate a smooth saturation characteristic with desirable accuracy.
- We provide extensive studies comparing the proposed saturable VBR model and several established models to verify that the new model accurately accounts for the qd-axis cross saturation in steady state and transients. We also show that numerical accuracy of the new model is well preserved even for large time steps, which represents an advantage over commonly used existing EMTP models.

II. Synchronous-Machine Saturation Representation

Without loss of generality, a three-phase salient-pole synchronous machine with one field winding, fd, and one damper winding kd in the d-axis, and two damper windings kq1 and kq2 in the q-axis, is considered in this paper. The round-rotor machines can be obtained as a special case with equal magnetizing inductances in q and d axes. It is assumed that the reader is familiar with all relevant equations describing voltages, flux linkages, and torque for the magnetically-linear qd and VBR models which can be found in [19] and [31].

A. Saliency Factor Approach

For the purpose of this paper, we assume the saliency factor (SF) approach [10], [11]. The d-axis saturation characteristic is typically considered readily available as it can be measured using an open-circuit test [16]. However, the saturation characteristic along the q-axis is generally not measurable through simple experiments, and instead it is approximated. The approach makes use of such indirect approximation. In particular, the saliency factor S_F is defined as

$$S_F = \sqrt{\frac{L_{mqu}}{L_{mdu}}} = \sqrt{\frac{L_{mqs}}{L_{mds}}} \tag{1}$$

where L_{mqu}, L_{mdu}, L_{mqs}, and L_{mds} are the unsaturated and saturated qd axes magnetizing inductances, respectively.

Furthermore, it is assumed that the saliency factor S_F remains constant at all saturation levels. This commonly-used assumption allows the anisotropic salient-pole machine to be converted into an equivalent isotropic machine. Thereby, the main flux vector λ_m and the magnetizing current i_m of the equivalent isotropic machine are defined as [10], [11]

$$\lambda_m = \sqrt{\left(\frac{\lambda_{mq}}{S_F}\right)^2 + \lambda_{md}^2} \tag{2}$$

$$i_m = \sqrt{(S_F i_{mq})^2 + i_{md}^2}. \tag{3}$$

A similar approach has also been used in [24] for salient-pole synchronous machines in EMTP, where the equivalent magnetizing flux λ_m was expressed by $\lambda_m = \sqrt{\lambda_{mq}^2 + \lambda_{md}^2}$ instead of (2). This suggests that the SF approach was not correctly implemented in [24], as it was pointed out in [10].

B. Piecewise Linear Method

Numerous methods were proposed to integrate magnetic saturation into the qd machine models. For example, in the case of the state-variable-based models, the flux correction methods [5], [19], and the generalized flux space vector approaches [9]–[11] are often used to integrate the saturation into machine's state-space equations. However, in EMTP software packages [20]–[23], the machine equations have to be discretized for interfacing with the nodal equations of the external circuit-network, which generally assumes a linear circuit within a given time-step Δt in order to achieve a non-iterative network solution [18]. For this purpose, the piecewise linear methods of representing the nonlinear elements are widely used and preferred for computational efficiency.

Thus, it is assumed that the d-axis magnetic saturation characteristic is represented by a nonlinear monotonic saturation function Λ_d. Based on the SF approach, the main-flux magnetic saturation characteristic of an equivalent isotropic machine may also be represented by the d-axis characteristic as follows:

$$\lambda_m = \Lambda_d(i_m). \tag{4}$$

Furthermore, calculating the partial derivative of (4) with respect to magnetizing current i_m, the so-called dynamic or incremental inductance is obtained as

$$L = \frac{\partial \lambda_m}{\partial i_m}. \tag{5}$$

For the EMTP solution, this partial differentiation is then approximated within a given integration time step Δt as

$$L_D = \frac{\Delta \lambda_m}{\Delta i_m}. \tag{6}$$

Approximation of the nonlinear characteristic L in (5) by (6) is also shown in Fig. 1, and it assumes a very small range Δi_m. Therefore, the nonlinear saturation function (4) may be represented within a time step Δt by the following linear relationship:

$$\lambda_m = L_D i_m + \lambda_{res} \tag{7}$$

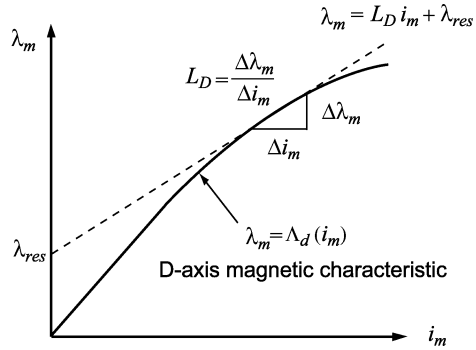

Fig. 1. Piecewise linear representation of magnetic saturation.

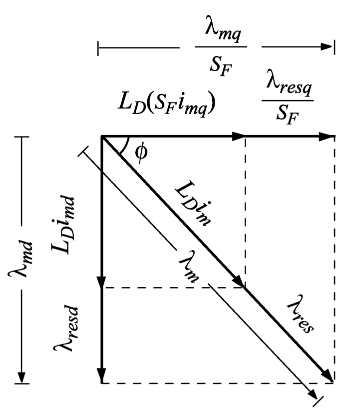

Fig. 2. Projections of the magnetizing and residual fluxes onto qd axes.

where λ_{res} is the residual flux, and L_D is the local slope, the incremental inductance, as defined by (6).

As the machine equations are expressed in qd coordinates, the linear relationship (7) is projected into the qd axes as shown in Fig. 2. Thus, according to Fig. 2, the projections of the main magnetizing flux is found as

$$\frac{\lambda_{mq}}{S_F} = L_D(S_F i_{mq}) + \frac{\lambda_{resq}}{S_F} \tag{8}$$

$$\lambda_{md} = L_D i_{md} + \lambda_{resd} \tag{9}$$

where residual fluxes are also found as

$$\lambda_{resq} = \lambda_{res} \cos\phi \cdot S_F \tag{10}$$

$$\lambda_{resd} = \lambda_{res} \sin\phi. \tag{11}$$

Here, the angle ϕ takes into account the position of the main flux within the rotor reference frame, qd coordinates.

III. Saturable Voltage-Behind-Reactance Model

The saturable VBR synchronous machine model is derived based on the qd model given in [29] (see [29, (17)–(26)]) and the piecewise linear representation of the saturation described in Section II. The magnetizing currents i_{mq} and i_{md} in (8) and (9) are first replaced by the corresponding winding currents as

$$\lambda_{mq} = L_D S_F^2 (i_{qs} + i_{kq1} + i_{kq2}) + \lambda_{resq} \tag{12}$$

$$\lambda_{md} = L_D (i_{ds} + i_{fd} + i_{kd}) + \lambda_{resd}. \tag{13}$$

Based on the qd model given in [29], the rotor currents are solved from the rotor flux linkage equations (see [29, (23) and (24)]) and substituting them into (12) and (13), the qd axes magnetizing flux linkages (12) and (13) are expressed in terms of stator currents and rotor flux linkages as

$$\lambda_{mq} = L''_{mq}\left(i_{qs} + \frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}}\right) + \frac{L''_{mq}}{L_D S_F^2}\lambda_{resq} \tag{14}$$

$$\lambda_{md} = L''_{md}\left(i_{ds} + \frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}}\right) + \frac{L''_{md}}{L_D}\lambda_{resd} \tag{15}$$

where

$$L''_{mq} = \left(\frac{1}{L_D S_F^2} + \frac{1}{L_{lkq1}} + \frac{1}{L_{lkq2}}\right)^{-1} \tag{16}$$

$$L''_{md} = \left(\frac{1}{L_D} + \frac{1}{L_{lfd}} + \frac{1}{L_{lkd}}\right)^{-1}. \tag{17}$$

Therefore, the stator flux linkages can be reformulated by substituting (14) and (15) into the stator flux linkage equations (see [29, (21) and (22)]), respectively, which gives

$$\lambda_{qs} = L''_q i_{qs} + \lambda''_q \tag{18}$$

$$\lambda_{ds} = L''_d i_{ds} + \lambda''_d \tag{19}$$

where

$$L''_q = L_{ls} + L''_{mq} \tag{20}$$

$$L''_d = L_{ls} + L''_{md} \tag{21}$$

and

$$\lambda''_q = L''_{mq}\left(\frac{\lambda_{kq1}}{L_{lkq1}} + \frac{\lambda_{kq2}}{L_{lkq2}} + \frac{\lambda_{resq}}{L_D S_F^2}\right) \tag{22}$$

$$\lambda''_d = L''_{md}\left(\frac{\lambda_{fd}}{L_{lfd}} + \frac{\lambda_{kd}}{L_{lkd}} + \frac{\lambda_{resd}}{L_D}\right). \tag{23}$$

Solving the rotor flux linkage equations (see [29, (23) and (24)]) for rotor currents and substituting them into the rotor voltage equation (see [29, (20)]), the rotor state-space equations are therefore formulated as

$$p\lambda_j = -\frac{r_j}{L_{lj}}(\lambda_j - \lambda_{mq}) + v_j; \quad j = kq1, kq2 \tag{24}$$

$$p\lambda_j = -\frac{r_j}{L_{lj}}(\lambda_j - \lambda_{md}) + v_j; \quad j = fd, kd. \tag{25}$$

Substituting (18) and (19) into the stator voltage equations (see [29, (17) and (18)]), respectively, the stator voltage equations can be rewritten as

$$v_{qs} = r_s i_{qs} + \omega_r L''_d i_{ds} + pL''_q i_{qs} + \omega_r \lambda''_d + p\lambda''_q \tag{26}$$

$$v_{ds} = r_s i_{ds} - \omega_r L''_q i_{qs} + pL''_d i_{ds} - \omega_r \lambda''_q + p\lambda''_d. \tag{27}$$

The terms $p\lambda''_q$ in (26) is calculated by taking the derivatives of (22) as

$$p\lambda''_q = \frac{L''_{mq}}{L_{lkq1}}p\lambda_{kq1} + \frac{L''_{mq}}{L_{lkq2}}p\lambda_{kq2} + \frac{L''_{mq}}{L_D S_F^2}p\lambda_{resq}. \tag{28}$$

In (28), $p\lambda_{kq1}$ and $p\lambda_{kq2}$ are eliminated by substituting the rotor state (24) and $p\lambda_{resq}$ is derived by calculating the derivative of (10). These manipulations lead to

$$p\lambda_q'' = \frac{L_{mq}'' r_{kq1}}{L_{lkq1}^2}\left(\lambda_q'' - \lambda_{kq1}\right) + \frac{L_{mq}'' r_{kq2}}{L_{lkq2}^2}\left(\lambda_q'' - \lambda_{kq2}\right) + L_{mq}''^2\left(\frac{r_{kq1}}{L_{lkq1}^2} + \frac{r_{kq2}}{L_{lkq2}^2}\right) i_{qs} - \frac{L_{mq}''}{L_D S_F}\lambda_{resd}\frac{d\phi}{dt}. \quad (29)$$

Similar procedure is applied to $p\lambda_d''$ which gives the following result:

$$p\lambda_d'' = \frac{L_{md}'' r_{fd}}{L_{lfd}^2}\left(\lambda_d'' - \lambda_{fd}\right) + \frac{L_{md}'' r_{kd}}{L_{lkd}^2}\left(\lambda_d'' - \lambda_{kd}\right) + L_{md}''^2\left(\frac{r_{fd}}{L_{lfd}^2} + \frac{r_{kd}}{L_{lkd}^2}\right) i_{ds} + \frac{L_{md}''}{L_D S_F}\lambda_{resq}\frac{d\phi}{dt} + \frac{L_{md}''}{L_{lfd}} v_{fd}. \quad (30)$$

The stator voltage equations are then expressed by substituting (29), (30) into (26) and (27), respectively, as

$$v_{qs} = r_s i_{qs} + \omega_r L_d'' i_{ds} + pL_q'' i_{qs} + v_q'' \quad (31)$$

$$v_{ds} = r_s i_{ds} - \omega_r L_q'' i_{qs} + pL_d'' i_{ds} + v_d'' \quad (32)$$

where the subtransient voltages are

$$v_q'' = \omega_r \lambda_d'' + \frac{L_{mq}'' r_{kq1}}{L_{lkq1}^2}\left(\lambda_q'' - \lambda_{kq1}\right) + \frac{L_{mq}'' r_{kq2}}{L_{lkq2}^2}\left(\lambda_q'' - \lambda_{kq2}\right) + L_{mq}''^2\left(\frac{r_{kq1}}{L_{lkq1}^2} + \frac{r_{kq2}}{L_{lkq2}^2}\right) i_{qs} - \frac{\omega_\phi L_{mq}''}{L_D S_F}\lambda_{resd} \quad (33)$$

$$v_d'' = -\omega_r \lambda_q'' + \frac{L_{md}'' r_{fd}}{L_{lfd}^2}\left(\lambda_d'' - \lambda_{fd}\right) + \frac{L_{md}'' r_{kd}}{L_{lkd}^2}\left(\lambda_d'' - \lambda_{kd}\right) + L_{md}''^2\left(\frac{r_{fd}}{L_{lfd}^2} + \frac{r_{kd}}{L_{lkd}^2}\right) i_{ds} + \frac{\omega_\phi L_{md}''}{L_D S_F}\lambda_{resq} + \frac{L_{md}''}{L_{lfd}} v_{fd} \quad (34)$$

with the rate of change of the magnetizing flux angle given as

$$\omega_\phi = \frac{d\phi}{dt}. \quad (35)$$

The stator voltage equations of the VBR model are obtained by transforming (31) and (32) back to abc coordinates as

$$\mathbf{v}_{abcs} = \mathbf{r}_s \mathbf{i}_{abcs} + p\left[\mathbf{L}_{abcsat}''(\theta_r)\mathbf{i}_{abcs}\right] + \mathbf{v}_{abcs}'' \quad (36)$$

where

$$\mathbf{L}_{abcsat}''(\theta_r) = \begin{bmatrix} L_S(2\theta_r) & L_M\left(2\theta_r - \frac{2\pi}{3}\right) & L_M\left(2\theta_r + \frac{2\pi}{3}\right) \\ L_M\left(2\theta_r - \frac{2\pi}{3}\right) & L_S\left(2\theta_r - \frac{4\pi}{3}\right) & L_M(2\theta_r) \\ L_M\left(2\theta_r + \frac{2\pi}{3}\right) & L_M(2\theta_r) & L_S\left(2\theta_r + \frac{4\pi}{3}\right) \end{bmatrix}. \quad (37)$$

Here, L_S and L_M are saturation dependent and have the similar form as the magnetically linear VBR model documented in [29]. The subtransient back emfs in abc coordinates are given as

$$\mathbf{v}_{abcs}'' = \left[\mathbf{K}_s^r(\theta_r)\right]^{-1}\left[v_q'' \quad v_d'' \quad 0\right]^T. \quad (38)$$

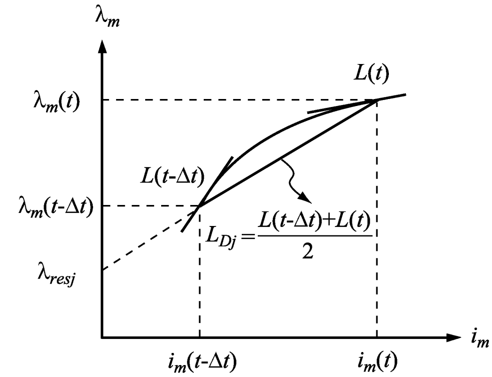

Fig. 3. Piecewise-linear representation of the saturation characteristic using implicit trapezoidal rule.

Therefore, the stator voltage (36)–(38), the rotor state equations (24) and (25) along with the magnetizing flux linkages given by (14)–(17), the back emf (33) and (34), and the mechanical equations [31, (1) and (2)] defined the saturable VBR model.

IV. Model Implementation

The implementation of saturable VBR synchronous machine model is similar to that of the original magnetically linear EMTP model [31] with additional challenges due to representation of nonlinear magnetic characteristic. Discretizing the stator voltage (31) using implicit trapezoidal rule with time step Δt, the stator equation becomes

$$\mathbf{v}_{abcs}(t) = \left(\mathbf{r}_s + \frac{2}{\Delta t}\mathbf{L}_{abcsat}''(t)\right)\mathbf{i}_{abcs}(t) + \mathbf{v}_{abcs}''(t) + \mathbf{e}_{sh}(t) \quad (39)$$

where the stator equivalent history voltage source is given as

$$\mathbf{e}_{sh}(t) = \left(\mathbf{r}_s - \frac{2}{\Delta t}\mathbf{L}_{abcsat}''(t-\Delta t)\right)\mathbf{i}_{abcs}(t-\Delta t) + \mathbf{v}_{abcs}''(t-\Delta t) - \mathbf{v}_{abcs}(t-\Delta t). \quad (40)$$

In contrast to the model in [31], the subtransient inductance matrix $\mathbf{L}_{abcsat}''$ in (39) is not only a function of the rotor displacement θ_r as shown in (37), but also depends on the magnetic saturation as indicated by (16) and (17). Therefore, the saturation characteristic (5) is also discretized using the implicit trapezoidal rule as

$$\frac{L(t) + L(t-\Delta t)}{2} = \frac{\lambda_m(t) - \lambda_m(t-\Delta t)}{i_m(t) - i_m(t-\Delta t)}. \quad (41)$$

This discretization step results in piecewise linear relationship shown in Fig. 3.

Based on (41) and (7), the piecewise linear saturation representation at the time point t can be expressed as

$$\lambda_m(t) = L_{Dj} i_m(t) + \lambda_{resj} \quad (42)$$

where

$$L_{Dj} = \frac{L(t) + L(t-\Delta t)}{2} \quad (43)$$

and

$$\lambda_{resj} = \lambda_m(t-\Delta t) - L_{Dj} i_m(t-\Delta t). \quad (44)$$

Here, the equivalent dynamic magnetizing inductance L_{Dj} represents the average of the magnetizing inductances $L(t-\Delta t)$ and $L(t)$ within one time step Δt. The residual flux λ_{resj} represents the contribution of the history terms $\lambda_m(t-\Delta t)$ and $i_m(t-\Delta t)$. The subscript j in (42)–(44) represents an arbitrary operating point corresponding to the discrete-time at time t.

It is noted in (43) that L_{Dj} is unknown for the solution at time t as the magnetizing inductance $L(t)$ depends on the magnetizing current or flux linkage at the same time step, which forms a nonlinear implicit relationship. To avoid solving the system of nonlinear equations using iterative methods, a linear extrapolation of the main flux linkage $\lambda_m(t)$ is used. Thus, the magnetizing inductance $L(t)$ can be calculated using the magnetic saturation characteristic (5).

The next step requires discretizing the rotor state equations (24) and (25) using implicit trapezoidal rule. After substituting λ_{mq} and λ_{md} in (24) and (25) with magnetizing flux linkages (14) and (15) and rearranging terms, we get

$$\begin{aligned}\begin{bmatrix}\lambda_{kq1}(t)\\ \lambda_{kq2}(t)\end{bmatrix} &= \mathbf{E}_1 i_{qs}(t) + \mathbf{E}_2\begin{bmatrix}\lambda_{kq1}(t-\Delta t)\\ \lambda_{kq2}(t-\Delta t)\end{bmatrix}\\ &\quad + \mathbf{E}_1 i_{qs}(t-\Delta t) + \mathbf{E}_3\lambda_{resjq}(t) \quad (45)\\ \begin{bmatrix}\lambda_{fd}(t)\\ \lambda_{kd}(t)\end{bmatrix} &= \mathbf{F}_1 i_{ds}(t) + \mathbf{F}_2\begin{bmatrix}\lambda_{fd}(t-\Delta t)\\ \lambda_{kd}(t-\Delta t)\end{bmatrix} + \mathbf{F}_1 i_{ds}(t-\Delta t)\\ &\quad + \mathbf{F}_3\lambda_{resjd}(t) + \mathbf{F}_4 v_{fd} \quad (46)\end{aligned}$$

where the coefficient matrices $\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3$, and $\mathbf{F}_4$ are defined for compactness similar to [31, Appendix A].

In order to calculate $\boldsymbol{\lambda}_{resjq}(t)$ and $\boldsymbol{\lambda}_{resjd}(t)$ in (45) and (46), the residual flux $\boldsymbol{\lambda}_{resj}(t)$ is projected along the qd axes as shown in Fig. 2. Here, $\boldsymbol{\lambda}_{resj}(t)$ is a known quantity calculated according to (44). The angle ϕ between λ_{mq} and λ_m is also required so that (10) and (11) may be calculated. This angle ϕ is calculated by discretizing (35) using trapezoidal rule as

$$\phi(t) = \phi(t-\Delta t) + \frac{\Delta t}{2}\left(\omega_\phi(t) + \omega_\phi(t-\Delta t)\right) \quad (47)$$

where the rotating speed $\omega_\phi(t)$ of the flux angle ϕ is predicted by the linear extrapolation.

The subtransient voltages are considered next. In particular, $\mathbf{v}''_{qd}$ are formulated by substituting (45) and (46) into (33) and (34) as

$$\mathbf{v}''_{qd}(t) = \begin{bmatrix}\mathbf{k}_1(\omega_r) & \mathbf{k}_2(\omega_r)\end{bmatrix}\mathbf{i}_{qds}(t) + \mathbf{h}_{qdr}(t). \quad (48)$$

Here, the coefficient vectors $\mathbf{k}_1(\omega_r)$, $\mathbf{k}_2(\omega_r)$, and $\mathbf{h}_{qdr}(t)$ are defined similarly to [31, Appendix A]. These subtransient voltages are then transformed back to abc coordinates, which gives the following result:

$$\mathbf{v}''_{abcs}(t) = \mathbf{K}(t)\mathbf{i}_{abcs}(t) + \mathbf{e}_r(t) \quad (49)$$

where

$$\mathbf{K}(t) = [\mathbf{K}_s^r(\theta_r)]^{-1}\begin{bmatrix}\mathbf{k}_1(\omega_r) & \mathbf{k}_2(\omega_r) & 0_{2\times 1}\\ 0 & 0 & 0\end{bmatrix}\mathbf{K}_s^r(\theta_r) \quad (50)$$

and

$$\mathbf{e}_r(t) = [\mathbf{K}_s^r(\theta_r)]^{-1}\begin{bmatrix}\mathbf{h}_{qdr}(t)\\ 0\end{bmatrix}. \quad (51)$$

Substituting (49) into the stator voltage (39), the saturable VBR model has the following standard form:

$$\mathbf{v}_{abcs}(t) = \mathbf{R}_{eq}(t)\mathbf{i}_{abcs}(t) + \mathbf{e}_h(t) \quad (52)$$

which is used for interfacing with the external network. Here, the final equivalent resistance matrix and the history terms are

$$\mathbf{R}_{eq}(t) = \mathbf{r}_s + \frac{2}{\Delta t}\mathbf{L}''_{abcsat}(t) + \mathbf{K}(t) \quad (53)$$

and

$$\mathbf{e}_h(t) = \mathbf{e}_r(t) + \mathbf{e}_{sh}(t). \quad (54)$$

The discretization of mechanical equations can be found in [31], and is not included here due to space considerations.

V. Case Studies

The proposed model has been implemented according to the EMTP solution methodology. For comparison purpose, a state-variable saturable qd synchronous machine model, based on the generalized flux space vector concept [9]–[11], has also been implemented in MATLAB/Simulink. Without loss of generality, the model using i_{ds}, i_{qs}, λ_{md}, λ_{mq}, i_{kd}, and i_{kq2} as state variables [10, Section 4.2] is chosen here. This model has been implemented without any approximation to include both steady-state and dynamic qd-axes cross saturation. This model is solved using fourth-order Runge-Kutta method with a very small time step of 1 μs to obtain very accurate numerical solutions, which are used as a reference. A synchronous-machine infinite-bus system is considered here. A salient-pole hydro turbine machine is used and the machine parameters are given in [19, Table 5.10-1].

A. Change of Excitation Voltage Without Load

A case of unloaded synchronous machine is considered first as to verify the model when the rotor is aligned with the main magnetizing flux. In addition to the proposed saturable VBR model and the reference model [10], the built-in synchronous machine model in EMTP-RV (linear and saturable) is used as well. As the salient-pole synchronous machine model in EMTP-RV uses two independent saturation representations for q and d axes, the qd cross saturation is ignored. For the purpose of model verification, without loss of generality, the two-slope saturation curve is used here for consistency with EMTP-RV's piecewise linear saturation representation. This makes it easier to define the same operating point for various models when

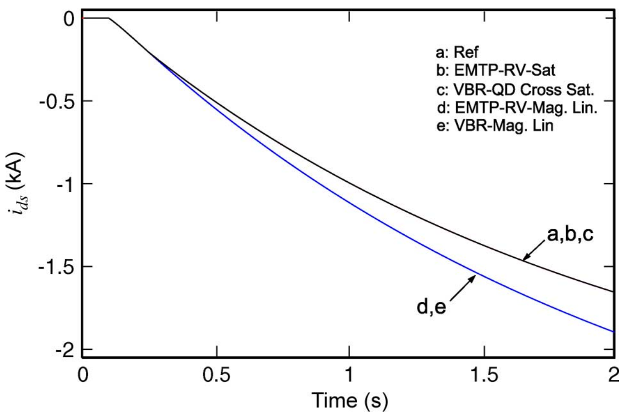

Fig. 4. Response of the stator d-axis currents i_{ds} to the change in excitation under no load conditions.

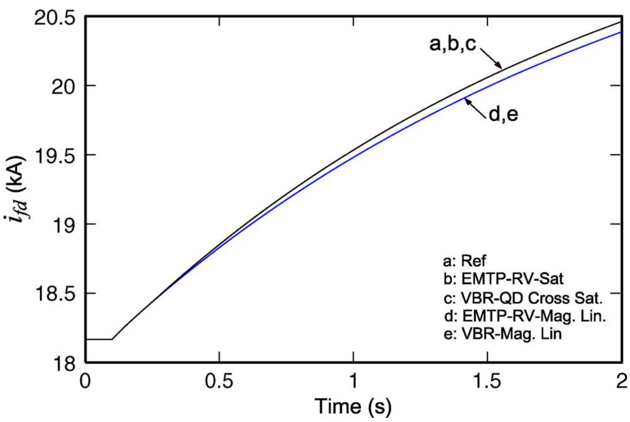

Fig. 5. Response of the field winding currents i_{fd} to the change in excitation under no load conditions.

machine operates in unsaturated region. The typical EMTP time step of 50 μs is used for all the subject models.

Initially, the synchronous machine is assumed to operate in steady state with the field excitation voltage E_f set to 1 pu, (i.e., 16.33 kV) and zero mechanical torque. At $t = 0.1$ s, the excitation voltage E_f is stepped up to 1.2 pu (i.e., 19.6 kV) so that the machine starts to operate in overexcited mode. As the machine operates without load, it is appropriate to look into the effect of d-axis saturation only. The predicted transient responses are shown in Figs. 4–6, respectively.

It is observed in Figs. 4–6 that initially all models (linear and saturable) predict the same operating condition. After $t = 0.1$ s, the saturable VBR (VBR-QD-CrossSat.) and EMTP-RV's qd models (EMTP-RV-Sat) give the identical transient responses that are consistent with the reference model [10]. At the same time, the two linear models (VBR-Mag.Lin) and (EMTP-RV-Mag.Lin) also predict consistent and matching transient responses that visibly deviate from the saturable models as shown in Figs. 4–6. It is also noted that to obtain the results of Figs. 4–6 using the EMTP-RV, the stator current i_{ds} and the magnetizing flux λ_{md} had to be rescaled by a factor of $\sqrt{2/3}$ since this program uses a power invariant transformation matrix [23].

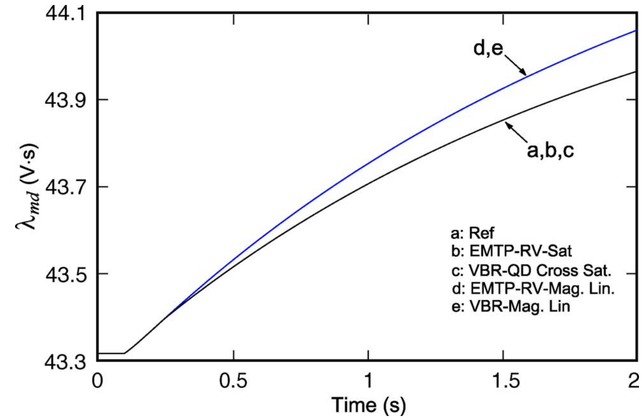

Fig. 6. Response in magnetizing flux in d-axis λ_{md} to the change in excitation under no load conditions.

B. Change of Load Torque

In this subsection, the impact of including the q-axis and qd-axis cross saturation on the accuracy of machine modeling is studied. The saturable VBR model is implemented using the method described in Sections II–IV. The smooth saturation curve is used for the studies considered herein to demonstrate the generality and effectiveness of the proposed saturation modeling method. The linear VBR model is also considered for comparison. In addition to that, a VBR model that includes the saturation along the d-axis only has also been implemented. Such model can represent the PSCAD built-in model, where only the d-axis saturation is included.

In the following transient study, the machine initially operates under no-load steady state with the field excitation E_f set to 1 pu (i.e., 16.33 kV). As the smooth saturation curve is used, this operating condition corresponds to under-excited mode $(i_{ds} < 0)$ due to the reduced d-axis magnetizing flux λ_{md}. To achieve the same initial operating point for the magnetically linear model, the modified-air-gap-line with $L_{md} = 2.2044$ mH is used. Similar approach has been used in [10] for achieving the same steady-state operating condition for the unsaturated model. At $t = 0.1$ s, a mechanical load torque $T_m = 1$ pu is applied, bringing the machine into the motor operation. As the rotor angle δ becomes significant, the q-axis currents, and the importance of including q-axis and qd-axes cross saturation become apparent. The transient responses of the stator currents i_{qs} and i_{ds}, the magnetizing flux linkages λ_{mq} and λ_{md}, and the rotor angle δ predicted by various models are shown in Figs. 7–11.

It is seen in Figs. 7 and 9 that the q-axis current i_{qs} and the flux linkage λ_{mq} are increased significantly due to the applied load torque. At the same time, the d-axis magnetizing flux linkage is slightly reduced as shown in Fig. 10. Therefore, it can be verified that the main flux of the equivalent isotropic machine, represented by (2), has shifted its direction and now has significant projections onto both d and q axes. This new operating condition corresponds to a different saturation level as well. It is also observed in Figs. 7–11 that the saturable VBR model (VBR-QD-CrossSat.) including the qd-axis cross saturation predicted transient responses identical the reference model [10]. However, the d-axis saturable (Sat. D-axis only) and magnetically linear

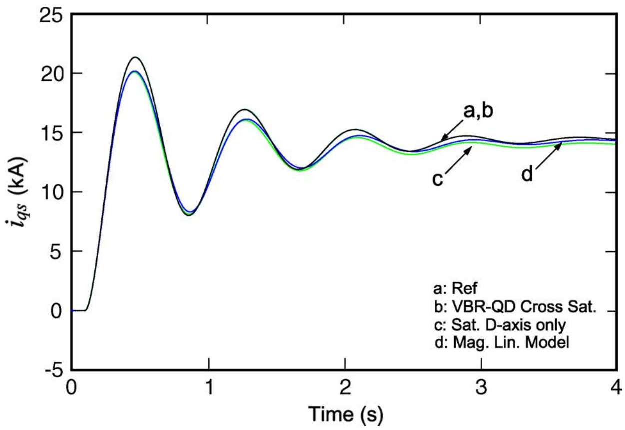

Fig. 7. Response in stator q-axis currents i_{qs} to change in load torque.

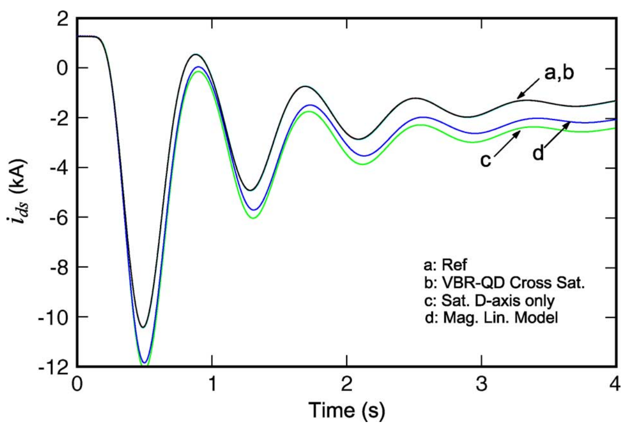

Fig. 8. Response in stator d-axis currents i_{ds} to change in load torque.

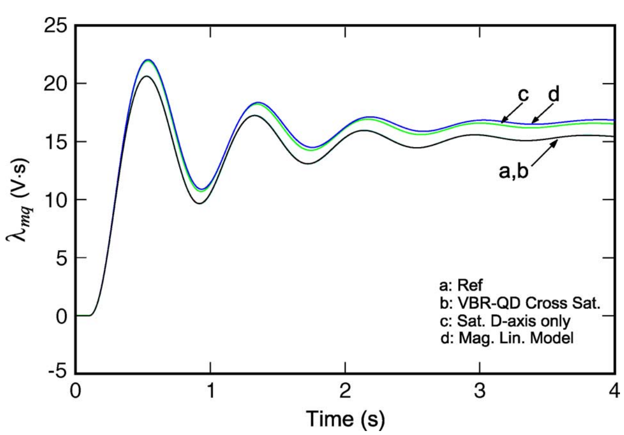

Fig. 9. Response in magnetizing flux linkages along q-axis λ_{mq}.

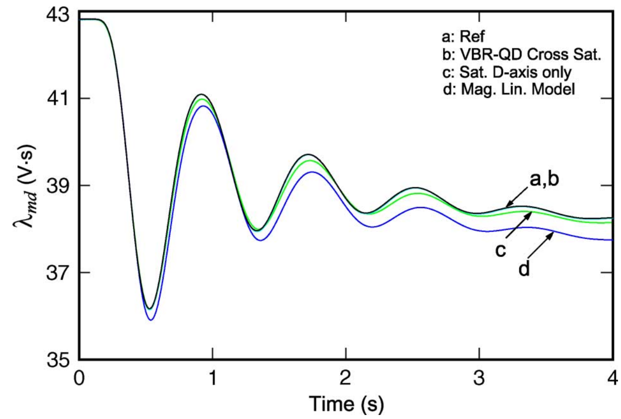

Fig. 10. Response in magnetizing flux linkages along d-axis λ_{md}.

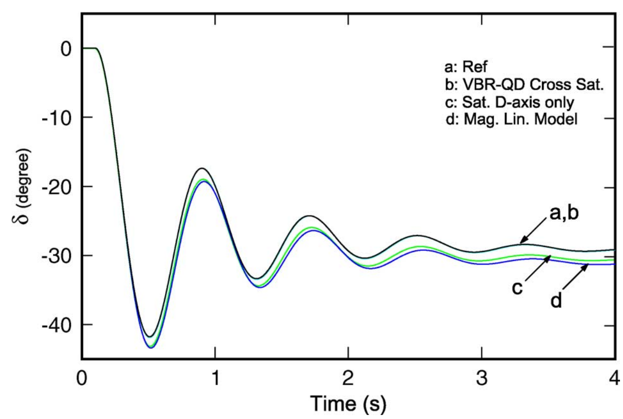

Fig. 11. Response in rotor angles δ to the change in load torque.

(Mag. Lin. Model) models predict visibly different transient responses compared with the reference. In fact, the performance of the d-axis saturable model is very close to the magnetically linear model in terms of predicting the q-axis current and flux linkage in Figs. 7 and 9, respectively. For predicting the d-axis current and flux, the d-axis saturable (Sat. D-axis only) is also less accurate than the full saturable VBR model. The omission of the q-axis saturation (Sat. D-axis only) may lead to a noticeable error in predicting the rotor angle δ as shown in Fig. 11.

C. Change of Stator Terminal Voltage Under Full Load

In order to verify the accuracy of the proposed saturation modeling approach for large integration time steps, the same saturable VBR model has been implemented with the time step of 50 μs and 500 μs in this subsection. The synchronous machine is initially operated at full load, $T_m = 1$ pu, with the rated field voltage, $E_f = 1$ pu. For the purpose of comparison, the magnetically linear VBR model is implemented again, wherein the modified-air-gap-line magnetizing inductances $L_{md} = 2.169$ mH and $L_{mq} = 1.07$ mH are used so that the same initial steady-state operating point is established for all machine models. The smooth saturation characteristic is used here for all saturable models. At $t = 0.2$ s, the stator terminal voltages are increased from 1 pu to 1.05 pu. Similar case studies have been also used in [10] and [11] for investigating synchronous machine saturation modeling and are considered very appropriate and informative.

Figs. 12–14 show the transient responses of the stator currents i_{qs} and i_{ds}, and the rotor angle δ, as predicted by various models considered in this subsection. As it can be clearly seen in Figs. 12–14, the transient responses produced by the saturable models are visibly different from those produced by the magnetically linear model (which converges to a slightly different operating point). At the same time, the responses produced by the saturable VBR model, even with a large time step of 500 μs, closely follow the solutions produced by the reference model

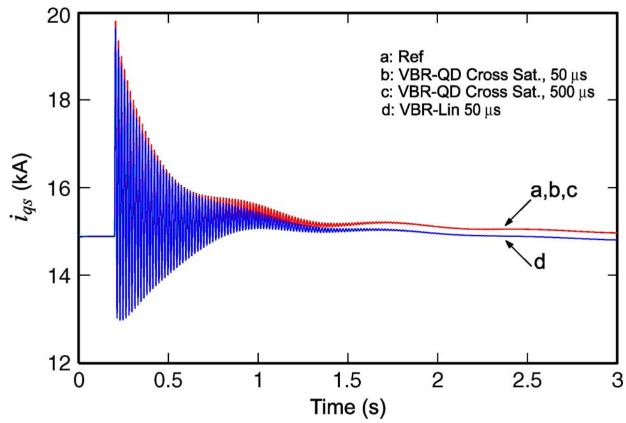

Fig. 12. Response in stator q-axis currents i_{qs} under full load conditions.

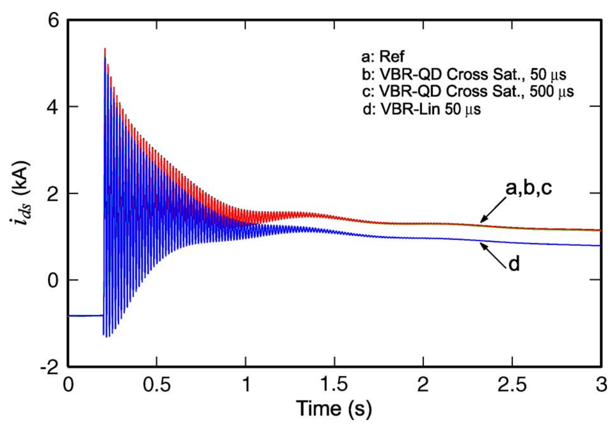

Fig. 13. Response in stator d-axis currents i_{ds} under full load conditions.

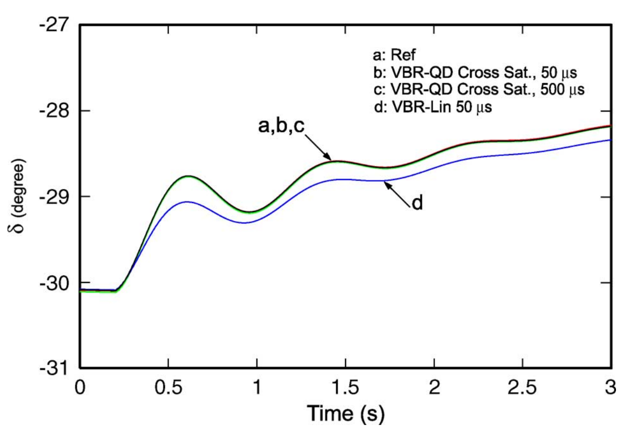

Fig. 14. Response in rotor angles δ under full load conditions.

[10]. This confirms the ability of the proposed saturable VBR model to use a large time step, which is consistent with what has been demonstrated previously for the magnetically linear VBR models [31], [33].

VI. Conclusion

This paper extends the previous work and proposes a magnetically saturable VBR synchronous machine model that achieves a direct interface for EMTP-type solution. The model is very flexible and can be applied to synchronous machines with salient or round rotors. The piecewise linear representation of the saturation curve is used, which provides a convenient means to implement the nonlinear magnetic characteristic in a non-iterative EMTP-type solution. A saliency factor approach is applied, which enables proper representation of the qd-axis cross saturation in steady state and transients with the accuracy sufficient for most power systems studies and is an improvement over considering d-axis saturation only. Case studies demonstrate that proposed saturable VBR synchronous machine model also maintains its numerical accuracy at large time steps. This should further extend the range of application of machine models used in EMTP in terms of maintaining good accuracy for systems studies where a use of large time step may be desirable.

Appendix

D-axis saturation data for the arctangent function representation [13]: $\lambda_T = 43.3962\ \mathrm{V\cdot s}$, $\tau_T = 0.3\ (1/\mathrm{V\cdot s})$, $M_a = 730.58\ (1/\mathrm{H})$, $M_d = 331.165\ (1/\mathrm{H})$.

D-axis saturation data for the two-slope piecewise linear representation: $I_{sat} = 18.2\ \mathrm{kA}$, $L_{mdsat} = 0.9538\ \mathrm{mH}$.

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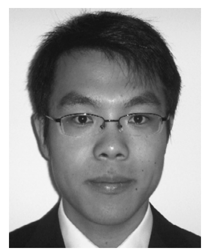

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A Phase-Domain Synchronous Machine Model with Constant Equivalent Conductance Matrix for EMTP-Type Solution

Invited Paper

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***Abstract*—Interfacing machine models in either nodal analysis-based (EMTP-like) or state variable-based transient simulation programs plays an important role in numerical accuracy and computational performance of the overall simulation. As an advantageous alternative to the traditional qd models, a number of advanced phase-domain (PD) and voltage-behind-reactance (VBR) machine models have been recently introduced. However, the rotor-position-dependent conductance matrix in the machine-network interface complicates the use of such models in EMTP. This paper focuses on achieving constant and efficient interfacing circuit for the PD synchronous machine model. It is shown that the machine conductance matrix can be formulated into a constant sub-matrix plus a time-variant sub-matrix. Eliminating numerical saliency from the second term results in a constant conductance matrix of the proposed PD model, which is a very desirable property for the EMTP solution since the re-factorization of the network conductance matrix at every time step is avoided. Case studies demonstrate that the proposed PD model represents a significant improvement over other established models used in EMTP while preserving the accuracy of the original/classical PD model.**

***Index Terms*—Constant conductance matrix, EMTP, G matrix, phase-domain model, qd model, saliency elimination, synchronous machine, voltage-behind-reactance model.**

NOMENCLATURE

Throughout this paper, bold font is used to denote matrix and vector quantities, and italic non-bold font is used to denote scalar quantities.

$\mathbf{v}_{abcs}$, $\mathbf{v}_{qdr}$ stator and rotor voltage vectors

$\mathbf{i}_{abcs}$, $\mathbf{i}_{qdr}$ stator and rotor current vectors

$\boldsymbol{\lambda}_{abcs}$, $\boldsymbol{\lambda}_{qdr}$ stator and rotor flux linkage vectors

$\mathbf{R}_s$, $\mathbf{R}_r$ stator and rotor diag. resistance matrices

$\mathbf{L}_s(\theta_r)$, $\mathbf{L}_r$ stator and rotor self-inductance matrices

$\mathbf{L}_{sr}(\theta_r)$, $\mathbf{L}_{rs}(\theta_r)$ stator and rotor mutual-inductance matrices

θ_r rotor position angle

Δt discretization time-step

$\mathbf{R}_{eq}^{pd}$, $\mathbf{G}_{eq}$ equivalent resistance and conductance matrices

$\mathbf{e}_h^{pd}$, $\mathbf{i}_h^{pd}$ equivalent history voltage and current sources

$\mathbf{e}_{sh}^{pd}$, $\mathbf{e}_{rh}^{pd}$ stator and rotor equivalent history sources

Z''_q, Z''_d equivalent q and d-axis subtransient impedances

Z_{mq}, Z_{md} equivalent q and d-axis magnetizing impedances

Z_{lj}, $j = kq1,\cdots kqM, fd, kd1,\cdots kdN$ equivalent field and damper winding impedances

Z'''_q PD model modified q-axis subtransient impedance with added damper winding.

R_{kqM+1}, L_{lkqM+1} added damper winding resistance and leakage inductance

Z_{lkqM+1} added damper winding impedance

τ_{QM+1} added damper winding pole time constant

f_{fit} fitting frequency

$L_{critical}$ critical leakage inductance for the added damper winding.

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I. INTRODUCTION

MACHINE models for studying the power systems transients are generally based on the qd-axes model formulation. Such models have received extensive attention in the literature and are widely available in many nodal analysis-based electro-magnetic transient programs (i.e. EMTP-type [1]) and state variable-based (e.g. Simulink [2]) simulation packages as built-in library components that are extensively used by many practicing engineers and researchers in industry and academia. Since such general purpose machine models are widely used, improving their numerical efficiency and accuracy can have significant impact and improve many simulation packages.

To improve the interface between the machine model and the external power system network, which is typically represented in physical abc phase coordinates, the so called phase-domain (PD) [3]–[6] and voltage-behind-reactance (VBR) [7]–[9] models have been introduced. For numerically-efficient interfacing of induction and synchronous machine models in state-variable transient simulation programs, a unified constant parameter *RL*-branch equivalent circuit has been recently proposed in [10].

Improving the numerical efficiency and accuracy of the synchronous machine models for the EMTP-type programs has been attractive for a long time. Numerous machine models have been proposed and implemented in various software packages including MicroTran [11], ATP/EMTP [12], PSCAD/EMTDC [13], and EMTP-RV [14], where the qd models are typically used. The main advantage of these methods is that they result in a constant machine conductance sub-matrix. However, predicting relatively fast electrical variables introduces interfacing errors that significantly reduce the numerical accuracy of such models, and may potentially cause convergence problem [5], [8], [9]. In EMTP implementation, the advantages of PD and VBR synchronous machine models generally come at the price of having a rotor-position/speed-dependent machine conductance sub-matrix, which generally requires re-factorization of the entire network conductance $\mathbf{G}$ matrix at every time step as the rotor position changes. A PD and VBR induction machine models that achieves constant conductance sub-matrix have been recently proposed by the authors in [15]. Achieving the same property for the VBR synchronous machine is very challenging due to the saliency and structure of equations [8]. Instead, the main focus and goal of this paper is to propose a new PD synchronous machine model that also achieves a constant conductance sub-matrix in EMTP solution, which to the best of the authors' knowledge, has not been proposed before. Interested reader will find particularly useful the related work summarized in [8],[10],[15]. The properties of the proposed PD model and the additional contributions of this paper are summarized as follows:

- This paper shows that an equivalent resistance sub-matrix of the discretized PD synchronous machine model can be expressed as a constant plus a rotor-position-dependent term.
- This property is used for deriving a model that has constant machine conductance sub-matrix as in [15] (assuming a magnetically linear machine) using approximation of saliency technique [16].
- The proposed constant conductance (CC-PD) model can be achieved for either salient-pole or round rotor machines with appropriately fitted parameters for required accuracy and discretization time step.
- The proposed CC-PD model is shown to maintain very good accuracy even at large time step similar to the classical PD model.

II. Coupled-Circuit Phase-Domain Machine Model

For the purpose of power systems transients, a three-phase electrical machine can be modeled by lumped-parameter coupled circuits in physical variables and abc coordinates, known in EMTP community as the PD model. Without loss of generality, this paper considers a three-phase synchronous machine with one field winding fd, and M and N damper windings in the q and d axes, respectively. The parameters of the example machine considered in the case studies are summarized in Appendix A. All rotor windings are assumed to be referred to the stator side by appropriate turn ratios. Motor convention is used so that the stator currents flowing into the machine have a positive sign in the voltage equations. The flux linkage of each winding is assumed to have the same sign as the current flowing in that winding.

A. Continuous-Time Phase-Domain Model

To set the stage for the further derivations, the PD model in continuous time is briefly included here. The voltage equation in physical variables and phase coordinates is [17]

$$\begin{bmatrix} \mathbf{v}_{abcs} \\ \mathbf{v}_{qdr} \end{bmatrix} = \mathbf{R}\begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix} + p\begin{bmatrix} \boldsymbol{\lambda}_{abcs} \\ \boldsymbol{\lambda}_{qdr} \end{bmatrix} \tag{1}$$

where the resistance matrix is given as

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_s & \\ & \mathbf{R}_r \end{bmatrix} \tag{2}$$

The flux linkage equation is expressed as

$$\begin{bmatrix} \boldsymbol{\lambda}_{abcs} \\ \boldsymbol{\lambda}_{qdr} \end{bmatrix} = \mathbf{L}(\theta_r)\begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}_{qdr} \end{bmatrix} \tag{3}$$

where the stator and rotor self and mutual inductance matrix may be expressed as

$$\mathbf{L}(\theta_r) = \begin{bmatrix} \mathbf{L}_s(\theta_r) & \mathbf{L}_{sr}(\theta_r) \\ \mathbf{L}_{rs}(\theta_r) & \mathbf{L}_r \end{bmatrix} \tag{4}$$

The detailed expressions of the resistance and inductance sub-matrices are given in Appendix B.

The induced electromagnetic torque is calculated in terms of physical variables (currents) as follows:

$$T_e = \frac{P}{2}\left[\frac{1}{2}\mathbf{i}_{abcs}^T \frac{\partial}{\partial\theta_r}\left(\mathbf{L}_s - L_{ls}\mathbf{I}\right)\mathbf{i}_{abcs} + \mathbf{i}_{abcs}^T \frac{\partial \mathbf{L}_{sr}}{\partial\theta_r}\mathbf{i}_{qdr}\right] \tag{5}$$

which can be written in expanded form given in Appendix B, (B9) for simplified implementation instead of the involved matrix product (5). The mechanical equations can be found in [17] and [8], and are not included here due to space limitation.

B. Discretized Phase-Domain Model for EMTP Solution

In order to obtain numerical solution within the EMTP, the PD machine model is discretized using implicit trapezoidal rule [1]. To interface the machine, the discretized model is first expressed in the following general form [8]:

$$\mathbf{v}_{abcs}(t) = \mathbf{R}_{eq}^{pd}\,\mathbf{i}_{abcs}(t) + \mathbf{e}_h^{pd}(t) \tag{6}$$

where the equivalent resistance matrix $\mathbf{R}_{eq}^{pd}$ is expressed as

$$\mathbf{R}_{eq}^{pd}(t) = \mathbf{R}_s + \frac{2}{\Delta t}\mathbf{L}_s(t) - \frac{4}{\Delta t^2}\mathbf{L}_{sr}(t)\left(\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r\right)^{-1}\mathbf{L}_{rs}(t)\,. \quad (7)$$

The history term $\mathbf{e}_h^{pd}(t)$ is represented as

$$\mathbf{e}_h^{pd}(t) = \mathbf{e}_{sh}^{pd}(t) + \frac{2}{\Delta t}\mathbf{L}_{sr}(t)\left(\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r\right)^{-1}\left(\mathbf{v}_{qdr}(t) - \mathbf{e}_{rh}^{pd}(t)\right). \quad (8)$$

where

$$\begin{aligned}\mathbf{e}_{sh}^{pd}(t) &= \left(\mathbf{R}_s - \frac{2}{\Delta t}\mathbf{L}_s(t-\Delta t)\right)\mathbf{i}_{abcs}(t-\Delta t) \\ &\quad - \frac{2}{\Delta t}\mathbf{L}_{sr}(t-\Delta t)\mathbf{i}_{qdr}(t-\Delta t) - \mathbf{v}_{abcs}(t-\Delta t)\end{aligned} \quad (9)$$

and

$$\begin{aligned}\mathbf{e}_{rh}^{pd}(t) &= \left(\mathbf{R}_r - \frac{2}{\Delta t}\mathbf{L}_r\right)\mathbf{i}_{qdr}(t-\Delta t) \\ &\quad - \frac{2}{\Delta t}\mathbf{L}_{rs}(t-\Delta t)\mathbf{i}_{abcs}(t-\Delta t) - \mathbf{v}_{qdr}(t-\Delta t)\end{aligned} \quad (10)$$

The rotor currents can be expressed as

$$\begin{aligned}\mathbf{i}_{qdr}(t) &= \left(\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r\right)^{-1}\Big(\mathbf{v}_{qdr}(t) \\ &\quad - \frac{2}{\Delta t}\mathbf{L}_{rs}(t)\mathbf{i}_{abcs}(t) - \mathbf{e}_{rh}^{pd}(t)\Big)\end{aligned} \quad (11)$$

In the machine interface equation (6), the equivalent resistance matrix $\mathbf{R}_{eq}^{pd}$ is of our particular interest. Based on (7), it is observed that $\mathbf{R}_{eq}^{pd}$ is the sum of three terms: the stator resistance matrix $\mathbf{R}_s$, the stator inductance matrix term $\frac{2}{\Delta t}\mathbf{L}_s(t)$, and the third term: the triple matrix product of $-\frac{4}{\Delta t^2}\mathbf{L}_{sr}(t)\left(\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r\right)^{-1}\mathbf{L}_{rs}(t)$. It is shown in [15] that for induction machines, due to machine's symmetry, the second and third terms are independent of the rotor displacement θ_r. Therefore, matrix $\mathbf{R}_{eq}^{pd}$ is constant for symmetrical induction machines (assuming magnetic saturation is ignored).

In the case of synchronous machines, the stator inductance matrix $\mathbf{L}_s(t)$ is rotor-position-dependent due to salient-pole rotor. The third term, $\frac{4}{\Delta t^2}\mathbf{L}_{sr}(t)\left(\mathbf{R}_r + \frac{2}{\Delta t}\mathbf{L}_r\right)^{-1}\mathbf{L}_{rs}(t)$, in (7) is also rotor-position-dependent since the machine's geometry and winding parameters in *q*- and *d*-axis are no longer the same as in induction machines.

However, as it turns out (see Appendix C, Theorem 1) that for the synchronous machine with salient or round rotor, the equivalent resistance matrix (7) can be expressed as a constant plus a rotor-position-dependent part, as follows:

$$\mathbf{R}_{eq}^{pd}(t) = \mathbf{R}_0 + \frac{1}{3}\left(Z_q'' - Z_d''\right)\begin{bmatrix} \cos 2\theta_r & \cos 2(\theta_r - \frac{\pi}{3}) & \cos 2(\theta_r + \frac{\pi}{3}) \\ \cos 2(\theta_r - \frac{\pi}{3}) & \cos 2(\theta_r - \frac{2\pi}{3}) & \cos 2\theta_r \\ \cos 2(\theta_r + \frac{\pi}{3}) & \cos 2\theta_r & \cos 2(\theta_r + \frac{2\pi}{3}) \end{bmatrix} \quad (12)$$

where $\mathbf{R}_0$ represents the constant part. Moreover, in (12) the equivalent discrete-time *q* and *d*-axis subtransient impedances Z_q'' and Z_d'' are defined as

$$Z_q'' = \left(\left(Z_{mq}\right)^{-1} + \sum_{i=1}^{M}\left(Z_{lkqi}\right)^{-1}\right)^{-1} \quad (13)$$

$$Z_d'' = \left(\left(Z_{md}\right)^{-1} + \left(Z_{lfd}\right)^{-1} + \sum_{i=1}^{N}\left(Z_{lkdi}\right)^{-1}\right)^{-1} \quad (14)$$

with

$$Z_{mq} = \frac{2}{\Delta t}L_{mq} \quad \text{and} \quad Z_{md} = \frac{2}{\Delta t}L_{md} \quad (15)$$

and

$$Z_{lj} = R_j + \frac{2}{\Delta t}L_{lj} \ \text{ where } \ j = kq1, \cdots kqM,\ fd, kd1, \cdots kdN. \quad (16)$$

The equivalent circuits depicting the discrete-time domain subtransient impedances Z_q'' and Z_d'' are illustrated in Fig. 1. In general, the term $Z_q'' - Z_d''$ in (12) is non zero for either round or salient rotor machines and it is not to be confused with dynamic or rotor saliency of the machine. However, this numerical saliency, $Z_q'' \neq Z_d''$ in (12), makes matrix $\mathbf{R}_{eq}^{pd}$ also rotor-position-dependent. By analogy, the unequal *q*- and *d*-axes subtransient inductances/reactances make the equivalent inductance matrix rotor-position-dependent in the continuous time synchronous machine VBR model [7].

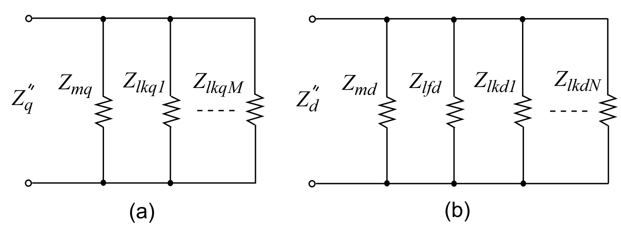

Fig. 1. Equivalent subtransient impedances in discrete-time domain: (a) *q*-axis, Z_q''; and (b) *d*-axis, Z_d''.

III. Constant Equivalent Conductance Matrix

To interface the machine model with the external EMTP network, the machine branch voltages (6) are replaced by the nodal equation as

$$\mathbf{G}_{eq}\mathbf{v}_{abcs} = \mathbf{i}_{abcs} + \mathbf{i}_h^{pd} \quad (17)$$

Here, the equivalent conductance matrix is calculated as

$$\mathbf{G}_{eq} = \left[\mathbf{R}_{eq}^{pd}\right]^{-1} \quad (18)$$

This 3-by-3 sub-matrix $\mathbf{G}_{eq}$ is then inserted into the overall network conductance matrix $\mathbf{G}$, which forms the standard nodal equation, $\mathbf{G}\mathbf{V}_n = \mathbf{I}_h$. The machine history current sources, $\mathbf{i}_h^{pd} = \mathbf{G}_{eq}\mathbf{e}_h^{pd}$, is then injected into the nodes corresponding to the machine terminals (appropriately included into $\mathbf{I}_h$). Moreover, for efficient EMTP solution, it is desirable to have matrix $\mathbf{G}$ constant [1]. The main focus of this Section is to achieve a constant machine equivalent conductance matrix $\mathbf{G}_{eq}$ for the conventional PD models [4],[5],[8].

A. Elimination of Numerical Saliency

The approach of removing this numerical saliency in (12) is based on a pioneering technique introduced in [16], which adds one additional artificial damper winding to one of the axis with the purpose of achieving a certain numerical property. Otherwise, this additional winding has no physical significance or relationship to the actual machine parameters. In particular, depending on the relative values of Z''_q and Z''_d, the artificial winding should be added to the axis with the highest discrete-time domain subtransient impedance. Observing that the winding parameters in (13) and (14) are asymmetrical, it is typical that the d-axis has the lowest impedance (see [10], Section VII). Without loss of generality, it is assumed that a new $M+1$ damper winding with resistance R_{kqM+1} and leakage inductance L_{lkqM+1} is added to the q-axis. To distinguish the subtransient quantities of the new model that uses the ($M+1$)-th winding, and for consistency with [10], we use the triple-prime sign (‴) instead of double-prime sign (″). The new equivalent subtransient impedance in the q-axis, including the added winding, becomes

$$Z'''_q = \left(\left(Z_{mq}\right)^{-1} + \sum_{i=1}^{M} \left(Z_{lkqi}\right)^{-1} + \left(Z_{lkqM+1}\right)^{-1} \right)^{-1} \tag{19}$$

where

$$Z_{lkqM+1} = R_{kqM+1} + \frac{2}{\Delta t} L_{lkqM+1} \tag{20}$$

The parameters R_{kqM+1} and L_{lkqM+1} should be carefully selected such that

$$Z'''_q = Z''_d . \tag{21}$$

Condition (21) removes the rotor-position-dependent term in (12) and gives the desired result:

$$\mathbf{G}_{eq} = \left[\mathbf{R}_{eq}^{pd}\right]^{-1} = \left[\mathbf{R}_0\right]^{-1} = const. \tag{22}$$

The new constant conductance phase-domain (CC-PG) model is then implemented similar to the conventional PD model [8], except that $\mathbf{R}_{eq}^{pd}$ and $\mathbf{G}_{eq}$ are calculated only once using (7) and (22).

B. Calculation of Added Winding Parameters

It is noted that discrete-time domain impedances Z''_q and Z''_d are dependent on the machine parameters (resistances and inductances) as well as the discretization time step Δt [see (13) - (16)]. Therefore, based on (13), (14) and (20), (21), the added winding equivalent impedance in discrete time can be calculated as

$$Z_{lkqM+1} = \left[\left(Z''_d\right)^{-1} - \left(Z''_q\right)^{-1}\right]^{-1} \tag{23}$$

However, (23) and (20) do not uniquely define the finding parameters for a given time step Δt. Moreover, as it has been shown in [16], adding this extra damper winding also changes the q-axis operational impedance by adding a corresponding pole (and zero) to the respective transfer function [17, Chap. 7]. In order to preserve a good accuracy in predicting the machine's transients, it is important to select the additional winding parameters such that the low frequency response is preserved, and the effect of extra winding may appear visible only in high frequencies that are above the range of interest - fit frequency, $f_{fit} \approx 120Hz$ [16]. Assuming the same guidelines as in [16], the added pole time constant should be placed about ten times higher than the fit frequency. That is,

$$\tau_{QM+1} \le \frac{1}{2\pi \cdot 10 f_{fit}} . \tag{24}$$

Furthermore, using the analysis of machine time constants [17, Chap. 7], the corresponding time constant can be also calculated as

$$\tau_{QM+1} = \frac{L_{lkqM+1}}{R_{kqM+1}} + \frac{1}{R_{kqM+1}} \left(\left(L_{mq}\right)^{-1} + \sum_{i=1}^{M} \left(L_{lkqi}\right)^{-1} \right)^{-1} \tag{25}$$

Combining (24) and (25), we get

$$R_{kqM+1} \ge 20\pi \cdot f_{fit} \cdot L_{lkqM+1} + + 20\pi \cdot f_{fit} \cdot \left(\left(L_{mq}\right)^{-1} + \sum_{i=1}^{M} \left(L_{lkqi}\right)^{-1} \right)^{-1} \tag{26}$$

Substituting (26) into (20), we can define a critical value for the leakage inductance as

$$L_{critical} = \frac{Z_{lkqM+1} - 20\pi \cdot f_{fit} \cdot \left(\left(L_{mq}\right)^{-1} + \sum_{i=1}^{M} \left(L_{lkqi}\right)^{-1} \right)^{-1}}{20\pi \cdot f_{fit} + 2/\Delta t} \tag{27}$$

Thereafter, the additional winding leakage inductance and resistance are calculated as follows:

$$L_{lkqM+1} \le L_{critical} \tag{28}$$

$$R_{kqM+1} = Z_{lkqM+1} - \frac{2}{\Delta t} L_{lkqM+1} \tag{29}$$

For the purpose of this paper, it was assumed that $L_{lkqM+1} = L_{critical}$. The corresponding parameters calculated for different time steps are included in Appendix A.

IV. Model Computational Efficiency Analysis

In order to achieve fastest possible simulation speed, the numerical implementation of the machine model has to be carefully optimized. For the conventional PD and CC-PD models considered in this paper, we consider all possible computational savings, including efficient calculation of trigonometric functions and utilization of symmetrical properties of machine matrices. It is observed in Section II and Appendix B that PD models require evaluation of trigonometric functions for the calculation of resistance matrix, history terms and electromagnetic torque. More specifically, the models need evaluating $\cos(2\theta_r)$ and $\cos(2\theta_r \pm 2\pi/3)$ terms in $\mathbf{L}_s(\theta_r)$; then terms $\sin(\theta_r)$, $\sin(\theta_r \pm 2\pi/3)$, $\cos(\theta_r \pm 2\pi/3)$ in $\mathbf{L}_{sr}(\theta_r)$; and $\sin(2\theta_r)$ in $T_e(\theta_r)$. The evaluation of trigonometric functions depends on many factors, including the CPU hardware and low-level software implementation of trigonometric functions on a specific computer platform. Usually high order polynomial approximation or look-up table methods are used to evaluate these functions [19]. For example, the studies conducted on a typical Intel-based PC with Microsoft Windows Operating System show that evaluation of a single trigonometric function may cost about twenty equivalent flops, where one flop is defined as one addition, subtraction, multiplication, or division of two floating-point numbers [20].

As the evaluation of trigonometric functions is in general more expensive than performing simple additions and multiplications, efficient use of trigonometric functions is important. In this paper, an efficient calculation of the required trigonometric functions is implemented similar to the method used in [21]. Specifically, instead of calculating these functions directly by calling $\cos(\cdot)$ and $\sin(\cdot)$ each time for different arguments, only two functions $\cos(\theta_r)$ and $\sin(\theta_r)$ for a given angle θ_r are evaluated. Thereafter, the remaining trigonometric functions are calculated indirectly using identities:

$$\cos(2\theta_r) = 2\cos^2\theta_r - 1 \tag{30}$$

$$\sin(2\theta_r) = 2\cos\theta_r \sin\theta_r \tag{31}$$

$$\cos(2\theta_r \pm 2\pi/3) = -1/2\cos 2\theta_r \mp \sqrt{3}/2\sin 2\theta_r \tag{32}$$

$$\cos(\theta_r \pm 2\pi/3) = -1/2\cos\theta_r \mp \sqrt{3}/2\sin\theta_r \tag{33}$$

$$\sin(\theta_r \pm 2\pi/3) = -1/2\sin\theta_r \pm \sqrt{3}/2\cos\theta_r \tag{34}$$

Evaluation of trigonometric functions using (30) - (34) requires much less computational efforts compared with the direct evaluation of these functions. For example, the required number of equivalent flops using both direct and indirect approaches is summarized in Table I. Since indirect calculation achieves significant computational savings, it is therefore considered in the final implementation of both conventional PD and CC-PD models.

Next, the computational costs of PD and CC-PD models are compared in terms of flop counts for a single time-step. A typical salient-pole hydro turbine synchronous machine is used wherein the machine parameters are given in Appendix A and [17, see Table 5.10-1]. The considered generator model has the field winding *fd*, and two damper windings, *kd* and *kq1*, respectively. The CC-PD model is formulated according to Section III with one additional damper winding *kq2* in *q*-axis. Thus, the dimension of the rotor matrices and variables in the CC-PD model is higher than those in the original PD model. To maximize the numerical efficiency of both models, all constant terms and coefficient are pre-calculated outside of the main time stepping loop of the simulation. For example, the constant coefficients in the machine resistance matrix (12) are pre-calculated before the time-step loop. For efficient calculation of electromagnetic torque in PD models, instead of the direct calculation using (5), the expanded form of torque equation (B9) in Appendix B is used. This reduces the number of flops since the two triple matrix products in (5) are avoided and minimizes the additional flops due to the added damper winding (see Table II).

After utilization of the efficient method for evaluation of trigonometric functions described in this Section, the total numbers of flops required by the PD and CC-PD models are summarized in Table II. For better comparison and understanding of the computational costs, the flops are roughly assigned to several terms of the respective models. It is observed in Table II that the CC-PD model requires no flops for the evaluation of $\mathbf{R}_{eq}$ in time stepping loop. However, the calculation of history term $\mathbf{e}_h$ and rotor current $\mathbf{i}_{qdr}$ make the constant CC-PD model slightly more costly than the conventional PD model due to the higher dimension of the rotor matrices and one extra rotor current. As a result, the CC-PD model requires more flops per time step compared to the conventional PD model (295 flops vs. 251 flops).

TABLE I:
COMPARISON OF TRIG FUNCTIONS IMPLEMENTATIONS

Method	Direct	Indirect
Functions	$\cos\theta$, $\sin\theta$ $\cos(2\theta)$, $\sin(2\theta)$ $\cos(2\theta \pm 2\pi/3)$ $\cos(\theta \pm 2\pi/3)$ $\sin(\theta \pm 2\pi/3)$	$\cos\theta$, $\sin\theta$ plus additional operations $10\otimes + 7\oplus$
Total cost	10 trigs + 5 flops	2 trigs + 17 flops
Equiv. flops	205	57

Here $\otimes$ and $\oplus$ denote multiplication and addition/subtraction operations.

TABLE II
FLOPS AND TRIG FUNCTIONS COUNT PER TIME STEP

Conventional PD Model			CC-PD Model		
$\mathbf{L}_s(\theta_r)$ $\mathbf{L}_{sr}(\theta_r)$ $\mathbf{L}_{rs}(\theta_r)$	flop: 33	trig: 2	$\mathbf{L}_s(\theta_r)$ $\mathbf{L}_{sr}(\theta_r)$ $\mathbf{L}_{rs}(\theta_r)$	flop: 33	trig: 2
$\mathbf{R}_{eq}^{PD}$	9	-	$\mathbf{R}_{eq}^{CC-PD}$	0	-
$\mathbf{e}_h^{PD}$	90	-	$\mathbf{e}_h^{CC-PD}$	130	-
$\mathbf{i}_{qdr}^{PD}$	29	-	$\mathbf{i}_{qdr}^{CC-PD}$	41	-
T_e^{PD}	50	-	T_e^{CC-PD}	51	-
Total	211	2	Total	255	2
Equiv. flops	251		Equiv. flops	295	

V. Case Studies

Since the main idea of this paper is about the new CC-PD machine model rather than the power network solution, and due to the limited space, the presented studies and discussion are focused on a synchronous-machine infinite-bus system. The proposed model has been implemented according to the EMTP solution methodology. For comparison purpose, a state variable $qd0$ synchronous machine model [17] has also been implemented in MATLAB/Simulink. This model is solved using fourth-order Runge-Kutta method with a very small time step of $1\mu s$ to obtain very accurate numerical solutions, which are used as reference. To demonstrate the improved numerical accuracy of the proposed CC-PD model over the conventional EMTP-type $qd0$ model, we used the build-in EMTP-RV's synchronous machine model as a benchmark case.

The salient-pole hydro turbine generator with parameters given in Appendix A has been used in all studies. To investigate the accuracy of the proposed PD model, one may consider a number of case studies, e.g., short-circuit of a fully loaded machine, a case of machine with large dynamic saliency, pole-slipping events, etc. However, such transient events excite mostly the low frequency modes where the approximation introduced by additional fictitious damper winding should not have any visible impact. The transient studies of a three-phase fault have been previously considered in [8], and are not included here due to space limitations. Therefore, for the purpose of this paper, a single-phase-to-ground fault study is chosen as this even excites higher frequencies. In the following transient study, the generator initially operates with no load. At $t = 0.002s$, a single-phase-to-ground fault is applied at the machine terminals (phase a). The transient responses produced by various models are plotted in Figs. 2 through 5. Such studies have been previously considered in [9] and may be very useful for benchmarking and/or validating various models.

A. Model Verification using Small Time-Step

The considered study of an unbalanced fault is first simulated using a relatively small time step, $\Delta t = 50\mu s$. The resulting transients of the field current i_{fd}, stator currents i_{as}, i_{bs}, and electromagnetic torque T_e are depicted in Fig. 2. The simulation results obtained by the EMTP-RV, the PD model [8] and the proposed CC-PD models are overlaid with the reference solutions. As can be observed from Figs. 2, the transient responses produced by all models coincide and converge to the reference solutions. This clearly demonstrates that these models are all equivalent and predict the unbalanced operations with acceptable accuracy for the given time step, which is an expected result.

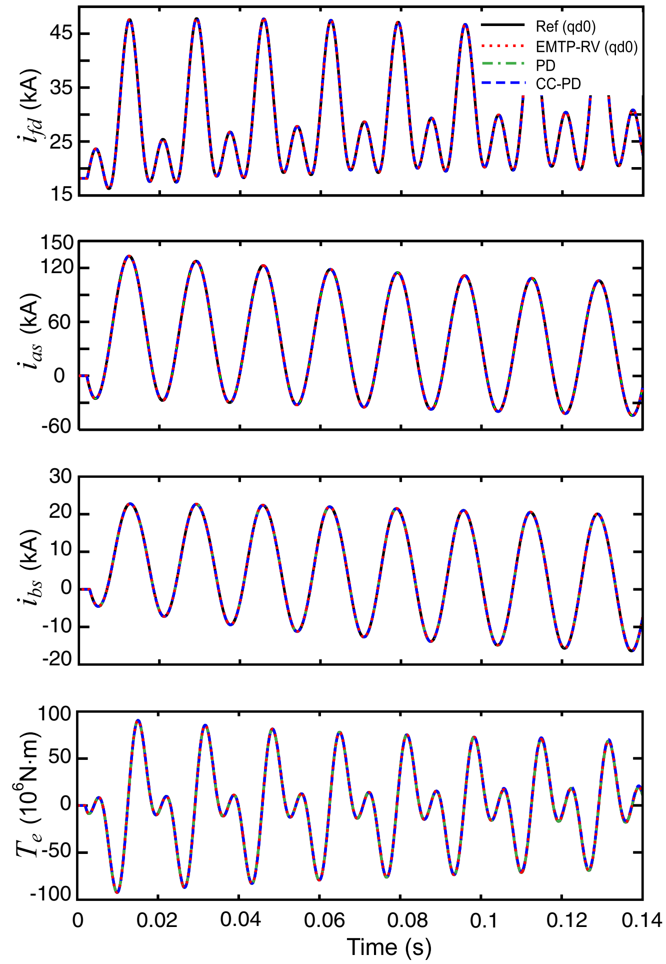

Fig. 2. Field and phase currents, and electromagnetic torque for the single-phase fault study using time-step of $50\mu s$, as predicted by various models.

B. Model Verification using Large Time-Step

To further compare the numerical properties and robustness of the new CC-PD model, a larger time step, $\Delta t = 1ms$, is applied to the same test case. The same variables are plotted in Figs. 3 – 5. As can be observed in Fig. 3, the results produced by the various models visibly diverge from the reference solution. The transient responses obtained by EMTP-RV(qd0) model (red dotted line) diverge from the reference solutions with the largest error among the considered models. The problem of convergence and accuracy of qd models is attributed to their interface with the external network, and is discussed in detail in [8], [9], [22]. At the same time, the PD and CC-PD models still produce reasonably accurate and convergent simulation results.

A more detailed fragment of the transient observed in i_{fd} is shown in Fig. 4, which is a magnified view of Fig. 3 (first subplot). A magnified view of the phase current i_{bs} is shown in Fig. 5, which is a detailed view of Fig. 3 (third subplot). As Figs. 4 and 5 show, the point-by-point solutions obtained by the PD and CC-PD models are almost identical and are reasonably close to the reference solution. The detailed analysis of other variables shows similar result.

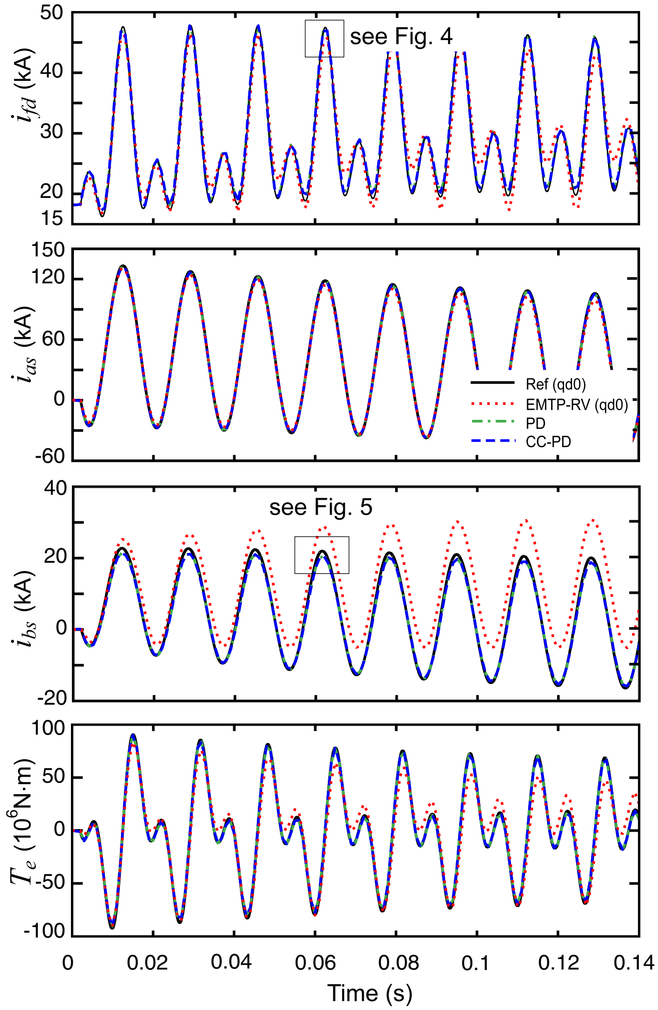

Fig. 3. Field and phase currents, and torque for the single-phase fault study using time-step of $1ms$, as predicted by various models.

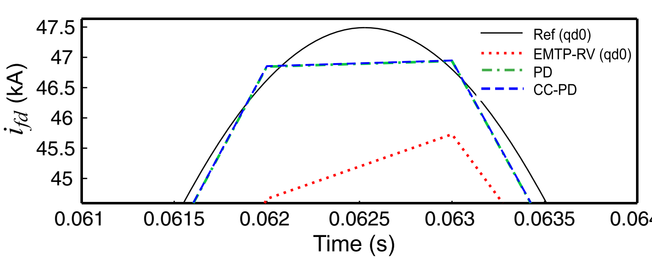

Fig. 4. Magnified view of field current for the single-phase fault study using time-step of $1ms$, as predicted by various models.

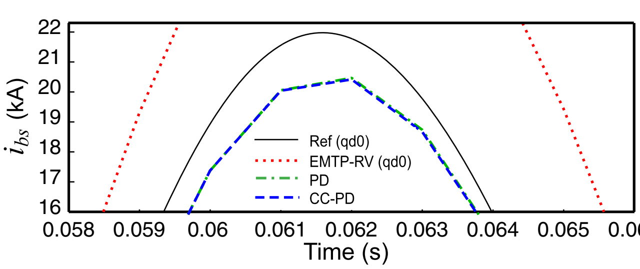

Fig. 5. Magnified view of phase b current for the single-phase fault study using time-step of $1ms$, as predicted by various models.

VI. Discussion

A. Choice of Added Winding Parameters

The idea of adding a fictitious damper winding to eliminate the dynamic saliency was originated in [16]. This approach is used here for the proposed CC-PD model for EMTP-type solution. It is noted that Z_{lkqM+1} in (20) consists of two variables, R_{kqM+1} and L_{lkqM+1}, that need to be determined. These parameters should be determined subject to satisfying the constraint (23). At the same time, the q-axis frequency response in the low-frequency range should remains the same as it was in the original machine. Observing (20), one can see that for a given value of Z_{lkqM+1} [since (23) is a constraint], choosing smaller L_{lkqM+1} results in larger R_{kqM+1}. Also, as can be observed from (25), the larger winding resistance R_{kqM+1} makes the additional pole time constant τ_{QM+1} very small, which pushes the corner frequency of this pole far away from the fitting frequency f_{fit}. Therefore, satisfying the inequality (28), and then choosing R_{kqM+1} according to (29), ensures that the added winding does not affect the machine's response in frequency-domain as well as in time-domain in the range of interest.

Interested reader can find very insightful discussions in [16] and [10] regarding the added winding parameter choice for implementing the synchronous machine model (in continuous time-domain) for state-variable-based simulation packages. Therein, increasing the winding resistance R_{kqM+1}, although improves the model accuracy, also makes the machine model numerically stiff and possibly hard to solve/simulate, if explicit integration method is used. However, for the EMTP solution, this does not represent a problem since the equations are discretized at the branch level using the implicit (stiffly stable) trapezoidal rule. So, in general one can choose small L_{lkqM+1} to satisfy the constraint (28), and then choose very large resistance R_{kqM+1} so that the added winding effectively appear open-circuited, which minimizes its impact.

B. Model Accuracy

The PD model in general improves the numerical accuracy compared with the conventional $qd0$ models that are implemented in EMTP-RV, PSCAD, ATP/EMTP, etc. This is the result of the direct interface of the machine’s stator circuit in abc phase coordinates with EMTP network solution. The proposed CC-PD model possesses the same interfacing accuracy as the conventional PD model. The detailed accuracy comparisons of various machine models, such as $qd0$, PD and VBR, are documented in references [8], [9], [21], [22]. The numerical accuracy of the proposed CC-PD model is also verified by the simulation results in Figs. 2 through 5. Moreover, since the added damper winding has no effect (error) in steady state and only minimally impacts the transient response in very high frequencies due to relatively large value of the winding resistance, the proposed CC-PD model is expected to give very good results for most practical cases. The winding approximation is expected to give very good results for larger machines and the studies with predominantly slow transients that do not have high frequencies (e.g. three-phase faults and pole slipping of large generators, etc.). The study presented in this paper demonstrates a single-phase-to-ground fault (which is richer in higher frequencies), wherein the proposed CC-PD model produces results that are almost identical to the conventional PD model even at relatively large time step (see Figs. 4 and 5). However, potentially, the area where the proposed CC-PD model may be less accurate could be for example small machines driven by switching converters (i.e. wherever the high-frequency transients are excited). For such cases, the fitting frequency f_{fit} will have to be increased according the needs of the study.

C. Network Solution Impact

Although the CC-PD model has slightly higher flop count per step (295 flops vs. 251 flops, see Table II) than conventional PD, its major advantage is that it enables very efficient EMTP time stepping network solution. For the conventional PD model, the time-variant machine conductance sub-matrix forces the entire network **G** matrix to be re-factorized at each time step. Although, the use of sparse matrix techniques such as partial matrix factorization and optimal node ordering methods can alleviate the impact of the time-varying **G** matrix [23], [24], a constant conductance matrix is always preferred for efficient EMTP network solution. Interested reader can find case studies documented in [5] that compare the calculation times and the numbers of triangulations when using the conventional PD model and the classical *qd* model (Type-59 synchronous machine model in EMTP) with constant G-matrix. Therein, for a 190-bus 3-generator system, a ratio of 308.3/70.1 (about 4.4 times) of increased calculation time due to re-triangulation was reported [5]. However, with the proposed CC-PD model, the need for re-triangulation of the network **G** matrix is removed. At the same time, just like the conventional PD model, the CC-PD model preserves the direct interface of the machine electrical variables within the EMTP network solution. It is also understood that the computational speed-up due to avoiding the network **G** matrix re-triangulation will be system and case dependent. In general, the larger the power system network is, the better simulation speedup from using the proposed CC-PD over the conventional PD model may be anticipated. These are indeed very interesting issues that would also shift the focus of publication toward power systems. Therefore, in the near future, the authors would like to take this issue to a separate dedicated publication in appropriate journal.

VII. Conclusions

In this paper, we have presented a new phase-domain synchronous machine model for EMTP-type solution. The new model has a constant equivalent conductance matrix in the discrete-time machine-network interface equation, which is achieved by eliminating the dynamic saliency existing in the equivalent resistance matrix of the original phase-domain model. This constant equivalent conductance matrix is desirable since it avoids the re-factorization of the network conductance matrix at every time step. This feature may make the new model very attractive for possible use in various EMTP packages. A case study of salient-pole synchronous machine has been performed to demonstrate the effectiveness of the proposed constant equivalent conductance phase-domain model. The large time step study has shown that the numerical accuracy of the proposed model is well preserved compared with the original PD model and represents significant improvement over the conventional *qd* models implemented in commercial EMTP-type software packages such as EMTP-RV.

Appendix A

Parameters of synchronous hydro turbine generator [17]: 325 MVA; 20 kV; 0.85 pf; 64 poles; 112.5 r/min; $H = 7.5s$; $J = 35.1\times 10^6 kg\cdot m^2$; $\omega_b = 376.99 rad/s$.

Nominal circuit parameters: $R_s = 0.00234\Omega$; $X_{ls} = 0.1478\Omega$; $X_q = 0.5911\Omega$; $X_d = 1.0467\Omega$;
$R_{fd} = 0.0005\Omega$; $X_{lfd} = 0.2523\Omega$;
$R_{kd} = 0.01736\Omega$; $X_{lkd} = 0.1970\Omega$;
$R_{kq1} = 0.01675\Omega$; $X_{lkq1} = 0.1267\Omega$;

Additional winding parameters:
a) fitted at $\Delta t = 50\mu s$: $R_{kq2} = 1{,}827.7\Omega$; $X_{lkq2} = 91.29\Omega$.
b) fitted at $\Delta t = 1000\mu s$: $R_{kq2} = 38.13\Omega$; $X_{lkq2} = 1.8081\Omega$.

Appendix B

Matrices and block sub-matrices used in the model:

$$\mathbf{R}_s = diag\left(R_s \quad R_s \quad R_s\right) \tag{B1}$$

$$\mathbf{R}_r = diag\left[\mathbf{R}_{rq} \quad \mathbf{R}_{rd}\right] = diag\left[diag\left(R_{kq1} \quad \cdots \quad R_{kqM}\right) \quad diag\left(R_{fd} \quad R_{kd1} \quad \cdots \quad R_{kdN}\right)\right] \tag{B2}$$

$$\mathbf{L}_s(\theta_r)=\begin{bmatrix} L_{ls}+L_A-L_B\cos 2\theta_r & -\frac{1}{2}L_A-L_B\cos 2(\theta_r-\frac{\pi}{3}) & -\frac{1}{2}L_A-L_B\cos 2(\theta_r+\frac{\pi}{3}) \\ -\frac{1}{2}L_A-L_B\cos 2(\theta_r-\frac{\pi}{3}) & L_{ls}+L_A-L_B\cos 2(\theta_r-\frac{2\pi}{3}) & -\frac{1}{2}L_A-L_B\cos 2(\theta_r+\pi) \\ -\frac{1}{2}L_A-L_B\cos 2(\theta_r+\frac{\pi}{3}) & -\frac{1}{2}L_A-L_B\cos 2(\theta_r+\pi) & L_{ls}+L_A-L_B\cos 2(\theta_r+\frac{2\pi}{3}) \end{bmatrix} \tag{B3}$$

where $L_A=\dfrac{L_{md}+L_{mq}}{3}$ and $L_B=\dfrac{L_{md}-L_{mq}}{3}$ (B4)

Mutual inductance matrices:

$$\mathbf{L}_{sr}(\theta_r)=\left\lfloor \mathbf{L}_{srq}(\theta_r) \quad \mathbf{L}_{srd}(\theta_r)\right\rfloor \text{ and } \mathbf{L}_{rs}(\theta_r)=\frac{2}{3}\left(\mathbf{L}_{sr}(\theta_r)\right)^T=\frac{2}{3}\begin{bmatrix}\mathbf{L}_{srq}(\theta_r)\\ \mathbf{L}_{srd}(\theta_r)\end{bmatrix} \tag{B5}$$

where $\mathbf{L}_{srq}(\theta_r)=\begin{bmatrix} L_{mq}\cos\theta_r & \cdots & L_{mq}\cos\theta_r \\ L_{mq}\cos(\theta_r-\frac{2\pi}{3}) & \cdots & L_{mq}\cos(\theta_r-\frac{2\pi}{3}) \\ L_{mq}\cos(\theta_r+\frac{2\pi}{3}) & \cdots & L_{mq}\cos(\theta_r+\frac{2\pi}{3}) \end{bmatrix}_{3\times M}$ and $\mathbf{L}_{srq}(\theta_r)=\begin{bmatrix} L_{md}\sin\theta_r & \cdots & L_{md}\sin\theta_r \\ L_{md}\sin(\theta_r-\frac{2\pi}{3}) & \cdots & L_{md}\sin(\theta_r-\frac{2\pi}{3}) \\ L_{md}\sin(\theta_r+\frac{2\pi}{3}) & \cdots & L_{md}\sin(\theta_r+\frac{2\pi}{3}) \end{bmatrix}_{3\times(N+1)}$ (B6)

Rotor inductance matrix and its blocks:

$$\mathbf{L}_r=diag\left(\mathbf{L}_{rq} \quad \mathbf{L}_{rd}\right) \tag{B7}$$

where $\mathbf{L}_{rq}=\begin{bmatrix} L_{lkq1}+L_{mq} & L_{mq} & \cdots & L_{mq} \\ L_{mq} & L_{lkq2}+L_{mq} & L_{mq} & \vdots \\ \vdots & L_{mq} & \ddots & L_{mq} \\ L_{mq} & \cdots & L_{mq} & L_{lkqM}+L_{mq} \end{bmatrix}_{M\times M}$ and $\mathbf{L}_{rd}=\begin{bmatrix} L_{lfd}+L_{md} & L_{md} & \cdots & L_{md} \\ L_{md} & L_{lkd1}+L_{md} & L_{md} & \vdots \\ \vdots & L_{md} & \ddots & L_{md} \\ L_{md} & \cdots & L_{md} & L_{lkdN}+L_{md} \end{bmatrix}_{(N+1)\times(N+1)}$ (B8)

$$T_e=\frac{P}{2}\left\{\begin{array}{l} \frac{L_{md}-L_{mq}}{3}\left[\left(i_{as}^2-\frac{1}{2}i_{bs}^2-\frac{1}{2}i_{cs}^2-i_{as}i_{bs}-i_{as}i_{cs}-i_{bs}i_{cs}\right)\sin(2\theta_r)+\frac{\sqrt{3}}{2}\left(i_{bs}^2-i_{cs}^2-2i_{as}i_{bs}+2i_{as}i_{cs}\right)\cos(2\theta_r)\right] \\ -L_{mq}\left(\sum_{x=1}^{M}i_{kqx}\right)\left[\left(i_{as}-\frac{1}{2}i_{bs}-\frac{1}{2}i_{cs}\right)\sin(\theta_r)-\frac{\sqrt{3}}{2}(i_{bs}-i_{cs})\cos(\theta_r)\right]+L_{md}\left(i_{fd}+\sum_{x=1}^{N}i_{kdx}\right)\left[\left(i_{as}-\frac{1}{2}i_{bs}-\frac{1}{2}i_{cs}\right)\cos(\theta_r)+\frac{\sqrt{3}}{2}(i_{bs}-i_{cs})\sin(\theta_r)\right] \end{array}\right\} \tag{B9}$$

APPENDIX C

Theorem 1: For synchronous machine with arbitrary number of damper windings, the equivalent resistance matrix $\mathbf{R}_{eq}^{pd}$ can be expressed in the form of (12):

Proof: Without loss of generality, a synchronous machine with two damping windings *kq1*, *kq2* in *q* axis, one field winding *fd* and one damping winding *kd* in *d* axis is assumed. The proof goes through the following five steps:

Step 1:

The triple matrix product in (7) is reformulated using the machine resistance and inductance matrices (B2), (B5)–(B8) into two separate parts, for *q* and *d* axis components as

$$\mathbf{L}_{sr}(t)\left(\mathbf{R}_r+\frac{2}{\Delta t}\mathbf{L}_r\right)^{-1}\mathbf{L}_{rs}(t)=\mathbf{L}_{srq}(t)\left(\mathbf{R}_{rq}+\frac{2}{\Delta t}\mathbf{L}_{rq}\right)^{-1}\mathbf{L}_{rsq}(t)+\mathbf{L}_{srd}(t)\left(\mathbf{R}_{rd}+\frac{2}{\Delta t}\mathbf{L}_{rd}\right)^{-1}\mathbf{L}_{rsd}(t) \tag{C1}$$

Step 2:

The matrix inverse in the first term (the *q*-axis term) of (C1) can be expressed as

$$\left(\mathbf{R}_{rq}+\frac{2}{\Delta t}\mathbf{L}_{rq}\right)^{-1}=\begin{bmatrix} Z_{lkq1}+Z_{mq} & Z_{mq} \\ Z_{mq} & Z_{lkq2}+Z_{mq} \end{bmatrix}^{-1} \tag{C2}$$

The matrix inverse of (C2) can be further solved using Theorem 2 in Appendix D as

$$\begin{bmatrix} Z_{lkq1}+Z_{mq} & Z_{mq} \\ Z_{mq} & Z_{lkq2}+Z_{mq} \end{bmatrix}^{-1} = \begin{bmatrix} \dfrac{1+Z_{mq}Z_{lkq2}^{-1}}{Z_{lkq1}D_{2q}} & -\dfrac{Z_{mq}}{Z_{lkq1}Z_{lkq2}D_{2q}} \\ -\dfrac{Z_{mq}}{Z_{lkq2}Z_{lkq1}D_{2q}} & \dfrac{1+Z_{mq}Z_{lkq1}^{-1}}{Z_{lkq2}D_{2q}} \end{bmatrix} \quad \text{(C3)}$$

where $D_{2q} = 1+\dfrac{Z_{mq}}{Z_{lkq\Sigma 2}}$ and $Z_{lkq\Sigma 2} = \left(\left(Z_{lkq1}\right)^{-1} + \left(Z_{lkq2}\right)^{-1}\right)^{-1}$ (C4)

Using (C2)–(C4), the first term of (C1) can be expressed as:

$$\mathbf{L}_{srq}(t)\left(\mathbf{R}_{rq}+\frac{2}{\Delta t}\mathbf{L}_{rq}\right)^{-1}\mathbf{L}_{rsq}(t) = \frac{2}{3}\frac{L_{mq}^2}{Z_{kq\Sigma 2}+Z_{mq}} \begin{bmatrix} \cos^2\theta_r & \cos\theta_r\cos(\theta_r-\frac{2\pi}{3}) & \cos\theta_r\cos(\theta_r+\frac{2\pi}{3}) \\ \cos(\theta_r-\frac{2\pi}{3})\cos\theta_r & \cos^2(\theta_r-\frac{2\pi}{3}) & \cos(\theta_r-\frac{2\pi}{3})\cos(\theta_r+\frac{2\pi}{3}) \\ \cos(\theta_r+\frac{2\pi}{3})\cos\theta_r & \cos(\theta_r+\frac{2\pi}{3})\cos(\theta_r-\frac{2\pi}{3}) & \cos^2(\theta_r+\frac{2\pi}{3}) \end{bmatrix} \quad \text{(C5)}$$

Step 3:
The similar procedure is applied for the second term of (C1) (the *d*-axis term). The matrix inverse of the second term in (C1) is expressed as

$$\left(\mathbf{R}_{rd}+\frac{2}{\Delta t}\mathbf{L}_{rd}\right)^{-1} = \begin{bmatrix} Z_{lfd}+Z_{md} & Z_{md} \\ Z_{md} & Z_{lkd}+Z_{md} \end{bmatrix}^{-1} = \begin{bmatrix} \dfrac{1+Z_{md}Z_{lkd}^{-1}}{Z_{lfd}D_{2d}} & -\dfrac{Z_{md}}{Z_{lfd}Z_{lkd}D_{2d}} \\ -\dfrac{Z_{md}}{Z_{lkd}Z_{lfd}D_{2d}} & \dfrac{1+Z_{md}Z_{lfd}^{-1}}{Z_{lkd}D_{2d}} \end{bmatrix} \quad \text{(C6)}$$

where $D_{2d} = 1+\dfrac{Z_{md}}{Z_{lkd\Sigma 2}}$ and $Z_{lkd\Sigma 2} = \left(\left(Z_{lfd}\right)^{-1} + \left(Z_{lkd}\right)^{-1}\right)^{-1}$ (C7)

Using (C6) and (C7), the second term of (C1) can be expressed as:

$$\mathbf{L}_{srd}(t)\left(\mathbf{R}_{rd}+\frac{2}{\Delta t}\mathbf{L}_{rd}\right)^{-1}\mathbf{L}_{rsd}(t) = \frac{2}{3}\frac{L_{md}^2}{Z_{kd\Sigma 2}+Z_{md}} \begin{bmatrix} \sin^2\theta_r & \sin\theta_r\sin(\theta_r-\frac{2\pi}{3}) & \sin\theta_r\sin(\theta_r+\frac{2\pi}{3}) \\ \sin(\theta_r-\frac{2\pi}{3})\sin\theta_r & \sin^2(\theta_r-\frac{2\pi}{3}) & \sin(\theta_r-\frac{2\pi}{3})\sin(\theta_r+\frac{2\pi}{3}) \\ \sin(\theta_r+\frac{2\pi}{3})\sin\theta_r & \sin(\theta_r+\frac{2\pi}{3})\sin(\theta_r-\frac{2\pi}{3}) & \sin^2(\theta_r+\frac{2\pi}{3}) \end{bmatrix} \quad \text{(C8)}$$

Step 4:
For the purpose of model derivation, the triple matrix product in the equivalent resistance matrix (7) is defined as a new matrix $\mathbf{A}$.

$$\mathbf{L}_{sr}(t)\left(\mathbf{R}_{r}+\frac{2}{\Delta t}\mathbf{L}_{r}\right)^{-1}\mathbf{L}_{rs}(t) = \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \text{(C9)}$$

Using (C5) and (C8), the elements of $\mathbf{A}$ can be represented in terms of machine parameters. Without loss of generality, the first element a_{11} is expressed here as an example,

$$a_{11} = \frac{2}{3}\frac{L_{mq}^2}{Z_{kq\Sigma 2}+Z_{mq}}\cos^2\theta_r + \frac{2}{3}\frac{L_{md}^2}{Z_{kd\Sigma 2}+Z_{md}}\sin^2\theta_r \quad \text{(C10)}$$

Substituting the trigonometric identity, $\sin^2\theta_r = 1-\cos^2\theta_r$, the coefficient a_{11} can be expressed as

$$a_{11} = \frac{2}{3}\left(\frac{L_{mq}^2}{Z_{kq\Sigma 2}+Z_{mq}} - \frac{L_{md}^2}{Z_{kd\Sigma 2}+Z_{md}}\right)\cos^2\theta_r + \frac{2}{3}\frac{L_{md}^2}{Z_{kd\Sigma 2}+Z_{md}} \quad \text{(C11)}$$

Step 5:
The constant and rotor-position-dependent terms in the equivalent resistance matrix $\mathbf{R}_{eq}^{pd}(t)$ can be separated by substituting (B3) and (C9) into (7). The rotor-position-dependent terms can be combined and further simplified. Without loss of generality, as an example, the rotor-position-dependent term $r_{11}(t)$ in the first element of $\mathbf{R}_{eq}^{pd}(t)$ is expressed as

$$R_{11}(t)=-\frac{2}{\Delta t}L_B\cos 2\theta_r-\frac{2}{3}\left(\frac{2}{\Delta t}\right)^2\left(\frac{L_{mq}^2}{Z_{kq\Sigma 2}+Z_{mq}}-\frac{L_{md}^2}{Z_{kd\Sigma 2}+Z_{md}}\right)\cos^2\theta_r \tag{C12}$$

Substituting the trigonometric identity $\cos^2\theta_r=\frac{1+\cos 2\theta_r}{2}$ into (C12), $R_{11}(t)$ can be expressed as

$$R_{11}(t)=-\frac{2}{\Delta t}\frac{L_{md}-L_{mq}}{3}\cos 2\theta_r-\frac{2}{3}\left(\frac{2}{\Delta t}\right)^2\left(\frac{L_{mq}^2}{Z_{kq\Sigma 2}+Z_{mq}}-\frac{L_{md}^2}{Z_{kd\Sigma 2}+Z_{md}}\right)\frac{1+\cos 2\theta_r}{2} \tag{C13}$$

The rotor-position-dependent terms in (C13) can be collected as

$$\begin{aligned}R'_{11}(t)&=-\frac{2}{\Delta t}\frac{L_{md}-L_{mq}}{3}\cos 2\theta_r-\frac{1}{3}\left(\frac{2}{\Delta t}\right)^2\left(\frac{L_{mq}^2}{Z_{kq\Sigma 2}+Z_{mq}}-\frac{L_{md}^2}{Z_{kd\Sigma 2}+Z_{md}}\right)\cos 2\theta_r\\&=\frac{1}{3}\left(\frac{Z_{kq\Sigma 2}Z_{mq}}{Z_{kq\Sigma 2}+Z_{mq}}-\frac{Z_{kd\Sigma 2}Z_{md}}{Z_{kd\Sigma 2}+Z_{md}}\right)\cos 2\theta_r=\frac{1}{3}\left(Z''_q-Z''_d\right)\cos 2\theta_r\end{aligned} \tag{C14}$$

It is observed in (C14) that $R'_{11}(t)$ is the same as the rotor-position-dependent term in the first element of the equivalent resistance matrix (12). The same procedure and conclusion apply to all other elements in the variable term in (12), which completes the proof.

APPENDIX D

Theorem 2: The analytical inverse of a special matrix (that has self inductances in diagonal and the same mutual inductances in all off diagonal entries) has the following form:

$$\begin{bmatrix} Z_{lkq1}+Z_{mq} & Z_{mq} & \cdots & Z_{mq}\\ Z_{mq} & Z_{lkq2}+Z_{mq} & \cdots & Z_{mq}\\ \vdots & \vdots & \ddots & \vdots\\ Z_{mq} & Z_{mq} & \cdots & Z_{lkqN}+Z_{mq}\end{bmatrix}_{M\times M}^{-1}=\begin{bmatrix}\frac{1+Z_{mq}\left(Z_{lkq\Sigma M}^{-1}-Z_{lkq1}^{-1}\right)}{Z_{lkq1}D_{Mq}} & -\frac{Z_{mq}}{Z_{lkq1}Z_{lkq2}D_{Mq}} & \cdots & -\frac{Z_{mq}}{Z_{lkq1}Z_{lkqM}D_{Mq}}\\ -\frac{Z_{mq}}{Z_{lkq2}Z_{lkq1}D_{Mq}} & \frac{1+Z_{mq}\left(Z_{lkq\Sigma M}^{-1}-Z_{lkq2}^{-1}\right)}{Z_{lkq2}D_{Mq}} & \cdots & -\frac{Z_{mq}}{Z_{lkq2}Z_{lkqM}D_{Mq}}\\ \vdots & \vdots & \ddots & \vdots\\ -\frac{Z_{mq}}{Z_{lkqM}Z_{lkq1}D_{Mq}} & -\frac{Z_{mq}}{Z_{lkqM}Z_{lkq2}D_{Mq}} & \cdots & \frac{1+Z_{mq}\left(Z_{lkq\Sigma M}^{-1}-Z_{lkqM}^{-1}\right)}{Z_{lkqM}D_{Mq}}\end{bmatrix}_{M\times M} \tag{D1}$$

where $D_{Mq}=1+\frac{Z_{mq}}{Z_{lkq\Sigma M}}$ and $Z_{lkq\Sigma M}=\left(\sum_{j=1}^{N}\left(Z_{lkqj}\right)^{-1}\right)^{-1}$ (D2)

Note: It can be shown that using step-by-step row and column reductions one can obtain the result (D1). The detailed proof is not included in Appendix D due to space limitation.

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Parametric Average-Value Model of Synchronous Machine-Rectifier Systems

Juri Jatskevich, *Member, IEEE*, Steven D. Pekarek, *Member, IEEE*, and Ali Davoudi, *Student Member, IEEE*

***Abstract*—A new average-value model of a rectifier circuit in a synchronous-machine-fed rectifier system is set forth. In the proposed approach, a proper state model of the synchronous machine in the *qd*-rotor reference frame is used, whereas the rectifier/dc-link dynamics are represented using a suitable proper transfer function and a set of nonlinear algebraic functions that are obtained from the detailed model using numerical averaging. The new model is compared to a detailed simulation as well as to measured data and is shown to be very accurate in predicting the large-signal time-domain transients as well as small-signal frequency-domain characteristics.**

***Index Terms*—Average-value model (AVM), impedance characterization, line-commutated rectifier, synchronous machine.**

I. Introduction

SYNCHRONOUS machines/converters, such as those depicted in Fig. 1, are commonly used in the electrical subsystems of aircraft, ships, automotive, and ground vehicles, brushless excitation systems of larger generators, wind power generators, etc. In many of these applications, the overall power system also includes multiple power-electronic loads that exhibit negative impedance characteristics. The small- and large-displacement stability of power-electronics-based systems is an important issue [1], [2].

There are various techniques for investigating the stability of power-electronic-based systems and the design of controllers that are based upon frequency-domain characteristics. In these approaches, the small-signal input/output (I/O) impedance characteristics of each source and load are determined over a range of operating points and used to investigate small-signal dynamic interactions. The impedance characteristics can be extracted from a detailed model of the systems (wherein the switching of each power-electronic valve/device is represented) or from a hardware prototype. The traditional methods of extracting impedance information include frequency sweep techniques [3] and the injection of nonsinusoidal and possibly spike-like signals [4]. The later methods work well with linear systems, and the frequency sweep methods work well with

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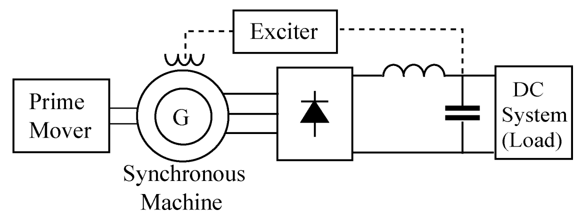

Fig. 1. Electromechanical system with a synchronous machine and dc link.

both linear and nonlinear systems with or without switching. To calculate the impedance of the system depicted in Fig. 1 using frequency sweep, small-signal disturbances are injected in the output of the generator–rectifier. The ratio of the change in output voltage to the chance in output current represents the source impedance at a given frequency and loading condition. However, since the detailed models are computationally intensive, determining the impedance over a wide range of frequencies is a very time-consuming procedure, particularly when it includes obtaining data points at very low frequencies (<10 Hz).

These challenges led to the development of so-called average-value models (AVMs) wherein the effects of fast switching are neglected or "averaged" with respect to the prototypical switching interval and the respective state variables are constant in the steady-state. Although the resulting models are often nonlinear and only approximate the longer-term dynamics of the original systems, the AVMs are continuous and, therefore, can be linearized about a desired operating point. Thereafter, obtaining local transfer-function and/or frequency-domain characteristics becomes a straightforward and almost instantaneous procedure. Many simulation programs offer automatic linearization and subsequent state-space and/or frequency-domain analysis tools [5], [6]. Additionally, the AVMs typically execute orders of magnitude faster than the corresponding detailed models, making them ideal for representing the respective components in system-level studies.

The analytical derivation of accurate AVMs for synchronous machine-converter systems is challenging. Initial steps in this direction can be traced back to the late 1960s. In particular, in [7] and [8], the rectifier circuit is represented using algebraic expressions to relate transformed ac source voltages and rectifier dc variables for a system modeled using a constant reactance behind a voltage source to represent the generator. The reduced-order models in which the stator dynamics are neglected have been presented in [9] and [10]. Although accurate in the steady-state of a single operating mode, these models are inaccurate for predicting the output impedance at higher frequencies [11]. A dynamic AVM has been derived in [11], wherein a very

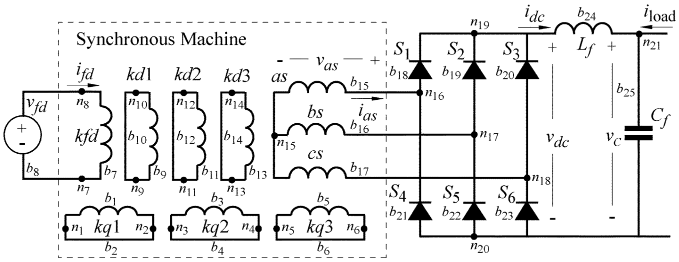

Fig. 2. Circuit diagram for the detailed model.

good match in time and in frequency domains with the detailed simulation and hardware is reported. This work was extended to model an inductorless rectifier-machine system in [12]. In each case ([7]–[11]), the analytical development is based upon a single switching pattern (conduction and commutation intervals) and, therefore, is valid only for that operating mode. An approach similar to [8] has recently been used in [13] and [14], wherein the parameters of the rectifier AVM were obtained from a detailed simulation. However, in [14], the rectifier AVM parameters are not dependent on operating condition, which, as will be shown herein, results in significant error when predicting the source impedance.

The method presented in this paper extends the work of [8] and [14] by which it has been inspired. The approach initially requires a detailed simulation for extracting the essential AVM parameters. In the resulting AVM, the dynamics of the rectifier/dc link are represented using a suitable proper transfer function and a set of nonlinear algebraic functions. The proposed method is based upon a full-order classical (Park's) state model of the synchronous machine expressed in the qd-rotor reference frame. The new model is shown to be accurate in different rectifier modes over a wide range of frequencies. An attractive feature of the proposed AVM is that it does not require extensive analytical derivations and is therefore readily extendable to more complex machine-converter configurations.

II. Case System Detailed Model

A circuit diagram of the synchronous machine system considered in this paper is depicted in Fig. 2. This system has been studied previously by other research and is well documented in the literature [11], [15]. For consistency, the system parameters including the dc-link filter are summarized in the Appendix [see a) and b)]. There are numerous simulation languages and programs that can be used to create a detailed model of a machine/converter system. Herein, the model is created using Simulink, which is a state-variable-based simulation language. The machine is modeled in terms of physical (abc) variables. Using physical variables, the machine model is developed using simple circuit elements: voltage sources, resistors, and coupled inductors to represent the magnetic coupling of the respective windings. The corresponding voltage equations in physical-machine variables can be written as

$$\begin{bmatrix}\mathbf{v}_{abcs}\\ \mathbf{v}_{qdr}\end{bmatrix} = \begin{bmatrix}\mathbf{r}_s & 0\\ 0 & \mathbf{r}_r\end{bmatrix}\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}_{qdr}\end{bmatrix} + \frac{d}{dt}\begin{bmatrix}\boldsymbol{\lambda}_{abcs}\\ \boldsymbol{\lambda}_{qdr}\end{bmatrix} \tag{1}$$

where $\mathbf{v}_{abcs}, \mathbf{i}_{abcs}$, and $\boldsymbol{\lambda}_{abcs}$ are vectors consisting of the stator phase voltage, current, and flux linkage, respectively; $\mathbf{v}_{qdr}, \mathbf{i}_{qdr}$, and $\boldsymbol{\lambda}_{qdr}$ are vectors consisting of the rotor winding voltage, current, and flux linkage; and $\mathbf{r}_s$ and $\mathbf{r}_r$ are matrices that contain the stator and rotor winding resistances. The flux linkages can be expressed in terms of currents as

$$\begin{bmatrix}\boldsymbol{\lambda}_{abcs}\\ \boldsymbol{\lambda}_{qdr}\end{bmatrix} = \begin{bmatrix}\mathbf{L}_s(\theta_r) & \mathbf{L}_{sr}(\theta_r)\\ \mathbf{L}_{rs}(\theta_r) & \mathbf{L}_r\end{bmatrix}\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}_{qdr}\end{bmatrix}. \tag{2}$$

In (2), the matrices $\mathbf{L}_s(\theta_r)$ and $\mathbf{L}_{sr}(\theta_r)$ represent the stator windings self- and mutual-inductances and the stator-to-rotor mutual inductances, respectively. The matrices $\mathbf{L}_r(\theta_r)$ and $\mathbf{L}_{rs}(\theta_r)$ contain the rotor winding self- and mutual inductances and the rotor-stator mutual inductances, respectively. Expressions for the inductance matrices can be found in [16].

The stator windings together with the rectifier represent a switched network. For each topological instance of the system depicted in Fig. 2, there exists a corresponding state equation that can be assembled by partitioning the overall circuit graph into a spanning tree and link branches, and selecting the inductive link currents and capacitor tree voltages as the state variables. However, due to rectifier switching, analytically establishing a state model for all potential topologies is very challenging. To overcome this challenge, an algorithm for generating the state equations has been developed in [15], [17], and [18]. Utilizing this approach, a circuit can be defined by a branch list composed of statements such as

$$\mathbf{L\_branch}(\mathbf{bn},\ \mathbf{pn},\ \mathbf{nn},\ \mathbf{r},\ \mathbf{L},\ \mathbf{e},\ \mathbf{Iic})$$

which defines an inductive branch, for example. Here, $\mathbf{bn}$ is the branch number; $\mathbf{pn}$ and $\mathbf{nn}$ are the positive and negative nodes; $\mathbf{r}$, $\mathbf{L}$, and $\mathbf{e}$ are the branch series resistance, inductance, and the voltage source; and $\mathbf{Iic}$ is the initial inductor current, respectively. A mutual inductance can be specified using a statement

$$\mathbf{L\_mutual}(\mathbf{b1},\ \mathbf{b2},\ \mathbf{Lm})$$

where $\mathbf{b1}$ and $\mathbf{b2}$ are the inductive branch numbers and $\mathbf{Lm}$ is the respective mutual inductance. Other circuit branches may be defined using similar syntax. For consistency, the branch numbering is also shown in Fig. 2. As the circuit switches, a state equation is automatically generated and updated for each new topology [17]. In particular, the state equation for the ith topological state has the following implicit form:

$$\mathbf{M}^i(\theta_r)\frac{d\mathbf{x}^i}{dt} = \mathbf{F}^i(\mathbf{x}^i, \theta_r) + \mathbf{g}^i(\mathbf{u}) \tag{3}$$

where $\mathbf{M}(\theta_r)$ is a positive-definite mass matrix, $\mathbf{F}(\mathbf{x}, \theta_r)$ is a term that contains state-self dynamics, and the forcing term $\mathbf{g}(\mathbf{u})$ accounts for external inputs from independent sources. In order to establish an overall transient response, the initial condition for a subsequent topology is established in such a way that the currents through inductors and voltage across capacitors are continuous according to circuits laws.

A rectifier with nonzero inductances on the source and on the dc sides may operate with a wide range of loads. In this paper, the two most commonly encountered rectifier modes are considered. Under light load (mode 1), there are two distinct intervals;

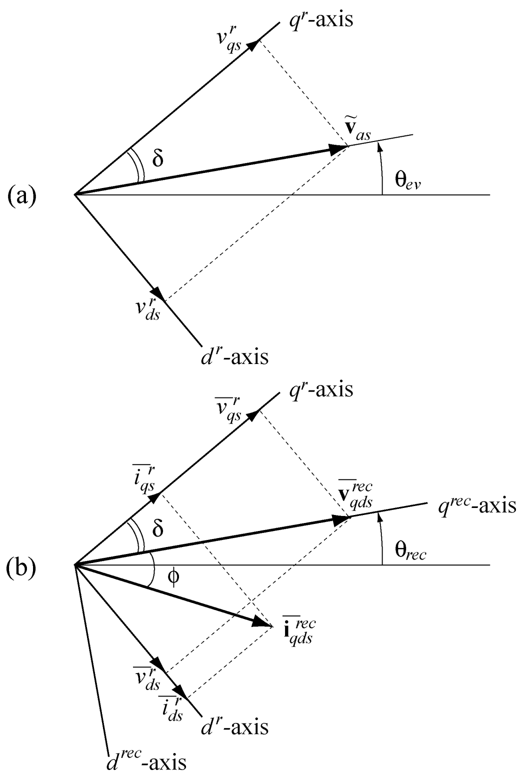

Fig. 3. Relationship between generator and rectifier variables.

a conduction interval (two diodes conducting) and a commutation interval (three diodes conducting). When operating under heavy load (mode 2), the conduction interval disappears (three diodes conducting at all instances). Within mode 2, there is a one diode turning "on" and one turning "off" at each switching instance. In either case, a complete prototypical switching interval is $\pi/3$ and for the 60-Hz base frequency, the interval equals to 1/360 s [16]. Moreover, the simulation method considered here [15], [17], [18] automatically implements any operating mode.

III. Average Value Modeling

In contrast to the detailed model, development of the AVM is achieved using the synchronous machine equations expressed in the rotor frame of reference. For consistency, the corresponding equations are summarized in the Appendix [see c)]. Specifically, using Park's transformation (A.15), (1) and (2) are transformed to the rotor reference frame (A.1)–(A.14). The dynamic averaging of the rectifier circuit relies on establishing a relationship between the dc-link variables on one side and the ac variables transferred to a suitable reference frame on the other side.

To facilitate further development, it is instructive to recall the rotor reference frame, which is depicted in Fig. 3(a) for consistency. For sinusoidal voltages, the phase-a voltage phasor $\tilde{v}_{as}$ is related to the rotor reference frame components v_{qs}^r and v_{ds}^r through the angle δ which depends on the load. In the case of a synchronous generator being connected to a rectifier, the voltage and current waveforms are highly distorted and the load may be changing dynamically. To overcome these difficulties, the ac (abc) variables must be transferred to a synchronous reference frame and then averaged. In particular, it is convenient to consider a reference frame in which the averaged d-axis component of the ac input voltage is zero [8]. A diagram relating the so-called rectifier reference frame and the rotor reference frame is shown in Fig. 3(b). In Fig. 3(b), the rectifier reference frame with a transformation angle $\theta_{\rm rec}$ is selected to ensure $\bar{v}_{ds}^{\rm rec} = 0$. Here, the subscript (superscript) "rec" denotes the quantities in the rectifier reference frame, and the bar symbol denotes the so-called fast average evaluated over a prototypical switching interval. A similar reference frame is used for dynamic averaging of a three-phase line-commutated converter [16, chap. 11]. From Fig. 3(b), it can be seen that the $q^{\rm rec}$-axis is synchronized with the peak of the phase a rectifier input (abc) voltages. The averaged generator voltages expressed in the rotor reference frame are represented by variables $\bar{v}_{qs}^r$ and $\bar{v}_{ds}^r$. The relationship between the respective voltages in the rotor and rectifier reference frames can be expressed

$$\begin{bmatrix} \bar{v}_{qs}^{\rm rec} \\ 0 \end{bmatrix} = \begin{bmatrix} \cos(\delta) & \sin(\delta) \\ -\sin(\delta) & \cos(\delta) \end{bmatrix} \begin{bmatrix} \bar{v}_{qs}^r \\ \bar{v}_{ds}^r \end{bmatrix}. \tag{4}$$

If the generator supplies real power to a load, the rotor reference frame leads the terminal voltages by the rotor angle δ used in (4). Under inductive load, the fundamental component of the rectifier current has a lagging power factor. Therefore, in Fig. 3(b), the current $\bar{\mathbf{i}}_{qds}^{\rm rec}$ lags the voltage $\bar{\mathbf{v}}_{qds}^{\rm rec}$ by an angle denoted by ϕ.

After defining the rectifier frame of reference, the next step is to relate the averaged rectifier dc current $\bar{i}_{\rm dc}$ to the currents $\bar{\mathbf{i}}_{qds}^{\rm rec}$, and the averaged output voltage $\bar{v}_{\rm dc}$ to the voltages $\bar{\mathbf{v}}_{qds}^{\rm rec}$, respectively. Assuming that rectifier does not contain energy-storing components, it is reasonable to approximate these relationships as

$$\left\|\bar{\mathbf{v}}_{qds}^{\rm rec}\right\| = \alpha(\cdot)\bar{v}_{\rm dc} \tag{5}$$

$$\bar{i}_{\rm dc} = \beta(\cdot)\left\|\bar{\mathbf{i}}_{qds}^{\rm rec}\right\| \tag{6}$$

where $\alpha(\cdot)$ and $\beta(\cdot)$ are algebraic functions of the loading conditions. In order to completely describe the rectifier, it is necessary to establish the angle between the vectors $\bar{\mathbf{v}}_{qds}^{\rm rec}$ and current $\bar{\mathbf{i}}_{qds}^{\rm rec}$. From Fig. 3(b), this angle can be expressed

$$\phi(\cdot) = \arctan\left(\frac{\bar{i}_{ds}^{\rm r}}{\bar{i}_{qs}^{\rm r}}\right) - \delta = \arctan\left(\frac{\bar{i}_{ds}^{\rm r}}{\bar{i}_{qs}^{\rm r}}\right) - \arctan\left(\frac{\bar{v}_{ds}^{\rm r}}{\bar{v}_{qs}^{\rm r}}\right). \tag{7}$$

The approach taken here utilizes a detailed simulation to obtain the functions $\alpha(\cdot)$, $\beta(\cdot)$, and $\phi(\cdot)$ numerically. The generator and the AVM of the rectifier are depicted in Fig. 4. In Fig. 4(a), the rectifier is represented by an algebraic block that outputs the averaged dc voltage and rectifier currents. The inputs to the model are the generator voltages and the dc-link current. The configuration of Fig. 4(a) is noniterative provided the filter inductor L_f is nonzero and both filter components L_f and C_f are modeled using a proper state model. However, difficulties arise if L_f is identically zero. Additionally, on the ac side of the rectifier, implementation of Fig. 4(a) requires that the generator voltages $\bar{\mathbf{v}}_{qds}^{\rm r}$ are outputs of the generator model. Using the generator voltages as the output may be accomplished if one uses differentiation (instead of integration) to solve the corresponding stator voltage equations (A.1) and (A.2). This results in a nonproper generator state model and is highly undesirable for many reasons including numerical noise, convergence, stability, etc. [19]. A similar approach to the model configuration shown in Fig. 4(a) has been suggested in [14] with what would be equivalent to fixed (load independent) values for α and β. An important aspect of the model derived herein is to show that α and β cannot be considered constant, but instead must be modeled as a function of load to yield accurate results over a wide range of operating conditions.

In order to avoid the mentioned above disadvantages, the method considered in this paper is based upon the configuration depicted in Fig. 4(b), wherein the generator is represented by a proper state model with stator currents as outputs and stator voltages as inputs. In this way, the generator can be readily modeled using classical Park's equations (A.1)–(A.14) [16]. This, in turn, suggests that the input and output of the rectifier AVM are $\bar{v}_{\mathrm{dc}}$ and $\bar{i}_{\mathrm{dc}}$, respectively. On one hand, the last condition is readily accommodated if L_f is identically zero, in which case, the capacitor voltage $v_C = v_{\mathrm{dc}}$ (which is also a state variable).

On the other hand, when $L_f \neq 0$, the effects of the filter inductor L_f are incorporated into the rectifier AVM. In particular, the steady-state effect due to resistive voltage drop across L_f is absorbed into $\alpha(\cdot)$ (as this function is determined numerically from detailed simulation). The dynamic effect of L_f does have an impact on the output impedance, particularly in the range of higher frequencies. In order to account for this effect, the rectifier dc voltage is related to the capacitor voltage in the frequency domain as

$$v_{\mathrm{dc}} = v_C + H_L(s) i_{\mathrm{dc}}. \tag{8}$$

In order to avoid the numerical differentiation when implementing (8) in the time domain, the $H_L(s)$ must be proper. Therefore, in order to represent the dynamics of the inductor L_f in a range up to the rectifier switching frequency, it is assumed that

$$H_L(s) = \frac{L_f s}{\tau s + 1} \tag{9}$$

where τ is a time constant small enough so that its effect at the switching frequency is negligible (1e-5 has been used herein).

The functions α, β, and ϕ depend upon the loading conditions that may be specified in terms of an impedance. For the purpose of this paper, such impedance can be conveniently defined based upon the detailed simulation of Fig. 2 as the operation point

$$z = \frac{\overline{v}_C}{\left\| \bar{\mathbf{i}}^{\mathrm{r}}_{qds} \right\|}. \tag{10}$$

The selection of variables in (10) ensures availability of voltage and current from the respective detailed and averaged state models without introducing algebraic loops. In particular, when using the detailed model, the stator abc currents are used to compute $\bar{\mathbf{i}}^{\mathrm{r}}_{qds}$; and when using the AVM, the vector norm of $\bar{\mathbf{i}}^{\mathrm{r}}_{qds}$ is readily computed from the generator qd model.

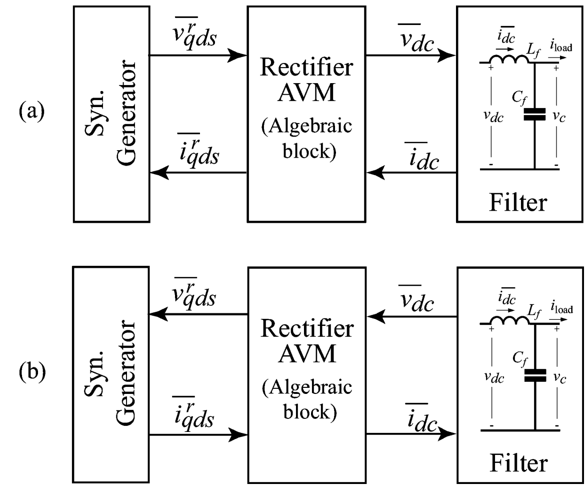

Fig. 4. Average-value modeling. (a) Rectifier is represented as an algebraic block with generator voltages as inputs. (b) Rectifier is represented as an algebraic block with generator currents as inputs.

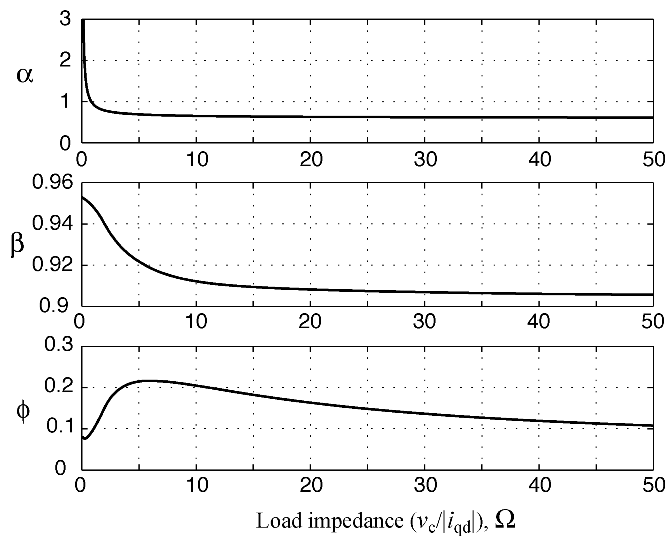

Fig. 5. Functions α, β, and ϕ obtained from the detailed model.

The detailed model described in Section II has been used for computing $\alpha(z)$, $\beta(z)$, and $\phi(z)$ according to (5)–(7). The resulting functions are plotted in Fig. 5. The variables in (5)–(7) were obtained by averaging the respective currents and voltages over the rectifier switching interval. The system of Fig. 2 has been connected to a resistive load that was varied in a wide range in order to obtain $\alpha(z)$, $\beta(z)$, and $\phi(z)$ that are valid for various operating conditions. It can be observed in Fig. 5 that these functions are nonlinear particularly at heavy loads.

The functions depicted in Fig. 5 may be stored and fitted into a spline or a lookup table for example. The support point that are sufficient to reproduce these functions using the Matlab function $spline()$ [20] are given in the Appendix [see d)]. It should be noted that for better accuracy, more data points are given for the region of low load impedance where the functions are very nonlinear. Once these functions are available, the proposed AVM is implemented according to the block diagram shown in Fig. 6. In particular, the filter capacitor C_f is modeled using a first-order state equation $(H_C(s) = 1/C_s)$. The impedance z is computed

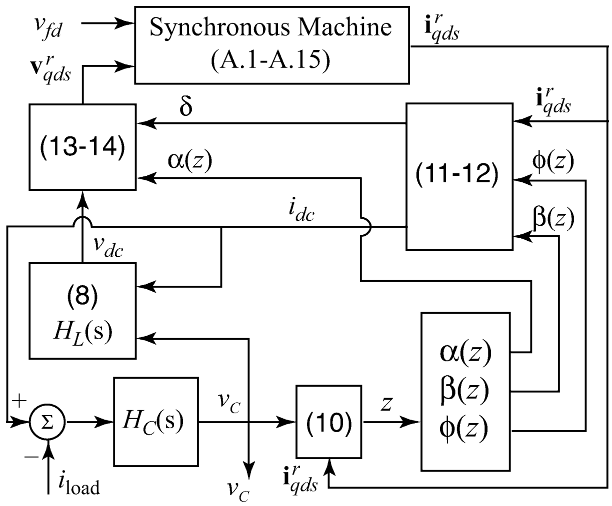

Fig. 6. Model structure of the proposed AVM.

according to (10) and the functions α, β, and ϕ are evaluated for a given value of z. Based on ϕ, the rotor angle is computed using

$$\delta = \arctan\left(\frac{i_{ds}^{\mathrm{r}}}{i_{qs}^{\mathrm{r}}}\right) - \phi(z). \tag{11}$$

The dc-link current is computed using

$$i_{\mathrm{dc}} = \beta(z)\sqrt{i_{qs}^2 + i_{ds}^2}. \tag{12}$$

Equation (8) is then used to compute the rectifier dc voltage v_{dc}. The generator voltages are expressed using the vector relationships depicted in Fig. 3(b) and 5 as

$$v_{qs}^{\mathrm{r}} = \alpha(z)v_{dc}\cos(\delta) \tag{13}$$
$$v_{ds}^{\mathrm{r}} = \alpha(z)v_{dc}\sin(\delta). \tag{14}$$

It is noted that the resistive loss due to the filter inductor L_f is included in $\alpha(z)$. However, the dynamic effect of L_f is included when (8) is used to compute v_{dc} for (13) and (14).

IV. Computer Studies

The detailed state model of the synchronous machine rectifier system described in Section II has been implemented in Matlab/Simulink as a masked CMEX S-function. The details of the implementation as well as the user interface are described in detail in [17]. The system of Fig. 2 is defined in terms of branches with rotor-position-dependent inductances, whereas the appropriate switching logic is implemented to model the rectifier circuit in valve-by-valve detail assuming idealized ON/OFF switching characteristics. The resulting detailed model was used as a benchmark in subsequent studies.

The proposed AVM depicted in Fig. 6 has also been implemented in Simulink using standard library blocks. In order to fully compare the AVM against the detailed simulation, the respective models were compared first in the time domain and then in the frequency domain. In all cases, a constant generator speed $\omega_r = 120\pi$ rad/s was assumed.

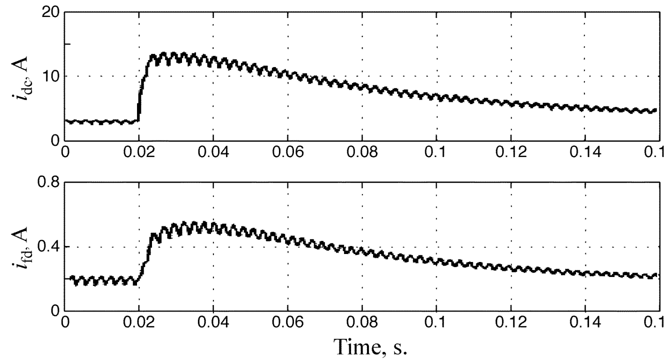

Fig. 7. Measured dc and field currents.

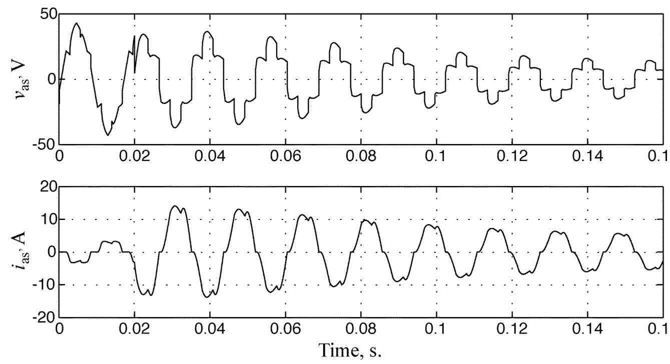

Fig. 8. Simulated response to a load step change in mode 1; ac voltage and current.

A. Time-Domain Studies

In the following study, the system starts up with initial conditions corresponding to steady-state operation with a constant excitation of 19.5 V and a load resistance of 21 Ω connected to the dc filter output. At time $t = 0.02$ s, a 4.04-Ω resistor is connected in parallel, resulting in a load resistance of 3.64 Ω. The time-domain comparison between the detailed model and the corresponding responses measured in the hardware are presented in [9] and [15], wherein it is shown that the detailed model portrays the response of the actual hardware system with acceptable accuracy. For consistency, the measured dc and field currents are plotted in Fig. 7. The computer-generated response of the detailed model and AVMs are depicted in Figs. 8–10. In particular, in Fig. 8, the transient observed in the phase a generator voltage and current is predicted using the detailed simulation. Therein, it is shown that the current i_{as} is discontinuous and, thus, the rectifier operates in mode 1. In Fig. 9, the detailed simulation is compared with the response generated by the AVM, wherein the parameters α, β, and ϕ were kept constant using the values corresponding to a 21-Ω load. As can be seen in Fig. 9, the AVM response follows the general transient very well with the exception of some initial overshoot in dc current i_{dc} and output voltage v_C. An improved dynamic response is depicted in Fig. 10, wherein the AVM is implemented with α, β, and ϕ dependent on the impedance defined in (10). Here, the response of the averaged model follows almost exactly the trace produced by the detailed model during the entire transient and is closer to the measured results depicted in Fig. 7. Although the mismatch in the transient response in Fig. 9 is not that large, one

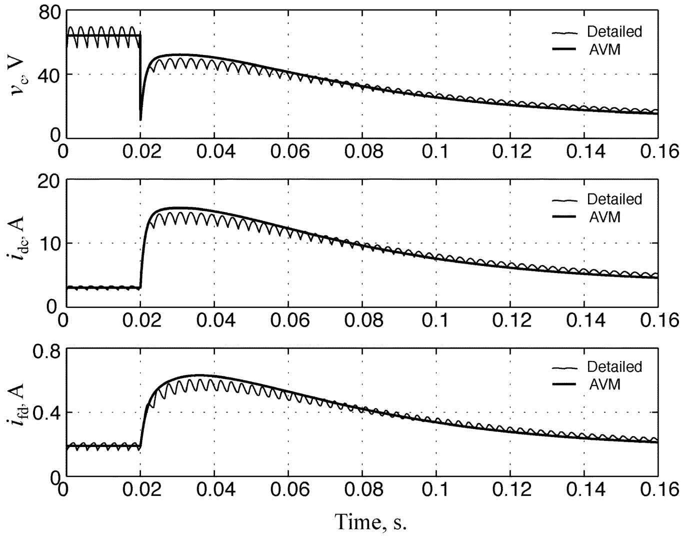

Fig. 9. System response to a load step change; detailed and AVM with constant parameters.

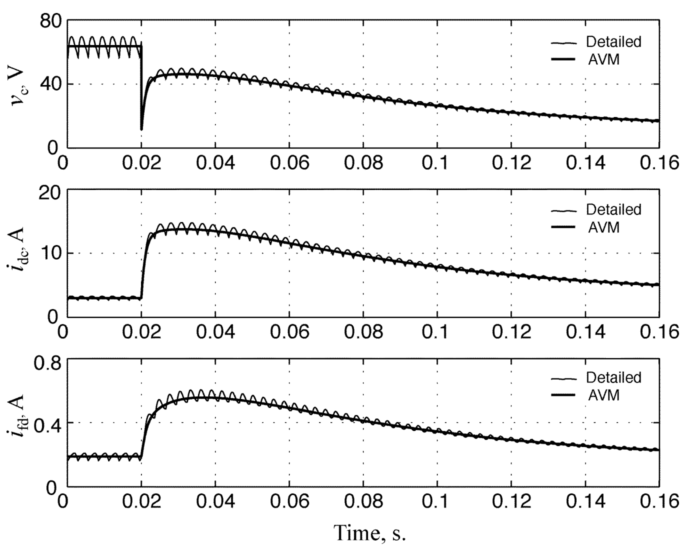

Fig. 10. System response to a load step change; detailed and AVM with variable parameters.

still can conclude that for better accuracy, the parameters α, β, and ϕ must be implemented as functions of loading condition.

In order to drive the machine-rectifier into mode 2, at $t = 0.62$ s, the load resistance was decreased further to 2 Ω. The simulated phase a generator voltage and current are shown in Fig. 11. Therein, it is shown that i_{as} becomes continuous at the zero crossing which indicates the mode 2 operation. In this mode, there are three diodes conducting at any switching interval. The corresponding dynamic response of the dc quantities is shown in Fig. 12, wherein the traces produced by the detailed simulation are overlaid with respective traces obtained from the AVMs with constant and variable parameters, respectively. As can be seen, the AVM with variable α, β, and ϕ matches the detailed response almost exactly, whereas the accuracy of the AVM with constant parameters has decreased even further.

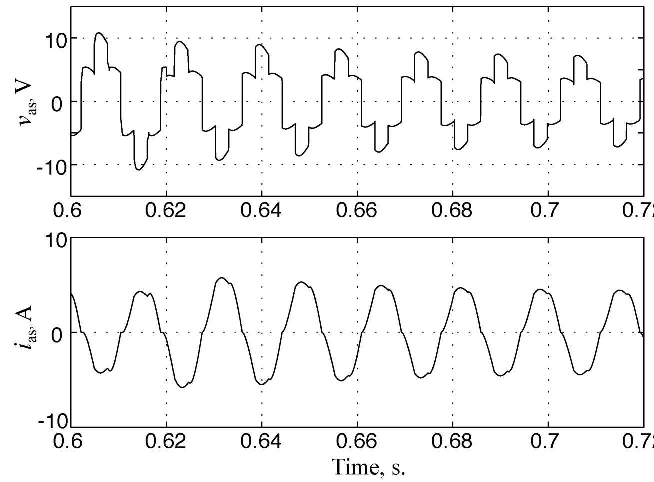

Fig. 11. Simulated ac voltage and current for operation in mode 2.

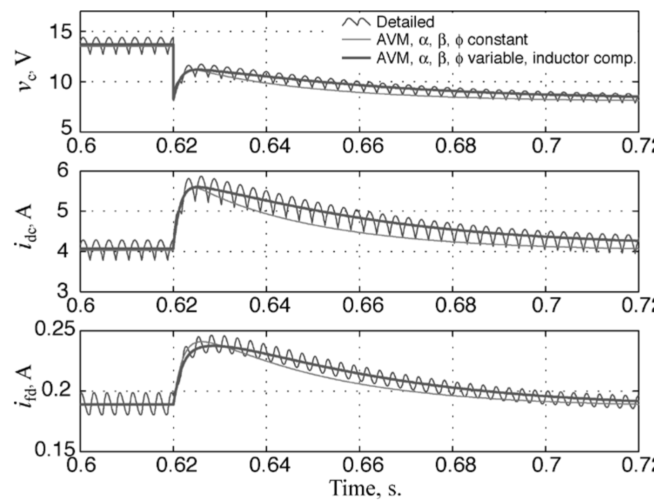

Fig. 12. System response to a load step change in mode 2; detailed model and AVMs with constant and variable parameters.

B. Impedance Characterization

The developed AVM should exhibit the same frequency-domain characteristics as the original system. The output impedance of the system described in Fig. 2 has also been considered for verification of the analytically derived averaged model [11], wherein an excellent agreement among the impedance curves obtained from the measured data, the detailed model, and the analytically derived averaged model were reported. For consistency, the same load of 10.74 Ω has been assumed herein for the frequency-domain comparisons. Under the given load, the rectifier operates in mode 1. The measured output impedance phase and magnitude are plotted in Fig. 13. The measured impedance curves are overlaid and compared with the results obtained from the detailed model. Since the detailed model contains switching and is discontinuous, the small-signal injection and subsequent frequency sweep method has been implemented in the same Simulink model and used to extract the required impedance information. The measured and simulated impedance curves are very close as can be seen in Fig. 13. Because the AVM only approaches the detailed model

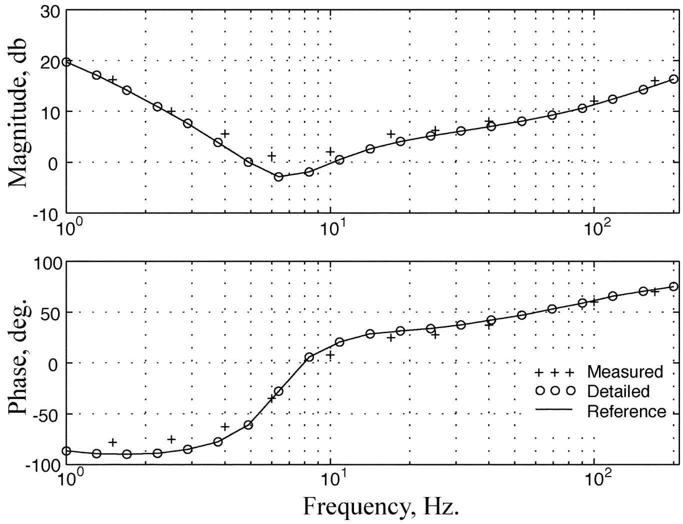

Fig. 13. Output impedance in mode 1; measured and detailed simulation. The curves obtained from detailed simulation are taken as reference.

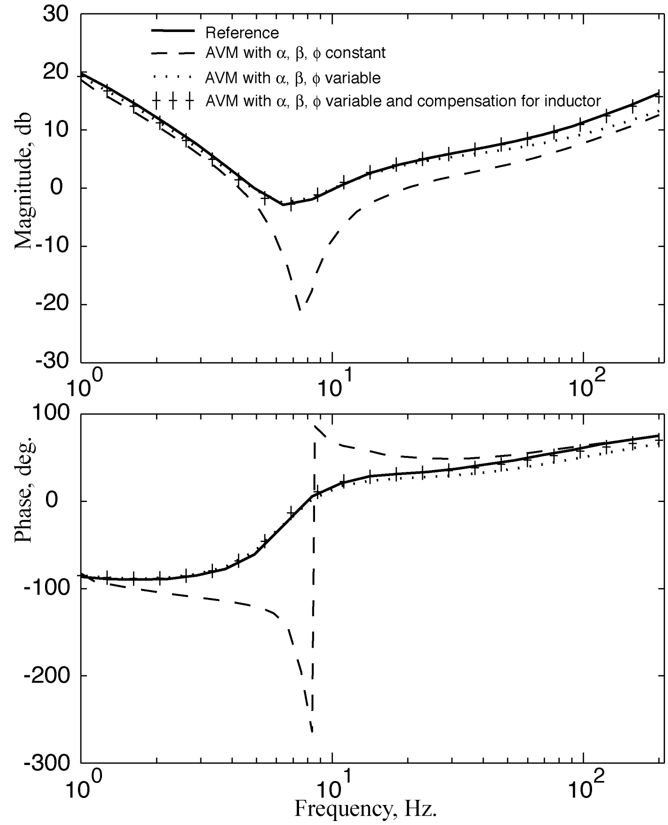

Fig. 14. Comparison of output impedance for the mode 1 predicted by various AVMs.

in terms of accuracy, the impedance curves produced by the detailed model are considered as a reference for the subsequent comparisons with AVMs. The reference impedance is plotted in Fig. 13 using the solid line.

The impedances are evaluated in the frequency range from 1 to 200 Hz. Closer to the rectifier switching frequency, the results become distorted due to the interaction of the injected signal with the rectifier switching. In general, considering frequencies close to and above the switching frequency have limited use for the average model since the basic assumptions of averaging are no longer valid.

The AVM with the same load of 10.74 Ω has been implemented. Since the model is continuous, the required output impedance can be extracted using the linearization technique as well as the frequency sweep (both yielding identical results). The corresponding impedance curves are compared with the results from the detailed model in Fig. 14. Here, several variations of the AVM have been considered. In particular, the first one (dashed line) wherein the parameters α, β, and ϕ were kept constant and the filter inductor L_f was neglected. As can be observed in Fig. 14, very significant errors exist in both phase and magnitude, especially in the range from 6 to 10 Hz. Since α, β, and ϕ were constant, their slope (Fig. 5) was not captured in the respective linearized model. In the second study (dotted line), α, β, and ϕ were implemented as functions of the loading impedance (10). This clearly eliminates the significant discrepancy around 7 Hz and improves the overall match up to around 20 Hz. However, at higher frequencies, the effect of inductor L_f becomes more pronounced, which explains a somewhat higher impedance magnitude and phase of the reference system in the range from 20 to 200 Hz. The last study (crossed line) corresponds to the case when the AVM was implemented with variable α, β, and ϕ as well as compensation for the filter inductor according to (8). As can be seen in Fig. 14, the final AVM matches the reference impedance almost exactly over the entire frequency range considered.

Measuring impedance of the hardware system in mode 2 (heavy mode) using a frequency sweep technique, for example, is undesirable. Also, since the commutation/conduction pattern is different from the mode 1, any analytically developed averaged model [11] must be re-derived specifically for this new operating mode. However, the proposed averaged model is based upon functions $\alpha(z)$, $\beta(z)$, and $\phi(z)$ that are defined over a wide range of loading conditions up to a short circuit (Fig. 5). Fig. 15 shows the comparisons of the output impedance as seen by a 2.0-Ω load resistor predicted by various models. As before, the impedance predicted by detailed simulation is taken as reference. As can be seen in Fig. 15, the proposed AVM with functional representation of rectifier and proper compensation for the filter inductor matches the reference impedance with an excellent level of accuracy; whereas the AVM with constant rectifier parameters significantly underestimates the impedance magnitude at higher frequencies.

C. Case Study With Dynamic Load

Despite the fact that the functions $\alpha(z)$, $\beta(z)$, and $\phi(z)$ were computed under resistive load, the developed average-value model should correctly predict the system behavior under other types of loads with possible nonlinearities and/or dynamics. In order to verify this, a load composed of an induction machine drive supplied of the dc bus is considered herein. In particular, the drive system considered is shown in Fig. 16 and consists of an indirect field-oriented control (IFOC), a current-source inverter (CSI), and an induction motor (IM). The motor parameters are summarized in the Appendix [see e)]. The CSI operates in a hysteresis-delta modulation mode with a 50-kHz sampling rate. The classical IFOC and CSI are implemented

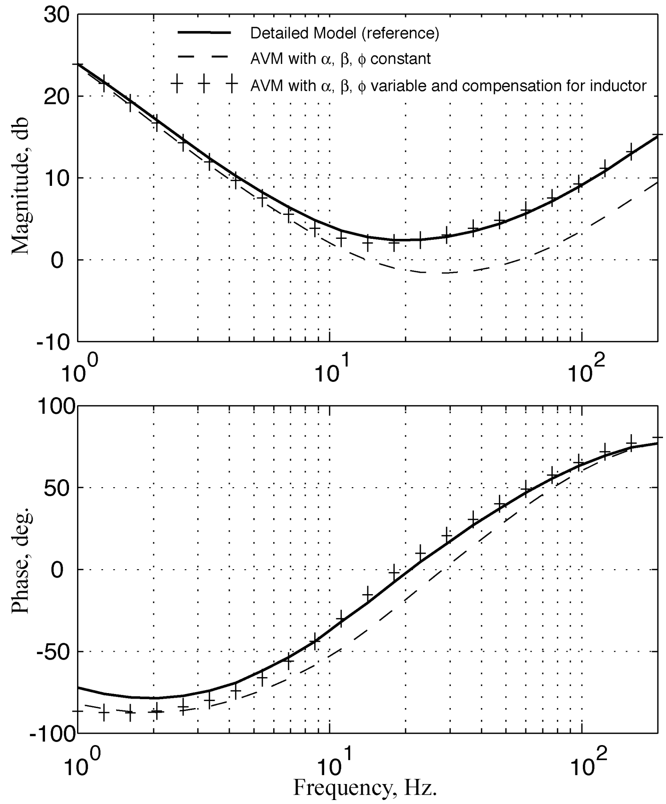

Fig. 15. Comparison of output impedance for the mode 2 predicted by various AVMs.

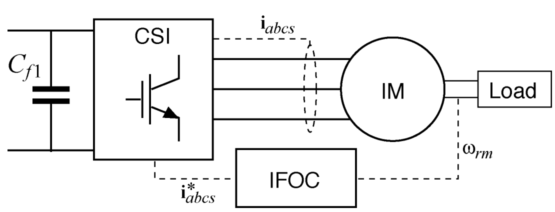

Fig. 16. Induction machine drive load.

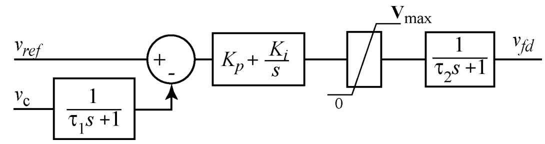

Fig. 17. Voltage regulator–exciter.

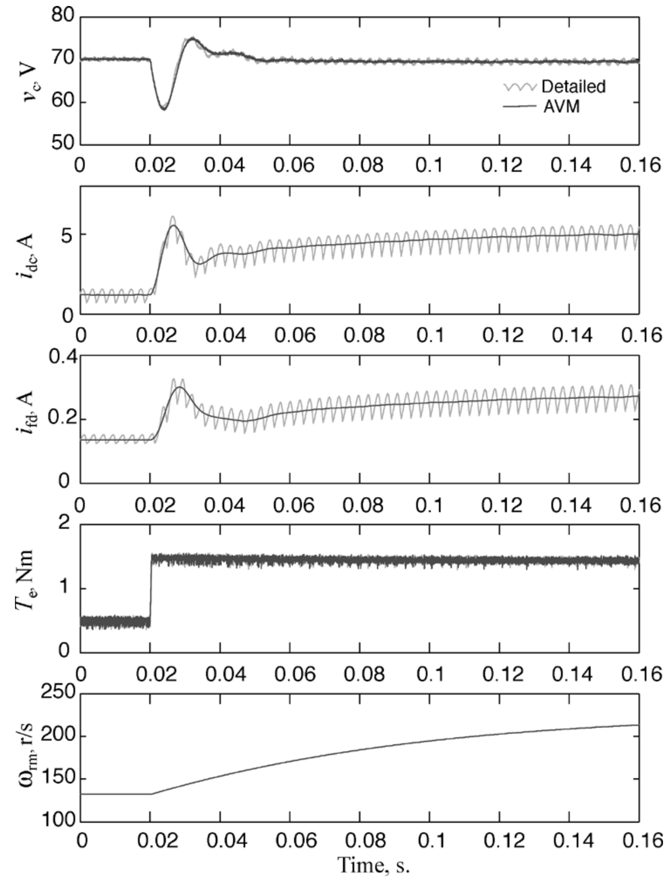

Fig. 18. System response to a step change in torque command; detailed and AVM models.

according to the methodology described in [16, Chap. 13 and 14], which is not repeated here due to space constraints. The IM is implemented according to [16, Chap. 4]. The torque-speed characteristic of the mechanical load considered is given in the Appendix [see f)]. Utilizing the field-oriented control (also known as vector control) together with the CSI, it is possible to regulate the developed electromagnetic (EM) torque very rapidly. For power quality as well as for dynamic stability [2], the CSI dc bus contains an additional capacitor $C_{f1} = 500\ \mu$F.

The motor drive load requires a dc supply of 70 V. As shown in Fig. 1, an excitation system is required for the synchronous-generator-diode-rectifier in order to regulate the dc-link voltage. A block diagram of the voltage regulator–exciter considered here is depicted in Fig. 17. The regulator–exciter consists of proportional plus integral (PI) controller with two filters, and its parameters are summarized in Appendix (g).

The described above motor drive load of Fig. 16 and the regulator–exciter of Fig. 17 were implemented in Simulink using standard library blocks. These models were then used in a transient study with the detailed model and the average-value model of the synchronous generator–rectifier system under consideration. In the following study, it is assumed that the motor initially operates in a steady-state, driving the load at 132 r/s which corresponds to a torque command of $0.4947\ \mathrm{N{\cdot}m}$. At $\mathrm{t} = 0.02\ \mathrm{s}$, the torque command is step-changed to $1.4841\ \mathrm{N{\cdot}m}$. The resulting transient is depicted in Fig. 18. As can be seen in Fig. 18, following an almost instantaneous response in EM torque T_e, the rotor mechanical speed ω_{rm} begins to increase. The plots of T_e and ω_{rm} produced using the proposed AVM of generator-rectifier lay on the top of the responses produced using the detailed model, without noticeable difference. The dc-link voltage v_c has some transient after $\mathrm{t} = 0.02\ \mathrm{s}$ and very quickly stabilizes at 70 V due to the voltage regulator–exciter action. As can be seen in Fig. 18, the v_c obtained from the detailed model contains the 360-Hz ripple due to rectifier operation as well as the high-frequency harmonics due to the CSI switching. At the same time, the v_c obtained from the AVM, contains only the high-frequency noise due to the CSI switching of the drive load, and it follows the overall transient envelope very well. The other two variables of interest, namely the inductor current i_{dc} and the generator field current i_{fd}, are also predicted by the AVM with excellent agreement during the entire transient. As is seen in Fig. 18, the solid lines of i_{dc} and i_{fd}, produced by the proposed AVM, pass through the 360-Hz rectifier ripple produced by the detailed model.

V. CONCLUSION

In this paper, numerical averaging of a rectifier circuit for a synchronous machine–rectifier system has been presented. In the proposed averaging method, the synchronous machine is implemented in a proper state model form using the classical qd formulation, and the parameters defining the relationship between the averaged dc-link variables and the generator currents and voltages viewed in the rotor reference frame vary dynamically depending on the loading condition. Although establishing the correct averaged model requires running the detailed simulation in a wide range of loading conditions, once established, the resulting model is continuous and valid for large-signal time-domain studies as well as for linearization and small-signal impedance characterization for a large range of operating conditions of the overall system. The proposed formulation also allows accurate representation of the rectifier output inductor and capacitor when such components are required. The resulting nonlinear-averaged model is verified in time and in frequency domain against detailed simulation as well as measured data.

APPENDIX

a) Synchronous machine parameters: U.S. Electrical Motors, 5 hp, 230 V, 215 T Frame, 1800 r/min, rated field current 1.05 A, custom made for university lab.

TABLE A.1

$\omega_b = 2 \cdot \pi \cdot 60$ rad/s	$P = 4$ (four poles)
$r_s = 0.382\ \Omega$	$x_{ls} = 0.4222\ \Omega$
$x_{mq} = 9.3871\ \Omega$	$x_{md} = 14.8158\ \Omega$
$r'_{kq1} = 5.07\ \Omega$	$x'_{lkq2} = 1.3195\ \Omega$
$r'_{kq2} = 1.06\ \Omega$	$x'_{lkq2} = 1.3195\ \Omega$
$r'_{kq3} = 0.447\ \Omega$	$x'_{lkq3} = 9.8772\ \Omega$
$r'_{kd1} = 140\ \Omega$	$x'_{lkd1} = 3.7209\ \Omega$
$r'_{kd2} = 1.19\ \Omega$	$x'_{lkd2} = 1.8510\ \Omega$
$r'_{kd3} = 1.58\ \Omega$	$x'_{lkd3} = 1.7002\ \Omega$
$r'_{fd} = 0.112\ \Omega$	$x'_{lfd} = 0.5768\ \Omega$
stator-to-field turns ratio	$N_s / N_{fd} = 0.0269$

b) DC-link filter parameters

$$L_f = 1.19\ \mathrm{mH} \qquad r_f = 0.32\ \Omega \qquad C_f = 2.28\ \mu\mathrm{F}.$$

c) Full-order qd synchronous machine equations used in the AVM

$$\frac{d\psi_{qs}}{dt} = \omega_b \left(-r_s i_{qs} - \frac{\omega_r}{\omega_b}\psi_{ds} + v_{qs}\right) \tag{A.1}$$

$$\frac{d\psi_{ds}}{dt} = \omega_b \left(-r_s i_{ds} + \frac{\omega_r}{\omega_b}\psi_{qs} + v_{ds}\right) \tag{A.2}$$

$$\frac{d\psi'_{kqj}}{dt} = -\,\omega_b r'_{kqj} i'_{kqj}, \qquad \text{for } j = 1,2,3 \tag{A.3}$$

$$\frac{d\psi'_{kdj}}{dt} = -\,\omega_b r'_{kdj} i'_{kdj}, \qquad \text{for } j = 1,2,3 \tag{A.4}$$

$$\frac{d\psi'_{fd}}{dt} = \omega_b \left(-r'_{fd} i'_{fd} + v'_{fd}\right) \tag{A.5}$$

$$i_{qs} = \frac{(\psi_{qs} - \psi_{mq})}{x_{ls}} \tag{A.6}$$

$$i_{ds} = \frac{(\psi_{ds} - \psi_{md})}{x_{ls}} \tag{A.7}$$

$$i'_{fd} = \frac{\left(\psi'_{fd} - \psi_{md}\right)}{x'_{lfd}} \tag{A.8}$$

$$i'_{kqj} = \frac{\left(\psi'_{kqj} - \psi_{mq}\right)}{x'_{lkqj}}, \qquad \text{for } j = 1,2,3 \tag{A.9}$$

$$i'_{kdj} = \frac{\left(\psi'_{kdj} - \psi_{md}\right)}{x'_{lkdj}}, \qquad \text{for } j = 1,2,3 \tag{A.10}$$

$$x_{aq} = \left(\frac{1}{x_{mq}} + \frac{1}{x_{ls}} + \sum_{j=1}^{3} \frac{1}{x'_{lkqj}}\right)^{-1} \tag{A.11}$$

$$x_{ad} = \left(\frac{1}{x_{mq}} + \frac{1}{x_{ls}} + \frac{1}{x'_{lfd}} + \sum_{j=1}^{3} \frac{1}{x'_{lkdj}}\right)^{-1} \tag{A.12}$$

$$\psi_{mq} = x_{aq} \left(\frac{\psi_{qs}}{x_{ls}} + \sum_{j=1}^{3} \frac{\psi'_{kqj}}{x'_{lkqj}}\right) \tag{A.13}$$

$$\psi_{md} = x_{ad} \left(\frac{\psi_{ds}}{x_{ls}} + \frac{\psi'_{fd}}{x'_{lfd}} + \sum_{j=1}^{3} \frac{\psi'_{kdj}}{x'_{lkdj}}\right) \tag{A.14}$$

$$\mathbf{K}_s = \frac{2}{3}\begin{bmatrix} \cos\theta_r & \cos\left(\theta_r - \frac{2\pi}{3}\right) & \cos\left(\theta_r + \frac{\pi}{3}\right) \\ \sin\theta_r & \sin\left(\theta_r - \frac{2\pi}{3}\right) & \sin\left(\theta_r + \frac{2\pi}{3}\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}. \tag{A.15}$$

d) Support points for the functions $\alpha(z)$, $\beta(z)$, and $\phi(z)$

TABLE A.2

z	α	β	ϕ
0.2	2.1819	0.9522	0.0767
0.6	1.1420	0.9505	0.0827
1.0	0.9391	0.9482	0.0983
2	0.7869	0.9399	0.1513
4	0.7097	0.9258	0.2073
7	0.6725	0.9163	0.2151
12	0.6484	0.9107	0.1958
25	0.6275	0.9075	0.1483
37	0.6202	0.9063	0.1236
50	0.6160	0.9056	0.1077

e) Induction machine parameters: Brook Hansen (S/N: P255 906), 0.25 hp, 34/68 V, 6.6/3.3 A, 1750 r/min, four poles, $J = 0.0017\ \mathrm{kg \cdot m^2}$; $r_s = 0.155\ \Omega$; $r'_r = 0.234\Omega$ $x_{ls} = 0.287\ \Omega$; $x'_{lr} = 0.287\ \Omega$; $x_m = 32.4\ \Omega$.

f) Torque-speed characteristic of the load (fan type)

$$T_L = \mathrm{sgn}(\omega_{rm}) \cdot \omega_{rm}^2 \cdot 2.7847e - 5.$$

g) Voltage regulator–exciter parameters

$$\tau_1 = 1e{-}3 \quad \tau_2 = 2e{-}2 \quad K_p = 10 \quad K_i = 100 \quad V_{\max} = 70.$$

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SIX PHASE SYNCHRONOUS MACHINE WITH AC AND DC STATOR CONNECTIONS, Part I: Equivalent Circuit Representation and Steady-State Analysis

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Abstract

This paper describes a detailed circuit representation of a six phase synchronous machine wherein mutual leakage couplings between the two sets of three phase stator windings are included. The main modes of power transfer of a six phase machine with sinusoidal inputs are examined. A harmonic phasor method of steady-state analysis to obtain the current, voltage and torque waveforms of a six phase machine with mixed AC-DC stator connections is described. It is shown that the output waveforms for various operating conditions from a digital computer program using the harmonic phasor method agree remarkably well with those obtained from an established detailed analog simulation.

INTRODUCTION

In the late 1920's, the six phase synchronous machine was used for power generation [1]; then the extra phases were needed to overcome the limitation imposed by the fault current interrupting capacity of circuit breakers. Presently, with increased demand for higher power drive systems, six phase stator machines are helping to overcome the current limitation imposed by semiconductor devices. These six phase drive systems have improved torque and MMF characteristics over those of standard three phase inverter drive systems [2].

A six phase synchronous machine with a controlled bridge rectifier connected to one set of stator windings is increasingly being used in novel AC-DC power supplies. With mechanical power input, the six phase machine is able to supply AC power from one stator set and DC power through the rectifier connected to the other stator set. Such AC-DC generators have been proposed as power supplies on aircrafts and ships because they not only cost and weigh less than alternative three phase generator-transformer-rectifier systems but also require less filtering. Besides, the rotor inertia helps to smoothen out fluctuations in the energy transfer. When only DC power input is available, the same configuration operated as an inverter can replace DC motor/AC generator sets for less weight and maintenance requirement. Such a six phase machine, inverter system has been proven effective for supplying AC power to air conditioners on board an electric mass transit train that obtain its DC supply from a third rail [3].

Previous steady state analysis [4] of the six phase machine with AC-DC stator connections divided the output cycle into basic intervals identified with the converter switchings, then analyzed each interval by the method of flux interlinkages in which the winding resistances were neglected. Transient analysis has been performed [5] with a little more complexity in that the winding resistances were included. Neither of these two methods, however, provide a quantitative account of the voltage or current harmonic content although this information could be indirectly extracted from the calculated waveforms.

This paper first describes a detailed circuit model of a six phase, salient pole synchronous machine that includes the stator mutual leakage inductances. Next, steady-state operations with sinusoidal voltage inputs are examined, using phasor diagrams to illustrate the transformer and motor modes of power transfer. Then, steady-state operations with AC-DC stator connections are considered. A Fourier analysis approach is taken to obtain the complex steady-state waveforms of the six phase machine with AC-DC stator connections. In this approach harmonic phasor components of the stator variables, transformed to the rotor reference frame, are applied to the d-q axis voltage equations in the frequency domain. The solution of these equations provides the voltage and current harmonics which are used to reconstruct the actual phase voltage, current and electromagnetic torque waveforms. The method is easily programmed on the digital computer. Aside from being a fresh approach to obtain the steady-state waveforms, it provides a convenient way of characterizing the voltage or current harmonic distortions and also the torque pulsation in terms of system parameters and operating conditions.

AN EQUIVALENT CIRCUIT REPRESENTATION

Voltage Equations

A schematic representation of the stator and rotor windings in a two pole, six phase, salient pole synchronous machine is given in Fig. 1. The six stator phases are normally divided into two wye-connected three-phase sets, labelled abc and xyz, whose magnetic axes are displaced by an arbitrary angle ξ. The windings of each three phase set are uniformly distributed and have axes that are displaced 120° apart. The three rotor windings are the Kq damper

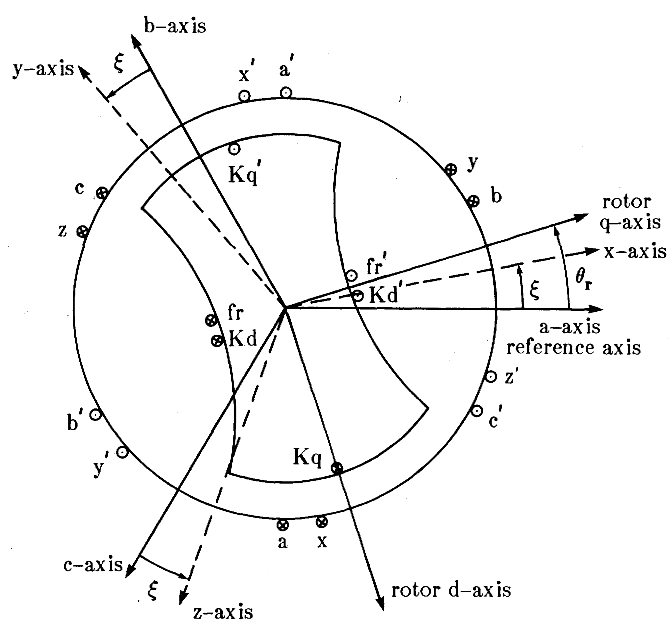

Figure 1. Two pole, salient pole, six phase synchronous machine.

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winding in the quadrature axis, the Kd damper and field (fr) windings in the direct axis.

The voltage equations for the six stator and three rotor windings can be written as:

$$\bar{v} = R\,\bar{i} + p\,\bar{\lambda} \tag{1a}$$

where the differential operator p denotes $\frac{d}{dt}$, the vectors $\bar{v}$, $\bar{i}$ and $\bar{\lambda}$ are of the form

$$\bar{f} = \left[f_a, f_b, f_c, f_x, f_y, f_z, f_{Kq}, f_{Kd}, f_{fr}\right]^t \tag{1b}$$

The derivation of the voltage equations and the transformation to the d-q-o rotor reference frame of a six phase machine have been reported by Fuchs and Rosenberg [1]. For the purpose of this paper, it is only necessary to elaborate a little on the leakage inductance terms, especially the mutual slot leakages, to identify their relationships to the displacement angle between the two stator winding sets and to the coupling between the d- and q- axis circuits. The leakage inductance terms can be grouped into the following submatrices

$$L_\ell = \begin{bmatrix} [L_{\ell s1}] & [L_{\ell 12}] & 0 \\ [L^t_{\ell 12}] & [L_{\ell s2}] & 0 \\ 0 & 0 & [L_{\ell r}] \end{bmatrix} \tag{2a}$$

where

$$[L_{\ell 12}] = \begin{bmatrix} L_{\ell ax} & L_{\ell ay} & L_{\ell az} \\ L_{\ell bx} & L_{\ell by} & L_{\ell bz} \\ L_{\ell cx} & L_{\ell cy} & L_{\ell cz} \end{bmatrix} \tag{2b}$$

$$[L_{\ell 1}] = \mathrm{diag}[L_{\ell s1}, L_{\ell s1}, L_{\ell s1}] \tag{2c}$$

$$[L_{\ell r}] = \mathrm{diag}[L_{\ell Kq}, L_{\ell Kd}, L_{\ell fr}] \tag{2d}$$

$$[L_{\ell 2}] = \mathrm{diag}[L_{\ell s2}, L_{\ell s2}, L_{\ell s2}] \tag{2e}$$

The subscripts s and r refer to the stator and rotor, while 1 and 2 refer to the abc and xyz windings, respectively. The mutual leakage inductances $[L_{\ell 12}]$ are due to the flux components which do not cross the air gap but couple the phase windings of the two stator sets. This coupling occurs when coil sides of abc and xyz windings share the same stator slots.

Since the abc and xyz windings are uniformly distributed about the stator, $[L_{\ell 12}]$ is cyclic, that is $L_{\ell ax}=L_{\ell by}=L_{\ell cz}$, $L_{\ell ay}=L_{\ell bz}=L_{\ell cx}$, and $L_{\ell az}=L_{\ell bx}=L_{\ell cy}$.

The voltage equations contain inductances that are a function of θ_r or time dependent; they are difficult to analyze and time consuming to solve. When the appropriate rotating reference frame transformation is applied to the stator variables, the resulting equations have constant inductances and are much easier to deal with [6]. For the six phase machine, this involves referring the stator variables (abc and xyz) to the common d-q reference frame of the rotor as indicated in Fig. 1. The required transformation is of the form

$$\bar{f}^r = \bar{K}(\theta_r)\,\bar{f} \tag{3a}$$

where

$$\bar{f}^r = [f^r_{q1}, f^r_{d1}, f_{01}, f^r_{q2}, f^r_{d2}, f_{02}, f_{kq}, f_{kd}, f_{fr}]^t \tag{3b}$$

$$\bar{K} = \text{block diag}[K(\theta_r),\ K(\theta_r - \xi), I_{3x3}] \tag{3c}$$

and

$$K(\theta_r) = \frac{2}{3}\begin{bmatrix} \cos\theta_r & \cos(\theta_r - 2\pi/3) & \cos(\theta_r + 2\pi/3) \\ \sin\theta_r & \sin(\theta_r - 2\pi/3) & \sin(\theta_r + 2\pi/3) \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \tag{3d}$$

The resultant rotor reference frame equations are similar to those for a pair of three phase synchronous machines with common magnetizing inductances except for some addition terms associated with the stator winding mutual leakage inductances $[L_{\ell 12}]$. The contribution due to the mutual leakage is

$$K(\theta_r)L_{\ell 12}K^{-1}(\theta_r - \xi) = \begin{bmatrix} L_{\ell m} & -L_{\ell dq} & 0 \\ L_{\ell dq} & L_{\ell m} & 0 \\ 0 & 0 & L_{\ell ax}+L_{\ell ay}+L_{\ell az} \end{bmatrix} \tag{4a}$$

where

$$L_{\ell m} = L_{\ell ax}\cos\xi + L_{\ell ay}\cos(\xi + 2\pi/3) + L_{\ell az}\cos(\xi - 2\pi/3) \tag{4b}$$

$$L_{\ell dq} = L_{\ell ax}\sin\xi + L_{\ell ay}\sin(\xi + 2\pi/3) + L_{\ell az}\sin(\xi - 2\pi/3) \tag{4c}$$

The voltage equations of the six phase synchronous machine in the rotor reference frame are

$$v^r_{q1} = r_1 i^r_{q1} + \omega_r\lambda^r_{d1} + p\lambda^r_{q1} \tag{5a}$$

$$v^r_{d1} = r_1 i^r_{d1} - \omega_r\lambda^r_{q1} + p\lambda^r_{d1} \tag{5b}$$

$$v_{01} = r_1 i_{01} + p\lambda_{01} \tag{5c}$$

$$v'^r_{q2} = r'_2 i'^r_{q2} + \omega_r\lambda'^r_{d2} + p\lambda'^r_{q2} \tag{5d}$$

$$v'^r_{d2} = r'_2 i'^r_{d2} - \omega_r\lambda'^r_{q2} + p\lambda'^r_{d2} \tag{5e}$$

$$v_{02} = r_2 i_{02} + p\lambda_{02} \tag{5f}$$

$$v'_{Kq} = r'_{Kq} i'_{Kq} + p\lambda'_{Kq} \tag{5g}$$

$$v'_{Kd} = r'_{Kd} i'_{Kd} + p\lambda'_{Kd} \tag{5h}$$

$$v'_{fr} = r'_{fr} i'_{fr} + p\lambda'_{fr} \tag{5i}$$

where

$$\lambda^r_{q1} = L_{\ell 1} i^r_{q1} + L'_{\ell m}(i^r_{q1} + i'^r_{q2}) - L'_{\ell dq} i'^r_{d2} + L_{mq}(i^r_{q1} + i'^r_{q2} + i'_{Kq}) \tag{5j}$$

$$\lambda^r_{d1} = L_{\ell 1} i^r_{d1} + L'_{\ell m}(i^r_{d1} + i'^r_{d2}) + L'_{\ell dq} i'^r_{q2} + L_{md}(i^r_{d1} + i'^r_{d2} + i'_{Kd} + i'_{fr}) \tag{5k}$$

$$\lambda_{01} = L_{01} i_{01} + (L_{\ell ax} + L_{\ell ay} + L_{\ell az})(i_{01} + i_{02}) \tag{5l}$$

$$\lambda'^r_{q2} = L_{\ell 2} i'^r_{q2} + L'_{\ell m}(i^r_{q1} + i'^r_{q2}) + L'_{\ell dq} i^r_{d1} + L_{mq}(i^r_{q1} + i'^r_{q2} + i'_{Kq}) \tag{5m}$$

$$\lambda'^r_{d2} = L_{\ell 2} i'^r_{d2} + L'_{\ell m}(i^r_{d1} + i'^r_{d2}) - L'_{\ell dq} i^r_{q1} + L_{md}(i^r_{d1} + i'^r_{d2} + i'_{Kd} + i'_{fr}) \tag{5n}$$

$$\lambda_{02} = L_{02} i_{02} + (L_{\ell ax} + L_{\ell ay} + L_{\ell az})(i_{01} + i_{02}) \tag{5o}$$

$$\lambda'_{Kq} = L'_{\ell Kq} i'_{Kq} + L_{mq}(i^r_{q1} + i'^r_{q2} + i'_{Kq}) \tag{5p}$$

$$\lambda'_{Kd} = L'_{\ell Kd} i'_{Kd} + L_{md}(i^r_{d1} + i'^r_{d2} + i'_{Kd} + i'_{fr}) \tag{5q}$$

$$\lambda'_{fr} = L'_{\ell fr} i'_{fr} + L_{md}(i^r_{d1} + i'^r_{d2} + i'_{Kd} + i'_{fr}) \tag{5r}$$

and

$$L_{\ell 1} = L_{\ell s1} - L'_{\ell m} \tag{5s}$$

$$L_{\ell 2} = L'_{\ell s2} - L'_{\ell m} \tag{5t}$$

$$L_{01} = L_{\ell s1} - L_{\ell ax} - L_{\ell ay} - L_{\ell az} \tag{5u}$$

$$L_{02} = L_{\ell s2} - L_{\ell ax} - L_{\ell ay} - L_{\ell az} \tag{5v}$$

$$L'_{\ell m} = \frac{N_1}{N_2} L_{\ell m} \tag{5w}$$

$$L'_{\ell dq} = \frac{N_1}{N_2} L_{\ell dq} \tag{5x}$$

The effective number of turns of the abc and xyz windings are N_1 and N_2 respectively. Here, the prime quantities are referred to the abc windings. These voltage and flux linkage equations suggest the equivalent circuit shown in Fig. 2 where $\pm L'_{\ell dq}$ represent the mutual leakage coupling between the stator windings.

Torque Equation

Assuming a conservative field, the electromagnetic torque (T_{em}) for the $\bar{P}$ pole, six phase synchronous machine can be expressed in terms of the rotor reference frame variables.

$$T_{em} = \frac{3}{2}\,\frac{\bar{P}}{2}\Big[(i^r_{q1} + i'^r_{q2})L_{md}(i^r_{d1} + i'^r_{d2} + i'_{Kd} + i'_{fr}) - (i^r_{d1} + i'^r_{d2})L_{mq}(i^r_{q1} + i'^r_{q2} + i'_{Kq})\Big] \tag{6}$$

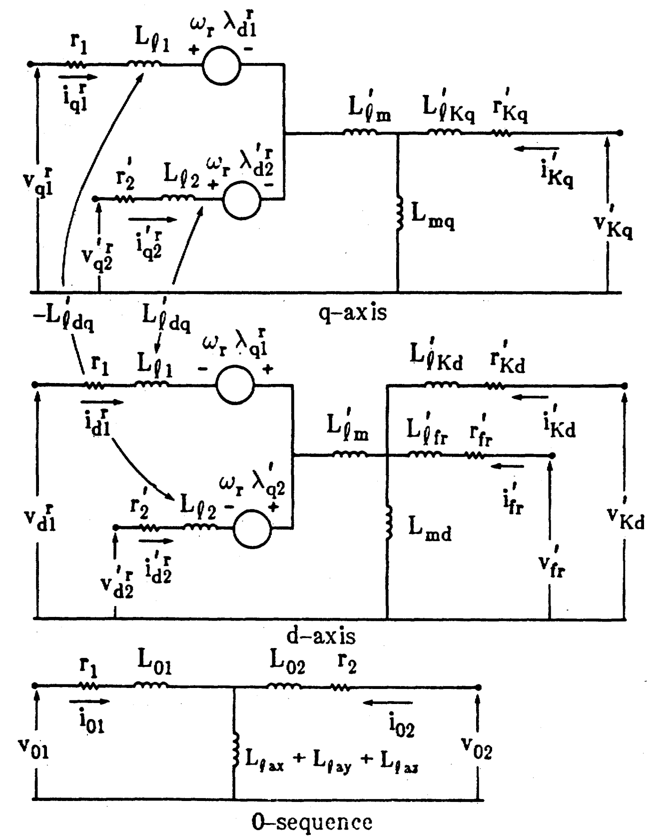


Figure 2. An equivalent circuit of a six phase synchronous machine.

STEADY STATE ANALYSIS WITH SINUSOIDAL VOLTAGE INPUTS

Steady state analysis is conducted under the assumption that each three phase stator voltage set is made up of balanced sinusoids displaced by $\frac{2\pi}{3}$ radians. For convenience, time zero (t=0) is chosen so that

$$v_a = V_1 \cos \omega_e t \tag{7a}$$

$$v'_x = V'_2 \cos(\omega_e t - \gamma) \tag{7b}$$

where

V_1 = peak value of the abc voltages

V'_2 = peak value of the xyz voltages referred to the abc windings

ω_e = synchronous frequency in radians per second

γ = phase difference between the a and x voltages

Under steady state operation with the rotor rotating at synchronous speed ω_e, the rotor angle can be written as $\theta_r = \omega_e t + \delta$ where δ is constant. Assuming three wire connections, there is no zero sequence component present in the voltages and currents of both stator winding sets. In the rotor reference frame, the stator d-q voltages become constant voltages:

$$V^r_{q1} = V_1 \cos \delta \tag{8a}$$

$$V^r_{d1} = V_1 \sin \delta \tag{8b}$$

$$V'^r_{q2} = V'_2 \cos(\delta - \xi - \gamma) \tag{8c}$$

$$V'^r_{d2} = V'_2 \sin(\delta - \xi - \gamma) \tag{8d}$$

These voltages are related to the synchronously rotating phasors V_a and V_x by:

$$\tilde{V}_a e^{-j\delta} = V^r_{q1} - jV^r_{d1} \tag{9a}$$

$$\tilde{V}'_x e^{-j(\delta-\xi)} = V'^r_{q2} - jV'^r_{d2} \tag{9b}$$

Similarly, the steady state rotor reference frame currents I^r_{q1}, I^r_{d1}, I'^r_{q2} and I'^r_{d2} are related to stator current phasors $\tilde{I}_a$ and $\tilde{I}_x$ by:

$$\tilde{I}_a e^{-j\delta} = I^r_{q1} - jI^r_{d1} \tag{10a}$$

$$\tilde{I}'_x e^{-j(\delta-\xi)} = I'^r_{q2} - jI'^r_{d2} \tag{10b}$$

where

$$\tilde{I}_a = I_1 \angle{-\phi_1} \tag{11a}$$

$$\tilde{I}'_x = I'_2 \angle{-\gamma-\phi_2} \tag{11b}$$

and

ϕ_1 = power factor angle on the abc windings

ϕ_2 = power factor angle on the xyz windings

I_1 = peak value of the abc winding currents

I'_2 = peak value of the xyz winding currents referred to the abc windings

For steady state operation with sinusoidal voltage inputs, the damper winding currents and all the time derivative terms are zero, the voltage equations in phasor form are

$$\tilde{V}_a = \left[r_1 + j\frac{\omega_e}{\omega_b}(X_{\ell 1} + X'_{\ell m} + X_{mq})\right]\tilde{I}_a + \frac{\omega_e}{\omega_b}\left[X'_{\ell dq} + j(X'_{\ell m} + X_{mq})\right]\tilde{I}'_x e^{j\xi} + \tilde{E}_q \tag{12a}$$

$$\tilde{V}'_x e^{j\xi} = \left[r'_2 + j\frac{\omega_e}{\omega_b}(X_{\ell 2} + X'_{\ell m} + X_{mq})\right]\tilde{I}'_x e^{j\xi} + \frac{\omega_e}{\omega_b}\left[-X'_{\ell dq} + j(X'_{\ell m} + X_{mq})\right]\tilde{I}_a + \tilde{E}_q \tag{12b}$$

where ω_b is the base frequency,

$$\tilde{E}_q = \left[\frac{\omega_e}{\omega_b}(X_{md} - X_{mq})(I^r_{d1} + I'^r_{d2}) + E'_{fr}\right]e^{j\delta} \tag{12c}$$

and the open-circuit generated voltage

$$E'_{fr} = \frac{\omega_e}{\omega_b} X_{md} I'_{fr} \tag{12d}$$

The phasor diagram corresponding to these equations is shown in Fig. 3.

In terms of the stator voltages and the angle δ, the steady state torque, with stator resistances neglected, is

$$\begin{aligned} T_{em} = \frac{P}{2}\,\frac{3}{2}\,\frac{1}{\omega_b}\left(\frac{\omega_b}{\omega_e}\right)^2 \Bigg\{ & \frac{-E'_{fr}}{(X_{\ell 1}+X_{\ell 2})X_d}\Big[V_1(X_{\ell 2}\sin\delta + X'_{\ell dq}\cos\delta) + V'_2(X_{\ell 1}\sin(\delta-\xi-\gamma) - X'_{\ell dq}\cos(\delta-\xi-\gamma))\Big] \\ & + \frac{1}{(X_{\ell 1}+X_{\ell 2})^2}\left(\frac{1}{X_q} - \frac{1}{X_d}\right) \Big[\frac{V_1^2}{2}((X'^2_{\ell dq} - X^2_{\ell 2})\sin 2\delta - X_{\ell 2}X'_{\ell dq}\cos 2\delta) \\ & + \frac{V'^2_2}{2}((X'^2_{\ell dq} - X^2_{\ell 1})\sin 2(\delta-\xi-\gamma) + X_{\ell 1}X'_{\ell dq}\cos 2(\delta-\xi-\gamma)) + V_1V'_2 \\ & (X'_{\ell dq}(X_{\ell 2}-X_{\ell 1})\cos(2\delta-\xi-\gamma) - (X_{\ell 1}X_{\ell 2} + X'^2_{\ell dq})\sin(2\delta-\xi-\gamma))\Big]\Bigg\} \end{aligned} \tag{13a}$$

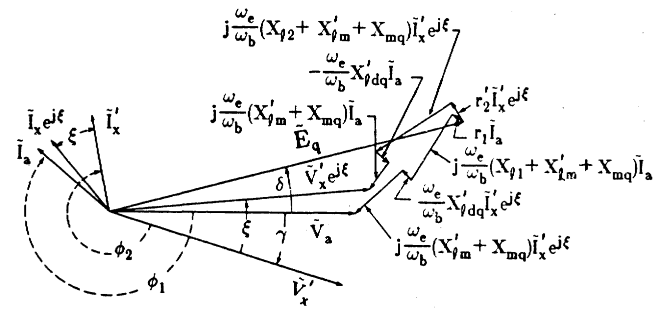


Figure 3. Steady state phasor diagram for a six phase synchronous machine.

where

$$X_d = X'_{\ell m} + \frac{X_{\ell 1}X_{\ell 2} - X'^{2}_{\ell dq}}{(X_{\ell 1} + X_{\ell 2})} + X_{md} \qquad 13b$$

$$X_q = X'_{\ell m} + \frac{X_{\ell 1}X_{\ell 2} - X'^{2}_{\ell dq}}{(X_{\ell 1} + X_{\ell 2})} + X_{mq} \qquad 13c$$

This torque expression reduces to that for a standard three phase synchronous machine when either stator winding set is open circuited [7].

Phasor diagrams

Phasor diagrams for various operating conditions are obtained by solving the steady state voltage and torque equations simultaneously. Assuming that the specified terminal constraints are the stator terminal voltage magnitudes, the real and reactive power flowing into the xyz windings, and the electromagnetic torque, the solution provides the values of the phasor voltages $\tilde{V}_a$, $\tilde{V}_x e^{j\xi}$ and $\tilde{E}_q$, the phasor currents $\tilde{I}_a$ and $\tilde{I}_x e^{j\xi}$, the excitation voltage (E'_{fr}), and the air gap voltage ($\tilde{E}_{ag}$). The air gap voltage is calculated from

$$\tilde{E}_{ag} = \tilde{E}_q + j\frac{\omega_e}{\omega_b}X_{mq}(\tilde{I}_a + \tilde{I}_x e^{j\xi}) \qquad 14$$

As an example, the 240 volt, 100 kva six phase machine described in the Appendix III, with specified inputs of $|\tilde{V}_a| = 200$ volts, $|\tilde{V}_x| = 200$ volts, $|P_{xyz}| = 25$ kw, and $|Q_{xyz}| = 18$ kvar is considered. Here, power flowing into the stator terminals and motoring torque are considered positive. The phasor diagrams for three different shaft loads and with negative Q_{xyz} are given in Fig. 4a. The phase displacement between adjacent phasors in these diagrams is exaggerated so that their phase relationship can be easily seen; relative phasor magnitudes, however, are preserved. The actual values of the terminal powers and excitation voltage are given in the caption below each phasor diagram. From these phasor diagrams two methods of power transfer in a six phase machine can be identified. The first method, called transformer action, occurs exclusively when there is no mechanical load: power flows from the xyz terminals through the air gap to the abc terminals with no power transfer to the shaft. This condition is indicated by $\tilde{V}_x e^{j\xi}$ leading $\tilde{E}_{ag}$, $\tilde{V}_a$ lagging $\tilde{E}_{ag}$, and $\tilde{E}_q$ coinciding with $\tilde{E}_{ag}$. The second method of power flow, called motor action, is most obvious when the machine is motoring. Here power from both stator terminal sets flows through the air gap and onto the machine shaft. This condition is indicated by an $\tilde{E}_{ag}$ that leads $\tilde{E}_q$. No net power transfer through transformer action occurs as $\tilde{E}_{ag}$ is not positioned between the stator voltage phasors. However, when the machine is generating, the rotor q axis leads the air gap voltage as indicated by $\tilde{E}_q$ leading $\tilde{E}_{ag}$.

Phasor diagrams for operating conditions with reactive power flowing into the xyz winding terminals of the machine are given in Fig. 4b. Again, power flows by motor and transformer actions are evident from the phase relationships between the air gap voltage, the stator voltages and the rotor q-axis. Since part of the reactive power required by the machine is now supplied externally from the xyz terminals, the field winding is underexcited.

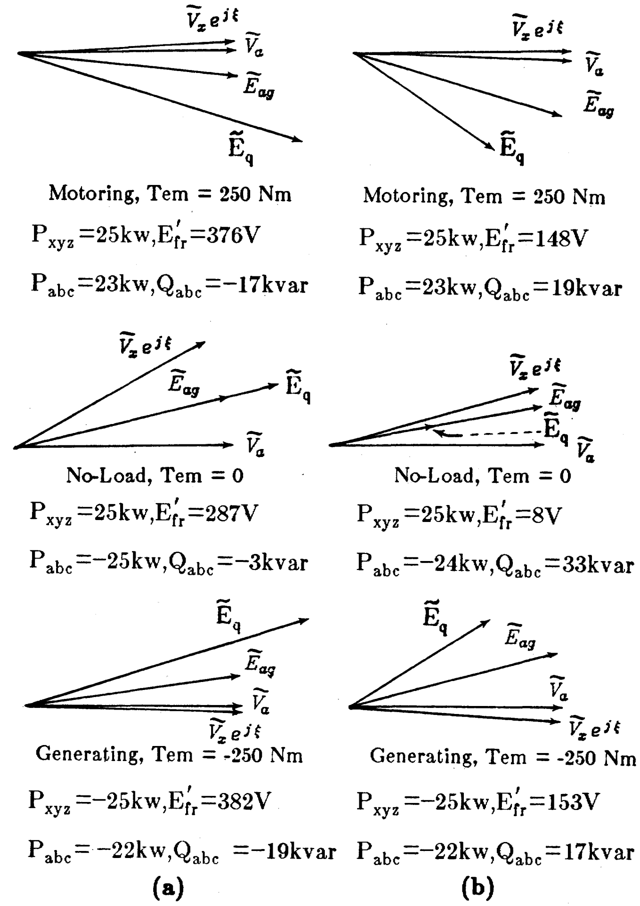


Figure 4. Phasor diagrams for reactive power flowing a) out of the xyz windings ($Q_{xyz} < 0$) and b) into the xyz windings ($Q_{xyz} > 0$).

STEADY-STATE ANALYSIS WITH AC-DC TERMINAL CONNECTIONS

The analysis described in the following text can readily be extended to a form which is applicable to a variety of configurations of terminal excitation. Just to illustrate the method, the system shown in Fig. 5 is chosen. The operation of the bridge converter introduces quasi-square currents into the abc stator windings. The system response in the presence of the current harmonics can be determined using a piece-wise solution technique [4] or a dq harmonic balance technique [8]. The approach described here is based on Fourier analysis, using harmonic phasor forms. A simplified flow chart of computational steps involved is given in Fig. 6.

Assuming that the DC voltage V_{DC}, AC frequency ω_e, and the voltage magnitude $|V_{xyz}|$ and power flows, P_{xyz} and Q_{xyz}, of the xyz windings are known or specified, iteration for the fundamental steady-state condition using a Newton-

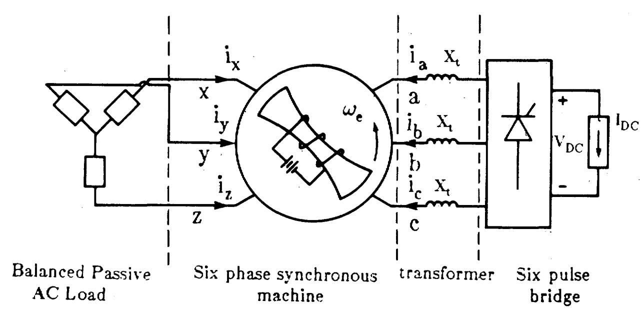
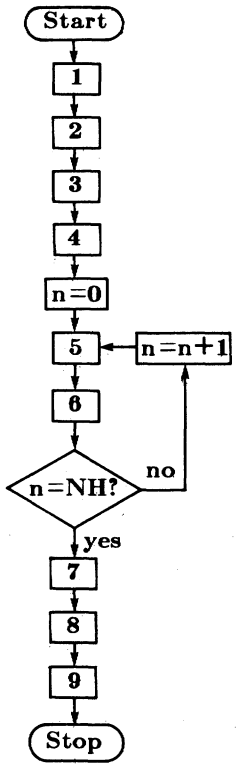


Figure 5. Six phase machine with AC and DC stator connections.



1) Input. Calculate machine parameters and bridge constants.
2) Obtain steady state solution for desired shaft torque and terminal power constraints using a Newton Raphson iterative method.
3) Calculate abc current Fourier coefficients to the NHth harmonics based on calculated values of bridge delay, commutation and extinction angles, and I_{DC}.
4) Determine rotor reference frame Fourier coefficients of abc currents (i^r_{q1} and i^r_{d1}) to the NHth harmonics.
5) Calculate the nth harmonic phasors, $\tilde{I}^r_{q2}$, $\tilde{I}^r_{d2}$, $\tilde{V}^r_{q1}$, $\tilde{V}^r_{d1}$, $\tilde{V}^r_{q2}$, $\tilde{V}^r_{d2}$, $\tilde{I}_{kd}$, $\tilde{I}_{kq}$ and $\tilde{I}_{fr}$.
6) Convert the nth harmonic phasor to Fourier series components.
7) Calculate electromagnetic torque Fourier coefficients using point by point multiplication and an FFT.
8) Transform rotor reference frame Fourier series solutions to abc-xyz voltage and current Fourier series.
9) Calculate harmonic distortion. Print coefficients of all waveforms. Plot voltage, current and electromagnetic torque waveforms.

Figure 6. Simplified flow chart for the harmonic phasor method.

Raphson technique is made with progressive estimates of the DC current I_{DC} until the desired mechanical load torque is obtained. The calculated magnitude of the fundamental air gap voltage, along with the values of the commutating reactance, V_{DC} and I_{DC}, is then used to determine the bridge delay and commutation (or extinction) angles, and the abc phase currents. Fourier series of abc phase currents are phase shifted so that their fundamental components satisfy phase relationships of the desired operating point then transformed to the rotor reference frame to produce harmonic phasor components of i^r_{q1} and i^r_{d1}. These harmonic phasor currents in the rotor reference frame are applied to the respective frequency domain representation of the machine obtained by replacing the time derivative p with $jn\omega_e$. The voltage equation for the nth harmonic phasor components is

$$\begin{bmatrix} \tilde{V}^r_{q1}(n\omega_e) \\ \tilde{V}^r_{d1}(n\omega_e) \\ \tilde{V}'^r_{q2}(n\omega_e) \\ \tilde{V}'^r_{d2}(n\omega_e) \\ \tilde{V}'_{fr}(n\omega_e) \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} r_1 + jn(X_{\ell 1} + X'_{\ell m} + X_{mq}) & X_{\ell 1} + X'_{\ell m} + X_{md} & jn(X'_{\ell m} + X_{mq}) + X'_{\ell dq} & -jnX'_{\ell dq} + (X'_{\ell m} + X_{md}) & X_{md} & jnX_{mq} & X_{md} \\ -(X_{\ell 1} + X'_{\ell m} + X_{mq}) & r_1 + jn(X_{\ell 1} + X'_{\ell m} + X_{md}) & jnX'_{\ell dq} - (X'_{\ell m} + X_{mq}) & jn(X'_{\ell m} + X_{md}) + X'_{\ell dq} & jnX_{md} & -X_{mq} & jnX_{md} \\ jn(X'_{\ell m} + X_{mq}) - X'_{\ell dq} & jnX'_{\ell dq} + (X'_{\ell m} + X_{md}) & r'_2 + jn(X_{\ell 2} + X'_{\ell m} + X_{mq}) & X_{\ell 2} + X'_{\ell m} + X_{md} & X_{md} & jnX_{mq} & X_{md} \\ -jnX'_{\ell dq} - (X'_{\ell m} + X_{mq}) & jn(X'_{\ell m} + X_{md}) - X'_{\ell dq} & -(X_{\ell 2} + X'_{\ell m} + X_{mq}) & r'_2 + jn(X_{\ell 2} + X'_{\ell m} + X_{md}) & jnX_{md} & -X_{mq} & jnX_{md} \\ 0 & jnX_{md} & 0 & jnX_{md} & r'_{fr} + jn(X'_{\ell fr} + X_{md}) & 0 & jnX_{md} \\ jnX_{mq} & 0 & jnX_{mq} & 0 & 0 & r'_{Kq} + jn(X'_{\ell Kq} + X_{mq}) & 0 \\ 0 & jnX_{md} & 0 & jnX_{md} & jnX_{md} & 0 & r'_{Kd} + jn(X'_{\ell Kd} + X_{md}) \end{bmatrix} \times \begin{bmatrix} \tilde{I}^r_{q1}(n\omega_e) \\ \tilde{I}^r_{d1}(n\omega_e) \\ \tilde{I}'^r_{q2}(n\omega_e) \\ \tilde{I}'^r_{d2}(n\omega_e) \\ \tilde{I}'_{fr}(n\omega_e) \\ \tilde{I}'_{Kq}(n\omega_e) \\ \tilde{I}'_{Kd}(n\omega_e) \end{bmatrix} \quad 15$$

On the xyz windings the relationship between the nth harmonic phasor components of the voltage and current as determined by the external load characteristic, is

$$\begin{bmatrix} \tilde{V}'^r_{q2}(n\omega_e) \\ \tilde{V}'^r_{d2}(n\omega_e) \end{bmatrix} = \begin{bmatrix} -r' - jnX'_L & -X'_L \\ X'_L & -r' - jnX'_L \end{bmatrix} \begin{bmatrix} \tilde{I}'^r_{q2}(n\omega_e) \\ \tilde{I}'^r_{d2}(n\omega_e) \end{bmatrix} \quad 16$$

where r' and X'_L are the resistive and reactive components of the xyz load referred to the abc windings.

Simultaneous solution of Eqs. 15 and 16 yields the phasor components $\tilde{V}^r_{q1}(n\omega_e)$, $\tilde{V}^r_{d1}(n\omega_e)$, $\tilde{I}^r_{q2}(n\omega_e)$, $\tilde{I}^r_{d2}(n\omega_e)$, $\tilde{V}^r_{q2}(n\omega_e)$, $\tilde{V}^r_{d2}(n\omega_e)$, $\tilde{I}_{Kq}(n\omega_e)$, $\tilde{I}_{Kd}(n\omega_e)$ and $\tilde{I}_{fr}(n\omega_e)$. With constant speed operation, Eqs. 15 and 16 are linear and the principle of superposition applies, the harmonic components can be added to produce Fourier series solutions. Instantaneous values of the electromagnetic torque at discrete time intervals are calculated from the currents and the result processed by an FFT to give the Fourier components of the torque. The abc and xyz voltages and currents are obtained by transforming the rotor reference frame stator voltages and currents to the stationary reference frames.

Steady State Waveforms

This section presents a comparison of the output waveforms from the digital calculations with those from an analog simulation of the 240 V, 125 kva, four pole, salient pole six phase machine given in Appendix III with a winding displacement angle of 30° and stator connections as shown in Fig. 5. Since the purpose of the waveform comparison is to verify the approach used in the digital calculation using the analog simulation, effort was made to ensure that the waveforms from the digital simulation contain sufficient spectral components to reproduce the fine details on the commutation notches and the torque pulsations. This was achieved by including harmonics up to 75th in the digital calculations.

The analog simulation used was a detailed dynamic model of the system shown in Fig. 5. The analog simulation of the six phase synchronous machine was based on the integral equations of the six phase synchronous machine given in Appendix II and was implemented in the same manner as for a standard three phase synchronous machine [9]. Stator winding arrangements with different displacement angles could be readily studied by making appropriate changes to those parameters dependent on the value of the displacement angle. A standard excitation system of the IEEE Type 1 was simulated for the purpose of regulating the terminal voltages of the xyz windings. In the analog simulation of the converter, each thyristor was simulated by an idealized switch controlled by a logic signal from an inverse cosine firing signal generator.

The first condition investigated was for inverter operation with a 37.5 kva, 0.8 lagging power factor load on the xyz windings. The output waveforms from the digital and analog simulations for this operating condition are shown in Fig. 7. The output waveforms from the two simulations appear alike, down to such details like the exact location and size of voltage dips caused by the commutation process. For inverter operation, the phase current $-I_a$ lags V_a by an angle between 90° and 180°. Voltage dips appear in V_a when I_a is commutated These dips appear to a lesser extent in V_x indicating some filtering effect. Note that V_x lags V_a by about 30°, corresponding to the winding displacement angle chosen. The current, I_x, contains very little harmonic distortion as the inductive load tended to restrict the higher harmonic currents from flowing.

The electromagnetic torque (T_{em}) and the rotor currents (I'_{kq} and I'_{kd}) have large sixth harmonic components. The sixth harmonic currents in the rotor windings are caused by the fifth and seventh harmonic stator current components of the six pulse bridge. The fifth harmonic stator current induces negatively rotating sixth harmonic rotor current, while the seventh harmonic stator current induces positively rotating sixth harmonic rotor current. Since the products of the stator and rotor currents make up the electromagnetic torque, T_{em} has a large sixth harmonic pulsating component

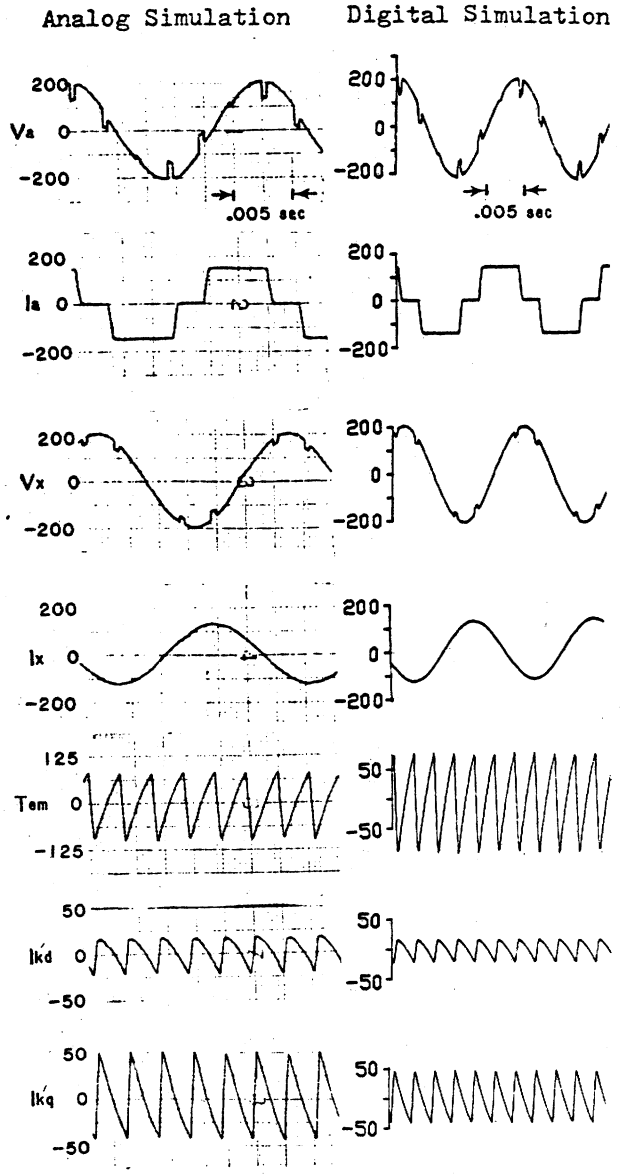


Figure 7. Steady state waveforms for $\xi = 30^\circ$, 0.65 pitch, 0.8 lagging power factor AC load and bridge inverter operation.

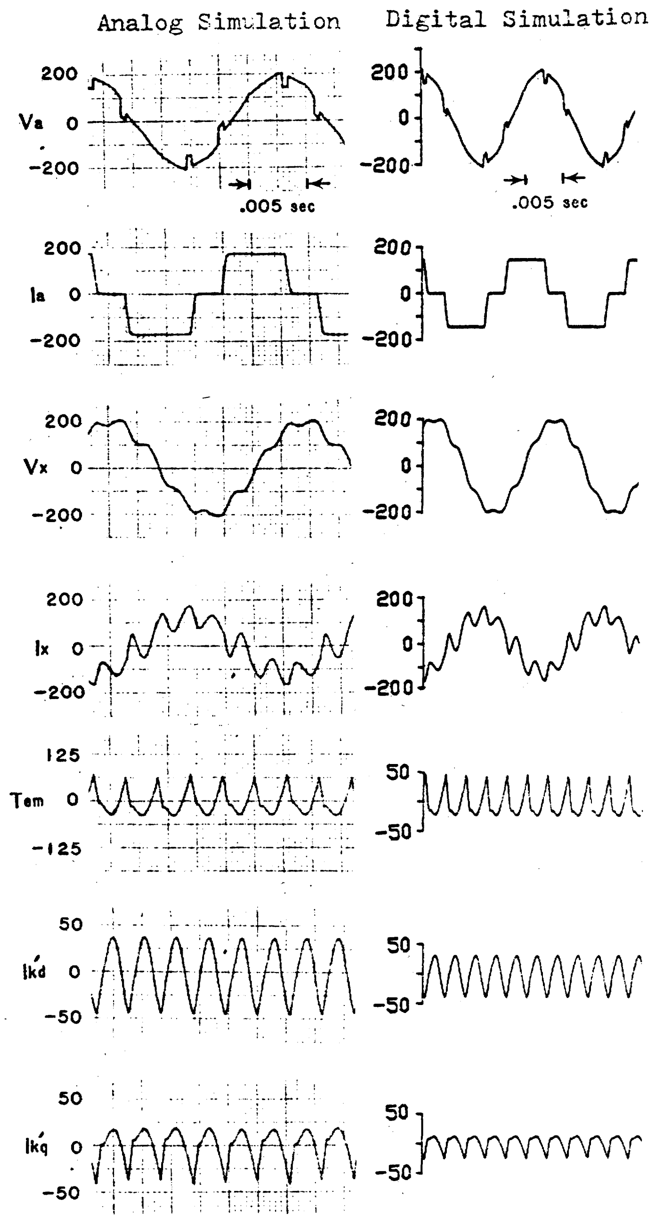


Figure 8. Steady state waveforms for $\xi = 30^\circ$, 0.65 pitch, 0.8 leading power factor AC load and bridge inverter operation.

as shown. The average electromagnetic torque is zero since the load torque is zero.

The second operating condition investigated was that of the bridge operating as an inverter and a 37.5 kva, 0.8 leading power factor load on the xyz terminals. Good agreement between the output waveforms from the analog and digital simulations is evident from Fig. 8. Phase relationships between V_a, I_a, V_x and I_x are as expected with inverter operation and a leading power factor load. A large seventh harmonic component is evident in V_x and I_x due to capacitive reactance. T_{em} takes on similar characteristics as stator q-axis currents since the largest component of harmonic torque pulsations comes from the product of q axis stator currents and the large DC field current.

Figure 9 shows output waveforms for bridge rectifier operation with an a 37.5 kva, 0.8 lagging power factor load on the xyz windings. Again, the agreement between the output waveforms from the analog and digital simulations is good. With rectifier operation, the current $-I_a$ lags V_a by less than 90°. Stator phase voltages and currents have similar waveshapes like those for inverter operation with a lagging power factor load. Since the machine was generating, the electromagnetic torque had a negative average value.

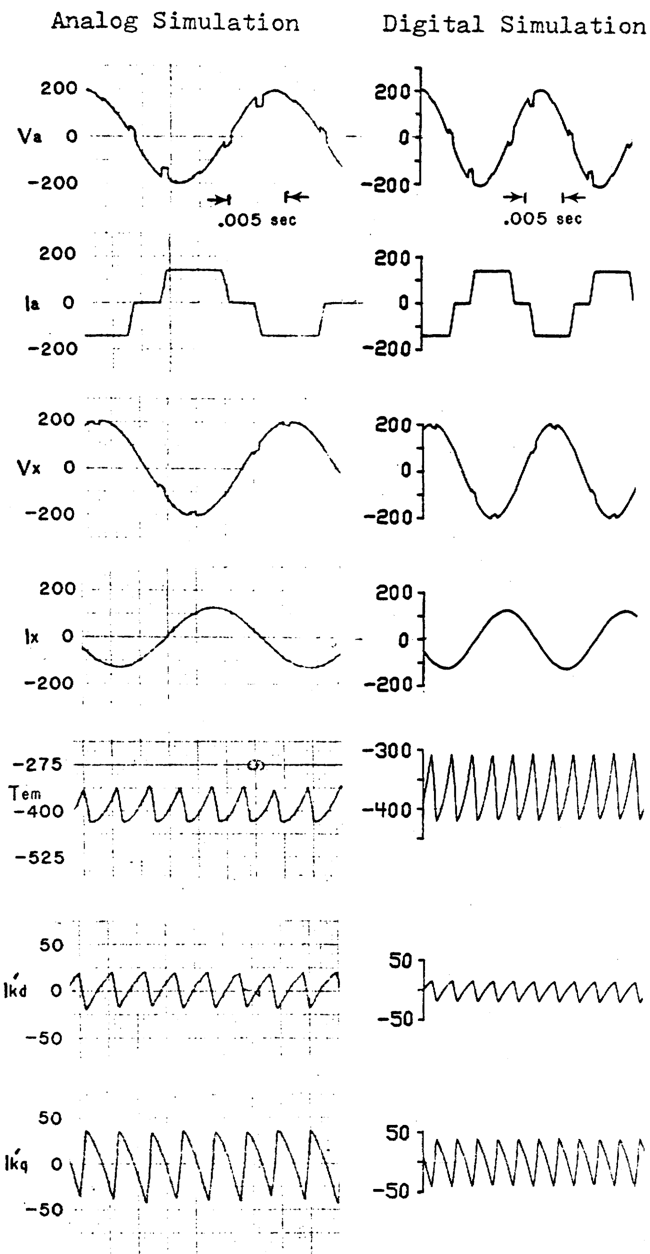


Figure 9. Steady state waveforms for $\xi = 30^\circ$, 0.65 pitch, 0.8 lagging power factor and bridge rectifier operation.

CONCLUSION

A harmonic phasor method is presented for the steady-state analysis of a six phase synchronous machine with AC-DC stator connections. The method has been shown to be accurate in comparison with a detailed analog simulation.

APPENDIX I
WINDING INDUCTANCES

The idealized model of the six phase machine can be regarded to have equivalent sinusoidally distributed windings. The calculations of inductances are made by neglecting saturation and space harmonics and assuming sinusoidally distributed, radial flux pattern. In addition, the airgap is assumed to be sinusoidally distributed so that its inverse at an angle of ϕ can be written as a function of ϕ and θ_r, the angular displacement of the q-axis of the rotor from the same reference as for ϕ. Thus

$$g^{-1}(\phi) = h_A - h_B\cos 2(\theta_r - \phi)$$

where

$$h_A = \frac{1}{2}\left(\frac{1}{g_{min}} + \frac{1}{g_{max}}\right), \qquad h_B = \frac{1}{2}\left(\frac{1}{g_{min}} - \frac{1}{g_{max}}\right)$$

and g_{min}, g_{max} are the minimum and maximum air gap widths.

The self and mutual inductances between two arbitrary windings m and n with N_m and N_n equivalent turns and magnetic axes displaced α_m and α_n from the reference axis are given by

$$L_{mn} = L_{nm} = L_{Amn}\cos(\alpha_m - \alpha_n) - L_{Bmn}\cos(2\theta_r - \alpha_m - \alpha_n)$$

where

$$L_{Amn} = N_m N_n\left(\frac{P_{gq} + P_{gd}}{2}\right), \qquad L_{Bmn} = N_m N_n\left(\frac{P_{gq} - P_{gd}}{2}\right)$$

$$P_{gq} = \mu_o \frac{r\ell\pi}{4}\left(h_A - \frac{h_B}{2}\right), \qquad P_{gd} = \mu_o \frac{r\ell\pi}{4}\left(h_A + \frac{h_B}{2}\right),$$

r and ℓ are the mean radius of the air-gap and effective length of the stator core respectively.

Using the above expression, the self and mutual inductances of the stator, stator/rotor and rotor windings can be written as follow:

Stator Self and Mutual Inductances

$L_{aa} = L_{A11} - L_{B11}\cos 2\theta r$

$L_{ab} = -\frac{1}{2}L_{A11} - L_{B11}\cos(2\theta r - 2\pi/3)$

$L_{ac} = -\frac{1}{2}L_{A11} - L_{B11}\cos(2\theta r + 2\pi/3)$

$L_{bb} = L_{A11} - L_{B11}\cos 2(\theta r - 2\pi/3)$

$L_{bc} = -\frac{1}{2}L_{A11} - L_{B11}\cos 2\theta r$

$L_{cc} = L_{A11} - L_{B11}\cos 2(\theta r + 2\pi/3)$

$L_{xx} = L_{A22} - L_{B22}\cos 2(\theta r - \xi)$

$L_{xy} = -\frac{1}{2}L_{A22} - L_{B22}\cos(2\theta r - 2\xi - 2\pi/3)$

$L_{xz} = L_{zx} = -\frac{1}{2}L_{A22} - L_{B22}\cos(2\theta r - 2\xi + 2\pi/3)$

$L_{yy} = L_{A22} - L_{B22}\cos 2(\theta r - \xi - 2\pi/3)$

$L_{yz} = -\frac{1}{2}L_{A22} - L_{B22}\cos(2\theta r - 2\xi)$

$L_{zz} = L_{A22} - L_{B22}\cos 2(\theta r - \xi + 2\pi/3)$

$L_{ax} = L_{A12}\cos\xi - L_{B12}\cos(2\theta r - \xi)$

$L_{ay} = L_{A12}\cos(\xi + 2\pi/3) - L_{B12}\cos(2\theta r - \xi - 2\pi/3)$

$L_{az} = L_{A12}\cos(\xi - 2\pi/3) - L_{B12}\cos(2\theta r - \xi + 2\pi/3)$

$L_{bx} = L_{A12}\cos(\xi - 2\pi/3) - L_{B12}\cos(2\theta r - \xi - 2\pi/3)$

$L_{by} = L_{A12}\cos\xi - L_{B12}\cos(2\theta r - \xi + 2\pi/3)$

$L_{bz} = L_{A12}\cos(\xi + 2\pi/3) - L_{B12}\cos(2\theta r - \xi)$

$L_{cx} = L_{A12}\cos(\xi + 2\pi/3) - L_{B12}\cos(2\theta r - \xi + 2\pi/3)$

$L_{cy} = L_{A12}\cos(\xi - 2\pi/3) - L_{B12}\cos(2\theta r - \xi)$

$L_{cz} = L_{A12}\cos\xi - L_{B12}\cos(2\theta r - \xi - 2\pi/3)$

Stator/Rotor Mutual Inductances

$L_{aKq} = L_{1q}\cos\theta_r$ $\qquad L_{xKq} = L_{2q}\cos(\theta_r - \xi)$

$L_{bKq} = L_{1q}\cos(\theta_r - 2\pi/3)$ $\qquad L_{yKq} = L_{2q}\cos(\theta_r - \xi - 2\pi/3)$

$L_{cKq} = L_{1q}\cos(\theta_r + 2\pi/3)$ $\qquad L_{zKq} = L_{2q}\cos(\theta_r - \xi + 2\pi/3)$

$L_{aKd} = L_{1d}\sin\theta_r$ $\qquad L_{xKd} = L_{2d}\sin(\theta_r - \xi)$

$L_{bKd} = L_{1d}\sin(\theta_r - 2\pi/3)$ $\qquad L_{yKd} = L_{2d}\sin(\theta_r - \xi - 2\pi/3)$

$L_{cKd} = L_{1d}\sin(\theta_r + 2\pi/3)$ $\quad L_{zKd} = L_{2d}\sin(\theta_r - \xi + 2\pi/3)$

$L_{afr} = L_{1f}\sin\theta_r$ $\quad L_{xfr} = L_{2f}\sin(\theta_r - \xi)$

$L_{bfr} = L_{1f}\sin(\theta_r - 2\pi/3)$ $\quad L_{yfr} = L_{2f}\sin(\theta_r - \xi - 2\pi/3)$

$L_{cfr} = L_{1f}\sin(\theta_r + 2\pi/3)$ $\quad L_{zfr} = L_{2f}\sin(\theta_r - \xi + 2\pi/3)$

where

$$L_{nq} = N_n N_{kq} P_{gq}, \qquad L_{nd} = N_n N_{kd} P_{gd}$$
$$L_{nf} = N_n N_f P_{gd}$$

Rotor Self and Mutual Inductances

$L_{KqKq} = N_{kq}^2 P_{gq}$ $\quad L_{KdKd} = N_{kd}^2 P_{gd}$

$L_{KqKd} = 0$ $\quad L_{Kdfr} = N_{kd} N_f P_{gd}$

$L_{Kqfr} = 0$ $\quad L_{frfr} = N_f^2 P_{gd}$

APPENDIX II
ANALOG SIMULATION

The detailed analog simulation of a six phase machine is based on the integral form of the machine's voltage and torque equations with flux linkages per second and speed as state variables, winding currents as output variables, applied voltages and load torque as input variables. The following steps show how the equations of the six phase machine can be manipulated into the desired form. The flux linkages per second are expressed in terms of the currents as follows:

$$\psi_{q1}^r - \psi_{mq} = X_{\ell 1} i_{q1}^r + X'_{\ell m}(i_{q1}^r + i'^r_{q2}) - X'_{\ell dq} i'^r_{d2}$$
$$\psi_{d1}^r - \psi_{md} = X_{\ell 1} i_{d1}^r + X'_{\ell m}(i_{d1}^r + i'^r_{d2}) + X'_{\ell dq} i'^r_{q2}$$
$$\psi'^r_{q2} - \psi_{mq} = X_{\ell 2} i'^r_{q2} + X'_{\ell m}(i_{q1}^r + i'^r_{q2}) + X'_{\ell dq} i_{d1}^r$$
$$\psi'^r_{d2} - \psi_{md} = X_{\ell 2} i'^r_{d2} + X'_{\ell m}(i_{d1}^r + i'^r_{d2}) - X'_{\ell dq} i_{q1}^r$$
$$\psi'_{Kq} - \psi_{mq} = X'_{\ell Kq} i'_{Kq}$$
$$\psi'_{Kd} - \psi_{md} = X'_{\ell Kd} i'_{Kd}$$
$$\psi'_{fr} - \psi_{md} = X'_{\ell fr} i'_{fr}$$

where

$$\psi_{mq} = X_{mq}(i_{q1}^r + i'^r_{q2} + i'_{Kq})$$
$$\psi_{md} = X_{md}(i_{d1}^r + i'^r_{d2} + i'_{Kd} + i'_{fr})$$

Solving for the currents gives:

$$i_{q1}^r = \frac{1}{X_{\ell 6}}\Big[(X_{\ell 2} + X'_{\ell m})\psi_{q1}^r - X'_{\ell m}\psi'^r_{q2} - X_{\ell 2}\psi_{mq} + X'_{\ell dq}(\psi'^r_{d2} - \psi_{md})\Big]$$

$$i_{d1}^r = \frac{1}{X_{\ell 6}}\Big[(X_{\ell 2} + X'_{\ell m})\psi_{d1}^r - X'_{\ell m}\psi'^r_{d2} - X_{\ell 2}\psi_{md} - X'_{\ell dq}(\psi'^r_{q2} - \psi_{mq})\Big]$$

$$i'^r_{q2} = \frac{1}{X_{\ell 6}}\Big[-X'_{\ell m}\psi_{q1}^r + (X_{\ell 1} + X'_{\ell m})\psi'^r_{q2} - X_{\ell 1}\psi_{mq} - X'_{\ell dq}(\psi_{d1}^r - \psi_{md})\Big]$$

$$i'^r_{d2} = \frac{1}{X_{\ell 6}}\Big[-X'_{\ell m}\psi_{d1}^r + (X_{\ell 1} + X'_{\ell m})\psi'^r_{d2} - X_{\ell 1}\psi_{md} + X'_{\ell dq}(\psi_{q1}^r - \psi_{mq})\Big]$$

$$i'_{Kq} = \frac{1}{X'_{\ell Kq}}\Big[\psi'_{Kq} - \psi_{mq}\Big]$$

$$i'_{Kd} = \frac{1}{X'_{\ell Kd}}\Big[\psi'_{Kd} - \psi_{md}\Big]$$

$$i'_{fr} = \frac{1}{X'_{\ell fr}}\Big[\psi'_{fr} - \psi_{md}\Big]$$

where

$$X_{\ell 6} = X_{\ell 1} X_{\ell 2} + (X_{\ell 1} + X_{\ell 2}) X'_{\ell m} - X'^2_{\ell dq}$$

Back substituting these currents into the voltage equations and rewriting them as integral equations yields:

$$\psi_{q1}^r = \frac{\omega_b}{p}\left\{v_{q1}^r - \frac{\omega_r}{\omega_b}\psi_{d1}^r - \frac{r_1}{X_{\ell 6}}\Big[(X_{\ell 2} + X'_{\ell m})\psi_{q1}^r - X'_{\ell m}\psi'^r_{q2} - X_{\ell 2}\psi_{mq} + X'_{\ell dq}(\psi'^r_{d2} - \psi_{md})\Big]\right\}$$

$$\psi_{d1}^r = \frac{\omega_b}{p}\left\{v_{d1}^r + \frac{\omega_r}{\omega_b}\psi_{q1}^r - \frac{r_1}{X_{\ell 6}}\Big[(X_{\ell 2} + X'_{\ell m})\psi_{d1}^r - X'_{\ell m}\psi'^r_{d2} - X_{\ell 2}\psi_{md} - X'_{\ell dq}(\psi'^r_{q2} - \psi_{mq})\Big]\right\}$$

$$\psi'^r_{q2} = \frac{\omega_b}{p}\left\{v'^r_{q2} - \frac{\omega_r}{\omega_b}\psi'^r_{d2} - \frac{r'_2}{X_{\ell 6}}\Big[-X'_{\ell m}\psi_{q1}^r + (X_{\ell 1} + X'_{\ell m})\psi'^r_{q2} - X_{\ell 1}\psi_{mq} - X'_{\ell dq}(\psi_{d1}^r - \psi_{md})\Big]\right\}$$

$$\psi'^r_{d2} = \frac{\omega_b}{p}\left\{v'^r_{d2} + \frac{\omega_r}{\omega_b}\psi'^r_{q2} - \frac{r'_2}{X_{\ell 6}}\Big[-X'_{\ell m}\psi_{d1}^r + (X_{\ell 1} + X'_{\ell m})\psi'^r_{d2} - X_{\ell 1}\psi_{md} + X'_{\ell dq}(\psi_{q1}^r - \psi_{mq})\Big]\right\}$$

$$\psi'_{Kq} = \frac{\omega_b}{p}\left\{v'_{Kq} - \frac{r'_{Kq}}{X'_{\ell Kq}}\Big[\psi'_{Kq} - \psi_{mq}\Big]\right\}$$

$$\psi'_{Kd} = \frac{\omega_b}{p}\left\{v'_{Kd} - \frac{r'_{Kd}}{X'_{\ell Kd}}\Big[\psi'_{Kd} - \psi_{md}\Big]\right\}$$

$$\psi'_{fr} = \frac{\omega_b}{p}\left\{v'_{fr} - \frac{r'_{fr}}{X'_{\ell fr}}\Big[\psi'_{fr} - \psi_{md}\Big]\right\}$$

ψ_{mq} and ψ_{md} can be expressed as functions of the state variables,

$$\psi_{mq} = X^*_{mq}\left[\frac{X_{\ell 2}\psi_{q1}^r + X_{\ell 1}\psi'^r_{q2} + X'_{\ell dq}(\psi'^r_{d2} - \psi_{d1}^r)}{X_{\ell 6}} + \frac{\psi'_{Kq}}{X'_{\ell Kq}}\right]$$

$$\psi_{md} = X^*_{md}\left[\frac{X_{\ell 2}\psi_{d1}^r + X_{\ell 1}\psi'^r_{d2} + X'_{\ell dq}(\psi_{q1}^r - \psi'^r_{q2})}{X_{\ell 6}} + \frac{\psi'_{Kd}}{X'_{\ell Kd}} + \frac{\psi'_{fr}}{X'_{\ell fr}}\right]$$

where

$$X^*_{mq} = \left[\frac{1}{X_{mq}} + \frac{1}{X'_{\ell Kq}} + \frac{X_{\ell 1} + X_{\ell 2}}{X_{\ell 6}}\right]^{-1}$$

$$X^*_{md} = \left[\frac{1}{X_{md}} + \frac{1}{X'_{\ell Kd}} + \frac{1}{X'_{\ell fr}} + \frac{X_{\ell 1} + X_{\ell 2}}{X_{\ell 6}}\right]^{-1}$$

The electromagnetic torque equation can be written in terms of ψ_{mq} and ψ_{md} as

$$T_{em} = \frac{3}{2}\,\frac{P}{2}\,\frac{1}{\omega_b}\Big[(i_{q1}^r + i'^r_{q2})\psi_{md} - (i_{d1}^r + i'^r_{d2})\psi_{mq}\Big]$$

and the rotor dynamics equation can be rewritten in integral form as

$$\frac{\omega_r}{\omega_b} = \frac{1}{p}\left[\frac{1}{\omega_b}\,\frac{P}{2}\,\frac{1}{J}(T_{em} - T_\ell)\right]$$

where T_{em} is positive for motor action, T_ℓ is the load torque and J is the inertia of the rotor assembly.

APPENDIX III

Parameters for a 100 kva, 240 volt, four pole, salient pole, six phase synchronous machine with unity pitch, 30° displaced stator winding are given below.

$X_{mq} = 0.5210\ \Omega$	$r'_{Kq} = 0.00250\ \Omega$
$X_{md} = 1.128\ \Omega$	$r'_{Kd} = 0.00237\ \Omega$
$X'_{\ell Kq} = 0.06760\ \Omega$	$r'_{fr} = 0.00160\ \Omega$
$X'_{\ell Kd} = 0.05290\ \Omega$	$r_1 = 0.01660\ \Omega$
$X'_{\ell fr} = 0.04684\ \Omega$	$r'_2 = 0.01660\ \Omega$
$X_{\ell 1} = X_{\ell 2} = .0558\ \Omega$	$X'_{\ell m} = 0$
$X'_{\ell dq} = 0$	$J = 5.728\ kg - m^2$

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Efficient Modeling of Six-Phase PM Synchronous Machine-Rectifier Systems in State-Variable-Based Simulation Programs

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***Abstract*—Many advanced energy conversion systems utilize multiphase (e.g., six or more phases) electrical machines that feed high-count-pulse (e.g., 12 or more pulses) rectifiers for supplying dc power. Efficient simulation of such power systems in commonly-used state-variable-based programs requires fast and accurate models of electrical machines and power electronic converters with compatible interfaces. As an alternative to the existing qd and voltage-behind-reactance (VBR) machine models, this paper develops a constant-parameter VBR model for six-phase permanent magnet synchronous machines to offer computationally efficient interface. For system-level studies where the switching details of rectifiers can be neglected, a multiple-reference-frame parametric-average-value model is developed for the 12-pulse rectifiers that provides fast simulation and preserves the dominant ac harmonics of interest. The accuracy and numerical efficiency of the proposed machine-converter models are verified using the detailed and alternative existing models.**

***Index Terms*—Constant parameter voltage behind reactance (CPVBR), parametric average-value modeling (PAVM), simulation, six-phase permanent magnet synchronous machine (PMSM), 12-pulse rectifiers.**

I. Introduction

MULTI-PHASE electrical machines (e.g., with six or more phases) and line-commutated rectifiers are often considered in advanced ac–dc energy conversion systems such as vehicles [1], ships [2], aircraft [3]–[5], wind generation [6], classic HVDC systems [7], distributed energy resources [8], exciters [9], etc., due to their increased reliability and high power transfer capability [10]. An example generic configuration considered in this paper is depicted in Fig. 1, where a PM synchronous machine (PMSM) is feeding a 12-pulse rectifier connected to a dc subsystem (load). In more conventional configurations, a three-phase generator followed by a transformer with delta/wye secondary windings feeding 12-pulse rectifiers are used to supply dc power with low ripple while reducing the ac-side harmonics [11]. Design and analysis of such power systems require conducting numerous simulations that rely on accurate and numerically efficient models of various components, including electrical machines and power electronic converters.

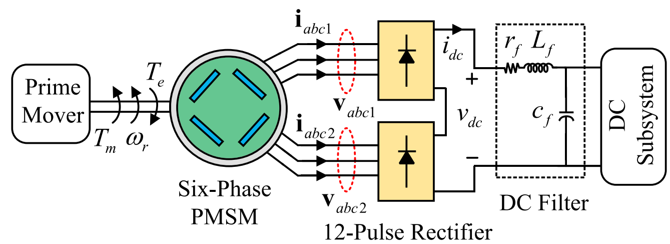

Fig. 1. A generic machine-converter ac–dc power system composed of a six-phase PM synchronous generator connected to a 12-pulse diode rectifier.

Power electronic converters can be accurately represented using detailed switching models, which are typically computationally expensive. For efficient modeling of line-commutated converters, the so-called average-value models (AVMs) have been presented in [4], [5], [12]–[21]. Using AVMs, the computational burden associated with discrete switching states of all semiconductor devices is reduced by averaging the variables (currents and voltages) within a prototypical switching interval, which makes such models suitable for system-level studies where the switching details of converters can be neglected. The developed parametric AVMs (PAVMs) [5], [17]–[21] have been demonstrated to provide numerically efficient and accurate results for rectifiers over a wide range of operating conditions. PAVMs, which are developed in synchronously rotating qd coordinates, can be interfaced with external network using either controlled voltage or controlled current sources (depending on the formulation). A multiple-reference-frame PAVM (MRF-PAVM) has been presented in [22] for six-pulse diode rectifiers, which can also include several ac harmonics in addition to the fundamental components. Recently, a generalized PAVM (GPAVM) has been developed in [23] for thyristor-controlled rectifiers which considers the ac harmonics and vari-

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able frequency operation. In MRF-PAVM [22] and GPAVM [23], the ac-side subsystem is assumed to be a three-phase network.

For modeling six-phase electrical machines, conventionally, the $qd12$ models that are expressed in two sets of qd rotor reference frames/coordinates have been used [24]. However, since the $qd12$ models are interfaced with external network using controlled current sources, interfacing them with the detailed switching models of power electronic converters in state-variable-based simulation programs [25]–[27] requires artificial snubbers which make the system stiff [28]. This may considerably reduce the simulation speed and necessitates the use of stiff/implicit solvers in the simulation programs. In addition to the inaccuracy introduced by the fictitious snubber circuits, the numerical instability of the simulation becomes a challenge. It has been also shown in [29] that inaccurate synchronous machine models can negatively influence power system stabilizer (PSS) and damping controller performance.

As an alternative to the $qd12$ model, the so-called voltage-behind-reactance (VBR) model has been recently presented for six-phase wound-rotor synchronous machines in [30] and six-phase PM synchronous machines in [31]. The VBR formulation [32]–[34] eliminates the need of snubbers for interfacing with external networks and increases the simulation speed even with non-stiff/explicit solvers. However, similar to coupled-circuit phase domain (CCPD) synchronous machine models [16, Ch. 5], the interfacing inductance matrix of the conventional VBR model is rotor-position-dependent due to rotor dynamic saliency [30]–[32], and it has to be re-computed at every time-step. This requires special considerations and additional computational effort in most simulation programs, for example, PLECS [27]. Recently, a constant-parameter VBR (CPVBR) formulation has been presented in [35] for six-phase wound-rotor synchronous machines, which increases the simulation speed by keeping a constant interfacing inductance matrix.

It has also been envisioned in [36]–[38] that computationally efficient models of machines and converters are essential for application in real-time simulators to allow using larger time-steps and avoid numerical instability. Interested reader is referred to publications by the IEEE Task Force on Interfacing Techniques for Simulation Tools, such as [28] and [39], for more information on interfacing.

It is noted that the VBR and CPVBR models are developed based on the well-established classical qd models. Such classical qd models have been used and verified experimentally in many publications, for example in [40], [41]. The work presented in this paper assumes that the reference $qd12$ model (without including main and leakage flux saturations) has sufficient accuracy for the system-level studies, while there are numerous methodologies for including magnetic saturation, losses, etc., for example as in [30], [42]–[44] that can be considered as necessary.

This paper extends the prior works [22], [23] and [31] on machine-converter modeling with the focus on achieving numerically efficient interfacing of the machine and converter subsystems. The presented machine-converter models include several ac harmonics of interest that are pertained to the six-phase 12-pulse system depicted in Fig. 1, and makes the following additional contributions:

- Using the approach set forth in [35] for six-phase wound-rotor synchronous machines, a CPVBR model is proposed for six-phase PMSMs. The explicit formulation of the proposed six-phase PMSM CPVBR model achieves a constant 6×6 interfacing inductance matrix, which eliminates the need for re-calculating inductances at each time-step.
- An extended MRF-PAVM is proposed for 12-pulse rectifiers considering six-phase ac-side power system where both lower-order harmonics (5^{th} and 7^{th}) and also higher-order harmonics (11^{th} and 13^{th}) are present and can be dominant depending on the operating condition and system parameters. The formulation of the proposed extended MRF-PAVM allows direct interfacing with six-phase machine models.
- Using comprehensive steady-state and transient studies, the advantages of the proposed machine-converter models in terms of accuracy and numerical efficiency in different combinations of models/interfaces are demonstrated.
- It is shown that the proposed PMSM CPVBR model outperforms the existing/alternative $qd12$ and VBR counterparts and has the best simulation performance when either stiff or non-stiff ODE solvers are used.
- It is also demonstrated that in system-level studies, where only the fundamental components and several dominant harmonics of ac waveforms are required, the proposed MRF-PAVM can be very effective by increasing the simulation speed, either with stiff or non-stiff ODE solvers.

Depending on the study objectives, interfacing requirements, the required details of waveforms, and the type of ODE solver used, the user may choose the best combination of the proposed machine and rectifier models.

II. Six-Phase PMSM Modeling

The six-phase PMSM considered in Fig. 1 is described in detail in [31]. This machine has two sets of three-phase windings on stator with a spatial shift equal to ξ and a damper cage on the rotor. For the modeling purpose, first, the machine variables in $abc12$ coordinates (i.e., stator voltages $\mathbf{v}_{abc12}$ and stator currents $\mathbf{i}_{abc12}$) are transformed to $qd12$ coordinates as

$$\mathbf{v}_{sqd012} = \bar{\mathbf{K}}(\theta_r)\mathbf{v}_{abc12}, \quad \mathbf{i}_{sqd012} = \bar{\mathbf{K}}(\theta_r)\mathbf{i}_{abc12}, \qquad (1)$$

where θ_r is the rotor angular position and $\bar{\mathbf{K}}(\theta_r)$ is a modified Park's transformation matrix defined as (A5) in Appendix A. Due to floating neutrals of the stator windings, the zero sequence variables in (1) for the system of Fig. 1 are not considered. Also, generator sign convention is considered for machine variables.

A. $qd12$ Model

The equivalent $qd12$ circuit of six-phase PMSM is demonstrated in Fig. 2 [24]. Therein, the rotor damper cage is modeled by two damper windings, one on $q12$-axis and one on $d12$-axis. For notational compactness, the stator and rotor voltages,

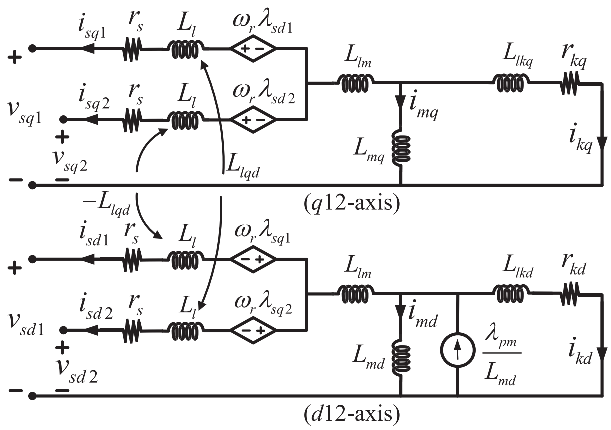

Fig. 2. The equivalent $qd12$ circuit of a six-phase PMSM.

currents, and fluxes are defined in matrix form as

$$\mathbf{v}_{sqd12} = \begin{bmatrix} v_{sq1} & v_{sq2} & v_{sd1} & v_{sd2} \end{bmatrix}^T, \tag{2}$$

$$\boldsymbol{\lambda}_{sqd12} = \begin{bmatrix} \lambda_{sq1} & \lambda_{sq2} & \lambda_{sd1} & \lambda_{sd2} \end{bmatrix}^T, \tag{3}$$

$$\mathbf{i}_{sqd12} = \begin{bmatrix} i_{sq1} & i_{sq2} & i_{sd1} & i_{sd2} \end{bmatrix}^T, \tag{4}$$

$$\boldsymbol{\lambda}_{kqd} = \begin{bmatrix} \lambda_{kq} & \lambda_{kd} \end{bmatrix}^T, \quad \mathbf{i}_{kqd} = \begin{bmatrix} i_{kq} & i_{kd} \end{bmatrix}^T. \tag{5}$$

Here, the subscripts 's' and 'k' denote stator and damper winding variables, respectively [24]. Also, the resistances and leakage inductances of damper windings and stator leakage inductances on q- and d-axes are defined as

$$\mathbf{r}_{kqd} = \begin{bmatrix} r_{kq} & 0 \\ 0 & r_{kd} \end{bmatrix}, \quad \mathbf{L}_{lkqd} = \begin{bmatrix} L_{lkq} & 0 \\ 0 & L_{lkd} \end{bmatrix}, \tag{6}$$

$$\mathbf{L}_{lm} = \begin{bmatrix} L_l + L_{lm} & L_{lm} & 0 & -L_{lqd} \\ L_{lm} & L_l + L_{lm} & L_{lqd} & 0 \\ 0 & L_{lqd} & L_l + L_{lm} & L_{lm} \\ -L_{lqd} & 0 & L_{lm} & L_l + L_{lm} \end{bmatrix}, \tag{7}$$

where

$$L_{lm} = L_{a1a2}\cos(\xi) + L_{a1b2}\cos\left(\xi + \frac{2\pi}{3}\right) + L_{a1c2}\cos\left(\xi - \frac{2\pi}{3}\right), \tag{8}$$

$$L_{lqd} = L_{a1a2}\sin(\xi) + L_{a1b2}\sin\left(\xi + \frac{2\pi}{3}\right) + L_{a1c2}\sin\left(\xi - \frac{2\pi}{3}\right), \tag{9}$$

with L_{a1a2}, L_{a1b2}, L_{a1c2} as the stator mutual leakage inductances. The magnetizing inductances and currents in q- and d-axes are defined as

$$\mathbf{L}_{mqd} = \begin{bmatrix} L_{mq} & 0 \\ 0 & L_{md} \end{bmatrix}, \quad \mathbf{i}_{mqd} = \begin{bmatrix} i_{mq} \\ i_{md} \end{bmatrix} = \begin{bmatrix} -i_{sq1} - i_{sq2} - i_{kq} \\ -i_{sd1} - i_{sd2} - i_{kd} + \lambda_{pm}/L_{md} \end{bmatrix}, \tag{10}$$

where λ_{pm} is the flux of the PMSM's permanent magnet. Based on circuits of Fig. 2, stator and rotor fluxes can be expressed as

$$\boldsymbol{\lambda}_{sqd12} = \begin{bmatrix} \mathbf{1}_{2\times1} & \mathbf{0}_{2\times1} \\ \mathbf{0}_{2\times1} & \mathbf{1}_{2\times1} \end{bmatrix}_{4\times2} \mathbf{L}_{mqd}\mathbf{i}_{mqd} - \mathbf{L}_{lm}\mathbf{i}_{sqd12}, \tag{11}$$

$$\boldsymbol{\lambda}_{kqd} = \mathbf{L}_{mqd}\mathbf{i}_{mqd} - \mathbf{L}_{lkqd}\mathbf{i}_{kqd}. \tag{12}$$

The stator and rotor voltages can also be written as

$$\mathbf{v}_{sqd12} = -r_s\mathbf{i}_{sqd12} + \omega_r \begin{bmatrix} \mathbf{0}_{2\times2} & \mathbf{I}_{2\times2} \\ -\mathbf{I}_{2\times2} & \mathbf{0}_{2\times2} \end{bmatrix}_{4\times4} \boldsymbol{\lambda}_{sqd12} + p\boldsymbol{\lambda}_{sqd12}, \tag{13}$$

$$\mathbf{0}_{2\times1} = -\mathbf{r}_{kqd}\mathbf{i}_{kqd} + p\boldsymbol{\lambda}_{kqd}, \tag{14}$$

where r_s is the resistance of stator windings, and $p = d/dt$ is the Heaviside's operator.

In the $qd12$ model, the stator and rotor fluxes (i.e., $\boldsymbol{\lambda}_{sqd12}$ and $\boldsymbol{\lambda}_{kqd}$) are chosen as the state-variables, stator voltages $\mathbf{v}_{sqd12}$ are the inputs, and stator and rotor currents (i.e., $\mathbf{i}_{sqd12}$ and $\mathbf{i}_{kqd}$) are the outputs of the model, respectively [24]. Therefore, the state-space equations of the $qd12$ model can be compactly written as

$$p\begin{bmatrix} \boldsymbol{\lambda}_{sqd12} \\ \boldsymbol{\lambda}_{kqd} \end{bmatrix} = \mathbf{A}^{qd12}\begin{bmatrix} \boldsymbol{\lambda}_{sqd12} \\ \boldsymbol{\lambda}_{kqd} \end{bmatrix} + \mathbf{B}_1^{qd12}\lambda_{pm} + \mathbf{B}_2^{qd12}\mathbf{v}_{sqd12}, \tag{15}$$

$$\begin{bmatrix} \mathbf{i}_{sqd12} \\ \mathbf{i}_{kqd} \end{bmatrix} = \mathbf{C}^{qd12}\begin{bmatrix} \boldsymbol{\lambda}_{sqd12} \\ \boldsymbol{\lambda}_{kqd} \end{bmatrix} + \mathbf{D}_1^{qd12}\lambda_{pm} + \mathbf{D}_2^{qd12}\mathbf{v}_{sqd12}, \tag{16}$$

where the state-space matrices are derived in (A1)–(A4) in Appendix A. The electromagnetic torque of the machine can also be calculated using (A6).

Equations (15)–(16) form the state-space model of the six-phase PMSM in the $qd12$ reference frame. Implementation of this model is illustrated in Fig. 3(a) with its interface to an arbitrary network in $abc12$ coordinates. Since the model computes currents as its outputs, controlled current sources are used in the interfacing circuit. Moreover, in order to interconnect with an external network that may be inductive or include switching power electronics, the shunt snubbers are required for establishing the terminal/interfacing voltages. This interfacing is sometimes referred to as indirect [28, Sec. IV-A].

B. VBR Model

In the VBR model, the rotor fluxes $\boldsymbol{\lambda}_{kqd}$ are chosen as the state variables. The stator currents $\mathbf{i}_{abc12}$ (or equivalently $\mathbf{i}_{sqd12}$) are the inputs to the VBR model, which are calculated along with the

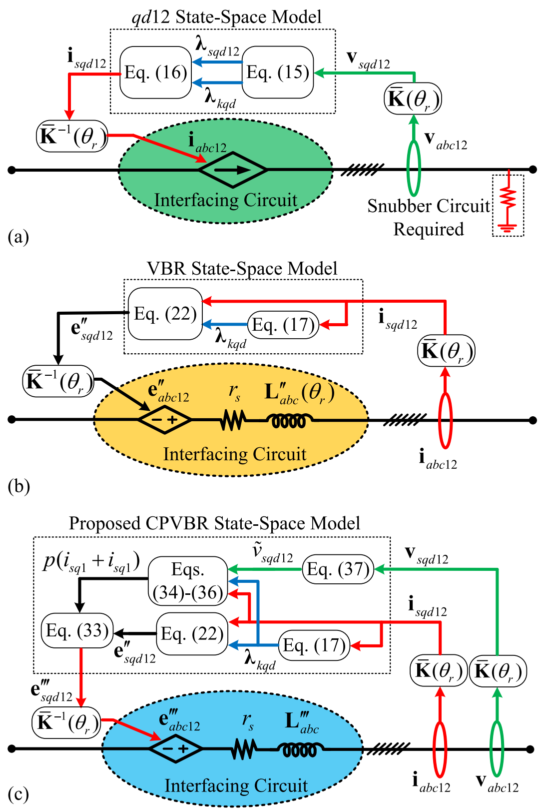

Fig. 3. Implementation of the subject models and their interfacing with an arbitrary ac network in $abc12$ coordinates: (a) $qd12$ model with indirect interfacing using current sources and snubber circuits; (b) VBR model with direct interfacing using voltage sources; (c) proposed CPVBR model with direct interfacing using voltage sources.

stator voltages $\mathbf{v}_{abc12}$(or equivalently $\mathbf{v}_{sqd12}$) by the simulation program. Therefore, replacing $\mathbf{i}_{kqd}$ in (14) using (10) and (12), the state equation for rotor fluxes are obtained as

$$p\boldsymbol{\lambda}_{kqd} = \mathbf{A}\boldsymbol{\lambda}_{kqd} + \mathbf{B}_1\lambda_{pm} + \mathbf{B}_2\mathbf{i}_{sqd12}, \tag{17}$$

$$\mathbf{A} = \begin{bmatrix} \frac{-r_{kq}}{(L_{lkq}+L_{mq})} & 0 \\ 0 & \frac{-r_{kd}}{(L_{lkd}+L_{md})} \end{bmatrix}, \tag{18}$$

$$\mathbf{B}_1 = \begin{bmatrix} 0 \\ \frac{r_{kd}}{(L_{lkd}+L_{md})} \end{bmatrix},$$

$$\mathbf{B}_2 = -\begin{bmatrix} r_{kq}\frac{L''_{mq}}{L_{lkq}}\mathbf{1}_{1\times2} & \mathbf{0}_{1\times2} \\ \mathbf{0}_{1\times2} & r_{kd}\frac{L''_{md}}{L_{lkd}}\mathbf{1}_{1\times2} \end{bmatrix}_{2\times4}, \tag{19}$$

where L''_{mq} and L''_{md} are the sub-transient magnetizing inductances defined as

$$L''_{mq} = \left(\frac{1}{L_{lkq}} + \frac{1}{L_{mq}}\right)^{-1}, \quad L''_{md} = \left(\frac{1}{L_{lkd}} + \frac{1}{L_{md}}\right)^{-1}. \tag{20}$$

Replacing $\boldsymbol{\lambda}_{sqd12}$ in (13) using (10)–(11), the stator voltages can be expressed as

$$\mathbf{v}_{sqd12} = -r_s\mathbf{i}_{sqd12} - \mathbf{L}''p\mathbf{i}_{sqd12} + \omega_r\begin{bmatrix} \mathbf{0}_{2\times2} & -\mathbf{I}_{2\times2} \\ \mathbf{I}_{2\times2} & \mathbf{0}_{2\times2} \end{bmatrix}_{4\times4}\mathbf{L}''\mathbf{i}_{sqd12} + \mathbf{e}''_{sqd12}. \tag{21}$$

Here, $\mathbf{e}''_{sqd12}$ consists of the sub-transient back-emfs defined as

$$\mathbf{e}''_{sqd12} = \begin{bmatrix} e''_{q1} & e''_{q2} & e''_{d1} & e''_{d2} \end{bmatrix}^T = \mathbf{C}\boldsymbol{\lambda}_{kqd} + \mathbf{D}_1\lambda_{pm} + \mathbf{D}_2\mathbf{i}_{sqd12}, \tag{22}$$

$$\mathbf{C} = -\begin{bmatrix} \frac{L''_{mq}}{L_{lkq}}\frac{r_{kq}}{(L_{lkq}+L_{mq})}\mathbf{1}_{2\times1} & -\omega_r\frac{L''_{md}}{L_{lkd}}\mathbf{1}_{2\times1} \\ \omega_r\frac{L''_{mq}}{L_{lkq}}\mathbf{1}_{2\times1} & \frac{L''_{md}}{L_{lkd}}\frac{r_{kd}}{(L_{lkd}+L_{md})}\mathbf{1}_{2\times1} \end{bmatrix}_{4\times2}, \tag{23}$$

$$\mathbf{D}_1 = \begin{bmatrix} \omega_r\frac{L''_{md}}{L_{md}}\mathbf{1}_{2\times1} \\ \frac{L''_{md}}{L_{lkd}}\frac{r_{kd}}{(L_{lkd}+L_{md})}\mathbf{1}_{2\times1} \end{bmatrix}_{4\times1}, \tag{24}$$

$$\mathbf{D}_2 = -\begin{bmatrix} r_{kq}\left(\frac{L''_{mq}}{L_{lkq}}\right)^2\mathbf{1}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times2} & r_{kd}\left(\frac{L''_{md}}{L_{lkd}}\right)^2\mathbf{1}_{2\times2} \end{bmatrix}_{4\times4}. \tag{25}$$

Also, $\mathbf{L}''$ is the sub-transient inductance matrix defined as

$$\mathbf{L}'' = \begin{bmatrix} L''_q & L''_q - L_l & 0 & -L_{lqd} \\ L''_q - L_l & L''_q & L_{lqd} & 0 \\ 0 & L_{lqd} & L''_d & L''_d - L_l \\ -L_{lqd} & 0 & L''_d - L_l & L''_d \end{bmatrix}, \tag{26}$$

where

$$L''_q = L_l + L_{lm} + L''_{mq}, \quad L''_d = L_l + L_{lm} + L''_{md}. \tag{27}$$

In order to interface the VBR model with an arbitrary network, (21) should be transformed back to $abc12$ coordinates using the inverse of modified Park's matrix $\bar{\mathbf{K}}^{-1}(\theta_r)$, which yields

$$\mathbf{v}_{abc12} = -r_s\mathbf{i}_{abc12} - p\left(\mathbf{L}''_{abc}(\theta_r)\mathbf{i}_{abc12}\right) + \mathbf{e}''_{abc12}, \tag{28}$$

where the interfacing inductance matrix $\mathbf{L}''_{abc}(\theta_r)$ is

$$\mathbf{L}''_{abc}(\theta_r) = \bar{\mathbf{K}}^{-1}(\theta_r)\mathbf{L}''\bar{\mathbf{K}}(\theta_r). \tag{29}$$

When the stator voltages (21) are transformed to $abc12$ coordinates, having unequal inductances in $q12$- and $d12$-axes in $\mathbf{L}''$ (26) (which is generally the case due to machine salient structure) results in a 6×6 variable (rotor-position-dependent) interfacing matrix $\mathbf{L}''_{abc}(\theta_r)$ in (29) [30], [31]. Equations (17)–(20) with (22)–(29) form the VBR model, which can be directly interfaced (without snubbers) with an arbitrary network in $abc12$

coordinates as illustrated in Fig. 3(b). Therein, controlled voltage sources are used to inject the sub-transient back emfs. The interfacing equivalent circuit has to be implemented using six branches composed of resistors and variable coupled inductors that have to be available in the simulation programs as basic circuit elements (e.g., [27] permits such elements, but [26] does not). The branch currents are then calculated by the simulation program and passed to the VBR model as the inputs.

C. Proposed CPVBR Model

To achieve a constant-parameter VBR formulation (with constant inductances in $abc12$ coordinates), $\mathbf{L}''$ in (26) is split into two parts as

$$\mathbf{L}'' = \mathbf{L}''' + \left(L''_{mq} - L''_{md}\right)\begin{bmatrix}\mathbf{1}_{2\times2} & \mathbf{0}_{2\times2}\\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2}\end{bmatrix}_{4\times4}, \qquad (30)$$

where

$$\mathbf{L}''' = \begin{bmatrix} L''_d & L''_d - L_l & 0 & -L_{lqd}\\ L''_d - L_l & L''_d & L_{lqd} & 0\\ 0 & L_{lqd} & L''_d & L''_d - L_l\\ -L_{lqd} & 0 & L''_d - L_l & L''_d\end{bmatrix}. \qquad (31)$$

The new matrix $\mathbf{L}'''$ (31) has equal inductances in $q12$- and $d12$-axes. Then, (21) is rewritten by replacing $\mathbf{L}''$ with (30), resulting in

$$\begin{aligned}\mathbf{v}_{sqd12} = & -r_s\mathbf{i}_{sqd12} - \mathbf{L}'''p\mathbf{i}_{sqd12}\\ & + \omega_r\begin{bmatrix}\mathbf{0}_{2\times2} & -\mathbf{I}_{2\times2}\\ \mathbf{I}_{2\times2} & \mathbf{0}_{2\times2}\end{bmatrix}_{4\times4}\mathbf{L}'''\mathbf{i}_{sqd12} + \mathbf{e}'''_{sqd12}, \end{aligned}\qquad (32)$$

where

$$\begin{aligned}\mathbf{e}'''_{sqd12} = & \mathbf{e}''_{sqd12} + \omega_r\left(L''_{mq} - L''_{md}\right)\\ & \times\begin{bmatrix}\mathbf{0}_{2\times2} & \mathbf{0}_{2\times2}\\ \mathbf{1}_{2\times2} & \mathbf{0}_{2\times2}\end{bmatrix}_{4\times4}\mathbf{i}_{sqd12}\\ & - \left(L''_{mq} - L''_{md}\right)p\left(i_{sq1} + i_{sq2}\right)\begin{bmatrix}\mathbf{1}_{2\times1}\\ \mathbf{0}_{2\times1}\end{bmatrix}_{4\times1}. \end{aligned}\qquad (33)$$

Here, the derivative of the sum of i_{sq1} and i_{sq2} in (33) is provided in (34)–(36), shown at the bottom of the page. Therein, the derivate expression contains the term $[(v_{sq1}+v_{sq2}) + (v_{sd2} - v_{sd1})\frac{L_{lqd}}{L_l}]$ which makes $\mathbf{v}_{sqd12}$ in (32) implicitly dependent on itself. This implicit dependency of (32)–(34) on $\mathbf{v}_{sqd12}$ creates an algebraic loop in the model [35]. To relax (break) this algebraic loop, the term corresponding to $qd12$ voltages in (34), is approximated (predicted) using a low-pass filter as

$$\tilde{v}_{sqd12} = \left(\frac{1}{1+\tau s}\right)\left[\left(v_{sq1}+v_{sq2}\right) + \left(v_{sd2} - v_{sd1}\right)\frac{L_{lqd}}{L_l}\right]. \qquad (37)$$

After that, the output voltage of filter ($\tilde{v}_{sqd12}$) will be used in (34) instead. A detailed procedure of selecting the low-pass filter time-constant τ in (37) can be found in [45].

Transforming the stator voltages in (32) to $abc12$ coordinates using $\bar{\mathbf{K}}^{-1}(\theta_r)$, the interfacing equation of the proposed CPVBR model is obtained as

$$\mathbf{v}_{abc12} = -r_s\mathbf{i}_{abc12} - p\left(\mathbf{L}'''_{abc}\mathbf{i}_{abc12}\right) + \mathbf{e}'''_{abc12}, \qquad (38)$$

where the interfacing inductance matrix $\mathbf{L}'''_{abc}$ is

$$\mathbf{L}'''_{abc} = \bar{\mathbf{K}}^{-1}(\theta_r)\mathbf{L}'''\bar{\mathbf{K}}(\theta_r). \qquad (39)$$

Here, $\mathbf{L}'''_{abc}$ is a 6×6 constant matrix.

Implementation of the proposed CPVBR model is illustrated in Fig. 3(c). According to Fig. 3(c), the machine stator terminals are interfaced with the external arbitrary ac network in $abc12$ coordinates using six branches with constant RL elements and dependent voltage sources. The branch currents and voltages are calculated by the simulation program and passed to the CPVBR model as inputs.

III. 12-Pulse Rectifier Average-Value Modeling

In the PAVM methodology [19], the relationships between the dc and ac (expressed in qd coordinates) variables of the rectifier are defined with respect to their averaged values within the prototypical switching interval as

$$\overline{x(t)} = \frac{1}{T}\int_{t-T_{sw}}^{t} x(\tau)d\tau. \qquad (40)$$

$$p\left(i_{sq1}+i_{sq2}\right) = \left(\frac{L_l}{{L_l}^2 + 2L_l\left(L_{lm}+L''_{mq}\right) - {L_{lqd}}^2}\right)$$
$$\left\{\begin{aligned}&\left[\left(v_{sq1}+v_{sq2}\right)+\left(v_{sd2}-v_{sd1}\right)\frac{L_{lqd}}{L_l}\right] - r_s\left[i_{sq1}+i_{sq2}+\left(i_{sd2}-i_{sd1}\right)\frac{L_{lqd}}{L_l}\right]\\ &-\omega_r\left[\lambda_{sd1}+\lambda_{sd2}-\left(\lambda_{sq2}-\lambda_{sq1}\right)\frac{L_{lqd}}{L_l}\right] - \frac{2L''_{mq}r_{kq}}{{L_{lkq}}^2}\left[\left(\frac{L''_{mq}}{L_{lkq}}-1\right)\lambda_{kq} - L''_{mq}\left(i_{sq1}+i_{sq2}\right)\right]\end{aligned}\right\}, \qquad (34)$$

$$\left(\lambda_{sd1}+\lambda_{sd2}\right) = -\left(2L''_{md}+2L_{lm}+L_l\right)\left(i_{sd1}+i_{sd2}\right) + \left(\frac{2L''_{md}}{L_{lkd}}\right)\lambda_{kd} + \left(\frac{2L''_{md}}{L_{md}}\right)\lambda_{pm} + L_{lqd}\left(i_{sq1}-i_{sq2}\right), \qquad (35)$$

$$\left(\lambda_{sq2}-\lambda_{sq1}\right) = L_l\left(i_{sq1}-i_{sq2}\right) - L_{lqd}\left(i_{sd1}+i_{sd2}\right). \qquad (36)$$

Here, x denotes currents and voltages, T_{sw} is the switching period equal to $1/(6f_e)$ and f_e is the line frequency. Then, the relationships between the averaged ac (in $qd12$ coordinates) and dc variables can be captured using the following parametric functions as

$$w_{i,1}(.) = \frac{\overline{i_{dc}}}{\left\|\overline{\mathbf{i}_{sqd1}^1}\right\|}, \quad w_{i,2}(.) = \frac{\overline{i_{dc}}}{\left\|\overline{\mathbf{i}_{sqd2}^1}\right\|}, \tag{41}$$

$$w_{v,1}^1(.) = \frac{\left\|\overline{\mathbf{v}_{sqd1}^1}\right\|}{\overline{v_{dc}}}, \quad w_{v,2}^1(.) = \frac{\left\|\overline{\mathbf{v}_{sqd2}^1}\right\|}{\overline{v_{dc}}}. \tag{42}$$

Also, the angle between ac voltages and currents are captured by defining the parametric functions as

$$\varphi_1(.) = \tan^{-1}\left(\frac{\overline{v_{sd1}^1}}{\overline{v_{sq1}^1}}\right) - \tan^{-1}\left(\frac{\overline{i_{sd1}^1}}{\overline{i_{sq1}^1}}\right), \tag{43}$$

$$\varphi_2(.) = \tan^{-1}\left(\frac{\overline{v_{sd2}^1}}{\overline{v_{sq2}^1}}\right) - \tan^{-1}\left(\frac{\overline{i_{sd2}^1}}{\overline{i_{sq2}^1}}\right). \tag{44}$$

The parametric functions (41)–(44) form the key relationships of the PAVM of 12-pulse rectifier where only the dc and fundamental components of ac variables are considered.

In order to reproduce the ac harmonics, an MRF-PAVM is proposed, where additional parametric functions are used to capture and represent several dominant harmonics of interest. The relationship between the magnitude of harmonic voltages and the 12-pulse rectifier dc voltage are captured using the parametric functions defined as

$$w_{v,1}^n(.) = \frac{\left\|\overline{\mathbf{v}_{sqd1}^n}\right\|}{\overline{v_{dc}}}, \quad w_{v,2}^n(.) = \frac{\left\|\overline{\mathbf{v}_{sqd2}^n}\right\|}{\overline{v_{dc}}}, \tag{45}$$

where n is the harmonic order. Also, the phase angles of harmonic voltages are captured with respect to the fundamental components of voltages of each set. Since the angular frequency of harmonics are n times faster than the fundamental components, in order to lock the harmonic voltages to their corresponding fundamental components, the following parametric functions are defined as

$$\theta_{v,1}^n(.) = n\tan^{-1}\left(\frac{\overline{v_{sd1}^1}}{\overline{v_{sq1}^1}}\right) - \tan^{-1}\left(\frac{\overline{v_{sd1}^n}}{\overline{v_{sq1}^n}}\right), \tag{46}$$

$$\theta_{v,2}^n(.) = n\tan^{-1}\left(\frac{\overline{v_{sd2}^1}}{\overline{v_{sq2}^1}}\right) - \tan^{-1}\left(\frac{\overline{v_{sd2}^n}}{\overline{v_{sq2}^n}}\right). \tag{47}$$

The operating condition of the rectifier may be specified by the dynamic impedances, herein defined as

$$z_{d,1} = \frac{\overline{v_{dc}}}{\left\|\overline{\mathbf{i}_{sqd1}^1}\right\|}, \quad z_{d,2} = \frac{\overline{v_{dc}}}{\left\|\overline{\mathbf{i}_{sqd2}^1}\right\|}. \tag{48}$$

The parametric functions (41)–(47) are computed numerically by running the detailed simulation over the desired range of operating conditions. These parametric functions are saved in look-up tables in terms of $z_{d,1}$ and $z_{d,2}$ (48).

In the case of 12-pulse rectifier symmetry as shown in Fig. 1, the parametric functions associated with fundamental components of ac variables and dc variables (41)–(44) would be equal

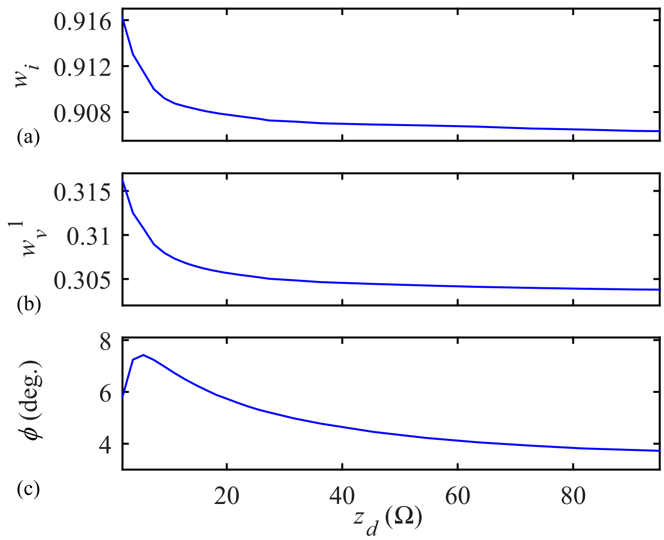

Fig. 4. Parametric functions corresponding to dc and fundamental components of ac variables of the 12-pulse diode rectifier over a wide range of operating conditions: (a) $w_i(.)$, (b) $w_v^1(.)$, (c) $\varphi(.)$.

for both sets as

$$w_i(.) = w_{i,1}(.) = w_{i,2}(.), \tag{49}$$

$$w_v^1(.) = w_{v,1}^1(.) = w_{v,2}^1(.), \tag{50}$$

$$\varphi(.) = \varphi_1(.) = \varphi_2(.). \tag{51}$$

Similarly, the parametric functions corresponding to harmonic voltages (45)–(47) would also become equal as

$$w_v^n(.) = w_{v,1}^n(.) = w_{v,2}^n(.), \tag{52}$$

$$\theta_v^n(.) = \theta_{v,1}^n(.) = \theta_{v,2}^n(.). \tag{53}$$

Moreover, since the dynamic impedances (48) would also be equal, the operating condition of the rectifier can be captured by only one dynamic impedance as

$$z_d = z_{d,1} = z_{d,2}. \tag{54}$$

It is worth mentioning that in this paper the ac currents and dc voltage of rectifier are the inputs to the AVMs, and therefore these variables are used for defining the dynamic impedances. However, depending on the interfacing requirements and the corresponding formulations, different pairs of dc and ac variables can be appropriately selected for defining the dynamic impedances or admittance (e.g., [46], where dc current and ac voltages are used due to interfacing requirements).

For the symmetric power system of Fig. 1 (with parameters summarized in Appendix B), the parametric functions corresponding to the dc and fundamental components of ac variables are shown in Figs. 4(a)–(c) for $w_i(.)$,$w_v^1(.)$ and $\varphi(.)$, respectively. As it can be observed in Fig. 4, the non-linear relationships between the dc and fundamental components of the ac variables are tabulated in one-dimensional lookup tables with respect to dynamic impedance z_d. Also, the parametric functions corresponding to the dominant ac harmonics (up to 13$^{\text{th}}$) are illustrated in Figs. 5(a)–(b) over a wide range of operating conditions for magnitude (w_v^n) and phase (θ_v^n) of harmonic voltages, respectively. As it can be observed in Fig. 5(a), for the

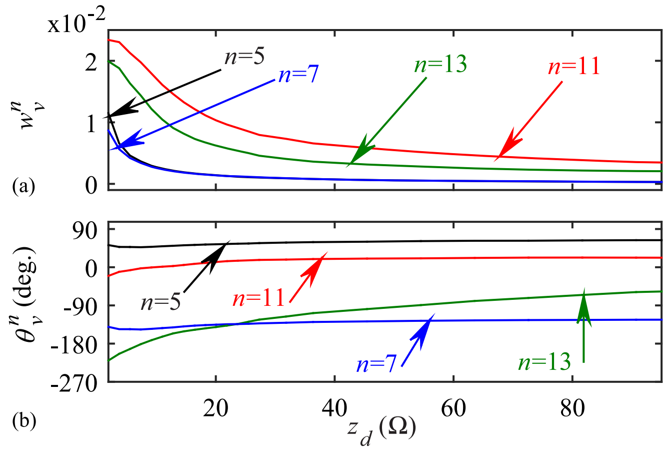

Fig. 5. Parametric functions depicting the magnitude and phase angle of various harmonics; respectively, in ac voltages of the 12-pulse rectifier over a wide range of operating conditions: (a) $w_v^n(.)$, (b) $\theta_v^n(.)$.

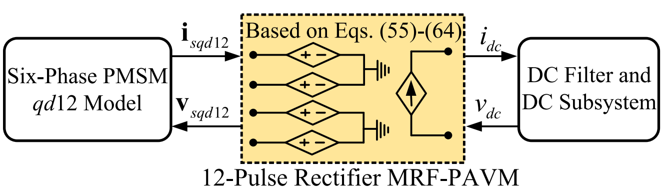

Fig. 6. Implementation of the proposed extended MRF-PAVM and its interfacing with the dc subsystem and ac subsystem in $qd12$ coordinates.

considered 12-pulse rectifier system connected to the six-phase machine, both lower-order (i.e., 5$^{\text{th}}$ and 7$^{\text{th}}$) and higher-order (i.e., 11$^{\text{th}}$ and 13$^{\text{th}}$) harmonics are present on the ac voltages and their corresponding parametric functions are included in the proposed extended MRF-PAVM model; hence $n \in \{5, 7, 11, 13\}$.

To implement the developed MRF-PAVM, dependent voltage and current sources are used to interface with the ac and dc subsystems as depicted in Fig. 6. Therein, the ac currents $\mathbf{i}_{sqd12}$ and the dc voltage v_{dc} are the inputs to the MRF-PAVM, where $z_{d,1}$ and $z_{d,2}$ are computed based on (48) and used as inputs to the lookup tables. The fundamental components of the ac voltages are calculated as

$$v_{sq1}^1 = w_{v,1}^1(.)v_{dc}\cos(\delta_1), \quad v_{sd1}^1 = w_{v,1}^1(.)v_{dc}\sin(\delta_1), \tag{55}$$

$$v_{sq2}^1 = w_{v,2}^1(.)v_{dc}\cos(\delta_2), \quad v_{sd2}^1 = w_{v,2}^1(.)v_{dc}\sin(\delta_2), \tag{56}$$

where

$$\delta_1 = \tan^{-1}\left(i_{sd1}^1 / i_{sq1}^1\right) + \varphi_1(.), \tag{57}$$

$$\delta_2 = \tan^{-1}\left(i_{sd2}^1 / i_{sq2}^1\right) + \varphi_2(.). \tag{58}$$

The harmonic voltages for the first set are calculated based on (45) and (46) as

$$v_{sq1}^n = w_{v,1}^n(.)v_{dc}\cos\left(n\delta_1 - \theta_{v,1}^n(.)\right), \tag{59}$$

$$v_{sd1}^n = w_{v,1}^n(.)v_{dc}\sin\left(n\delta_1 - \theta_{v,1}^n(.)\right). \tag{60}$$

Similar equations can be written for the second set based on (45) and (47) as

$$v_{sq2}^n = w_{v,2}^n(.)v_{dc}\cos\left(n\delta_2 - \theta_{v,2}^n(.)\right), \tag{61}$$

$$v_{sd2}^n = w_{v,2}^n(.)v_{dc}\sin\left(n\delta_2 - \theta_{v,2}^n(.)\right). \tag{62}$$

Finally, the interfacing ac voltages and dc current that become the outputs of the MRF-PAVM are calculated as

$$\mathbf{v}_{sqd12} = \mathbf{v}_{sqd12}^1 + \sum_n \left\{\bar{\mathbf{K}}\left[(n-1)\theta_r\right]\right\}^{-1}\mathbf{v}_{sqd12}^n, \tag{63}$$

and

$$i_{dc} = w_{i,1}(.)\left\|\mathbf{i}_{sqd1}^1\right\| = w_{i,2}(.)\left\|\mathbf{i}_{sqd2}^1\right\|. \tag{64}$$

For interfacing the current source i_{dc} with the dc subsystem, the filter inductor L_f is modeled using a transfer function as in [19, see eq. (9)] with a small time-constant ($\tau_f = 2 \times 10^{-5}$ given in Appendix B).

It is worth mentioning that in case thyristor switches are used in the rectifier of Fig. 1, the dynamics of thyristor firing angle can be considered in the presented MRF-PAVM by appropriately incorporating the firing angle in the parametric functions (41)–(47), and subsequently extending the lookup tables using the approach set forth in [23]. The variable frequency operation of rectifier can also be considered using the approach presented in [23].

IV. Computer Studies

The machine-converter system of Fig. 1 has been implemented in MATLAB/Simulink [25] using PLECS toolbox [27] with system parameters summarized in Appendix B. For constructing the detailed model of 12-pulse rectifier, the standard library switching diode bridges are used. Three versions of detailed switching machine-rectifier models are considered: $qd12$-Detailed, VBR-Detailed, and CPVBR-Detailed, where the six-phase PMSM is modeled using the $qd12$, VBR, and CPVBR formulations as presented in Sections II-A, II-B and II–C, respectively. For interfacing the $qd12$ model with the detailed model of 12-pulse rectifier, parallel snubber resistors of 350 Ω are selected to ensure maximum error in stator current to be on the order of 1% at nominal loading condition. The VBR and CPVBR models have a direct interface of the stator circuit terminals to the rectifier bridges and do not require snubbers. In order to limit the solution error of the CPVBR model to less than 1%, the time constant of the low-pass filter in (37) is selected to be $\tau = 0.17$ [45].

Also, two versions of the average-value models are considered: $qd12$-PAVM and $qd12$-MRF-PAVM, where the $qd12$ model of six-phase PMSM is interfaced with the PAVM and the proposed extended MRF-PAVM models of the 12-pulse rectifier using (55)–(58), (64) and (55)–(64), respectively. Without loss of generality, in all the models the dc subsystem has been represented as an equivalent variable resistive load r_l. Also, the parametric functions of the two AVMs (i.e., PAVM and MRF-PAVM) have been obtained using the fast procedure presented in [20] by simulating the detailed model of the power system in Fig. 1 with a fast sweep of the load resistance over a wide range of operating conditions (from $r_l = 1$ Ω to $r_l = 1000$ Ω). It

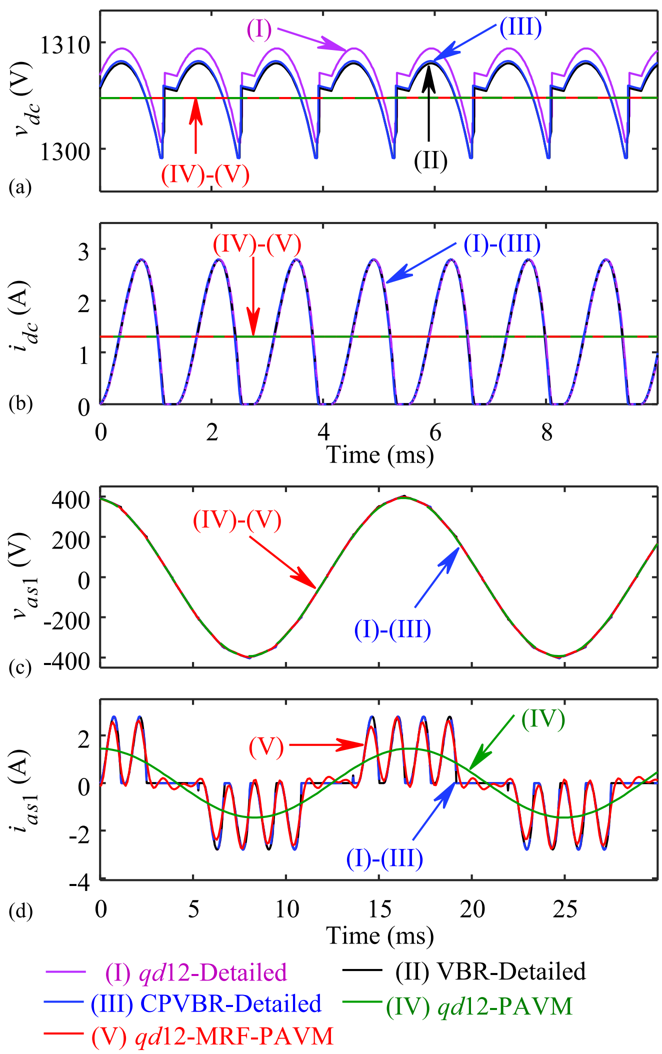

Fig. 7. Rectifier dc and ac variables in DCM as predicted by the subject models: (a) dc voltage, (b) dc current, (c) phase $a1$ voltage, (d) phase $a1$ current.

should also be noted that this is a one-time process, and once the parametric functions are calculated and saved in lookup tables, they can be used for various studies including transients. In the subsequent subsections, the performance of all the considered models is verified in steady-state and in transients.

A. Steady-State Studies

Here, the generator speed is kept constant corresponding to the line frequency $f_e = 60$ Hz. The steady-state waveforms of rectifier variables predicted by the subject models are shown in Figs. 7 and 9 for DCM condition (with $r_l = 1000\ \Omega$) and CCM nominal condition (with $r_l = 10\ \Omega$) [47], respectively; with their associated harmonic contents depicted in Figs. 8 and 10.

As it can be observed in Figs. 7–10, the proposed CPVBR-Detailed model accurately follows the steady-state results of the $qd12$-Detailed and VBR-Detailed models. Also, using artificial snubbers [28] in the $qd12$-Detailed model results in error as can be seen in Figs. 7(a) and 9(a). Meanwhile, the proposed $qd12$-MRF-PAVM provides an excellent accuracy in predicting

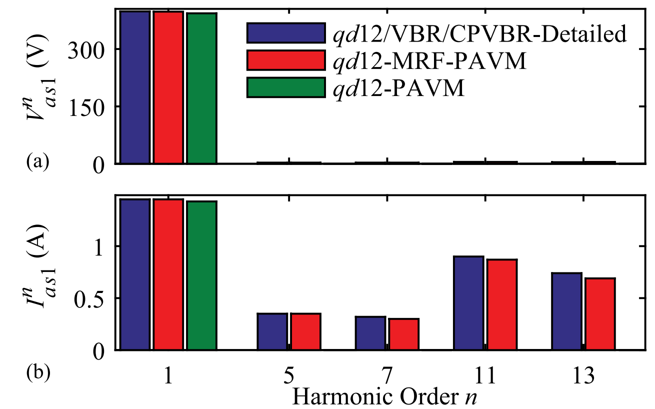

Fig. 8. Harmonic content of rectifier ac phase $a1$ voltage and phase $a1$ current as predicted by the subject models in DCM condition.

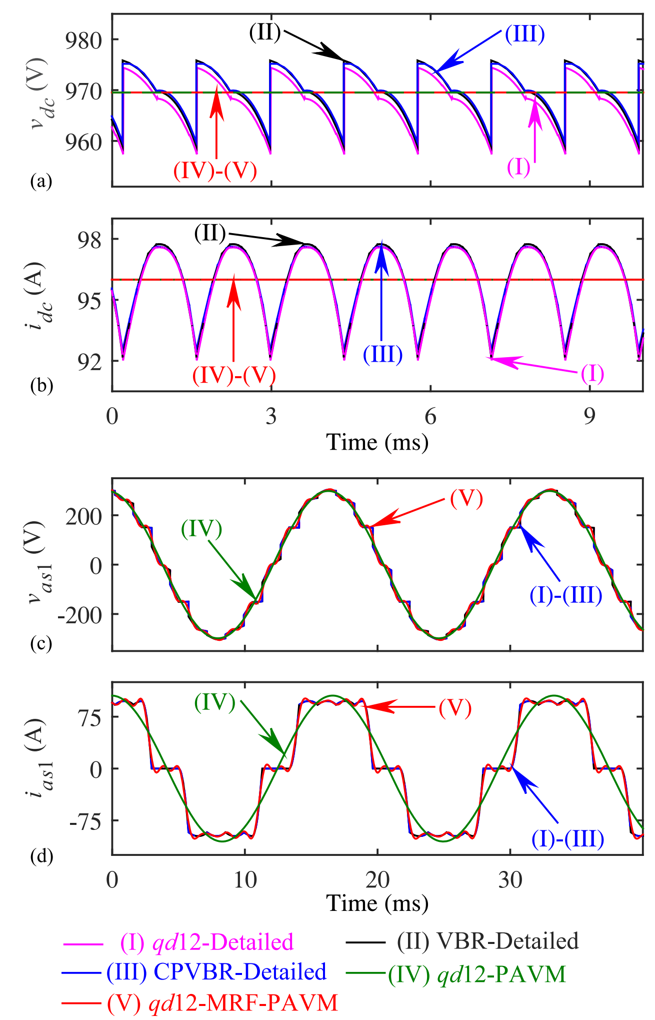

Fig. 9. Rectifier dc and ac variables in CCM as predicted by the subject models: (a) dc voltage, (b) dc current, (c) phase $a1$ voltage, (d) phase $a1$ current.

the average values of the dc variables as well as the fundamental components and the harmonics of the ac variables as compared to the rectifier detailed switching models. However, the $qd12$-PAVM can only predict the average values of dc and the fundamental components of ac variables.

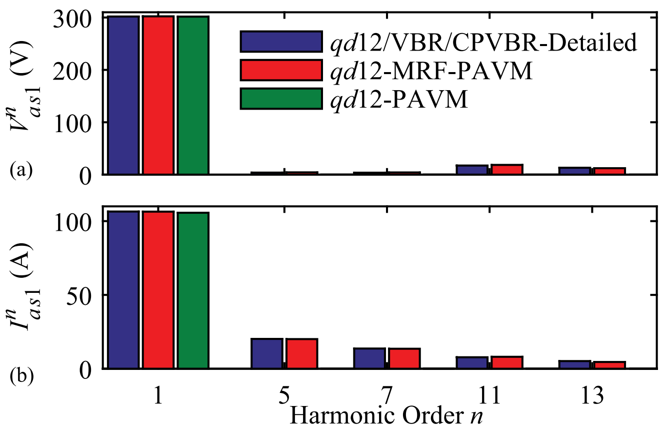

Fig. 10. Harmonic content of rectifier ac phase $a1$ voltage and phase $a1$ current as predicted by the subject models in CCM condition.

It should also be noted that in Fig. 10, the 11th and 13th harmonics are dominant in the voltage (compared to the 5th and 7th harmonics), while the 5th and 7th harmonics are larger in the current (compared to the 11th and 13th harmonics). This is due to the fact that the impedance of the electrical machine increases very rapidly with the harmonic frequency.

B. Transient Studies

Here, the dynamic performance of the proposed machine-converter models is investigated for the system transient by changing r_l from 1000 Ω to 100 Ω (DCM to CCM) at $t = 0.5$ s. The transient responses of several system variables are shown in Fig. 11 for all the subject models. As it can be observed in Fig. 11, the proposed CPVBR-Detailed model accurately follows the transient results of system variables compared to the qd12-Detailed and VBR-Detailed models. Moreover, the error in generator torque predicted by the qd12-Detailed model, as can be observed in Fig. 11(c), is a result of the error introduced to the stator currents due to the artificial snubbers.

Meanwhile, the proposed qd12-MRF-PAVM very accurately follows the transient behavior of detailed switching models, as can be observed in Figs. 11(a)–(b) for dc variables and also in Fig. 11(d) for the ac variables including the harmonics. Moreover, the ac-side voltage and current harmonics predicted by the proposed qd12-MRF-PAVM result in an accurate prediction of the generator torque ripple (compared to VBR/CPVBR-Detailed switching models), as demonstrated in Fig. 11(c), where the qd12-PAVM only predicts the average value of the torque.

C. Computational Performance Comparison

To benchmark the subject models, the following 5-second transient study has been carried out. The system initially operates with half of the nominal load (with $r_l = 20$ Ω). At $t = 2.5$ s the load is switched to its nominal value corresponding to $r_l = 10$ Ω. For consistency, all the models are run on a PC with Intel Core i7-4510U @ 2.00 GHz processor. To evaluate the effect of different interfacing methods and numerical stiffness, the explicit (non-stiff) $ode45$ and implicit (stiff) $ode23tb$ solvers have been used for all the subject models with relative and absolute tolerances set to 10^{-3}. The maximum time-step is also

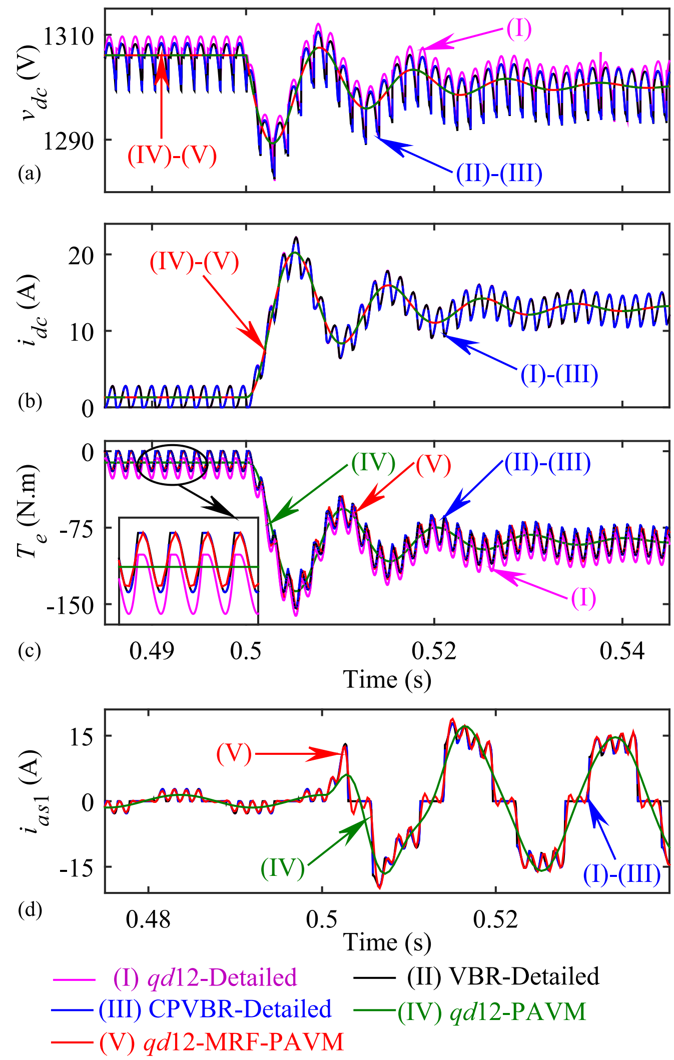

Fig. 11. Transient response due to load change from DCM to CCM as predicted by the subject models: (a) rectifier dc voltage, (b) rectifier dc current, (c) generator torque, (d) generator phase $a1$ current.

set to 0.5 ms for the $qd12/\mathrm{VBR}/\mathrm{CPVBR}$-Detailed models, and 1.0 ms and 0.2 ms for the qd12-PAVM and qd12-MRF-PAVM models, respectively.

1) Numerical Accuracy of Detailed Models

To verify the accuracy of the proposed CPVBR model of the six-phase PMSM compared to the existing qd12 and VBR models, the solutions obtained by the $qd12/\mathrm{VBR}/\mathrm{CPVBR}$-Detailed models are first compared with a reference solution. The reference solution is obtained using the VBR model ran with a very small maximum time-step as well as absolute and relative errors, all set to 10^{-5}. The VBR model does not introduce a solution approximation (error) such as the qd12 model due to artificial snubbers or the CPVBR model due to algebraic loop relaxation filter. Therefore, its solution trajectory obtained with a very small time-step is considered to be the reference. The cumulative 2-norm errors [48] for the subject models computed for several variables are summarized in Table I for the 5-second transient study.

As it can be observed in Table I, the VBR model ran with larger time-step and relaxed tolerances introduces very small

TABLE I
2-NORM ERROR IN SEVERAL VARIABLES OF THE SUBJECT DETAILED MODELS FOR THE CONSIDERED 5-SECOND TRANSIENT STUDY

Model	2-Norm Error		
Machine - Rectifier	torque T_e	stator current i_{as1}	dc current i_{dc}
VBR-Detailed	0.08 %	0.07 %	0.07 %
$qd12$-Detailed	0.85 %	1.15 %	0.18 %
CPVBR-Detailed	0.21 %	0.23 %	0.18 %

TABLE II
COMPUTATIONAL PERFORMANCE OF THE SUBJECT DETAILED AND AVERAGED MODELS FOR THE CONSIDERED 5-SECOND TRANSIENT STUDY WITH THE NON-STIFF SOLVER $ode45$

Model / Machine - Rectifier	Number of steps	CPU Time (sec)
$qd12$-Detailed	8,666,686	663.19 (~11 min)
VBR-Detailed	24,850	23.66
CPVBR-Detailed	38,741	7.06
$qd12$-MRF-PAVM	14,649	2.92
$qd12$-PAVM	12,851	0.81

error compared to the reference ($\sim$0.08%). The errors of the $qd12$ model in torque and stator current are on the order of 1%, which are consistent with the selected interfacing snubbers. Meanwhile, the proposed CPVBR model offers higher accuracy compared to the $qd12$ model in torque (0.21% vs. 0.85%) and stator current (0.23% vs. 1.15%).

2) Performance with Non-Stiff Solver

Here, the computational performance of the subject models with non-stiff ($ode45$) solver is investigated, and the results are summarized in Table II. As it can be observed in Table II, the $qd12$-Detailed model is the slowest ($\sim$11 min) due to the stiffness introduced by the snubber circuit. The VBR-Detailed model improves the simulation speed (23.66 s). Meanwhile, the proposed CPVBR-Detailed model outperforms both the $qd12$-Detailed and VBR-Detailed models in terms of the CPU time (7.06 s CPVBR vs. 23.66 s VBR vs. 663.19 s $qd12$).

Moreover, the CPVBR-Detailed and VBR-Detailed models take almost the same number of steps (38,741 and 24,850 respectively) which are much less than the number of steps required by the $qd12$ model (8,666,686).

3) Performance with Stiff Solver

Here, the computational performance of the subject models with stiff ($ode23$tb) solver is investigated and the results are summarized in Table III. As it can be observed in Table III, using stiff solver reduces the computational burden caused by the large snubber resistors in the $qd12$-Detailed model, which leads to faster simulation speed compared to VBR-Detailed model (13.42 s $qd12$ vs. 33.92 s VBR). This is expected since unlike the VBR-Detailed model with variable inductors, the $qd12$-Detailed model has constant parameters. Meanwhile, the proposed CPVBR-Detailed model outperforms both $qd12$- and VBR-Detailed models in terms of the CPU time (8.49 s CPVBR vs. 13.42 s $qd12$ vs. 33.92 s VBR). The faster performance of the CPVBR-Detailed model is due to its constant parameter interfacing circuit (compared to VBR-Detailed) and avoiding

TABLE III
COMPUTATIONAL PERFORMANCE OF THE SUBJECT DETAILED AND AVERAGED MODELS FOR THE CONSIDERED 5-SECOND TRANSIENT STUDY WITH THE STIFF SOLVER $ode23$TB

Model / Machine - Rectifier	Number of steps	CPU Time (sec)
$qd12$-Detailed	137,345	13.42
VBR-Detailed	54,966	33.92
CPVBR-Detailed	56,891	8.49
$qd12$-MRF-PAVM	16,760	2.40
$qd12$-PAVM	5,029	0.21

snubber circuits that introduce stiffness (compared to $qd12$-Detailed). Moreover, the CPVBR-Detailed and VBR-Detailed models take almost the same number of steps (56,891 and 54,966 respectively) which are less than the number of steps required by the $qd12$ model (137,345).

As it can be concluded based on Tables II and III, in applications where the switching details of the rectifier have to be considered, the proposed CPVBR-Detailed machine-rectifier model achieves the best performance with either stiff or non-stiff solvers, assuming that the accuracy of CPVBR model of PMSM is acceptable.

4) Performance of AVMs vs Detailed Models

As it can be observed in Table II, the CPU times associated with $qd12$-MRF-PAVM and $qd12$-PAVM models with $ode45$ non-stiff solver are 2.92 s (in 14,649 steps) and 0.81 s (in 12,851 steps), respectively. Also, Table III shows that it takes 2.40 s (in 16,760 steps) and 0.21 s (in 5,029 steps) to run the $qd12$-MRF-PAVM and $qd12$-PAVM models with $ode23$tb stiff solver, respectively. Overall, the $qd12$-PAVM and $qd12$-MRF-PAVM models offer much faster simulations compared to the three detailed models (i.e., $qd12$/VBR/CPVBR-Detailed) by eliminating the discrete switching states of the rectifier. However, the $qd12$-PAVM model is only capable of predicting the dc and fundamental components of ac variables. Meanwhile, the proposed $qd12$-MRF-PAVM also captures the dominant ac harmonics that constitute much of the details of the ac waveforms.

It is also worth mentioning that the lookup tables of the presented MRF-PAVM have been constructed using a 14 s simulation of the power system of Fig. 1 over the desired range of loading conditions (see Section III) using the CPVBR-Detailed model. With the software settings presented in Section IV-C, it took approximately 49 s of CPU time to obtain the parametric functions. This time may be taken into consideration when choosing between the presented MRF-PAVM and the detailed models. Since computing parameter functions are a one-time process, the difference between the required CPU times for the detailed model and MRF-PAVM in simulations can determine the minimum number of studies that make the MRF-PAVM preferable in terms of simulation time (which depends on the specific studies and length of each simulation). Therefore, for system-level studies with many long transient studies, wherein capturing the system fast dynamics and accurate waveforms (without switching) is sufficient, the proposed $qd12$-MRF-PAVM of the machine-converter system may be the best choice with either stiff or non-stiff solvers.

V. Discussion

In this paper, efficient models for six-phase machine rectifier systems have been developed for application in state-variable-based simulation programs (as emphasized in the title). Interfacing the presented AVMs in nodal-analysis-based EMTP-type packages requires special attention and considerations [49]–[51]. In EMTP-type programs, typically the fixed-step trapezoidal integration is used. When the power system becomes larger and contains multiple converters and electrical machines, fairly small time-step sizes are typically required to ensure sufficient accuracy and numerical stability. At the same time, the modelling approach presented in this paper removes the switching from the models of converters, which can potentially use larger time-steps and reduce the computational burden.

The power system of Fig. 1 has been selected as an example topology for the purpose of presenting the modeling methodologies in an unobstructed way. However, the presented techniques can be used for more complicated system-level studies where many electrical machines and converters may exist, where the presented models can be used very effectively in conducting multiple studies for designing and optimization of system controllers [52], etc. When needed, thyristor operation of rectifiers can be considered by including the dynamics of thyristor firing angle in the presented formulations using the approach set forth in [23].

It is noted that due to the non-linear nature of commutation in line-commutated rectifiers, deriving analytical AVMs is complicated, and that is where the PAVM technique can be an effective alternative. Since obtaining analytical AVMs for classical 2-level voltage-source-converters (VSCs) is straightforward due to forced commutations, the AVMs for such VSCs can be readily constructed using the well-established relationships between the average dc and transformed ac variables [16, Chap. 12].

In addition to the computational advantages (i.e., larger time steps, less CPU time, etc.) of the presented PAVM and MRF-PAVM compared to the detailed model, the presented AVMs are continuous and linearizable, which may be very beneficial for some analysis. However, since the parametric functions of these models are obtained with certain values of system parameters, they may need to be updated (recalculated) if the system parameters change, which is a relative disadvantage.

It should also be clarified that the studies presented in this paper do not focus on unbalanced operation of the system. It has been demonstrated in [51], [53] that the PAVMs of line-commutated rectifier (which do not include harmonics) provide reasonable prediction of distorted waveforms under moderately unbalanced conditions introduced in the source voltages. However, more investigations are required to evaluate the MRF-PAVMs (that include harmonics) under unbalanced conditions, which are outside of scope for this paper and represents a future research.

VI. Conclusions

In this paper, a CPVBR model has been developed for six-phase PMSMs for simulation of machine-converter systems. The CPVBR model achieves a constant inductance matrix and interfacing circuit which eliminates the need for matrix recalculation at each time-step and increases the simulation speed. The superior computational efficiency of the proposed CPVBR model has been verified against the existing/alternative $qd12$ and VBR models with the detailed switching models of the 12-pulse rectifier in ac–dc energy conversion systems. The relaxation (approximation) used in the CPVBR model does not make the system numerically very stiff, and the resulting model has been shown to outperform both $qd12$ and VBR models when solved with either explicit or implicit solvers.

For the system-level studies of electrical machines feeding high-pulse rectifiers, only the fundamental components and several dominant harmonics of ac variables as well as the dc variables are typically required and considered sufficient. Therefore, the MRF-PAVM of 12-pulse rectifiers has been developed in this paper which can reconstruct several dominant ac harmonics (up to 13$^{\text{th}}$) of interest in addition to the fundamental component of ac variables and dc variables. The accuracy of MRF-PAVM has also been verified against the detailed models and previous PAVM (which considers only the fundamental components). Including harmonics requires additional computational resources and makes the MRF-PAVM slower compared to the PAVM, but it is still much more efficient than the detailed switching model.

It has been verified that, when the switching of converters need to be considered, the proposed six-phase PMSM CPVBR model achieves the best simulation performance. Also, the proposed extended MRF-PAVM of 12-pulse rectifier interfaced with the $qd12$ model of six-phase PMSM has the best performance when the switching details of rectifier can be neglected in system-level studies.

Appendix A

$$\mathbf{A}^{qd12} = \omega_r \begin{bmatrix} \mathbf{0}_{2\times2} & -\mathbf{I}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{I}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \\ \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} & \mathbf{0}_{2\times2} \end{bmatrix}_{6\times6} - \begin{bmatrix} r_s\mathbf{I}_{4\times4} & \mathbf{0}_{4\times2} \\ \mathbf{0}_{2\times4} & \mathbf{r}_{kqd} \end{bmatrix}_{6\times6} \mathbf{L}_a^{-1}, \tag{A1}$$

$$\mathbf{B}_1^{qd12} = \begin{bmatrix} r_s\mathbf{I}_{4\times4} & \mathbf{0}_{4\times2} \\ \mathbf{0}_{2\times4} & \mathbf{r}_{kqd} \end{bmatrix}_{6\times6} \mathbf{L}_a^{-1} \begin{bmatrix} \mathbf{0}_{2\times1} \\ \mathbf{1}_{2\times1} \\ 0 \\ 1 \end{bmatrix}_{6\times1},$$

$$\mathbf{B}_2^{qd12} = \begin{bmatrix} \mathbf{I}_{4\times4} \\ \mathbf{0}_{2\times4} \end{bmatrix}_{6\times4}, \tag{A2}$$

$$\mathbf{C}^{qd12} = -\mathbf{L}_a^{-1}, \quad \mathbf{D}_1^{qd12} = \mathbf{L}_a^{-1} \begin{bmatrix} \mathbf{0}_{2\times1} \\ \mathbf{1}_{2\times1} \\ 0 \\ 1 \end{bmatrix}_{6\times1}, \quad \mathbf{D}_2^{qd12} = \mathbf{0}_{6\times4}, \tag{A3}$$

$$\mathbf{L}_a = \begin{bmatrix} L_l + L_{lm} + L_{mq} & L_{lm} + L_{mq} & 0 & -L_{lqd} & L_{mq} & 0 \\ L_{lm} + L_{mq} & L_l + L_{lm} + L_{mq} & L_{lqd} & 0 & L_{mq} & 0 \\ 0 & L_{lqd} & L_l + L_{lm} + L_{md} & L_{lm} + L_{md} & 0 & L_{md} \\ -L_{lqd} & 0 & L_{lm} + L_{md} & L_l + L_{lm} + L_{md} & 0 & L_{md} \\ L_{mq} & L_{mq} & 0 & 0 & L_{lkq} + L_{mq} & 0 \\ 0 & 0 & L_{md} & L_{md} & 0 & L_{lkd} + L_{md} \end{bmatrix}_{6\times 6}. \quad \text{(A4)}$$

$$\bar{\mathbf{K}}(\theta_r) = \frac{2}{3}\begin{bmatrix} \cos(\theta_r) & \cos\left(\theta_r - \frac{2\pi}{3}\right) & \cos\left(\theta_r + \frac{2\pi}{3}\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\theta_r - \xi) & \cos\left(\theta_r - \frac{2\pi}{3} - \xi\right) & \cos\left(\theta_r + \frac{2\pi}{3} - \xi\right) \\ \sin(\theta_r) & \sin\left(\theta_r - \frac{2\pi}{3}\right) & \sin\left(\theta_r + \frac{2\pi}{3}\right) & 0 & 0 & 0 \\ 0 & 0 & 0 & \sin(\theta_r - \xi) & \sin\left(\theta_r - \frac{2\pi}{3} - \xi\right) & \sin\left(\theta_r + \frac{2\pi}{3} - \xi\right) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}_{6\times 6}. \quad \text{(A5)}$$

$$T_e = \frac{3P}{4}\left\{L_{md}\left(i_{sq1} + i_{sq2}\right)\left(i_{sd1} + i_{sd2} + i_{kd} - \frac{\lambda_{pm}}{L_{md}}\right) - L_{mq}\left(i_{sd1} + i_{sd2}\right)\left(i_{sq1} + i_{sq2} + i_{kq}\right)\right\}. \quad \text{(A6)}$$

where $\mathbf{L}_a$ is defined as (A4), shown at the top of the page. See also (A5)–(A6) at the top of the page.

Appendix B

Six-phase machine parameters [24], [35]:

Rated phase voltage: 240 V, rated power: 100 kVA, poles: 4, rated speed: 1800 rpm, PM flux: $\lambda_{pm} = 1.05$ wb, $\xi = \pi/6$,

$$r_s = 16\ \text{m}\Omega, r_{kd} = 2.37\ \text{m}\Omega, r_{kq} = 2.5\ \text{m}\Omega,$$

$$L_{a1a2} = 43\ \mu\text{H}, L_{a1b2} = -43\ \mu\text{H}, L_{a1c2} = 0\ \mu\text{H}, L_l = 75\ \mu\text{H},$$

$$L_{lkd} = 140\ \mu\text{H}, L_{lkq} = 180\ \mu\text{H}, L_{md} = 3\ \text{mH}, L_{mq} = 1.4\ \text{mH}.$$

DC filter parameters:

$$r_f = 0.1\ \Omega, L_f = 0.5\ \text{mH}, C_f = 1000\ \mu\text{F}, \tau_f = 2\times 10^{-5}.$$

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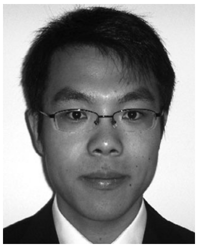

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Constant-Parameter Voltage-Behind-Reactance Modeling of Five-Phase Synchronous Machines With Air-Gap Flux Harmonics

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***Abstract*—Five-phase electric machines possess several distinct features and are sometimes considered in vehicular power systems and various special-purpose energy sources. Design and analysis of such machine-converter systems are highly dependent on simulation software programs, where accurate and numerically efficient dynamic models of five-phase synchronous machines are essential. Depending on levels of fidelity and interfacing compatibility, the traditional models for five-phase machines include the coupled-circuit-phase-domain (CCPD) and *qd*0 models. To achieve a computationally-efficient model suitable for many simulation programs, in this paper, a new constant-parameter voltage-behind-reactance (CPVBR) model is proposed for five-phase synchronous machines where a simple interfacing circuit is achieved with constant *RL* branches. The proposed model is validated by experimental measurements and simulation studies. The proposed CPVBR model is demonstrated to have superior numerical efficiency and simulation speed compared to the existing conventional models.**

***Index Terms*—Constant-parameter voltage behind reactance, five-phase synchronous machine, modeling, numerical efficiency, simulation.**

I. Introduction

FIVE-PHASE electric machines are being increasingly considered in machine-converter systems of aircraft [1], railway locomotion [2], electric vehicles [3], and other applications [4]–[8]. Compared with conventional three-phase machines, the five-phase machines offer interesting features such as improved torque utilization, reduced power per-phase, fault-tolerant operation [6]–[8], etc. Particularly, the five-phase synchronous machines are able to utilize low-order current harmonics (e.g., 3rd harmonic) to increase the electromagnetic torque, which results in higher torque per-current-peak compared to conventional three-phase machines [8].

Design, study and analysis of machine-converter systems including five-phase machines are highly dependent on available simulation programs and their machine models. These software tools are available as state-variable-based simulation packages (e.g., MATLAB/Simulink [9], Simscape Electrical [10], PLECS [11]), and nodal analysis-based packages (e.g., PSCAD/EMTDC [12], EMTP-RV [13]), as well as real-time simulation packages (e.g., RT-LAB [14], Typhoon HIL [15]). In all these packages, the development of numerically efficient and accurate models of electric machines, particularly five-phase synchronous machines, is critical in obtaining reliable simulation studies and facilitating practical design decision-making.

Five-phase electric machines can be modeled in a multitude of ways. Conventionally, the coupled-circuit phase-domain (CCPD) approach has been used to model five-phase electrical machines based on the magnetically-coupled electric circuit model of the machine's windings [16]–[18]. The resulting model has an interface consisting of *RL* branches with variable rotor position-dependent and coupled inductive terms. Interfacing inductances can also contain harmonic terms in case of non-sinusoidal air-gap and winding distribution in the machine. The CCPD model of five-phase machines can be easily interfaced with an arbitrary external network (including switching converters). However, the variable interfacing inductances require recalculation of the network impedances at each time-step, which degrades simulation performance creating a computational bottleneck [19]. Also, the variable inductances limit the model implementation only to simulation programs that offer this feature (e.g., PLECS [11]).

Alternatively, the classical *qd*0 model [20] has been developed for five-phase machines using multi-reference-frame five-phase transformation [16] including air-gap flux harmonics up to third component [16]–[18]. The *qd*0 models offer constant parameter equivalent circuit and are simple to implement in most commercial simulation packages. However, one major disadvantage of *qd*0 models is that their interfacing with external circuits is done using controlled current sources [21]. This requires the use of artificial snubber circuits when the machine is interfaced with inductive or power-electronic networks [21]. Snubber circuits are shown to introduce numerical error as well as numerical stiffness to the model, resulting in reduced simulation performance [22].

To overcome this limitation, the voltage-behind-reactance (VBR) models have been developed for electrical machines [19], [23] to achieve a direct machine-network interface similar to the CCPD model. In addition, VBR models are shown to provide accurate results while formulating a smaller interfacing matrix compared to the CCPD model [23]. However, the variable interfacing inductances still exist in the VBR models

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[19] when modeling salient rotor machines, which consequently slow down the simulation speed [24]. This limitation has led to the development of the so-called constant-parameter VBR (CPVBR) models [25]–[30] that utilize approximation methods (such as auxiliary rotor windings [25], [26] and algebraic-loop relaxation [27]–[31]) to formulate a constant interfacing circuit (to achieve high computational efficiency) while providing acceptable accuracy.

To the best of authors' knowledge, the VBR models of five-phase machines have so far been developed for only round rotor induction machines [32]. In this paper, the VBR modeling approach is further extended for a salient rotor five-phase synchronous machine, thus making following additional contributions:

- A variable-parameter VBR model of salient rotor five-phase synchronous machines is developed for the first time, which considers air-gap flux harmonics up to third component. The new model also considers rotor circuits, including damper and field windings.
- Based on the developed variable-parameter VBR model, a numerically efficient CPVBR model of five-phase synchronous machines is proposed using algebraic-loop relaxation method [27]. This model offers a simple and constant-parameter *RL* branch interfacing circuit in *abcde* phase coordinates.
- The modeling approach considers cross-couplings between the fundamental and third harmonic components of the machine inductances in its formulation, which improves the ability of the model to predict machine dynamics accurately with respect to experimental measurements.

The proposed CPVBR model of five-phase synchronous machine is validated by extensive simulation studies and experimental measurements, and is shown to offer accurate results with superior simulation performance compared to the existing CCPD, *qd*0 and the proposed VBR models. It is envisioned that the proposed CPVBR model will become a valuable asset in many simulation programs for design and analysis of machine-converter systems that include five-phase synchronous machines (e.g., [8]).

II. Five-Phase Synchronous Machine Modeling

In this paper, a salient pole five-phase synchronous machine is considered where the stator phases are displaced by 72 electrical degrees. Without loss of generality, the machine rotor is assumed to have one damper winding in each of the *q*- and *d*-axes as well as a field winding in the *d*-axis. A cross-section of the subject five-phase machine is depicted in Fig. 1. In addition, up to 3rd harmonic component for the stator winding functions and air-gap functions are considered, which is proven to provide acceptable accuracy [16]. It is also assumed that the 3rd harmonic component of air-gap and winding functions may not be aligned with the fundamental component. Throughout this paper, all machine parameters are referred to the stator side and motor sign convention is used. For consistency with [20] and prior literature (e.g., [27]), matrices, vectors and instantaneous quantities/variables are denoted using bold upper-case, bold lower-case, and lower case italic letters, respectively.

A. Coupled-Circuit Phase-Domain Model

The CCPD model of the five-phase synchronous machine is derived by expressing the machine stator and rotor voltages

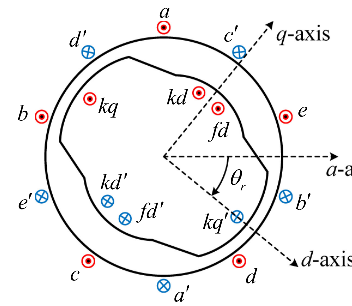

Fig. 1. A simplified cross-section of a salient pole five-phase synchronous machine.

in terms of the machine currents using the self and mutual inductances between the stator and rotor windings and their corresponding resistances as [16]

$$\begin{bmatrix}\mathbf{v}_{sabcde}\\ \mathbf{v}_{rqd}\end{bmatrix}_{8\times1}=\begin{bmatrix}\mathbf{R}_s & 0\\ 0 & \mathbf{R}_r\end{bmatrix}_{8\times8}\begin{bmatrix}\mathbf{i}_{sabcde}\\ \mathbf{i}_{rqd}\end{bmatrix}_{8\times1}+p\begin{bmatrix}\boldsymbol{\lambda}_{sabcde}\\ \boldsymbol{\lambda}_{rqd}\end{bmatrix}_{8\times1}, \tag{1}$$

$$\begin{bmatrix}\boldsymbol{\lambda}_{sabcde}\\ \boldsymbol{\lambda}_{rqd}\end{bmatrix}_{8\times1}=\begin{bmatrix}\mathbf{L}_s(\theta_r) & \mathbf{L}_{sr}(\theta_r)\\ \mathbf{L}_{rs}(\theta_r) & \mathbf{L}_r\end{bmatrix}_{8\times8}\begin{bmatrix}\mathbf{i}_{sabcde}\\ \mathbf{i}_{rqd}\end{bmatrix}_{8\times1}. \tag{2}$$

Here, $\mathbf{R}_s$ and $\mathbf{R}_r$ are the stator and rotor resistance matrices

$$\mathbf{R}_s=r_s\mathbf{I}_{5\times5},\quad \mathbf{R}_r=\begin{bmatrix}r_{kq} & 0 & 0\\ 0 & r_{kd} & 0\\ 0 & 0 & r_{fd}\end{bmatrix}, \tag{3}$$

where $\mathbf{I}$ is identity matrix. Also, in (1) lower-case p is the Heaviside's operator used to denote the time derivative *d/dt*. Moreover, in (2) the stator inductance matrices are obtained using the approach in [16], considering all coupling and cross-coupling terms between the machine air-gap and winding functions for both fundamental and 3rd harmonic components. The stator inductance matrix in (2) is expressed as

$$\mathbf{L}_s(\theta_r)=\begin{bmatrix}L_a(\theta_r) & L_{ab}(\theta_r) & L_{ac}(\theta_r) & L_{ad}(\theta_r) & L_{ae}(\theta_r)\\ L_{ab}(\theta_r) & L_b(\theta_r) & L_{bc}(\theta_r) & L_{bd}(\theta_r) & L_{be}(\theta_r)\\ L_{ac}(\theta_r) & L_{bc}(\theta_r) & L_c(\theta_r) & L_{cd}(\theta_r) & L_{ce}(\theta_r)\\ L_{ad}(\theta_r) & L_{bd}(\theta_r) & L_{cd}(\theta_r) & L_d(\theta_r) & L_{de}(\theta_r)\\ L_{ae}(\theta_r) & L_{be}(\theta_r) & L_{ce}(\theta_r) & L_{de}(\theta_r) & L_e(\theta_r)\end{bmatrix}, \tag{4}$$

where $L_a(\theta_r)$, $L_{ab}(\theta_r)$ and $L_{ac}(\theta_r)$ are defined in (5) shown at the bottom of next page, and the inductances of other stator phases are defined as

$$\begin{aligned}
L_b(\theta_r)&=L_a(\theta_r-2\pi/5),\quad L_c(\theta_r)=L_a(\theta_r-4\pi/5),\\
L_d(\theta_r)&=L_a(\theta_r-6\pi/5),\quad L_e(\theta_r)=L_a(\theta_r-8\pi/5),\\
L_{ad}(\theta_r)&=L_{ac}(\theta_r-\pi/5),\quad L_{ae}(\theta_r)=L_{ab}(\theta_r-3\pi/5),\\
L_{bc}(\theta_r)&=L_{ab}(\theta_r-2\pi/5),\quad L_{bd}(\theta_r)=L_{ac}(\theta_r-2\pi/5),\\
L_{be}(\theta_r)&=L_{ac}(\theta_r-3\pi/5),\quad L_{cd}(\theta_r)=L_{ab}(\theta_r-4\pi/5),\\
L_{ce}(\theta_r)&=L_{ac}(\theta_r-4\pi/5),\quad L_{de}(\theta_r)=L_{ab}(\theta_r-6\pi/5).
\end{aligned} \tag{6}$$

In (4), the fundamental component of the machine self-inductance in stator phase *a* is assumed to be aligned with the rotor angle θ_r. Using a similar approach, the mutual inductances between rotor and stator windings in (2) are also defined as

$$\mathbf{L}_{sr}(\theta_r) = \begin{bmatrix} L_{akq}(\theta_r) & L_{bkq}(\theta_r) & L_{ckq}(\theta_r) & L_{dkq}(\theta_r) & L_{ekq}(\theta_r) \\ L_{akd}(\theta_r) & L_{bkd}(\theta_r) & L_{ckd}(\theta_r) & L_{dkd}(\theta_r) & L_{ekd}(\theta_r) \\ L_{afd}(\theta_r) & L_{bfd}(\theta_r) & L_{cfd}(\theta_r) & L_{dfd}(\theta_r) & L_{efd}(\theta_r) \end{bmatrix}^T, \quad (7)$$

$$\mathbf{L}_{rs}(\theta_r) = \frac{2}{5}[\mathbf{L}_{sr}(\theta_r)]^T, \quad (8)$$

where $L_{akq}(\theta_r)$ and $L_{akd}(\theta_r)$ are defined in (9) shown at the bottom of this page, and are used to derive mutual inductances between the rotor windings and stator phases *b*, *c*, *d* and *e* as

$$\begin{aligned} &L_{bx}(\theta_r) = L_{ax}(\theta_r - 2\pi/5), \quad L_{cx}(\theta_r) = L_{ax}(\theta_r - 4\pi/5), \\ &L_{dx}(\theta_r) = L_{ax}(\theta_r - 6\pi/5), \quad L_{ex}(\theta_r) = L_{ax}(\theta_r - 8\pi/5), \\ &\quad x = (kq, kd), \\ &L_{yfd}(\theta_r) = L_{ykd}(\theta_r), \quad y = (a, b, c, d, e). \end{aligned} \quad (10)$$

In (5), (9) and (12) L_{ls}, L_{lkq}, L_{lkd} and L_{lfd} are the stator and rotor leakage inductances. Moreover, the following relations between inductances in (5), (9) are defined as

$$\begin{aligned} &L_{A1} = 0.2\,(L_{md1} + L_{mq1}), \quad L_{B1} = 0.2\,(L_{md1} - L_{mq1}), \\ &L_{A3} = 0.2\,(L_{md3} + L_{mq3}), \quad L_{B3} = 0.2\,(L_{md3} - L_{mq3}), \\ &L_{C3} = 0.4L_{mqd3}, \quad L_{D13} = 0.4L_{m13}, \quad L_{E13} = 0.4L_{mqd13}. \end{aligned} \quad (11)$$

In (11), L_{A1} and L_{B1} are the average and saliency level of the machine inductance corresponding to the fundamental component, and L_{A3} and L_{B3} denote similar quantities (i.e., average and saliency level) corresponding to the 3rd harmonic component, respectively. Also, L_{C3}, L_{D13}, and L_{E13} are inductive terms corresponding to cross-coupling for 3rd harmonic component, direct coupling between fundamental and 3rd harmonic components, and cross-coupling between fundamental and third harmonic components, respectively. Finally, the rotor inductances are defined as

$$\mathbf{L}_r = \begin{bmatrix} L_{kq} & 0 & 0 \\ 0 & L_{kd} & L_{md} \\ 0 & L_{md} & L_{fd} \end{bmatrix}, \quad (12)$$

where

$$\begin{aligned} &L_{kq} = L_{lkq} + 2L_{m13} + L_{mq1} + L_{mq3}, \\ &L_{md} = 2L_{m13} + L_{md1} + L_{md3}, \\ &L_{kd} = L_{lkd} + L_{md}, \; L_{fd} = L_{lfd} + L_{md}. \end{aligned} \quad (13)$$

The electromagnetic torque of the five-phase machine is defined as

$$T_e(\theta_r) = \left(\frac{P}{2}\right)\left[-\frac{1}{2}(\mathbf{i}_{sabcde})^T\left(\frac{\partial \mathbf{L}_s(\theta_r)}{\partial \theta_r} - L_{ls}\mathbf{I}_{5\times5}\right) \times \mathbf{i}_{sabcde} + (\mathbf{i}_{sabcde})^T\frac{\partial \mathbf{L}_{sr}(\theta_r)}{\partial \theta_r}\mathbf{i}_{rqd}\right], \quad (14)$$

where the upper-case *P* is the number of machine poles. The equations (1)–(12) and (14) constitute the CCPD model of the five-phase synchronous machine which can be implemented and interfaced directly to any arbitrary network. However, it is noted that the interfacing inductance matrix in (2) is variable and rotor-position-dependent (function of θ_r), which creates a significant computational burden.

B. *qd*0 Model

The *qd*0 model of the subject five-phase synchronous machine is derived by transforming the variables in CCPD model to the rotor reference frame as

$$\mathbf{f}_{qd130} = \mathbf{K}_{5ph}(\theta_r)\mathbf{f}_{abcde}, \quad (15)$$

where $\mathbf{K}_{5ph}(\theta_r)$ is Park's five-phase transformation matrix [16] defined in (16) shown at the bottom of next page, and

$$\mathbf{f}_{abcde} = [f_a, f_b, f_c, f_d, f_e]^T, \quad (17)$$

$$\mathbf{f}_{qd130} = [f_{q1}, f_{d1}, f_{q3}, f_{d3}, f_0]^T. \quad (18)$$

Here, $\mathbf{f}_{abcde}$ and $\mathbf{f}_{qd130}$ may represent voltage, current, or flux vectors in stator phase coordinates and rotor reference frame, respectively. It is noted that *q*1-*d*1 frame corresponds to fundamental component of variables, while *q*3-*d*3 frame corresponds to the 3rd harmonic. For clarity and distinction from conventional three-phase machine model, the *qd*0 model of the five-phase synchronous machine in this paper is denoted as *qd*130 in order to indicate coordinates in both fundamental and third harmonic components.

Transforming the machine voltage and flux equations in (1) and (2) using (15), the fundamental and 3rd harmonic components of stator voltage can be expressed in rotor reference frame

$$\begin{aligned} L_a(\theta_r) &= L_{ls} + L_{A1} + L_{A3} + L_{B1}\cos(2\theta_r) + L_{B3}\cos(6\theta_r) - L_{C3}\sin(6\theta_r) + 2L_{D13}\cos(2\theta_r) - 2L_{E13}\sin(2\theta_r), \\ L_{ab}(\theta_r) &= L_{A1}\cos\left(\frac{2\pi}{5}\right) + L_{A3}\cos\left(\frac{6\pi}{5}\right) + L_{B1}\cos\left(2\theta_r - \frac{2\pi}{5}\right) + L_{B3}\cos\left(6\theta_r - \frac{6\pi}{5}\right) - L_{C3}\sin\left(6\theta_r - \frac{6\pi}{5}\right) \\ &\quad + L_{D13}\left[\cos\left(2\theta_r - \frac{6\pi}{5}\right) + \cos\left(2\theta_r + \frac{2\pi}{5}\right)\right] - L_{E13}\left[\sin\left(2\theta_r - \frac{6\pi}{5}\right) + \sin\left(2\theta_r + \frac{2\pi}{5}\right)\right], \\ L_{ac}(\theta_r) &= L_{A1}\cos\left(\frac{4\pi}{5}\right) + L_{A3}\cos\left(\frac{12\pi}{5}\right) + L_{B1}\cos\left(2\theta_r - \frac{4\pi}{5}\right) + L_{B3}\cos\left(6\theta_r - \frac{12\pi}{5}\right) - L_{C3}\sin\left(6\theta_r - \frac{12\pi}{5}\right) \\ &\quad + L_{D13}\left[\cos\left(2\theta_r - \frac{12\pi}{5}\right) + \cos\left(2\theta_r + \frac{4\pi}{5}\right)\right] - L_{E13}\left[\sin\left(2\theta_r - \frac{12\pi}{5}\right) + \sin\left(2\theta_r + \frac{4\pi}{5}\right)\right]. \end{aligned} \quad (5)$$

$$\begin{aligned} L_{akq}(\theta_r) &= -(L_{m13} + L_{mq1})\sin(\theta_r) - (L_{m13} + L_{mq3})\sin(3\theta_r) + L_{mqd13}\cos(\theta_r) + (L_{mqd3} - L_{mqd13})\cos(3\theta_r), \\ L_{akd}(\theta_r) &= (L_{m13} + L_{md1})\cos(\theta_r) + (L_{m13} + L_{md3})\cos(3\theta_r) + L_{mqd13}E(\theta_r) - (L_{mqd3} + L_{mqd13})\sin(3\theta_r). \end{aligned} \quad (9)$$

as

$$v_{sq1} = r_s i_{sq1} + p\lambda_{sq1} + \omega_r \lambda_{sd1}, \tag{19}$$

$$v_{sd1} = r_s i_{sd1} + p\lambda_{sd1} - \omega_r \lambda_{sq1}, \tag{20}$$

and

$$v_{sq3} = r_s i_{sq3} + p\lambda_{sq3} + 3\omega_r \lambda_{sd3}, \tag{21}$$

$$v_{sd3} = r_s i_{sd3} + p\lambda_{sd3} - 3\omega_r \lambda_{sq3}, \tag{22}$$

$$v_{s0} = r_s i_{s0} + p\lambda_{s0}, \tag{23}$$

where ω_r is the rotor synchronous speed in rad/s. Also, the transformed rotor voltages can be expressed as

$$\begin{aligned} 0 &= r_{kq} i_{kq} + p\lambda_{kq}, \\ 0 &= r_{kd} i_{kd} + p\lambda_{kd}, \\ v_{fd} &= r_{fd} i_{fd} + p\lambda_{fd}. \end{aligned} \tag{24}$$

In (19)–(24), the stator fluxes are defined as

$$\lambda_{sq1} = \lambda_{mq1} + L_{ls} i_{sq1}, \quad \lambda_{sd1} = \lambda_{md1} + L_{ls} i_{sd1}, \tag{25}$$

$$\lambda_{sq3} = \lambda_{mq3} + L_{ls} i_{sq3}, \quad \lambda_{sd3} = \lambda_{md3} + L_{ls} i_{sd3}, \tag{26}$$

$$\lambda_{s0} = L_{ls} i_{s0}, \tag{27}$$

and the rotor fluxes are

$$\lambda_{kq} = \lambda_{mq1} + \lambda_{mq3} + L_{lkq} i_{kq}, \tag{28}$$

$$\begin{aligned} \lambda_{kd} &= \lambda_{md1} + \lambda_{md3} + L_{lkd} i_{kd}, \\ \lambda_{fd} &= \lambda_{md1} + \lambda_{md3} + L_{lfd} i_{fd}. \end{aligned} \tag{29}$$

In (25)–(29), the magnetizing flux components are

$$\begin{bmatrix} \lambda_{mq1} \\ \lambda_{md1} \\ \lambda_{mq3} \\ \lambda_{md3} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_{m11} & \mathbf{L}_{m13} \\ \mathbf{L}_{m13}^T & \mathbf{L}_{m33} \end{bmatrix}_{4\times 4} \begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \end{bmatrix} + \begin{bmatrix} \mathbf{L}_{m11} + \mathbf{L}_{m13} \\ \mathbf{L}_{m13}^T + \mathbf{L}_{m33} \end{bmatrix}_{4\times 2} \boldsymbol{\alpha} \begin{bmatrix} i_{kq} \\ i_{kd} \\ i_{fd} \end{bmatrix}, \tag{30}$$

where

$$\mathbf{L}_{m11} = \begin{bmatrix} L_{mq1} & 0 \\ 0 & L_{md1} \end{bmatrix}, \mathbf{L}_{m13} = \begin{bmatrix} L_{m13} & L_{mqd13} \\ -L_{mqd13} & L_{m13} \end{bmatrix}, \tag{31}$$

$$\mathbf{L}_{m33} = \begin{bmatrix} L_{mq3} & L_{mqd3} \\ L_{mqd3} & L_{md3} \end{bmatrix}, \boldsymbol{\alpha} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}. \tag{32}$$

The electromagnetic torque can be derived using the machine stator currents and magnetizing fluxes as [16], [32]

$$T_e = \frac{5}{2}\frac{P}{2}\left(i_{sq1}\lambda_{md1} - i_{sd1}\lambda_{mq1} + 3i_{sq3}\lambda_{md3} - 3i_{sd3}\lambda_{mq3}\right). \tag{33}$$

The machine voltage and flux equations in (19)–(32) form the *qd*130 model of the five-phase synchronous machine, where its equivalent circuit in rotor reference frame is depicted in Fig. 2.

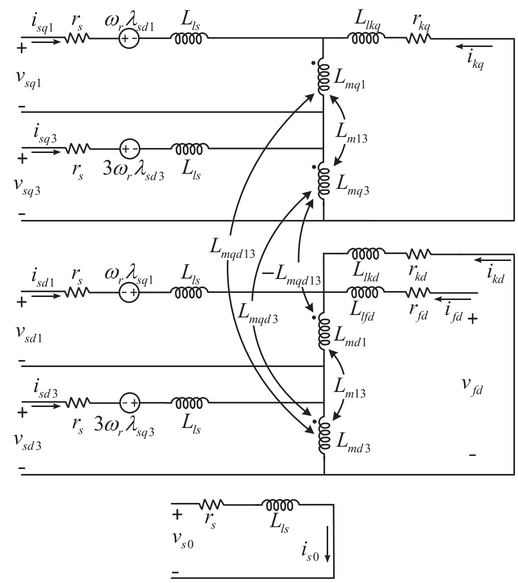

Fig. 2. The *qd*130 equivalent circuit of the five-phase synchronous machine in rotor reference frame, including coupling and cross-coupling between fundamental and third harmonic components.

As seen in Fig. 2, the *qd*130 model [(19)–(33)] can be readily implemented using stator and rotor fluxes as the state variables, stator and rotor voltages as inputs, and stator and rotor currents as outputs. Therefore, this model is interfaced with external circuits using controlled current sources.

C. Proposed Variable-Parameter VBR Model

Based on the *qd*130 equivalent circuit, the variable-parameter VBR model of the synchronous five-phase machine is derived using a standard procedure [19]. Therein, the stator circuit is replaced by an equivalent interfacing circuit with sub-transient inductances and back-emf voltage sources, while rotor subsystem is formulated in terms of fluxes as the state variables and stator currents and rotor voltages as the inputs.

To obtain the rotor subsystem state equation, the rotor flux derivatives are expressed in terms of rotor currents, field voltage, and magnetizing fluxes based on (24), as

$$p\begin{bmatrix} \lambda_{kq} \\ \lambda_{kd} \\ \lambda_{fd} \end{bmatrix} = -\mathbf{R}_r \boldsymbol{\Gamma}_{lr} \left(\begin{bmatrix} \lambda_{kq} \\ \lambda_{kd} \\ \lambda_{fd} \end{bmatrix} - \boldsymbol{\beta} \begin{bmatrix} \lambda_{mq1} \\ \lambda_{md1} \\ \lambda_{mq3} \\ \lambda_{md3} \end{bmatrix} \right) + \begin{bmatrix} 0 \\ 0 \\ v_{fd} \end{bmatrix}, \tag{34}$$

where

$$\boldsymbol{\Gamma}_{lr} = \begin{bmatrix} 1/L_{lkq} & 0 & 0 \\ 0 & 1/L_{lkd} & 0 \\ 0 & 0 & 1/L_{lfd} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}. \tag{35}$$

$$\mathbf{K}_{5ph}(\theta_r) = \frac{2}{5}\begin{bmatrix} \cos(\theta_r) & \cos\left(\theta_r - \frac{2\pi}{5}\right) & \cos\left(\theta_r - \frac{4\pi}{5}\right) & \cos\left(\theta_r + \frac{4\pi}{5}\right) & \cos\left(\theta_r + \frac{2\pi}{5}\right) \\ \sin(\theta_r) & \sin\left(\theta_r - \frac{2\pi}{5}\right) & \sin\left(\theta_r - \frac{4\pi}{5}\right) & \sin\left(\theta_r + \frac{4\pi}{5}\right) & \sin\left(\theta_r + \frac{2\pi}{5}\right) \\ \cos(3\theta_r) & \cos\left(3\theta_r - \frac{6\pi}{5}\right) & \cos\left(3\theta_r - \frac{2\pi}{5}\right) & \cos\left(3\theta_r + \frac{2\pi}{5}\right) & \cos\left(3\theta_r + \frac{6\pi}{5}\right) \\ \sin(3\theta_r) & \sin\left(3\theta_r - \frac{6\pi}{5}\right) & \sin\left(3\theta_r - \frac{2\pi}{5}\right) & \sin\left(3\theta_r + \frac{2\pi}{5}\right) & \sin\left(3\theta_r + \frac{6\pi}{5}\right) \\ 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}. \tag{16}$$

Substituting rotor currents in (30) from (28), (29), the magnetizing flux vector in (34) can be written in terms of rotor fluxes and stator currents as

$$\begin{bmatrix}\lambda_{mq1}\\ \lambda_{md1}\\ \lambda_{mq3}\\ \lambda_{md3}\end{bmatrix} = \mathbf{L}''_m \begin{bmatrix}i_{sq1}\\ i_{sd1}\\ i_{sq3}\\ i_{sd3}\end{bmatrix} + \mathbf{L}_A \begin{bmatrix}\lambda_{kq}\\ \lambda_{kd}\\ \lambda_{fd}\end{bmatrix}, \tag{36}$$

where the sub-transient inductance matrix $\mathbf{L}''_m$ is defined in (37) shown at the bottom of this page, and $\mathbf{L}_A$ is a 4×3 matrix expressed as

$$\mathbf{L}_A = \left(\mathbf{I}_{4\times4} + \begin{bmatrix}\mathbf{L}_{m11} + \mathbf{L}_{m13}\\ \mathbf{L}^T_{m13} + \mathbf{L}_{m33}\end{bmatrix} \boldsymbol{\alpha}\boldsymbol{\Gamma}_{lr}\boldsymbol{\beta}\right)^{-1} \times \begin{bmatrix}\mathbf{L}_{m11} + \mathbf{L}_{m13}\\ \mathbf{L}^T_{m13} + \mathbf{L}_{m33}\end{bmatrix} \boldsymbol{\alpha}\boldsymbol{\Gamma}_{lr}. \tag{38}$$

By writing the magnetizing flux vector in (34) in terms of state variables and inputs using (36), the state equation is derived as

$$p\begin{bmatrix}\lambda_{kq}\\ \lambda_{kd}\\ \lambda_{fd}\end{bmatrix} = -\mathbf{R}_r\boldsymbol{\Gamma}_{lr}\left((\mathbf{I}_{3\times3} + \boldsymbol{\beta}\mathbf{L}_A)\begin{bmatrix}\lambda_{kq}\\ \lambda_{kd}\\ \lambda_{fd}\end{bmatrix} - \boldsymbol{\beta}\mathbf{L}''_m\begin{bmatrix}i_{sq1}\\ i_{sd1}\\ i_{sq3}\\ i_{sd3}\end{bmatrix}\right) + \begin{bmatrix}0\\ 0\\ v_{fd}\end{bmatrix}. \tag{39}$$

Substituting the stator flux in (19)–(23) with the ones defined in (25)–(27), the stator voltage equations in *q*- and *d*-axes for fundamental and third harmonic components can be written as

$$\begin{bmatrix}v_{sq1}\\ v_{sd1}\\ v_{sq3}\\ v_{sd3}\\ v_{s0}\end{bmatrix} = r_s\begin{bmatrix}i_{sq1}\\ i_{sd1}\\ i_{sq3}\\ i_{sd3}\\ i_{s0}\end{bmatrix} + L_{ls}\mathbf{I}_{5\times5}p\begin{bmatrix}i_{sq1}\\ i_{sd1}\\ i_{sq3}\\ i_{sd3}\\ i_{s0}\end{bmatrix} + \omega_r\begin{bmatrix}\lambda_{sd1}\\ -\lambda_{sq1}\\ \lambda_{sd3}\\ -\lambda_{sq3}\\ 0\end{bmatrix} + p\begin{bmatrix}\lambda_{mq1}\\ \lambda_{md1}\\ \lambda_{mq3}\\ \lambda_{md3}\\ 0\end{bmatrix}, \tag{40}$$

where stator fluxes are calculated using (25)–(26).

In (40), the magnetizing flux derivatives in *q*- and *d*-axes can be calculated using (36) as

$$p\begin{bmatrix}\lambda_{mq1}\\ \lambda_{md1}\\ \lambda_{mq3}\\ \lambda_{md3}\end{bmatrix} = \mathbf{L}''_m p\begin{bmatrix}i_{sq1}\\ i_{sd1}\\ i_{sq3}\\ i_{sd3}\end{bmatrix} + \mathbf{L}_A p\begin{bmatrix}\lambda_{kq}\\ \lambda_{kd}\\ \lambda_{fd}\end{bmatrix}. \tag{41}$$

Furthermore, replacing magnetizing flux derivatives in (40) with (41), and after some algebraic manipulations, the stator voltage equations can be written in the VBR form as

$$\begin{bmatrix}v_{sq1}\\ v_{sd1}\\ v_{sq3}\\ v_{sd3}\\ v_{s0}\end{bmatrix} = r_s\begin{bmatrix}i_{sq1}\\ i_{sd1}\\ i_{sq3}\\ i_{sd3}\\ i_{s0}\end{bmatrix} + \begin{bmatrix}L_{ls}\mathbf{I}_{4\times4} + \mathbf{L}''_m & \mathbf{0}_{4\times1}\\ \mathbf{0}_{1\times4} & L_{ls}\end{bmatrix}_{5\times5} \times p\begin{bmatrix}i_{sq1}\\ i_{sd1}\\ i_{sq3}\\ i_{sd3}\\ i_{s0}\end{bmatrix} + \begin{bmatrix}e''_{q1}\\ e''_{d1}\\ e''_{q3}\\ e''_{d3}\\ 0\end{bmatrix} + \omega_r\begin{bmatrix}\lambda''_{d1}\\ -\lambda''_{q1}\\ \lambda''_{d3}\\ -\lambda''_{q3}\\ 0\end{bmatrix}, \tag{42}$$

where the sub-transient fluxes are defined as

$$\begin{bmatrix}\lambda''_{q1}\\ \lambda''_{d1}\\ \lambda''_{q3}\\ \lambda''_{d3}\end{bmatrix} = (L_{ls}\mathbf{I}_{4\times4} + \mathbf{L}''_m)\begin{bmatrix}i_{sq1}\\ i_{sd1}\\ i_{sq3}\\ i_{sd3}\end{bmatrix}, \tag{43}$$

and the sub-transient back-emfs are

$$\begin{bmatrix}e''_{q1}\\ e''_{d1}\\ e''_{q3}\\ e''_{d3}\end{bmatrix} = \omega_r\begin{bmatrix}\lambda_{sd1} - \lambda''_{d1}\\ -\lambda_{sq1} + \lambda''_{q1}\\ \lambda_{sd3} - \lambda''_{d3}\\ -\lambda_{sq3} + \lambda''_{q3}\end{bmatrix} + \mathbf{L}_A p\begin{bmatrix}\lambda_{kq}\\ \lambda_{kd}\\ \lambda_{fd}\end{bmatrix}. \tag{44}$$

In (44), the rotor flux derivatives are calculated using the state equation (39) in terms of state variables (rotor fluxes) and inputs (stator currents and rotor field voltage).

Next, applying Park's inverse five-phase transformation matrix defined as

$$\mathbf{f}_{abcde} = \mathbf{K}^{-1}_{5ph}(\theta_r)\mathbf{f}_{qd130},$$

$$\mathbf{K}^{-1}_{5ph}(\theta_r) = \frac{5}{2}\mathbf{K}^T_{5ph}(\theta_r), \tag{45}$$

the stator interfacing equation of the VBR model can be finally derived in *abcde* phase coordinates as

$$\mathbf{v}_{sabcde} = r_s\mathbf{i}_{sabcde} + \mathbf{L}''_{abcde}(\theta_r)p\mathbf{i}_{sabcde} + \mathbf{e}''_{abcde}, \tag{46}$$

where the variable-parameter sub-transient inductance matrix is defined as

$$\mathbf{L}''_{abcde}(\theta_r) = \mathbf{K}^{-1}_{5ph}(\theta_r)\begin{bmatrix}L_{ls}\mathbf{I}_{4\times4} + \mathbf{L}''_m & \mathbf{0}_{4\times1}\\ \mathbf{0}_{1\times4} & L_{ls}\end{bmatrix}_{5\times5}\mathbf{K}_{5ph}(\theta_r). \tag{47}$$

The electromagnetic torque in the VBR model is identical to the *qd*130 model and can be directly calculated using (33). The rotor subsystem state model (39)–(44), the stator interfacing equation (46), and (33) constitute the proposed variable-parameter VBR model of five-phase synchronous machine. This model can be implemented as shown in Fig. 3, and be directly interfaced with arbitrary external network without requiring any snubber circuits. However, the variable-parameter inductance matrix in (47) is still rotor-position-dependent.

$$\mathbf{L}''_m = \left(\mathbf{I}_{4\times4} + \begin{bmatrix}\mathbf{L}_{m11} + \mathbf{L}_{m13}\\ \mathbf{L}^T_{m13} + \mathbf{L}_{m33}\end{bmatrix}_{4\times2}\boldsymbol{\alpha}\boldsymbol{\Gamma}_{lr}\boldsymbol{\beta}\right)^{-1}\begin{bmatrix}\mathbf{L}_{m11} & \mathbf{L}_{m13}\\ \mathbf{L}^T_{m13} & \mathbf{L}_{m33}\end{bmatrix}_{4\times4} = \begin{bmatrix}L''_{mq1} & L''_{mqd1} & L''_{mq13} & L''_{mqd13}\\ L''_{mdq1} & L''_{md1} & L''_{mdq13} & L''_{md13}\\ L''_{mq31} & L''_{mqd31} & L''_{mq3} & L''_{mqd3}\\ L''_{mdq31} & L''_{md31} & L''_{mdq3} & L''_{md3}\end{bmatrix}. \tag{37}$$

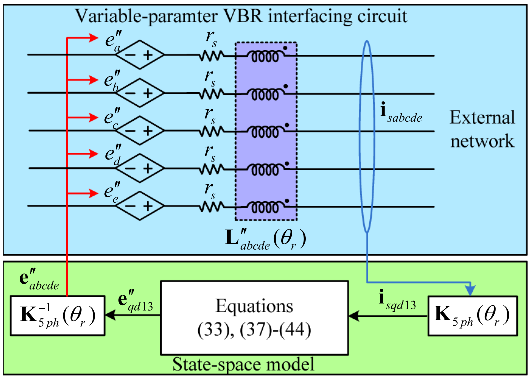

Fig. 3. Implementation of the proposed variable-parameter VBR model of the five-phase synchronous machine with coupled and rotor-position-dependent inductances, directly interfaced with arbitrary external network.

D. Proposed Constant-Parameter VBR Model

To achieve a more efficient VBR model, a constant-parameter interfacing circuit is preferred. It is observed that the variable-parameter inductances in phase coordinates are due to the rotor saliency and coupling terms in the machine inductance matrix in the rotor qd coordinates [i.e., see (37)]. Thus, to achieve a constant parameter interfacing, these terms have to be separated from the machine inductance matrix in rotor qd coordinates, and their effect has to be considered as voltage terms and included in the sub-transient back emf voltage sources. To achieve this separation, the sub-transient inductance matrix in (37) is partitioned into two parts as

$$\mathbf{L}''_m = \mathbf{L}'''_m + \mathbf{L}''_{m\_res}, \tag{48}$$

where the non-salient and magnetically decoupled inductance matrix $\mathbf{L}'''_m$ is defined based on d-axis inductances as

$$\mathbf{L}'''_m = \begin{bmatrix} L''_{md1} & 0 & 0 & 0 \\ 0 & L''_{md1} & 0 & 0 \\ 0 & 0 & L''_{md3} & 0 \\ 0 & 0 & 0 & L''_{md3} \end{bmatrix}, \tag{49}$$

and the residual inductance matrix $\mathbf{L}''_{m\_res}$ (including saliency and coupling terms) is defined as

$$\mathbf{L}''_{m\_res} = \begin{bmatrix} L''_{mq1} - L''_{md1} & L''_{mqd1} & L''_{mq13} & L''_{mqd13} \\ L''_{mdq1} & 0 & L''_{mdq13} & L''_{md13} \\ L''_{mq31} & L''_{mqd31} & L''_{mq3} - L''_{md3} & L''_{mqd3} \\ L''_{mdq31} & L''_{md31} & L''_{mdq3} & 0 \end{bmatrix}. \tag{50}$$

The non-salient and decoupled inductance matrix $\mathbf{L}'''_m$ is used to derive a new constant-parameter interfacing inductance matrix in phase coordinates, and the residual inductance matrix $\mathbf{L}''_{m\_res}$ is represented as modified voltage sources and included into the sub-transient back-emf voltages. For this purpose, by substituting $\mathbf{L}''_m$ in (42) with the one in (48) using (49) and (50), the stator voltage equation in (42) is rewritten as

$$\begin{bmatrix} v_{sq1} \\ v_{sd1} \\ v_{sq3} \\ v_{sd3} \\ v_{s0} \end{bmatrix} = r_s \begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \\ i_{s0} \end{bmatrix} + \begin{bmatrix} L_{ls}\mathbf{I}_{4\times4} + \mathbf{L}'''_m & \mathbf{0}_{4\times1} \\ \mathbf{0}_{1\times4} & L_{ls} \end{bmatrix} p \begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \\ i_{s0} \end{bmatrix} + \omega_r \begin{bmatrix} \lambda'''_{d1} \\ -\lambda'''_{q1} \\ \lambda'''_{d3} \\ -\lambda'''_{q3} \\ 0 \end{bmatrix} + \begin{bmatrix} e''_{q1} \\ e''_{d1} \\ e''_{q3} \\ e''_{d3} \\ 0 \end{bmatrix} + \omega_r \begin{bmatrix} \lambda''_{d1} - \lambda'''_{d1} \\ -\lambda''_{q1} + \lambda'''_{q1} \\ \lambda''_{d3} - \lambda'''_{d3} \\ -\lambda''_{q3} + \lambda'''_{q3} \\ 0 \end{bmatrix} + \begin{bmatrix} \mathbf{I}_{4\times4} + \mathbf{L}''_{m\_res} & \mathbf{0}_{4\times1} \\ \mathbf{0}_{1\times4} & 0 \end{bmatrix} p \begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \\ 0 \end{bmatrix}, \tag{51}$$

where the new sub-transient fluxes are

$$\begin{bmatrix} \lambda'''_{q1} \\ \lambda'''_{d1} \\ \lambda'''_{q3} \\ \lambda'''_{d3} \end{bmatrix} = (L_{ls}\mathbf{I}_{4\times4} + \mathbf{L}'''_m) \begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \end{bmatrix}. \tag{52}$$

By defining new sub-transient back-emf voltages as

$$\begin{bmatrix} e'''_{q1} \\ e'''_{d1} \\ e'''_{q3} \\ e'''_{d3} \end{bmatrix} = \begin{bmatrix} e''_{q1} \\ e''_{d1} \\ e''_{q3} \\ e''_{d3} \end{bmatrix} + \mathbf{L}''_{m\_res} p \begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \end{bmatrix} + \omega_r \begin{bmatrix} \lambda''_{d1} - \lambda'''_{d1} \\ -\lambda''_{q1} + \lambda'''_{q1} \\ \lambda''_{d3} - \lambda'''_{d3} \\ -\lambda''_{q3} + \lambda'''_{q3} \end{bmatrix}, \tag{53}$$

the stator voltage equation can be written as

$$\begin{bmatrix} v_{sq1} \\ v_{sd1} \\ v_{sq3} \\ v_{sd3} \\ v_{s0} \end{bmatrix} = r_s \begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \\ i_{s0} \end{bmatrix} + \begin{bmatrix} L_{ls}\mathbf{I}_{4\times4} + \mathbf{L}'''_m & \mathbf{0}_{4\times1} \\ \mathbf{0}_{1\times4} & L_{ls} \end{bmatrix} \times p \begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \\ i_{s0} \end{bmatrix} + \begin{bmatrix} e'''_{q1} \\ e'''_{d1} \\ e'''_{q3} \\ e'''_{d3} \\ 0 \end{bmatrix} + \omega_r \begin{bmatrix} \lambda'''_{d1} \\ -\lambda'''_{q1} \\ \lambda'''_{d3} \\ -\lambda'''_{q3} \\ 0 \end{bmatrix}. \tag{54}$$

As can be seen in (53), calculating the new sub-transient back-emfs requires stator current derivatives in q- and d-axes for both fundamental and third harmonic components which can be calculated using (42) as

$$p\begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \end{bmatrix} = (L_{ls}\mathbf{I}_{4\times4} + \mathbf{L}''_m)^{-1}\left(\begin{bmatrix} v_{sq1} \\ v_{sd1} \\ v_{sq3} \\ v_{sd3} \end{bmatrix} - r_s\begin{bmatrix} i_{sq1} \\ i_{sd1} \\ i_{sq3} \\ i_{sd3} \end{bmatrix}\right.$$
$$\left. -\omega_r\begin{bmatrix} \lambda''_{d1} \\ -\lambda''_{q1} \\ \lambda''_{d3} \\ -\lambda''_{q3} \end{bmatrix} - \begin{bmatrix} e''_{q1} \\ e''_{d1} \\ e''_{q3} \\ e''_{d3} \end{bmatrix}\right). \tag{55}$$

It is observed that the stator current derivatives in (55) have a direct algebraic relation with the stator voltages. Also, it is noted that the stator voltages in (54) have a direct algebraic relation with the new sub-transient back-emfs defined in (53), which are themselves calculated using stator current derivatives in (55). As a result, an algebraic-loop is created and the stator voltage formulation becomes implicit. In order to achieve an explicit formulation, the relaxation method [34] is applied where a tuned filter is used to provide accurate approximation of the stator voltages as

$$\mathbf{v}'_s = \begin{bmatrix} v'_{sq1} \\ v'_{sd1} \\ v'_{sq3} \\ v'_{sd3} \end{bmatrix} = F(s)\begin{bmatrix} v_{sq1} \\ v_{sd1} \\ v_{sq3} \\ v_{sd3} \end{bmatrix}, \tag{56}$$

which is used to approximate the stator voltages in (55) for calculating stator current derivatives. The pole selection and tuning of the filter $F(s)$ is done according to [33], [34].

Finally, the stator interfacing equation in *abcde* phase coordinates is derived by applying the inverse Park's transformation (45) to voltage equation (54) which results in

$$\mathbf{v}_{sabcde} = r_s\mathbf{i}_{sabcde} + \mathbf{L}'''_{abcde}p\mathbf{i}_{sabcde} + \mathbf{e}'''_{abcde}. \tag{57}$$

Here, the sub-transient interfacing inductance matrix in phase coordinates $\mathbf{L}'''_{abcde}$ is constant (as a result of $\mathbf{L}'''_m$ being non-salient and decoupled) and is defined as

$$\mathbf{L}'''_{abcde} = \mathbf{K}^{-1}_{5ph}(\theta_r)\begin{bmatrix} L_{ls}\mathbf{I}_{4\times4} + \mathbf{L}'''_m & \mathbf{0}_{4\times1} \\ \mathbf{0}_{1\times4} & L_{ls} \end{bmatrix}_{5\times5}\mathbf{K}_{5ph}(\theta_r). \tag{58}$$

The proposed CPVBR model of the five-phase synchronous machine can be implemented using the rotor state equation (39), the new sub-transient back-emfs (53), and the stator interfacing equation (57), as illustrated in Fig. 4. Similar to the VBR implementation of Fig. 3, the CPVBR has coupled inductive branches that can be interfaced with an arbitrary external network as shown in Fig. 4. However, the parameters of these interfacing inductive branches are now constant.

III. Model Validation and Computer Studies

In order to validate the proposed models, a 12-pole five-phase synchronous machine with wye-connected stator windings and one rotor field winding is considered. The experimental laboratory setup and its simplified schematic are demonstrated in Fig. 5. Therein, the five-phase machine is used as a generator, which supplies a ten-pulse diode rectifier load through five identical lines represented with r_{line} resistances. The mechanical power is provided using a dc motor-prime-mover which operates at constant mechanical speed. The system parameters including

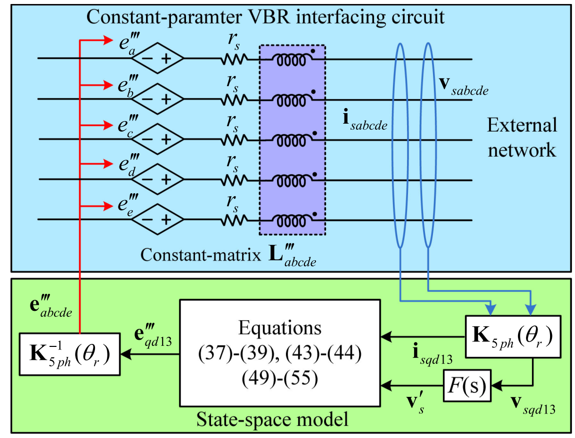

Fig. 4. Implementation of the proposed CPVBR model of the five-phase synchronous machine with coupled and constant *RL* branches, directly interfaced with arbitrary external network.

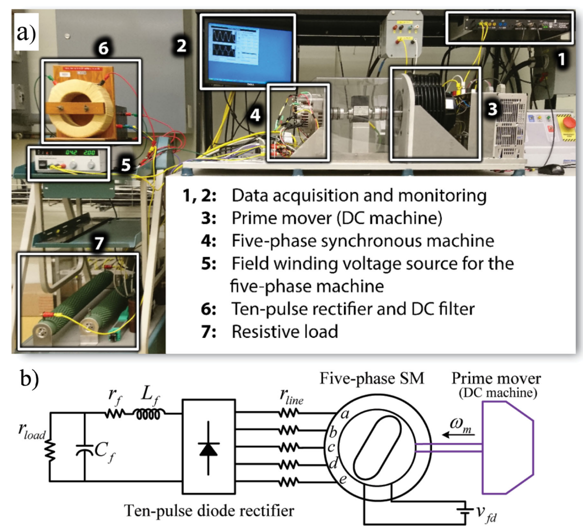

Fig. 5. The considered case study system: (a) experimental laboratory setup, (b) its corresponding circuit schematic.

the five-phase machine, line and diode-rectifier are summarized in Appendix.

For the purpose of benchmarking, the CCPD, the conventional *qd*130, the variable-parameter VBR and CPVBR models presented in Sections II-A–II-D, respectively, have been implemented in MATLAB/Simulink [9] using PLECS toolbox [11]. For consistency, all simulations are conducted on a PC with Intel Corei7-8700K @ 3.7 GHz processor using the variable time-step solver *ode*45 with both absolute and relative error tolerances set to 10^{-4}. For comparison purposes, a reference solution is obtained by simulating the CCPD model using a very small time-step of 1 µs.

To interface the *qd*130 model with the diode rectifier, the snubber resistors are connected in parallel to each of the stator phases. The snubbers are chosen to be 160 Ω, which limits the snubber branch current to approximately 1% of the machine phase current in the considered studies.

In the proposed CPVBR model, the filter *F(s)* in (56) consists of a low-pass filter and a band-pass filter as presented in [33],

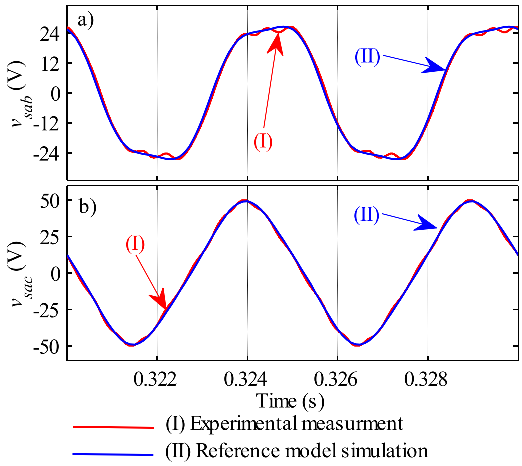

Fig. 6. No-load steady state line voltages obtained from experimental measurement and simulated reference model under no-load operating condition: (a) v_{sab} and (b) v_{sac}.

[34]. Since the machine is connected to a ten-pulse five-phase rectifier, the 7th, 9th, 11th and 13th harmonics are dominantly present on the machine ac variables. These harmonics result in oscillations corresponding to $10\omega_r$ in the variables transformed to rotor reference frame. The filter parameters have been chosen according to [33], [34], with the band-pass filter resonance frequency equal to $10\omega_r$ (here, chosen as $10\omega_r = 12{,}566$ rad/s corresponding to a rotor speed of 2000 rpm) and be critically damped (i.e., damping factor equal to 0.7), and the low-pass filter pole of –1300.

A. Steady-State Analysis and Model Verification

To validate the considered models, the reference solution is compared to experimental measurements from a prototype five-phase synchronous machine. Comparisons are carried out under no-load and loaded conditions.

In the following no-load study, the five-phase machine is rotating at 2000 rpm (200 Hz stator frequency) and the field winding is supplied by a 4.24 V voltage source (0.53 V when referred to stator side).

The waveforms of line voltages v_{sab} and v_{sac} obtained from experimental measurements and simulations of the reference model are presented in Fig. 6. As it can be seen in Fig. 6, the reference model is able to produce accurate results very much consistent with experimental measurements. Specifically, it is observed that the five-phase machine provides line voltages with two different amplitudes as expected in a balanced five-phase system. It is also worth mentioning that the waveforms obtained from the experiment contain higher order harmonics which are not considered in the subject models. Such high-order harmonics have much smaller amplitudes compared to the dominant fundamental and third harmonic (thus having small impact on the waveforms as seen in Fig. 6), and are generally often neglected [8].

To compare the simulation with experimental results under loading conditions, the machine's rectifier is loaded by a 1 Ω resistor resulting in a total delivered power of 510 W. The corresponding measured and simulated waveforms of the line voltages v_{sab} and v_{sac} and line current i_{sa} are shown in Fig. 7. As observed in Fig. 7, the reference model is able to provide highly accurate prediction with respect to experimental measurements

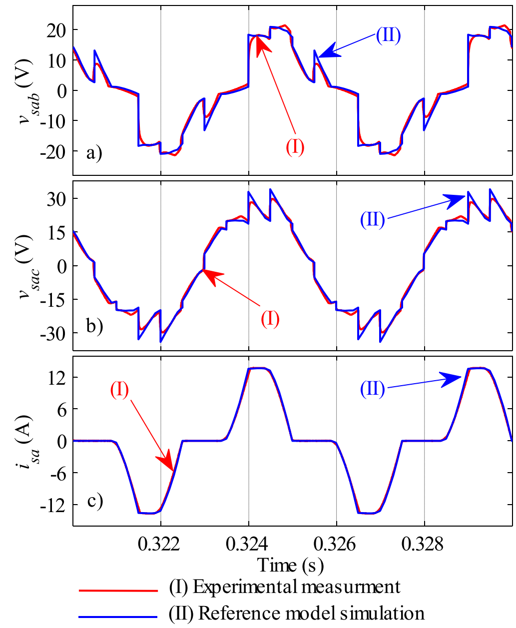

Fig. 7. Steady-state results obtained from experimental measurement and the reference model for the machine line voltages and current: (a) v_{sab}, (b) v_{sac} and (c) i_{sa}, when the rectifier is loaded by a 1 Ω resistor.

even under non-linear loading condition by considering the fundamental and third harmonic (and their cross-couplings) in the five-phase machine model.

B. Transient Case-Study

For the purpose of comparison between the subject models (presented in Section II-A–II-D) in terms of numerical accuracy and computational efficiency, a 2-second transient simulation study is conducted here. The machine is assumed to be under the same prime mover and excitation conditions as in Section III-A, and rectifier is initially supplying a dc load corresponding to $r_{load} = 15\,\Omega$ (which results in approximately 160 W). At $t =$ 1 s, the rectifier load is increased to 510 watts (corresponding to $r_{load} = 1\,\Omega$).

The fragment of transient responses in the stator line voltages v_{sab} and v_{sac}, line current i_{sa} and electromagnetic torque T_e, obtained by the subject models are presented in Fig. 8. As seen in Fig. 8, increasing the rectifier load leads to more distorted stator voltages [see Fig. 8(a)–(b)], increased rectifier current [see Fig. 8(c)] and torque [see Fig. 8(d)]. Fig. 8 also shows that all subject models are able to produce similar and consistent results with respect to the reference solution.

In order to provide a more detailed comparison, a magnified view of stator line current i_{sa} and electromagnetic torque T_e after load change are presented in Fig. 9. It can be seen in Fig. 9 that the *qd*130 model has a visible deviation from the reference solution, which is caused by the current flowing into the snubber branches. The CCPD and the proposed variable-parameter VBR models both are able to produce nearly identical results to the reference solution.

In addition, Fig. 9 shows that the proposed CPVBR model can obtain highly accurate results compared to the reference solution, with negligible visible error due to the filter approximation (56).

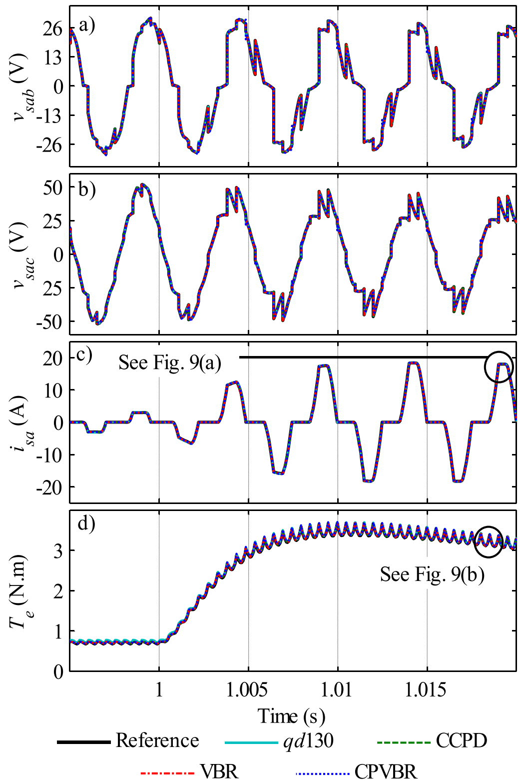

Fig. 8. Fragment of the transient response of several machine variables during load change as predicted by the subject models: (a) line voltage v_{sab}, (b) line voltage v_{sac}, (c) line current i_{sa}, and (d) electromagnetic torque.

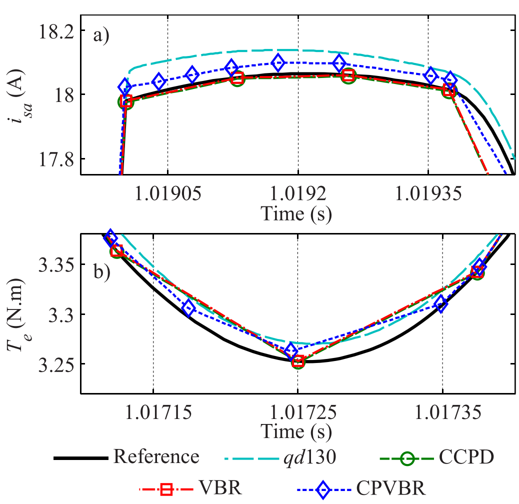

Fig. 9. Magnified view of selected variables obtained by the subject models: (a) line current i_{sa}, and (b) electromagnetic torque.

C. Numerical Accuracy Comparison

To quantitatively assess the models' accuracy, the 2-norm numerical errors [35] in trajectories of the stator current and electromagnetic torque as predicted by the subject models with respect to the reference solutions are calculated and summarized in Table I.

As can be seen in Table I, the *qd*130 model has the largest numerical error (0.97% in stator current and 0.85% in torque), which is expected due to presence of snubbers. The CCPD and the proposed variable-parameter VBR models both are shown to have much smaller errors (0.008% and 0.007% in stator current, respectively, and 0.006% in torque). These errors are mainly associated with solver discretization. Table I also shows that the proposed CPVBR model has good accuracy with small errors (0.26% in stator current and 0.20% in torque), which are due to filter approximation.

TABLE I
COMPARISON OF 2-NORM ERRORS FOR THE 2-SECOND TRANSIENT STUDY AS OBTAINED BY THE SUBJECT MODELS

Model	Stator phase *a* current (i_{sa})	Electromagnetic torque (T_e)
*qd*130 model (with snubbers)	0.97 %	0.85 %
CCPD model (with variable inductances)	0.008 %	0.006 %
Proposed VBR model (with variable inductances)	0.007 %	0.006 %
Proposed CPVBR model	0.26 %	0.20 %

TABLE II
COMPUTATIONAL PERFORMANCE COMPARISONS OF THE SUBJECT MODELS FOR THE 2-SECOND TRANSIENT STUDY

Model	Largest eigenvalue	Number of steps	CPU time per step	CPU Time
*qd*130 model (with snubbers)	-1.11×10^{6}	685,185	72 µs	49.6 s
CCPD model (with variable inductances)	–262	26,616	725 µs	19.3 s
Proposed VBR model (with variable inductances)	–260	26,546	682 µs	18.1 s
Proposed CPVBR model	–8,745	45,156	140 µs	6.3 s

D. Computational Performance

The computational performance of the subject models is compared as summarized in Table II. As seen in Table II, the *qd*130 model has the largest eigenvalue (-1.11×10^6, i.e., numerically stiff) associated with the use of snubbers, which results in the largest number of time steps (685,185). This leads to a prohibitive total CPU time (49.6 s), despite the relatively small average CPU time per-step (72 µs).

Table II also shows that both the CCPD and variable-parameter VBR models have significantly smaller eigenvalues (-262 and -260, respectively), which enables the simulation results with much fewer time steps (26,616 and 26,546 respectively). It is also observed that both CCPD and variable-parameter VBR models take significantly larger average CPU time per-step (725 µs and 682 µs, respectively). This is due to the fact that these two models have variable-parameter interfacing inductances. However, the total required CPU times (19.3 s and 18.1 s, respectively) are still smaller than that for the *qd*310 model.

Furthermore, it is seen in Table II that the proposed CPVBR model has a larger eigenvalue ($-8{,}745$ which is due to the filter approximation) compared to the CCPD and variable-parameter VBR models, and as a result it requires 45,156 time steps. However, it is noted that the constant-parameter interfacing in the CPVBR model significantly reduces the complexity and the required CPU time per step (140 µs). Therefore, this model yields the smallest total required CPU time (6.3 s) resulting in the fastest simulation. Considering the small numerical error (less than 0.3%) shown in Table I, it can be concluded that the proposed CPVBR model possess an advantageous combination of simulation performance and numerical accuracy among the subject models.

IV. Conclusion

In this paper, the VBR model has been proposed for the first time for five-phase wound-rotor synchronous machines considering the rotor saliency, air-gap flux harmonics up to third component, as well as their cross-couplings with the fundamental component. Both variable-parameter VBR and constant-parameter CPVBR models have been derived. The proposed models have been validated experimentally on a laboratory prototype of a five-phase generator-rectifier system and are shown to predict the machine's behavior under various operating conditions with good accuracy.

The CPVBR model possesses a simple and easy-to-implement interfacing circuit consisting of constant-parameter *RL* branches, which is achieved using approximation. Rigorous computer simulations have been conducted to validate the accuracy and computational performance of the CPVBR model against the existing conventional models of five-phase synchronous machines. It is demonstrated that the proposed CPVBR model is able to provide accurate results, while significantly outperforming conventional *qd*0, CCPD, and VBR models in terms of numerical efficiency and simulation speed.

It is envisioned that the proposed CPVBR model can be effectively utilized in many commonly-used state-variable-based transient simulation packages for design, study and control of five-phase synchronous machine-converter systems.

Appendix

Five-phase machine:

TABLE III
Laboratory Five-Phase Synchronous Machine Parameters

Machine parameters measured at i_{fd} = 16 A (referred to stator) (determined using locked-rotor measurements over 360 electrical degrees of rotor rotation)	
Parameters	**Value**
L_{mq1}, L_{md1} , L_{mq3}, L_{md3}	0.4 mH , 1.35 mH , 90 µH , 54 µH
L_{m13} , L_{mqd3} , L_{mqd13}	–84 µH , –6.1 µH , 13.2 µH
L_{ls} , L_{lfd}	150 µH , 120 µH
r_s , r_{fd}	0.19 Ω , 0.033 Ω

Line resistance:

$$r_{line} = 0.04\,\Omega.$$

Ten-pulse diode rectifier:
Harris semiconductor: RHRG7560,
$Ron = 0.091\ \Omega$, $Von = 0.637$ V.
Rectifier dc filter:

$$L_f = 12\,\mathrm{mH}, r_f = 0.6\,\Omega, C_f = 470\,\mu\mathrm{F}.$$

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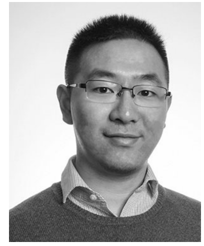

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